

CHAPTER 1

AIRCRAFT GENERAL

CHAPTER OVERVIEW

SECTION	TITLE
1-1	General Information
1-2	Equipment Description and Data
1-3	Servicing
1-4	Cleaning
1-5	Lubrication
1-6	Ground Handling
1-7	Periodic/Special Inspections
1-8	Retirement Schedule

SECTION 1-1

GENERAL INFORMATION

SECTION OVERVIEW

PARAGRAPH	TITLE
1-1-1	General

1-1-1 GENERAL.

1-1-1.1 Scope.

This manual contains complete description information and maintenance procedures.

1-1-1.2 Maintenance Forms, Records, and Reports.

Forms, records, and reports are to be used per local command directives.

1-1-1.3 Preparation for Storage or Shipment.

Refer to Appendix E for instructions on preparation for storage or shipment.

1-1-1.4 Connection of Electric Connectors.

- a. Self-locking threaded connectors, MIL-C-5015 and MIL-DTL-83723, will not connect with receptacles unless properly torqued. Coupling rings must be hand-torqued until tabs on locking devices align with center of groove.
- b. Bayonet coupling connectors, MIL-C-26482 and MIL-C-38999, will not connect with receptacles unless properly torqued. Coupling rings must be hand-torqued until bayonet pins can be seen, heard, and felt snapping into holes of coupling rings.
- c. Threaded connectors not using a locking mechanism will not connect with receptacles unless properly torqued. These connectors are identified by a 1/2-inch red dot located on the structure adjacent to connector receptacles. Connectors of this type must be hand-torqued and lockwired.

1-1-1.5 Routing and Clamping of Wire Harnesses.

Upon installation of components, such as left and right relay panels, No. 3 relay panel, and No. 1 and

No. 2 junction boxes, check for adequate clearance between these components and wire harnesses routed close to them. All wiring should be checked for proper clamping and for installation of sufficient harness ties to avoid wire chafing.

1-1-1.6 Airworthiness of Components.

WARNING

Injury to personnel and damage to equipment will result if components from helicopters involved in crashes/accidents are installed on helicopter. Components from helicopters involved in crashes/accidents may be offered for sale and use on airworthy helicopters. Please be advised, when a part is subjected to stresses, thermal effects, or damage outside of normal operating parameters, some adverse change in that part may occur. Such change may not be visually detectable and may therefore require destructive testing to determine airworthiness. Unless stresses, thermal effects, and other damage to which parts are subjected as a result of a crash/accident can be evaluated and determined to be within design parameters, any part that may affect safe operation of the helicopters shall be assumed to be damaged beyond acceptable limits and shall not be reused on airworthy helicopters. It is necessary that the integrity of crash-damaged parts be assessed by the operator before flight.

SECTION 1-2

EQUIPMENT DESCRIPTION AND DATA

SECTION OVERVIEW

PARAGRAPH	TITLE
1-2-1	Aircraft General
1-2-2	Flight Safety Critical Aircraft Parts (FSCAP) Program

1-2-1/(1-2-2 Blank)

1-2-1 AIRCRAFT GENERAL.

1-2-1.1 Aircraft Description.

- a. Helicopters are twin turboshaft engine aircraft. The helicopters are powered by two T700-GE-700, T700-GE-701A, or T700-GE-701C engines mounted above the mid-fuselage. Their primary mission is the transportation of passengers, litter patients, and equipment. The crew consists of a pilot, copilot, and crew chief.

The main rotor group consists of a four-bladed, fully-articulated, elastomeric rotor. The tail rotor group consists of a canted cross-beam tail rotor with two continuous composite spars running from blade tip to blade tip, crossing each other at the hub to form the four tail rotor blades.

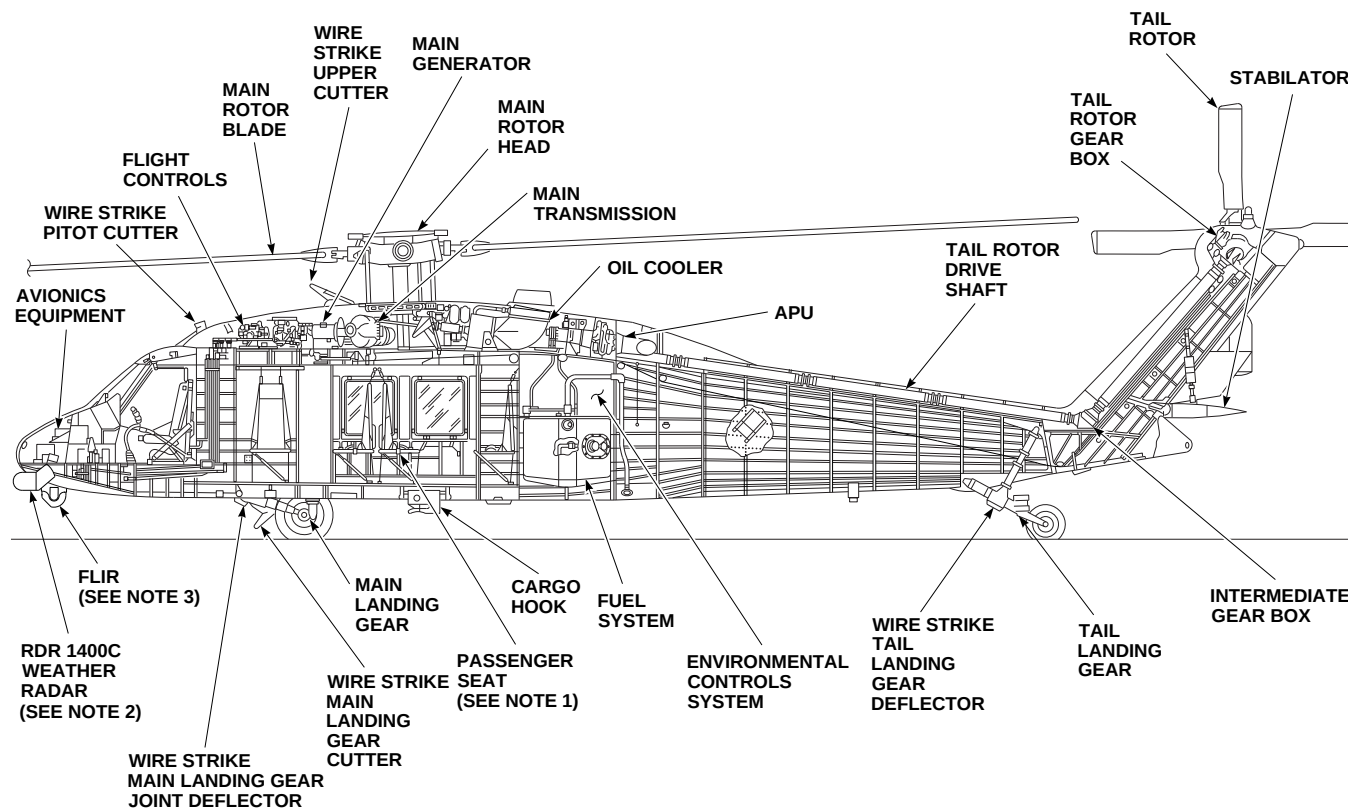
Forward, rear, lateral, and vertical flight is done by the main rotor system, while the tail rotor system counteracts torque from the main rotor and provides directional control. Power to drive the main rotor is supplied from engine torque transmitted by drive shafting to the input module of the main transmission. The tail rotor is driven by drive shafting extending from the main module of the main transmission through the intermediate gear box to the tail gear box.

Three separate hydraulic systems are used in the helicopter. The No. 1 and No. 2 hydraulic systems provide power for the main rotor servos and the pilot-assist servos. The No. 3 (or backup) hydraulic system provides backup power for the No. 1 and No. 2 hydraulic systems and recharges the auxiliary power unit (APU) start subsystem. The No. 3 hydraulic system also provides power for ground checks without operation of the main transmission.

Basic electrical power is supplied by two AC generators mounted on the accessory module. AC power is converted to DC power for operation of certain systems.

- b. Kit installations for helicopter consist of external stores support system (ESSS), rescue hoist, medical evacuation, blade deicing, engine air particle separator (EAPS), blade erosion protection, winterization, blackout devices, helicopter infrared suppression system (HIRSS), and range extension tanks.
- c. **EMEP** Electrical wiring and components are hardened for electromagnetic environment protection (EMEP) for eliminating abnormal responses to external electromagnetic radiation. Electrical wiring for selected systems and components of EMEP helicopters are equipped with pin filter adapters, connectors, and receptacles where necessary. Components susceptible to electromagnetic radiation are internally modified with filters and gaskets. The instrument panel and other mounting surfaces for EMEP components are metal with a chemical conversion coating conforming to MIL-C-5541, Class 3, or use bonding straps to provide low resistance to airframe ground. The following list identifies those components having pin filter adapters and pin filter connectors:
 - (1) Blade Deice Test Panel (pin filter adapter).
 - (2) Blade Deice Controller (pin filter adapter).
 - (3) Copilot's HSI (pin filter adapter).
 - (4) Pilot's HSI (pin filter adapter).
 - (5) SAS/FPS Computer (pin filter adapter).
 - (6) Rate Gyro (pin filter adapter).
 - (7) No. 1 Generator Control Unit (pin filter adapter).
 - (8) No. 2 Generator Control Unit (pin filter adapter).
 - (9) Copilot's VSI (pin filter adapter).
 - (10) Pilot's VSI (pin filter adapter).

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|--|--|
| (11) Caution/Advisory Panel (pin filter adapter). | (23) No. 1 Signal Data Converter (pin filter adapter). |
| (12) APU Generator Control Unit (pin filter adapter). | (24) No. 2 Signal Data Converter (pin filter adapter). |
| (13) No. 1 Lateral Accelerometer (pin filter adapter). | (25) No. 1 Stabilator Actuator (pin filter adapter). |
| (14) No. 2 Lateral Accelerometer (pin filter adapter). | (26) No. 2 Stabilator Actuator (pin filter adapter). |
| (15) 70583-702374 Battery Conditioner/Analyzer (pin filter adapter). ◀ | (27) Roll Trim Servo (pin filter adapter). |
| (16) Pilot's HSI/VSI Mode Select Panel (pin filter adapter). | (28) No. 1 Stabilator Amplifier (pin filter adapter). |
| (17) Copilot's HSI/VSI Mode Select Panel (pin filter adapter). | (29) No. 2 Stabilator Amplifier (pin filter adapter). |
| (18) Yaw Trim Servo (pin filter connector). | (30) VOR/ILS Receiver (pin filter adapter). |
| (19) Pilot's Master Warning Panel (pin filter connector). | (31) CIS Processor (pin filter adapter). |
| (20) Copilot's Master Warning Panel (pin filter connector). | (32) Air Data Transducer (pin filter adapter). |
| (21) Radar Altimeter Receiver/Transmitter (pin filter adapter). | (33) Compass Control Panel (pin filter adapter). |
| (22) Airspeed Transducer (pin filter connector). | (34) Doppler Signal Data Converter (pin filter adapter). |
| | (35) Doppler Computer Display Unit (pin filter adapter). |



NOTES

1. PASSENGER SEATS CAN BE REMOVED TO ACCOMMODATE LITTER KIT INSTALLATIONS.
2. WEATHER RADAR NOT INSTALLED ON ALL HELICOPTERS.
3. FLIR NOT INSTALLED ON ALL HELICOPTERS.
4. **S70-I**
ENGINES ARE T700-GE-700, 701A.
5. **S70-II**
ENGINES ARE T700-GE-701C.
6. AUXILIARY SEARCHLIGHT NOT INSTALLED ON ALL HELICOPTERS.

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FIGURE 1-2-1-1. GENERAL ARRANGEMENT (SHEET 1 OF 2)

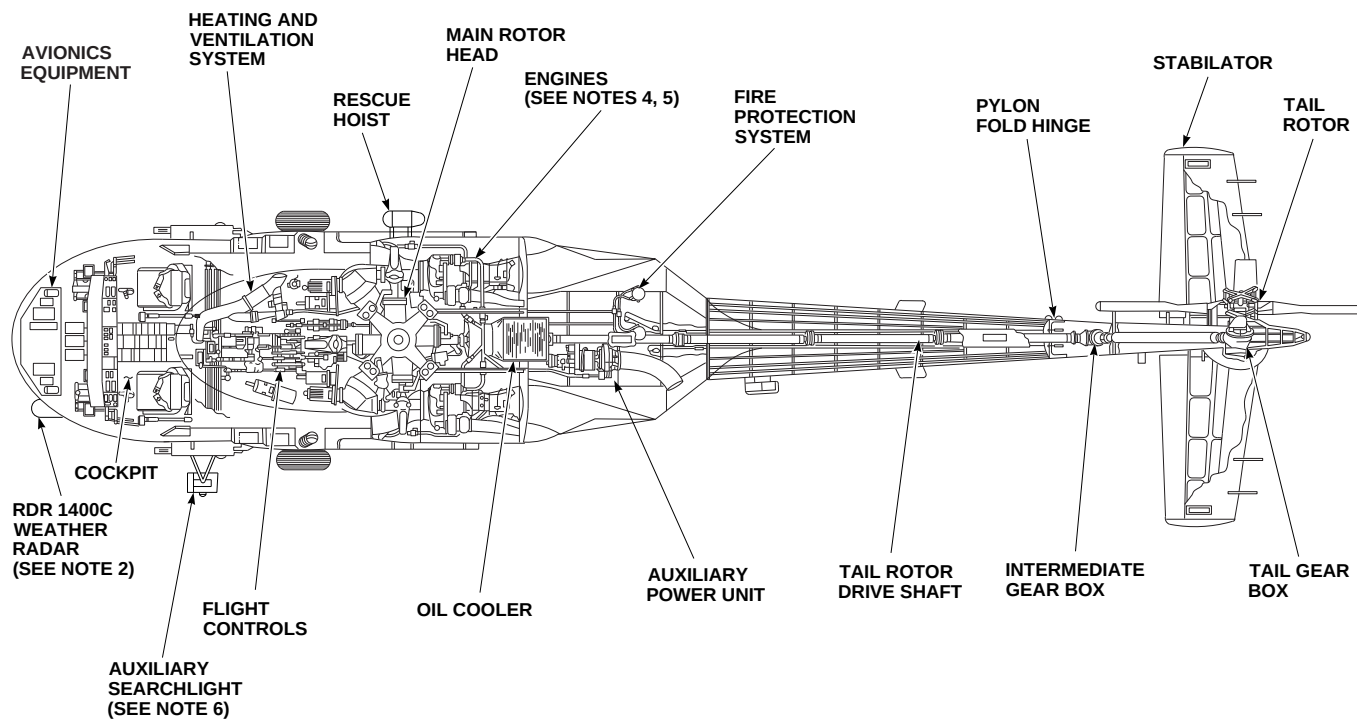
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FIGURE 1-2-1-1. GENERAL ARRANGEMENT (SHEET 2 OF 2)

1-2-1.2 Principal Dimensions.

- a. Principal dimensions of the helicopter are shown in Figure 1-2-1-2.
- b. Stations, waterlines, and buttlines (in inches) are used as an accurate method of locating or removing/installing equipment in the airframe. See Figure 1-2-1-3 for stations, waterlines, and buttlines for this helicopter.

1-2-1.3 Access and Inspection Provisions.

Access and inspection provisions consist of access doors, covers, panels, platforms, screens, and openings used for maintenance, inspection, and servicing of the helicopter and its components. Principal access and inspection openings are shown in Figure 1-2-1-4 and Table 1-2-1-1.

Table 1-2-1-1 Access Panels and Fairing

PANEL NO. (See Note 2)	TYPE OF PANEL	SIZE (INCHES)	TYPE OF FASTENERS	ACCESS TO
1B-2B-1	Nose Door (Hinged)	37 ¼ x 70	2 Latches	Electronics
4B-26	Access Panel	6 x 7	1 Latch	External Power
3T-3 4T-4	Access Panel (Hinged)	13 ½ x 33 ½	2 Latches 12 Screws	Shock Strut
3T-3 4T-4	Access Panel (Hinged)	13 ½ x 33 ½	2 Latches 12 Screws	Shock Strut
3B-30 4B-31	Access Panel (2 Parts)	23 x 25	18 Screws	Shock Strut
3T-4T-2	Sliding Cover (On Tracks)	64 x 81	2 Camlocs 12 Dzus	Controls/Accessories
3T-7 4T-8	Engine Cowling (Hinged)	26 x 50	2 Latches	Engine
3T-5 4T-6	Access Panel	6 x 15	4 Screws	Anti-Icing Valve
4B-24	Access Cover			Cargo Hook
3T-25	Access Panel (Hinged)	14 x 22	2 Latches	Oil Cooler
3T-27	Access Panel	12 x 40	2 Latches	APU
4T-28	Access Panel (Hinged)	12 x 40	2 Latches	Fire Extinguishers
3T-22 4T-22	Access Panel (Hinged)	8 x 10	1 Latches	Gravity Fuel
3T-3B-23	Access Panel (Hinged)	12 x 26	2 Latches	Pressure Fuel
3T-3B-21	Access Panel (Hinged)	7 x 10	Latch	Pneumatic Ground Start
3B-3T-20	Access Panel	7 x 10	10 Screws	Magnetic Flux Valve

Table 1-2-1-1 Access Panels and Fairing (Cont)

PANEL NO. (See Note 2)	TYPE OF PANEL	SIZE (INCHES)	TYPE OF FASTENERS	ACCESS TO
5T-6T-9	Access Panel (Hinged)	9 x 80	4 Dzus	Drive Shaft
5T-6T-10	Access Panel (Hinged)	9 x 79	4 Dzus	Drive Shaft
5T-6T-17	Intermediate Gear Box Fairing	19 x 32	12 Screws	Intermediate Gear Box
5T-6T-11	Pylon Drive Shaft Cover/# 1 FM Antenna	14 x 79	9 Dzus	Drive Shaft
5T-5B-18 6T-6B-32	Access Panel		12 Screws	Pylon Fold Hinge Bolts
5T-6T-12	Tail Gear Box Fairing		1 Flat-Tip Fastener 11 Phillips-Head	Tail Gear Box
5T-6T-15	Fairing	20 x 23	26 Screws	Pylon Trailing Edge
5B-6B-16	Fairing		16 Screws 4 Bolts	Lower Pylon
5B-45 6B-46	Cover	4 x 7	4 Screws	Stabilator Attachment Fittings
5B-5T-19 6B-6T-19	VOR/LOC Antenna		4 Screws	
5T-6T-13	Portable VHF FM Antenna		31 Screws	
4B-33	Access Panel (Hinged)	20 x 27	2 Latches	Transition Avionics Compartment
3T-33 4T-34	Fairing Platform (2-Part Hinged) (See Note 1)		6 Camlocs 19 Screws	ESSS Maintenance Crane Attachment

Table 1-2-1-1 Access Panels and Fairing (Cont)

PANEL NO. (See Note 2)	TYPE OF PANEL	SIZE (INCHES)	TYPE OF FASTENERS	ACCESS TO
3T-35 4T-36	Cap Fairing (See Note 1)		14 Camlocs	ESSS
3T-3B-37 4T-4B-38	Access Panel (Hinged) (See Note 1)		3 Camlocs 7 Screws	Step/Shock Strut
3B-39 4B-40	Access Panel (See Note 1)		14 Screws	Shock Strut Landing Gear
3B-41	Lower Fairing STA 295.0 (See Note 1)		6 Camlocs	ESSS
3B-42	Lower Fairing STA 308.0 (See Note 1)		4 Camlocs	ESSS
4B-43	Lower Fairing STA 295.0 (See Note 1)		6 Camlocs	ESSS
4B-44	Lower Fairing STA 308.0 (See Note 1)		4 Camlocs	ESSS
4T-47	Access Panel (Hinged)		2 Latches	Oil Cooler

NOTES

1. **ESSS**
2. Number codes for panels and fairings:
 - 1 - left side cockpit
 - 2 - right side cockpit
 - 3 - left side cabin
 - 4 - right side cabin
 - 5 - left side tail cone and pylon

Table 1-2-1-1 Access Panels and Fairing (Cont)

6 - right side tail cone and pylon

T - Top above WL 232.0

B - Bottom below WL 232.0

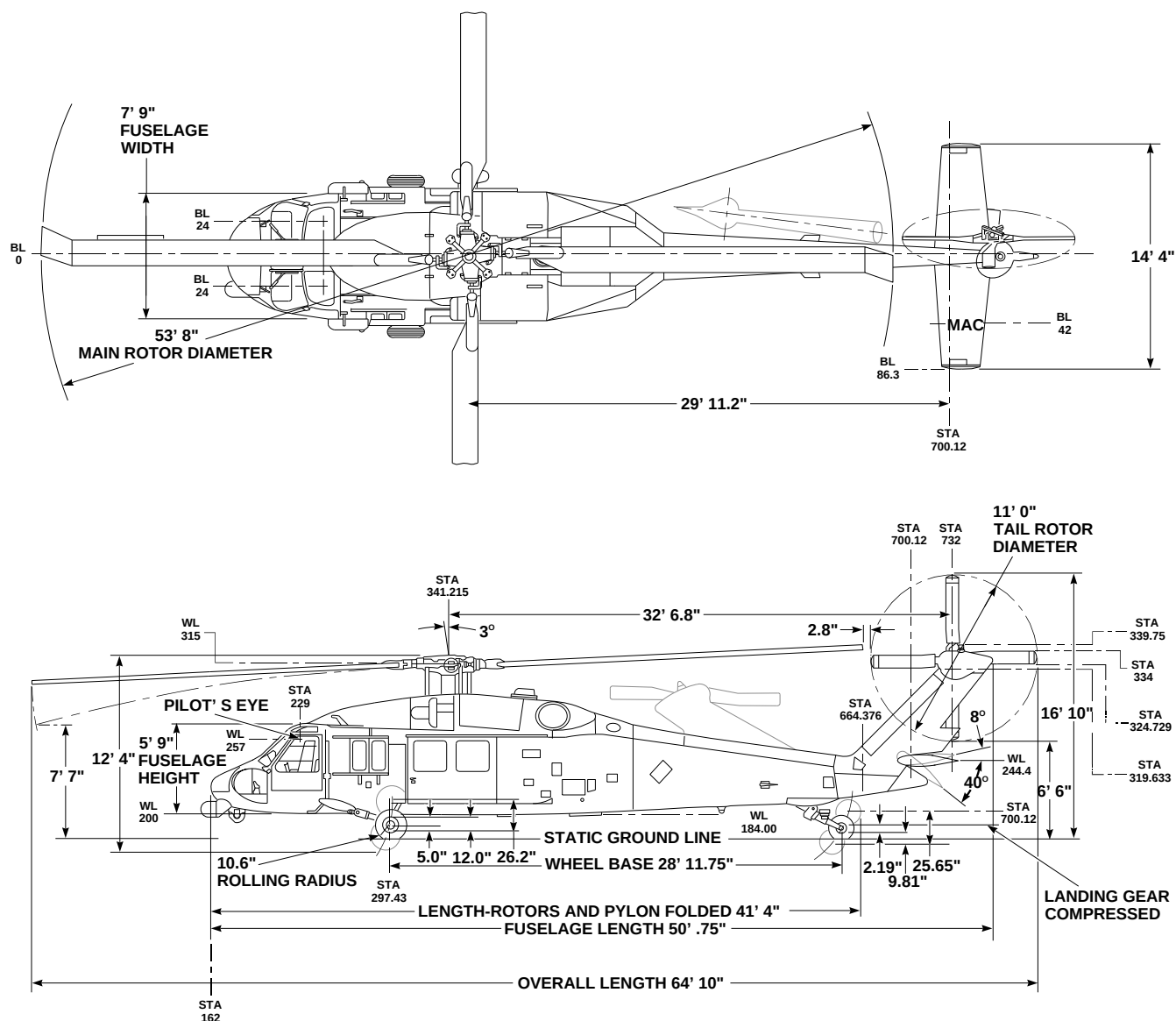
**1-2-1.4 Steps, Handholds, and Walkways/
Work Areas.**

Steps, handholds, and walkways that aid in inspection and maintenance of the helicopter are shown in Figure 1-2-1-5. Work areas are shown in Figure 1-2-1-6.

roundings when working on or around helicopter. Danger areas while rotors are turning are shown in Figure 1-2-1-7.

**1-2-1.5 Danger Zones While Rotors are
Turning.**

Main and tail rotors pose great danger while turning. Personnel must always be mindful of their sur-



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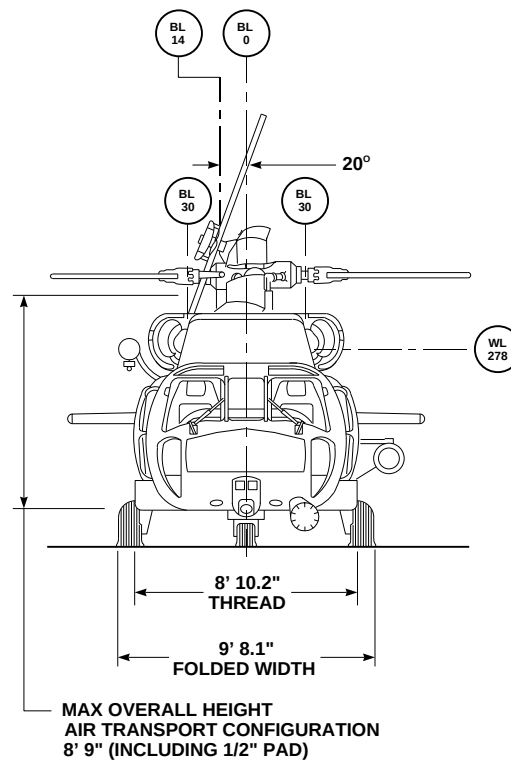
NOTE

WEATHER RADAR AND FLIR NOT
INSTALLED ON ALL HELICOPTERS.

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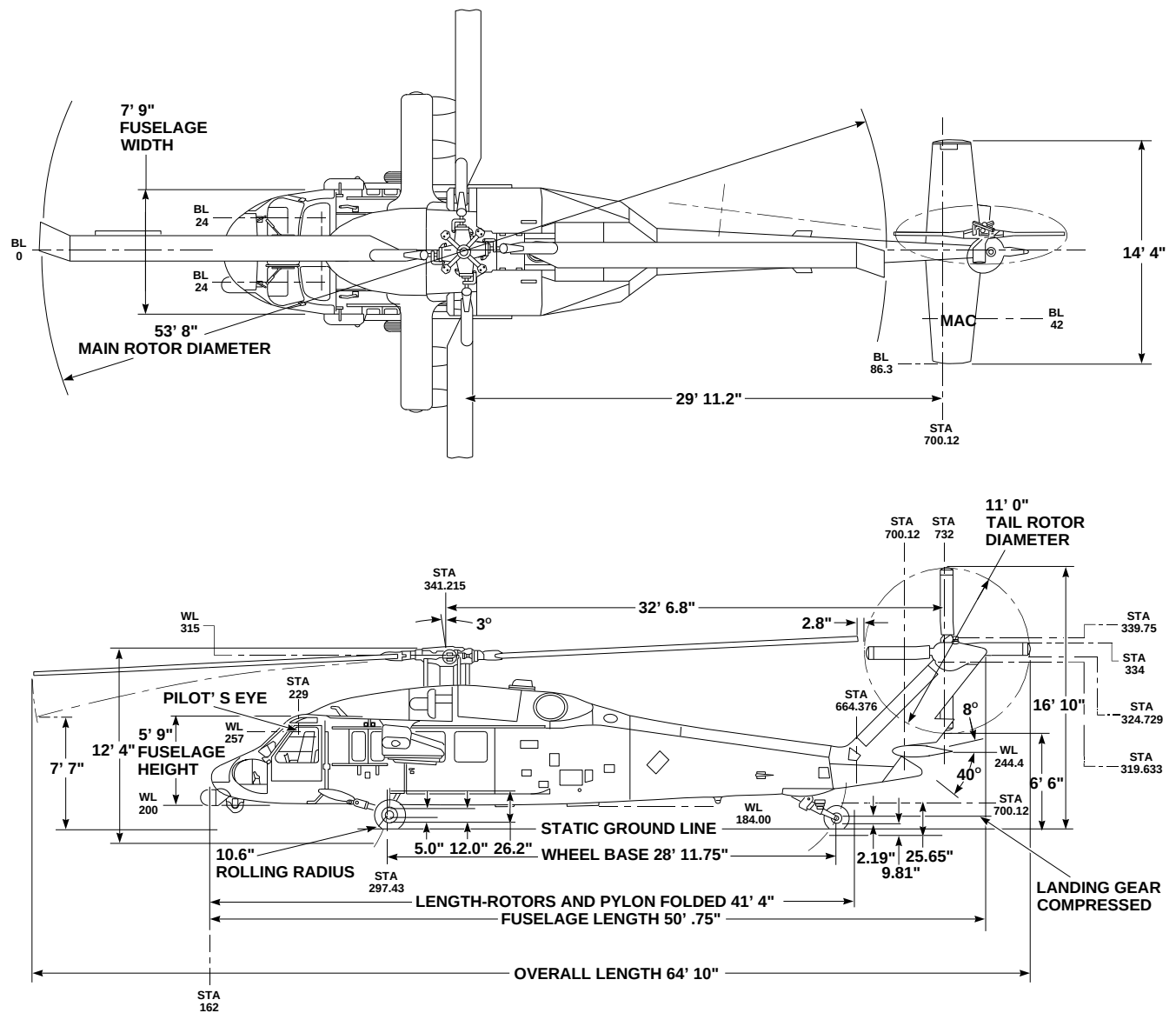
FIGURE 1-2-1-2. PRINCIPAL DIMENSIONS (SHEET 1 OF 4)

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FIGURE 1-2-1-2. PRINCIPAL DIMENSIONS (SHEET 2 OF 4)



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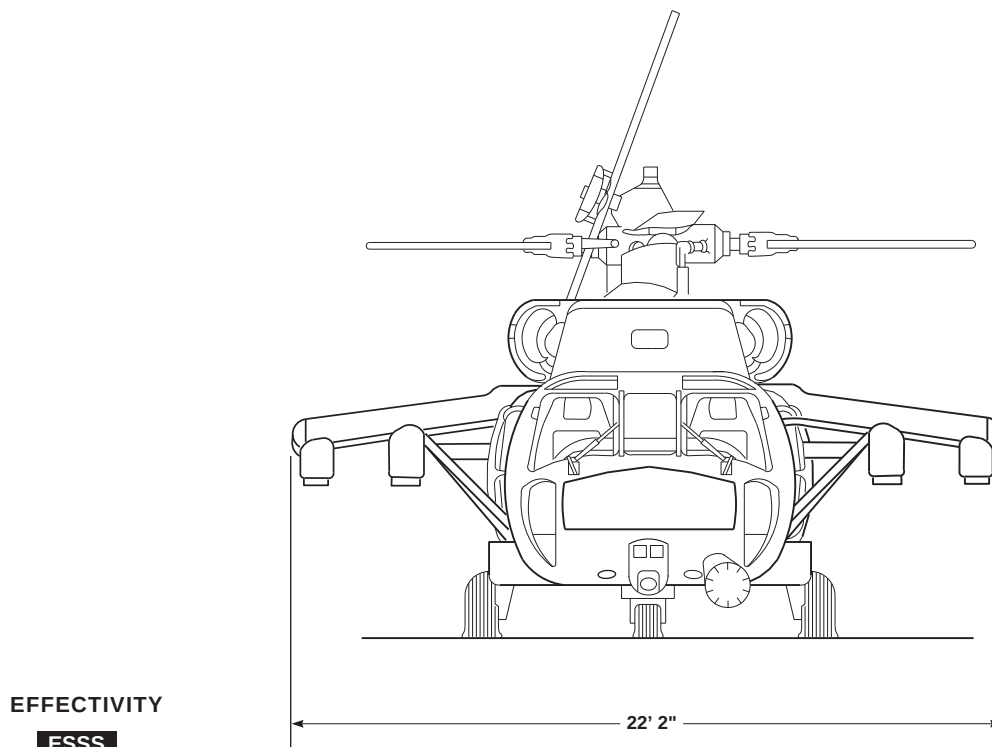
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FIGURE 1-2-1-2. PRINCIPAL DIMENSIONS (SHEET 3 OF 4)

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FIGURE 1-2-1-2. PRINCIPAL DIMENSIONS (SHEET 4 OF 4)

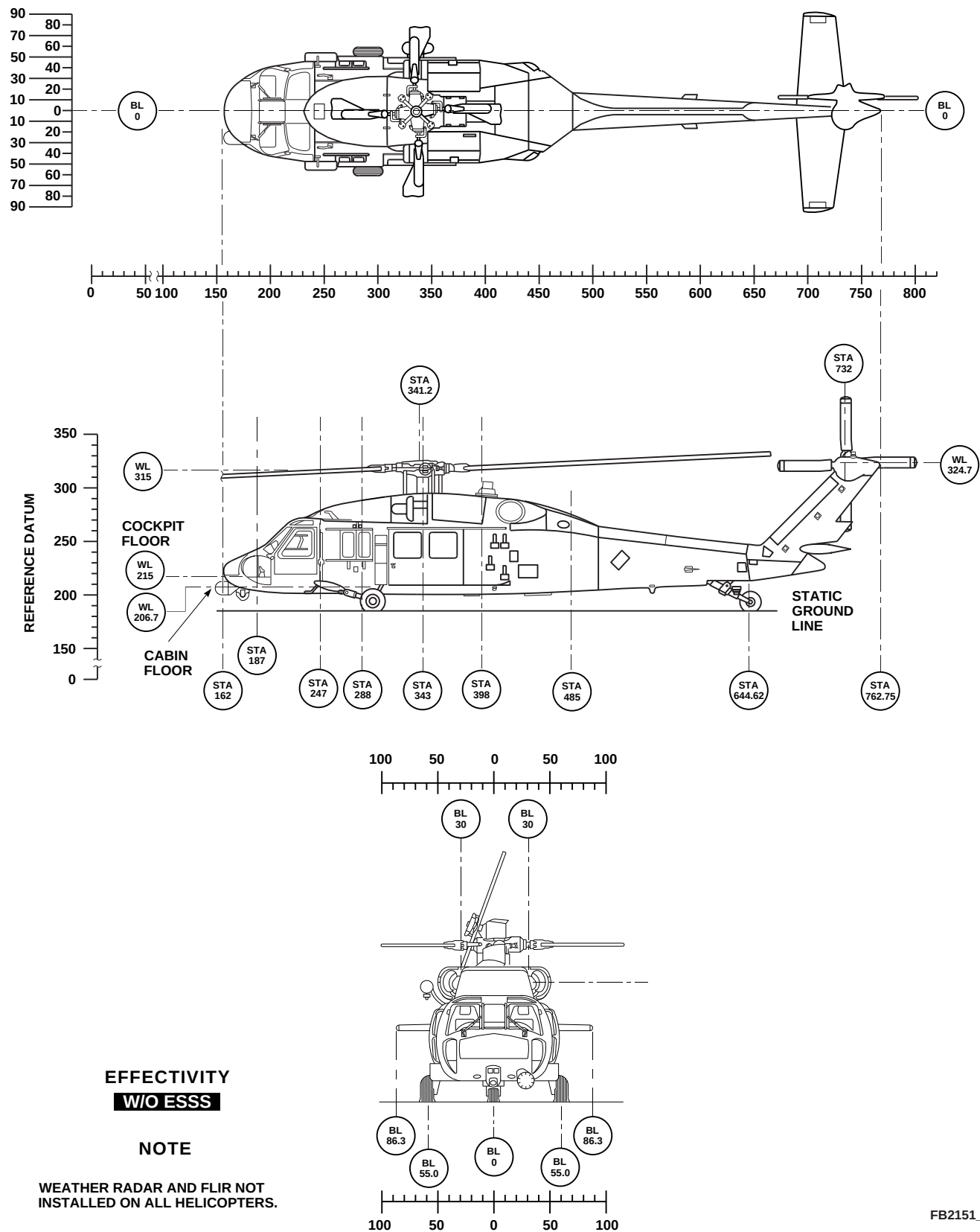


FIGURE 1-2-1-3. STATIONS, WATERLINES, AND BUTTLINES (SHEET 1 OF 2)

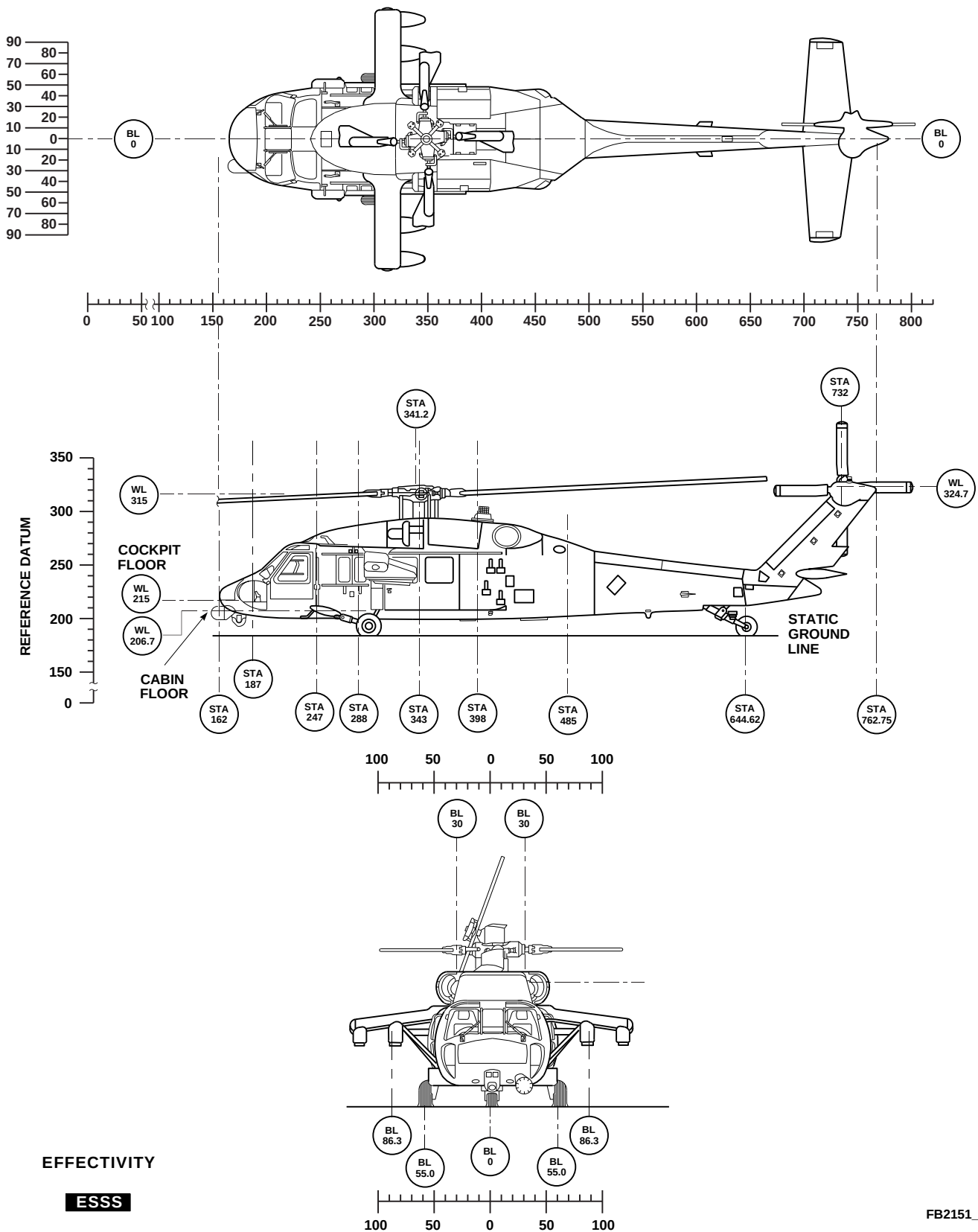


FIGURE 1-2-1-3. STATIONS, WATERLINES, AND BUTTLINES (SHEET 2 OF 2)

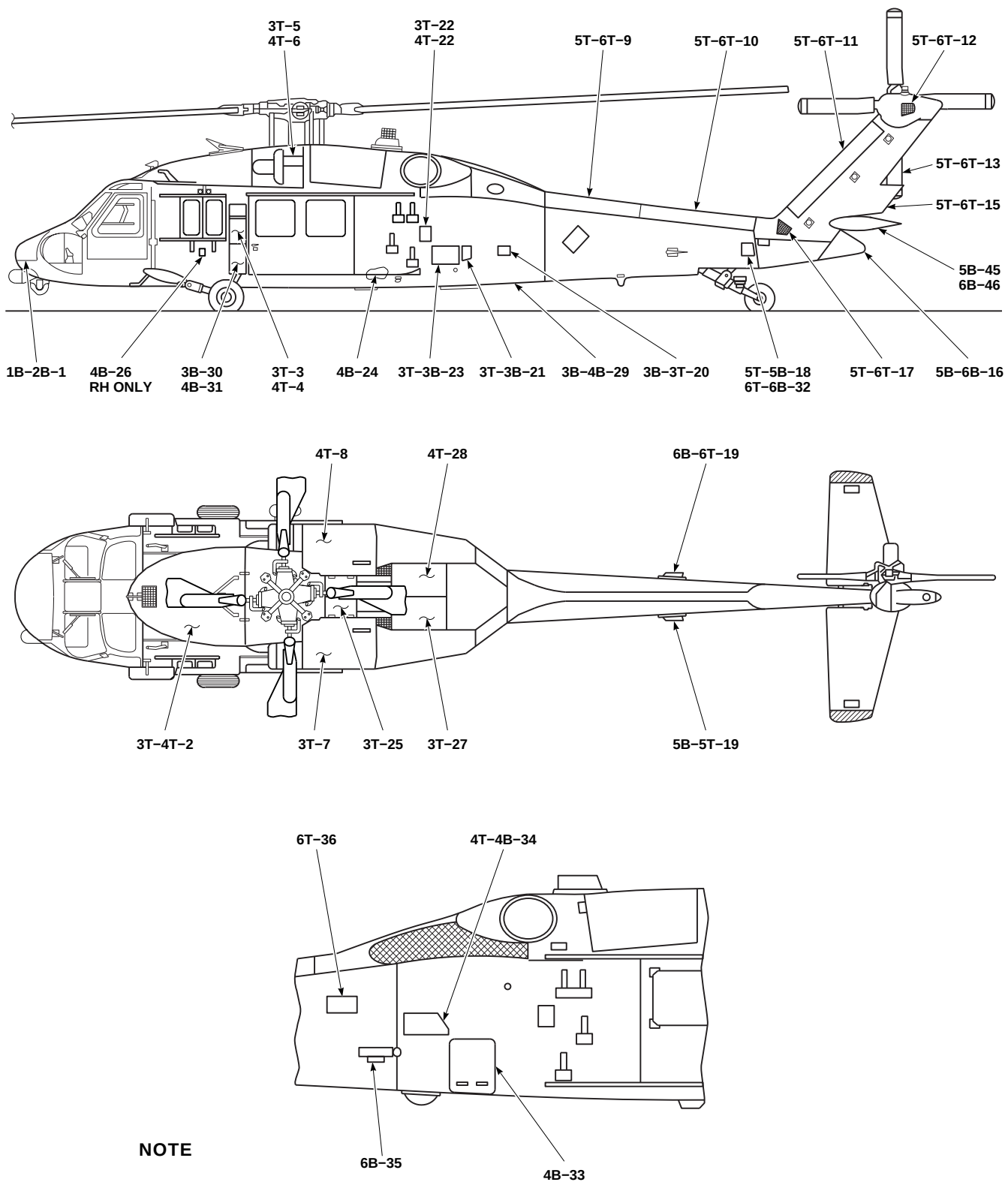


FIGURE 1-2-1-4. ACCESS AND INSPECTION PROVISIONS (SHEET 1 OF 2)

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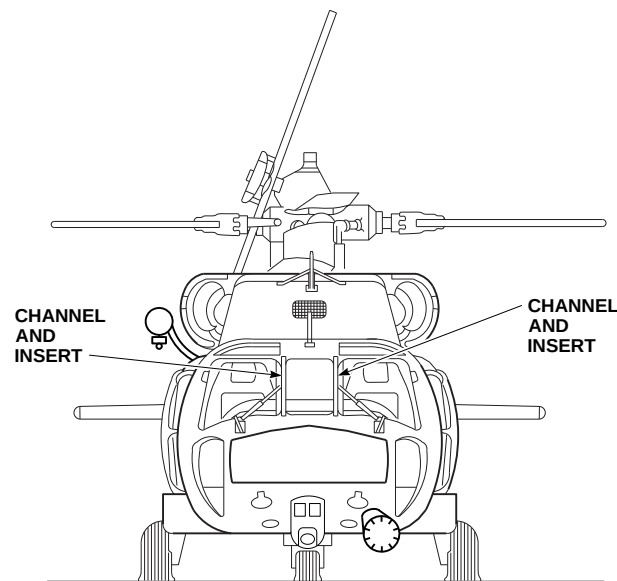
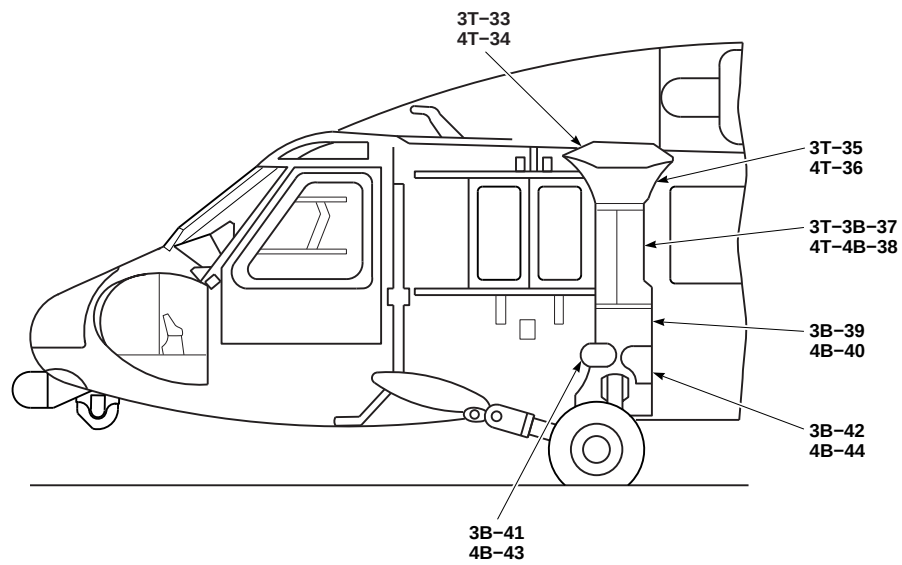
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FIGURE 1-2-1-4. ACCESS AND INSPECTION PROVISIONS (SHEET 2 OF 2)

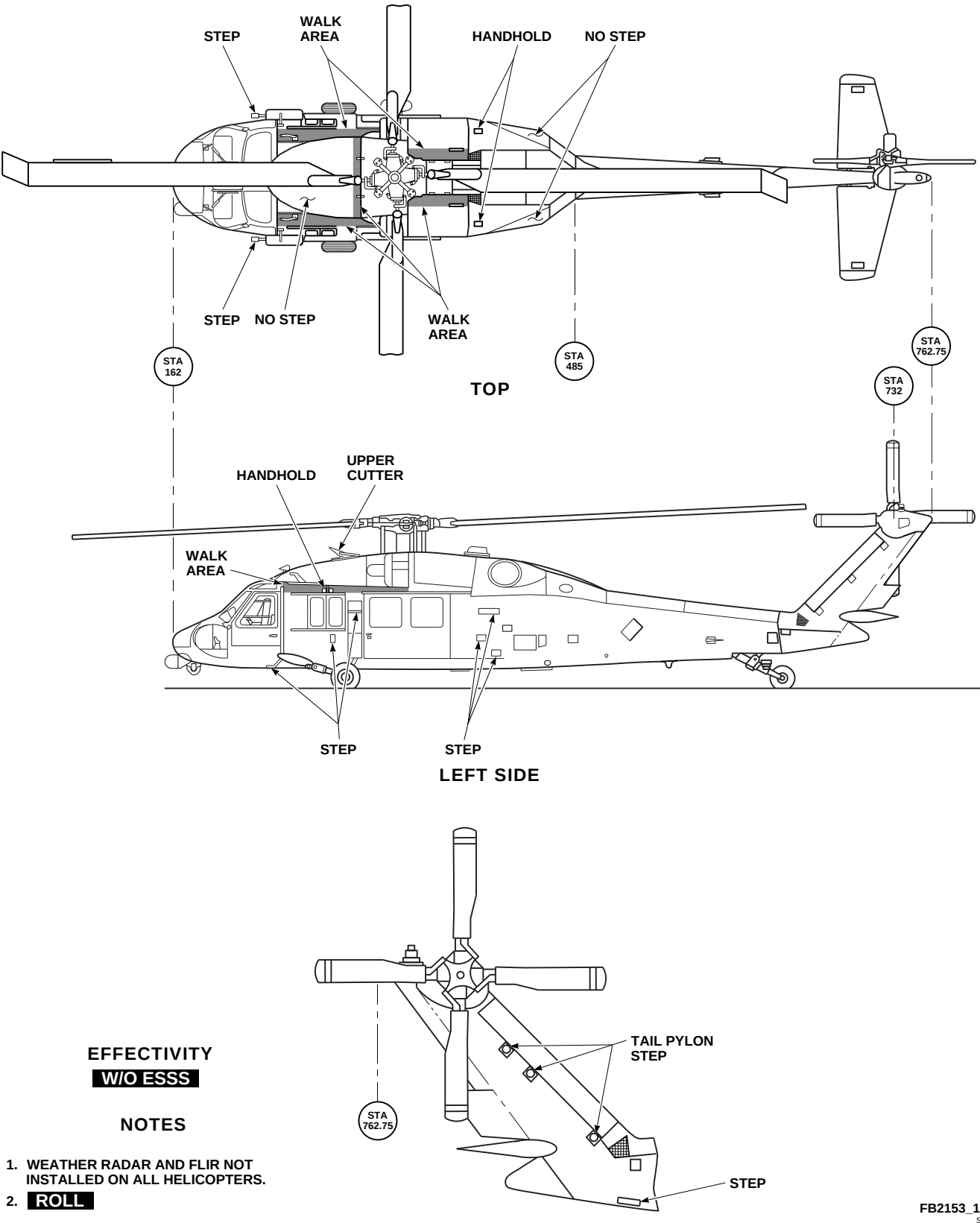
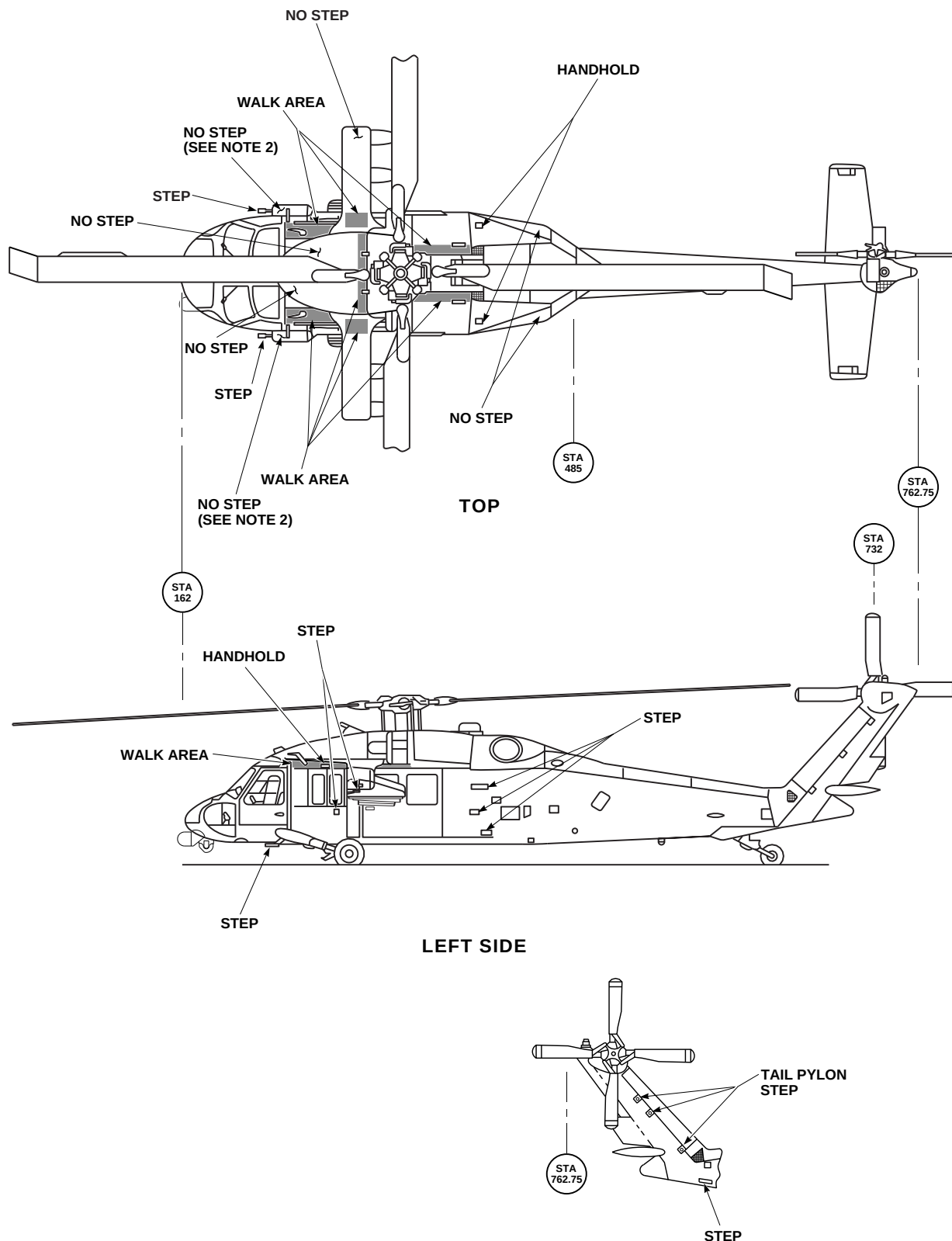


FIGURE 1-2-1-5. STEPS, HANDHOLDS, AND WALKWAYS (SHEET 1 OF 2)

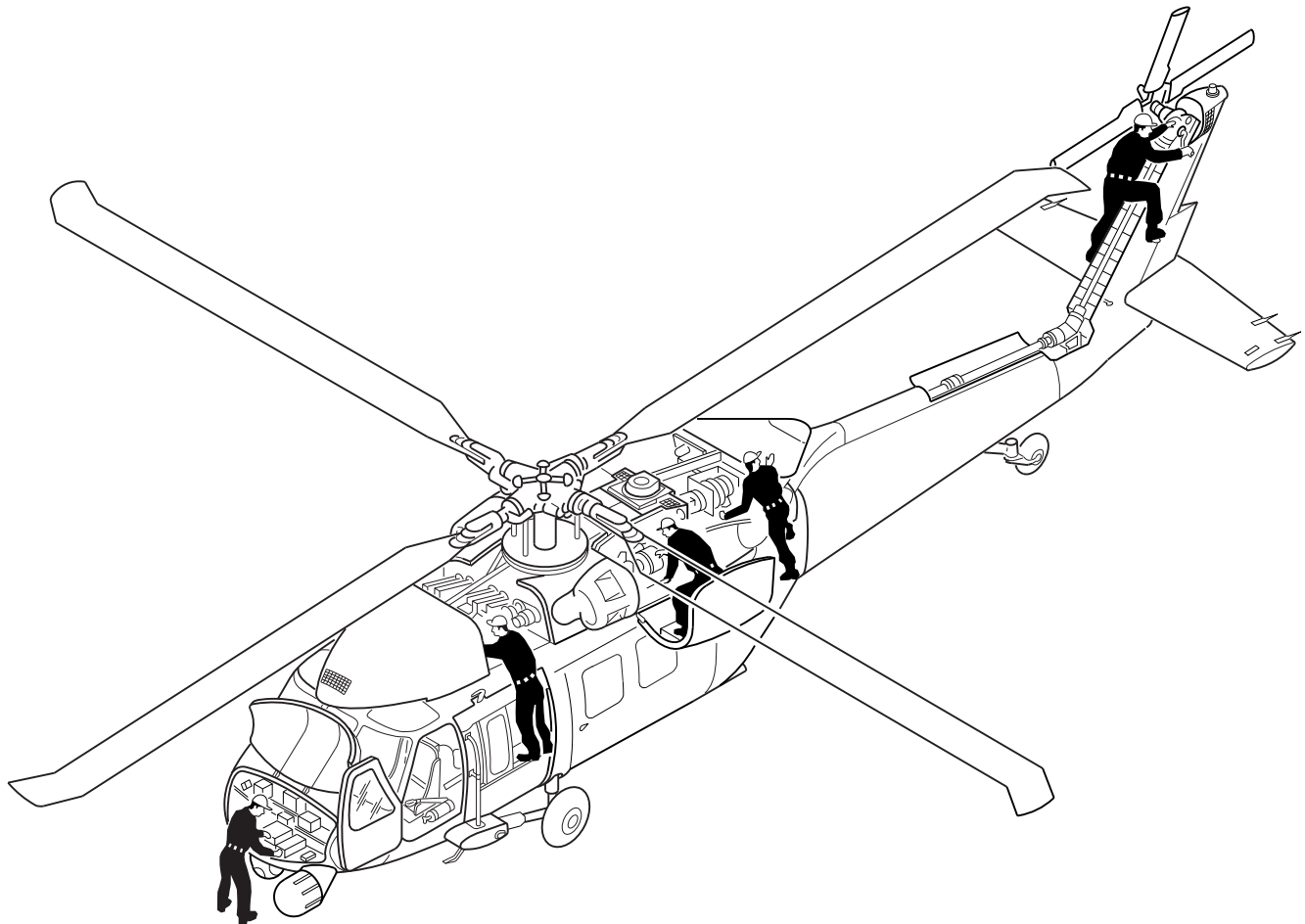


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FIGURE 1-2-1-5. STEPS, HANDHOLDS, AND WALKWAYS (SHEET 2 OF 2)

1-2-1-18 Change 9



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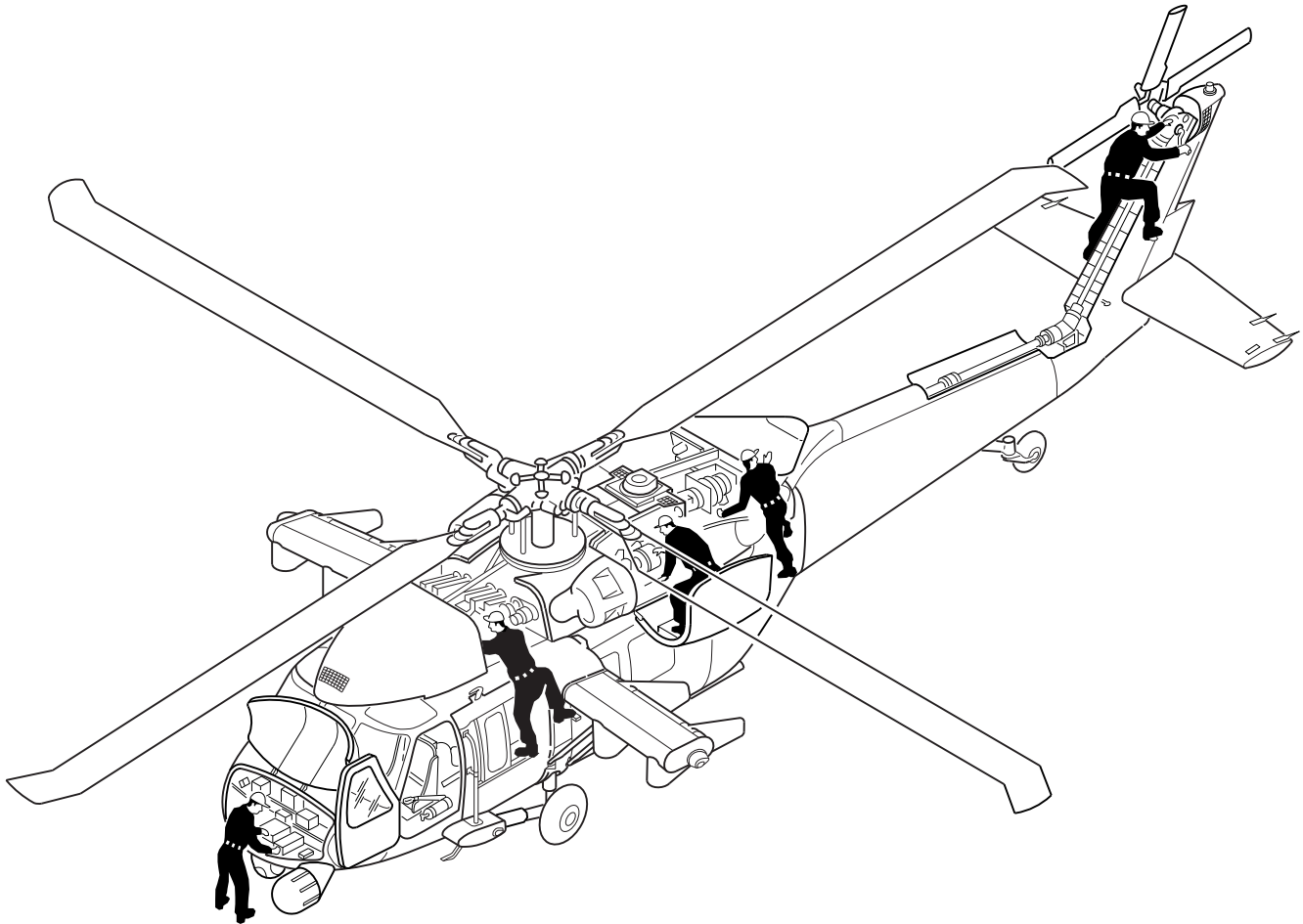
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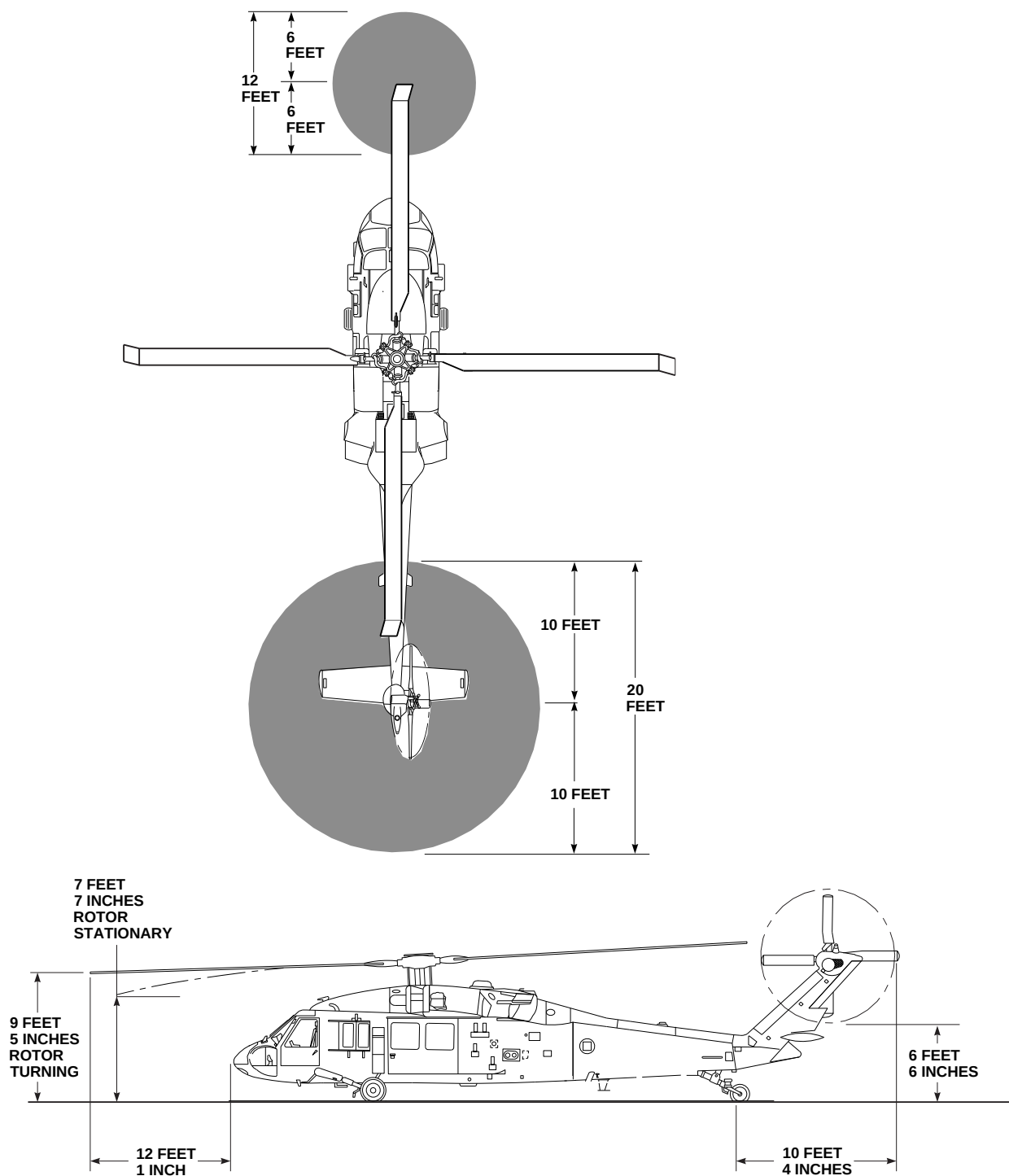
NOTE

WEATHER RADAR AND FLIR NOT
INSTALLED ON ALL HELICOPTERS.

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FIGURE 1-2-1-6. WORK AREAS (SHEET 1 OF 2)

**EFFECTIVITY****ESSS**FB2154_2A
SA**FIGURE 1-2-1-6. WORK AREAS (SHEET 2 OF 2)**



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FIGURE 1-2-1-7. DANGER ZONES WHILE ROTORS ARE TURNING

1-2-2 FLIGHT SAFETY CRITICAL AIRCRAFT PARTS (FSCAP) PROGRAM.

1-2-2.1 Flight Safety Critical Aircraft Parts (FSCAP) Program.

- a. A flight safety part (FSP) or a flight safety critical aircraft part (FSCAP) is defined as a part, assembly, or installation procedure with one or more critical characteristics that, if not conforming to the design data or quality requirements, could result in the serious injury or death of crew members and/or serious damage to the helicopter. It should be noted that references to the FSP or FSCAP are interchangeable. The FSP program is the predecessor of the FSCAP program.
- b. A critical characteristic is any dimension, tolerance, finish, material manufacturing, assembly, or inspection process, or other feature which, if nonconforming or missing, could cause failure or malfunction of the FSCAP.
- c. Critical characteristics produced during the manufacturing process are termed "manufacturing critical characteristics". Critical characteristics not introduced during manufacturing but critical in terms of assembly/installation (e.g. proper torque) are termed "installation critical characteristics".
- d. Table 1-2-2-1 identifies parts, assemblies, or installations under the FSCAP program requiring special handling during maintenance. Throughout the maintenance procedures warnings appear emphasizing critical instructions to be followed. These warnings are identified as "FSCAP" warnings and are inserted whenever and wherever necessary.
- e. Torque stripes are used to identify all fasteners with critical characteristic torque or installation procedures. Torque stripes are a painted on stripe that is applied to either end of a fastener. Torque stripes are similar to slippage marks, however, they are not used to show relative motion of fasteners. Refer to PARA 2-6-5 for the correct application of torque stripes.

Table 1-2-2-1. Flight Safety Critical Aircraft Parts

COMPONENT	PART NUMBER
Main Rotor Spindle	38023-10371-101
Main Rotor Spindle and Liner Assy	38023-10374-041, -042
Horizontal Stabilator Instl	70000-07050-011
Rotor Head/Transmission Instl	70000-08000-016, -017, -018
Rotor Head/Transmission Assy	70000-08000-042, -043, -044
Engine Instl	70000-10100-011, -012, -013, -014, -017, -018, -021, -022
Engine Instl	70000-10101-011, -012, -013, -014, -015, -016
Tail Rotor Instl	70000-11100-011, -012, -013, -014, -015
Main Rotor Spindle Assy	70070-10030-041, -042, -043, -044, -045, -046, -047
Slotted Oil Cooler Sleeve, Mod Kit	70070-35139-011

Table 1-2-2-1. Flight Safety Critical Aircraft Parts (Cont)

COMPONENT	PART NUMBER
Main Rotor Shaft, Mod Kit	70070-35141-011, -012, -014
Sleeve, Rad and Fan, Mod Kit	70070-35143-011
Trim and Balance, Mod Kit	70070-35145-011
Mixer Links, Retrofit Kit	70070-40033-011, -041, -042, -043, -044
Shaft Instl, Mod Kit	70070-40036-011, -012
Linkage Instl, Mod Kit	70070-40037-015, -016, -042
Mixer Assy, Mod Kit	70070-40046-011
Linkage Instl, Mod Kit	70070-40053-014, -015, -016, -017, -042
Spindle Assy, Mod Kit	70083-10001-011, -041
Hub Assy, Mod Kit	70083-10001-012, -042
Main Rotor Spindle Assy	70083-10001-045
Shaft and Oil Cooler Instl	70083-35003-011
Rotor Brake Instl, Mod Kit	70083-35006-012
Rod End	70101-08202-041, -043
Tail Rotor Blade Assy	70101-11200-042, -043, -044
Spar Assy, Tail Rotor Blade	70101-11201-041, -042
Main Rotor Horn	70102-08012-105, -106
Main Rotor Spindle Forging	70102-08101-001
Main Rotor Spindle Detail	70102-08101-103
Main Rotor Spindle Nut	70102-08105-102
Main Rotor Horn Assy	70102-08111-043, -044, -045, -046, -047
Main Rotor Spindle Assy	70102-08200-041, -042, -043, -044, -045, -046, -053, -054, -055, -056, -057, -063, -065, -068, -069
Main Rotor Spindle and Liner Assy	70102-08216-041, -042, -045, -046

1-2-2-2 Change 3

Table 1-2-2-1. Flight Safety Critical Aircraft Parts (Cont)

COMPONENT	PART NUMBER
Main Rotor Hub Forging	70103-08101-001, -002
Main Rotor Hub	70103-08101-101, -104
Main Rotor Blade Expandable Pin	70103-08107-102
Main Rotor Blade Solid Pin	70103-08108-041
Main Rotor Hub Sub-Assy	70103-08112-041, -042, -047, -048
Main Rotor Hub Assy	70103-08200-041
Swashplate Assy	70104-08000-043, -044, -045, -046, -049, -050
Main Rotor Swashplate	70104-08002-102
Stationary Swashplate	70104-08011-103
Swashplate Expandable Pin	70104-08015-102
Scissors Assembly	70104-08020-044, -045
Main Rotor Blade Assy	70150-09100-041, -043
Main Rotor Blade Sub-Assy	70150-09101-041, -043
Main Rotor Blade Spar	70150-09103-101, -103
Tip Cap Assy, Main Rotor Blade	70150-09107-041, -042, -046, -049, -051, -054, -055
Tip Cap Fairing, Main Rotor Blade	70150-09107-101
Cuff Assy, Main Rotor Blade	70150-09109-041
Cuff, Main Rotor Blade	70150-09109-101
Spar Tube, Main Rotor Blade	70150-09122-102
Spar Ground Sheet, Main Rotor Blade	70150-09123-101
Rolled Plate, Main Rotor Blade Rolled Plate	70150-09124-001
Cuff Forging, Main Rotor Blade	70150-09127-001
Pylon/Tail Cone Joining Instl	70200-06051-011, -012

Table 1-2-2-1. Flight Safety Critical Aircraft Parts (Cont)

COMPONENT	PART NUMBER
External Stores Support System Instl	70200-42400-011, -015, -016, -017, -018, -019
External Stores Support System Assy	70200-42400-051, -052
External Fuel System	70200-42404-011
Volcano Rack Assy	70200-42500-041, -042, -050, -051
Stabilator Plating Instl	70203-27000-014
Spar Fitting Assy, TRB Stabilator Attach	70209-06052-041, -043, -044, -046
Spar Fitting, TRB Stabilator Attach	70209-06052-101, -103
Support Fitting Assy, TRB Stabilator Actuator	70209-06054-044
Support Fitting Assy, Stabilator Actuator	70209-07053-043, -044
Support Fitting, Stabilator Actuator	70209-07053-103
Right Attach Bolts, Stabilator	70209-07057-101
Right Attach Bolts, Folding Stabilator	70209-07058-101
Servo Beam Fitting Assy	70209-22103-049, -050, -051, -052, -053, -054, -055, -056, -065, -067, -069, -071
Servo Beam Fitting	70209-22103-105, -106, -107, -108
Stabilator Centerbox Fwd Spar Attach Fitting Assy	70209-27001-045
Spar Attach Fitting , Stabilator Centerbox Fwd	70209-27001-105
Spar Attach Fitting Assy, Stabilator Centerbox Aft	70209-27002-042
Spar Attach Fitting, Stabilator Centerbox Aft	70209-27002-102
Actuator Attach Fitting Assy, Stabilator Centerbox	70209-27003-044, -045
Actuator Attach Fitting, Stabilator Centerbox	70209-27003-102
Folding Stabilator Attach Bolts	70209-27007-102, -104
Servo Beam Fitting Assy	70219-02134-045, -046, -047, -048, -049, -050, -051, -052

Table 1-2-2-1. Flight Safety Critical Aircraft Parts (Cont)

COMPONENT	PART NUMBER
Servo Beam Fitting	70219-02134-105, -106, -107, -108
Bulkhead Fitting Assy, Tail Cone Canted Hinge	70219-05001-043, -044, -045
Bulkhead Fitting, Tail Cone Canted Hinge	70209-05001-105, -107
Volcano Strut Assy	70219-42505-041, -042
Main Landing Gear Drag Beam	70250-12062-101, -102, -104
Shaft and Cooler Instl	70350-03000-012, -013
Tail Drive Shaft Instl	70350-05000-012, -013
Transmission Instl, Pylon	70350-06000-012, -014
Input Module, Main Gear Box	70351-08001-043, -045, -046, -047, -049
Main Module Assy, Main Gear Box	70351-08100-053, -054, -055, -056, -057, -058, -059, -060, -061, -062, -063, -064, -065, -066, -067, -068, -069, -070, -071, -072, -073, -074
MGB Housing Assy, Matched Set	70351-08110-044, -045
MGB Main Housing Casting	70351-08111-002
MGB Main Housing	70351-08111-103
Main Rotor Shaft	70351-08131-044, -045, -046, -047, -048, -102, -103, -107, -108
Shaft Extension Forging	70351-08186-002
Shaft Extension Assy	70351-08186-043, 044
Shaft Extension	70351-08186-103
Input Flange Assy	70351-08206-041
Input Flange	70351-08206-101, -102
Swashplate Guide	70351-08228-101
Main Gear Box Dowel Pin	70351-08404-101, -102, -103, -104
MGB Housing Assy, Matched Set	70351-28110-043, -044

Table 1-2-2-1. Flight Safety Critical Aircraft Parts (Cont)

COMPONENT	PART NUMBER
MGB Main Housing	70351-28111-002
Take-Off Pinion	70351-28151-101
Dowel Pin, MGB	70351-28404-101
Rotor Brake Instl	70351-28500-011, -013
MGB Housing Assy, Matched Set	70351-38110-043, -044, -045
MGB Main Housing Casting	70351-38111-002
Main Rotor Shaft Assy	70351-38131-042
Main Rotor Shaft	70351-38131-102
Dowel Pin, MGB	70351-38404-101
Main Module Assy	70351-48100-041
Main Module Assy	70351-78100-041, -043, -046, -048
Output Assy, Tail Gear Box	70358-06603-041, -042, -043, -045
Center Housing, Tail Gear Box	70358-06608-001
Inboard Retention Plate Forging, Tail Gear Box	70358-06612-001, -002
Inboard Retention Plate, Tail Gear Box	70358-06612-102
Output Gear, Tail Gear Box	70358-06620-101, -102
Tail Rotor Gear Box and Servo Assy	70358-06800-041, -042, -043, -047
Tail Drive Section I Shaft Assy	70361-03001-044
Axial Fan, Main Gear Box	70361-03005-104, -105, -106, -107
Axial Fan, Main Gear Box	70361-03700-101
Tail Drive Section II Shaft Assy	70361-05001-049, -052
Tail Drive Section III Shaft Assy	70361-05002-044
Tail Drive Section III Shaft Assy	70361-06002-044, -048

Table 1-2-2-1. Flight Safety Critical Aircraft Parts (Cont)

COMPONENT	PART NUMBER
Engine Drive Shaft Assy	70361-08004-043
Directional Cable Instl	70400-00510-011, -013
Cockpit Controls Instl	70400-01100-011
Cockpit Cyclic Instl	70400-01200-012
Cockpit Push Rod Assy	70400-01250-052, -053, -054
Cockpit Push Rod Assy	70400-01255-103, -104
Collective Control Instl, Cockpit	70400-01400-012
Directional Instl, Cockpit	70400-01600-012, -013
Pedal Pivot Assy	70400-01601-042, -043, -044, -046
Pedal Adjustor Assy	70400-01613-049, -050
Controls Instl, Center Section	70400-02000-012
Torque Shaft Instl	70400-02108-011, -013
Push Rod Instl	70400-02108-012
Push Rod Assy	70400-02252-120, -123, -124, -128, -129, -130, -131, -132, -133, -134
Push Rod Assy	70400-02258-048
Push Rod Assy	70400-02264-051, -060
Push Rod Tube	70400-02265-102, -103, -104, -105, -106, -112, -114, -115, -116, -117
Mixer Assy	70400-02300-041, -043, -044, -045, -046, -048, -050
Controls Instl, Lower Cabin	70400-02500-012
Control Assy, Lower Cabin	70400-02500-041, -042
Directional Lever Instl	70400-02625-011
Controls Instl, Upper Deck	70400-02700-011, -012, -013

Table 1-2-2-1. Flight Safety Critical Aircraft Parts (Cont)

COMPONENT	PART NUMBER
Tail Rotor Servo/Quadrant Instl	70400-06100-011
Tail Rotor Servo Coupling Assy	70400-06100-041, -042
Horizontal Stabilator Actuator Instl	70400-06636-012, -013, -014
Actuator Assy, Horizontal Stabilator	70400-06637-043, -044, -045, -046, -047, -049, -051, -052
Push Rod Tube	70400-06700-101, -102, -103, -106
Link Assy, Tail Rotor Servo	70400-06701-042
Link, Tail Rotor Servo	70400-06701-102
Stabilator Position Sensor Instl	70400-06712-011
Bolt, Horizontal Stabilizer Actuator	70400-06802-101
Bellcrank Assy, Forward Longitudinal	70400-08101-044, -045, -046, -047
Bellcrank, Forward Longitudinal	70400-08101-104
Bellcrank Assy, Aft Longitudinal	70400-08102-044, -045
Bellcrank, Aft Longitudinal	70400-08102-105
Walking Beam Assy, Aft Longitudinal	70400-08104-044, -045, -046, -047, -048, -049
Walking Beam, Aft Longitudinal	70400-08104-106, -107
Swashplate Link Assy, Forward Longitudinal	70400-08110-047, -053, -060
Swashplate Link Assy, Aft Longitudinal	70400-08110-048, -054, -061
Forward Longitudinal Swashplate Link	70400-08110-108
Swashplate Link, Aft Longitudinal	70400-08110-109
Tie Rod Assy, Aft Longitudinal	70400-08111-041, -042, -043, -044
Tie Rod, Aft Longitudinal	70400-08111-101, -103
Support Fitting Assy, Aft Longitudinal Tie Rod	70400-08112-044, -045, -046, -047
Support Fitting, Aft Longitudinal Tie Rod	70400-08112-107, -109

1-2-2-8 Change 3

Table 1-2-2-1. Flight Safety Critical Aircraft Parts (Cont)

COMPONENT	PART NUMBER
Right Tie Rod Assy	70400-08114-047, -048, -049, -050, -051, -052
Right Tie Rod	70400-08114-111
Left Tie Rod Assy	70400-08115-043, -045, -046, -047
Left Tie Rod	70400-08115-103
Support Assy, Forward Swashplate Bellcrank	70400-08116-045, -046, -047, -048
Support, Forward Swashplate Bellcrank	70400-08116-115
Support Assy, Aft Bellcrank	70400-08117-046, -047, -048, -049, -050, -051
Support, Aft Bellcrank	70400-08117-103, -113
Lateral Bellcrank Assy	70400-08150-043, -044, -045, -046
Lateral Bellcrank	70400-08150-103
Link Assy, Lateral Swashplate	70400-08151-045, -046, -047, -048, -050, -060, -061
Link, Lateral Swashplate	70400-08151-106
Rod Assy, Forward Main Rotor Servo Output	70400-08155-049, -053
Push Rod Assy, Aft Longitudinal	70400-08155-050, -054
Rod Assy, Aft Main Rotor Servo Output	70400-08155-051, -055
Rod Assy, Lateral Main Rotor Servo Output	70400-08155-052, -056
Rod, Forward Main Rotor Servo Output	70400-08155-109
Push Rod, Aft Longitudinal	70400-08155-110
Rod, Aft Main Rotor Servo Output	70400-08155-111
Rod, Lateral Main Rotor Servo Output	70400-08155-112
Support, Aft Longitudinal	70400-08158-101, -102
Bolt, Special, Self-Retaining	70400-08159-101, -102, -103, -104, -105, -106, -107, -108
Support Assy, Forward Swashplate Bellcrank	70400-08162-041, -042

Table 1-2-2-1. Flight Safety Critical Aircraft Parts (Cont)

COMPONENT	PART NUMBER
Support, Forward Swashplate Bellcrank	70400-08162-101
Support Assy, Aft Longitudinal Bellcrank	70400-08163-043, -044
Support, Aft Bellcrank	70400-08163-102
Right Tie Rod Assy	70400-08165-042, -043
Right Tie Rod	70400-08165-101
Lateral Bellcrank Assy	70400-08166-041, -042
Lateral Bellcrank	70400-08166-101
Controls Instl, Swashplate	70400-08200-011, -012, -013
Controls Instl, Swashplate	70400-08200-041, -042, -043, -044
Pedal Adjustor Assy	70400-21613-041, -042
Push Rod Assy	70400-22258-043, -044
Actuator Assy, Roll SAS	70410-02450-043, -048, -049
Actuator, Pitch Trim	70410-02561-102, -104, -105
Actuator Assy, Pitch Trim	70410-02561-112, -114
Servo Assy, Pitch Trim	70410-02760-044, -045, -046, -047
Primary Servo, Main Rotor	70410-02800-105, -106, -107, -111, -112, -113, -114, -116
Primary Servo Assy, Main Rotor	70410-02820-044, -045, -046, -050, -051, -052, -053, -054
Servo, Yaw Boost	70410-02900-126
Servo Assy, Yaw Boost	70410-02910-047, -051, -053, -059, -060, -065, -067
Servo, Tail Rotor	70410-06500-105, -107, -108, -110
Servo Assy, Tail Rotor	70410-06520-043, -044, -045, -046
Bearing Assy, Main Rotor Shaft	70952-08557-043

Table 1-2-2-1. Flight Safety Critical Aircraft Parts (Cont)

COMPONENT	PART NUMBER
Bearing, Tail Rotor Drive Shaft	SB1138-101, -102
Bearing	SB1162-102
Bearing	SB5201-1

SECTION 1-3

SERVICING

SECTION OVERVIEW

PARAGRAPH	TITLE
1-3-1	Servicing Requirements
1-3-2	Service Main Landing Gear Shock Struts
1-3-3	Service Main Landing Gear Tires
1-3-4	Service Tail Landing Gear Shock Strut
1-3-5	Service Tail Landing Gear Tire
1-3-6	Service/Drain Engine
1-3-7	Service Engine Starter
1-3-8	Service Hydraulic Pump Modules
1-3-9	Service/Drain APU
1-3-10	Service APU Accumulator
1-3-11	Service Winterization Kit APU Accumulator
1-3-12	Service/Drain Main Transmission
1-3-13	Service/Drain Intermediate Gear Box
1-3-14	Service/Drain Tail Gear Box
1-3-15	Service Main Rotor Head Damper
1-3-16	Service Tail Rotor Drive Shaft Viscous Damper Bearings
1-3-17	Pressure Refuel Main Fuel Cells
1-3-18	Gravity Refuel Main Fuel Cells
1-3-19	Closed Circuit Refuel
1-3-20	Pressure Defuel Main Fuel Cells

	PARAGRAPH	TITLE
I	1-3-21	Gravity Defuel Main Fuel Cells
	1-3-22	Transfer Defuel Extended Range Tanks
I	1-3-23	Preserve Main Fuel Cell
	1-3-24	Gravity Refuel Extended Range Tanks
	1-3-25	Adapter Defuel Extended Range Tanks
	1-3-26	Clean and Preserve Extended Range Tanks
	1-3-27	Environmental Control System - Service
	1-3-28	Environmental Control System Compressor - Service
	1-3-29	Service Main Rotor Blade BIM® System
	1-3-30	Service Hydraulic Rescue Hoist
	1-3-31	Service Electric Rescue Hoist

1-3-1 SERVICING REQUIREMENTS.

1-3-1.1 Scope.

This section contains servicing procedures. Servicing intervals may be dependent upon environmental conditions or specific time intervals. When unusual local conditions of environment, utilization, mission, experience of flight crew and maintenance personnel, or periods of inactivity are encountered, the scope and/or frequency of servicing may need to increase to make sure of safe flight.

imum weight, engine cowlings can support including equipment, is 400 pounds. No more than two people are permitted on engine cowling at any one time, with no one person weighing over 250 pounds. Do not apply load to vertical surface.

Work platforms located on engine cowlings provide access to the engines. Each platform is capable of supporting a static weight of 400 pounds.

1-3-1.2 Work Platforms.

CAUTION

Damage can occur if limits noted on engine cowlings are exceeded. Maxi-

1-3-2 SERVICE MAIN LANDING GEAR SHOCK STRUTS.

PARAGRAPH BREAKDOWN

		PAGE
1-3-2.1	Service Main Landing Gear Shock Struts	1-3-2-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Beaker, 800 ml
- Container, 1 Gallon
- Hydraulic Fluid Dispenser, AF5
- Nitrogen Servicing Unit, 901002
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Leak Detection Fluid, Item 189, Appendix D
- Lockwire, Item 197, Appendix D

- Machinery Towel, Item 211, Appendix D
- Nitrogen, Item 221, Appendix D
- Petrolatum, Item 234, Appendix D
- Rags
- Tubing, Nonmetallic, Item 344, Appendix D
- Union

References

- Appendix D
- PARA 1-6-3
- PARA 2-4-95
- PARA 2-4-96
- PARA 3-4-7
- PARA 3-4-9
- PARA 16-4-54

1-3-2.1 Service Main Landing Gear Shock Struts.

NOTE

- If both upper and lower stage struts require servicing, service upper stage strut first.
- Servicing of left and right main landing gear shock struts is the same.

1-3-2.1.1 Service Upper Stage Strut (with Helicopter on Jacks).

CAUTION

- To prevent damage to helicopter, refer only to procedure in this manual for proper charge pressure. Disregard all other labels on helicopter concerning charge pressure.
 - An upper stage strut needing service should be carefully checked for hydraulic fluid leakage. Repeated nitrogen servicing of a leaking strut without servicing with hydraulic fluid can lead to a completely nitrogen-inflated strut, causing internal damage.
- a. If installed, remove ESSS 230 gallon external fuel tanks (PARA 16-4-54).
 - b. Remove lower and upper main landing gear fairings (PARA 2-4-95 and PARA 2-4-96).

WARNING

Injury to personnel and damage to equipment will result if main landing gear is not jacked before servicing upper stage strut. Helicopter can settle up to 16-inches if upper stage strut is depressurized and helicopter is unsupported. Damage will occur to external brake lines on drag beam. Rear lower hard

point will contact and damage drag beam.

NOTE

Improper servicing will result if tire is allowed to touch the ground.

- c. Jack front of helicopter until main landing gear is fully extended and tires clear the ground (PARA 1-6-3).
- d. Remove cap from upper air valve (Figure 1-3-2-1, Sheet 1, Detail B).
- e. Release nitrogen pressure from upper stage strut as follows:
 - (1) Slide nonmetallic tubing, Item 344, Appendix D, over end of upper air valve. Place free end in 1 gallon container to catch hydraulic fluid discharge, if any, when valve is opened.

WARNING

Injury to personnel and damage to equipment will result if all nitrogen pressure is not released prior to removing upper air valve. Shock strut is under high pressure. Make sure all nitrogen pressure is released prior to removing valve.

- (2) Slowly turn out hexnut counterclockwise until nitrogen hissing is heard from upper air valve.
- (3) When nitrogen pressure is completely released from upper stage strut, measure collected hydraulic fluid, if any, using 800 ml beaker. Proceed as follows:
 - (a) **IF HYDRAULIC FLUID DISCHARGE IS GREATER THAN 200 MILLILITERS (CUBIC CENTIMETERS), REPLACE MAIN LANDING GEAR SHOCK STRUT (PARA 3-4-7).**

- (b) **IF HYDRAULIC FLUID DISCHARGE IS BETWEEN 100 AND 200 MILLILITERS (CUBIC CENTIMETERS), STRUT CAN BE CONTINUED IN SERVICE, BUT REQUIRES INSPECTION AT NEXT 200 FLIGHT HOURS.**
- (c) **IF HYDRAULIC FLUID DISCHARGE IS LESS THAN 100 MILLILITERS (CUBIC CENTIMETERS), STRUT CAN BE CONTINUED IN SERVICE, BUT REQUIRES INSPECTION AT NEXT 500 FLIGHT HOURS.**
- (4) Remove nonmetallic tubing from upper air valve.
- (5) Turn inner hexnut counterclockwise, and remove upper air valve and packing. Discard packing.
- f. Bottom floating piston in upper stage strut, using hydraulic fluid dispenser, as follows:

NOTE

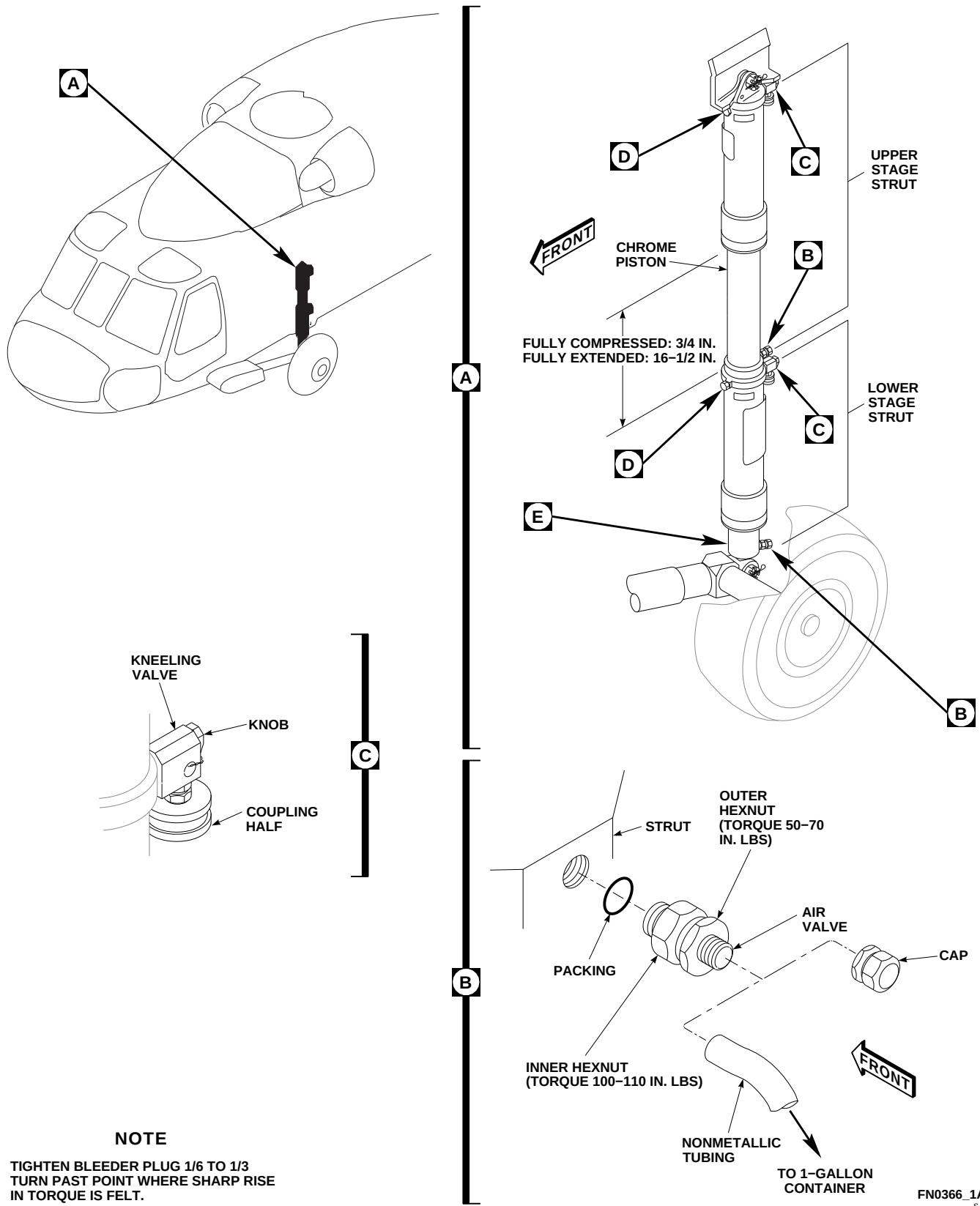
Under emergency conditions, hydraulic fluid, Item 155, Appendix D, may be used to service struts. However, reserve struts with hydraulic fluid, Item 154, Appendix D, as soon as convenient.

- (1) Service dispenser with hydraulic fluid, Item 154, Appendix D.
- (2) Connect dispenser to coupling half on upper stage kneeling valve (Figure 1-3-2-1, Sheet 1, Detail C).
- (3) Turn knob on upper stage kneeling valve to OPEN.

NOTE

- If dispenser pressure drops below 10 psi, recharge dispenser.
- Floating piston is bottomed when a dull thud is heard.

- (4) Slowly release pressurized hydraulic fluid from dispenser into upper stage strut until floating piston bottoms. **DO NOT EXCEED 100 PSI HYDRAULIC PRESSURE.**
- (5) Release pressure from dispenser.
- g. Bleed hydraulic fluid from upper stage strut as follows:
 - (1) Place machinery towels, Item 211, Appendix D, under upper bleeder plug to catch hydraulic fluid when upper bleeder plug is removed.
 - (2) Remove upper bleeder plug and packing from bleed port in upper stage strut (Figure 1-3-2-1, Sheet 2, Detail D). Retain packing for reuse.
 - (3) Install union with packing (previously removed) in bleeder port.
 - (4) Connect overflow line to union. Place free end in 1 gallon container to catch hydraulic fluid overflow.
 - (5) Slowly operate jack, fully compressing upper stage strut until no more than $\frac{3}{4}$ -inch of chrome piston is visible. While compressing upper stage strut, watch fluid flowing from overflow line. If fluid contains bubbles, perform the following:
 - (a) Stop jacking.
 - (b) Remove overflow line, union, and packing from bleeder port. Retain packing for reuse.
 - (c) Install upper bleeder plug and packing (previously removed). **TIGHTEN BLEEDER PLUG $\frac{1}{6}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
 - (d) Lower helicopter and remove jack from under jacking pad on drag beam (PARA 1-6-3).



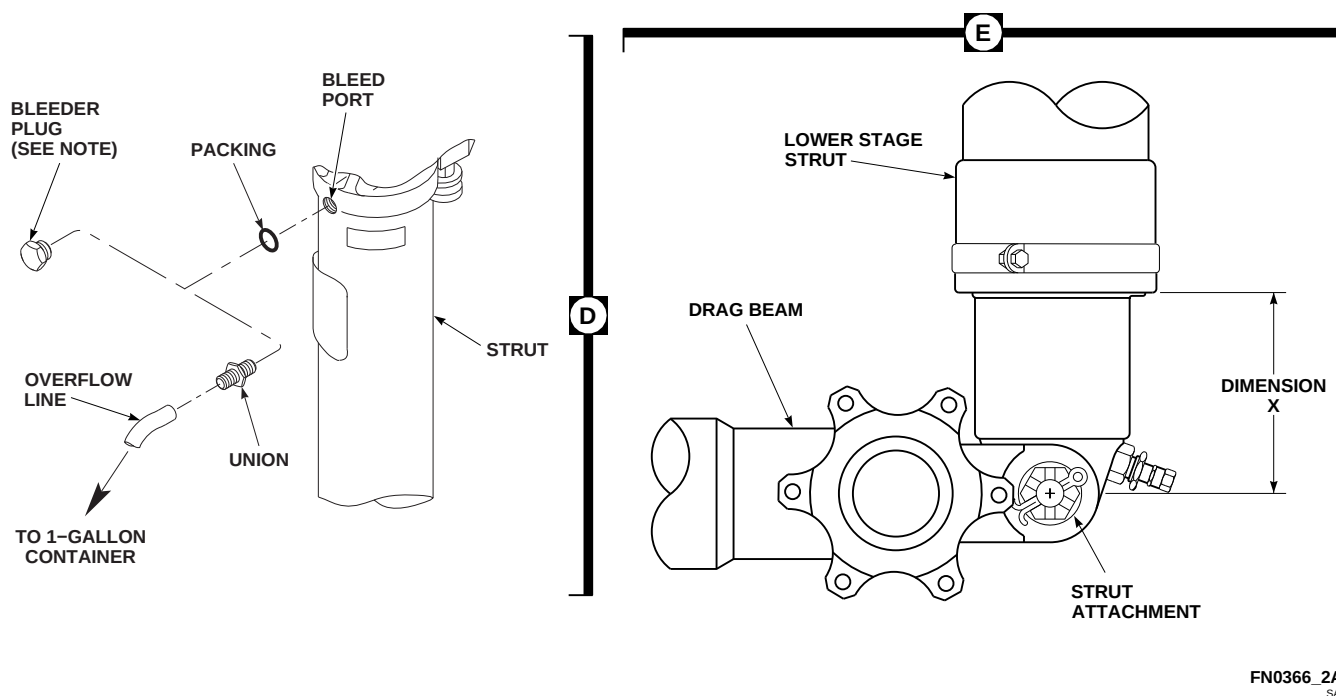


FIGURE 1-3-2-1. SERVICE MAIN LANDING GEAR SHOCK STRUTS (SHEET 2 OF 2)

- (e) Slowly release pressurized hydraulic fluid from fluid dispenser into upper stage strut until strut is fully extended. When strut is fully extended, a sharp rise in pressure will occur. **DO NOT EXCEED 100 PSI HYDRAULIC PRESSURE.**
- (f) Repeat step f. (5) through step g. (5) until bubbles are no longer visible in hydraulic fluid and upper stage strut is fully compressed.
- (6) Turn knob on upper stage kneeling valve to CLOSE.
- (7) Remove overflow line, union, and packing from upper bleeder plug port. Discard packing.
- (8) Install new packing lubricated with petrolatum, Item 234, Appendix D, on upper bleeder plug.
- (9) Install upper bleeder plug and packing into upper stage strut. **TIGHTEN BLEEDER PLUG HANDTIGHT.**
- (10) Turn knob on upper stage kneeling valve to OPEN.
- (11) Lower helicopter and remove jack from under jacking pad on drag beam.
- (12) Slowly release pressurized hydraulic fluid from fluid dispenser into upper stage strut until strut begins to extend. **DO NOT EXCEED 100 PSI HYDRAULIC PRESSURE.**
- (13) Release pressure from dispenser.
- (14) Place rags around upper bleeder plug. Loosen bleeder plug.
- (15) Place jack underneath jack pad on drag beam (PARA 1-6-3).
- (16) Slowly operate jack, fully compressing upper stage strut until no more than $\frac{3}{4}$ inch of chrome piston is visible.
- (17) Slowly release pressurized hydraulic fluid from fluid dispenser into upper stage strut and monitor flow of oil until air bubbles are no longer visible.

- (18) Stop pumping fluid. **TIGHTEN BLEEDER PLUG HANDTIGHT.**
 - (19) Turn knob on upper stage kneeling valve to CLOSE.
 - (20) Using lockwire, Item 197, Appendix D, lockwire knob to hole in upper stage kneeling valve housing.
 - (21) **TIGHTEN BLEEDER PLUG $\frac{1}{6}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
 - (22) Using lockwire, Item 197, Appendix D, lockwire bleeder plug to upper stage strut.
 - (23) Disconnect dispenser from coupling half on upper stage kneeling valve (Figure 1-3-2-1, Sheet 1, Detail C).
 - (24) Install new packing lubricated with petrolatum, Item 234, Appendix D, on upper air valve (Figure 1-3-2-1, Sheet 1, Detail B).
 - (25) Screw upper air valve and packing into upper stage strut. **TORQUE INNER HEXNUT TO 100 - 110 INCH POUNDS.**
 - (26) Lower helicopter and remove jack from under jacking pad on drag beam (PARA 1-6-3).
- h. Service upper stage strut with nitrogen as follows:
- (1) Attach servicing chuck from a regulated nitrogen servicing unit to upper air valve.
- (2) Slowly apply pressure using nitrogen, Item 221, Appendix D, to fully extend upper stage strut. Continue until gage on nitrogen cylinder assembly reads 730 - 750 psig. **DO NOT EXCEED 1000 PSIG MAXIMUM PRESSURE.**
 - (3) Measure from half-way point of upper air valve to upper portion of exposed chrome piston. Verify that full-extend measurement on upper stage strut is 16 $\frac{1}{2}$ inches as shown in Figure 1-3-2-1, Sheet 1, Detail A.
 - (4) Verify that tire is not touching ground when upper stage strut is fully extended.
 - (5) **TORQUE OUTER HEXNUT TO 50 - 70 INCH POUNDS.**
 - (6) Disconnect servicing chuck from upper air valve.
- i. Apply leak detection fluid, Item 189, Appendix D, to upper air valve and make sure valve does not leak. If bubbles appear, perform the following:
- (1) Determine if packing or air valve is leaking.
 - (2) Replace main landing gear shock strut air valve or packing as required (PARA 3-4-9).
 - (3) Repeat step h. and step i.
- j. Install cap on upper air valve.
- k. If lower stage is going to be serviced, proceed to PARA 1-3-2.1.2.

CAUTION

Damage to strut will result if overpressurized during servicing. Internal damage or strut failure will result. Do not overpressurize strut during servicing.

NOTE

Improper servicing will result if tire is allowed to touch the ground.

NOTE

If lower stage strut is to be serviced, do not perform step l.

- l. If lower stage will not be serviced, perform the following:
- (1) Lower helicopter and remove jack from under jacking pad on drag beam.

- (2) Install lower and upper main landing gear fairings (PARA 2-4-95 and PARA 2-4-96).
- (3) If required, install ESSS 230 gallon external fuel tanks (PARA 16-4-54).

1-3-2.1.2 Service Lower Stage Strut with Helicopter on Jacks (Preferred Method).

CAUTION

- To prevent damage to helicopter, refer only to procedure in this manual for proper charge pressure. Disregard all other labels on helicopter concerning charge pressure.
 - A lower stage strut needing service should be carefully checked for hydraulic fluid leakage. Repeated nitrogen servicing of a leaking strut without servicing with hydraulic fluid can lead to a completely nitrogen-inflated strut, causing internal damage.
- a. If installed, remove ESSS 230 gallon external fuel tanks (PARA 16-4-54).
 - b. Remove lower and upper main landing gear fairings (PARA 2-4-95 and PARA 2-4-96).

NOTE

Improper servicing will result if tire is allowed to touch the ground.

- c. Jack front of helicopter until main landing gear is fully extended and tires clear the ground (PARA 1-6-3).
- d. Remove cap from lower air valve (Figure 1-3-2-1, Sheet 1, Detail B).
- e. Release nitrogen pressure from lower stage strut as follows:
 - (1) Slide nonmetallic tubing, Item 344, Appendix D, over end of lower air valve.

Place free end in 1 gallon container to catch hydraulic fluid discharge, if any, when valve is opened.

WARNING

Injury to personnel and damage to equipment will result if all nitrogen pressure is not released prior to removing lower air valve. Shock strut is under high pressure. Make sure all nitrogen pressure is released prior to removing valve.

- (2) Slowly turn outer hexnut counterclockwise until nitrogen hissing is heard from lower air valve.
- (3) When nitrogen pressure is completely released from lower stage strut, measure collected hydraulic fluid, if any, using 800 ml beaker. Proceed as follows:
 - (a) **IF HYDRAULIC FLUID DISCHARGE IS GREATER THAN 200 MILLILITERS (CUBIC CENTIMETERS), REPLACE MAIN LANDING GEAR SHOCK STRUT (PARA 3-4-7).**
 - (b) **IF HYDRAULIC FLUID DISCHARGE IS BETWEEN 100 AND 200 MILLILITERS (CUBIC CENTIMETERS), STRUT CAN BE CONTINUED IN SERVICE, BUT REQUIRES INSPECTION AT NEXT 200 FLIGHT HOURS.**
 - (c) **IF HYDRAULIC FLUID DISCHARGE IS LESS THAN 100 MILLILITERS (CUBIC CENTIMETERS), STRUT CAN BE CONTINUED IN SERVICE, BUT REQUIRES INSPECTION AT NEXT 500 FLIGHT HOURS.**
- (4) Remove nonmetallic tubing from lower air valve.

- (5) Turn inner hexnut counterclockwise, and remove lower air valve and packing. Discard packing.
- f. Bottom floating piston in lower stage strut, using hydraulic fluid dispenser, as follows:

NOTE

Under emergency conditions, hydraulic fluid, Item 155, Appendix D, may be used to service struts. However, reserve struts with hydraulic fluid, Item 154, Appendix D, as soon as convenient.

- (1) Service dispenser with hydraulic fluid, Item 154, Appendix D.
- (2) Connect dispenser to coupling half on lower stage kneeling valve (Figure 1-3-2-1, Sheet 1, Detail C).
- (3) Turn knob on lower stage kneeling valve to OPEN.

NOTE

- If dispenser pressure drops below 10 psi, recharge dispenser.
- Floating piston is bottomed when a dull thud is heard.

- (4) Slowly release pressurized hydraulic fluid from dispenser into lower stage strut until floating piston bottoms. **DO NOT EXCEED 100 PSI HYDRAULIC PRESSURE.**

- (5) Release pressure from dispenser.

- g. Bleed hydraulic fluid from lower stage strut as follows:

- (1) Place machinery towels, Item 211, Appendix D, under lower bleeder plug to catch hydraulic fluid when lower bleeder plug is removed.
- (2) Remove lower bleeder plug and packing from bleed port in lower stage strut (Figure 1-3-2-1, Sheet 2, Detail D). Retain packing for reuse.

- (3) Install union with packing (previously removed) in bleeder port.

- (4) Connect overflow line to union. Place free end in 1 gallon container to catch hydraulic fluid overflow.

- (5) Slowly operate jack, fully compressing lower stage until no more than ½-inch of chrome piston is visible. While compressing lower stage, watch fluid flowing from overflow line. If fluid contains bubbles, perform the following:

- (a) Stop jacking.

- (b) Install lower bleeder plug and packing (previously removed). **TIGHTEN BLEEDER PLUG ⅙ TO ⅓ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

- (c) Lower helicopter and remove jack from under jacking pad on drag beam (PARA 1-6-3).

- (d) Perform step f. (4) through step g. (5).

- (6) Stop when bubbles are no longer seen in fluid and strut is fully compressed. Turn knob on lower stage kneeling valve to CLOSE. Using lockwire, Item 197, Appendix D, lockwire knob to hole in lower stage kneeling valve housing.

- (7) Disconnect fluid dispenser from coupling half on lower stage kneeling valve (Figure 1-3-2-1, Sheet 1, Detail C).

- (8) Remove overflow line, union, and packing from lower bleeder plug port. Discard packing.

- (9) Install new packing lubricated with petrolatum, Item 234, Appendix D, on lower bleeder plug (Figure 1-3-2-1, Sheet 2, Detail D).

- (10) Install lower bleeder plug and packing into lower stage strut. **TIGHTEN**

BLEEDER PLUG ¼ TO ⅓ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT. Using lockwire, Item 197, Appendix D, lockwire bleeder plug to lower cylinder.

- (11) Install new packing lubricated with petrolatum, Item 234, Appendix D, on lower air valve (Figure 1-3-2-1, Sheet 1, Detail B).
 - (12) Screw lower air valve and packing into lower stage strut. **TORQUE INNER HEXNUT TO 100 - 110 INCH POUNDS.**
 - (13) Lower helicopter and remove jack from under jacking pad on drag beam (PARA 1-6-3).
- h. Service lower stage strut with nitrogen as follows:
- (1) Attach servicing chuck from a regulated nitrogen servicing unit to lower air valve.



- Damage to strut will result if overpressurized during servicing. Internal damage or strut failure will result. Do not overpressurize strut during servicing.
- To prevent damage to helicopter, check part number on shock strut before servicing.

NOTE

Improper servicing will result if tire is allowed to touch the ground.

- (2) Slowly apply pressure, using nitrogen, Item 221, Appendix D, to fully extend lower stage strut. Continue until gage on nitrogen cylinder assembly reads as follows:

SHOCK STRUT PART NUMBER	PRESSURE (PSIG)
----------------------------	-----------------

70250-12051-042	275 - 325	
70250-12051-043, -044, or -045	190 - 240	

- (3) Verify that tire is not touching ground when lower stage strut is fully extended.
 - (4) **TORQUE OUTER HEXNUT TO 50 - 70 INCH POUNDS.**
 - (5) Disconnect servicing chuck from lower air valve.
- i. Apply leak detection fluid, Item 189, Appendix D, to lower air valve and make sure valve does not leak. If bubbles appear, perform the following:
- (1) Determine if packing or air valve is leaking.
 - (2) Replace main landing gear shock strut air valve or packing as required (PARA 3-4-9).
 - (3) Repeat step g. and step h.
- j. Install cap on lower air valve.
- k. Using lockwire, Item 197, Appendix D, lockwire lower air valve to piston.
- l. Remove jacks from drag beam supports (PARA 1-6-3).
- m. Install lower and upper main landing gear fairings (PARA 2-4-95 and PARA 2-4-96).
- n. If required, install ESSS 230 gallon external fuel tanks (PARA 16-4-54).

1-3-2.1.3 Service Lower Stage Strut on Ground (Alternate Method).

CAUTION

- To prevent damage to helicopter, refer only to procedure in this manual for proper charge pressure. Disregard all other labels on helicopter concerning charge pressure.
 - A lower stage strut needing service should be carefully checked for hydraulic fluid leakage. Repeated nitrogen servicing of a leaking strut without servicing with hydraulic fluid can lead to a completely nitrogen-inflated strut, causing internal damage.
- a. If installed, remove ESSS 230 gallon external fuel tanks (PARA 16-4-54).
 - b. Remove lower and upper main landing gear fairings (PARA 2-4-95 and PARA 2-4-96).
 - c. Set parking brake.
 - d. Remove cap from lower air valve (Figure 1-3-2-1, Sheet 1, Detail B).
 - e. Release nitrogen pressure from lower stage strut as follows:

WARNING

Injury to personnel and damage to equipment will result if area under helicopter is not clear prior to servicing landing gear. Helicopter will settle up to 7 inches. Look under helicopter and make sure area is clear of personnel and equipment.

- (1) Slide nonmetallic tubing, Item 344, Appendix D, over end of lower air valve. Place free end in 1 gallon container to catch hydraulic fluid discharge, if any, when valve is opened.

WARNING

Injury to personnel and damage to equipment will result if all nitrogen pressure is not released prior to removing lower air valve. Shock strut is under high pressure. Make sure all nitrogen pressure is released prior to removing valve.

- (2) Slowly turn outer hexnut counterclockwise until nitrogen hissing is heard from lower air valve.
- (3) When nitrogen pressure is completely released from lower stage strut, measure collected hydraulic fluid, if any, using 800 ml beaker. Proceed as follows:
 - (a) **IF HYDRAULIC FLUID DISCHARGE IS GREATER THAN 200 MILLILITERS (CUBIC CENTIMETERS), REPLACE MAIN LANDING GEAR SHOCK STRUT (PARA 3-4-7).**
 - (b) **IF HYDRAULIC FLUID DISCHARGE IS BETWEEN 100 AND 200 MILLILITERS (CUBIC CENTIMETERS), STRUT CAN BE CONTINUED IN SERVICE, BUT REQUIRES INSPECTION AT NEXT 200 FLIGHT HOURS.**
 - (c) **IF HYDRAULIC FLUID DISCHARGE IS LESS THAN 100 MILLILITERS (CUBIC CENTIMETERS), STRUT CAN BE CONTINUED IN SERVICE, BUT REQUIRES INSPECTION AT NEXT 500 FLIGHT HOURS.**
- (4) Remove nonmetallic tubing from lower air valve.
- (5) Turn inner hexnut counterclockwise, and remove lower air valve and packing. Discard packing.

- f. Bottom floating piston in lower stage strut, using hydraulic fluid dispenser, as follows:

NOTE

Under emergency conditions, hydraulic fluid, Item 155, Appendix D, may be used to service struts. However, reserve struts with hydraulic fluid, Item 154, Appendix D, as soon as convenient.

- (1) Service dispenser with hydraulic fluid, Item 154, Appendix D.
- (2) Connect dispenser to coupling half on lower stage kneeling valve (Figure 1-3-2-1, Sheet 1, Detail C).
- (3) Turn knob on lower stage kneeling valve to OPEN.

NOTE

- If dispenser pressure drops below 10 psi, recharge dispenser.
 - Floating piston is bottomed when a dull thud is heard.
- (4) Slowly release pressurized hydraulic fluid from dispenser into lower stage strut until floating piston bottoms. **DO NOT EXCEED 100 PSI HYDRAULIC PRESSURE.**
 - (5) Release pressure from dispenser.

- g. Bleed hydraulic fluid from lower stage strut as follows:

- (1) Place machinery towels, Item 211, Appendix D, under lower bleeder plug to catch hydraulic fluid when lower bleeder plug is removed.
- (2) Remove lower bleeder plug and packing from bleed port in lower stage strut (Figure 1-3-2-1, Sheet 2, Detail D). Retain packing for reuse.
- (3) Install union with packing (previously removed) in bleeder port.

- (4) Connect overflow line to union. Place free end in 1 gallon container to catch hydraulic fluid overflow.

- (5) Release pressurized hydraulic fluid into lower stage strut and watch fluid flowing from overflow line. Stop when bubbles are no longer seen in fluid.

- (6) Turn knob on lower stage kneeling valve to CLOSE.

- (7) Using lockwire, Item 197, Appendix D, lockwire knob to hole in lower stage kneeling valve housing.

- (8) Disconnect dispenser from coupling half on lower stage kneeling valve.

- (9) Remove overflow line, union, and packing from lower bleeder plug port. Discard packing.

- (10) Install new packing lubricated with petrolatum, Item 234, Appendix D, on lower bleeder plug.

- (11) Install lower bleeder plug and packing into lower stage strut. **TIGHTEN BLEEDER PLUG $\frac{1}{6}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

- (12) Using lockwire, Item 197, Appendix D, lockwire bleeder plug to lower stage strut.

- (13) Install new packing lubricated with petrolatum, Item 234, Appendix D, on lower air valve (Figure 1-3-2-1, Sheet 1, Detail B).

- (14) Screw lower air valve and packing into lower stage strut. **TORQUE INNER HEXNUT TO 100 - 110 INCH POUNDS.**

- h. Service lower stage strut with nitrogen as follows:

- (1) Attach servicing chuck from a regulated nitrogen cylinder assembly, Item 221, Appendix D, to lower air valve.

NITROGEN SERVICE CHART EFFECTIVITY: HELICOPTERS WITH SHOCK STRUT, 70250-12051-042	
GAGE PRESSURE TOLERANCE BETWEEN: -25 TO 25 PSI	DIMENSION X TOLERANCE BETWEEN: -0.12 TO 0.12 IN. (1/8 IN.)
1255	4.2 (4-1/4)
1115	4.7 (4-3/4)
978	5.2 (5-1/4)
880	5.7 (5-3/4)
800	6.2 (6-1/4)
733	6.7 (6-3/4)
670	7.2 (7-1/4)
627	7.7 (7-3/4)
585	8.2 (8-1/4)
548	8.7 (8-3/4)
515	9.2 (9-1/4)
486	9.7 (9-3/4)
460	10.2 (10-1/4)
436	10.7 (10-3/4)
415	11.2 (11-1/4)
298	FULL EXTENSION

NITROGEN SERVICE CHART EFFECTIVITY: HELICOPTERS WITH SHOCK STRUT, 70250-12051-043, -044, -045	
GAGE PRESSURE TOLERANCE BETWEEN: -25 TO 25 PSI	DIMENSION X TOLERANCE BETWEEN: -0.12 TO 0.12 IN. (1/8 IN.)
914	4.25 (4-1/4)
799	4.75 (4-3/4)
710	5.25 (5-1/4)
638	5.75 (5-3/4)
579	6.25 (6-1/4)
530	6.75 (6-3/4)
488	7.25 (7-1/4)
453	7.75 (7-3/4)
422	8.25 (8-1/4)
395	8.75 (8-3/4)
371	9.25 (9-1/4)
349	9.75 (9-3/4)
330	10.25 (10-1/4)
313	10.75 (10-3/4)
298	11.25 (11-1/4)
212	FULL EXTENSION

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FIGURE 1-3-2-2. MAIN LANDING GEAR NITROGEN SERVICE CHART

- (2) With lower stage strut fully compressed (wheel on ground), perform the following:

- (a) Measure dimension X on lower stage strut (Figure 1-3-2-1, Sheet 2, Detail E).

CAUTION

Damage to strut will result if overpressurized during servicing. Internal damage or strut failure will result. Do not overpressurize strut during servicing.

- (b) Slowly pressurize lower stage strut with nitrogen. As soon as strut starts to extend, shut off pressure. **DO NOT EXCEED MAXIMUM PRESSURE FOR SHOCK STRUT AS FOLLOWS:**

SHOCK STRUT PART NUMBER	MAXIMUM PRESSURE
----------------------------	---------------------

70250-12051-042	1300 PSIG
70250-12051-043, -044, or -045	1000 PSIG

- (c) Note pressure on nitrogen cylinder assembly gage.
- (d) Refer to GAGE PRESSURE column of NITROGEN SERVICE CHART. Find pressure closest to pressure taken from nitrogen cylinder assembly gage (Figure 1-3-2-2).
- (e) Read across chart to DIMENSION X column.
- (f) Add nitrogen to extend lower stage strut until dimension X is obtained.

- (3) **TORQUE OUTER HEXNUT TO 50 - 70 INCH POUNDS** (Figure 1-3-2-1, Sheet 1, Detail B).
- (4) Disconnect servicing chuck from lower air valve.
- i. Apply leak detection fluid, Item 189, Appendix D, to lower air valve and make sure valve does not leak. If bubbles appear, perform the following:
 - (1) Determine if packing or air valve is leaking.
 - (2) Replace air valve or packing as required (PARA 3-4-7).
- (3) Repeat step g. and step h.
- j. Install cap on lower air valve.
- k. Using lockwire, Item 197, Appendix D, lockwire lower air valve to piston.
- l. Install lower and upper main landing gear fairings (PARA 2-4-95 and PARA 2-4-96).
- m. If required, install ESSS 230 gallon external fuel tanks (PARA 16-4-54).
- n. Release parking brake.

1-3-3 SERVICE MAIN LANDING GEAR TIRES.

PARAGRAPH BREAKDOWN

		PAGE
1-3-3.1	Service Main Landing Gear Tires	1-3-3-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nitrogen Servicing Unit, 901002
- Tire Inflator

Materials/Parts

- Leak Detection Fluid, Item 189, Appendix D
- Nitrogen, Item 221, Appendix D

References

- Appendix D
- PARA 3-4-11

1-3-3.1 Service Main Landing Gear Tires.

WARNING

- To prevent injury to personnel or damage to equipment, use a tire inflator kit containing a pressure relief valve or similar device. Before inflating tire, check tire inflator to make sure that nitrogen source is vented at 160 psi maximum.
- Tire inflation shall not be attempted if required equipment is inoperative or unavailable.
- To prevent injury to personnel, or damage to equipment, do not inflate tire above its rated inflation pressure.

- To prevent injury to personnel, do not use an unregulated nitrogen pressure source to service tire.
- To prevent injury to personnel, never stand facing sidewall of tire being serviced. Stand facing tire tread.

NOTE

Servicing of both main landing gear tires is the same.

- Use regulated nitrogen pressure source connected to tire inflator. **INFLATE TIRE TO 130 - 140 PSI** with nitrogen, Item 221, Appendix D.
- After a 5 to 10 minute waiting period, check air valve for leakage as follows:
 - Apply leak detection fluid, Item 189, Appendix D, to valve stem area.

- (2) If bubbles appear, replace valve packing, valve core, or valve stem as required (PARA 3-4-11).
- c. Install valve cap on valve stem.

1-3-4 SERVICE TAIL LANDING GEAR SHOCK STRUT.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-3-4.1	Service Tail Landing Gear Shock Strut (with Helicopter on Jacks)	1-3-4-2	1-3-4.2	Service Tail Landing Gear Shock Strut (with Helicopter on Ground)	1-3-4-9

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Beaker, 800 ml
- Container, 1-Quart
- Hydraulic Fluid Dispenser, AF5
- Nitrogen Servicing Unit, 901002
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Leak Detection Fluid, Item 189, Appendix D

- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Nitrogen, Item 221, Appendix D
- Petrolatum, Item 234, Appendix D
- Rags
- Tubing, Nonmetallic, Item 344, Appendix D
- Union

References

- Appendix D
- PARA 1-2-1
- PARA 1-6-3
- PARA 2-4-97
- PARA 3-4-22
- PARA 3-4-24

1-3-4.1 Service Tail Landing Gear Shock Strut (with Helicopter on Jacks).

NOTE

- Servicing tail landing gear strut with helicopter on jacks is the preferred method of servicing strut.
- If both upper and lower struts require servicing, service upper stage first.
- If installed, external fuel tanks must be defueled and removed prior to servicing struts.

1-3-4.1.1 Service Upper Stage Strut (with Helicopter on Jacks).

CAUTION

- To prevent damage to helicopter, refer only to procedure in this manual for proper charge pressure. Disregard all other labels attached to the helicopter concerning charge pressure.
 - A strut needing service should be carefully checked for hydraulic leakage. Repeated nitrogen servicing of a leaking strut without servicing with hydraulic fluid can lead to a completely nitrogen-inflated strut, causing internal damage.
- a. Remove tail cone access panels 5T-5B-18 and 6T-6B-32 (PARA 1-2-1).
 - b. Remove tail landing gear fairing (PARA 2-4-97).

NOTE

Improper servicing will result if tire is allowed to touch the ground.

- c. Jack rear of helicopter until tail landing gear is fully extended and tire clears the ground. Install jack under jack pad at bottom of rear of tail cone (PARA 1-6-3).

CAUTION

A strut requiring service should be carefully checked for hydraulic leakage. Repeated nitrogen servicing of a leaking strut without servicing with hydraulic fluid can lead to a completely nitrogen-inflated strut, causing internal damage.

- d. Remove cap from upper stage air valve (Figure 1-3-4-1, Detail B).

WARNING

- Injury to personnel and damage to equipment will result if area under helicopter is not clear prior to servicing landing gear.
 - Injury to personnel and damage to equipment will result if all nitrogen pressure is not released prior to removing air valve. Shock strut is under high pressure. Make sure all nitrogen pressure is released prior to removing valve.
- e. Release nitrogen pressure from upper stage strut as follows:
 - (1) Slip nonmetallic tubing, Item 344, Appendix D, over end of air valve. Place free end in 1-quart container to collect any hydraulic fluid discharge when valve is opened.
 - (2) Slowly turn outer hexnut counterclockwise until you hear nitrogen hissing from air valve.
 - (3) When nitrogen pressure is completely released from shock strut, measure collected hydraulic fluid, if any, using 800 ml beaker.
- (a) **IF DISCHARGE OF HYDRAULIC FLUID FROM VALVE IS GREATER THAN 80 MIL-**

LILITERS (CUBIC CENTIMETERS), REPLACE SHOCK STRUT (PARA 3-4-22).

(b) **IF DISCHARGE IS BETWEEN 40 AND 80 MILLILITERS (CUBIC CENTIMETERS), RESERVICE STRUT AND REINSPECT SHOCK STRUT AT 200 FLIGHT HOURS.**

(c) **IF DISCHARGE IS LESS THAN 40 MILLILITERS (CUBIC CENTIMETERS), RESERVICE STRUT AND REINSPECT SHOCK STRUT AT 500 FLIGHT HOURS.**

(4) Remove tube from air valve.

(5) Turn inner hexnut counterclockwise and remove upper air valve.

f. Using hydraulic fluid dispenser, bottom floating piston in upper stage strut as follows:

NOTE

Under emergency conditions, hydraulic fluid, Item 155, Appendix D, may be used to service struts. However, reservice struts with hydraulic fluid, Item 154, Appendix D, as soon as convenient.

(1) Service hydraulic fluid dispenser with hydraulic fluid, Item 154, Appendix D.

(2) Connect hydraulic fluid dispenser to upper stage kneeling valve quick-disconnect fitting (Figure 1-3-4-1, Detail A).

(3) Turn upper stage kneeling valve knob to OPEN.

NOTE

- If fluid dispenser pressure drops below 10 psi, recharge fluid dispenser.
- Floating piston is bottomed when a dull thud is heard.

(4) Slowly release pressurized fluid from fluid dispenser into shock strut until floating piston bottoms. **DO NOT EXCEED 100 PSI HYDRAULIC PRESSURE.**

(5) Release pressure from dispenser.

g. Bleed upper stage strut as follows:

NOTE

Place machinery towels, Item 211, Appendix D, under upper bleeder plug to catch surplus hydraulic fluid when upper bleeder plug is removed.

(1) Remove upper bleeder plug from upper stage strut.

(2) Install union with packing in bleeder port. Connect an overflow line to union.

(3) Place overflow line from upper bleeder port into 1-quart container to catch overflow of fluid.

(4) Place jack underneath tail landing gear fork (PARA 1-6-3).

(5) Slowly fully compress upper stage by jacking tail landing gear fork until no more than 1/2-inch of chrome piston is visible.

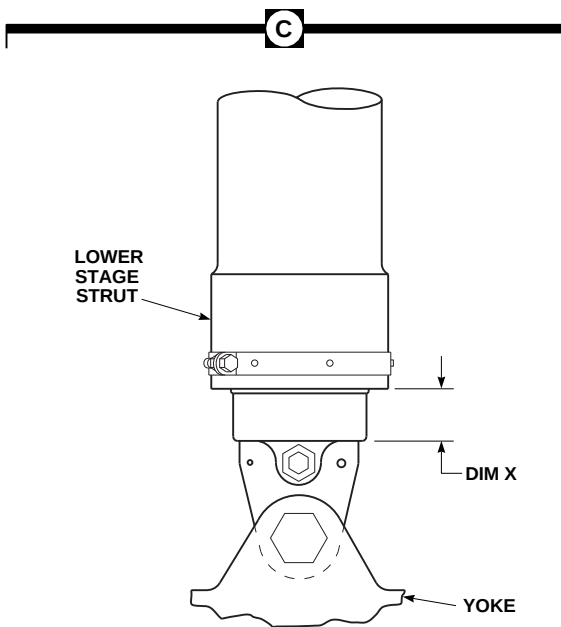
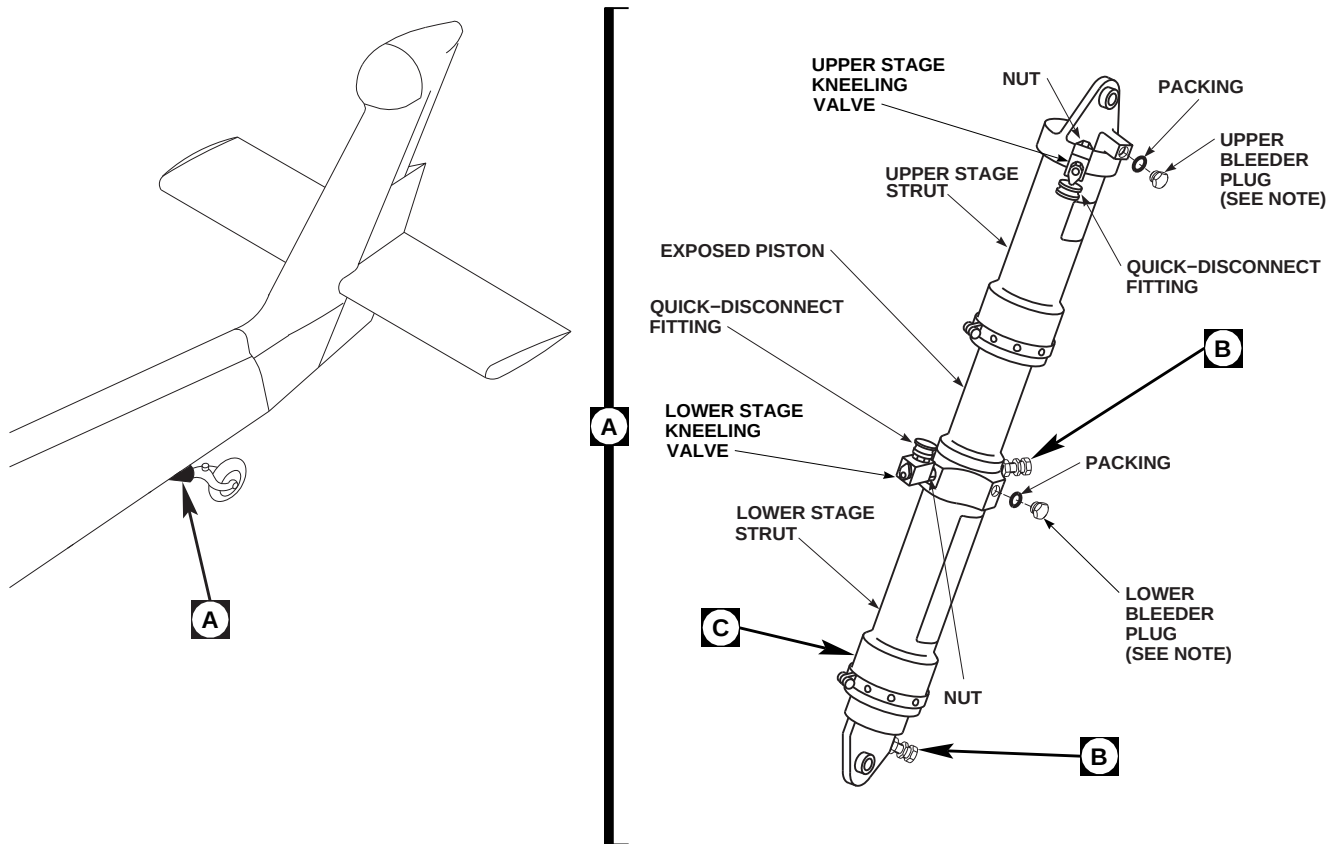
(6) Watch fluid flowing from overflow line.

(7) If fluid contains bubbles, perform the following:

(a) Install upper bleeder plug and packing. **TIGHTEN BLEEDER PLUG 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

(b) Remove jack from under tail landing gear fork (PARA 1-6-3).

(c) Slowly release pressurized fluid from fluid dispenser into shock strut until shock strut is fully extended.



NOTE

TIGHTEN BLEEDER PLUG 1/6 TO 1/3
TURN PAST POINT WHERE SHARP RISE
IN TORQUE IS FELT.

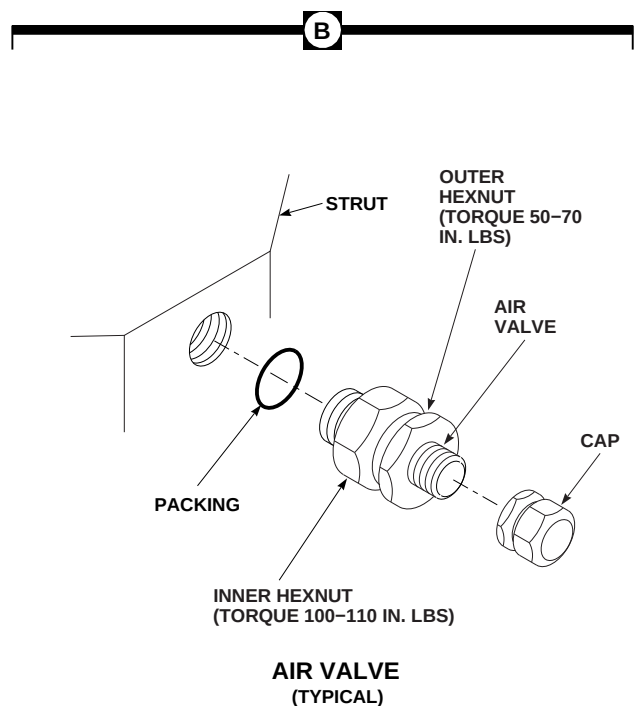


FIGURE 1-3-4-1. SERVICE TAIL LANDING GEAR

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When strut is fully extended, a sharp rise in pressure will occur. **DO NOT EXCEED 100 PSI HYDRAULIC PRESSURE.**

- (d) Repeat step f. (5) through step g. (6).
- (8) Stop when bubbles are no longer seen in fluid and strut is fully compressed. Turn upper stage kneeling valve knob to CLOSE.
- (9) Remove overflow line, union, and packing from upper bleeder plug port.
- (10) Install new packing lubricated with petrolatum, Item 234, Appendix D, on upper bleeder plug.
- (11) Install upper bleeder plug and packing into strut upper cylinder. **TIGHTEN BLEEDER PLUG HANDTIGHT.**
- (12) Turn upper stage kneeling valve knob to OPEN.
- (13) Remove jack from under tail landing gear fork (PARA 1-6-3).
- (14) Slowly release pressurized fluid from fluid dispenser into upper stage until strut begins to extend. **DO NOT EXCEED 100 PSI HYDRAULIC PRESSURE.**
- (15) Release pressure from fluid dispenser.
- (16) Place rags around upper bleeder plug. Loosen bleeder plug.
- (17) Place jack underneath tail landing gear fork (PARA 1-6-3).
- (18) Slowly fully compress upper stage by jacking tail landing gear fork until no more than 1/2-inch of chrome piston is visible.
- (19) Slowly release pressurized fluid from fluid dispenser into shock strut and monitor flow of oil until air bubbles are no longer seen.
- (20) Stop pumping fluid. **TIGHTEN BLEEDER PLUG HANDTIGHT.**
- (21) Turn upper stage kneeling valve knob to CLOSE. Using lockwire, Item 197, Appendix D, lockwire knob to hole in upper stage kneeling valve housing.
- (22) **TIGHTEN BLEEDER PLUG 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Using lockwire, Item 197, Appendix D, lockwire bleeder plug to upper cylinder.
- (23) Disconnect fluid dispenser from upper stage kneeling valve quick-disconnect fitting.
- (24) Install new packing lubricated with petrolatum, Item 234, Appendix D, on upper air valve (Figure 1-3-4-1, Detail B).
- (25) Screw upper air valve and packing into upper stage strut. **TORQUE INNER HEXNUT TO 100 - 110 INCH-POUNDS.**
- (26) Remove jack from under tail landing gear fork (PARA 1-6-3).
- h. Service upper stage strut with nitrogen as follows:
 - (1) Attach servicing chuck from a regulated nitrogen servicing unit to upper air valve.

CAUTION

- Damage to strut will result if overpressurized during servicing. Internal damage or strut failure will result. Do not overpressurize strut during servicing.
- To prevent damage to helicopter, check part number on shock strut before servicing; then refer to procedure in this manual for proper charge pressure. Disregard all other labels attached to the helicopter.

NOTE

Improper servicing will result if tire is allowed to touch the ground.

- (2) Slowly apply pressure using nitrogen, Item 221, Appendix D, to fully extend upper strut. Continue until gage on nitrogen servicing unit reads as follows:

SHOCK STRUT

PART NUMBER	PRESSURE (PSIG)
70250-13101-041	875 to 925
70250-13101-042, -043, -044	1000 to 1050

- (3) Tire must not touch ground when lower strut is fully extended.
- (4) **TORQUE OUTER HEXNUT TO 50 - 70 INCH-POUNDS.**
- (5) Disconnect servicing chuck from air valve.
- i. Apply leak detection fluid, Item 189, Appendix D, to upper air valve and make sure air valve does not leak.
- j. If bubbles appear, perform the following:
 - (1) Determine if packing or air valve is leaking.
 - (2) Replace tail landing gear shock strut air valve or packing as required (PARA 3-4-22).
 - (3) Repeat step h. and step i.
- k. Install air valve cap.
- l. If lower stage is being serviced, proceed to PARA 1-3-4.1.2.

NOTE

If lower stage is to be serviced, do not remove jack from bottom of rear of tail cone.

- m. If lower stage will not be serviced, remove jack from jack pad at bottom of rear of tail cone (PARA 1-6-3).
- n. If lower stage will not be serviced, install tail landing gear fairing (PARA 2-4-97).
- o. If lower stage will not be serviced, install tail cone access panels 5T-5B-18 and 6T-6B-32 (PARA 1-2-1).

1-3-4.1.2 Service Lower Stage Strut (with Helicopter on Jacks).**CAUTION**

- To prevent damage to helicopter, refer only to procedure in this manual for proper charge pressure. Disregard all other labels attached to the helicopter concerning charge pressure.
- A strut needing service should be carefully checked for hydraulic leakage. Repeated nitrogen servicing of a leaking strut without servicing with hydraulic fluid can lead to a completely nitrogen-inflated strut, causing internal damage.

NOTE

Servicing tail landing gear strut with helicopter on jacks is the preferred method of servicing strut.

- a. Remove tail cone access panels 5T-5B-18 and 6T-6B-32 (PARA 1-2-1).
- b. Remove tail landing gear fairing (PARA 2-4-97).

NOTE

Improper servicing will result if tire is allowed to touch the ground.

- c. Jack rear of helicopter until tail landing gear is fully extended and tire clears the ground. Install jack under jack pad at bottom of rear of tail cone (PARA 1-6-3).

- d. Remove cap from lower air valve (Figure 1-3-4-1, Detail B).

WARNING

Injury to personnel and damage to equipment will result if all nitrogen pressure is not released prior to removing air valve. Shock strut is under high pressure. Make sure all nitrogen pressure is released prior to removing valve.

- e. Release nitrogen pressure from lower stage strut as follows:

- (1) Slip nonmetallic tubing, Item 344, Appendix D, over end of air valve. Place free end in 1-quart container to collect any hydraulic fluid discharge when valve is opened.

- (2) Slowly turn outer hexnut counterclockwise until you hear nitrogen hissing from air valve.

- (3) When nitrogen pressure is completely released from shock strut, measure collected hydraulic fluid, if any, using 800 ml beaker.

- (a) **IF DISCHARGE OF HYDRAULIC FLUID FROM VALVE IS GREATER THAN 80 MILLILITERS (CUBIC CENTIMETERS), REPLACE SHOCK STRUT (PARA 3-4-22).**

- (b) **IF DISCHARGE IS BETWEEN 40 AND 80 MILLILITERS (CUBIC CENTIMETERS), RESERVICE STRUT AND REINSPECT SHOCK STRUT AT 200 FLIGHT HOURS.**

- (c) **IF DISCHARGE IS LESS THAN 40 MILLILITERS (CUBIC CENTIMETERS), RESERVICE STRUT AND REINSPECT SHOCK STRUT AT 500 FLIGHT HOURS.**

- (4) Remove tube from air valve.

- (5) Turn inner hexnut counterclockwise and remove lower air valve.

- f. Using hydraulic fluid dispenser, bottom floating piston in lower stage strut as follows:

NOTE

Under emergency conditions, hydraulic fluid, Item 155, Appendix D, may be used to service struts. However, reserve struts with hydraulic fluid, Item 154, Appendix D, as soon as convenient.

- (1) Service hydraulic fluid dispenser with hydraulic fluid, Item 154, Appendix D.

- (2) Connect hydraulic fluid dispenser to lower stage kneeling valve quick-disconnect fitting (Figure 1-3-4-1, Detail A).

- (3) Turn lower stage kneeling valve knob to OPEN.

NOTE

- If fluid dispenser pressure drops below 10 psi, recharge fluid dispenser.

- Floating piston is bottomed when a dull thud is heard.

- (4) Slowly release pressurized fluid from fluid dispenser into shock strut until floating piston bottoms. **DO NOT EXCEED 100 PSI HYDRAULIC PRESSURE.**

- (5) Release pressure from dispenser.

- g. Bleed lower stage strut as follows:

NOTE

Place machinery towels, Item 211, Appendix D, under lower bleeder plug to catch surplus hydraulic fluid when lower bleeder plug is removed.

- (1) Remove lower bleeder plug from lower stage strut.

- (2) Install union with packing in bleeder port. Connect an overflow line to union.
- (3) Place overflow line from lower bleeder port into 1-quart container to catch overflow of fluid.
- (4) Place jack underneath jack pad on tail landing gear fork (PARA 1-6-3).
- (5) Slowly fully compress lower stage by jacking tail landing gear until no more than $\frac{7}{16}$ -inch of chrome piston is visible.
- (6) Watch fluid flowing from overflow line.
- (7) If fluid contains bubbles, perform the following:
 - (a) Install lower bleeder plug and packing. **TIGHTEN BLEEDER PLUG $\frac{1}{6}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
 - (b) Remove jack from under tail landing gear fork (PARA 1-6-3).
 - (c) Slowly release pressurized fluid from fluid dispenser into shock strut until shock strut is fully extended. When strut is fully extended, a sharp rise in pressure will occur. **DO NOT EXCEED 100 PSI HYDRAULIC PRESSURE.**
 - (d) Repeat step f. (5) through step g. (6).
- (8) Stop when bubbles are no longer seen in fluid and strut is fully compressed. Turn lower stage kneeling valve knob to CLOSE.
- (9) Remove overflow line, union, and packing from lower bleeder plug port.
- (10) Install new packing lubricated with petrolatum, Item 234, Appendix D, on lower bleeder plug.
- (11) Install lower bleeder plug and packing into strut lower cylinder. **TIGHTEN BLEEDER PLUG HANDTIGHT.**
- (12) Turn lower stage kneeling valve knob to OPEN.
- (13) Remove jack from under tail landing gear fork (PARA 1-6-3).
- (14) Slowly release pressurized fluid from fluid dispenser into lower stage until strut begins to extend. **DO NOT EXCEED 100 PSI HYDRAULIC PRESSURE.**
- (15) Release pressure from fluid dispenser.
- (16) Place rags around lower bleeder plug. Loosen bleeder plug.
- (17) Place jack underneath tail landing gear fork (PARA 1-6-3).
- (18) Slowly fully compress lower stage by jacking tail landing gear fork until no more than $\frac{7}{16}$ -inch of chrome piston is visible.
- (19) Slowly release pressurized fluid from fluid dispenser into shock strut and monitor flow of oil until air bubbles are no longer seen.
- (20) Stop pumping fluid. **TIGHTEN BLEEDER PLUG HANDTIGHT.**
- (21) Turn lower stage kneeling valve knob to CLOSE. Using lockwire, Item 197, Appendix D, lockwire knob to hole in lower stage kneeling valve housing.
- (22) **TIGHTEN BLEEDER PLUG $\frac{1}{6}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Using lockwire, Item 197, Appendix D, lockwire bleeder plug to lower cylinder.
- (23) Disconnect fluid dispenser from lower stage kneeling valve quick-disconnect fitting.
- (24) Install new packing lubricated with petrolatum, Item 234, Appendix D, on lower air valve (Figure 1-3-4-1, Detail B).
- (25) Screw lower air valve and packing into lower stage strut. **TORQUE INNER HEXNUT TO 100 - 110 INCH-POUNDS.**

- (26) Remove jack from under tail landing gear fork (PARA 1-6-3).
- h. Service lower stage strut with nitrogen as follows:

(1) Attach servicing chuck from a regulated nitrogen servicing unit, Item 221, Appendix D, to lower air valve.



- Damage to strut will result if overpressurized during servicing. Internal damage or strut failure will result. Do not overpressurize strut during servicing.
- To prevent damage to helicopter, check part number on shock strut before servicing; then refer to procedure in this manual for proper charge pressure. Disregard all other labels attached to the helicopter.

NOTE

Improper servicing will result if tire is allowed to touch the ground.

- (2) Slowly apply pressure, using nitrogen, Item 221, Appendix D, to fully extend lower strut. Continue until gage on nitrogen servicing unit reads as follows:

SHOCK STRUT PART NUMBER	PRESSURE (PSIG)
70250-13101-041	170 to 220
70250-13101-042, -043, -044	250 to 300

- (3) Tire must not touch ground when lower strut is fully extended.

- (4) **TORQUE OUTER HEXNUT TO 50 - 70 INCH-POUNDS.**
- (5) Disconnect servicing chuck from air valve.

i. Apply leak detection fluid, Item 189, Appendix D, to lower air valve and make sure valve does not leak.

j. If bubbles appear, perform the following:

(1) Determine if packing or air valve is leaking.
 (2) Replace tail landing gear shock strut air valve or packing as required (PARA 3-4-24).
 (3) Repeat step h. and step i.

k. Using lockwire, Item 197, Appendix D, lockwire lower bleeder plug to lower strut.

l. Install lower air valve cap.

m. Using lockwire, Item 197, Appendix D, lockwire air valve to piston.

n. Make sure area is clean and free of foreign material.

o. Install tail cone access panels 5T-5B-18 and 6T-6B-32 (PARA 1-2-1).

p. Remove jack at bottom of rear of tail cone and lower rear of helicopter (PARA 1-6-3).

1-3-4.2 Service Tail Landing Gear Shock Strut (with Helicopter on Ground).

NOTE

- If both upper and lower struts require servicing, service upper stage first.
- If installed, external fuel tanks must be defueled and removed prior to servicing struts.

1-3-4.2.1 Service Upper Stage Strut (with Helicopter on Ground).

CAUTION

- To prevent damage to helicopter, refer only to procedure in this manual for proper charge pressure. Disregard all other labels attached to the helicopter concerning charge pressure.
 - A strut needing service should be carefully checked for hydraulic leakage. Repeated nitrogen servicing of a leaking strut without servicing with hydraulic fluid can lead to a completely nitrogen-inflated strut, causing internal damage.
- a. Remove tail cone access panels 5T-5B-18 and 6T-6B-32 (PARA 1-2-1).
 - b. Remove tail landing gear fairing (PARA 2-4-97).
 - c. Remove cap from upper stage air valve (Figure 1-3-4-1, Detail B).

WARNING

- Injury to personnel and damage to equipment will result if area under helicopter is not clear prior to servicing landing gear.
 - Helicopter will settle up to 13.6-inches. Look under helicopter and make sure area is clear of personnel and equipment.
 - Injury to personnel and damage to equipment will result if all nitrogen pressure is not released prior to removing air valve. Shock strut is under high pressure. Make sure all nitrogen pressure is released prior to removing valve.
- d. Release nitrogen pressure from upper stage strut as follows:

- (1) Slip nonmetallic tubing, Item 344, Appendix D, over end of air valve. Place free end in 1-quart container to collect any hydraulic fluid discharge when valve is opened.
- (2) Slowly turn outer hexnut counterclockwise until you hear nitrogen hissing from air valve.
- (3) When nitrogen pressure is completely released from shock strut, measure collected hydraulic fluid, if any, using 800 ml beaker.
 - (a) **IF DISCHARGE OF HYDRAULIC FLUID FROM VALVE IS GREATER THAN 80 MILLILITERS (CUBIC CENTIMETERS), REPLACE SHOCK STRUT (PARA 3-4-22).**
 - (b) **IF DISCHARGE IS BETWEEN 40 AND 80 MILLILITERS (CUBIC CENTIMETERS), RESERVICE STRUT AND REINSPECT SHOCK STRUT AT 200 FLIGHT HOURS.**
 - (c) **IF DISCHARGE IS LESS THAN 40 MILLILITERS (CUBIC CENTIMETERS), RESERVICE STRUT AND REINSPECT SHOCK STRUT AT 500 FLIGHT HOURS.**
- (4) Remove tube from air valve.
- (5) Turn inner hexnut counterclockwise and remove upper air valve.
- e. Using hydraulic fluid dispenser, bottom floating piston in upper stage strut as follows:

NOTE

Under emergency conditions, hydraulic fluid, Item 155, Appendix D, may be used to service struts. However, reservice struts with hydraulic fluid, Item 154, Appendix D, as soon as convenient.

- (1) Service hydraulic fluid dispenser with hydraulic fluid, Item 154, Appendix D.

- (2) Connect hydraulic fluid dispenser to upper stage kneeling valve quick-disconnect fitting (Figure 1-3-4-1, Detail A).
- (3) Turn upper stage kneeling valve knob to OPEN.

NOTE

- If fluid dispenser pressure drops below 10 psi, recharge fluid dispenser.
 - Floating piston is bottomed when a dull thud is heard.
- (4) Release pressurized fluid into shock strut until floating piston bottoms. **DO NOT EXCEED 100 PSIG.**
 - (5) Release pressure from dispenser.

- f. Bleed upper stage strut as follows:

NOTE

Place machinery towels, Item 211, Appendix D, under upper bleeder plug to catch surplus hydraulic fluid when upper bleeder plug is removed.

- (1) Remove upper bleeder plug from upper stage strut.
- (2) Install union with packing in bleeder port. Connect an overflow line to union.
- (3) Place overflow line from upper bleeder port into 1-quart container to catch overflow of fluid.
- (4) Release pressurized fluid into strut and watch fluid flowing from overflow line.
- (5) Stop when nitrogen bubbles are no longer seen in fluid. Turn upper stage kneeling valve knob to CLOSE. Using lockwire, Item 197, Appendix D, lockwire knob to hole in upper stage kneeling valve housing.
- (6) Disconnect coupler and hydraulic fluid dispenser from upper quick-disconnect fitting.

- (7) Remove overflow line, union, and packing from upper bleeder plug port.
- (8) Install new packing lubricated with petrolatum, Item 234, Appendix D, on upper bleeder plug.
- (9) Install upper bleeder plug and packing into strut upper cylinder. **TIGHTEN BLEEDER PLUG 1/4 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Using lockwire, Item 197, Appendix D, lockwire bleeder plug to upper cylinder.
- (10) Install new packing lubricated with petrolatum, Item 234, Appendix D, on upper air valve (Figure 1-3-4-1, Detail B).
- (11) Screw upper air valve and packing into upper stage strut. **TORQUE INNER HEXNUT TO 100 - 110 INCH-POUNDS.**

- g. Service upper stage strut with nitrogen as follows:

- (1) Attach servicing chuck from a regulated nitrogen servicing unit to upper air valve.

CAUTION

- Damage to strut will result if overpressurized during servicing. Internal damage or strut failure will result. Do not overpressurize strut during servicing.
 - To prevent damage to helicopter, check part number on shock strut before servicing; then refer to procedure in this manual for proper charge pressure. Disregard all other labels attached to the helicopter.
- (2) Slowly apply pressure using nitrogen, Item 221, Appendix D, to fully extend upper strut. Continue until gage on nitrogen servicing unit reads as follows:

SHOCK STRUT PART NUMBER	PRESSURE (PSIG)
70250-13101-041	875 to 925
70250-13101-042, -043, -044	1000 to 1050

(3) **TORQUE OUTER HEXNUT TO 50 - 70 INCH-POUNDS.**

(4) Disconnect servicing chuck from air valve.

h. Apply leak detection fluid, Item 189, Appendix D, to upper air valve and make sure valve does not leak.

i. If bubbles appear, perform the following:

(1) Determine if packing or air valve is leaking.

(2) Replace tail landing gear shock strut air valve or packing as required (PARA 3-4-24).

(3) Repeat step g. and step h.

j. Install air valve cap.

k. If lower stage is being serviced, proceed to PARA 1-3-4.2.2.

l. If lower stage will not be serviced, install tail landing gear fairing (PARA 2-4-97).

m. If lower stage will not be serviced, install tail cone access panels 5T-5B-18 and 6T-6B-32 (PARA 1-2-1).

1-3-4.2.2 Service Lower Stage Strut (with Helicopter on Ground).

CAUTION

- To prevent damage to helicopter, refer only to procedure in this manual for

proper charge pressure. Disregard all other labels attached to the helicopter concerning charge pressure.

- A strut needing service should be carefully checked for hydraulic leakage. Repeated nitrogen servicing of a leaking strut without servicing with hydraulic fluid can lead to a completely nitrogen-inflated strut, causing internal damage.

- Remove tail cone access panels 5T-5B-18 and 6T-6B-32 (PARA 1-2-1).
- Remove tail landing gear fairing (PARA 2-4-97).

WARNING

Injury to personnel and damage to equipment will result if area under helicopter is not clear prior to servicing landing gear. Helicopter will settle up to 7-inches. Look under helicopter and make sure area is clear of personnel and equipment.

- Remove cap from lower air valve (Figure 1-3-4-1, Detail B).

WARNING

Injury to personnel and damage to equipment will result if all nitrogen pressure is not released prior to removing air valve. Shock strut is under high pressure. Make sure all nitrogen pressure is released prior to removing valve.

- Release nitrogen pressure from lower stage strut as follows:
 - Slip nonmetallic tubing, Item 344, Appendix D, over end of air valve. Place free end in 1-quart container to collect any hydraulic fluid discharge when valve is opened.

- (2) Slowly turn outer hexnut counterclockwise until you hear nitrogen hissing from air valve.
- (3) When nitrogen pressure is completely released from shock strut, measure collected hydraulic fluid, if any, using 800 ml beaker.
 - (a) **IF DISCHARGE OF HYDRAULIC FLUID FROM VALVE IS GREATER THAN 80 MILLILITERS (CUBIC CENTIMETERS), REPLACE SHOCK STRUT (PARA 3-4-22).**
 - (b) **IF DISCHARGE IS BETWEEN 40 AND 80 MILLILITERS (CUBIC CENTIMETERS), RESERVICE STRUT AND REINSPECT SHOCK STRUT AT 200 FLIGHT HOURS.**
 - (c) **IF DISCHARGE IS LESS THAN 40 MILLILITERS (CUBIC CENTIMETERS), RESERVICE STRUT AND REINSPECT SHOCK STRUT AT 500 FLIGHT HOURS.**
- (4) Remove tube from air valve.
- (5) Turn inner hexnut counterclockwise and remove lower air valve.
- e. Using hydraulic fluid dispenser, bottom floating piston in lower stage strut as follows:

NOTE

Under emergency conditions, hydraulic fluid, Item 155, Appendix D, may be used to service struts. However, reservice struts with hydraulic fluid, Item 154, Appendix D, as soon as convenient.

- (1) Service hydraulic fluid dispenser with hydraulic fluid, Item 154, Appendix D.
- (2) Connect hydraulic fluid dispenser to lower stage kneeling valve quick-disconnect fitting (Figure 1-3-4-1, Detail A).

- (3) Turn lower stage kneeling valve knob to OPEN.

NOTE

- If fluid dispenser pressure drops below 10 psi, recharge fluid dispenser.
 - Floating piston is bottomed when a dull thud is heard.
- (4) Release pressurized fluid into shock strut until floating piston bottoms. **DO NOT EXCEED 100 PSIG.**
 - (5) Release pressure from dispenser.
 - f. Bleed lower stage strut as follows:

NOTE

Place machinery towels, Item 211, Appendix D, under lower bleeder plug to catch surplus hydraulic fluid when lower bleeder plug is removed.

- (1) Remove lower bleeder plug from lower stage strut.
- (2) Install union with packing in bleeder port. Connect an overflow line to union.
- (3) Place overflow line from lower bleeder port into 1-quart container to catch overflow of fluid.
- (4) Release pressurized fluid into strut and watch fluid flowing from overflow line.
- (5) Stop when nitrogen bubbles are no longer seen in fluid. Turn lower stage kneeling valve knob to CLOSE. Using lockwire, Item 197, Appendix D, lockwire knob to hole in lower stage kneeling valve housing.
- (6) Disconnect coupler and hydraulic fluid dispenser from lower quick-disconnect fitting.
- (7) Remove overflow line, union, and packing from lower bleeder plug port.

- (8) Install new packing lubricated with petrolatum, Item 234, Appendix D, on lower bleeder plug.
 - (9) Install lower bleeder plug and packing into strut lower cylinder. **TIGHTEN BLEEDER PLUG $\frac{1}{8}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Using lockwire, Item 197, Appendix D, lockwire bleeder plug to lower cylinder.
 - (10) Install new packing lubricated with petrolatum, Item 234, Appendix D, on lower air valve (Figure 1-3-4-1, Detail B).
 - (11) Screw lower air valve and packing into lower stage strut. **TORQUE INNER HEXNUT TO 100 - 110 INCH-POUNDS.**
- g. Service lower stage strut with nitrogen as follows:
- (1) Attach servicing chuck from a regulated nitrogen servicing unit to lower air valve.
 - (2) Look at Figure 1-3-4-1, Detail C, and see where DIM X is measured from.
- CAUTION**
- Damage to strut will result if overpressurized during servicing. Internal damage or strut failure will result. Do not overpressurize strut during servicing.
 - Damage to tail landing gear shock strut will result if tail gear contact is not maintained during servicing. Maintain tail gear ground contact at all times when servicing tail landing gear shock strut.
- (3) Slowly pressurize strut with nitrogen, Item 221, Appendix D. **DO NOT EXCEED 1200 PSIG MAXIMUM PRESSURE WHEN RAISING LOWER TAIL LANDING GEAR STRUT.** As soon as strut starts to extend, shut off pressure. If strut does not extend, rock tail back and forth to overcome possible friction.
 - (4) Check strut pressure.
 - (5) Look in GAGE PRESSURE column of NITROGEN SERVICE CHART and find pressure closest to pressure shown on nitrogen servicing unit gage (Figure 1-3-4-2).
 - (6) Read across chart to DIM X column. Add nitrogen pressure to extend lower strut until DIM X is obtained.
 - (7) **TORQUE OUTER HEXNUT TO 50 - 70 INCH-POUNDS** (Figure 1-3-4-1, Detail B).
 - (8) Disconnect servicing chuck from air valve.
- h. Apply leak detection fluid, Item 189, Appendix D, to lower air valve and make sure valve does not leak.
- i. If bubbles appear, perform the following:
- (1) Determine if packing or air valve is leaking.
 - (2) Replace air valve or packing as required (PARA 3-4-24).
 - (3) Repeat step g. and step h.
- j. Using lockwire, Item 197, Appendix D, lockwire lower bleeder plug to lower strut.
- k. Install lower air valve cap.
- l. Using lockwire, Item 197, Appendix D, lockwire air valve to piston.
- m. Make sure area is clean and free of foreign material.
- n. Install tail landing gear fairing (PARA 2-4-97).
- o. Install tail cone access panels 5T-5B-18 and 6T-6B-32 (PARA 1-2-1).

NITROGEN SERVICE CHART EFFECTIVITY: HELICOPTERS WITH SHOCK STRUT, 70250-13101-041	
GAGE PRESSURE TOLERANCE BETWEEN: -25 TO 25 PSI	DIMENSION X TOLERANCE BETWEEN: -0.12 TO 0.12 IN. (1/8 IN.)
1309	0.50 (1/2)
982	0.88 (7/8)
785	1.25 (1-1/4)
652	1.62 (1-5/8)
557	2.00 (2)
486	2.38 (2-3/8)
431	2.75 (2-3/4)
386	3.12 (3-1/8)
350	3.50 (3-1/2)
320	3.88 (3-7/8)
284	4.25 (4-1/4)
195	FULL EXTENSION

NITROGEN SERVICE CHART EFFECTIVITY: HELICOPTERS WITH SHOCK STRUT, 70250-13101-042, -043, -044	
GAGE PRESSURE TOLERANCE BETWEEN: -25 TO 25 PSI	DIMENSION X TOLERANCE BETWEEN: -0.12 TO 0.12 IN. (1/8 IN.)
1684	0.50 (1/2)
1293	0.88 (7/8)
1054	1.25 (1-1/4)
888	1.62 (1-5/8)
765	2.00 (2)
671	2.38 (2-3/8)
598	2.75 (2-3/4)
540	3.12 (3-1/8)
491	3.50 (3-1/2)
449	3.88 (3-7/8)
415	4.25 (4-1/4)
275	FULL EXTENSION

FIGURE 1-3-4-2. TAIL LANDING GEAR NITROGEN SERVICE CHART

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1-3-5 SERVICE TAIL LANDING GEAR TIRE.

PARAGRAPH BREAKDOWN

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1-3-5.1 Service Tail Landing Gear Tire	1-3-5-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nitrogen Servicing Unit, 901002
- Tire Inflator

Materials/Parts

- Leak Detection Fluid, Item 189, Appendix D
- Nitrogen, Item 221, Appendix D

References

- Appendix D
- PARA 3-4-28

1-3-5.1 Service Tail Landing Gear Tire.

WARNING

- To prevent injury to personnel or damage to equipment, use a tire inflator kit containing a pressure relief valve or similar device. Before inflating tire, check tire inflator to make sure that nitrogen source is vented at 145 psi maximum.
- Tire inflation shall not be attempted if required equipment is inoperative or unavailable.
- To prevent injury to personnel, or damage to equipment, do not inflate tire above its rated inflation pressure.

- To prevent injury to personnel, do not use a high-pressure nitrogen source, or an unregulated nitrogen pressure source.
- To prevent injury to personnel, never stand beside tire being serviced. Stand in front or behind tire.
 - Use regulated nitrogen pressure source connected to tire inflator. **INFLATE TIRE TO 115 - 125 PSI** with nitrogen, Item 221, Appendix D.
 - After a 5 to 10 minute waiting period, check air valve for leakage as follows:
 - Apply leak detection fluid, Item 189, Appendix D, to valve stem area.
 - If bubbles appear, deflate, disassemble, and correct at point of leakage (PARA 3-4-28).
 - Install valve cap on valve stem.

1-3-6 SERVICE/DRAIN ENGINE.

PARAGRAPH BREAKDOWN

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1-3-6.1 Service/Drain Engine	1-3-6-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 2-Gallon

Materials/Parts

- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 208, Appendix D

References

- Appendix D
- SEI 671
- TM 1-2840-248-23

1-3-6.1 Service/Drain Engine.

NOTE

Servicing/draining of both engines is the same, except as noted.

1-3-6.1.1 Service.

NOTE

- Helicopter must be level to get accurate oil tank readings. When helicopter is parked on a slope, downslope engine will read higher oil level than actual, and upslope engine will read lower.
 - Do not check engine oil level until at least 20 minutes after engine shutdown. This allows oil to drain back into oil tank.
 - An empty engine will require 7 quarts of oil.
- a. Look at oil level indicator on engine oil tank through engine air inlet scoop. If oil level is

below midpoint of sight glass, add oil in 1-quart quantities until level is at FULL line of indicator.

- b. Release latch and open engine cowling (Figure 1-3-6-1).
- c. Remove cap from oil fill port on inner side of No. 1 engine, or outer side of No. 2 engine (Figure 1-3-6-1, Detail A).

NOTE

- Do not use lubricating oil, Item 206, Appendix D, in engines. If lubricating oil, Item 206, Appendix D, is inadvertently added to the engines, drain and add the correct oil. There is no need to flush the system.
- Operation at temperatures below -29°F (-34°C) will require use of lubricating oil Item 208, Appendix D. If oils have been mixed, drain and change oil to Item 208, Appendix D.

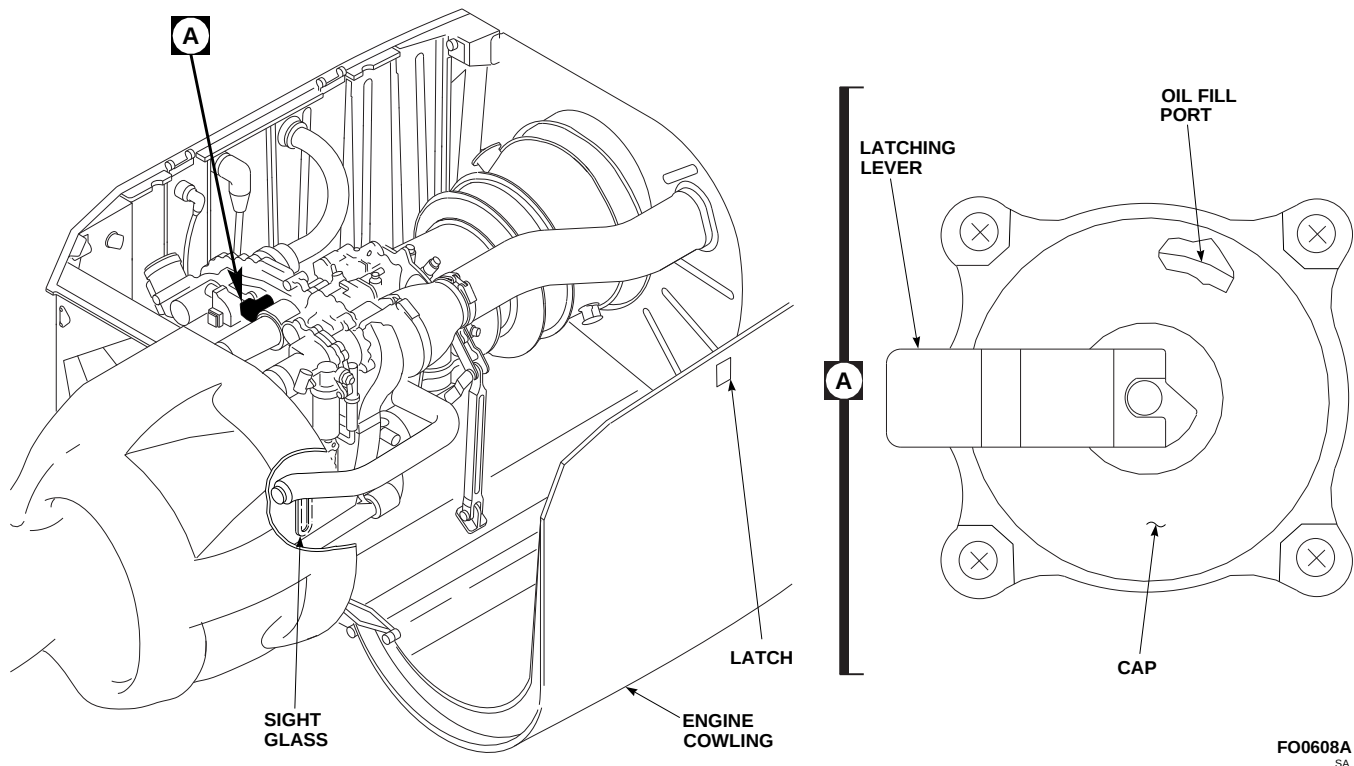


FIGURE 1-3-6-1. SERVICE/DRAIN ENGINE

- When replacing contaminated oil, or when changing oil from one oil specification to another, use instructions in SEI 671 and TM 1-2840-248-23.
- If using lubricating oil, Item 205, Appendix D, refer to General Electric SEI-671 manual for approved oils.
- d. Add lubricating oil, Item 205, Appendix D, or Item 208, Appendix D, to engine oil tank until oil level reaches FULL line of indicator. **DO NOT ADD OIL ABOVE FULL LINE.**

CAUTION

To prevent overboard loss of oil, properly secure the oil cap after servicing.

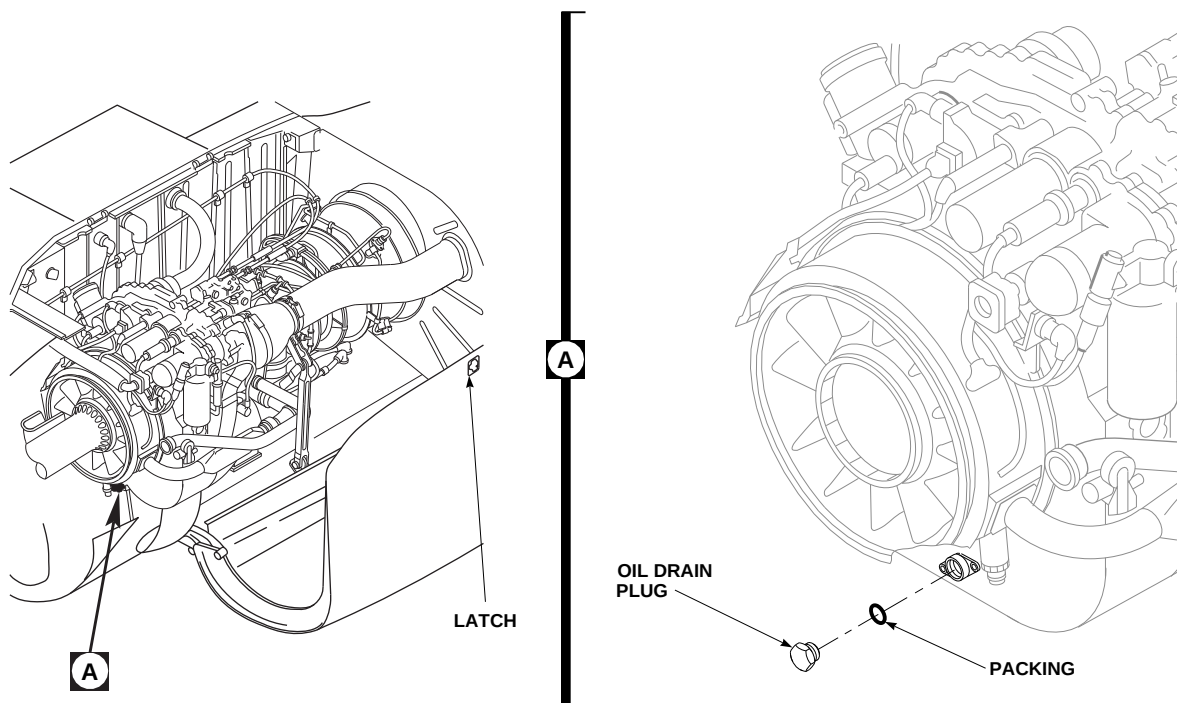
- e. To install cap on oil fill port, perform the following:

- (1) With latching lever turned full counterclockwise (toward OPEN), seat cap in tank opening.
- (2) Turn lever full clockwise (toward CLOSE). Do not press lever now.
- (3) Check cap security by pulling vigorously upward on lever with thumb and forefinger. Cap will remain in tank with locks engaged.
- (4) Press lever and make sure lugs on lever are flush with top of cap.

- f. Make sure area is clean and free of foreign material.
- g. Close and latch engine cowling (Figure 1-3-6-1).

1-3-6.1.2 Drain.

- a. Release latch and open engine cowling (Figure 1-3-6-2).



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FIGURE 1-3-6-2. DRAIN ENGINE OIL

- b. Place 2-gallon container under fuselage at drain line exit.

NOTE

Do not completely loosen and remove oil drain plug, as extra oil may overflow normal drain area and run overboard.

- c. Slowly loosen oil drain plug at the 6 o'clock position on engine main frame and allow oil to drain without running overboard.
- d. When largest amount of oil has been drained, remove drain plug and allow oil to drain for at least 10 minutes.
- e. Remove packing from oil drain plug (Figure 1-3-6-2, Detail A).

- f. Install packing on oil drain plug. Install oil drain plug. Tighten (15° wrench arc) plug as specified in SEI 671 and TM 1-2840-248-23.
- g. Service engine (PARA 1-3-6.1.1).
- h. Remove oil container from under fuselage.
- i. With qualified pilot at controls, run up engine and check for oil leaks.
- j. Make sure area is clean and free of foreign material.
- k. Close and latch engine cowling (Figure 1-3-6-2).

1-3-7 SERVICE ENGINE STARTER.

PARAGRAPH BREAKDOWN

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1-3-7.1 Service Engine Starter	1-3-7-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 208, Appendix D

- Machinery Towel, Item 211, Appendix D

References

- Appendix D
- SEI 671
- TM 1-2840-248-23

1-3-7.1 Service Engine Starter.

- d. Service as follows:

NOTE

- Make sure helicopter is on level ground when servicing starter on helicopter.
 - Do not service starter for at least 60 minutes after shutdown to allow for cooling.
- a. Release latch and open engine cowling if starter is to be serviced on helicopter.

CAUTION

If changing from one oil specification to another or when replacing contaminated oil, refer to SEI 671 and TM 1-2840-248-23

- b. Remove filler cap (Figure 1-3-7-1, Detail A). Discard packing.
- c. Place machinery towel, Item 211, Appendix D, under starter to catch any overflow.

NOTE

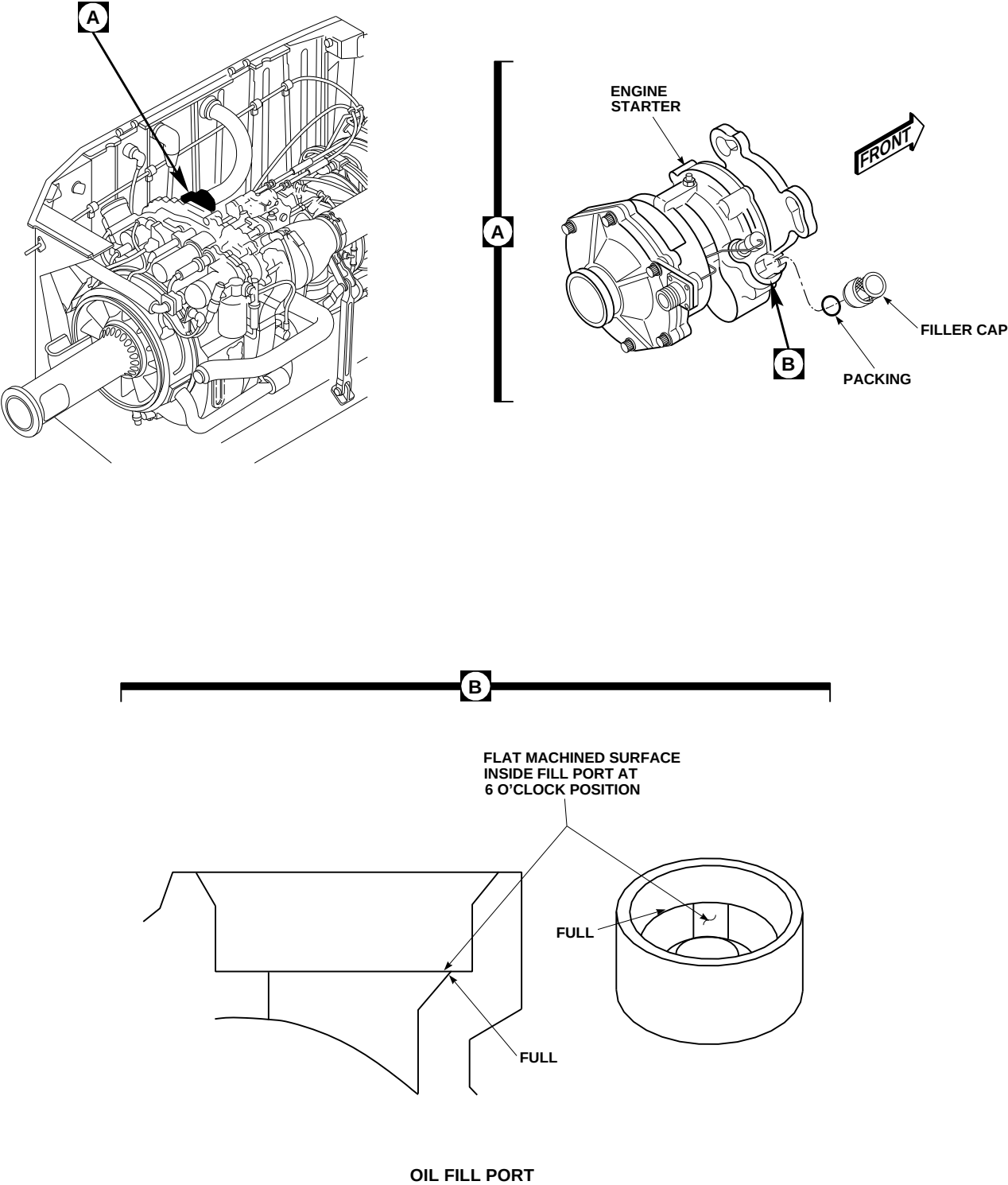
- Operations at temperatures below -40°F (-40°C) will require use of lubricating oil, Item 208, Appendix D.
 - Starter can be serviced while installed on helicopter or while removed.
 - If using lubricating oil, Item 205, Appendix D, refer to General Electric SEI-671 manual for approved oils.
 - When servicing new starter for first time, flush oil cavity with lubricating oil, Item 205, Appendix D, prior to final servicing.
- (1) If starter is to be serviced off helicopter, position starter similar to installed position. Rest starter on drain plug using starter vent as top-dead-center reference point.
- (2) Fill starter with lubricating oil, Item 205 or Item 208, Appendix D, until oil level is

I just above machined surface in fill port.
DO NOT OVER FILL STARTER
(Figure 1-3-7-1, Detail B).

- (3) Lubricate packing with lubricating oil, Item 205 or Item 208, Appendix D.
- (4) Install packing on filler cap (Figure 1-3-7-1, Detail A).

(5) Install filler cap on starter.

- e. Make sure area is clean and free of foreign objects.
- f. Close and latch engine cowling.



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FIGURE 1-3-7-1. SERVICE ENGINE STARTER

1-3-8 SERVICE HYDRAULIC PUMP MODULES.

PARAGRAPH BREAKDOWN

		PAGE
1-3-8.1	Service Hydraulic Pump Modules	1-3-8-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 7-2-1
- PARA 7-4-11

1-3-8.1 Service Hydraulic Pump Modules.

NOTE

- Wait at least 1 hour after operation to check reservoirs, so that fluid will be at outside air temperature.
- After operation, hot hydraulic fluid will expand, causing piston movement into overfill zone. Movement of piston up to

3/8-inch into overfill zone is acceptable. When piston moves more than 3/8-inch into overfill zone, bleed system until piston indicates 3/8-inch above full level.

- Servicing of both left and right hydraulic pump modules is the same.
 - a. Open main rotor pylon sliding cover.

CAUTION

Do not use unseparated gas-pressurized hydraulic service equipment to fill hydraulic handpump reservoir. This type of equipment pumps air/gas into the reservoir. Air/gas trapped in the hydraulic system can cause stiffened/sluggish flight control movement during certain maneuvers.

NOTE

- To prevent damage to helicopter during cold weather operations of -29°F (-34°C) or below, hydraulic fluid must be changed from Item 155, Appendix D to Item 154, Appendix D (PARA 7-4-11). If fluids have been mixed, drain and change to hydraulic fluid, Item 154, Appendix D.
 - Handpump reservoir holds 1.3 quarts of fluid. When fluid level is seen at refill mark in reservoir window, 0.3 quart of fluid is left in reservoir. Add a quart of fluid to fill handpump.
 - Reservoir cover has a can opener to puncture fluid cans.
- b. Service hydraulic pump module with hydraulic fluid, Item 155, Appendix D, as follows:
- (1) Unscrew handpump reservoir cover and pour in hydraulic fluid, Item 155, Appendix D, until reservoir is full (Figure 1-3-8-1). Make sure you can always see fluid in sight gage while servicing, or you will pump air into pump module's reservoir. Keep filling reservoir.
 - (2) Make sure handpump reservoir cover is clean; then screw cover on tight.
 - (3) Turn selector valve handle to desired reservoir to be filled (Figure 1-3-8-1, Detail A). FILL RESV 1 is left pump module, FILL RESV 2 is right pump module, FILL RESV 3 is backup pump module, and FILL RESV 4 is rotor brake system.
 - (4) Place suitable container under drain hose at left landing gear shock strut, STA 308.0.
 - (5) While holding selector valve handle down, crank pump handle on handpump clockwise. Fill desired hydraulic pump module until forward end of piston in reservoir window is past the green area of the reservoir decal and approximately 0.25-inch into the blue area of the reservoir decal (Figure 1-3-8-1, Detail B). Push in and turn bleed valve button, and bleed air from pump module. Release bleed valve button when air-free fluid shows, and the forward end of piston in reservoir window is at forward end of green decal on reservoir housing.
 - (6) Check that reservoirs stay full (forward end of piston at forward end of green decal) when fluid is at ambient temperature (1 hour after flight).
 - (7) Make sure area remains clean during procedure.
 - (8) Position pump handle in either 3 or 9 o'clock position.
- c. Turn on all electrical power (PARA 1-6-16).
- d. Check that all three RSVR LOW lights are off on caution/advisory panel.
- e. If any caution lights are on and fluid level window shows full, check fluid quantity switch (PARA 7-2-1).
- f. Turn off all electrical power (PARA 1-6-16).
- g. Make sure area is clean and free of foreign material.

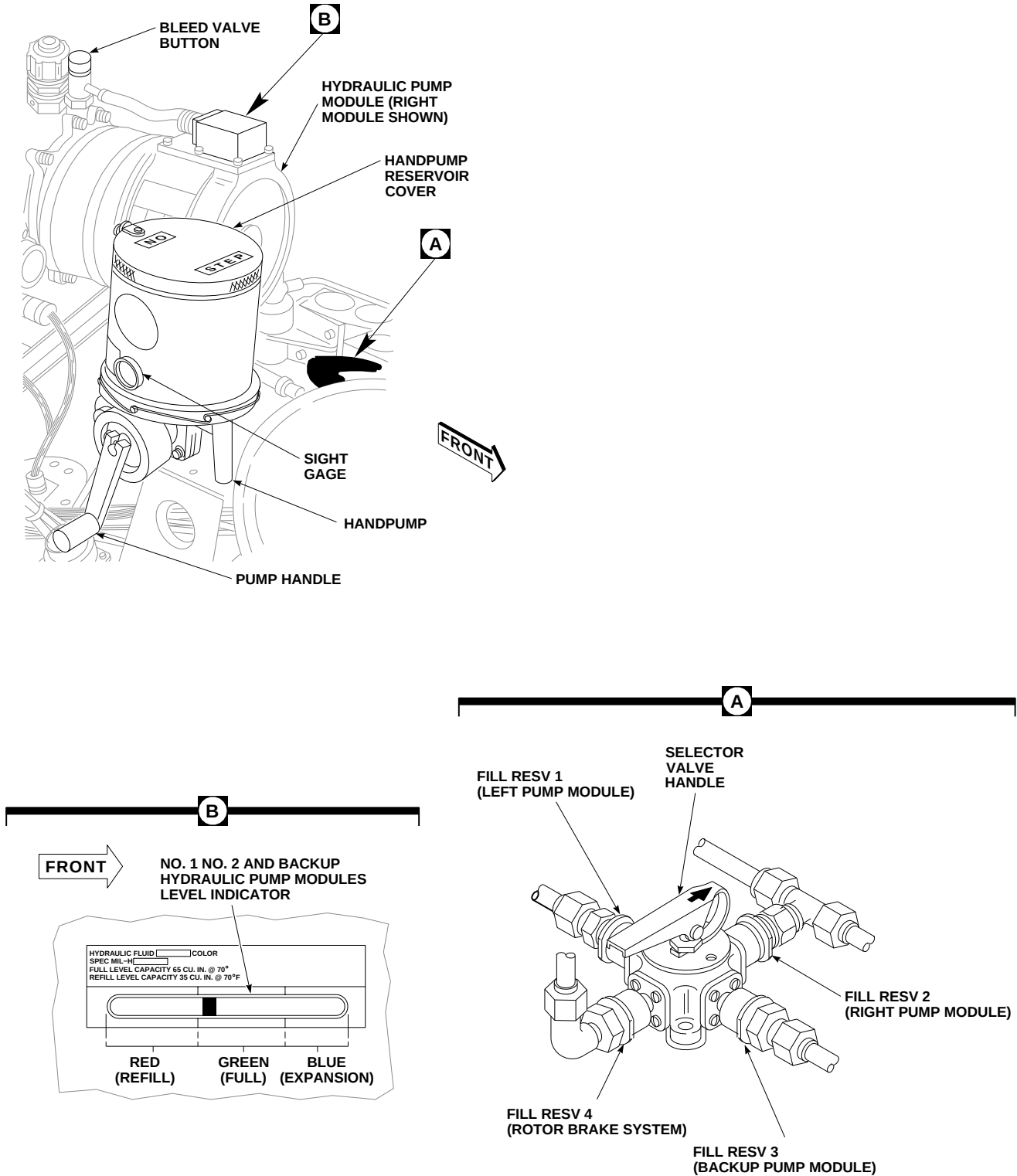


FIGURE 1-3-8-1. SERVICE HYDRAULIC PUMP MODULES

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Damage to hydraulic handpump handle will result if handle is not positioned

correctly before closing main rotor pylon sliding cover. Position pump handle in either 3 or 9 o'clock position before closing cover.

- h. Close and latch main rotor pylon sliding cover.

1-3-9 SERVICE/DRAIN APU.

PARAGRAPH BREAKDOWN

	PAGE
1-3-9.1 Service/Drain APU	1-3-9-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 2-Gallon
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 196, Appendix D
- Lubricating Oil, Item 205, Appendix D

- Lubricating Oil, Item 208, Appendix D

References

- Appendix D
- PARA 15-4-1

1-3-9.1 Service/Drain APU.

1-3-9.1.1 Service.

and check oil level on dipstick (Figure 1-3-9-1, Detail B).



Injury to personnel may result from servicing APU when hot.

- Open APU access doors.

NOTE

To determine correct oil level, the helicopter should be on normal three-point landing gear position on level ground.

- Press down on filler cap, turn ¼-turn counter-clockwise, remove from APU gear box sump,



- Lubricating oil, Item 205 and Item 208, Appendix D, shall not be mixed under any condition. Due to high detergent additives used in processing lubricating oils, any mixture of oils shall be drained as soon as possible. If contamination occurs, refill tank or sump with correct lubricating oil and then drain (this is now waste oil), refill tank or sump with new oil.
- Never overfill oil sump. An overfull sump will result in oil foaming, low oil pressure, and abnormal gear wear.

NOTE

Operation at temperatures below -29°F (-34°C) will require use of lubricating oil, Item 208, Appendix D.

- c. If required, fill sump with lubricating oil, Item 205 or Item 208, Appendix D, to proper oil level shown on filler cap/dipstick.
- d. Install filler cap. Press down on cap and turn ¼ turn clockwise.
- e. Make sure area is clean and free of foreign material.
- f. Close APU access doors.

1-3-9.1.2 Drain.

- a. Remove APU (PARA 15-4-1).

NOTE

Maximum capacity of lubricating oil for APU is three quarts.

- b. Position 2 gallon container to catch oil.
- c. Remove magnetic drain plug and packing on bottom of APU gear box sump and drain oil into container. Discard packing (Figure 1-3-9-1, Detail C).
- d. Check drained oil for any presence of debris. **NONE ALLOWED.** If debris is found, inspect system.
- e. Install magnetic drain plug and packing. **TORQUE DRAIN PLUG TO 45 INCH-POUNDS.**
- f. Using lockwire, Item 196, Appendix D, lockwire drain plug to valve.
- g. Install APU (PARA 15-4-1).

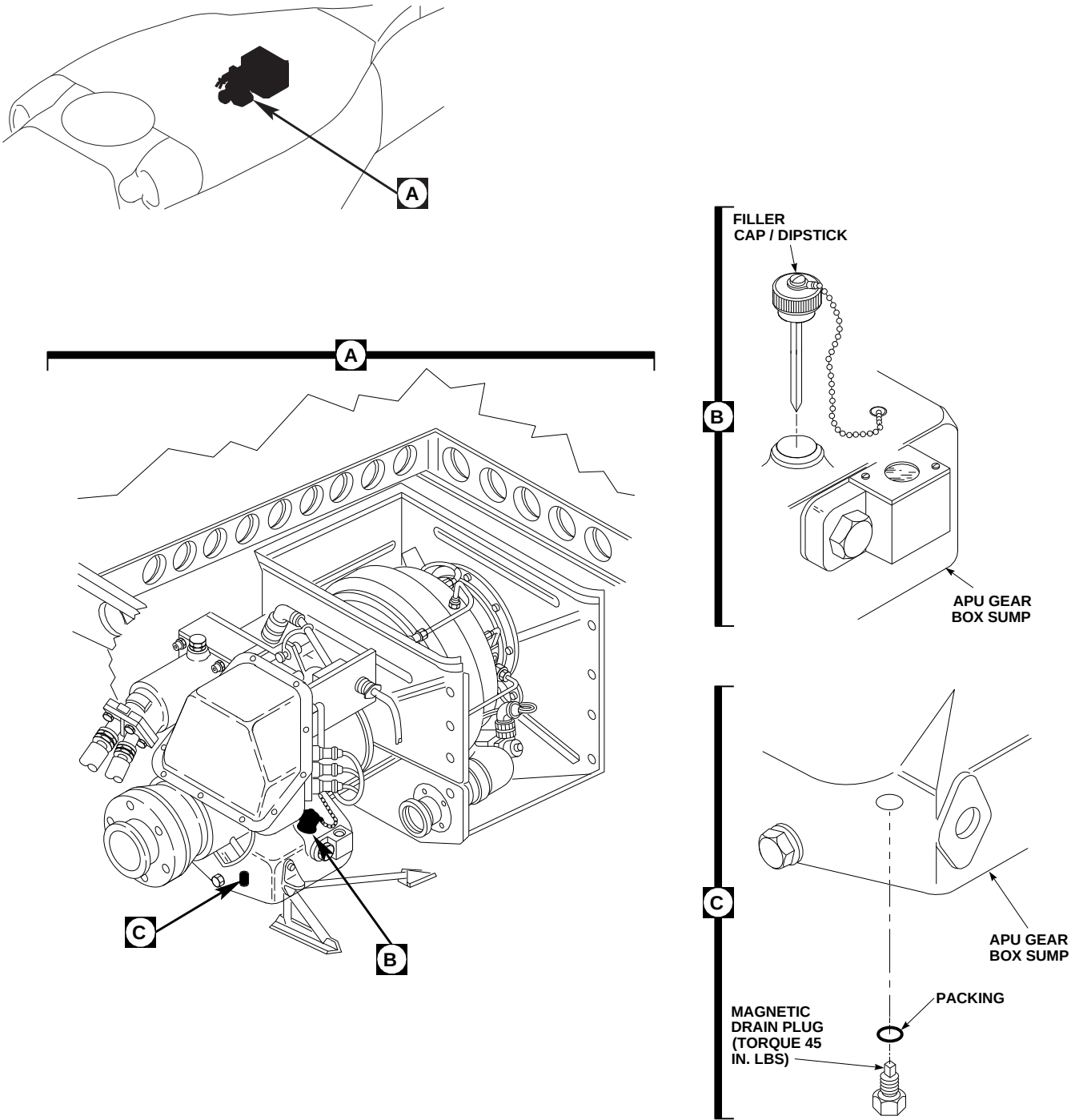


FIGURE 1-3-9-1. SERVICE/DRAIN APU

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1-3-10 SERVICE APU ACCUMULATOR.

PARAGRAPH BREAKDOWN

	PAGE
1-3-10.1 Service APU Accumulator	1-3-10-1

INITIAL SETUP	Nitrogen, Item 221, Appendix D
Tools	References
- Aircraft Mechanic’s Toolkit - Nitrogen Servicing Unit, 901002 - Torque Wrench, 30 - 150 in. lbs	- Appendix D - PARA 1-3-8 - PARA 1-3-11 - PARA 1-6-15 - PARA 1-6-16 - PARA 2-4-69
Materials/Parts	
- Hydraulic Fluid, Item 155, Appendix D - Leak Detection Fluid, Item 189, Appendix D	

1-3-10.1 Service APU Accumulator.

1-3-10.1.1 Check Precharge.



To avoid injury to personnel, make sure pump handle is firmly locked into handle receptacle before operating hand-pump.

NOTE

- APU handpump handle is in the left stowage compartment behind the rear crew compartment.
- If helicopter is equipped with winterization kit on APU, refer to PARA 1-3-11 for additional servicing information.

- Observe pressure gage (Figure 1-3-10-1, Detail A). If pressure gage does not read between 2700 - 3100 psig, use APU handpump to slowly pump accumulator charge to read approximately 2900 psi (Figure 1-3-10-1, Detail C). Allow five minutes for pressure to stabilize.

NOTE

In Graph A, Temperature/Tape Indicator Chart (Figure 1-3-10-2), the upper and lower dotted lines indicate the maximum and minimum value on the tape indicator for a given ambient temperature. The solid line indicates the ideal value on the tape indicator for a given ambient temperature.

- If tape indicator reads above 86% and gage reads below 2700 psig at an ambient temperature of 70°F (21°C), stop pumping. Precharge is too low and must be serviced. Use

Graph A, Temperature/Tape Indicator Chart (Figure 1-3-10-2) for correct indicator reading at ambient temperature other than 70°F (21°C).

- c. With charge at 2900 psig, compare actual tape reading with Graph A. If actual reading does not agree with Graph A, accumulator gas pre-charge is incorrect and must be serviced.

1-3-10.1.2 Service (Previously Installed Accumulator).

CAUTION

Do not use unseparated gas-pressurized hydraulic service equipment to fill hydraulic handpump reservoir. This type of equipment pumps air/gas into the reservoir. Air/gas trapped in the hydraulic system can cause stiffened/sluggish flight control movement during certain maneuvers.

- a. Open main rotor pylon sliding cover.
- b. Remove soundproofing panels from rear cabin ceiling (PARA 2-4-69).

WARNING

Make sure APU is clear when opening and closing lever on start valve solenoid.

NOTE

Accumulator is fully discharged when tape reading on indicator reads 0%.

- c. Manually open lever on start valve solenoid to fully discharge accumulator (Figure 1-3-10-1, Detail B).
- d. Check backup pump reservoir for proper service. Service backup pump module reservoir as required (PARA 1-3-8).
- e. Remove cap from servicing valve on pressure gage (Figure 1-3-10-1, Detail A).

NOTE

The APU accumulator nitrogen charge must be released prior to servicing the APU accumulator with hydraulic fluid.

- f. Loosen locknut on servicing valve and slowly release nitrogen charge.

WARNING

To avoid damage to equipment or injury to personnel, do not service accumulator with free air or oxygen.

- g. Connect nitrogen bottle, Item 221, Appendix D, to servicing valve.
- h. Fully open locknut on nitrogen servicing valve.
- i. Slowly charge accumulator with nitrogen to 1000 psi.
- j. Manually open APU start valve, to make sure accumulator floating piston is bottomed. Indicator to show 0% (Figure 1-3-10-1, Detail A and Detail B).

NOTE

- Floating piston is considered bottomed when APU will no longer motor with start valve open and 1000 psi nitrogen applied.
- The following 1450 psi pressure and tape reading is based on a 70°F (21°C) temperature day. Use the Graph B Temperature/Precharge Pressure Chart (Figure 1-3-10-2) for correct pressure.
- k. Add nitrogen to bring precharge to 1450 psi. Tighten locknut to close servicing valve (Figure 1-3-10-1, Detail A).
- l. Close nitrogen bottle valve and disconnect nitrogen bottle from servicing valve.
- m. **TORQUE LOCKNUT TO 50 - 60 INCH-POUNDS.**

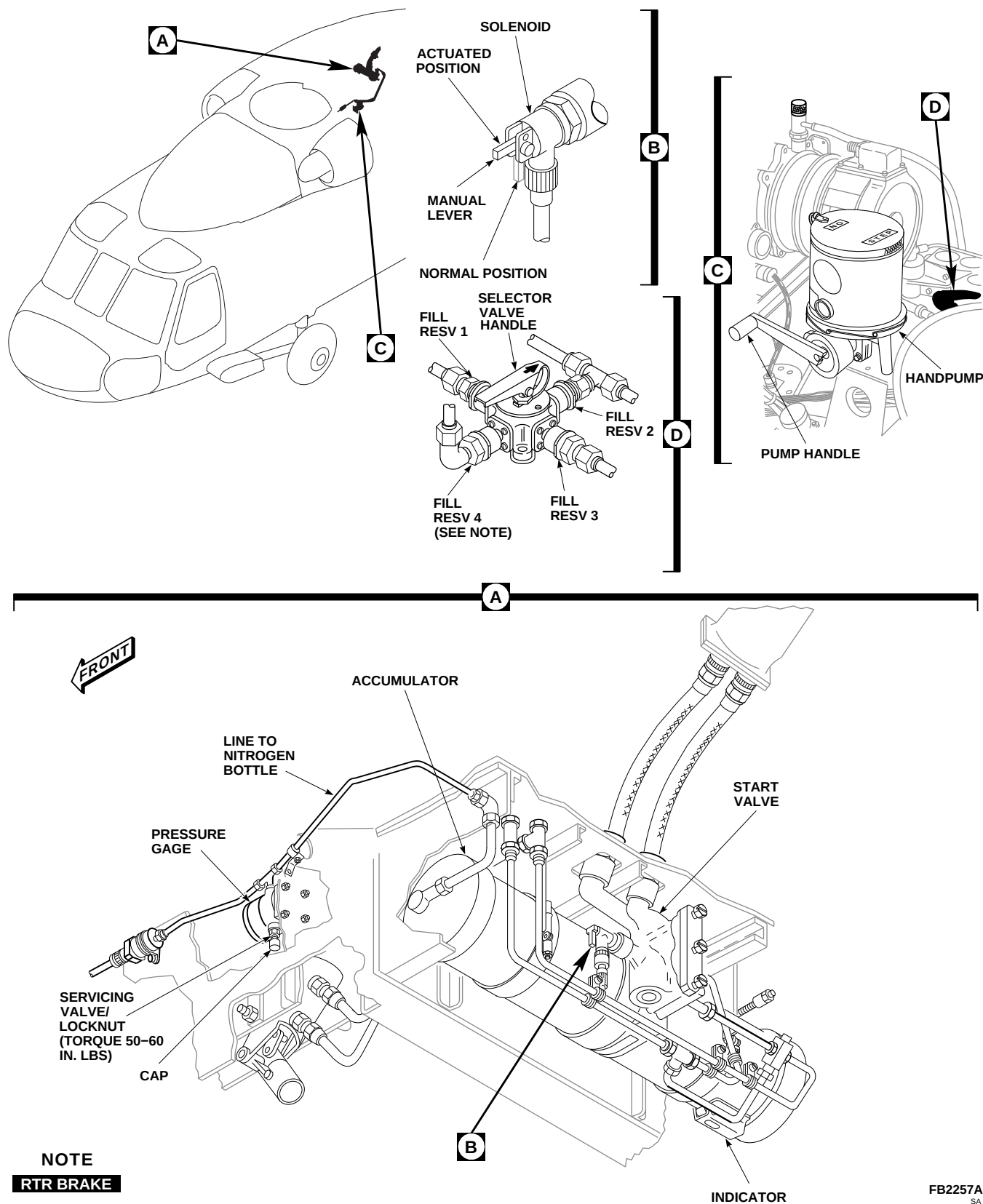
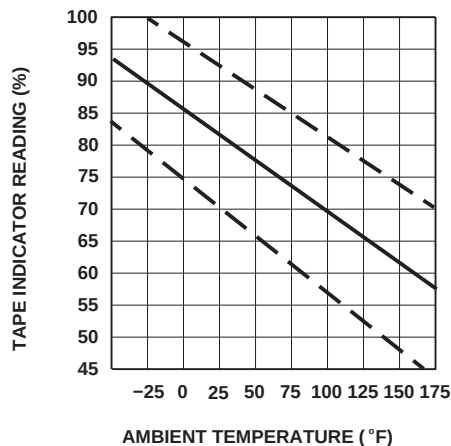
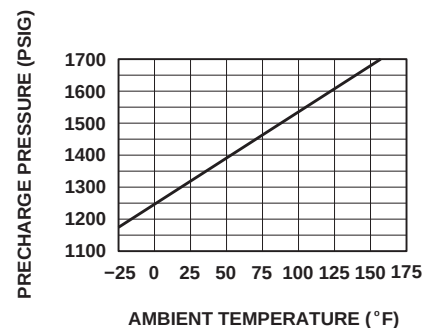


FIGURE 1-3-10-1. SERVICE APU ACCUMULATOR



GRAPH A

TEMPERATURE/TAPE INDICATOR CHART



GRAPH B

TEMPERATURE/PRECHARGE PRESSURE CHART

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FIGURE 1-3-10-2. APU ACCUMULATOR SERVICING CHARTS

- n. Check servicing valve for leaks with leak detection fluid, Item 189, Appendix D.
- o. Install valve cap on servicing valve.

WARNING

To avoid injury to personnel, make sure pump handle is firmly locked into handle receptacle before operating hand-pump.

NOTE

APU handpump handle is in the left stowage compartment behind the rear crew compartment.

- p. Hydraulically charge accumulator using APU handpump to approximately 2900 psi reading

on the accumulator nitrogen gage or operate the backup hydraulic system pump using external electrical power (PARA 1-6-16) until backup pump automatically shuts off. Indicator tape reading to be between 63 and 86. Use Graph A (Figure 1-3-10-2) for correct indicator reading at ambient temperature other than 70°F (21°C).

- q. Check backup pump reservoir for proper service. Service backup pump module reservoir as required (PARA 1-3-8).
- r. Make sure area is clean and free of foreign material.
- s. Install soundproofing panels in rear cabin ceiling (PARA 2-4-69).
- t. Make sure area is clean and free of foreign material.

CAUTION

Damage to hydraulic hand pump handle will result if handle is not positioned correctly before closing main rotor pylon sliding cover. Position pump handle in either 3 or 9 o'clock position before closing cover.

- u. Close and latch main rotor pylon sliding cover.

1-3-10.1.3 Service (Newly Installed Accumulator).

CAUTION

Do not use unseparated gas-pressurized hydraulic service equipment to fill hydraulic handpump reservoir. This type of equipment pumps air/gas into the reservoir. Air/gas trapped in the hydraulic system can cause stiffened/sluggish flight control movement during certain maneuvers.

NOTE

Air must be purged from a new accumulator prior to servicing. The use of external hydraulic power is recommended as the fluid capacity of the accumulator exceeds the fluid available in the backup pump reservoir. If external hydraulic power is not available backup pump may be used but care must be taken to avoid depleting backup pump reservoir.

- a. Open main rotor pylon sliding cover.
- b. Remove soundproofing panels from rear cabin ceiling (PARA 2-4-69).
- c. Service accumulator with external hydraulic power as follows:
 - (1) Remove cap from servicing valve on pressure gage (Figure 1-3-10-1, Detail A).

- (2) Loosen locknut on servicing valve and slowly release nitrogen charge.
- (3) Connect external hydraulic power to backup hydraulic system (PARA 1-6-15).
- (4) With servicing valve locknut loose, slowly fill accumulator with hydraulic fluid, Item 155, Appendix D, until a tape reading of 100% is obtained.
- (5) Reduce hydraulic supply pressure to zero.
- (6) Connect nitrogen bottle, Item 221, Appendix D, to servicing valve.
- (7) Fully open locknut on servicing valve.
- (8) Slowly charge accumulator with nitrogen to 1000 psi.
- (9) Manually open APU start valve. Indicator to show 0% (Figure 1-3-10-1, Detail A and Detail B).

NOTE

- Floating piston is considered bottomed when APU will no longer motor with start valve open and 1000 psi nitrogen applied.
- The following 1450 psi pressure and tape reading is based on a 70°F (21°C) temperature day. Use the Graph B Temperature/Precharge Pressure Chart (Figure 1-3-10-2) for correct pressure.
- (10) Add nitrogen to bring precharge to 1450 psi. Tighten locknut to close servicing valve (Figure 1-3-10-1, Detail A).
- (11) Close nitrogen bottle valve and disconnect nitrogen bottle from servicing valve.
- (12) **TORQUE LOCKNUT TO 50 - 60 INCH-POUNDS.**
- (13) Check servicing valve for leaks with leak detection fluid, Item 189, Appendix D.
- (14) Install valve cap on servicing valve.

WARNING

To avoid injury to personnel, make sure pump handle is firmly locked into handle receptacle before operating hand-pump.

NOTE

APU handpump handle is in the left stowage compartment behind the rear crew compartment.

- (15) Hydraulically charge accumulator using external hydraulic power or APU hand-pump to approximately 2900 psi reading on the accumulator nitrogen gage. Indicator tape reading to be between 63 and 86. Use Graph A (Figure 1-3-10-2) for correct indicator reading at ambient temperature other than 70°F (21°C).
 - (16) Manually open and close lever on start valve solenoid (Figure 1-3-10-1, Detail B).
 - (17) Repeat step c. (15).
 - (18) Disconnect external hydraulic power to backup hydraulic system (PARA 1-6-15).
 - (19) Check backup pump reservoir for proper service. Service backup pump module reservoir as required (PARA 1-3-8).
- d. Service accumulator with backup pump as follows:
- (1) Check backup pump reservoir for proper service. Service backup pump module reservoir as required (PARA 1-3-8).
 - (2) Remove cap from servicing valve on pressure gage (Figure 1-3-10-1, Detail A).
 - (3) Loosen locknut on servicing valve and slowly release nitrogen charge.

NOTE

Backup pump reservoir will need to be serviced per PARA 1-3-8 multiple times before a tape reading of 100% is obtained due to the limited capacity of the backup pump reservoir.

- (4) With nitrogen locknut loose and while watching backup pump reservoir, run backup pump intermittently to slowly fill hydraulic accumulator. When backup pump reservoir is depleted, turn backup pump off and service reservoir (PARA 1-3-8).
- (5) Repeat step d. (4) until a tape reading of 100% is obtained.
- (6) Turn off backup pump.

WARNING

To avoid damage to equipment or injury to personnel, do not service accumulator with free air or oxygen.

- (7) Connect nitrogen bottle, Item 221, Appendix D, to servicing valve.
- (8) Fully open locknut on servicing valve.
- (9) Slowly charge accumulator with nitrogen to 1000 psi.

WARNING

Make sure APU is clear when opening and closing lever on start valve solenoid.

- (10) Manually open APU start valve. Indicator to show 0% (Figure 1-3-10-1, Detail A and Detail B).

NOTE

- Floating piston is considered bottomed when APU will no longer motor with start valve open and 1000 psi nitrogen applied.
 - The following 1450 psi pressure and tape reading is based on a 70°F (21°C) temperature day. Use the Graph B Temperature/Precharge Pressure Chart (Figure 1-3-10-2) for correct pressure.
- (11) Add nitrogen to bring precharge to 1450 psi. Tighten locknut to close servicing valve (Figure 1-3-10-1, Detail A).
 - (12) Close nitrogen bottle valve and disconnect nitrogen bottle from servicing valve.
 - (13) **TORQUE LOCKNUT TO 50 - 60 INCH-POUNDS.**
 - (14) Check servicing valve for leaks with leak detection fluid, Item 189, Appendix D.
 - (15) Install valve cap on servicing valve.

NOTE

- The reservoir capacity of a single accumulator exceeds the reservoir capacity of the backup pump. Once started, the backup pump will run for 90 seconds before shutting off. The backup pump reservoir, however, will be empty in less than 90 seconds. For this reason, it will be required to pull the backup pump power circuit breaker before the reservoir pump runs dry.
- Backup pump power circuit breaker is located on No. 1 DC PRI BUS circuit breaker panel.
- It may be necessary to refill backup pump reservoir 2 to 3 times before accumulator is adequately charged.

- (16) Hydraulically charge accumulator by operating the backup hydraulic system pump using external electrical power (PARA 1-6-16). Shut down backup pump by pulling BACKUP PUMP PWR circuit breaker prior to backup pump reservoir depletion. Indicator tape reading is to be between 63 and 86. Use Graph A (Figure 1-3-10-2) for correct indicator reading at ambient temperature other than 70°F (21°C).
 - (17) Manually open and close lever on start valve solenoid (Figure 1-3-10-1, Detail B).
 - (18) Bleed air from backup pump reservoir and refill reservoir if needed (PARA 1-3-8).
 - (19) Repeat step d. (16).
 - (20) Check backup pump reservoir for proper service. Service backup pump module reservoir as required (PARA 1-3-8).
- e. Make sure area is clean and free of foreign material.
 - f. Install soundproofing panels in rear cabin ceiling (PARA 2-4-69).
 - g. Make sure area is clean and free of foreign material.

CAUTION

Damage to hydraulic hand pump handle will result if handle is not positioned correctly before closing main rotor pylon sliding cover. Position pump handle in either 3 or 9 o'clock position before closing cover.

- h. Close and latch main rotor pylon sliding cover.

1-3-11 SERVICE WINTERIZATION KIT APU ACCUMULATOR.

PARAGRAPH BREAKDOWN

	PAGE
1-3-11.1 Service Winterization Kit APU Accumulator	1-3-11-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nitrogen Servicing Unit, 901002
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Leak Detection Fluid, Item 189, Appendix D

- Nitrogen, Item 221, Appendix D

References

- Appendix D
- PARA 1-3-8
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-69
- PARA 7-4-11

1-3-11.1 Service Winterization Kit APU Accumulator.

1-3-11.1.1 Check Precharge.

- Open main rotor pylon sliding cover.

WARNING

To avoid injury to personnel, before operating handpump, make sure pump handle is firmly locked into receptacle.

NOTE

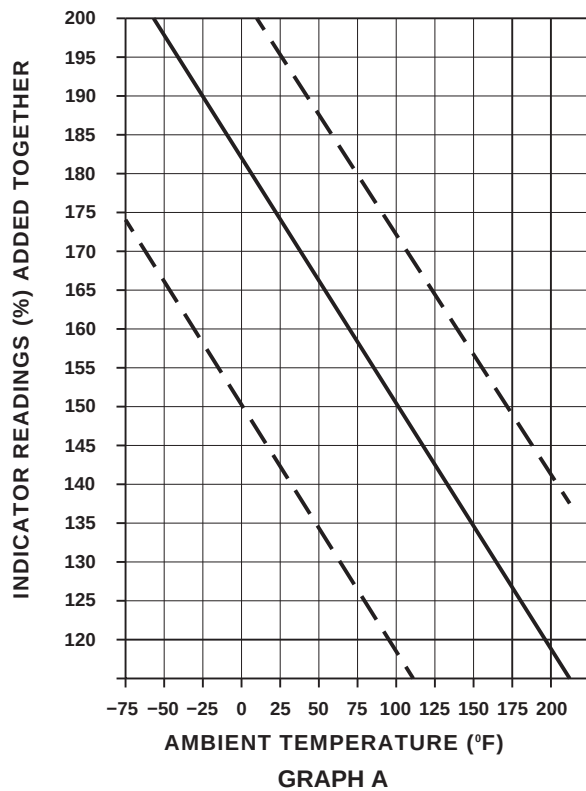
APU handpump handle is in the left stowage compartment behind the rear crew compartment.

- Observe pressure gage (Figure 1-3-11-2, Sheet 1, Detail A). If pressure does not read 2700 - 3100 psig, use APU handpump to slowly pump charge to 2900 psig (Figure 1-3-11-2, Sheet 2, Detail C). Allow several minutes for pressure to stabilize.

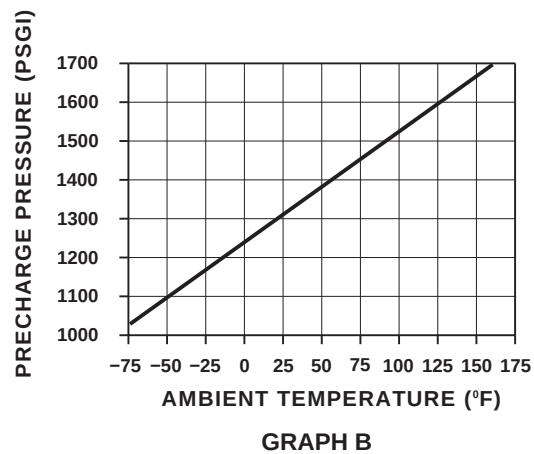
NOTE

In Graph A, Temperature/Tape Indicator Chart (Figure 1-3-11-1), the upper and lower dotted lines indicate the maximum and minimum value on the tape indicator for a given ambient temperature. The solid line indicates the ideal value on the tape indicator for a given ambient temperature.

- If tape indicators (added together) reach 180% before pressure gage reaches 2700 psig at



TEMPERATURE / TAPE INDICATOR CHART



TEMPERATURE / PRECHARGE PRESSURE

FIGURE 1-3-11-1. APU WINTERIZATION KIT SERVICING CHARTS

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ambient temperature of 70°F (21°C), stop pumping. Precharge is too low and must be serviced. Use Graph A (Figure 1-3-11-1) for correct indicator reading at ambient temperature other than 70°F (21°C).

- d. With charge at 2900 psig, compare actual tape readings (added together) with Graph A. If readings do not agree with Graph A, precharge is incorrect and must be serviced.

1-3-11.1.2 Service (Previously Installed Winterization Kit).

CAUTION

To prevent damage to helicopter during cold weather operations of -29°F (-34°C) or below, hydraulic fluid must be changed from Item 155, Appendix D to Item 154, Appendix D (PARA 7-4-11).

CAUTION

Do not use unseparated gas-pressurized hydraulic service equipment to fill hydraulic handpump reservoir. This type of equipment pumps air/gas into the reservoir. Air/gas trapped in the hydraulic system can cause stiffened/sluggish flight control movement during certain maneuvers.

- a. Open main rotor pylon sliding cover.
- b. Remove soundproofing panels from rear cabin ceiling (PARA 2-4-69).

WARNING

Make sure APU is clear when opening and closing lever on start valve solenoid.

NOTE

Accumulators are fully discharged when tape reading on indicator reads 0%.

- c. Manually open lever on start valve solenoid to fully discharge accumulators (Figure 1-3-11-2, Sheet 2, Detail B).
- d. Check backup pump reservoir for proper service. Service backup pump module reservoir as required (PARA 1-3-8).
- e. Remove cap from servicing valve on pressure gage (Figure 1-3-11-2, Sheet 1, Detail A).

NOTE

The APU accumulator nitrogen charge must be released prior to servicing the APU accumulators with hydraulic fluid.

- f. Loosen locknut on servicing valve and slowly release nitrogen charge.

WARNING

To avoid damage to equipment or injury to personnel, do not service accumulators with free air or oxygen.

- g. Connect nitrogen bottle, Item 221, Appendix D, to servicing valve.
- h. Fully open locknut on nitrogen servicing valve.
- i. Slowly charge accumulators with nitrogen to 1000 psi.
- j. Manually open APU start valve, to make sure accumulator floating pistons are bottomed. Indicators to show 0% (Figure 1-3-11-2, Sheet 1, Detail A and Sheet 2, Detail B).

NOTE

- Floating pistons are considered bottomed when APU will no longer motor with start valve open and 1000 psi nitrogen applied.
 - The following 1450 psi pressure and tape reading is based on a 70°F (21°C) temperature day. Use the Graph B Temperature/Precharge Pressure Chart (Figure 1-3-11-1) for correct pressure.
- k. Add nitrogen to bring precharge to 1450 psi. Tighten locknut to close servicing valve (Figure 1-3-11-2, Sheet 1, Detail A).
 - l. Close nitrogen bottle valve and disconnect nitrogen bottle from servicing valve.
 - m. **TORQUE LOCKNUT TO 50 - 60 INCH-POUNDS.**
 - n. Check servicing valve for leaks with leak detection fluid, Item 189, Appendix D.
 - o. Install valve cap on servicing valve.

WARNING

To avoid injury to personnel, make sure pump handle is firmly locked into handle receptacle before operating hand-pump.

NOTE

APU handpump handle is in the left stowage compartment behind the rear crew compartment.

- p. Hydraulically charge accumulator using APU handpump to approximately 2900 psi reading on the accumulator nitrogen gage or operate the backup hydraulic system pump using external electrical power (PARA 1-6-16) until backup pump automatically shuts off. Combined indicator tape readings to be between 130 and 180. Use Graph A (Figure 1-3-11-1) for correct indicator reading at ambient temperature other than 70°F (21°C).

- q. Check backup pump reservoir for proper service. Service backup pump module reservoir as required (PARA 1-3-8).
- r. Make sure area is clean and free of foreign material.
- s. Install soundproofing panels in rear cabin ceiling (PARA 2-4-69).
- t. Make sure area is clean and free of foreign material.

CAUTION

Damage to hydraulic hand pump handle will result if handle is not positioned correctly before closing main rotor pylon sliding cover. Position pump handle in either 3 or 9 o'clock position before closing cover.

- u. Close and latch main rotor pylon sliding cover.

1-3-11.1.3 Service (Newly Installed Winterization Kit).**CAUTION**

To prevent damage to helicopter during cold weather operations of -29°F (-34°C) or below, hydraulic fluid must be changed from Item 155, Appendix D to Item 154, Appendix D (PARA 7-4-11).

CAUTION

Do not use unseparated gas-pressurized hydraulic service equipment to fill hydraulic handpump reservoir. This type of equipment pumps air/gas into the reservoir. Air/gas trapped in the hydraulic system can cause stiffened/sluggish flight control movement during certain maneuvers.

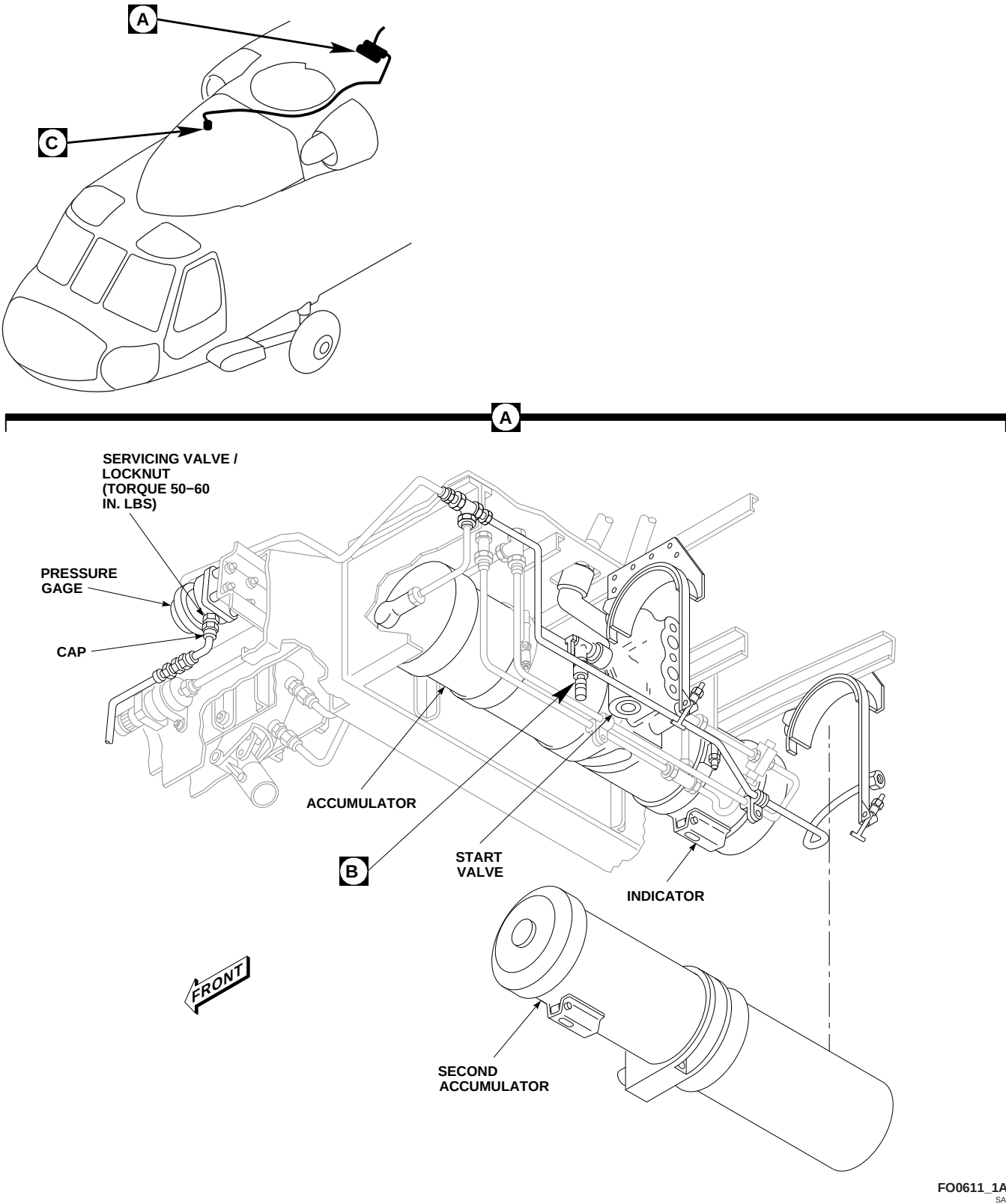


FIGURE 1-3-11-2. SERVICE WINTERIZATION KIT APU ACCUMULATOR (SHEET 1 OF 2)

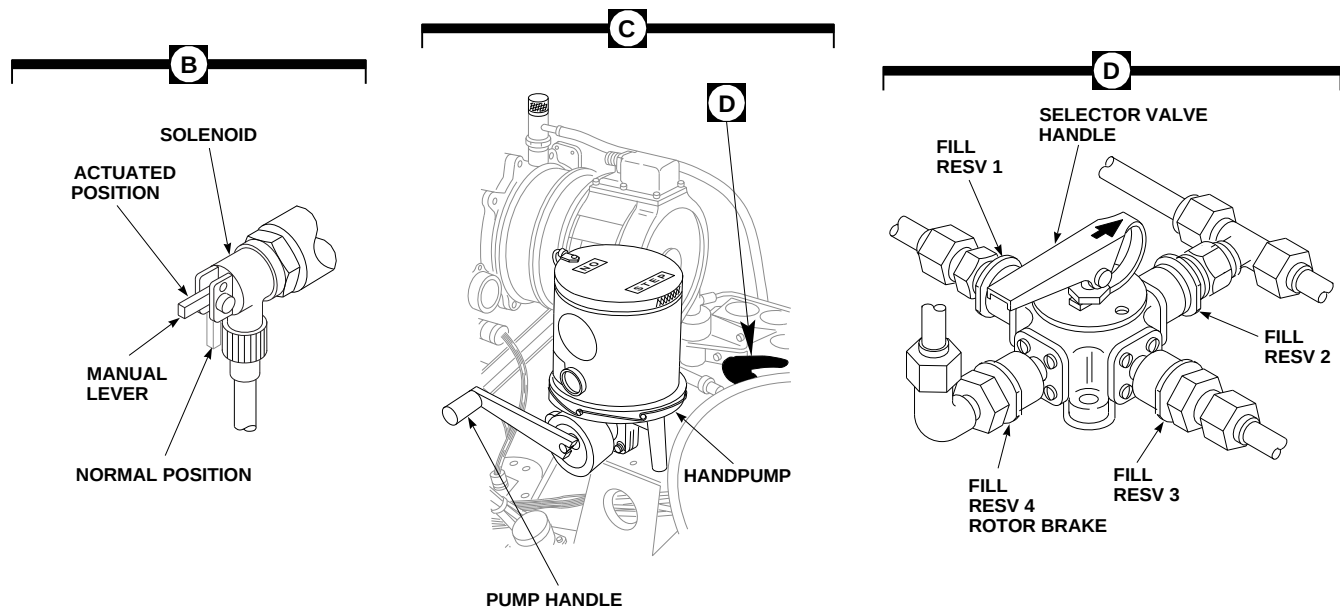
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FIGURE 1-3-11-2. SERVICE WINTERIZATION KIT APU ACCUMULATOR (SHEET 2 OF 2)

NOTE

Air must be purged from a new accumulator prior to servicing. The use of external hydraulic power is recommended as the fluid capacity of the accumulators exceeds the fluid available in the backup pump reservoir. If external hydraulic power is not available backup pump may be used but care must be taken to avoid depleting backup pump reservoir.

- a. Open main rotor pylon sliding cover.
- b. Remove soundproofing panels from rear cabin ceiling (PARA 2-4-69).
- c. Service accumulators with external hydraulic power as follows:
 - (1) Remove cap from servicing valve on pressure gage (Figure 1-3-11-2, Sheet 1, Detail A).
 - (2) Loosen locknut on servicing valve and slowly release nitrogen charge.
 - (3) Connect external hydraulic power to backup hydraulic system (PARA 1-6-15).
 - (4) With the servicing valve locknut loose, slowly fill accumulators until both accumulator tapes read 100%.
 - (5) Reduce hydraulic supply pressure to zero.
 - (6) Connect nitrogen bottle, Item 221, Appendix D, to servicing valve.
 - (7) Fully open locknut on servicing valve.
 - (8) Slowly charge accumulators with nitrogen to 1000 psi.
 - (9) Manually open APU start valve. Indicators to show 0% (Figure 1-3-11-2, Sheet 1, Detail A and Sheet 2, Detail B).

NOTE

- Floating pistons are considered bottomed when APU will no longer motor with start valve open and 1000 psi nitrogen applied.
 - The following 1450 psi pressure and tape reading is based on a 70°F (21°C) temperature day. Use the Graph B Temperature/Precharge Pressure Chart (Figure 1-3-11-1) for correct pressure.
- (10) Add nitrogen to bring precharge to 1450 psi. Tighten locknut to close servicing valve (Figure 1-3-11-2, Sheet 1, Detail A).
 - (11) Close nitrogen bottle valve and disconnect nitrogen bottle from servicing valve.
 - (12) **TORQUE LOCKNUT TO 50 - 60 INCH-POUNDS.**
 - (13) Check servicing valve for leaks with leak detection fluid, Item 189, Appendix D.
 - (14) Install valve cap on servicing valve.

WARNING

To avoid injury to personnel, make sure pump handle is firmly locked into handle receptacle before operating hand-pump.

NOTE

APU handpump handle is in the left stowage compartment behind the rear crew compartment.

- (15) Hydraulically charge accumulators using external hydraulic power or APU hand-pump to approximately 2900 psi reading on the accumulator nitrogen gage. Combined indicator tape readings to be between 130 and 180. Use Graph A (Figure 1-3-11-1) for correct indicator reading at ambient temperature other than 70°F (21°C).

- (16) Manually open and close lever on start valve solenoid (Figure 1-3-11-2, Sheet 2, Detail B).
 - (17) Repeat step c. (15).
 - (18) Disconnect external hydraulic power to backup hydraulic system (PARA 1-6-15).
 - (19) Check backup pump reservoir for proper service. Service backup pump module reservoir as required (PARA 1-3-8).
- d. Service accumulators with backup pump as follows:
- (1) Check backup pump reservoir for proper service. Service backup pump module reservoir as required (PARA 1-3-8).
 - (2) Remove cap from servicing valve on pressure gage (Figure 1-3-11-2, Sheet 1, Detail A).
 - (3) Loosen locknut on servicing valve and slowly release nitrogen charge.

NOTE

Backup pump reservoir will need to be serviced per PARA 1-3-8 multiple times before a tape reading of 100% is obtained on both accumulators, due to the limited capacity of the backup pump reservoir.

- (4) With the nitrogen locknut loose and while watching the backup pump reservoir, run the backup pump intermittently to slowly fill the hydraulic accumulators. When backup pump reservoir is depleted, turn backup pump off and service reservoir (PARA 1-3-8).
- (5) Repeat step d. (4) until tape readings of 100% are obtained on both accumulators.
- (6) Turn off backup pump.

WARNING

To avoid damage to equipment or injury to personnel, do not service accumulators with free air or oxygen.

- (7) Connect nitrogen bottle, Item 221, Appendix D, to servicing valve.
- (8) Fully open locknut on servicing valve.
- (9) Slowly charge accumulators with nitrogen to 1000 psi.

WARNING

Make sure APU is clear when opening and closing lever on start valve solenoid.

- (10) Manually open APU start valve. Indicators to show 0% (Figure 1-3-11-2, Sheet 1, Detail A and Sheet 2, Detail B).

NOTE

- Floating pistons are considered bottomed when APU will no longer motor with start valve open and 1000 psi nitrogen applied.
 - The following 1450 psi pressure and tape reading is based on a 70°F (21°C) temperature day. Use the Graph B Temperature/Precharge Pressure Chart (Figure 1-3-11-1) for correct pressure.
- (11) Add nitrogen to bring precharge to 1450 psi. Tighten locknut to close servicing valve (Figure 1-3-11-2, Sheet 1, Detail A).
 - (12) Close nitrogen bottle valve and disconnect nitrogen bottle from servicing valve.
 - (13) **TORQUE LOCKNUT TO 50 - 60 INCH-POUNDS.**
 - (14) Check servicing valve for leaks with leak detection fluid, Item 189, Appendix D.

- (15) Install valve cap on servicing valve.

NOTE

- The reservoir capacity of a single accumulator exceeds the reservoir capacity of the backup pump. Once started, the backup pump will run for 180 seconds before shutting off. The backup pump reservoir, however, will be empty in less than 90 seconds. For this reason, it will be required to pull the backup pump power circuit breaker before the reservoir pump runs dry.
 - Backup pump power circuit breaker is located on No. 1 DC PRI BUS circuit breaker panel.
 - It may be necessary to refill backup pump reservoir 2 to 3 times before accumulator is adequately charged.
- (16) Hydraulically charge accumulators by operating the backup hydraulic system pump using external electrical power (PARA 1-6-16). Shut down backup pump by pulling BACKUP PUMP PWR circuit breaker prior to backup pump reservoir depletion. Combined indicator tape readings to be 160. Use Graph A (Figure 1-3-11-1) for correct indicator reading at ambient temperature other than 70°F (21°C).
 - (17) Manually open and close lever on start valve solenoid (Figure 1-3-11-2, Sheet 2, Detail B).
 - (18) Bleed air from backup pump reservoir and refill reservoir if needed.
 - (19) Repeat step d. (16).
 - (20) Check backup pump reservoir for proper service. Service backup pump module reservoir as required (PARA 1-3-8).
- e. Make sure area is clean and free of foreign material.
 - f. Install soundproofing panels in rear cabin ceiling (PARA 2-4-69).

- g. Make sure area is clean and free of foreign material.

handle in either 3 or 9 o'clock position before closing cover.



- h. Close and latch main rotor pylon sliding cover.

Damage to hydraulic hand pump handle will result if handle is not positioned correctly before closing main rotor pylon sliding cover. Position pump

1-3-12 SERVICE/DRAIN MAIN TRANSMISSION.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-3-12.1	Service/Drain Main Transmission	1-3-12-2	1-3-12.2	Flush Main Transmission/Change Oil Specification	1-3-12-3

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 10-Gallon
- Drain Line, D-730-L
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 197, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D

References

- Appendix D
- PARA 2-4-70

1-3-12.1 Service/Drain Main Transmission.**1-3-12.1.1. Service.****NOTE**

- Dipstick, 70351-08164-103, is calibrated for checking cold oil levels. Wait at least 2 hours after shutdown to check oil. If oil level must be checked when hot (immediately to ½ hour after shutdown), oil level will read about ½-inch low (halfway between full and add mark or ½-inch below add mark).

- Dipstick, 70351-08164-104, is calibrated for checking cold or hot oil levels. Use appropriate scale when checking oil level. Read hot side of dipstick when checking hot oil (immediately to ½ hour after shutdown), or cold side of dipstick when checking cold oil (at least 2 hours after shutdown).

- Remove oil level gage (dipstick) and check for proper oil level (Figure 1-3-12-1, Detail A).

NOTE

- Oil capacity of main transmission is about 7 gallons.
- Helicopter attitude should be taken into consideration when reading dipstick.
- Before adding oil determine whether main transmission contains lubricating oil, Item 205, Appendix D, Item 206, Appendix D, or Item 208, Appendix D. It is preferred to add the same type oil that is in the main transmission. For normal operation, if oils are mixed, it is not necessary to drain main transmission and refill with one type of oil. **NO MIXING IS ALLOWED FOR COLD WEATHER OPERATION.**
- Operation at temperature below -29°F (-34°C) will require use of lubricating oil, Item 208, Appendix D. If lubricating oils have been mixed, or another type of

oil is present, drain and refill with lubricating oil, Item 208, Appendix D.

- If necessary to add oil, hinge open filler tube cap and fill gear box with lubricating oil, Item 205, Appendix D, Item 206, Appendix D, or Item 208, Appendix D.
- Make sure area is clean and free of foreign material.
- Install oil level gage (dipstick).

1-3-12.1.2. Drain.

- Remove drain lines and main transmission drip pan (PARA 2-4-70), if required, or open access panel.
- Remove magnetic drain plug and packing from valve (Figure 1-3-12-2). Discard packing.
- Place end of drain line in empty 10-gallon container. Insert other end in valve on drain port. Turn fitting on drain line clockwise and drain oil into container.
- When oil stops flowing, turn fitting on drain line counterclockwise, and remove from valve.
- Inspect valve for damage. If damaged is found, perform the following:
 - Remove valve and packing from drain port on main transmission sump. Discard packing.
 - Lubricate packing with same oil main transmission is to be serviced with and install on replacement valve.
 - Install valve in drain port. **TORQUE VALVE 32 - 100 INCH-POUNDS.**
 - Lockwire, Item 197, Appendix D, valve to main transmission sump.
- Install magnetic drain plug as follows:
 - Lubricate packing with same oil main transmission is to be serviced with and install on plug.

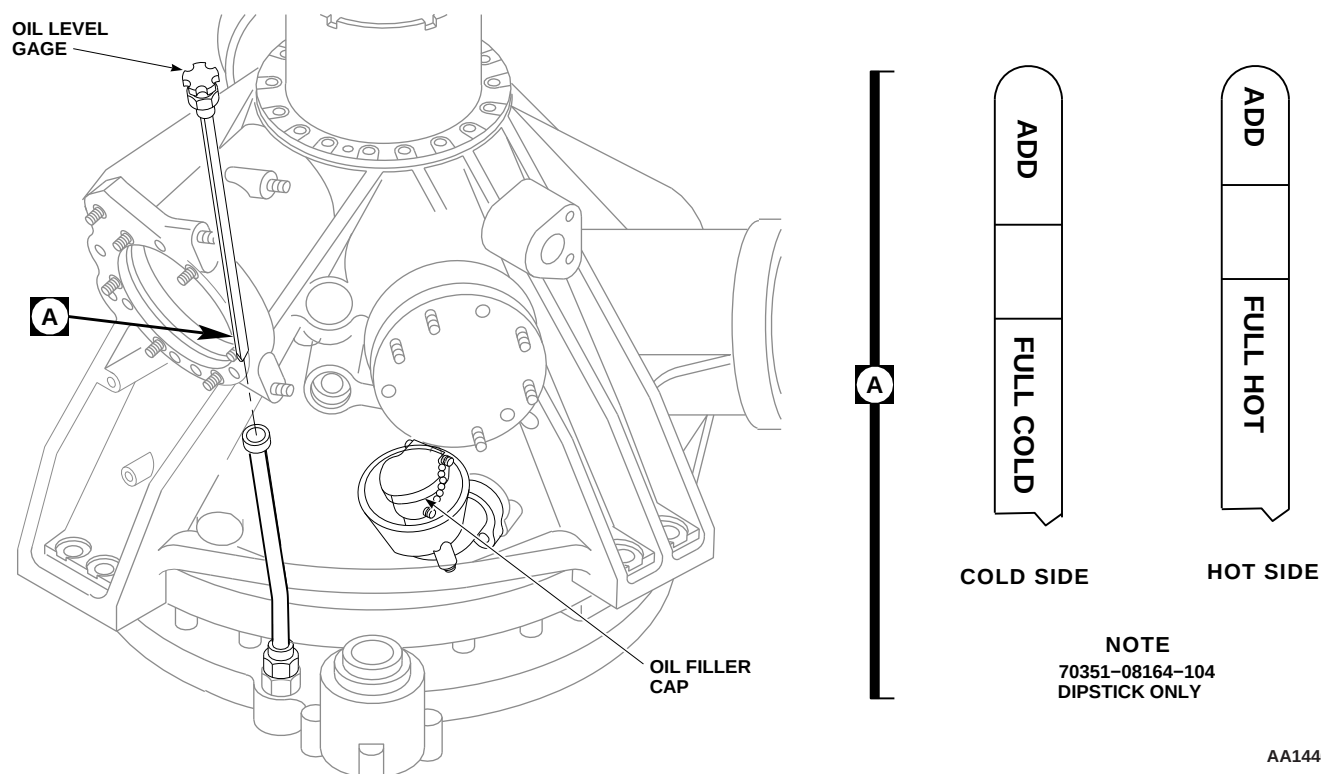


FIGURE 1-3-12-1. SERVICE MAIN TRANSMISSION

- (2) Install plug in valve. **TORQUE PLUG 20 - 60 INCH-POUNDS.**
- (3) Lockwire, Item 197, Appendix D, plug to valve.
- g. Make sure area is clean and free of foreign material.
- h. Install drain lines and main transmission drip pan (PARA 2-4-70), if required, or close access panel.

1-3-12.2 Flush Main Transmission/Change Oil Specification.

NOTE

- Operation at temperature below -29°F (-34°C) will require use of lubricating oil, Item 208, Appendix D. If lubricating oils have been mixed, or another type of oil is present, drain and refill with lubricating oil, Item 208, Appendix D.

- This procedure is essential when changing from one base oil to another (Item 206, Appendix D, to Item 208, Appendix D), and is desirable for all oil changes.
- There is no requirement to flush the main transmission when changing from lubricating oil, Item 205, Appendix D, to Item 206, Appendix D.
- a. Run main transmission until oil reaches normal operating temperature.
- b. Drain oil while still hot (PARA 1-3-12.1.2).
- c. Service main transmission (PARA 1-3-12.1.1).
- d. Run main transmission again until oil reaches normal operating temperature.
- e. Drain oil while still hot (PARA 1-3-12.1.2).
- f. Service main transmission (PARA 1-3-12.1.1).

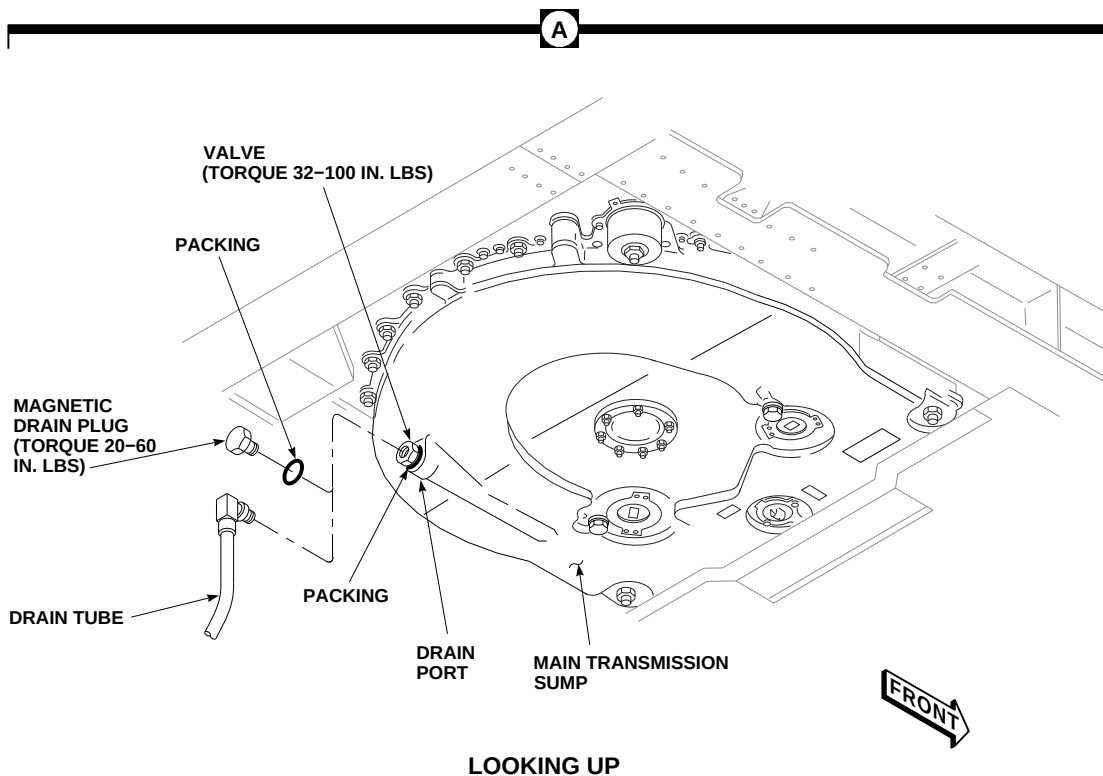
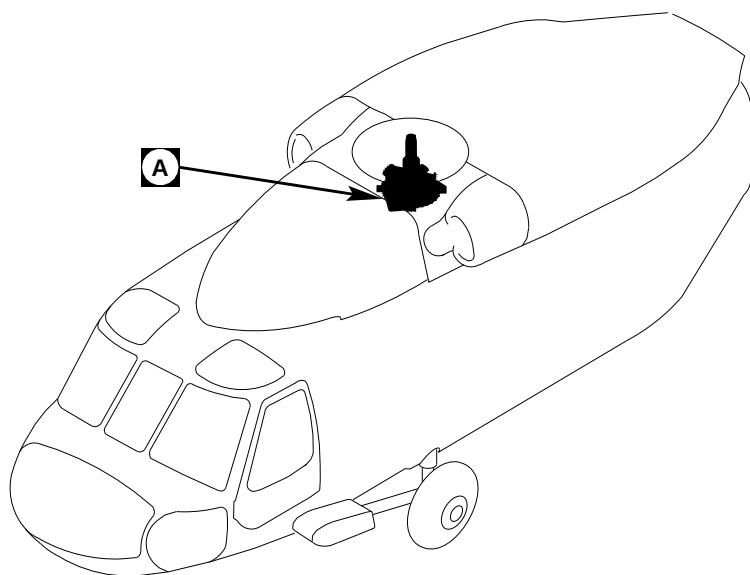


FIGURE 1-3-12-2. DRAIN MAIN TRANSMISSION

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1-3-13 SERVICE/DRAIN INTERMEDIATE GEAR BOX.

PARAGRAPH BREAKDOWN

	PAGE
1-3-13.1 Service/Drain Intermediate Gear Box	1-3-13-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Beaker, 2000 ml
- Container, 2-Quart
- Container, 2-Gallon
- Drain Line, H538A-3
- Funnel

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Lubricating Oil, Item 205, Appendix D

- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D
- PARA 2-4-139
- PARA 6-4-31
- PARA 6-4-34
- PARA 6-4-35
- PARA 6-4-37

1-3-13.1 Service/Drain Intermediate Gear Box.

1-3-13.1.1. Service.

- Remove intermediate gear box fairing (PARA 2-4-139).

NOTE

- Intermediate gear box sight plug will not give correct reading with tail rotor pylon folded.
- Visual oil level check will be performed with the inspector at eye level with the intermediate gear box sight plug.

- If intermediate gear box sight plug is clean and oil level is to top of bull’s-eye, lightly shake tail rotor pylon to make sure of accurate oil level (Figure 1-3-13-1, Detail A). Recheck oil level as follows:
 - If oil level remains at top of bull’s-eye, no action is required.
 - If oil level is low, service intermediate gear box per step e.
 - If oil level cannot be determined, drain oil into 2000 ml beaker (PARA 1-3-13.1.2, step a. through step e.), then proceed to step c.

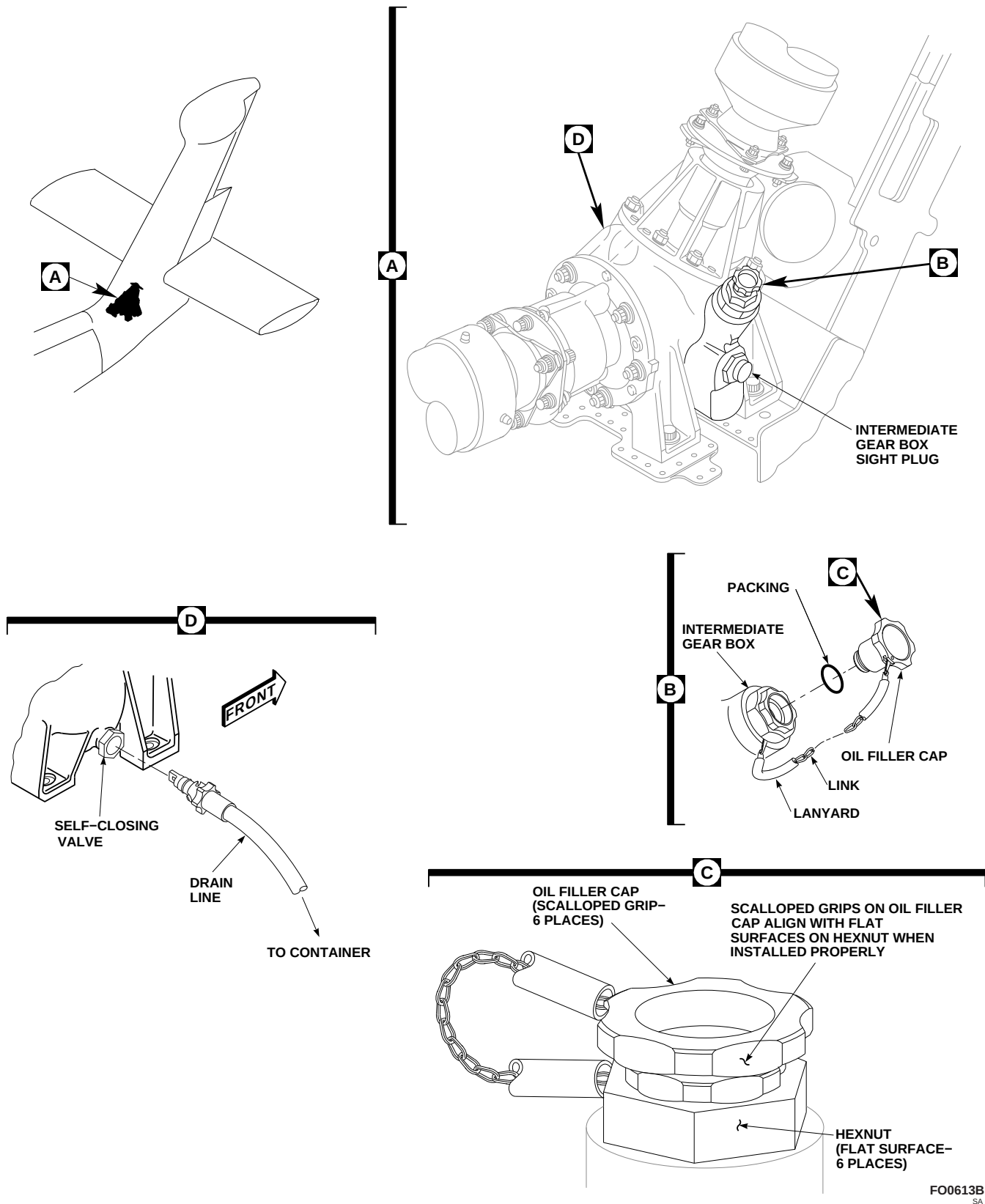


FIGURE 1-3-13-1. SERVICE/DRAIN INTERMEDIATE GEAR BOX

- c. If drained oil is 1200 ml (2.57 pints) or more, perform the following:
 - (1) Remove and inspect intermediate gear box sight plug (PARA 6-4-34).
 - (2) Clean intermediate gear box sight plug using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D.
 - (3) Install intermediate gear box sight plug (PARA 6-4-34).
 - (4) Service intermediate gear box per step e.
- d. If drained oil is less than 1200 ml (2.57 pints), replace intermediate gear box (PARA 6-4-31).
- e. Service intermediate gear box as follows:
 - (1) Determine whether gear box contains lubricating oil, Item 205, Item 206, or Item 208, Appendix D.

NOTE

Operation at temperatures below -29°F (-34°C) will require use of lubricating oil, Item 208, Appendix D. If lubricating oils have been mixed, or if another type of oil is present, drain and refill gear box with lubricating oil, Item 208, Appendix D.

- (2) If necessary to change from one oil specification to another, refer to PARA 1-3-13.1.3.
- (3) Remove oil filler cap (Figure 1-3-13-1, Detail B).
- (4) Inspect lanyard for broken link(s). If lanyard has broken link(s), replace intermediate gear box oil filler plug (PARA 6-4-35).

- (5) Inspect oil filler cap for damaged or missing packing. If damaged, discard packing and install new packing. If missing, install new packing.

NOTE

Servicing limit of 300 ml (0.63 pint) does not apply if servicing a completely drained intermediate gear box.

- (6) Add no more than 300 ml (0.63 pint) of lubricating oil, Item 205, Item 206, or Item 208, Appendix D.
- (7) If oil level comes within bull's-eye on intermediate gear box sight plug (Figure 1-3-13-1, Detail A), servicing is complete.

WARNING

Injury to personnel and damage to equipment will result if gear box is operated without oil or with too little oil. If gear box needs more than 300 ml (0.63 pint) to bring oil level to top of bull's-eye (and aircraft was operated at this low oil level), the intermediate gear box must be replaced.

- (8) If oil level remains low after adding 300 ml (0.63 pint) of oil, drain (PARA 1-3-13.1.2, step a. through step e.), and replace intermediate gear box (PARA 6-4-31).
- (9) Install oil filler cap (Figure 1-3-13-1, Detail B). Check for security as follows:
 - (a) Make sure scalloped grips on cap align with flat surfaces of hexnut (Figure 1-3-13-1, Detail C).

CAUTION

- Do not pull on cap with two fingers side by side when checking cap for security, cap will give false feeling of security.
- Do not press in on cap. Pressing in without turning cap clockwise at the same time will actually begin to loosen cap.

- (b) Using three or four fingers equally spaced around cap, pull out on cap to make sure it is seated in the locked position. Do not press cap inward. While pulling, twist cap clockwise and counterclockwise to make sure it is seated in locked position.

(10) Make sure area is clean and free of foreign material.

(11) Install intermediate gear box fairing (PARA 2-4-139).

1-3-13.1.2. Drain.

- a. Remove intermediate gear box chip detector/temperature sensor (PARA 6-4-37).
- b. Place open end of drain line into 2-quart container. Insert other end into self-closing valve by pushing in and turning clockwise (Figure 1-3-13-1, Detail D).
- c. Remove oil filler cap and drain oil into container (Figure 1-3-13-1, Detail B).
- d. Remove drain line from self-closing valve (Figure 1-3-13-1, Detail D).
- e. Install intermediate gear box chip detector/temperature sensor (PARA 6-4-37). Do not install intermediate gear box fairing at this time.
- f. Service intermediate gear box with new oil to top of bull's-eye on intermediate gear box sight plug (Figure 1-3-13-1, Detail A).

g. Install oil filler cap (Figure 1-3-13-1, Detail B). Check for security as follows:

- (1) Make sure scalloped grips on cap align with flat surfaces of hexnut (Figure 1-3-13-1, Detail C).

CAUTION

- Do not pull on cap with two fingers side by side when checking cap for security, cap will give false feeling of security.
- Do not press in on cap. Pressing in without turning cap clockwise at the same time will actually begin to loosen cap.
- (2) Using three or four fingers equally spaced around cap, pull out on cap to make sure it is seated in the locked position. Do not press cap inward. While pulling, twist cap clockwise and counterclockwise to make sure it is seated in locked position.

h. Make sure area is clean and free of foreign material.

i. Install intermediate gear box fairing (PARA 2-4-139).

1-3-13.1.3. Change Oil Specification.**NOTE**

This procedure is essential when changing from one base oil to another (Item 205, Item 206, Appendix D, to Item 208, Appendix D), and is desirable for all oil changes.

- a. Run intermediate gear box until oil reaches normal operating temperature.
- b. Drain oil while still hot (PARA 1-3-13.1.2, step a. through step g.).
- c. Run gear box again until oil reaches normal operating temperature.

1-3-13-4 Change 1

- d. Drain oil while still hot (PARA 1-3-13.1.2, step a. through step i.).

1-3-13.1.4. Flush Intermediate Gear Box.

- a. Run intermediate gear box until oil reaches normal operating temperature.
- b. Drain oil while still hot (PARA 1-3-13.1.2, step a. through step d.).
- c. Remove self-closing valve (PARA 6-4-37).
- d. Place funnel under chip detector self-closing valve opening (Figure 1-3-13-1, Detail D). Place 2-gallon container under funnel.
- e. Thoroughly flush gear box with clean oil until drained oil is clean with no trace of debris.
- f. Install self-closing valve (PARA 6-4-37).
- g. Service intermediate gear box with new oil to top of bull's-eye on intermediate gear box sight plug (Figure 1-3-13-1, Detail A).
- h. Install oil filler cap (Figure 1-3-13-1, Detail B). Check for security as follows:

- (1) Make sure scalloped grips on cap align with flat surfaces of hexnut (Figure 1-3-13-1, Detail C).



- Do not pull on cap with two fingers side by side when checking cap for security, cap will give false feeling of security.
- Do not press in on cap. Pressing in without turning cap clockwise at the same time will actually begin to loosen cap.
- (2) Using three or four fingers equally spaced around cap, pull out on cap to make sure it is seated in the locked position. Do not press cap inward. While pulling, twist cap clockwise and counterclockwise to make sure it is seated in locked position.
- i. Run intermediate gear box again until oil reaches normal operating temperature.
- j. Drain oil while still hot (PARA 1-3-13.1.2, step a. through step i.).

1-3-14 SERVICE/DRAIN TAIL GEAR BOX.

PARAGRAPH BREAKDOWN

	PAGE
1-3-14.1 Service/Drain Tail Gear Box	1-3-14-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Beaker, 2000 ml
- Container, 2-Quart
- Container, 2-Gallon
- Drain Line, H538A-3
- Funnel

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D

- Lubricating Oil, Item 208, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D
- PARA 2-4-137
- PARA 6-4-52
- PARA 6-4-55
- PARA 6-4-56
- PARA 6-4-57
- PARA 6-4-59
- PARA 11-4-113

1-3-14.1 Service/Drain Tail Gear Box.

1-3-14.1.1. Service.

- Remove tail gear box fairings (PARA 2-4-137).

NOTE

- Tail gear box sight plug will not give correct reading with tail rotor pylon folded.

- Visual oil level check will be performed with the inspector at eye level with the tail gear box sight plug.
- If tail gear box sight plug is clean and oil level is to top of bull’s-eye, lightly shake tail rotor pylon to make sure of accurate oil level (Figure 1-3-14-1, Detail D). Recheck oil level as follows:
 - If oil level remains at top of bull’s-eye, no action is required.

(2) If oil level is low, service tail gear box per step f.

(3) If oil level cannot be determined, drain oil into 2000 ml beaker (PARA 1-3-14.1.2, step a. through step e.), then proceed to step c.

c. Inspect drained oil for purple discoloration. If discoloration is present, perform the following:

(1) Remove tail rotor servo (PARA 11-4-113).

(2) Inspect tail gear box for corrosion (PARA 6-4-57). Pay particular attention to tail rotor servo bore area. Repair as required.

(3) Install tail rotor servo (PARA 11-4-113).

d. If drained oil is 1200 ml (2.57 pints) or more, perform the following:

(1) Remove and inspect tail gear box sight plug (PARA 6-4-55).

(2) Clean tail gear box sight plug using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D.

(3) Install tail gear box sight plug (PARA 6-4-55).

(4) Service tail gear box per step h.

e. If drained oil is less than 1200 ml (2.57 pints), replace tail gear box (PARA 6-4-52).

f. Service tail gear box as follows:

(1) Determine whether gear box contains lubricating oil, Item 205, Item 206, or Item 208, Appendix D.

NOTE

Operation at temperatures below -29°F (-34°C) will require use of lubricating oil, Item 208, Appendix D. If lubricating oils have been mixed, or if another type of oil is present, drain and refill gear box with lubricating oil, Item 208, Appendix D.

(2) If necessary to change from one oil specification to another, refer to PARA 1-3-13.1.3.

(3) Remove oil filler cap (Figure 1-3-14-1, Detail B).

(4) Inspect lanyard for broken link(s). If lanyard has broken link(s), replace tail gear box oil filler plug (PARA 6-4-56).

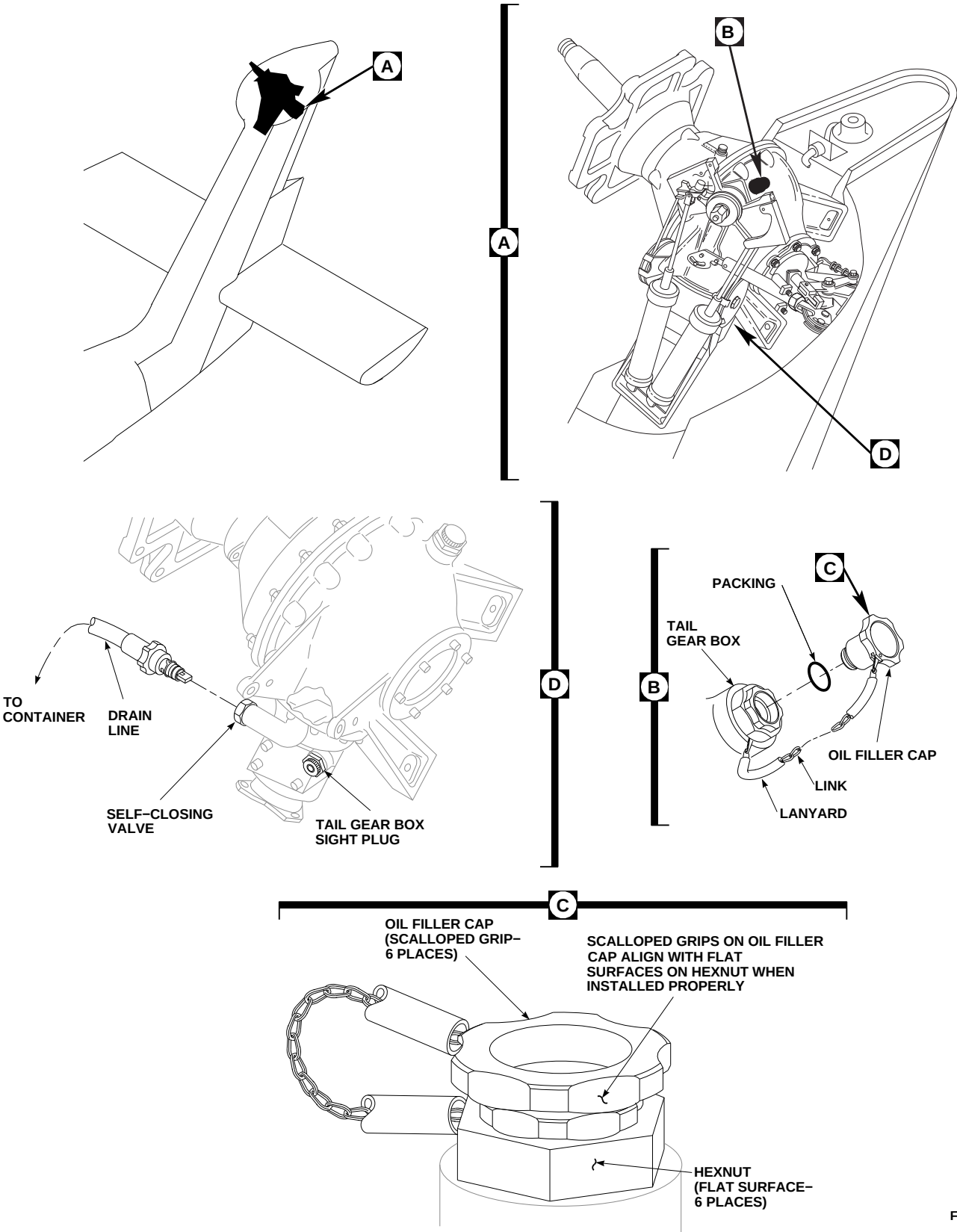
(5) Inspect oil filler cap for damaged or missing packing. If damaged, discard packing and install new packing. If missing, install new packing.

NOTE

Servicing limit of 400 ml (0.85 pint) does not apply if servicing a completely drained tail gear box.

(6) Add no more than 400 ml (0.85 pint) of lubricating oil, Item 205, Item 206, or Item 208, Appendix D.

(7) If oil level comes within bull's-eye on tail gear box sight plug (Figure 1-3-14-1, Detail D), servicing is complete.



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FIGURE 1-3-14-1. SERVICE/DRAIN TAIL GEAR BOX

WARNING

Injury to personnel and damage to equipment will result if tail gear box is operated without oil or with too little oil. If tail gear box needs more than 400 ml (0.85 pint) to bring oil level to top of bull's-eye (and aircraft was operated at this low oil level), the tail gear box must be replaced.

NOTE

Pitch change shaft may remain in service if inspected and found serviceable.

- (8) If oil level remains low after adding 400 ml (0.85 pint) of oil, drain and replace tail gear box and inspect pitch change shaft (PARA 1-3-14.1.2, step a. through step e., PARA 6-4-52 and PARA 11-4-113).
- (9) Install oil filler cap (Figure 1-3-14-1, Detail B). Check for security as follows:
 - (a) Make sure scalloped grips on cap align with flat surfaces of hexnut (Figure 1-3-14-1, Detail C).

CAUTION

- Do not pull on cap with two fingers side by side when checking cap for security, cap will give false feeling of security.
- Do not press in on cap. Pressing in without turning cap clockwise at the same time will actually begin to loosen cap.
 - (b) Using three or four fingers equally spaced around cap, pull out on cap to make sure it is seated in the locked position. Do not press cap inward. While pulling, twist cap clockwise and counterclockwise to make sure it is seated in locked position.

- (10) Make sure area is clean and free of foreign material.
- (11) Install tail gear box fairings (PARA 2-4-137).

1-3-14.1.2. Drain.

- a. Remove tail gear box chip detector/temperature sensor (PARA 6-4-59).
- b. Place open end of drain line into 2-quart container. Insert other end into self-closing valve by pushing in and turning clockwise (Figure 1-3-14-1, Detail D).
- c. Remove oil filler cap and drain oil into container (Figure 1-3-14-1, Detail B).
- d. Remove drain line from self-closing valve (Figure 1-3-14-1, Detail D).
- e. Install tail gear box chip detector/temperature sensor (PARA 6-4-59). Do not install tail gear box fairings at this time.
- f. Service tail gear box with new oil to top of bull's-eye on tail gear box sight plug.
- g. Install oil filler cap (Figure 1-3-14-1, Detail B). Check for security as follows:
 - (1) Make sure scalloped grips on cap align with flat surfaces of hexnut (Figure 1-3-14-1, Detail C).

CAUTION

- Do not pull on cap with two fingers side by side when checking cap for security, cap will give false feeling of security.
- Do not press in on cap. Pressing in without turning cap clockwise at the same time will actually begin to loosen cap.
 - (2) Using three or four fingers equally spaced around cap, pull out on cap to make sure

it is seated in the locked position. Do not press cap inward. While pulling, twist cap clockwise and counterclockwise to make sure it is seated in locked position.

- h. Make sure area is clean and free of foreign material.
- i. Install tail gear box fairings (PARA 2-4-137).

1-3-14.1.3. Change Oil Specification.

NOTE

This procedure is essential when changing from one base oil to another (Item 205, or Item 206, Appendix D, to Item 208, Appendix D), and is desirable for all oil changes.

- a. Run tail gear box until oil reaches normal operating temperature.
- b. Drain oil while still hot (PARA 1-3-14.1.2, step a. through step g.).
- c. Run tail gear box again until oil reaches normal operating temperature.
- d. Drain oil while still hot (PARA 1-3-14.1.2, step a. through step i.).

1-3-14.1.4. Flush Tail Gear Box.

NOTE

If possible the tail gear box should be flushed while the oil is at normal operating temperature. The tail gear box may be flushed with cold oil if the aircraft cannot be operated in order to bring the tail gear box oil to its normal operating temperature.

- a. Run tail gear box until oil reaches normal operating temperature.
- b. Drain oil while still hot (PARA 1-3-14.1.2, step a. through step d.).

- c. Remove self-closing valve (PARA 6-4-59).
- d. Place funnel under self-closing valve opening (Figure 1-3-14-1, Detail C). Place 2-gallon container under funnel.

NOTE

The tail gear box may be flushed with tail rotor servo installed. However, the flushing process will be more effective when performed with the tail rotor servo removed.

- e. If tail rotor servo is installed, flush gear box with clean lubricating oil. Repeat with clean oil until drained oil is clean with no trace of debris. Direct oil stream against inside of the gear box in order to flush out as much sludge and debris as possible.
- f. To flush the tail gear box with tail rotor servo removed, perform the following:

CAUTION

Damage to equipment will occur if foreign objects enter tail gear box. Foreign objects may enter tail gear box when tail rotor servo is removed. Make sure that no foreign objects enter the tail gear box during performance of this task.

- (1) Remove tail rotor servo (PARA 11-4-113).
- (2) Inspect inside of tail gear box, especially the inside diameter of the input bevel pinion, for buildup of sludge and debris accumulation. Remove sludge/debris using clean dry machinery towel, Item 211, Appendix D. Make sure no debris is left inside the tail gear box.

NOTE

Before flushing, determine whether tail gear box contains lubricating oil, Item 205, Item 206, or Item 208, Appendix D. Flush with appropriate lubricating oil.

(3) Flush tail gear box with clean lubricating oil Item 205, Item 206, or Item 208, Appendix D. Repeat using clean oil until oil drained from tail gear box is clean with no trace of debris. Make sure oil stream is directed against inside of tail gear box to make sure all debris is flushed from the tail gear box.

(4) Install tail rotor servo (PARA 11-4-113).

g. Install self-closing valve (PARA 6-4-59).

h. Service tail gear box with new oil to top of bull's-eye on tail gear box sight plug (Figure 1-3-14-1, Detail D).

i. Install oil filler cap (Figure 1-3-14-1, Detail B). Check for security as follows:

(1) Make sure scalloped grips on oil filler cap align with flat surfaces of receptacle hex-nut (Figure 1-3-14-1, Detail C).

CAUTION

- Do not pull on cap with two fingers side by side when checking cap for security, cap will give false feeling of security.

- Do not press in on cap. Pressing in without turning cap clockwise at the same time will actually begin to loosen cap.

(2) Using three or four fingers equally spaced around cap, pull out on cap to make sure it is seated in the locked position. Do not press cap inward. While pulling, twist cap clockwise and counterclockwise to make sure it is seated in locked position.

j. Run tail gear box again until oil reaches normal operating temperature.

k. Drain oil while still hot (PARA 1-3-14.1.2, step a. through step i.).

1-3-15 SERVICE MAIN ROTOR HEAD DAMPER.

PARAGRAPH BREAKDOWN

	PAGE
1-3-15.1 Service Main Rotor Head Damper	1-3-15-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Quart
- Damper Indicator Handle Retainer, Locally-Made, Figure H-215, Appendix H
- Hydraulic Fluid Dispenser, AF5
- Reducer
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D

- Insulation Sleeving, Heat-Shrinkable, Item 171, Appendix D

- Lockwire, Item 197, Appendix D

References

- Appendix D
- Appendix H
- PARA 1-6-18
- PARA 11-4-2

1-3-15.1 Service Main Rotor Head Damper.

- Locally make, damper indicator handle retainer (Figure H-215, Appendix H).



At temperatures below -40°F (-40°C), only hydraulic fluid, Item 154, Appendix D, shall be used.

NOTE

- At temperatures above -40°F (-40°C), hydraulic fluid, Item 155, Appendix D,

or Item 154, Appendix D, can be used when filling dampers. Mixing hydraulic fluids is allowed. Flushing is not required when switching from hydraulic fluid, Item 154, Appendix D, to hydraulic fluid, Item 155, Appendix D.

- Hydraulic fluid, Item 155, Appendix D, may be switched to hydraulic fluid, Item 154, Appendix D, without flushing.
- b. Service pressure hydraulic fluid dispenser with hydraulic fluid, Item 154, Appendix D, or Item 155, Appendix D.
- c. If not already done, engage gust lock (PARA 1-6-18).

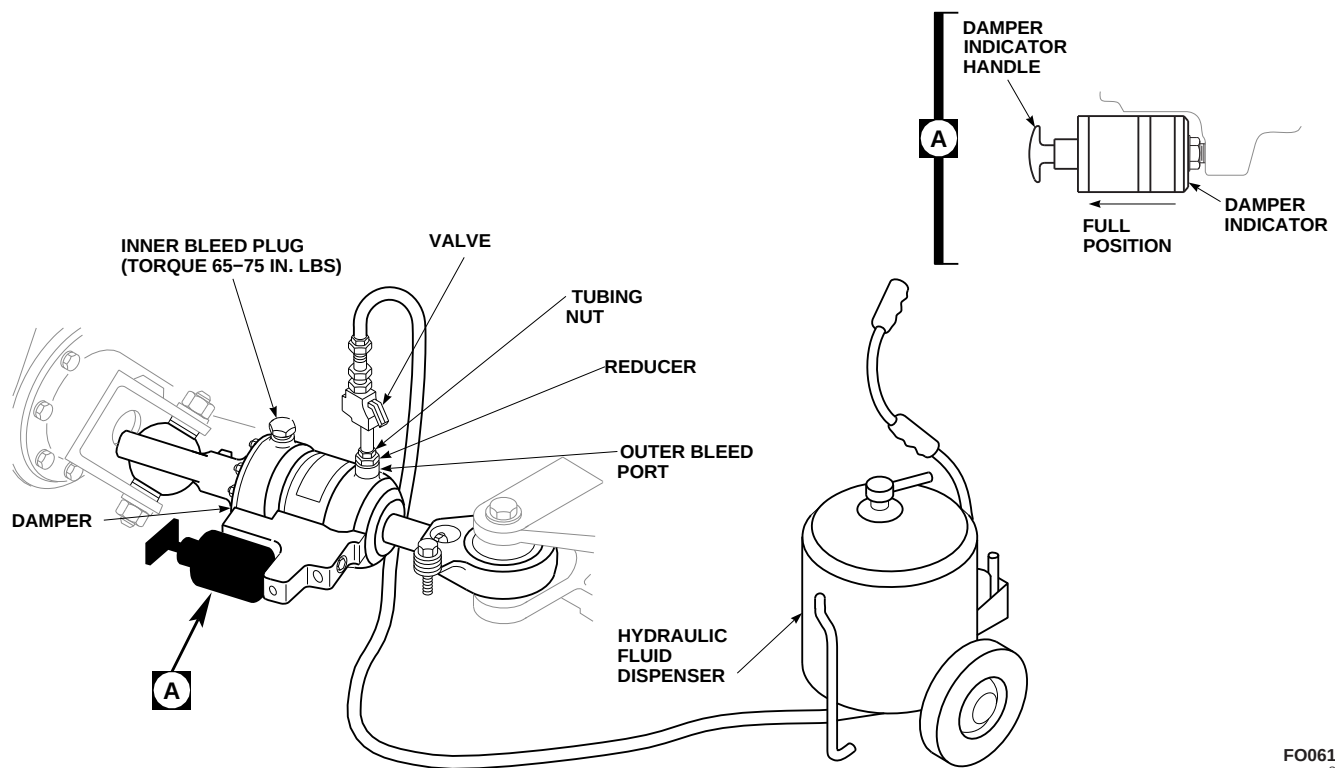


FIGURE 1-3-15-1. SERVICE MAIN ROTOR HEAD DAMPER

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- d. Install main rotor damper rigging blocks (PARA 11-4-2).
- e. Place 1-quart container below damper to catch any overflow of fluid.
- f. Slowly loosen outer bleed plug and bleed off hydraulic fluid until indicator is bottomed.
- g. Remove outer bleed plug and packing from outer bleed port. Discard packing. Install reducer (Figure 1-3-15-1).
- h. Connect nozzle and tubing of hydraulic fluid dispenser to reducer on damper.
- i. Loosen tubing nut, slowly press valve on nozzle, and bleed off any air trapped in dispenser.
- j. Release valve and tighten tubing nut to reducer.
- k. Loosen inner bleed plug, slowly press valve on nozzle, and bleed off air trapped in damper. Release valve and tighten inner plug.
- l. Press valve on nozzle, allowing indicator to move to full position (Figure 1-3-15-1, Detail A). Release valve and stop pressure flow. Loosen inner bleed plug and bleed off hydraulic fluid until indicator is bottomed. Repeat bleeding until all air is removed.
- m. Press valve on nozzle, allowing indicator to move to full position.
- n. Hold indicator in extended position using damper indicator handle retainer. Disconnect nozzle and remove reducer.
- o. Install bleed plug and packing in outer bleed port on damper.
- p. Release damper indicator handle.
- q. Check damper indicator for proper fluid level.
- r. **TORQUE BLEED PLUGS TO 65 - 75 INCH-POUNDS.**
- s. Using lockwire, Item 197, Appendix D, lockwire bleed plugs to each other and damper

- indicator. Cover lockwire with heat-shrinkable insulation sleeving, Item 171, Appendix D.
- t. Remove hydraulic fluid dispenser.
- u. Remove main rotor damper rigging blocks (PARA 11-4-2).
- v. Make sure area is clean and free of foreign material.
- w. If required, disengage gust lock (PARA 1-6-18).
- x. Record type of hydraulic fluid used to service damper in helicopter logbook.

1-3-16 SERVICE TAIL ROTOR DRIVE SHAFT VISCOUS DAMPER BEARINGS.

PARAGRAPH BREAKDOWN

		PAGE
1-3-16.1	Service Tail Rotor Drive Shaft Viscous Damper Bearings	1-3-16-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Syringe, 2 ounce - Catheter Tip
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Damping Fluid, Item 121, Appendix D
- Lockwire, Item 197, Appendix D

- Sealing Compound, Item 298, Appendix D

References

- Appendix D
- PARA 6-4-45

1-3-16.1 Service Tail Rotor Drive Shaft Viscous Damper Bearings.

NOTE

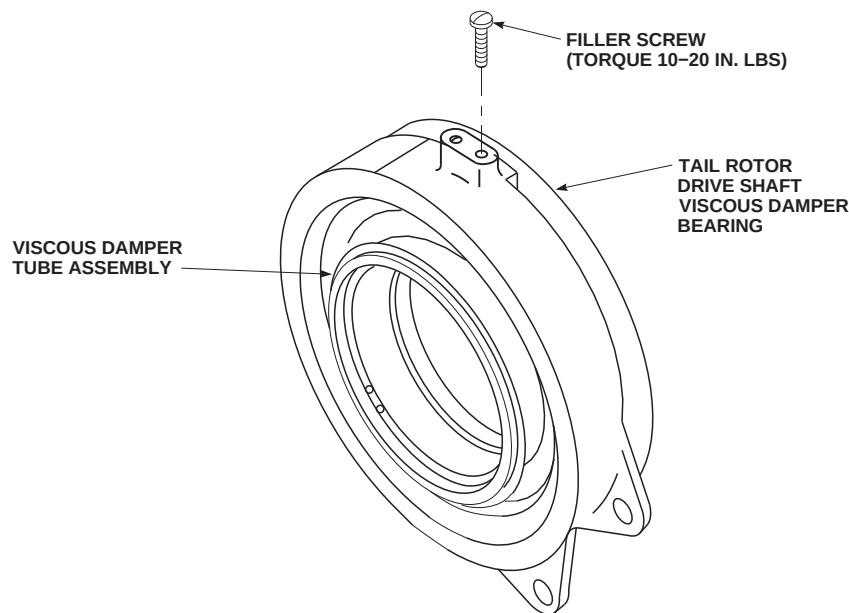
After reservicing viscous dampers, they must be inspected for leakage after next flight.

1-3-16.1.1 Service.

- Remove filler screws from viscous damper tube assembly (Figure 1-3-16-1).
- Fill tube assembly using syringe filled with damping fluid, Item 121, Appendix D. Move

drive shaft in tube assembly, while filling, to purge air bubbles from fluid. Add fluid until fluid starts to come out of adjacent filler hole. If tube is transparent, keep adding fluid.

- After servicing, check viscous damper tube assembly for air bubbles. An air bubble should be visible, but should not be more than ½-inch long, with support in vertical position.
- Coat threads of filler screws with sealing compound, Item 298, Appendix D, and install. **TORQUE SCREWS TO 10 - 20 INCH-POUNDS.** If screws have provisions for lockwire, lockwire, Item 197, Appendix D, screws.

FO0615B
SA**FIGURE 1-3-16-1. SERVICE TAIL ROTOR DRIVE SHAFT VISCIOUS DAMPER BEARINGS****1-3-16.1.2 Inspect.**

After servicing the viscous damper, inspect for leakage from the tube at completion of the first flight. If viscous damper tube assembly is transparent, leakage may be checked by making sure the air

bubble inside tube assembly is no more than 1-inch long. **IF LEAKAGE IS FOUND OR AIR BUBBLE IS MORE THAN 1-INCH LONG, REPLACE VISCIOUS DAMPER TUBE ASSEMBLY (PARA 6-4-45).**

1-3-17 PRESSURE REFUEL MAIN FUEL CELLS.

PARAGRAPH BREAKDOWN

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1-3-17.1 Pressure Refuel Main Fuel Cells	1-3-17-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Static Ground Cable

References

- PARA 1-6-16

1-3-17.1 Pressure Refuel Main Fuel Cells.

WARNING

Injury to personnel and damage to equipment will result if fuel servicing procedures are not followed exactly. Carelessness could cause an explosion. These procedures must be followed exactly and with extreme caution.

1-3-17.1.1. Prepare to Pressure Refuel.

- Turn off all electrical power (PARA 1-6-16).
- Park helicopter on level ground.
- Connect static ground cable to lower drag beam on axle.
- Connect static ground cable between servicing unit and ground.
- Connect static ground cable between helicopter and servicing unit.
- Plug fuel nozzle static ground into grounding receptacle on left side of helicopter fuselage (Figure 1-3-17-1, Detail A).

- Open pressure refuel door (Figure 1-3-17-1, Detail B).
- Remove filler cap from refuel/defuel adapter.
- Connect fuel nozzle to refuel/defuel adapter. Open valve by moving handle forward.

1-3-17.1.2. Pressure Refuel.

WARNING

To prevent injury to personnel or damage to equipment, refueling pumping system not to exceed a line pressure of 55 psi.

NOTE

Electrical power may be turned on to allow monitoring of fuel gages during refueling.

- Turn on all electrical power (PARA 1-6-16).
- Start refueling system and fuel helicopter.
- Turn off refueling system when no flow is indicated.

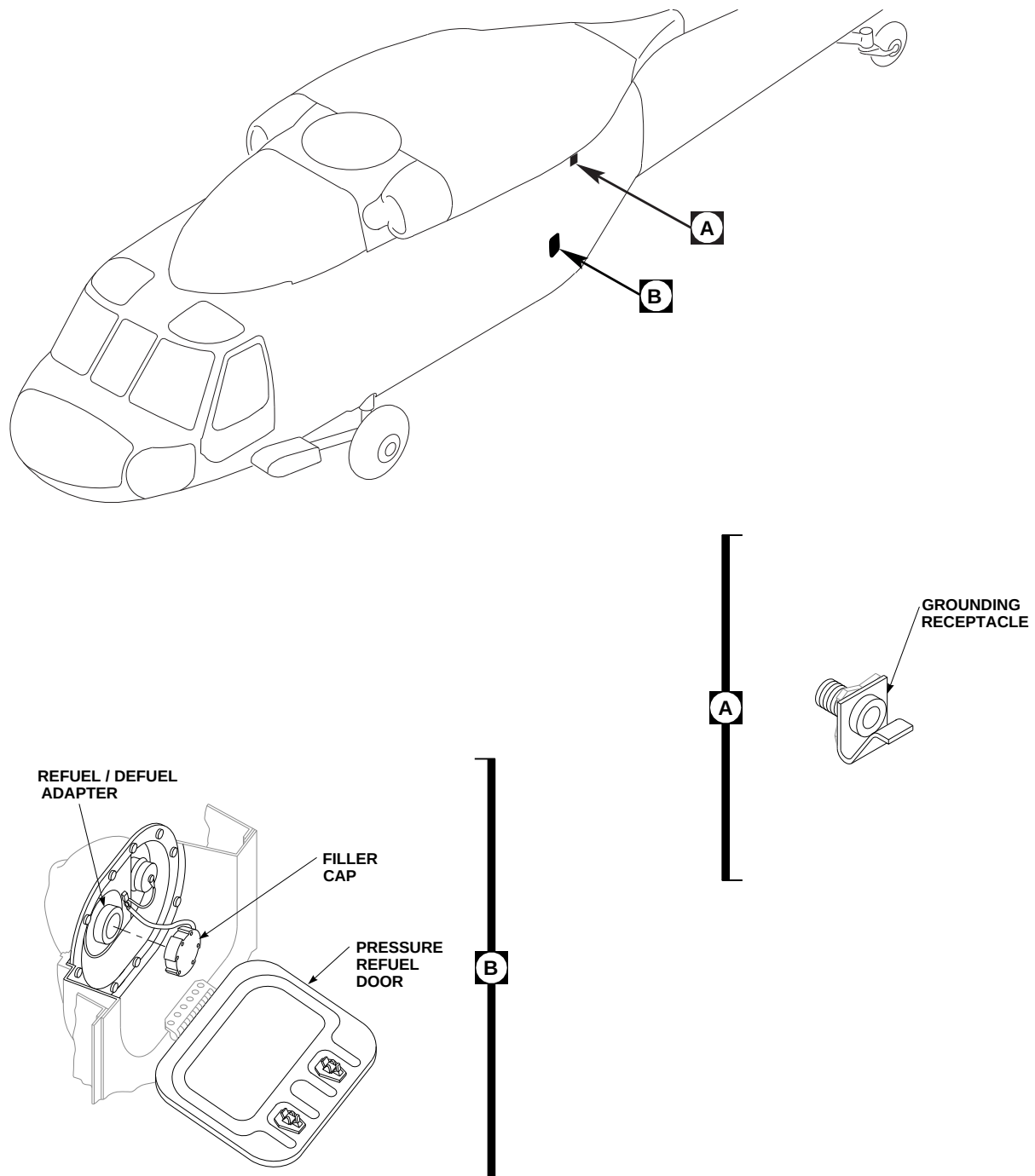


FIGURE 1-3-17-1. PRESSURE REFUEL MAIN FUEL CELLS

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- d. Remove fuel nozzle from refuel/defuel adapter.
- e. Install filler cap on refuel/defuel adapter (Figure 1-3-17-1, Detail B).
- f. Close pressure refuel door.
- g. Unplug fuel nozzle ground from ground receptacle on helicopter fuselage.
- h. Disconnect static ground cable between helicopter and servicing unit.
- i. Disconnect static ground cable between servicing unit and grounding receptacle (Figure 1-3-17-1, Detail A).
- j. Disconnect static ground cable from lower drag beam on axle.

1-3-18 GRAVITY REFUEL MAIN FUEL CELLS.

PARAGRAPH BREAKDOWN

	PAGE
1-3-18.1 Gravity Refuel Main Fuel Cells	1-3-18-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Static Ground Cable

References

- PARA 1-6-16

1-3-18.1 Gravity Refuel Main Fuel Cells.

WARNING

- Injury to personnel and damage to equipment will result if fuel servicing procedures are not followed exactly. Carelessness could cause an explosion. These procedures must be followed exactly and with extreme caution.
- Injury to personnel and damage to equipment will result if electrical power is applied to aircraft during gravity refueling. Stray electrical voltage could cause an explosion.

1-3-18.1.1. Prepare to Gravity Refuel.

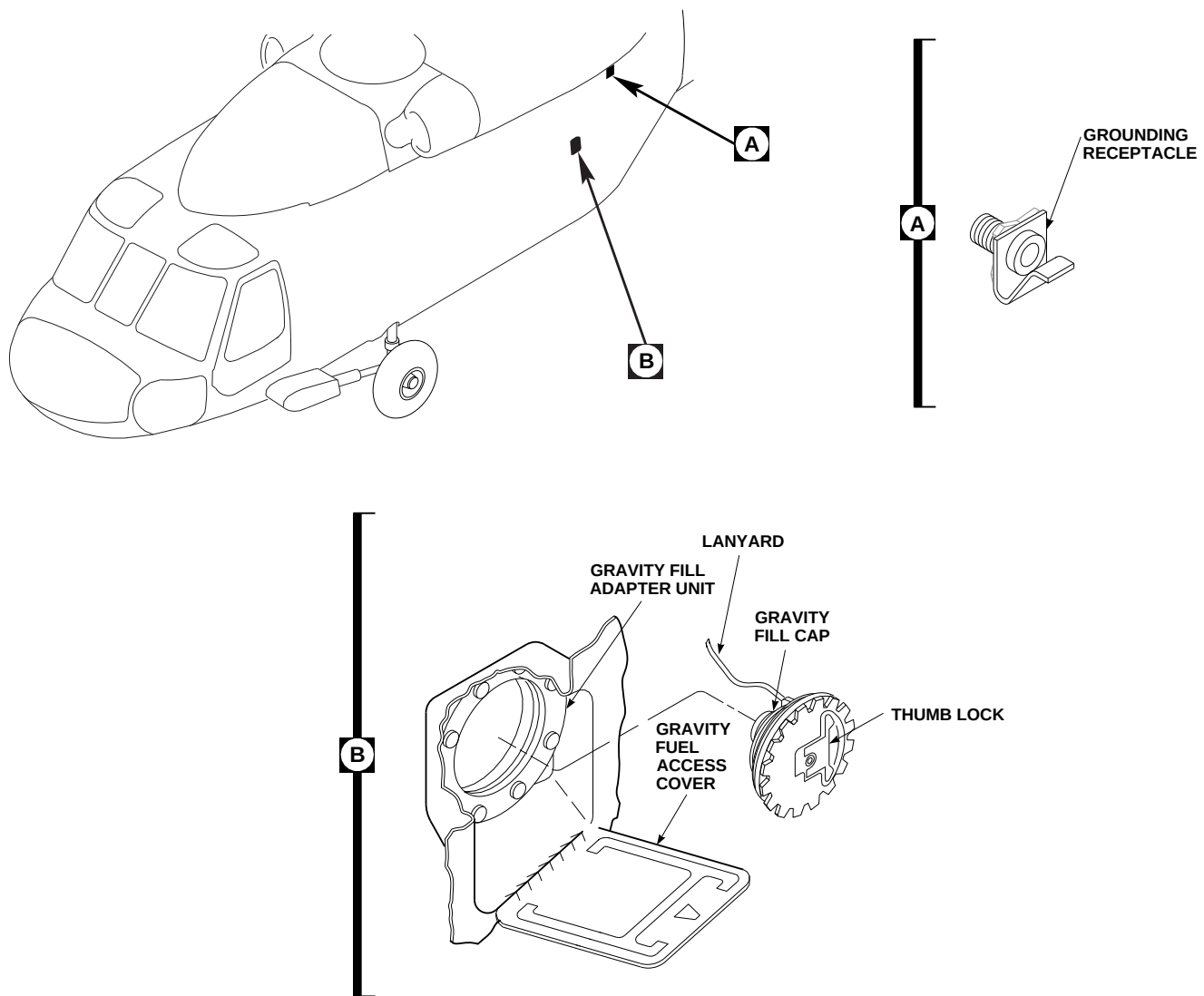
- Turn off all electrical power (PARA 1-6-16).
- Park helicopter on level ground.
- Connect static ground cable to lower drag beam on axle.
- Connect static ground cable between servicing unit and ground.

- Connect static ground cable between helicopter and servicing unit.
- Plug fuel nozzle static ground into grounding receptacle on left side of helicopter fuselage (Figure 1-3-18-1, Detail A).
- Open gravity fuel access cover (Figure 1-3-18-1, Detail B).
- Remove gravity fill cap as follows:
 - Release thumb lock on top of cap by pulling up on thumb lock.

CAUTION

Injury to personnel will result if main fuel cell pressure is not removed prior to removing filler cap. Before turning thumb lock, listen for escaping air and make sure no pressure remains in fuel cell.

- Turn thumb lock counterclockwise until it stops.
- Remove cap and let it hang by lanyard.

FR0612
SA**FIGURE 1-3-18-1. GRAVITY REFUEL MAIN FUEL CELLS**

- i. Place nozzle into fuel cell.

1-3-18.1.2. Gravity Refuel.

- a. Refuel helicopter.
- b. Stop refueling helicopter when fuel cells are full.
- c. Remove fuel nozzle from fuel cell.
- d. Install gravity fill cap as follows (Figure 1-3-18-1, Detail B):
 - (1) Place cap in housing and align forward-point arrow.
 - (2) Turn thumb lock clockwise to lock cap.
 - (3) Snap thumb lock closed to seal cap.
- e. Close gravity fuel access cover.
- f. Unplug fuel nozzle ground from grounding receptacle on helicopter fuselage (Figure 1-3-18-1, Detail A).
- g. Disconnect static ground cable between helicopter and servicing unit.
- h. Disconnect static ground cable between servicing unit and grounding receptacle.

- i. Disconnect static ground cable from lower drag beam on axle.

1-3-19 CLOSED CIRCUIT REFUEL.

PARAGRAPH BREAKDOWN

	PAGE
1-3-19.1 Closed Circuit Refuel	1-3-19-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Static Ground Cable

References

- PARA 1-6-16

1-3-19.1 Closed Circuit Refuel.

WARNING

Injury to personnel and damage to equipment will result if fuel servicing procedures are not followed exactly. Carelessness could cause an explosion. These procedures must be followed exactly and with extreme caution.

1-3-19.1.1 Prepare to Closed Circuit Refuel.

- Turn off all electrical power (PARA 1-6-16).
- Park helicopter on level ground.
- Connect static ground cable to lower drag beam on axle.
- Connect static ground cable between servicing unit and ground.
- Connect static ground cable between helicopter and servicing unit.
- Plug fuel nozzle static ground into ground receptacle on helicopter fuselage (Figure 1-3-19-1).

- Open pressure fuel access cover (Figure 1-3-19-1, Detail A).
- Remove cap from refuel adapter.
- Connect fuel nozzle to refuel adapter.

1-3-19.1.2 Closed Circuit Refuel.

NOTE

- Make sure refueling pump system provides a line pressure of 15 psi.
 - Electrical power may be turned on to allow monitoring of fuel gages during refueling (PARA 1-6-16).
- Refuel helicopter.
 - Stop refueling helicopter when tanks are full.
 - Remove fuel nozzle from refuel adapter (Figure 1-3-19-1, Detail A).
 - Install cap on refuel adapter.
 - Close pressure fuel access cover.
 - Unplug fuel nozzle static ground from ground receptacle on helicopter fuselage (Figure 1-3-19-1).

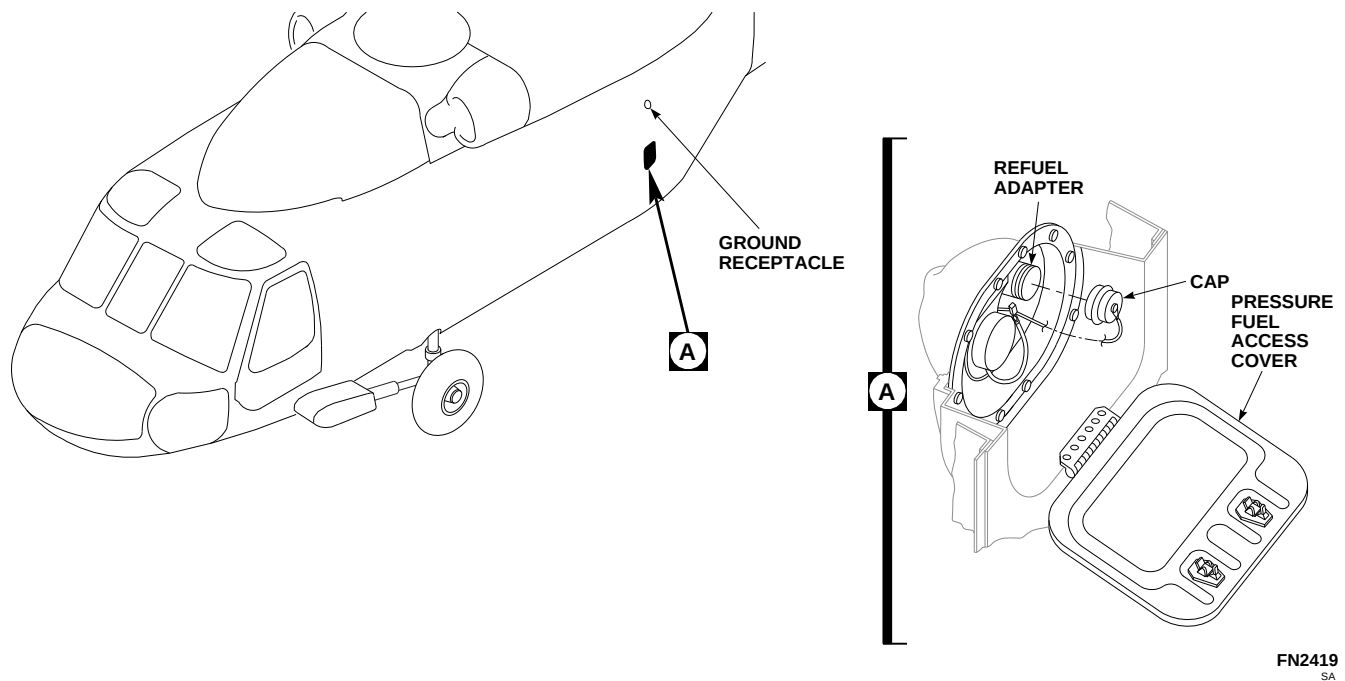


FIGURE 1-3-19-1. CLOSED CIRCUIT REFUEL

- g. Disconnect static ground cable between helicopter and servicing unit.
- h. Disconnect static ground cable between servicing unit and ground.
- i. Disconnect static ground cable from lower drag beam on axle.

1-3-20 PRESSURE DEFUEL MAIN FUEL CELLS.

PARAGRAPH BREAKDOWN

		PAGE
1-3-20.1	Pressure Defuel Main Fuel Cells	1-3-20-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Static Ground Cable

References

- PARA 1-3-21
- PARA 1-6-16

1-3-20.1 Pressure Defuel Main Fuel Cells.

WARNING

Injury to personnel and damage to equipment will result if fuel servicing procedures are not followed exactly. Carelessness could cause an explosion. These procedures must be followed exactly and with extreme caution.

1-3-20.1.1. Prepare to Pressure Defuel.

- Turn off all electrical power (PARA 1-6-16).
- Connect static ground cable to lower drag beam on axle.
- Connect static ground cable between servicing unit and ground.
- Connect static ground cable between helicopter and servicing unit.
- Plug fuel nozzle static ground into ground receptacle on left side of helicopter fuselage (Figure 1-3-20-1, Detail A).

- Open pressure refuel door (Figure 1-3-20-1, Detail B).
- Remove filler cap from refuel/defuel adapter.
- Connect fuel nozzle to refuel/defuel adapter.

1-3-20.1.2. Pressure Defuel.

CAUTION

To prevent damage to equipment, make sure defueling system does not exceed 3 psi negative pressure (suction).

- Defuel helicopter.

NOTE

When main fuel cells are defueled, some fuel will remain in fuel cell sumps.

- Stop defueling helicopter when fuel cells are empty.
- If necessary to remove remaining fuel, drain fuel from sumps (PARA 1-3-21).

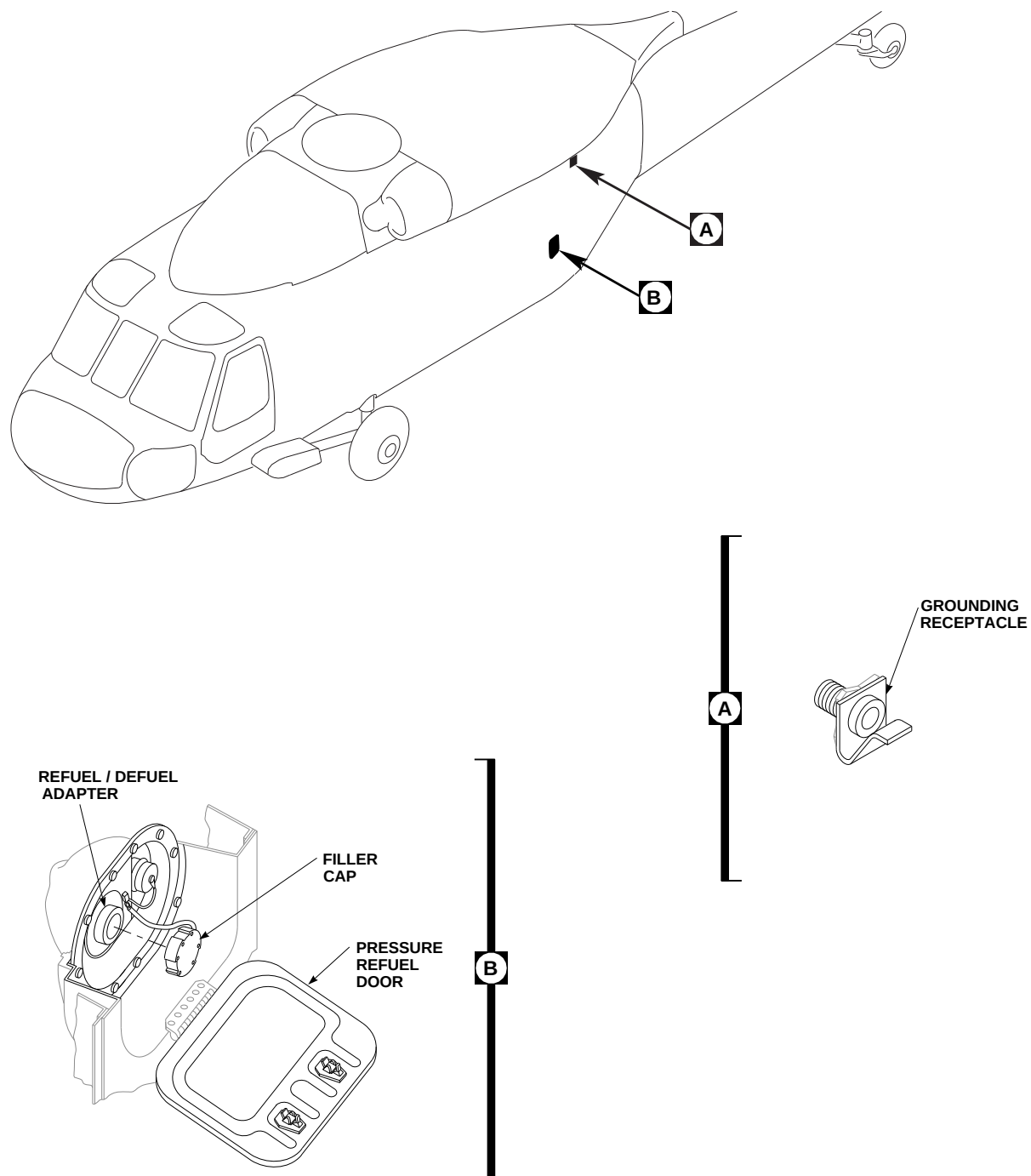


FIGURE 1-3-20-1. PRESSURE DEFUEL MAIN FUEL CELLS

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- d. Turn off defueling system.
- e. Remove fuel nozzle from refuel/defuel adapter.
- f. Install filler cap on refuel/defuel adapter (Figure 1-3-20-1, Detail B).
- g. Close pressure refuel door.
- h. Unplug fuel nozzle static ground from ground receptacle on helicopter fuselage (Figure 1-3-20-1, Detail A).
- i. Disconnect static ground cable between helicopter and servicing unit.
- j. Disconnect static ground cable between servicing unit and grounding receptacle.
- k. Disconnect static ground cable from lower drag beam on axle.

1-3-21 GRAVITY DEFUEL MAIN FUEL CELLS.

PARAGRAPH BREAKDOWN

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1-3-21.1 Gravity Defuel Main Fuel Cells	1-3-21-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Clamp, AN737-TW58 (or equivalent)
- Container, Variable

- Drain Valve, 70307-03018-102
- Flex Hose
- Static Ground Cable

References

- PARA 1-6-16

1-3-21.1 Gravity Defuel Main Fuel Cells.

WARNING

Injury to personnel and damage to equipment will result if fuel servicing procedures are not followed exactly. Carelessness could cause an explosion. These procedures must be followed exactly and with extreme caution.

NOTE

Gravity defueling of both main fuel cells is the same.

1-3-21.1.1 Prepare to Gravity Defuel.

- Turn off all electrical power (PARA 1-6-16).

WARNING

Injury to personnel will result if care is not taken when opening fuel sump drain

door. Fuel may be trapped above door. Take care when opening door.

- Unlock latches and turnlock fasteners, and open applicable fuel sump drain door (Figure 1-3-21-1, Detail A).
- Clamp length of flex hose to drain valve.
- Connect static ground cable between grounding receptacle on left side of helicopter fuselage and ground (Figure 1-3-21-1).

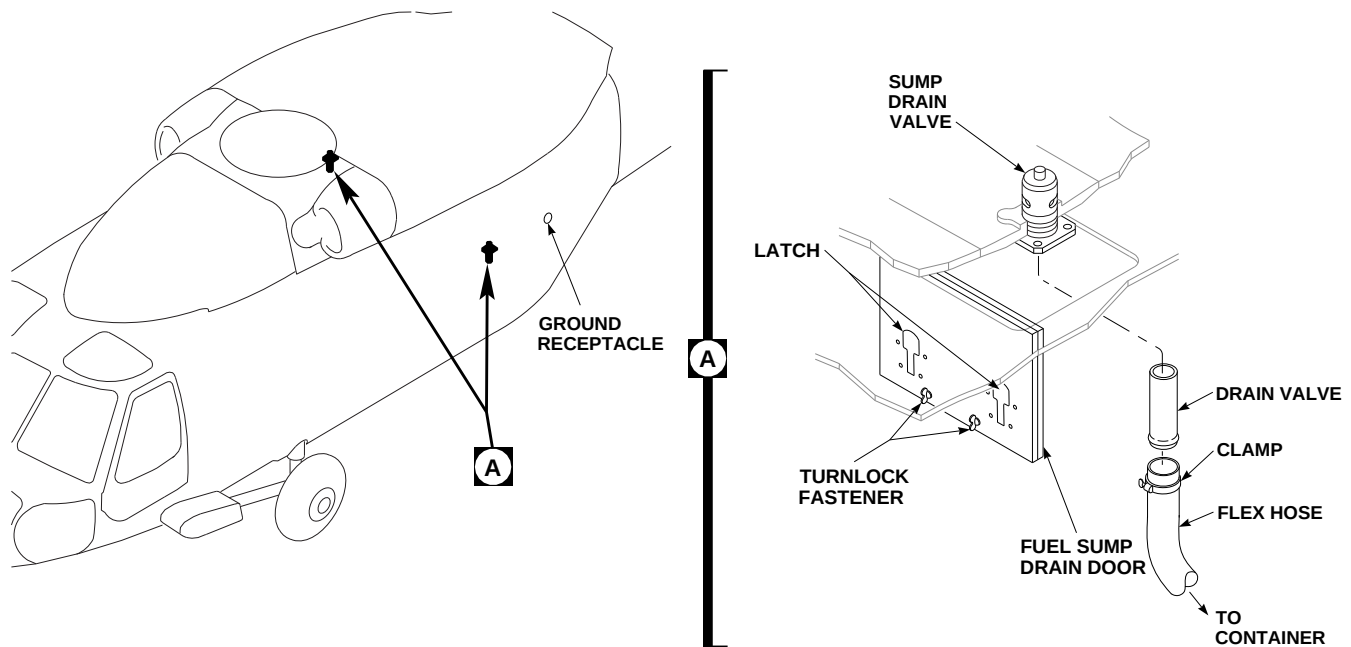
CAUTION

Make sure container is sufficiently large enough to contain all drained fuel.

- Place open end of flex hose in container to catch draining fuel.

1-3-21.1.2 Gravity Defuel.

- Position drain valve (clamped to flex hose) in sump drain valve, push up, and twist (Figure

FO0616A
SA**FIGURE 1-3-21-1. GRAVITY DEFUEL MAIN FUEL CELLS**

1-3-21-1, Detail A). Allow fuel to drain into container.

- b. Remove drain valve from sump drain valve when fuel cell is empty.
- c. Close fuel sump drain door, and lock latches and turnlock fasteners.

- d. Disconnect static ground cable from grounding receptacle on helicopter fuselage (Figure 1-3-21-1).

1-3-22 TRANSFER DEFUEL EXTENDED RANGE TANKS.

PARAGRAPH BREAKDOWN

	PAGE
1-3-22.1 Transfer Defuel Extended Range Tanks	1-3-22-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 25-Gallon

References

- Appropriate Flight Manual
- PARA 1-3-20
- PARA 1-3-21

1-3-22.1 Transfer Defuel Extended Range Tanks.

WARNING

Injury to personnel and damage to equipment will result if fuel servicing procedures are not followed exactly. Carelessness could cause an explosion. These procedures must be followed exactly and with extreme caution.

NOTE

Transfer defueling procedures apply to 230 gallon fuel tanks only.

1-3-22.1.1. Prepare to Transfer Defuel.

- Defuel helicopter internal tanks leaving approximately 20 gallons of fuel for APU operation (PARA 1-3-20 or PARA 1-3-21).

WARNING

Injury to personnel and damage to equipment will result if fuel transfer procedures are performed with refuel/defuel equipment connected. Make certain refuel/defuel equipment is disconnected prior to performing fuel transfer procedures.

- Set auxiliary fuel management control panel switches as follows (Figure 1-3-22-1, Detail A):
 - PRESS-OUTBD to OFF
 - PRESS-INBD to OFF
 - TANKS to OUTBD
 - MODE to OFF
 - MANUAL XFR-RIGHT to OFF

(6) MANUAL XFR-LEFT to OFF

(7) AUX FUEL QTY to TOTAL

- c. Make sure the following circuit breakers are pushed in on mission readiness circuit breaker panel:

(1) EXT FUEL RH

(2) NO. 2 XFER CONTROL

(3) EXT FUEL LH

(4) AUX FUEL QTY

- d. Make sure BACKUP HYD CONTR circuit breaker is pushed in on upper console.
- e. Pull out ESSS JTSN OUTBD and ESSS JTSN INBD circuit breakers on upper console.
- f. Pull out ESSS JTSN OUTBD, ESSS JTSN INBD, and BACKUP PUMP PWR circuit breakers on copilot's circuit breaker panel.

1-3-22.1.2. Transfer Defuel.

NOTE

More than one transfer may be required depending on amount of fuel in external tanks.

- a. Start APU (Appropriate Flight Manual).
- b. Place PRESS-OUTBD switch ON to pressurize outboard fuel tanks. Place PRESS-INBD switch ON to pressurize inboard fuel tanks.
- c. Place TANKS switch INBD or OUTBD to transfer the inboard or outboard fuel tanks.
- d. Place MODE switch to MANUAL.
- e. Place MANUAL XFR-RIGHT and MANUAL XFR-LEFT switches ON to transfer the right or left fuel tanks.

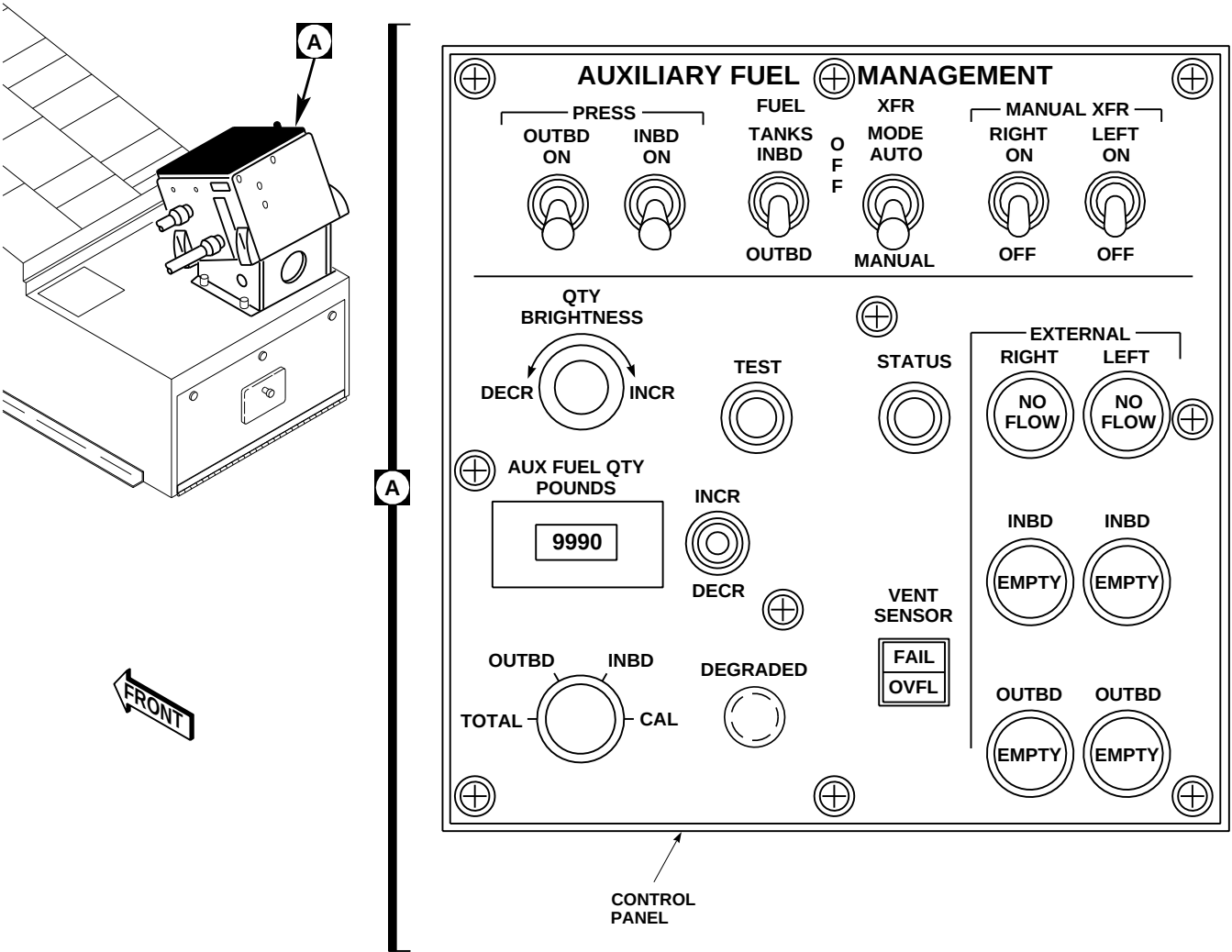
NOTE

- Time required for Center Display Unit TOTAL FUEL indication to increase in main fuel tanks will depend on fuel level in external tanks and time required to pressurize.
 - Auxiliary fuel management control panel EXTERNAL-RIGHT NO FLOW indicator will momentarily go on and then go off.
 - Auxiliary fuel management control panel AUX FUEL QTY indicator reading will decrease until internal tanks are full or external tanks are empty followed by RIGHT NO FLOW indicator going on.
- f. When transfer is complete, shut down APU (Appropriate Flight Manual).
- g. Repeat operation until desired fuel tanks are empty.
- h. Push in ESSS JTSN OUTBD, ESSS JTSN INBD, and BACKUP PUMP PWR circuit breakers on copilot's circuit breaker panel.
- i. Push in ESSS JTSN OUTBD and ESSS JTSN INBD circuit breakers on upper console.

WARNING

Injury to personnel and damage to equipment will result if residual fuel is not drained from fuel tank prior to removal. Residual fuel will add weight to fuel tank. Make certain all residual fuel is drained from fuel tank prior to removal.

- j. Drain all remaining fuel through drain valve on bottom of fuel tank using 25-gallon container.



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FIGURE 1-3-22-1. TRANSFER DEFUEL EXTENDED RANGE TANKS

1-3-23 PRESERVE MAIN FUEL CELL.

PARAGRAPH BREAKDOWN

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1-3-23.1 Preserve Main Fuel Cell	1-3-23-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Lubricating Oil, Item 207, Appendix D (or equivalent)
- Petrolatum, Item 234, Appendix D

References

- Appendix D
- PARA 1-3-20
- PARA 1-3-21
- PARA 2-4-45
- PARA 2-4-69
- PARA 10-4-40
- PARA 13-5-3
- PARA 13-5-6

1-3-23.1 Preserve Main Fuel Cell.

1-3-23.1.1 Fuel Cell Installed in Helicopter.

NOTE

Fuel cells installed in a helicopter that is being stored for less than 30 days require no special procedures. The following procedures apply if fuel cells are to remain drained for longer than 30 days.

- Defuel fuel cells (PARA 1-3-20 or PARA 1-3-21).
- Drain fuel cell sump (PARA 1-3-21).
- Gain access to main fuel cell compartment as follows:

- Remove rear passenger’s seats as required (PARA 2-4-45).
- Remove soundproofing panels from rear bulkhead covering left and right stowage compartments (PARA 2-4-69).
- Remove cargo netting from stowage compartments.
- If air conditioning is installed, remove ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
- Remove bolts, washers, and fuel cell access covers from main fuel cells. Remove packings from fuel cell access covers. Discard packings.
- Purge fuel cell of fumes (PARA 10-4-40).

NOTE

When coating cell, make sure that all puddled or pooled oil is removed immediately.

- f. Apply a light coating of lubricating oil, Item 207, Appendix D, or equivalent, to inside surfaces of each fuel cell.
- g. Lubricate packings with petrolatum, Item 234, Appendix D. Install packings on access covers.
- h. Coat threads of fuel cell access cover attachment bolts with antiseize compound, Item 74, Appendix D.
- i. Install fuel cell access covers on main fuel cells with bolts and washers. **TORQUE BOLTS TO 104 - 116 INCH-POUNDS.**
- j. Make sure area is clean and free of foreign material.
- k. Close fuel cell compartment as follows:
 - (1) If removed, install ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
 - (2) Install cargo netting.

- (3) Install soundproofing panels (PARA 2-4-69).
- (4) Install rear passenger's seats (PARA 2-4-45).

1-3-23.1.2 Fuel Cell not Installed in Helicopter.

- a. For fuel cell removed from helicopter for return to depot for repair, perform the following:
 - (1) Purge fuel cell of fumes (PARA 10-4-40).
 - (2) Scrub with warm, soapy water.
 - (3) Rinse in clean, clear water.
 - (4) Allow to air-dry.

NOTE

When coating cell, make sure all puddled or pooled oil is removed immediately.

- b. If fuel cell is to be stored for longer than 1 week, spray or swab a light coating of lubricating oil, Item 207, Appendix D, or equivalent, to inside surface of fuel cell.

1-3-24 GRAVITY REFUEL EXTENDED RANGE TANKS.

PARAGRAPH BREAKDOWN

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1-3-24.1 Gravity Refuel Extended Range Tanks	1-3-24-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Static Ground Cable

References

- PARA 1-6-16

1-3-24.1 Gravity Refuel Extended Range Tanks.

1-3-24.1.1. Prepare to Gravity Refuel.

WARNING

- Injury to personnel and damage to equipment will result if fuel servicing procedures are not followed exactly. Carelessness could cause an explosion. These procedures must be followed exactly and with extreme caution.
- Injury to personnel and damage to equipment will result if electrical power is applied to aircraft during gravity refueling. Stray electrical voltage could cause an explosion.

NOTE

Gravity refueling procedures apply to 230 gallon fuel tanks only.

- Turn off all electrical power (PARA 1-6-16).
- Park helicopter on level ground.

- Connect static ground cable between lower stage air valve on shock strut and ground (Figure 1-3-24-1, Detail A).
- Connect static ground cable to servicing unit.
- Connect static ground cable between helicopter and servicing unit.
- Plug fuel nozzle static ground into inboard ground receptacle on fuel tank (Figure 1-3-24-1).

WARNING

Injury to personnel will result if care is not taken when lifting up lock in center of fuel tank cap. Fuel tank may be pressurized. Stand clear of fuel tank cap when lifting up lock.

- Slowly lift up thumb lock in center of filler cap to relieve pressure in fuel tank (Figure 1-3-24-1, Detail B).
- Remove filler cap from fuel tank.
- Insert fuel nozzle into fuel tank.

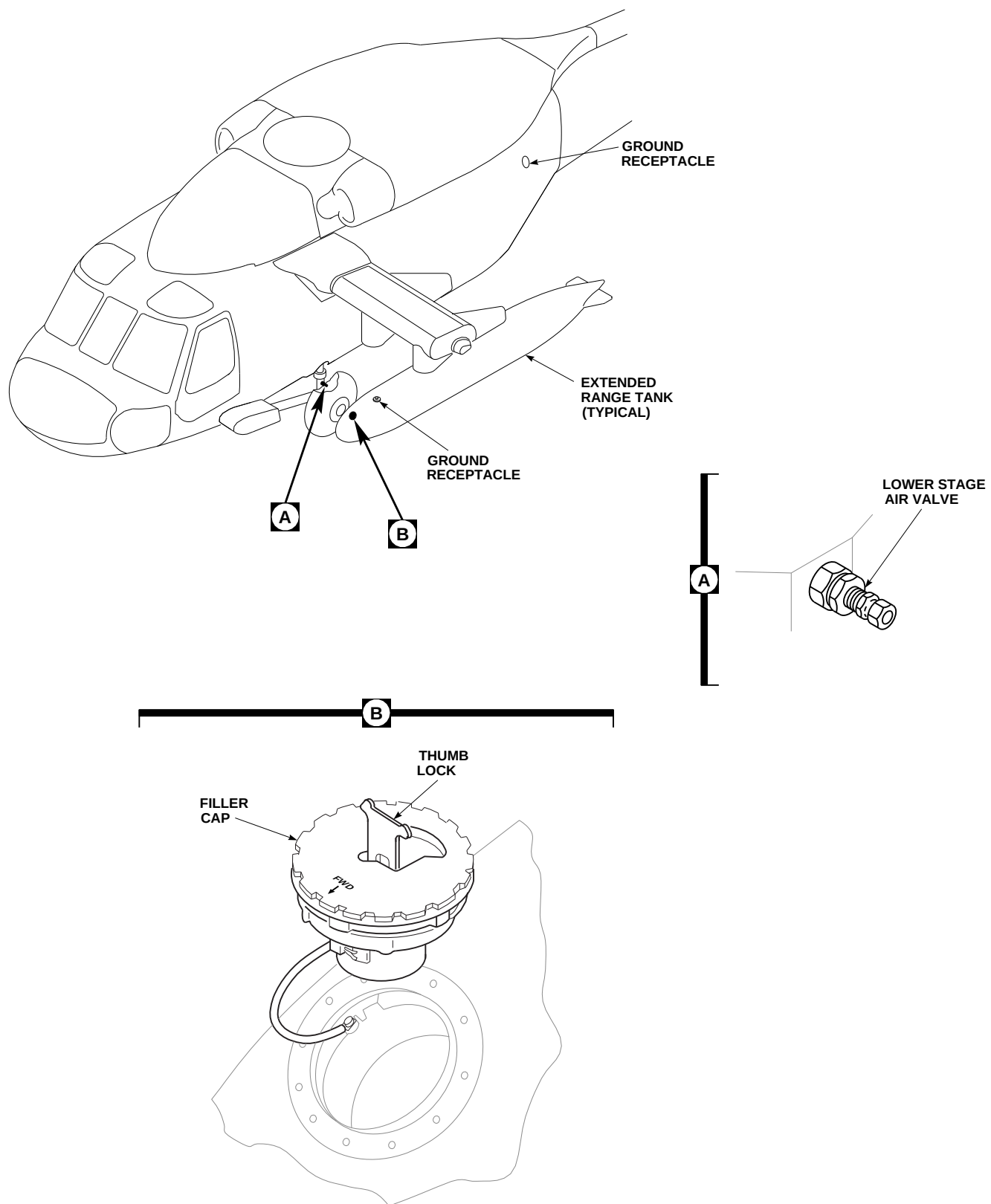


FIGURE 1-3-24-1. GRAVITY REFUEL EXTENDED RANGE TANKS

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1-3-24.1.2. Gravity Refuel.

- a. Refuel tanks.
- b. Stop refueling when tanks are full.
- c. Remove nozzle from tank and install filler cap in fuel tank with thumb lock in center of filler cap pointing to rear (Figure 1-3-24-1, Detail B).
- d. Unplug fuel nozzle static ground from ground receptacle on fuel tank (Figure 1-3-24-1).
- e. Disconnect static ground cable between helicopter and servicing unit.
- f. Disconnect static ground cable between servicing unit and ground.
- g. Disconnect static ground cable from lower stage air valve on shock strut (Figure 1-3-24-1, Detail A).

1-3-25 ADAPTER DEFUEL EXTENDED RANGE TANKS.

PARAGRAPH BREAKDOWN

	PAGE
1-3-25.1 Adapter Defuel Extended Range Tanks	1-3-25-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container
- Defuel Adapter, Locally-Made, Figure H-163, Appendix H
- Static Ground Cable

Materials/Parts

- Adapter, MS24392-8
- Petrolatum, Item 234, Appendix D

References

- Appendix D
- Appendix H
- PARA 1-6-16

1-3-25.1 Adapter Defuel Extended Range Tanks.

1-3-25.1.1. Prepare to Adapter Defuel.

- Locally make defuel adapter (Figure H-163, Appendix H).

WARNING

Injury to personnel and damage to equipment will result if fuel servicing procedures are not followed exactly. Carelessness could cause an explosion. These procedures must be followed exactly and with extreme caution.

NOTE

Adapter defueling procedures apply to 230 gallon fuel tanks only.

- Turn off all electrical power (PARA 1-6-16).
- Connect static ground cable to lower stage air valve on shock strut (Figure 1-3-25-1, Sheet 1, Detail A).
- Connect static ground cable to servicing unit.
- Connect static ground cable between helicopter and servicing unit.
- Connect fuel nozzle static ground into grounding receptacle on fuel tank (Figure 1-3-25-1, Sheet 2, Detail B).

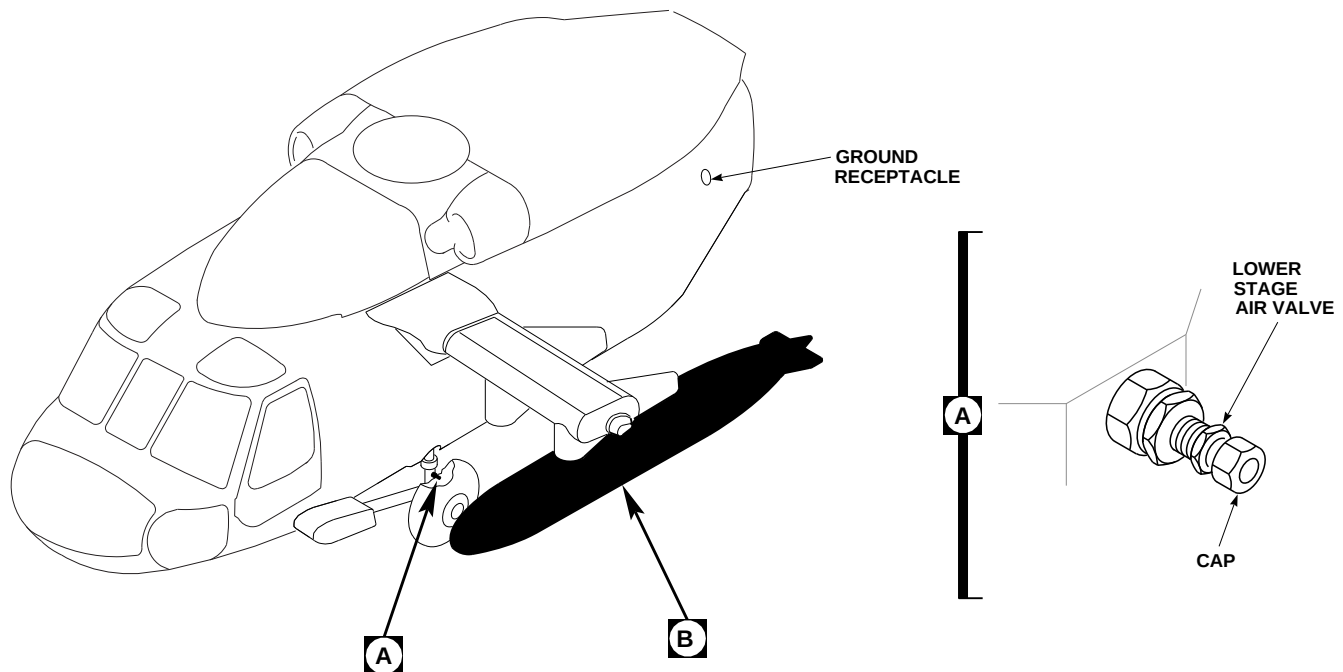
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FIGURE 1-3-25-1. ADAPTER DEFUEL EXTENDED RANGE TANKS (SHEET 1 OF 2)

WARNING

Injury to personnel will result if care is not taken when lifting up thumb lock in center of filler cap. Fuel tank may be pressurized. Stand clear of filler cap when lifting up thumb lock.

- g. Slowly lift up thumb lock in center of filler cap to relieve pressure in fuel tank (Figure 1-3-24-1, Sheet 2, Detail C).
- h. Remove plug and packing closest to rear suspension lug (Figure 1-3-25-1, Sheet 2, Detail B).
- i. Lubricate packing with petrolatum, Item 234, Appendix D.
- j. Install packing and fuel tank adapter, into fuel tank.
- k. Install defuel adapter to fuel tank adapter.

- l. Connect defuel nozzle to defuel adapter.

1-3-25.1.2. Adapter Defuel.**NOTE**

Make sure defueling system does not exceed 3 psi negative pressure (suction).

- a. Start defueling helicopter fuel tank.
- b. Stop defueling when tank is empty.
- c. Disconnect defuel nozzle from defuel adapter.
- d. Remove defuel adapter.
- e. Remove fuel tank adapter and packing (Figure 1-3-25-1, Sheet 2, Detail B). Discard packing.
- f. Lubricate packing with petrolatum, Item 234, Appendix D.
- g. Install plug and packing in fuel tank.
- h. Close thumb lock in center of filler cap, making sure it is pointing to rear (Figure 1-3-25-1, Sheet 2, Detail C).

1-3-25-2

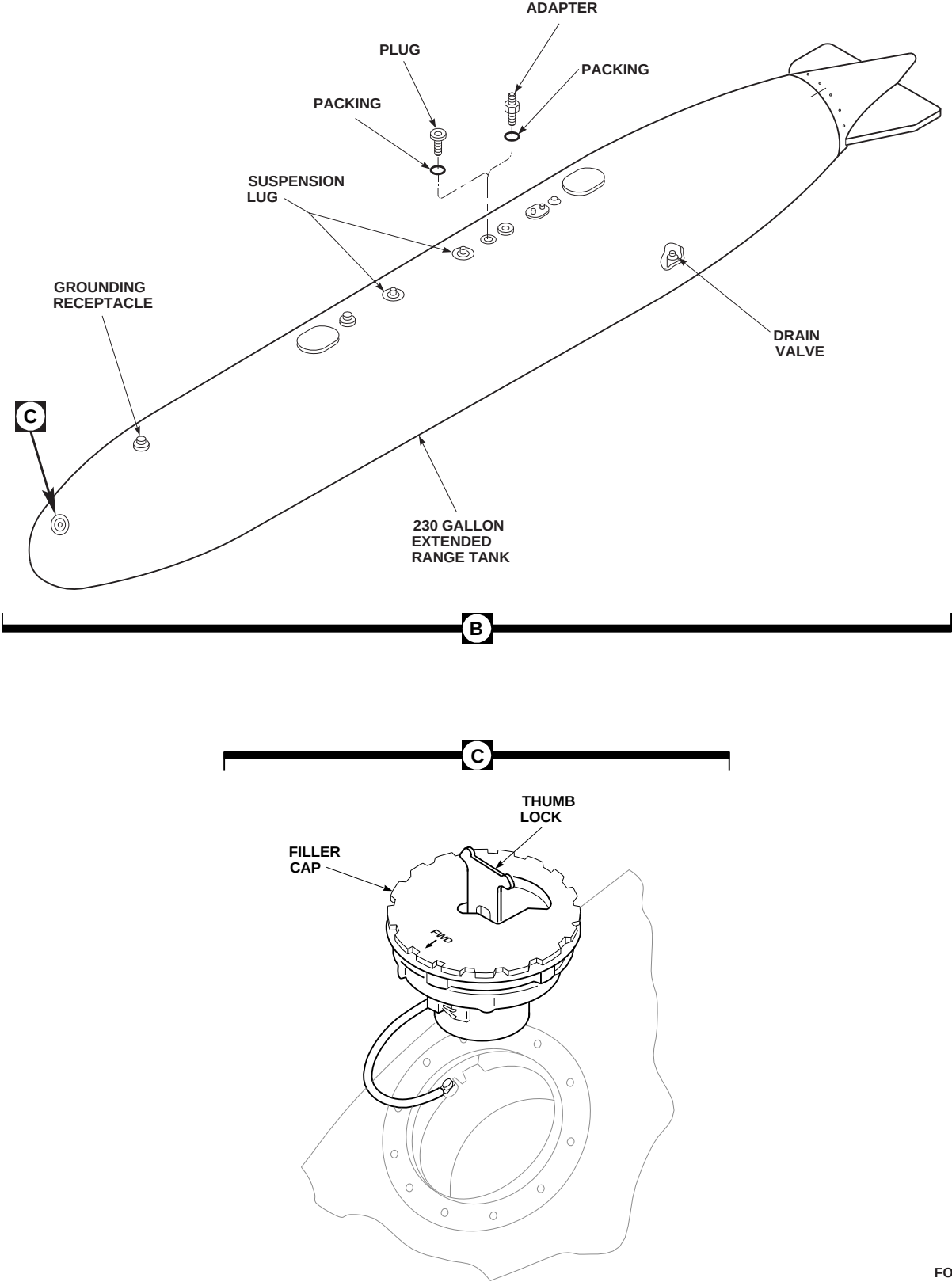


FIGURE 1-3-25-1. ADAPTER DEFUEL EXTENDED RANGE TANKS (SHEET 2 OF 2)

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- i. Disconnect defuel nozzle static ground from grounding receptacle on fuel tank (Figure 1-3-25-1, Sheet 2, Detail B).
 - j. Disconnect static ground cable between helicopter and servicing unit.
 - k. Disconnect static ground cable from servicing unit.
 - l. Disconnect static ground cable from lower stage air valve on shock strut (Figure 1-3-25-1, Sheet 1, Detail A).
 - m. Drain all remaining fuel through drain valve on bottom of fuel tank in suitable container (Figure 1-3-25-1, Sheet 2, Detail B).
- not drained from fuel tank prior to removal. Residual fuel will add weight to fuel tank. Make certain all residual fuel is drained from fuel tank prior to removal.

WARNING

Injury to personnel and damage to equipment will result if residual fuel is

1-3-26 CLEAN AND PRESERVE EXTENDED RANGE TANKS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE	
1-3-26.1	Clean Extended Range Tanks	1-3-26-1	1-3-26.3	Depreserve Extended Range Tanks	1-3-26-3	
1-3-26.2	Preserve Extended Range Tanks	1-3-26-2	1-3-26.4	Store Extended Range Tanks	1-3-26-3	

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Combustible Gas Indicator, 51-8018
- Fire Extinguisher, CO2-type
- Flexible Hose, ¼-inch-diameter
- Shipping Cradle, 24ATC005-501
- Static Ground Cable
- Wooden Stick, 3-feet long, Locally-Made

Materials/Parts

- Aircraft Fuel, 10-Gallons
- Cheesecloth, Item 90, Appendix D

- Cleaning Compound, Item 97A, Appendix D
- Lubricating Oil, Item 207, Appendix D
- Nitrogen, Item 221, Appendix D
- Tags

References

- Appendix D
- PARA 16-4-54
- PARA 16-4-55
- PARA 16-4-63

1-3-26.1 Clean Extended Range Tanks.

- Rinse outside of fuel tank with clean water to wash out foreign particles and contamination.
- Slowly lift up thumb lock in center of filler cap to relieve pressure in fuel tank.
- Remove access doors (PARA 16-4-63).
- Rinse inside of fuel tank with clean water to wash out foreign particles and contamination.
- Remove drain plug and all other plugs and caps. Position tank at a 4° nose-up attitude to allow water to drain from tank during cleaning process.
- Clean fuel tank inside and out using cleaning compound, Item 97A, Appendix D.
- Rinse tank with clean water.
- Completely dry inside and outside of tank using clean, lint-free, cheesecloth, Item 90, Appendix D.

- i. Install access doors (PARA 16-4-63).
- j. Close thumb lock in center of filler cap, making sure it is pointing to rear.

1-3-26.2 Preserve Extended Range Tanks.

WARNING

Injury to personnel and damage to equipment will result if extended range tanks are improperly purged. Tanks could explode. Observe all warnings, cautions, and notes while purging extended range tanks.

- a. Drain all fuel from extended range tank. Make sure drain valve is closed when tank is defueled.
- b. Remove tank from helicopter (PARA 16-4-54). Place tank in suitable cradle or on padded supports.
- c. Connect static ground cable to ground receptacle on fuel tank.
- d. Pour 5 gallons of lubricating oil, Item 207, Appendix D, in tank. Rotate tank 360° to make sure all internal surfaces of tank are coated with oil.
- e. Drain lubricating oil from tank.
- f. Remove drain valve (PARA 16-4-55) and open filler cap.
- g. Locally make wooden stick, such as a broom stick, to a length of 3 feet.

NOTE

Insert wooden stick just enough to open flapper valve.

- h. Carefully push wooden stick in filler neck to open flapper valve.
- i. Connect 1/4-inch diameter flexible hose to a regulated air source and insert the open end in drain valve hole.

- j. Purge tank with regulated air for one hour at 50 psi.
- k. Remove air hose from hole. Install drain valve (PARA 16-4-55).

WARNING

To avoid generating static electrical charge when using a carbon dioxide fire extinguisher, remove horn from hose before inserting into filler neck. Ground fire extinguisher to helicopter.

NOTE

- Use fire extinguisher as a source of carbon dioxide.
- Nitrogen, Item 221, Appendix D, or other inert gas may be used in place of carbon dioxide.
- l. Attach one end of flexible hose to the nitrogen or inert gas source and carefully insert the open end into the tank filler neck. Insert hose far enough to open and extend past filler valve.
- m. Discharge nitrogen or inert gas into tank at a rate of not more than one pound of gas per minute.
- n. Use at least 3 pounds of nitrogen or inert gas to fill/purge tank.
- o. Remove nitrogen or inert gas hose from filler neck and allow tank to set for 3 hours.
- p. Test fuel tank with a combustible gas indicator for presence of fuel vapor. If fuel vapor is detected, repeat steps f. through p. until no vapors are detected.
- q. If tank is to be stored for more than 45 days, preserve fuel tank by coating all interior surfaces with lubricating oil, Item 207, Appendix D.
- r. Attach a tag to filler cap stating: "THIS TANK HAS BEEN PRESERVED IN AC -

CORDANCE WITH TM 1-70-23 AND IS GAS FREE. REMOVE THIS TAG UPON FILLING TANK WITH FUEL." Sign and date tag.

1-3-26.3 Depreserve Extended Range Tanks.

- a. Slowly lift up thumb lock in center of filler cap to relieve pressure in fuel tank.
- b. Remove access doors (PARA 16-4-63).
- c. Inspect the inside of the tank for fungus contamination.
- d. If fungus contamination is present, clean tank (PARA 1-3-26.1).
- e. Install access doors (PARA 16-4-63).

WARNING

Fuel tank weighs approximately 143 pounds empty. Aviation fuels weigh approximately 6.5 pounds per gallon. Make sure adequate number of personnel are utilized to lift and move the fuel tank.

- f. Pour approximately 10 gallons of aircraft fuel into the tank. Rotate the fuel tank 360° several times to flush the inside of the tank.
- g. Drain the flushing fuel from the tank.
- h. Close thumb lock in center of filler cap, making sure it is pointing to rear.

1-3-26.4 Store Extended Range Tanks.

- a. Remove extended range tank from helicopter (PARA 16-4-54).
- b. Clean and preserve tank (PARA 1-3-26.1 and PARA 1-3-26.2).
- c. Unlatch and remove top half of shipping cradle. Set top half aside.
- d. Place fuel tank into bottom half of shipping cradle. Place top half of shipping cradle onto tank making sure tank lugs fit into corresponding holes in cradle and that black arrow on top cradle half points toward the tank nose.
- e. Relatch cradle halves.
- f. Store tanks no more than three high in warm, dry area.

1-3-27 ENVIRONMENTAL CONTROL SYSTEM - SERVICE.

PARAGRAPH BREAKDOWN

	PAGE
1-3-27.1 Service/Deservice Environmental Control System	1-3-27-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 5-Gallon
- Electronic Leak Detector
- Multimeter, Clip-On, ME-489/U
- Protection, Eye, Skin, and Breathing
- Refrigerant Recovery and Recycling Station, 17500B
- Refrigeration Unit Service Toolkit

- Vacuum Gage

- Vacuum Pump, Rotary, 14008

Materials/Parts

- Nitrogen, Item 221, Appendix D
- Refrigerant, Item 261, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 13-5-6

1-3-27.1 Service/Deservice Environmental Control System.

1-3-27.1.1 Service.

- Open transition section avionics door.
- Connect low pressure (blue) hose of refrigerant recovery and recycling station to low side of ECS servicing manifold and high pressure (red) hose of refrigerant recovery and recycling station to high side of servicing manifold. Make sure manifold hand wheels are closed (Figure 1-3-27-1, Sheet 1, Detail A).

- Remove knurled brass covers from servicing valves.
- Connect other ends of hoses to color-coded servicing valves on helicopters.

NOTE

Blue hose will be on left side, red hose will be on right side.

- Connect yellow hose to refrigerant, Item 261, Appendix D, bottle and spare vacuum gage.

CAUTION

- To avoid damage to compressor, make sure freon bottle is positioned upright so only vapor will transfer.
 - Not purging or improper purging of air from service lines will allow air and water vapor to enter system causing damage and reduced efficiency.
- f. Open valve on refrigerant bottle and record bottle pressure.
 - g. Close valve on refrigerant bottle and disconnect yellow hose from spare vacuum gage.
 - h. Connect end of yellow hose to center supply port of servicing manifold.
 - i. Open refrigerant bottle valve.

WARNING

- To prevent injury to personnel, observe the following safety precautions when working on any part of air-conditioning system:
- Protect skin and eyes from possible contact with liquid refrigerant. Wear gloves and goggles. Keep your body as well protected as you can from possible contact with liquid refrigerant.
- If liquid refrigerant comes in contact with eyes or skin, get immediate medical aid.
- The refrigerant in a sealed system is always under pressure. Do not disconnect any line or component of pressurized cooling system without first carefully releasing pressure.
- Refrigerant will displace air and can cause suffocation, even though it is

nontoxic. It will produce highly toxic phosgene gas if exposed to open flame. Make sure work area is well ventilated, and that there are no open flames, or unshielded heaters in immediate area.

- The use of heat on any part of pressurized system will cause a pressure buildup and could cause an explosion. Do not try to solder or weld any part of the system while the system is charged with refrigerant.
- j. Purge yellow hose of air by loosening fitting at servicing manifold until a little gaseous freon comes out. Immediately tighten fitting.
 - k. Open low side and high side hand wheel until low pressure gage equals refrigerant bottle pressure.
 - l. Close low side, high side hand wheels and refrigerant bottle valve. Disconnect yellow hose from refrigerant bottle.
 - m. Close transition section avionics door.

1-3-27.1.2 Leak Check.**WARNING**

Injury to personnel and damage to equipment will result if system pressure exceeds 300 psig during leak test. Make sure that nitrogen pressure gages are properly calibrated and regulator adjusting screw is set for 0 psig. Close valves on servicing manifold prior to opening supply valve on nitrogen bottle.

- a. Open transition section avionics door.
- b. Unscrew regulator adjusting screw on regulated dry nitrogen source until it is loose. Connect yellow hose to regulated nitrogen, Item 221, Appendix D.
- c. Open valve on nitrogen source, set regulator to 5 psig.

- d. Purge yellow hose by slowly loosening fitting at servicing manifold until a small amount of air and nitrogen come out. Immediately tighten fitting.
- e. Increase regulator pressure until it equals refrigerant system pressure.
- f. Open high and low side hand wheel. Slowly increase refrigerant system pressure, indicated on servicing manifold high pressure gage, to 300 psig.
- g. Using electronic leak detector, check all fittings, connections, hoses, and components for leaks. Repair as necessary. Repeat until no leaks are found.
- h. Close high side and low side hand wheel on servicing manifold.
- i. Close nitrogen valve and unscrew regulator adjusting screw on nitrogen source. Disconnect yellow hose from nitrogen regulator fitting and place loose end into clean 5-gallon container, ¼-full of water.

NOTE

Too rapid release of pressure will remove oil from system.

- j. Slowly open high side and low side hand wheels until small amount of bubbles is evident in water. Drain system until no bubbles appear in container. Pressure gages should read 0 psig.
- k. Close high side and low side hand wheel and disconnect yellow hose.
- l. Close transition section avionics door.

1-3-27.1.3 Evacuating System.

- a. Open transition section avionics door.
- b. Connect yellow hose to vacuum pump and center supply port on ECS servicing manifold.
- c. Turn on vacuum pump and open low side and high side hand wheels. Operate pump for a minimum of 1.5 hours after low side gage reads between 29.5 and 29.9 in. Hg.

- d. Turn off vacuum pump and close low side and high side hand wheels.
- e. Disconnect yellow hose from vacuum pump.
- f. Close transition section avionics door.

1-3-27.1.4 Recharging System.

- a. Open transition section avionics door.
- b. Connect yellow hose to refrigerant bottle and center supply port side of ECS servicing manifold.

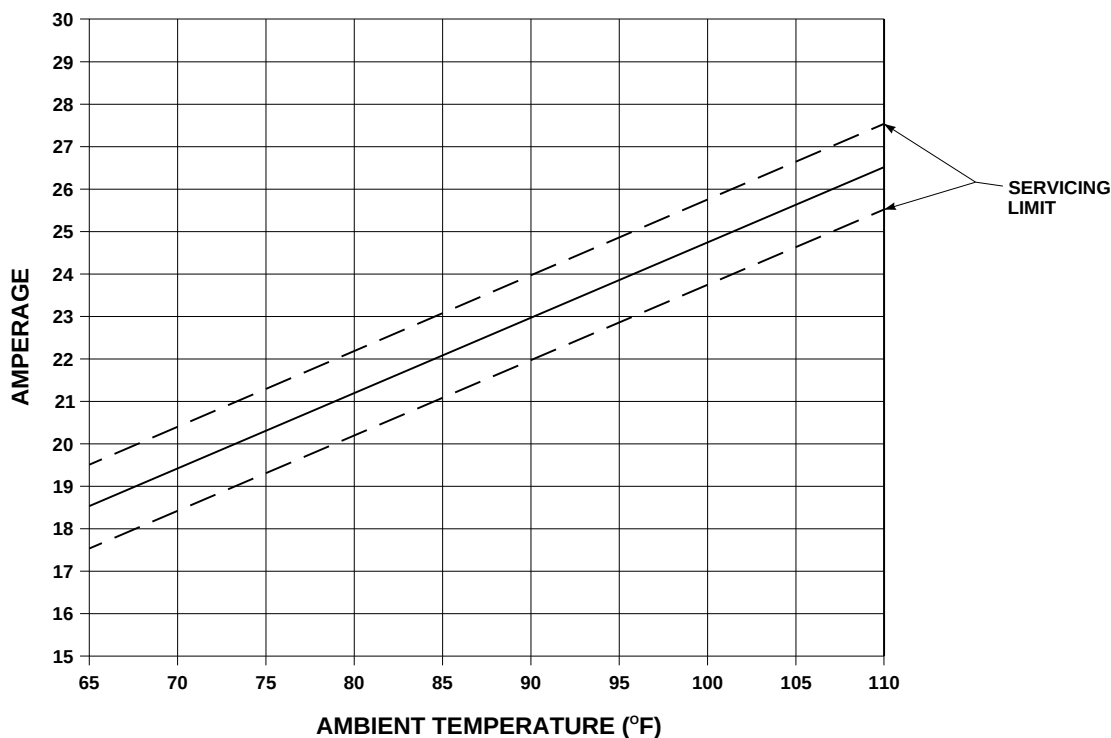
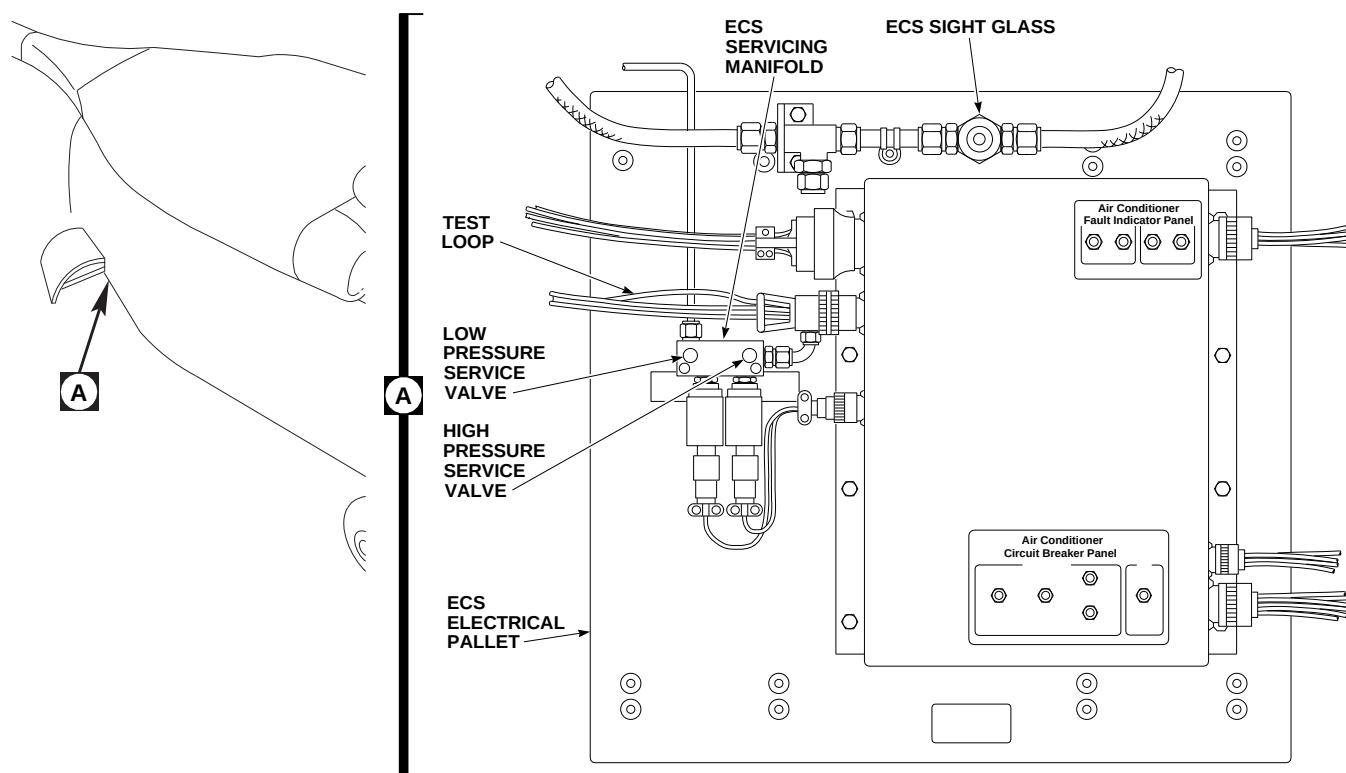
WARNING

- Refrigerant boils at -30°F (-34°C), and will cause frostbite if spilled on unprotected skin. To avoid personal injury, wear gloves and eye protection when purging or installing refrigerant servicing hoses.
- Refrigerant will displace air, and can cause suffocation, even though it is nontoxic. It will produce poisonous gas if exposed to open flame. Make sure work area is well ventilated, and there are no open flames or heating apparatus in immediate area.

CAUTION

Improper purging or not purging air from servicing lines will allow air and water vapor to enter system, causing damage and reduced efficiency.

- c. Purge yellow hose of air as follows:
 - (1) Open valve on refrigerant bottle.
 - (2) Loosen hose fitting at servicing manifold until a little gas comes out. Immediately tighten fitting.



AIR CONDITIONING SERVICE CHART - AMPERAGE / TEMPERATURE

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**FIGURE 1-3-27-1. SERVICE/DESERVICE ENVIRONMENTAL CONTROL SYSTEM
(SHEET 1 OF 2)**

CAUTION

Damage to compressor will occur if refrigerant bottle is not kept upright. Air or water vapor may enter system. Keep refrigerant bottle upright.

- d. Engage the following circuit breakers for the ECS:
 - (1) ECS PWR on NO. 1 DC PRI BUS
 - (2) ECS CONTR on NO. 2 DC PRI BUS
- e. Slowly open refrigerant, Item 261, Appendix D, bottle valve and low side and high side handwheels.
- f. When both gages are approximately equal, close both handwheels.
- g. Connect external electrical power cart to helicopter (PARA 1-6-16).
- h. Turn on EXT PWR switch.
- i. Turn on all electrical power (PARA 1-6-16).

NOTE

Backup pump may run to charge APU accumulators. ECS will not operate until back up pump stops.

- j. Connect the clip-on multimeter to one phase of the compressor/motor electrical harness. Use test loop provided on ECS electrical pallet.
- k. Position clean 5-gallon container under evaporator drain.
- l. When back up pump has shut off, turn on ECS by placing HTR switch to OFF and AIR COND switch to COOL (Figure 1-3-27-1, Sheet 2, Detail A). Set TEMP to full COOL.

NOTE

The evaporator blower will immediately start, the condenser blower will start

after 5 seconds, and after 10 seconds the compressor will start.

- m. Watch ECS sight glass and multimeter while servicing.

WARNING

Injury to personnel and damage to equipment will result if high pressure handwheel is used to meter refrigerant during recharging. Refrigerant supply bottle may explode. Do not use high pressure handwheel to meter refrigerant during recharging.

- n. Use low side handwheel to meter refrigerant into system until proper amperage value is achieved (Figure 1-3-27-1, Sheet 1).

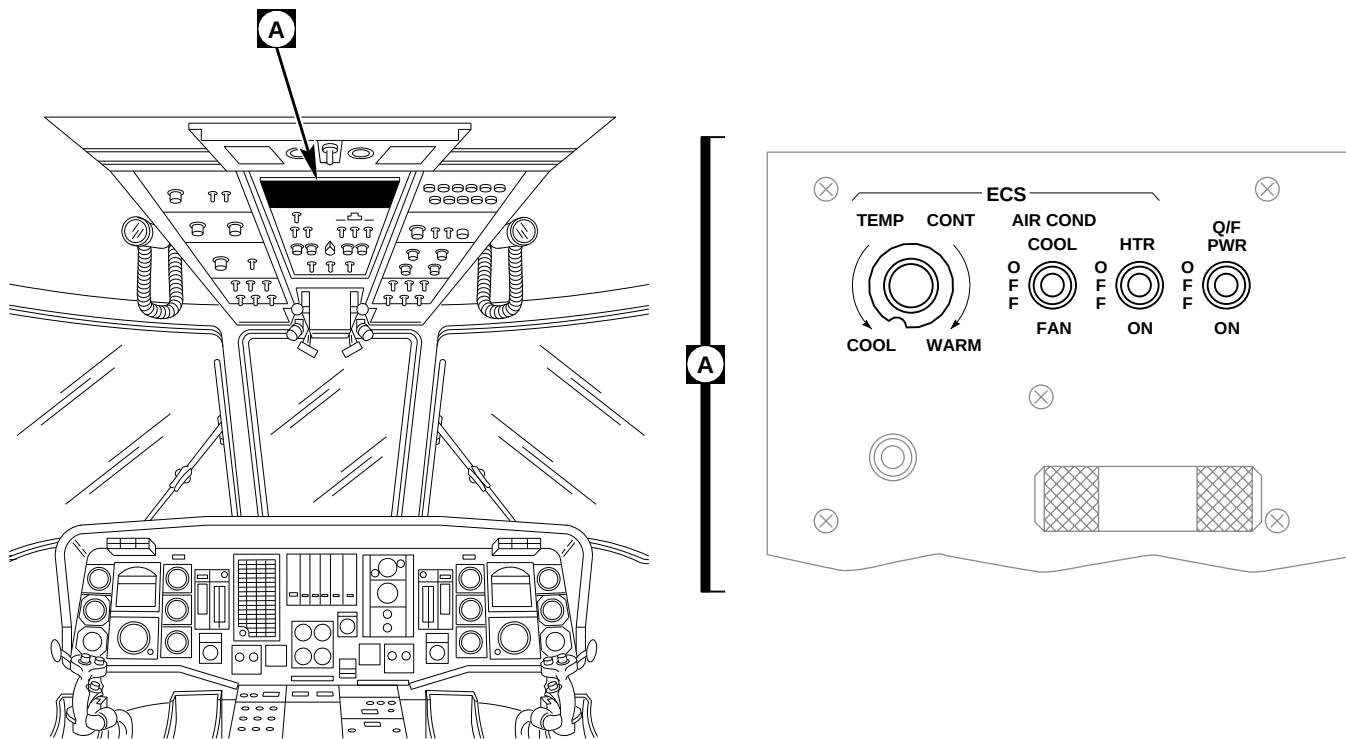
CAUTION

Compressor and motor will be damaged and cooling efficiency reduced if system is overserviced. Allow only enough refrigerant into system to allow for efficient operation.

NOTE

Optimum charge is reached at target amperage. Refer to servicing chart when servicing. Occasional bubbles in sight glass is acceptable.

- o. Compare ampmeter reading with servicing chart (Figure 1-3-27-1, Sheet 1). Close low side handwheel and refrigerant bottle valve when amperage is at low end of servicing limit. Sight glass should just be at point of clearing (Figure 1-3-27-1, Sheet 1, Detail A). An occasional flow of bubbles is allowed.
- p. Allow system to operate for 15 minutes minimum.
- q. Recheck multimeter reading.

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**FIGURE 1-3-27-1. SERVICE/DESERVICE ENVIRONMENTAL CONTROL SYSTEM
(SHEET 2 OF 2)**

- r. If amperage is below servicing limit, repeat steps m. and n.
- s. If amperage is above servicing limit, perform the following:
 - (1) Make sure low side hand wheel and refrigerant bottle valve are closed.
 - (2) Disconnect yellow hose from refrigerant bottle and connect to inlet filter on refrigerant recovery and recycling station.

WARNING

- To prevent injury to personnel, observe the following safety precautions when working on any part of air-conditioning system:
- Protect skin and eyes from possible contact with liquid refrigerant. Wear

gloves and goggles. Keep your body as well protected as you can from possible contact with liquid refrigerant.

- If liquid refrigerant comes in contact with eyes or skin, get immediate medical aid.
- Refrigerant will displace air and can cause suffocation, even though it is nontoxic. It will produce highly toxic phosgene gas if exposed to open flame. Make sure work area is well ventilated, and that there are no open flames, or unshielded heaters in immediate area.

NOTE

Do not open low side handwheel too fast. Oil from system will be released. Open low side handwheel slowly.

- (3) With system operating, slowly open low side handwheel to allow small amounts of refrigerant out of system.

- (4) Observe ampmeter. Close low side hand-wheel when amperage is within servicing limit.
- t. Observe indicator dot in center of sight glass on refrigerant recovery and recycling station. Indications are as follows:
 - (1) Yellow dot means moisture in system. Deservice, evacuate, and reservice system until dot turns green. If yellow dot does not disappear or turns green on second service, replace filter drier assembly (PARA 13-5-6).
 - (2) Green dot indicates normal operating conditions. No further servicing is necessary.
- u. Close low side and high side hand wheels and refrigerant bottle valve. Remove yellow hose.
- v. Place AIR COND switch to OFF position.
- w. Turn off EXT POWER switch.
- x. Turn off all external power (PARA 1-6-16).
- y. Record amount of refrigerant in system.
- z. Remove blue and red hoses from helicopter servicing valves. Install caps on valves.
- aa. Make sure area is clean and free of foreign material.
- ab. Preserve and store servicing equipment.
- ac. Close transition section avionics door.

1-3-27.1.5 Deservice.

WARNING

- To prevent injury to personnel, observe the following safety precautions when working on any part of air-conditioning system:
- Protect skin and eyes from possible contact with liquid refrigerant. Wear

gloves and goggles. Keep your body as well protected as you can from possible contact with liquid refrigerant.

- If liquid refrigerant comes in contact with eyes or skin, get immediate medical aid.
 - Refrigerant will displace air and can cause suffocation, even though it is nontoxic. It will produce highly toxic phosgene gas if exposed to open flame. Make sure work area is well ventilated, and that there are no open flames, or unshielded heaters in immediate area.
 - a. Open transition section avionics door.
 - b. Connect low pressure (blue) hose to low side of servicing manifold and high pressure (red) hose to high side of ECS servicing manifold. Make sure manifold hand wheels are closed.
 - c. Remove knurled brass covers from servicing valves.
 - d. Connect other end of hoses to color-coded servicing valves on helicopter.
- NOTE**
- Blue hose will be on left side, red hose will be on right side.
- e. Connect yellow hose to center supply port of servicing manifold, and to inlet filter on refrigerant recovery and recycling station.
 - f. Turn on all electrical power (PARA 1-6-16).
 - g. Turn on EXT PWR switch.

NOTE

Back up hydraulic pump may run to charge APU accumulator. ECS will not operate until back up pump stops.

- h. When back up pump has shut off, turn on ECS by placing HTR switch to OFF and AIR COND switch to COOL. Set TEMP CONT to full COOL. Operate ECS for at least 5 minutes (Figure 1-3-27-1, Sheet 2, Detail A).

- i. Set TEMP CONT to WARM. Operate ECS for at least 2 minutes.
- j. Place AIR COND switch to OFF.

WARNING

- Refrigerant boils at -30°F (-34°C), and will cause frostbite if spilled on unprotected skin. To prevent personnel injury, wear gloves and eye protection when purging refrigerant servicing hoses.
- Refrigerant will displace air and can cause suffocation, even though it is nontoxic. It will produce highly toxic phosgene gas if exposed to open flame. Make sure work area is well ventilated,

and that there are no open flames, or unshielded heaters in immediate area.

- k. Recover and recycle refrigerant using refrigerant recovery and recycling station per manufacturer's instructions.
- l. Turn off EXT PWR POWER switch.
- m. Turn off all external power (PARA 1-6-16).
- n. Remove blue and red hoses from helicopter servicing valves. Install caps on valves.
- o. Remove yellow hose from inlet filter on rear of recycling station.
- p. Make sure area is clean and free of foreign material.
- q. Preserve and store servicing equipment.
- r. Close transition section avionics door.

1-3-28 ENVIRONMENTAL CONTROL SYSTEM COMPRESSOR - SERVICE.

PARAGRAPH BREAKDOWN

		PAGE
1-3-28.1	Service Environmental Control System Compressor	1-3-28-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Lubricating Oil, Refrigerant Compressor, Item 209A, Appendix D

References

- Appendix D
- PARA 1-3-27
- PARA 2-4-69

1-3-28.1 Service Environmental Control System Compressor.

NOTE

- Should an unknown quantity of oil remain in a discharged or a shut down system, and there is no oil indication in sight glass, pallets, lines, and components should be flushed and compressor serviced to sight glass.
 - Environmental control system must not be operating when checking compressor oil level.
- Remove soundproofing panel from left side rear bulkhead (PARA 2-4-69).
 - Deservice ECS of refrigerant (PARA 1-3-27).

NOTE

- Acceptable oil level is achieved when oil level is shown in compressor site glass. If sight glass is completely covered with oil this indicates an overfill condition.
 - In event of an overfill condition, excess oil must be removed. Remove sight glass and drain oil.
- Loosen sight glass on compressor to eliminate trapped air in compressor pump (Figure 1-3-28-1, Detail A).
 - Disconnect discharge line from upper portion of tee fitting and turn discharge line upright. Temporarily cap tee to prevent moisture from entering system.

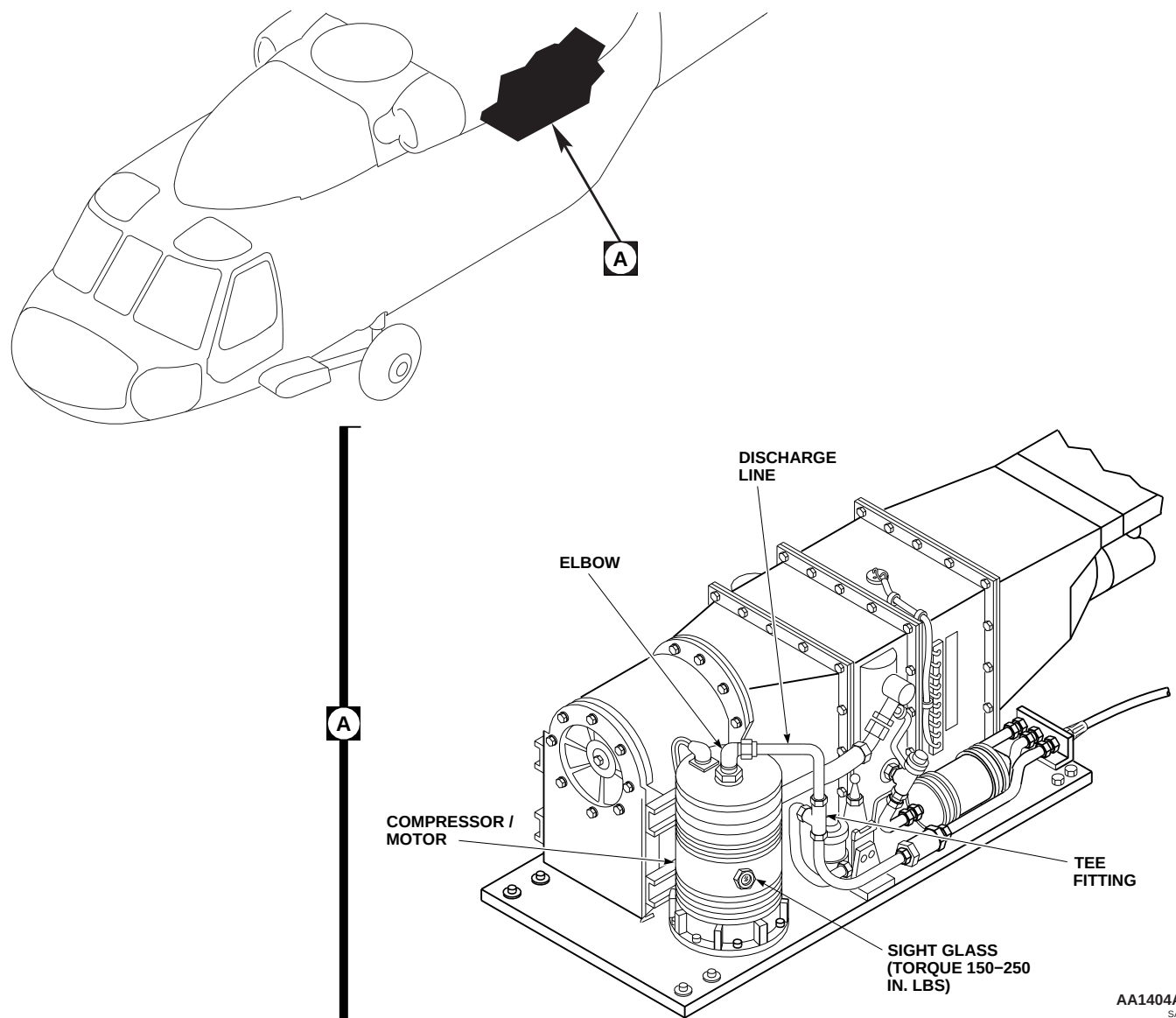


FIGURE 1-3-28-1. SERVICE ENVIRONMENTAL CONTROL SYSTEM COMPRESSOR

CAUTION

To prevent damage to compressor, under no circumstances should more than 1600 cc (54 oz) of oil be added at one servicing.

- e. Pour refrigerant compressor lubricating oil, Item 209A, Appendix D, in 4 ounce (118 cc) increments, allowing 2 minutes between each

filling, through discharge line into compressor. Fill until sight glass is covered half way.

- f. **TORQUE SIGHT GLASS TO 150 - 250 INCH-POUNDS.**
- g. Remove cap and reconnect discharge hose to tee fitting.
- h. Service ECS (PARA 1-3-27).
- i. Install soundproofing panel (PARA 2-4-69).

1-3-29 SERVICE MAIN ROTOR BLADE BIM® SYSTEM.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-3-29.1	Service Main Rotor Blade BIM® System	1-3-29-2	1-3-29.2	Test BIM® Pressure Indicator	1-3-29-6

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- BIM® Check and Fill Unit, 70700-77669-041
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Dishwashing Compound, Item 122, Appendix D
- Leak Detection Fluid, Item 189, Appendix D

- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Nitrogen, Item 221, Appendix D

References

- Appendix D
- PARA 5-4-31
- PARA 5-4-34

1-3-29.1 Service Main Rotor Blade BIM® System.

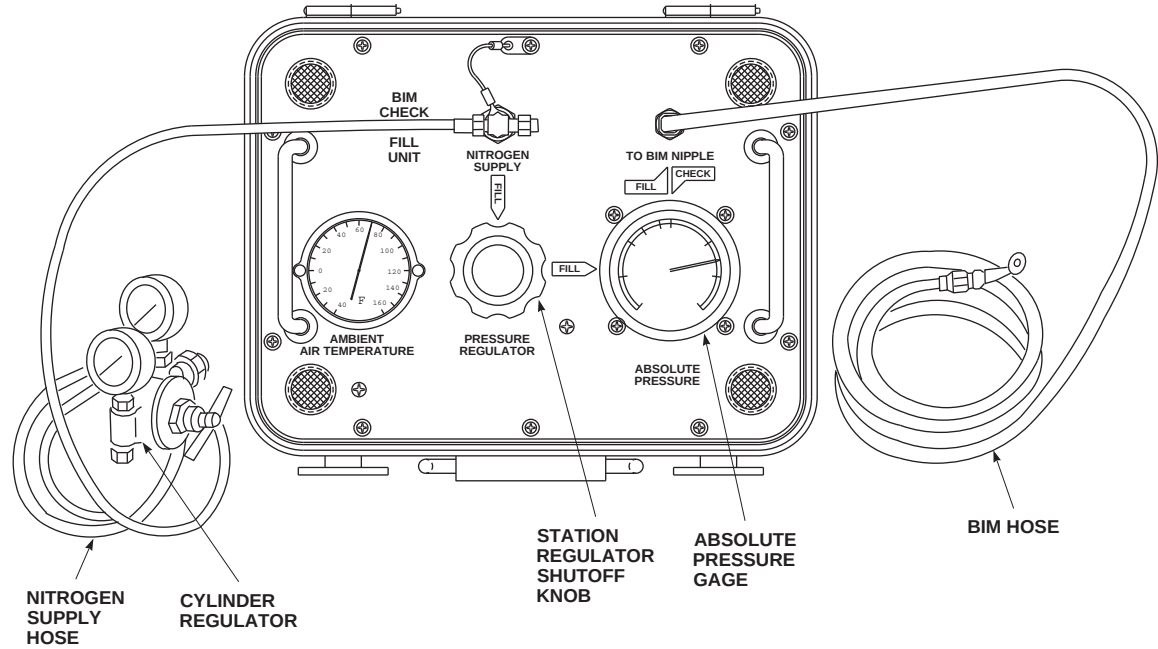
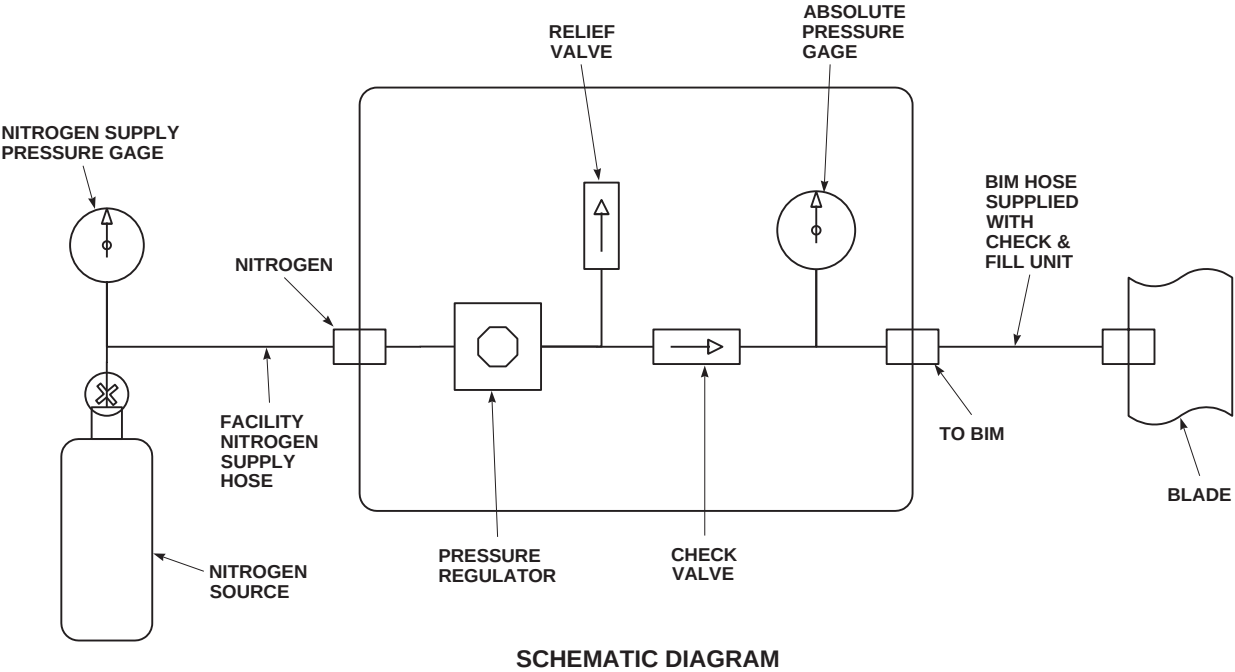
NOTE

- Blades are serviced with prepurified dry nitrogen, Item 221, Appendix D, using a BIM® check and fill unit.
- Pressure gage used in BIM® check and fill unit is an absolute pressure type. This means that gage, when not connected to blade, measures and indicates barometric pressure of the day in inches of mercury, but, when connected to blade, indicated in psi any pressure applied to it above that barometric pressure, up to a maximum of 20 psi(g). Pointer position may vary from day to day because of changes in barometric pressure. Pointer variance is a normal occurrence and should have no influence on use of checking and filling unit. No attempt should be made to zero pointer. The only requirement in servicing blade is to bring pressure indicated on gage up to pressure required for existing blade spar temperature, as shown in Table 1-3-29-1.
- If blade, equipment, or nitrogen tank is moved to an area of different temperature before servicing, for accurate servicing in accordance with measured blade spar temperature, allow enough time for temperature of these items to stabilize. For example, if temperature change is from 0° to 70°F (-17.8° to 21.1°C), allow 5 hours for temperature of nitrogen in blade, equipment, or tank to stabilize.
- Do not attempt to measure spar pressure with blade exposed to sunlight. Solar heating of blade can raise spar pressure by over 2 psi above free-air-temperature pressure, resulting in an inaccurate spar pressure measurement. Allow a minimum of 90 minutes after removing

blade from sunlight, for blade spar temperature to stabilize, before measuring pressure in spar.

- When helicopter is to be operated in extreme cold temperatures, generally below 0°F (-17.8°C), the blade may be serviced to 1 psi over the maximum allowable pressure for the ambient temperature at the time of servicing. Servicing must be performed in controlled climate environment otherwise the additional 1 psi should not be added. Allow time for temperature to stabilize when the blade has been exposed to outside air temperature.
- a. Allow unit to stand (with case open) near blades to be checked or filled for 5 minutes before proceeding.
 - b. Close check and fill units pressure regulator (turn knob fully counterclockwise position).
 - c. Connect nitrogen supply hose (Figure 1-3-29-1) to NITROGEN SUPPLY (Schrader) valve on unit. Open by loosening upper hexnut.
 - d. Remove BIM hose from lid of unit and attach quick-disconnect coupling to mating coupling on unit (labeled "TO BIM NIPPLE").
 - e. With other end of BIM hose ready to attach to Schrader valve on blade, provide 30 to 40 psig to unit.
 - f. Open pressure regulator to purge air from unit.
 - g. Attach hose to Schrader valve on blade.
 - h. Open blade Schrader valve by loosening upper hexnut. Observe reading on gage. This is pressure in the blade.
 - i. Note temperature reading on ambient air temperature dial.
 - j. Consult Blade Pressure Chart Table 1-3-29-1 to find correct blade pressure for that temperature.

1-3-29-2



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FIGURE 1-3-29-1. SERVICE MAIN ROTOR BLADE BIM® SYSTEM

Table 1-3-29-1. Blade Pressures Chart

Blade Spar Temperature (Note 1 and 5)		Permissible Blade Pressure (PSI) (Notes 2, 3, and 4)			
Degrees C	Degrees F	Minimum When Check- ing Installed Blade	Minimum When Install- ing Blade	Minimum When Servicing Blade	Maximum
70 to 72	158 to 162	12.50	14.00	14.25	14.50
67 to 69	153 to 157	12.25	13.75	14.00	14.25
64 to 66	148 to 152	12.00	13.50	13.75	14.00
61 to 63	143 to 147	11.75	13.25	13.50	13.75
59 to 60	138 to 142	11.50	13.00	13.25	13.50
56 to 58	133 to 137	11.25	12.75	13.00	13.25
53 to 55	128 to 132	11.00	12.50	12.75	13.00
50 to 52	123 to 127	10.75	12.25	12.50	12.75
48 to 49	118 to 122	10.50	12.00	12.25	12.50
42 to 47	108 to 117	10.25	11.75	12.00	12.25
39 to 41	103 to 107	10.00	11.50	11.75	12.00
36 to 38	98 to 102	9.75	11.25	11.50	11.75
34 to 35	93 to 97	9.50	11.00	11.25	11.50
31 to 33	88 to 92	9.25	10.75	11.00	11.25
28 to 30	83 to 87	9.00	10.50	10.75	11.00
25 to 27	78 to 82	8.75	10.25	10.50	10.75
23 to 24	73 to 77	8.50	10.00	10.25	10.50
20 to 22	68 to 72	8.50	9.75	10.00	10.25
17 to 19	63 to 67	8.25	9.50	9.75	10.00
14 to 16	58 to 62	8.00	9.25	9.50	9.75
12 to 13	53 to 57	7.75	9.00	9.25	9.50
9 to 11	48 to 52	7.50	8.75	9.00	9.25
6 to 8	43 to 47	7.25	8.50	8.75	9.00
3 to 5	38 to 42	7.00	8.25	8.50	8.75
1 to 2	33 to 37	6.75	8.00	8.25	8.50
-5 to 0	23 to 32	6.50	7.75	8.00	8.25
-7 to -6	18 to 22	6.25	7.50	7.75	8.00
-10 to -8	13 to 17	6.00	7.00	7.25	7.50
-15 to -14	3 to 7	5.50	6.75	7.00	7.25
-18 to -16	-2 to 2	5.25	6.50	6.75	7.00
-21 to -19	-7 to -3	5.00	6.25	6.50	6.75
-24 to -22	-12 to -8	4.75	6.00	6.25	6.50
-27 to -25	-17 to -13	4.50	5.75	6.00	6.25
-30 to -28	-22 to -18	4.50	5.50	5.75	6.00
-32 to -31	-27 to -23	4.25	5.25	5.50	5.75
-35 to -33	-32 to -28	4.00	5.00	5.25	5.50
-38 to -36	-37 to -33	3.75	4.75	5.00	5.25

1-3-29-4

Table 1-3-29-1. Blade Pressures Chart (Cont)

Blade Spar Temperature (Note 1 and 5)		Permissible Blade Pressure (PSI) (Notes 2, 3, and 4)			
Degrees C	Degrees F	Minimum When Check- ing Installed Blade	Minimum When Install- ing Blade	Minimum When Servicing Blade	Maximum
-41 to -39	-42 to -38	3.50	4.50	4.75	5.00
-46 to -42	-52 to -43	3.25	4.25	4.50	4.75
-49 to -47	-57 to -53	3.00	4.00	4.25	4.50
-52 to -50	-62 to -58	2.75	3.75	4.00	4.25
-55 to -53	-67 to -63	2.50	3.50	3.75	4.00

NOTES

1. This table is for measured blade spar temperature with nitrogen in blade spar at a stabilized temperature. If blade is moved to an area of different temperature before servicing or checking spar pressure, allow enough time for temperature of nitrogen in spar to stabilize. For example, if temperature change is from 0 - 70°F (-17.8 - 21.1°C), allow five hours for temperature of nitrogen in blade to stabilize. When servicing blade, also allow an equivalent period of time for temperature of equipment and nitrogen tank to stabilize.
2. Servicing pressures listed provide an allowance for normal nonsignificant leakage that can occur between servicing intervals, before blade pressure indicator will start to show red.
3. Pressure gage used in checking unit or BIM® check and fill unit is an absolute pressure type. This means that gage, when not connected to blade, measures and indicates barometric pressure of the day in inches of mercury, but, when connected to blade, indicates in psi any pressure applied to it above that barometric pressure, up to a maximum of 20 psi(g). Pointer position may vary from day to day because of changes in barometric pressure. Pointer variance is a normal occurrence and should have no influence on use of checking unit or BIM® check and fill unit. No attempt should be made to zero pointer.
4. It is not mandatory that spar pressure of each blade to be installed be checked against "MINIMUM WHEN INSTALLING BLADE" column if spar pressure has been checked within last 6 months. However, this practice is recommended to obtain maximum service from blade.
5. Round off measured blade temperature reading to nearest degree.

-
- k. To fill blade, open pressure regulator until gage indicates desired pressure.
 - l. After one minute has elapsed, close pressure regulator and observe gage reading after it stabilizes.

- m. Pressure in spar should be checked periodically by turning station valve shutoff knob fully clockwise and noting reading on gage at station regulator. When pressure in blade spar has reached specified limit, shut off nitrogen supply

at station valve by turning station valve shutoff knob fully clockwise. Check gage at station regulator for a reading within limits specified for measured blade spar temperature.

- n. Close blade Schrader valve. **TORQUE OUTER HEXNUT TO 50 - 70 INCH-POUNDS.** Lockwire, Item 197, Appendix D, outer hexnut to body of Schrader valve.
- o. Turn off nitrogen supply and remove hose from Schrader valve.
- p. Bleed residual pressure from unit.
- q. Disconnect hose from unit and stow in lid of unit.
- r. Close and cap nitrogen supply Schrader valve.
- s. Close pressure regulator.
- t. Check air valve for leakage, especially in stem area, by applying a liquid, such as leak detection fluid, Item 189, Appendix D, or dishwashing compound, Item 122, Appendix D.
- u. When check is complete and before test liquid dries, clean air thoroughly with clean, fresh water to remove all traces of liquid, then dry blade using machinery towel, Item 211, Appendix D.
- v. Test BIM® pressure indicator (PARA 1-3-29.2).

1-3-29.2 Test BIM® Pressure Indicator.

WARNING

Injury to personnel and damage to equipment will result if blade with indicator showing any red is continued in service. Any blade on which the pressure indicator shows any red color must be removed from service until the cause of the unsafe (red) indication is positively found and corrected. If the indicator is malfunctioning it must be

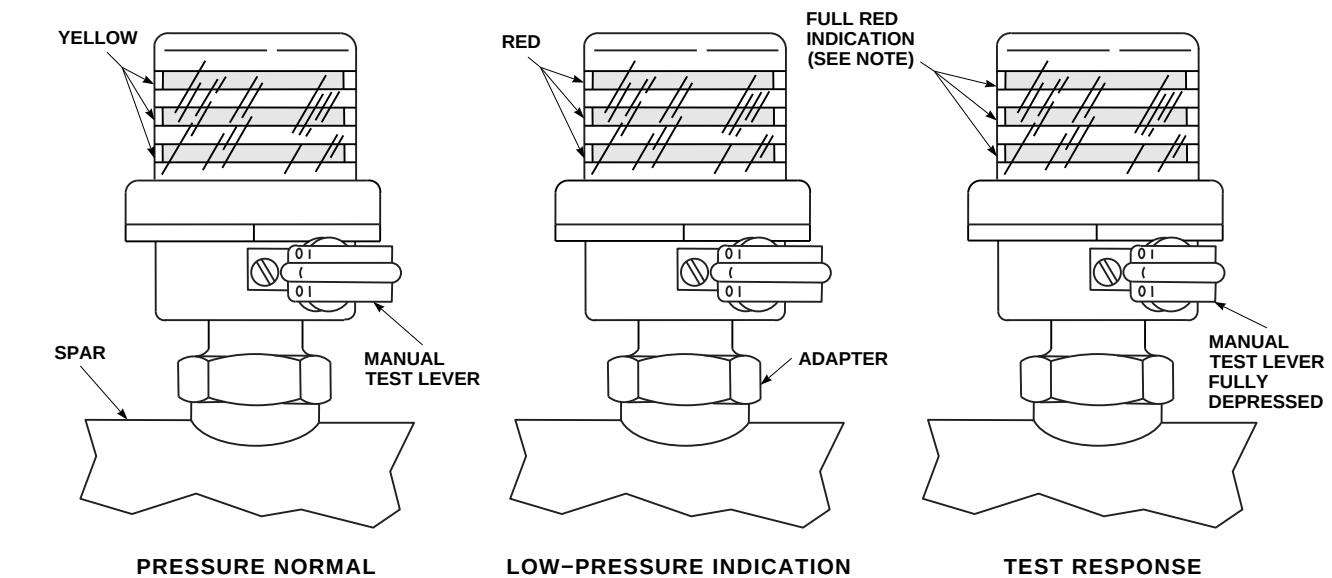
replaced, but only if spar pressure is within permissible limits.

CAUTION

Spar pressure loss may result if indicator lever is not depressed fully when testing indicator. A partly depressed plunger may cause loss of spar pressure and a slow indication, or no indication at all. When testing pressure indicator, valve plunger under manual lever must be pushed all the way down. This will shut off all spar pressure. It may be necessary to use both thumbs to do this. Also, press on ridged part of handle, not on smooth tip.

NOTE

- The pressure indicator compensates for changes in temperature and compares a reference pressure, built into the indicator, with the spar pressure. When the pressure in the blade is within the required limits, the slots in the cylinder show yellow, indicating that the blade is serviceable. If the pressure in the spar drops below the minimum permissible service limits, the indicator shows red, an unsafe indication.
 - Do not hold indicator, as heat of hand may change internal reference pressure and result in erroneous indicator reading.
- a. Press in and hold manual test lever (grenade-type handle) (Figure 1-3-29-2). Do not place hand on glass bulb.
 - b. If indicator is operating properly, a full red (unsafe) indication must show within 10 to 30 seconds. When lever is released, the red indication must snap back immediately, leaving an all yellow (safe) condition.
 - c. The 10 to 30-second time limit applies when temperature is 20°F (-6.7°C) or above. At any



NOTE
 ONLY FULL RED INDICATION IS ACCEPTABLE DURING TEST. INDICATOR MUST RETURN IMMEDIATELY TO NORMAL YELLOW INDICATION FOLLOWING TEST.

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FIGURE 1-3-29-2. TEST BIM® PRESSURE INDICATOR

lower temperature, extend upper limit to the corresponding time listed:

Temperature	Time
19° to 0°F (-7.2° to -17.8°C)	35 seconds
-1° to -20°F (-18.3° to -28.9°C)	40 seconds
-21° to -40°F (-29.4° to -40.0°C)	50 seconds
-41° to -60°F (-40.5° to -51.1°C)	60 seconds

- Make an entry in helicopter logbook stating reason for removal, whenever a blade is removed from service.
- If indicator does not meet these requirements and spar pressure is within permissible limits, replace indicator (PARA 5-4-34). If spar pressure is below permissible limit, replace blade (PARA 5-4-31). Forward blade with low spar pressure to BENCH for repair.

1-3-30 SERVICE HYDRAULIC RESCUE HOIST.

PARAGRAPH BREAKDOWN

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1-3-30.1 Service Hydraulic Rescue Hoist	1-3-30-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Lockwire, Item 195, Appendix D
- Lubricating Oil, Item 205, Appendix D

References

- Appendix D
- PARA 14-4-9

1-3-30.1 Service Hydraulic Rescue Hoist.

NOTE

Some rescue hoists have oil plugs while others have sight gages.

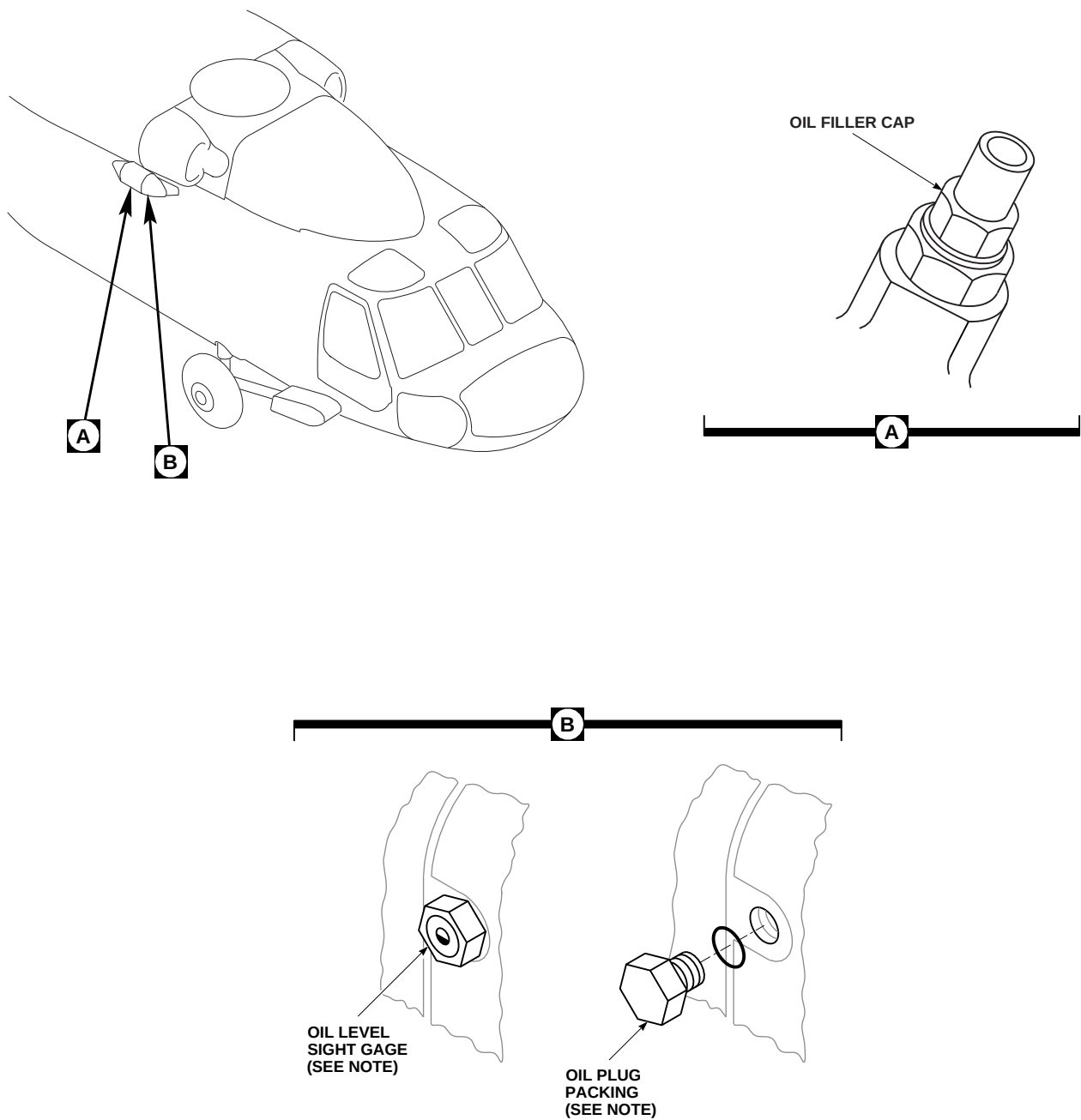
- Service rescue hoist with sight gage as follows:
 - Check sight gage to determine oil level (Figure 1-3-30-1, Detail B).
 - If oil is required, remove rescue hoist front cowling (PARA 14-4-9).
 - Remove oil filler cap (Figure 1-3-30-1, Detail A).
 - Fill to FULL mark on sight gage with lubricating oil, Item 205, Appendix D.
 - Install oil filler cap. Lockwire, Item 195, Appendix D, cap.
 - Make sure area is clean and free of foreign material.

- Install rescue hoist forward cowling (PARA 14-4-9).

- Service rescue hoist with oil plug as follows:

- Remove oil plug and packing from oil overflow port (Figure 1-3-30-1, Detail B).
- If oil level is below oil overflow port, service as follows:
 - Remove rescue hoist forward cowl-ing (PARA 14-4-9).
 - Remove oil filler cap (Figure 1-3-30-1, Detail A).
 - Fill with lubricating oil, Item 205, Appendix D, until oil flows from overflow port.
 - Install oil plug and packing into overflow port (Figure 1-3-30-1, Detail B). Lockwire, Item 195, Ap-pendix D, plug.
 - Make sure area is clean and free of foreign material.

- (f) Install rescue hoist forward cowling (PARA 14-4-9).



NOTE

SOME RESCUE HOISTS HAVE OIL PLUGS INSTALLED WHILE OTHERS HAVE LEVEL SIGHT GAGES.

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FIGURE 1-3-30-1. SERVICE HYDRAULIC RESCUE HOIST

1-3-31 SERVICE ELECTRIC RESCUE HOIST.

PARAGRAPH BREAKDOWN

		PAGE
1-3-31.1	Service Electric Rescue Hoist	1-3-31-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Syringe, 50cc

Materials/Parts

- Lockwire, Item 196, Appendix D
- Lubricating Oil, Item 205, Appendix D

References

- Appendix D
- PARA 14-4-9

1-3-31.1 Service Electric Rescue Hoist.



Damage to cable will occur if cable is lubricated. Cable lubrication is strictly prohibited. Cable must be completely clean to ensure proper tensioning.

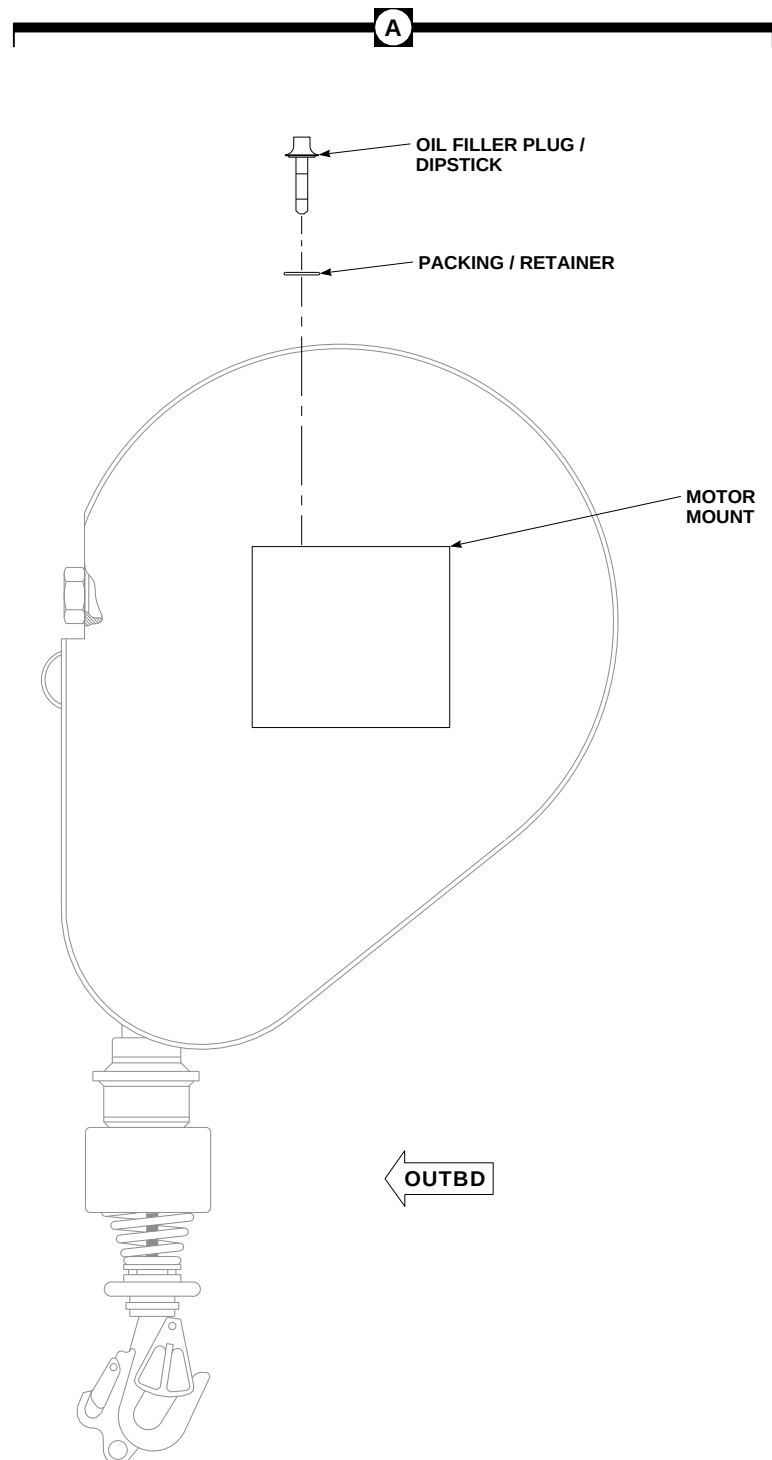
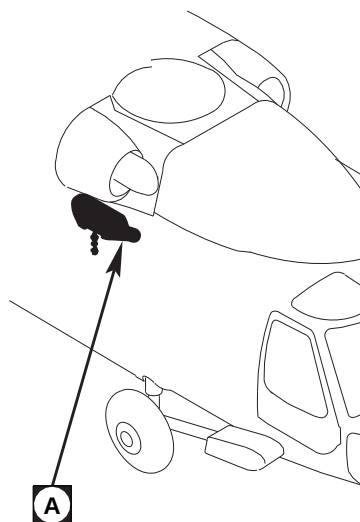
NOTE

Oil must be checked via an access plug at 6 month intervals of hoist operating time.

- Check oil level on oil dipstick at 6 month intervals of hoist operating time (Figure 1-3-31-1, Detail A). Hoist must be mounted in normal operating position. Allow 15 minutes

for oil setting before check. Use lubricating oil, Item 205, Appendix D, when servicing. To add lubricating oil (17 oz required), perform the following:

- Remove forward cowling (PARA 14-4-9).
- Remove oil filler plug/dipstick and packing/retainer from motor mount. Clean oil dipstick.
- Insert and thread dipstick to check oil level. Repeat if required.
- Using syringe, add lubricating oil, Item 205, Appendix D, as necessary.
- Install oil fill plug/dipstick, packing/retainer in motor mount. Lockwire, Item 196, Appendix D.
- Install forward cowling (PARA 14-4-9).



NOTE
VIEWED FROM FRONT.

FIGURE 1-3-31-1. SERVICE ELECTRIC RESCUE HOIST

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1-3-31-2

SECTION 1-4

CLEANING

SECTION OVERVIEW

PARAGRAPH	TITLE
1-4-1	Cleaning Requirements
1-4-2	Clean Flight Controls and Hydraulic Deck
1-4-3	Clean Helicopter Exterior
1-4-4	Clean Engine
1-4-5	Clean Main and Tail Rotor Blades
1-4-6	Clean Main Rotor Blade Pins
1-4-7	Clean Main Rotor Head
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1-4-9	Clean Cargo Hook
1-4-10	Clean Main and Tail Landing Gear
1-4-11	Clean Tail Wheel Lockpin
1-4-12	Clean Tail Wheel Tow Adapters
1-4-13	Clean Crew Chief's/Passenger's Seats
1-4-14	Clean Main Rotor Deice Slipring and Brush Block
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1-4-16	Clean Ejector Rack, BRU-22A/A
1-4-17	Clean Windshield Wiper Pivot Studs
1-4-18	Clean Cockpit Door Window
1-4-19	Clean Pilot's/Copilot's Seats Sliding Backframe
1-4-20	Clean Soundproofing Panels and Airframe Behind Panels

PARAGRAPH

TITLE

1-4-21	Clean Central Display Unit Faceplate
1-4-22	Clean Pilot's and Copilot's Pilot Display Unit Faceplate
1-4-23	Clean Blackout Curtains
1-4-24	Clean Exterior Surface of External Fuel Tank
1-4-25	Clean Hydraulic Rescue Hoist
1-4-26	Clean FLIR System
1-4-27	Clean APU Engine Air Particle Separator (EAPS)
1-4-28	Clean Electric Rescue Hoist

1-4-1 CLEANING REQUIREMENTS.

1-4-1.1 Scope.

This section contains cleaning procedures. Cleaning intervals may be dependent upon environmental conditions or specific time intervals. When unusual local conditions of environment, utilization, mission, experience of flight crew and maintenance personnel, or periods of inactivity are encountered, the scope and/or frequency of cleaning may need to increase to make sure of safe flight.

Maximum weight, engine cowlings can support including equipment, is 400 pounds. No more than two people are permitted on engine cowling at any one time, with no one person weighing over 250 pounds. Do not apply load to vertical surface.

Work platforms located on engine cowlings provide access to the engines. Each platform is capable of supporting a static weight of 400 pounds.

1-4-1.2 Work Platforms.

CAUTION

Damage can occur if limits noted on engine cowlings are exceeded. Maxi-

1-4-2 CLEAN FLIGHT CONTROLS AND HYDRAULIC DECK.

PARAGRAPH BREAKDOWN

		PAGE
1-4-2.1	Clean Flight Controls and Hydraulic Deck	1-4-2-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Alcohol, Ethyl, Item 71, Appendix D
- Hydraulic Fluid, Item 155, Appendix D

- Machinery Towel, Item 211, Appendix D

References

- Appendix D
- PARA 1-7-20

1-4-2.1 Clean Flight Controls and Hydraulic Deck.



To prevent buildup of static electricity, ground helicopter during cleaning operations.

- Open main rotor pylon sliding cover.
- Clean mixer, primary servos, and hydraulic deck with ethyl alcohol, Item 71, Appendix D.

- Wipe up remaining ethyl alcohol with a clean, dry machinery towel, Item 211, Appendix D.
- Wipe down exposed portion of servo pushrods with a clean machinery towel, Item 211, Appendix D, soaked in hydraulic fluid, Item 155, Appendix D.
- Inspect main transmission forward bridge drain hole (PARA 1-7-20).
- Close and latch main rotor pylon sliding cover.

1-4-3 CLEAN HELICOPTER EXTERIOR.

PARAGRAPH BREAKDOWN

	PAGE
1-4-3.1 Clean Helicopter Exterior	1-4-3-1

INITIAL SETUP Tools - Aircraft Mechanic’s Toolkit - Brush - Foam Generator, Spray, or Sponge Materials/Parts - Barrier Material, Item 80, Appendix D - Cleaning Compound, Solvent, Item 100, Appendix D - Cloth - Corrosion Preventive Compound, Item 114, Appendix D	- Corrosion Preventive Compound, Item 115, Appendix D - Dry-Cleaning Solvent, Item 124, Appendix D - Masking Tape, 1-inch, Item 212, Appendix D - Masking Tape, 1 ½-inches, Item 213, Appendix D - Masking Tape, 2-inches, Item 214, Appendix D - Masking Tape, 3-inches, Item 215, Appendix D References - Appendix D - PARA 1-2-1
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1-4-3.1 Clean Helicopter Exterior.

- a. Open main rotor pylon sliding cover.



Do not waterwash electric actuators (trim, stabilator, tailwheel, backup pump motor, main, and APU generators), fire

- extinguisher directional valve, or mixer bellcranks while cleaning helicopter.
- b. Cover electric actuators (trim, stabilator, and tailwheel), backup pump motor, main and APU generators, fire extinguisher directional valve, and mixer assembly with barrier material, Item 80, Appendix D, and secure with masking tape, Item 212 through Item 215, Appendix D.
 - c. Close main rotor pylon sliding cover.

- d. Close all doors, panels, covers, and hatches prior to washing (PARA 1-2-1).



- To prevent buildup of static electricity, ground helicopter during cleaning operations.
- To prevent damage to helicopter, do not use high-pressure water, steam-cleaning equipment, or high-pressure air.
- Damage to tail rotor drive shaft bearings and pitch control shaft bearing will result if exposed to cleaning solvents. Cleaning solvents can penetrate bearing seals and dissolve grease. Do not spray cleaning solvents on tail rotor drive shaft bearings and pitch control shaft bearing in tail gear box.

NOTE

- Use only approved cleaning compounds listed in this task.
- To prevent alkaline water-base cleaning solution from drying, have rinse water readily available. Cleaning will be less effective if alkaline water-base solution is allowed to dry.
- e. Clean helicopter per Table 1-4-3-1.
- f. Open main rotor pylon sliding cover.
- g. Remove barrier material from all components.
- h. If cleaning compounds contacted tail rotor pitch beam area, reapply corrosion preventive compound, Item 114 or Item 115, Appendix D, to exterior surfaces of pitch beam retaining nut and washer.
- i. Close main rotor pylon sliding cover.

Table 1-4-3-1. Cleaning

Type of Soil	Cleaning Agent or Compound	Mixing Directions	Cleaning Procedure
Light soils (dirt, dust, mud, salt, loose soot)	Solvent Cleaning Compound, Type II, Item 100, Appendix D or Solvent Cleaning Compound, Type I, Item 100, Appendix D	Type II - 1 part cleaning compound to 9 parts water. Type I - 1 part cleaning compound to 16 parts water.	Apply cleaning compound solution with foam generator, spray, sponge, or cloth. Scrub and then rinse with fresh water. Dry surface after rinse.

Table 1-4-3-1. Cleaning (Cont)

Type of Soil	Cleaning Agent or Compound	Mixing Directions	Cleaning Procedure
Moderate soils (hydraulic fluids, lubrication oils, light preservatives)	Solvent Cleaning Compound, Type II, Item 100, Appendix D or Solvent Cleaning Compound, Type I, Item 100, Appendix D	Type II - 1 part cleaning compound to 4 parts water. Type I - 1 part cleaning compound to 9 parts water.	Apply cleaning compound solution with foam generator, spray, sponge, or cloth. Rub gently with a circular motion for up to one minute. Rinse with fresh water. Dry surface after rinse.
Heavy soils (carbonized oil, aged preservatives, grease, gun blast, exhaust deposits)	Solvent Cleaning Compound, Type V, Item 100, Appendix D or Solvent Cleaning Compound, Type I, Item 100, Appendix D	Undiluted	Spray or brush on cleaning compound. Wait 5 to 10 minutes. Brush surface to remove soil. Rinse with fresh water. Dry surface after rinse.
	Dry-Cleaning Solvent, Type II or Type III, Item 124, Appendix D and Solvent Cleaning Compound, Type I or Type II, Item 100, Appendix D	Dry-Cleaning Solvent - Undiluted. Cleaning Compound, Type I - 1 part cleaning compound to 4 parts water or Cleaning Compound, Type II - 1 part cleaning compound to 1 part water.	Preclean surface by wiping or brushing soiled area with dry- cleaning solvent. Apply clean- ing compound solution with foam generator, spray, sponge, or cloth. Wait one minute. Scrub surface for one minute. Rinse with fresh water. Dry surface after rinse.

1-4-4 CLEAN ENGINE.

PARAGRAPH BREAKDOWN

	PAGE
1-4-4.1 Clean Engine	1-4-4-1

INITIAL SETUP

- TM 1-2840-248-23

References

- SEI 671

1-4-4.1 Clean Engine.

Refer to SEI 671 and TM 1-2840-248-23 for engine cleaning and washing instructions.

1-4-5 CLEAN MAIN AND TAIL ROTOR BLADES.

PARAGRAPH BREAKDOWN

	PAGE
1-4-5.1 Clean Main and Tail Rotor Blades	1-4-5-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Cleaning Compound, Item 94, Appendix D
- Corrosion Preventive Compound, Item 114, Appendix D

References

- Corrosion Preventive Compound, Item 115, Appendix D
- Hand Cleaner, Item 153, Appendix D
- Litmus Paper, Item 191, Appendix D
- Machinery Towel, Item 211, Appendix D
- Appendix D

1-4-5.1 Clean Main and Tail Rotor Blades.



NOTE

- Use only approved cleaning compounds listed in this task.
 - If water is not available, or temperature does not permit using water, proceed to step e.
 - To prevent alkaline water-base cleaning solution from drying, have rinse water readily available. Cleaning will be less effective if alkaline water-base solution is allowed to dry.
- Dilute 1 part cleaning compound, Item 94, Appendix D, with 9 parts water by volume. If parts being cleaned are heavily soiled, dilute 1 part cleaning compound with no less than 4 parts water by volume.
- Steam or water under pressure may enter and separate adhesive bond. Do not use steam cleaning equipment or high-pressure water.
 - Damage to pitch control shaft bearing will result if exposed to cleaning solvents. Cleaning solvents can penetrate bearing seal and dissolve grease. Do not spray cleaning solvents on pitch control shaft bearing in tail gear box.
- Wash rotor blades beginning at lowest point of part being cleaned.
 - Rinse parts beginning at highest point of part being cleaned.
 - Check cleaned parts after rinsing as follows:

- (1) Draw a piece of red litmus paper, Item 191, Appendix D, against wetted surface.
 - (2) If red litmus paper turns blue, repeat rinsing.
- e. If water is not available, or temperature does not permit using water, perform the following:
- (1) Clean main and tail rotor blades with hand cleaner, Item 153, Appendix D.
 - (2) Wipe off blades with a clean machinery towel, Item 211, Appendix D.
- f. Apply corrosion preventive compound, Item 114 or Item 115, Appendix D, to exterior surfaces of tail rotor pitch beam retaining nut and washer.

1-4-6 CLEAN MAIN ROTOR BLADE PINS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
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INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Acetone, Item 25, Appendix D
- Cotton Gloves

- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D
- PARA 5-3-9
- PARA 5-4-31

1-4-6.1 Clean Main Rotor Blade Expandable Pins.

- a. Release spring clip portion of cam handle from nut and open cam handle.
- b. Raise cam handle to the fully unlocked (open) position.
- c. Hand-clean or spray-wash lower ring, protruding below blade cuff, with dry-cleaning solvent, Item 124, Appendix D.
- d. Wipe up remaining solvent with clean machinery towel, Item 211, Appendix D.
- e. Perform spring clip/handle tension check (PARA 5-3-9).

1-4-6.2 Clean Main Rotor Blade Solid Pins.

- a. Remove main rotor blades (PARA 5-4-31).
- NOTE**
- Main rotor blade solid pins should be handled with cotton gloves following cleaning and during inspection. Do not touch pin shank with bare hands.
- b. Clean areas of bare metal using machinery towel, Item 211, Appendix D, dampened with acetone, Item 25, Appendix D. Allow to dry. After cleaning, handle pin with cotton gloves.
 - c. Inspect pin for wear (PARA 5-3-9).
 - d. Install main rotor blades (PARA 5-4-31).

1-4-7 CLEAN MAIN ROTOR HEAD.

PARAGRAPH BREAKDOWN

	PAGE
1-4-7.1 Clean Main Rotor Head	1-4-7-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Cleaning Compound, Item 94, Appendix D
- Cloth, Coated, Item 103, Appendix D
- Hand Cleaner, Item 153, Appendix D
- Litmus Paper, Item 191, Appendix D

- Machinery Towel, Item 211, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Masking Tape, 1 ½-inches, Item 213, Appendix D
- Masking Tape, 2-inches, Item 214, Appendix D
- Masking Tape, 3-inches, Item 215, Appendix D

References

- Appendix D

1-4-7.1 Clean Main Rotor Head.



To prevent damage to helicopter and components, do not use high-pressure water, steam-cleaning equipment, or high-pressure air.

- Cover inspection holes on main rotor hub using masking tape, Item 212 through Item 215, Appendix D. There are four, 2-inch holes on each hub arm.

NOTE

- Use only approved cleaning compounds listed in this task.

- If water is not available, or temperature does not permit using water, proceed to step f.
- To prevent alkaline water-base cleaning solution drying, have rinse water available. Cleaning will be less effective if alkaline water-base solution is allowed to dry.
- b. Dilute 1 part cleaning compound, Item 94, Appendix D, with 9 parts water by volume.
- c. Wash painted surfaces of main rotor head and attaching components, beginning at lowest point of part being cleaned.

CAUTION

Damage to elastomeric bearings will result if cleaning solution is not rinsed off. Cleaning solution will damage rubber. Do not allow cleaning solution to remain on elastomeric bearings.

d. Rinse parts beginning at highest point of part being cleaned.

e. Check cleaned parts after rinsing as follows:

- (1) Draw a piece of red litmus paper, Item 191, Appendix D, against wetted surface.

- (2) If red litmus paper turns blue, repeat rinsing.

f. If water is not available, or temperature does not permit using water, perform the following:

- (1) Clean main rotor head with hand cleaner, Item 153, Appendix D.

- (2) Wipe off main rotor head with a clean machinery towel, Item 211, Appendix D.

g. Remove dirt or grease from swashplate guide and bearing by wiping with coated cloth, Item 103, Appendix D.

1-4-8 CLEAN POWERTRAIN.

PARAGRAPH BREAKDOWN

	PAGE
1-4-8.1 Clean Powertrain	1-4-8-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Corrosion Preventive Compound, Item 113, Appendix D
- Corrosion Preventive Compound, Item 114, Appendix D

- Corrosion Preventive Compound, Item 115, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D
- PARA 1-2-1

1-4-8.1 Clean Powertrain.**CAUTION**

To prevent damage to powertrain, do not use high-pressure water, steam-cleaning equipment, or high-pressure air.

- a. Remove access panels and fairings (PARA 1-2-1).

CAUTION

Damage to tail drive shaft and oil cooler axial fan bearings will result if exposed to cleaning solvent. Cleaning solvent can penetrate bearing seals and dissolve grease. Do not spray cleaning solvents on tail drive shaft or cooler axial fan bearings.

- b. Hand-clean drive shafts with dry-cleaning solvent, Item 124, Appendix D.

CAUTION

Damage to tail drive shaft and oil cooler axial fan bearings will result if exposed to corrosion preventive compounds. Corrosion preventive compounds can

penetrate bearing seals and dissolve grease. Do not spray corrosion preventive compounds on tail drive shaft and cooler axial fan bearings.

- c. Apply corrosion preventive compound, Item 113, Appendix D, on special couplings and drive shaft hardware.

CAUTION

Damage to pitch control shaft bearing will result if exposed to cleaning solvents. Cleaning solvents can penetrate bearing seal and dissolve grease. Do not spray cleaning solvents on pitch control shaft bearing in tail gear box.

- d. Hand-clean or spray-wash main transmission, input and accessory modules, intermediate and tail gear boxes, with dry-cleaning solvent, Item 124, Appendix D.
- e. If dry-cleaning solvent contacted tail rotor pitch beam area, reapply corrosion preventive compound, Item 114 or Item 115, Appendix D, to exterior surface of pitch beam retaining nut and washer.
- f. Wipe up remaining solvent with clean machinery towel, Item 211, Appendix D.
- g. Install access panels and fairings (PARA 1-2-1).

1-4-9 CLEAN CARGO HOOK.

PARAGRAPH BREAKDOWN

	PAGE
1-4-9.1 Clean Cargo Hook	1-4-9-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Brush, Soft Bristle
- Protection, Eye

References

- PARA 12-4-57
- PARA 12-4-58

1-4-9.1 Clean Cargo Hook.

- Remove cargo hook explosive cartridge (PARA 12-4-58).
- Flush interior and exterior of hook with clean tap water at normal pressure.

CAUTION

To prevent damage to, or faulty operation of cargo hook, do not open electrical box. Seal must remain intact.

- While flushing interior of hook, manually operate release lever and load arm.
- While flushing exterior surfaces of hook, loosen dirt by wiping or brushing with a soft bristle brush.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- Dry interior and exterior surfaces of hook with compressed air (15 psig maximum).
- If after two successive cleanings, the hook still fails to operate properly, refer to normal maintenance procedures (PARA 12-4-57).
- Install cargo hook explosive cartridge (PARA 12-4-58).

1-4-10 CLEAN MAIN AND TAIL LANDING GEAR.

PARAGRAPH BREAKDOWN

	PAGE
1-4-10.1 Clean Main and Tail Landing Gear	1-4-10-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Dishwashing Compound, Item 122, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D

- Machinery Towel, Item 211, Appendix D

References

- Appendix D

1-4-10.1 Clean Main and Tail Landing Gear.

CAUTION

To prevent damage to equipment, do not water-wash wheels, piston cavities, brake parts, activators, switches, and other moisture sensitive equipment.

- Clean wheels and bearing covers thoroughly with dry-cleaning solvent, Item 124, Appendix D.
- Dry wheels and covers with a clean machinery towel, Item 211, Appendix D.

CAUTION

Do not use dry-cleaning solvent to clean tires.

- Clean tires with tap water and two ounces of dishwashing compound, Item 122, Appendix D, per gallon of water.
- Rinse parts, beginning at highest point of parts being cleaned.

1-4-11 CLEAN TAIL WHEEL LOCKPIN.

PARAGRAPH BREAKDOWN

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1-4-11.1 Clean Tail Wheel Lockpin	1-4-11-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Grease, Item 149, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D

1-4-11.1 Clean Tail Wheel Lockpin.

1-4-11.1.1. Every 500 Hours.

- Manually unlock tail wheel lockpin.
- Clean old grease from lockpin with a clean machinery towel, Item 211, Appendix D.

- Lightly coat lockpin with grease, Item 149, Appendix D. Actuate lockpin by hand to make sure it is completely coated with grease.

1-4-12 CLEAN TAIL WHEEL TOW ADAPTERS.

PARAGRAPH BREAKDOWN

		PAGE
1-4-12.1	Clean Tail Wheel Tow Adapters	1-4-12-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Corrosion Preventive Compound, Item 112, Appendix D

- Machinery Towel, Item 211, Appendix D

- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D

1-4-12.1 Clean Tail Wheel Tow Adapters.

NOTE

- Tail wheel tow adapters shall be cleaned after every 100 flight hours or if inspection shows that dirt or foreign material has built-up in the adapters.
- Tail wheel tow adapters are installed in both ends of the tail landing gear axle and are bonded in place.

- Clean tail wheel tow adapters using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
- Wipe up remaining methyl ethyl ketone, Item 217, Appendix D, with clean machinery towel, Item 211, Appendix D.
- Apply a coating of corrosion preventive compound, Item 112, Appendix D, to tail wheel tow adapters.

1-4-13 CLEAN CREW CHIEF’S/PASSENGER’S SEATS.

PARAGRAPH BREAKDOWN

	PAGE
1-4-13.1 Clean Crew Chief’s/ Passenger’s Seats	1-4-13-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Brush
- Sponge

Materials/Parts

- Cleaning Compound, Solvent, Item 100, Appendix D

References

- Appendix D

1-4-13.1 Clean Crew Chief’s/Passenger’s Seats.

- a. Brush loose particles from seats.
- b. Dilute 1 part solvent cleaning compound, Item 100, Appendix D, with 4 parts water by volume.

- c. Apply cleaning compound solution with sponge and scrub briskly. Rinse with clean dampened sponge using fresh water. Allow area to dry.
- d. Raise nap by brushing.

1-4-14 CLEAN MAIN ROTOR DEICE SLIPRING AND BRUSH BLOCK.

PARAGRAPH BREAKDOWN

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1-4-14.1	Clean Main Rotor Deice Slipring and Brush Block	1-4-14-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Brush, Item 83, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- PARA 12-4-26

1-4-14.1 Clean Main Rotor Deice Slipring and Brush Block.

- Remove slipring from distributor (PARA 12-4-26).
- Clean slipring with methyl ethyl ketone, Item 217, Appendix D.

- Remove brush assembly from slipring (PARA 12-4-26).
- Clean brush block with small brush, Item 83, Appendix D, to remove any debris.
- Install brush assembly onto slipring and install slipring (PARA 12-4-26).

1-4-15 CLEAN TAIL ROTOR DEICE SLIPRING.

PARAGRAPH BREAKDOWN

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1-4-15.1	Clean Tail Rotor Deice Slipring	1-4-15-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Brush, Sash
- Vacuum Cleaner

Materials/Parts

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- PARA 12-4-54

1-4-15.1 Clean Tail Rotor Deice Slipring.

- Remove tail rotor blade deice slipring stator brush assembly (PARA 12-4-54).
- Vacuum any loose debris from tail rotor assembly.
- Slowly turn tail rotor blades while using sash brush to clean slipring rotor. Vacuum all loose

debris. Make sure drain holes at bottom of rotor and stator are free of debris.

- Wipe rotor clean using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
- Install tail rotor blade deice slipring stator brush assembly (PARA 12-4-54).

1-4-16 CLEAN EJECTOR RACK, BRU-22A/A.

PARAGRAPH BREAKDOWN

	PAGE
1-4-16.1 Clean Ejector Rack, BRU-22A/A	1-4-16-1

INITIAL SETUP Tools - Aircraft Mechanic’s Toolkit - Breech Cap Plug Assembly Wrench and Go/No-Go Breech Cap Gage, Locally-Made, Figure H-135, Appendix H - Brush, Stiff Bristle - Nonmetallic Brush, Stiff Bristle - Protection, Eye - Soft Brass or Plastic Scraper - Torque Wrench, 30 - 150 in. lbs Materials/Parts - Cloth, Cleaning, Item 102, Appendix D - Corrosion Preventive Compound, Item 112, Appendix D	- Corrosion Preventive Compound, Item 114, Appendix D - Dishwashing Compound, Item 122, Appendix D - Dry-Cleaning Solvent, Item 124, Appendix D References - Appendix D - Appendix H - PARA 1-7-35 - PARA 16-2-4 - PARA 16-4-34 - PARA 16-4-36 - PARA 16-5-8
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1-4-16.1 Clean Ejector Rack, BRU-22A/A.

1-4-16.1.1 Clean Ejector Rack.

- Remove ejector rack, BRU-22A/A (PARA 16-4-34).
- Remove breech, gun, and piston assembly from ejector rack (PARA 16-4-36). Remove bearing ring from piston assembly only if rack has been fired or replacement of rings is required. Disassemble gun.



- Do not use abrasives on ejector components. Use soft brass or plastic scraper if particles cannot be removed by following procedure.
- Clean breech, gun, and piston assembly with stiff bristle brush and dry-cleaning solvent, Item 124, Appendix D. Reclean with dishwash-

ing compound, Item 122, Appendix D, and hot fresh water.

- d. Visually inspect breech, gun, and piston assembly (PARA 16-4-36).
- e. Inspect ejector rack for corrosion (PARA 16-4-36).
- f. Apply thin coating of corrosion preventive compound, Item 114, Appendix D, to above parts.

NOTE

Make certain breech is fully forward locked. Close hooks and leave latched at all times ejector unit is not in use.

- g. Assemble breech, gun, and piston assembly as required (PARA 16-4-36).
- h. Test ejector rack, BRU-22A/A (PARA 16-5-8).
- i. Install ejector rack, BRU-22A/A (PARA 16-4-34).
- j. Inspect ejector rack before loading stores (PARA 1-7-35).

1-4-16.1.2 Clean Firing Lead Cable.

NOTE

Firing lead cable may be removed without removing ejector rack.

- a. Remove firing lead cable and breech cap assembly from ejector rack breech (PARA 16-4-36).
- b. Locally make breech plug wrench and go/no-go breech cap gage (Figure H-135, Appendix H).
- c. Remove plug assembly from firing lead cable using locally-made breech cap plug wrench (Figure 1-4-16-1, Detail B).
- d. Remove contact assembly and spring from connector assembly.

- e. Clean breech cap, plug assembly, contact assembly, and spring with cleaning cloth, Item 102, Appendix D, moistened with dry-cleaning solvent, Item 124, Appendix D, or equivalent.
- f. Wipe parts with clean, dry cleaning cloth, Item 102, Appendix D.
- g. Using locally-made go/no-go breech cap gage, measure ground pin protrusion on plug assembly (Figure 1-4-16-1, Detail C). **MEASUREMENT MUST BE WITHIN 0.030 ± 0.010-INCH.** If measurement is not within noted tolerance, replace plug assembly.
- h. Assemble spring and contact assembly on cable (Figure 1-4-16-1, Detail B).
- i. Install plug assembly using breech cap plug wrench. **TORQUE PLUG TO 90 - 110 INCH-POUNDS.**
- j. Measure contact assembly protrusion with locally-made go/no-go breech cap gage (Figure 1-4-16-1, Detail C). **MEASUREMENT MUST BE WITHIN 0.045 (+0.020/-0.007)-INCH.** If measurement is not within noted tolerance, replace contact assembly spring. Remeasure, if still not within noted tolerance, replace contact assembly.
- k. Depress contact assembly and release to verify freedom of contact assembly movement.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- l. Clean coupler and connector using hot soapy water and stiff bristle nonmetallic brush

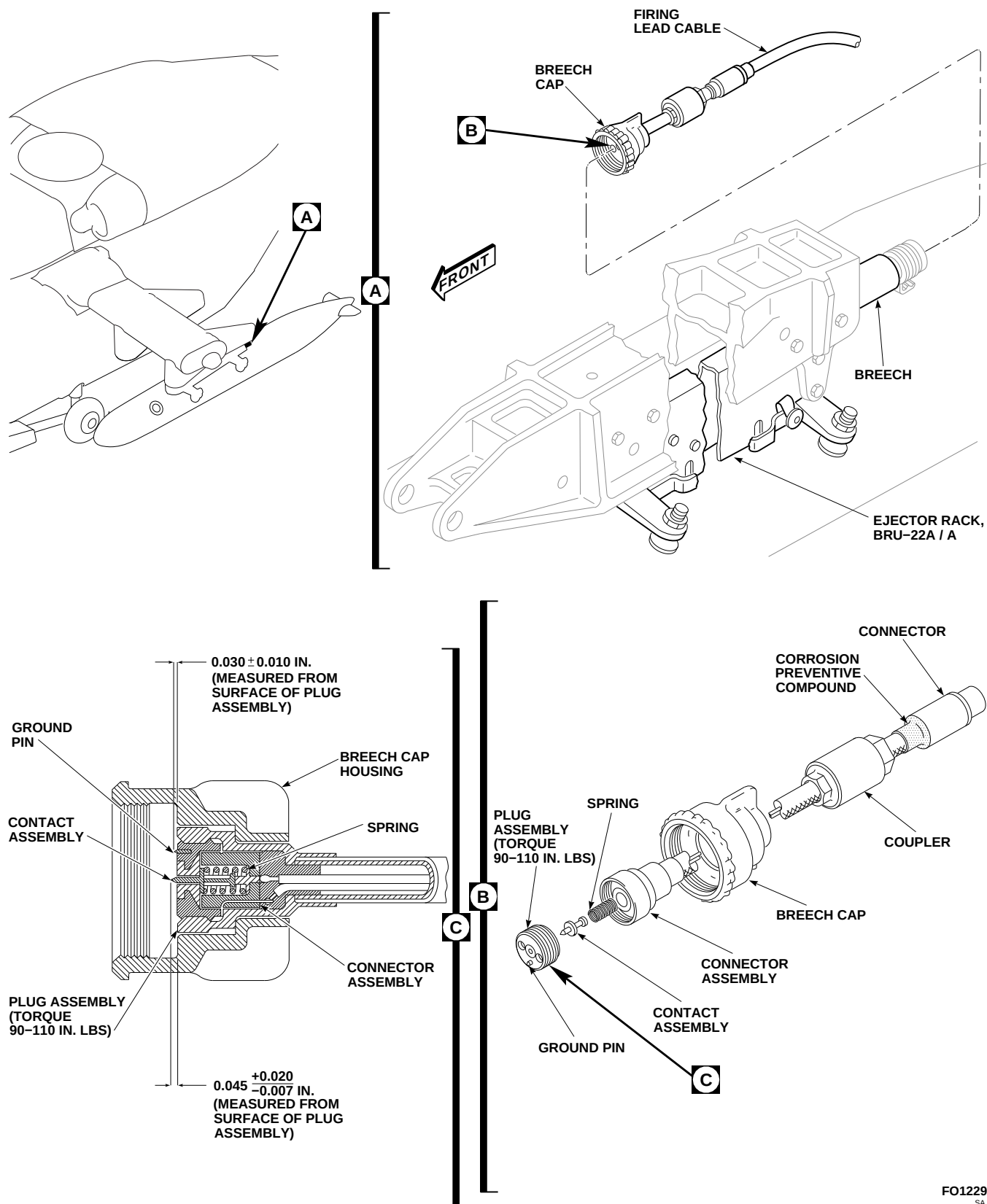


FIGURE 1-4-16-1. CLEAN FIRING LEAD CABLE

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(Figure 1-4-16-1, Detail B). Rinse thoroughly with fresh water and dry completely with compressed air.

- m. Apply corrosion preventive compound, Item 112, Appendix D, to outer surface of connector assembly and inner surface (except threads) of firing cable coupler.

- n. Perform continuity check on firing lead cable assembly (PARA 16-2-4).
- o. Apply thin coat of corrosion preventive compound, Item 112, Appendix D, to threads of adapter assembly receptacle and install firing lead cable breech cap assembly on ejector rack breech.

1-4-17 CLEAN WINDSHIELD WIPER PIVOT STUDS.

PARAGRAPH BREAKDOWN

		PAGE
1-4-17.1	Clean Windshield Wiper Pivot Studs	1-4-17-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D
- PARA 1-5-8
- PARA 12-4-19

1-4-17.1 Clean Windshield Wiper Pivot Studs.



To prevent damage to equipment, use only approved cleaning compounds in the task.

- Remove windshield wiper arm (PARA 12-4-19).

- Hand-clean wiper pivot studs with dry-cleaning solvent, Item 124, Appendix D.
- Wipe up remaining solvent with a clean machinery towel, Item 211, Appendix D.
- Lubricate wiper pivot studs (PARA 1-5-8).
- Install windshield wiper arm (PARA 12-4-19).

1-4-18 CLEAN COCKPIT DOOR WINDOW.

PARAGRAPH BREAKDOWN

	PAGE
1-4-18.1 Clean Cockpit Door Window	1-4-18-1

INITIAL SETUP	• Flannel Cloth, Item 140, Appendix D
Tools	• Maintenance Kit, Window, Item 211A, Appendix D
• Aircraft Mechanic’s Toolkit	• Soap
Materials/Parts	References
• Diaper Cloth (Alternate for Flannel Cloth) • Dishwashing Compound, Item 122, Appendix D	• Appendix D

1-4-18.1 Clean Cockpit Door Window.



Do not use the following materials for cleaning or permit their fumes to contact transparent plastics: gasoline, alcohol, benzene, toluene, hexane, kerosene, zylene, ketones (including acetone), carbon tetrachloride, fire extinguisher or deicing fluids, dope or lacquer thinners, perchloroethylene, paint stripping compounds, degreasing compounds, aromatic naphtha, or other than specifically authorized compounds. These materials can soften plastic and cause crazing. Do not apply pressure or scrub dry surfaces.

NOTE

Transparent acrylic plastic surfaces are free from effects of corrosion, but

surfaces are highly susceptible to scratches, optical distortion, and crazing. Special care must be taken to prevent destroying optical qualities of plastic. Transparent plastics should be maintained only as outlined below.

- Flush exterior of window with fresh water.
- Gently wash window with solution of mild soap and water using clean, soft cloth such as flannel cloth, Item 140, Appendix D, diaper cloth, or equivalent.

NOTE

Cloths should be kept in containers to keep them clean and grit free.

- Wipe window dry with clean, soft cloth such as flannel cloth, Item 140, Appendix D, diaper cloth, or equivalent.
- If fresh water supply is not available, or if window is greasy, clean window using soft

cloth and dishwashing compound, Item 122, Appendix D. Wipe dry after about 5 minutes.



Do not polish any window until surface has been thoroughly cleaned. If dirt, grit,

and sand are present during waxing or polishing, they would cause serious damage to window.

- e. If window is lightly scratched, polish window using window maintenance kit, Item 211A, Appendix D.

1-4-19 CLEAN PILOT’S/COPILOT’S SEATS SLIDING BACKFRAME.

PARAGRAPH BREAKDOWN

		PAGE
1-4-19.1	Clean Pilot’s/Copilot’s Seats Sliding Backframe	1-4-19-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Cloth, Cleaning, Item 102, Appendix D

- Corrosion Preventive Compound, Item 114, Appendix D

References

- Appendix D

1-4-19.1 Clean Pilot’s/Copilot’s Seats Sliding Backframe.

NOTE

Frequency of cleaning sliding backframe collars depends on operational environment such as extreme dust, etc.

- Spray a liberal amount of corrosion preventive compound, Item 114, Appendix D, around the top of the four collars on the sliding backframe.

WARNING

To prevent injury to personnel, do not pull forward on control lever unless someone is sitting in seat. The extension

springs are under load at all times. With seat at lowest position, the vertical preload on seat could be as high as 150 pounds. If no one is in seat and control lever is pulled forward, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes will be injured seriously.

- Cycle seat up and down several times by pulling forward on control lever on front right side.
- Wipe debris from columns with cleaning cloth, Item 102, Appendix D.
- Add additional compound to top of each collar and wipe excess from each column.
- Repeat operation as required for complete cleaning of collars and columns, resulting in proper movement of sliding backframe.

1-4-20 CLEAN SOUNDPROOFING PANELS AND AIRFRAME BEHIND PANELS.

PARAGRAPH BREAKDOWN

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1-4-20.1	Clean Soundproofing Panels	1-4-20-1
1-4-20.2	Cleaning Mildew from Airframe Behind Soundproofing Panels	1-4-20-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Alcohol, Isopropyl, Item 72, Appendix D
- Cloth
- Corrosion Preventive Compound, Item 109, Appendix D

- Corrosion Preventive Compound, Item 114, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D
- Soap, Mild
- Tribasic Sodium Phosphate, Item 340A, Appendix D

References

- Appendix D

1-4-20.1 Clean Soundproofing Panels.

- Clean lightly soiled panel surfaces with clean cloth, mild soap, and warm water. Rinse and wipe dry.
- Spot clean oil, grease, hydraulic oil, or badly soiled surfaces with dry-cleaning solvent, Item 124, Appendix D. Do not saturate panel. Wash surfaces with clean cloth, mild soap, and water. Rinse and wipe dry.

- Scrub mildew on surface with machinery towel, Item 211, Appendix D, dampened with tribasic sodium phosphate, Item 340A, Appendix D, water solution (2-1/2 ounces by weight per 10 gallons of water).
- Rinse the surface thoroughly with water.
- Dry with machinery towel, Item 211, Appendix D.

1-4-20.2 Cleaning Mildew from Airframe Behind Soundproofing Panels.

- For areas not treated with corrosion preventive compound, clean as follows:

- For areas treated with corrosion preventive compound, clean as follows:
 - Remove corrosion preventive compound using machinery towel, Item 211, Ap-

pendix D, dampened with isopropyl alcohol, Item 72, Appendix D.

- (2) Wipe dry with clean machinery towel, Item 211, Appendix D.

- c. To prevent recurrence of mildew, apply corrosion preventive compound as follows:

- (1) Spray corrosion preventive compound, Item 114, Appendix D, on area to remove any moisture.

- (2) Coat area with corrosion preventive compound, Item 109, Appendix D.

1-4-21 CLEAN CENTRAL DISPLAY UNIT FACEPLATE.

PARAGRAPH BREAKDOWN

		PAGE
1-4-21.1	Clean Central Display Unit Faceplate	1-4-21-1

INITIAL SETUP

Materials/Parts

- Alcohol, Isopropyl, Item 72, Appendix D
- Cloth, Cleaning, Item 102, Appendix D

References

- Appendix D

1-4-21.1 Clean Central Display Unit Faceplate.

- a. Dampen cleaning cloth, Item 102, Appendix D, with isopropyl alcohol, Item 72, Appendix D.
- b. Gently rub faceplate with cleaning cloth.

1-4-22 CLEAN PILOT’S AND COPILOT’S PILOT DISPLAY UNIT
FACEPLATE.

PARAGRAPH BREAKDOWN

		PAGE
1-4-22.1	Clean Pilot’s and Copilot’s Pilot Display Unit Faceplate	1-4-22-1

INITIAL SETUP

Materials/Parts

- Alcohol, Isopropyl, Item 72, Appendix D
- Cloth, Cleaning, Item 102, Appendix D

References

- Appendix D

1-4-22.1 Clean Pilot’s and Copilot’s Pilot
Display Unit Faceplate.

NOTE

Cleaning for both the pilot’s and
copilot’s pilot display unit faceplate is
the same.

- a. Dampen cleaning cloth, Item 102, Appendix D,
with isopropyl alcohol, Item 72, Appendix D.
- b. Gently rub faceplate with cleaning cloth.

1-4-23 CLEAN BLACKOUT CURTAINS.

PARAGRAPH BREAKDOWN

	PAGE
1-4-23.1 Clean Blackout Curtains	1-4-23-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Cloth (Alternate for Sponge)
- Dishwashing Compound, Item 122, Appendix D

- Sponge (Alternate for Cloth)

References

- Appendix D
- PARA 16-4-6

1-4-23.1 Clean Blackout Curtains.

- a. If installed, remove blackout curtains (PARA 16-4-6).
- b. Wash curtains with dishwashing compound, Item 122, Appendix D, using clean cloth or sponge.
- c. Rinse with clean dampened sponge or cloth using clean, fresh water.
- d. If required, install blackout curtains (PARA 16-4-6).

1-4-24 CLEAN EXTERIOR SURFACE OF EXTERNAL FUEL TANK.

PARAGRAPH BREAKDOWN

	PAGE
1-4-24.1 Clean Exterior Surface of External Fuel Tank	1-4-24-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Acetone, Item 25, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D

1-4-24.1 Clean Exterior Surface of External Fuel Tank.

NOTE

Seepage substance on exterior of tank will appear as a thin, shiny film which is sticky, has a slight amber coloring, and fuel odor.

- a. Inspect entire exterior surface of tank for indications of seepage. If seepage is found, note location(s).

NOTE

Do not saturate exterior of tank with acetone. Saturating exterior of tank will

cause additional seepage from tank surface.

- b. Moisten machinery towel, Item 211, Appendix D, with acetone, Item 25, Appendix D, and gently wipe exterior of tank.
- c. Allow minimum of 10 minutes for acetone to evaporate from exterior of tank.
- d. Inspect exterior of tank for seepage. If seepage substance is still indicated, repeat cleaning procedure until seepage substance is removed from tank.

1-4-25 CLEAN HYDRAULIC RESCUE HOIST.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-4-25.1	Clean Hydraulic Rescue Hoist	1-4-25-1	1-4-25.2	Clean Hydraulic Rescue Hoist Cable	1-4-25-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Protective Gloves

Materials/Parts

- Alcohol, Isopropyl, Item 72, Appendix D
- Barrier Material, Item 80, Appendix D
- Cheesecloth, Item 90, Appendix D
- Cloth, Cleaning, Item 102A, Appendix D

- Corrosion Preventive Compound, Item 114, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Gloves
- Lockwire, Item 195, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16

1-4-25.1 Clean Hydraulic Rescue Hoist.

- Wipe external surfaces with cleaning cloth, Item 102A, Appendix D, moistened with dry-cleaning solvent, Item 124, Appendix D. Dry thoroughly after cleaning with cleaning cloth, Item 102A, Appendix D, or dry compressed air no higher than 20 psig.
- Clean the complete length of hoist cable (PARA 1-4-25.2).

1-4-25.2 Clean Hydraulic Rescue Hoist Cable.

- Turn on external hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- On upper console, set RESCUE HOIST switch to ALL.



- If hoist system is under backup power, system limit switches will be bypassed. Check that system will operate in normal power before reeling cable.
- In cabin, check that cabin RESCUE HOIST control box NORMAL POWER light is on. If BACKUP CONTROL POWER switch light is on, open switch guard assembly, and press switch. BACKUP CONTROL POWER switch light should go off and NORMAL POWER light should come on. Close switch guard assembly and lockwire, Item 195, Appendix D, switch guard assembly to screws.

WARNING

- To prevent injury to personnel or damage to equipment, rescue hoist cable must be kept clear of all parts of the aircraft and free from other external obstacles. Cable abrasion during hoist operation can lead to cable failure. If cable contact or snagging occurs, inspect cable for damage immediately.
- Injury to personnel may occur if protective gloves are not worn when reeling cable in or out. Use protective gloves when reeling cable in or out.

CAUTION

Damage to cable may occur if cable is kinked when reeling cable in or out. Use care not to kink cable when reeling cable in or out.

- d. Place barrier material, Item 80, Appendix D, under hoist area to lay cable on during hoist operation.

NOTE

If hoist was used in salt water conditions, rinse cable with cold, clean fresh water.

- e. In cabin, operate rescue hoist and cargo hook pendant, while assistant walks out entire length of cable.

CAUTION

Do not run backup control with hoist in down position. Doing so will result in back winding cable on drum.

- f. Slowly reel hoist cable onto cable drum, while assistant applies firm drag on cable with gloved hand. On Breeze-Eastern hoist (BL27100 series), wipe cable with cheesecloth, Item 90, Appendix D, dampened with isopropyl alcohol, Item 72, Appendix D. On Lucas Aerospace hoist (42315 series), wipe cable with cheesecloth, Item 90, Appendix D, dampened with corrosion preventive compound, Item 114, Appendix D.
- g. Turn off all switches.
- h. Turn off external hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- i. Make sure area is clean and free of foreign material.

1-4-26 CLEAN FLIR SYSTEM.

PARAGRAPH BREAKDOWN

	PAGE
1-4-26.1 Clean FLIR System	1-4-26-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 1-4-3
- TM 11-70-23

1-4-26.1 Clean FLIR System.

NOTE

Make sure FLIR is in the STOW position when cleaning external surface.

- Clean external surface of FLIR (PARA 1-4-3).
- Clean FLIR germanium window (TM 11-70-23).

1-4-27 CLEAN APU ENGINE AIR PARTICLE SEPARATOR (EAPS).

PARAGRAPH BREAKDOWN

	PAGE
1-4-27.1 Clean APU Engine Air Particle Separator	1-4-27-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D

References

- Appendix D
- PARA 16-4-73

1-4-27.1 Clean APU Engine Air Particle Separator.

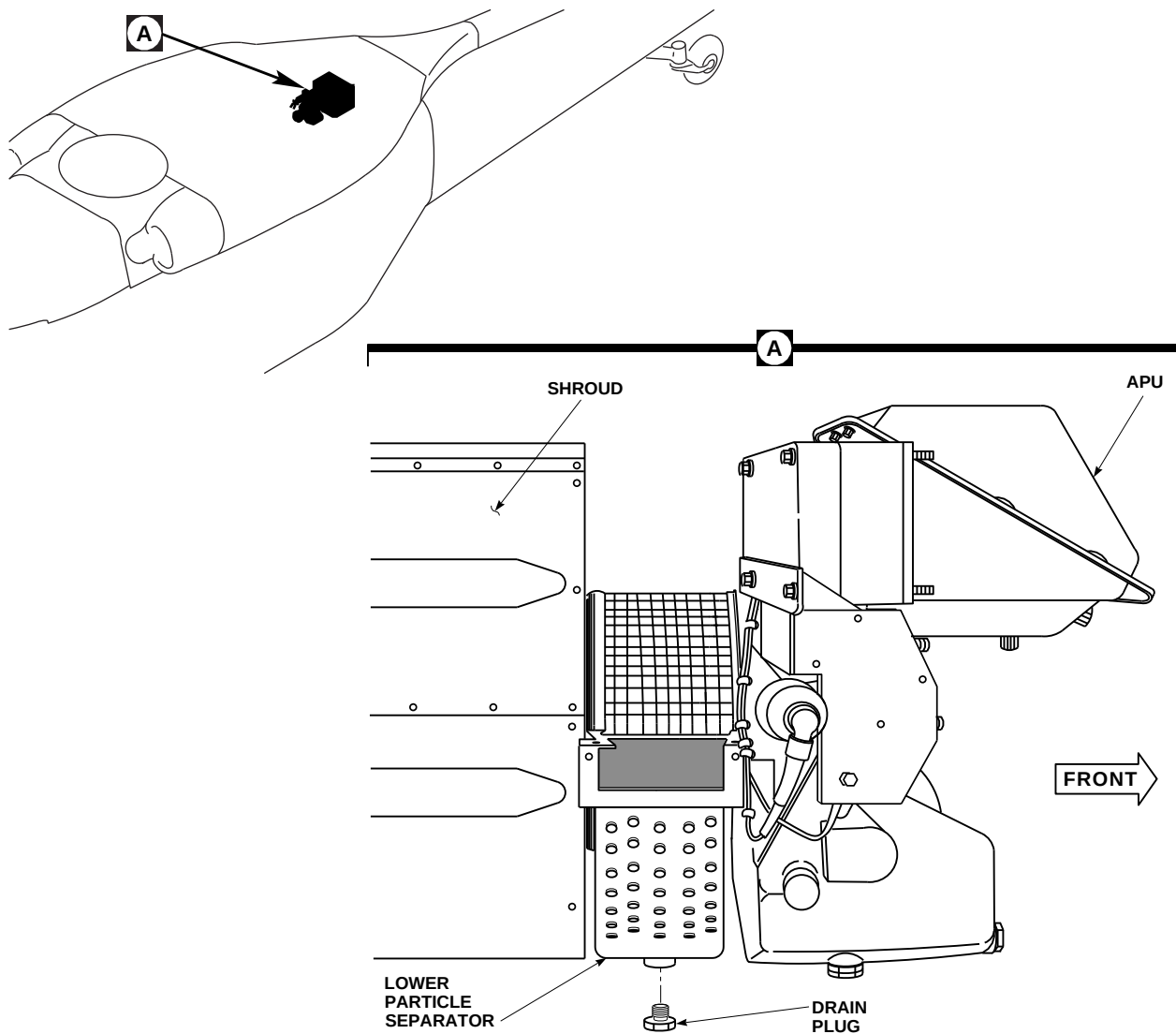
- Remove upper air particle separator (PARA 16-4-73).
- Check upper particle separator for dirt, grass, and other foreign objects.
- Remove grass and other foreign objects.
- Clean the upper particle separator with dry-cleaning solvent, Item 124, Appendix D.
- After cleaning, flush upper particle separator with clean water.
- Check upper particle separator for damage that would affect operation of APU.
- Flush collector box with clean water.
- If dirt that is built up on lower particle separator can be easily removed, do the following:
 - Check lower particle separator for dirt, grass and other foreign objects.
 - Remove grass and other foreign objects.

- After cleaning, flush lower particle separator with clean water.
- Remove drain plug and drain water from lower particle separator.
- Install drain plug.
- If dirt that is built up on lower particle separator cannot be easily removed, do the following:

NOTE

Removal of lower particle separator will require the APU to be removed.

- Remove lower particle separator (PARA 16-4-73).
- Clean the lower particle separator with dry-cleaning solvent, Item 124, Appendix D.
- After cleaning, flush lower particle separator with clean water.
- Drain water from lower particle separator.
- Make sure air inlet screen on APU is free of debris and other foreign objects.

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- k. If lower air particle separator was removed, reinstall lower air particle separator (PARA 16-4-73).
- l. Install upper air particle separator (PARA 16-4-73).

1-4-28 CLEAN ELECTRIC RESCUE HOIST.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-4-28.1	Clean Electric Rescue Hoist	1-4-28-1	1-4-28.2	Clean Electric Rescue Hoist Cable	1-4-28-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 55-gallon
- Nonmetallic Brush, Stiff-Bristle
- Protection, Eye

Materials/Parts

- Brush, Item 83, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D

References

- Appendix D

1-4-28.1 Clean Electric Rescue Hoist.



- Damage to threads will occur if a wire brush or scouring pad is used.
- Damage to electrical components will occur if components are immersed in dry-cleaning solvent.
- a. Loosen dirt and grease from gears, housings, and other metal parts using stiff-bristle nonmetallic brush, Item 83, Appendix D, and dry-cleaning solvent, Item 124, Appendix D.



- Injury to personnel will result if precautions are not taken while using compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.
- b. Air dry parts cleaned in preceding step, using 15 psi, clean compressed air, or by wiping with cleaning cloth, Item 102, Appendix D.

1-4-28.2 Clean Electric Rescue Hoist Cable.**WARNING**

Injury to personnel will occur if gloves are not worn when reeling out cable. leather gloves must be worn when reeling out cable.

NOTE

When reeling cable out for cleaning, do not drag on dirty floor or exceed 30° fleet angle.

- a. Reel cable out into a 55-gallon container filled with fresh water.
- b. Wash rescue hoist drum with cleaning cloth, Item 102, Appendix D, and fresh water and wipe dry with cleaning cloth, Item 102, Appendix D.

NOTE

When reeling cable in after cleaning, do not drag on dirty floor or exceed 30° fleet angle.

- c. Dry cable by passing it through dry, cleaning cloth, Item 102, Appendix D when reeling in.

SECTION 1-5

LUBRICATION

SECTION OVERVIEW

PARAGRAPH	TITLE
1-5-1	Lubrication Requirements
1-5-2	Lubricate Bifilar
1-5-3	Lubricate Swashplate
1-5-4	Lubricate Mixer Bellcranks
1-5-5	Lubricate Rotary Inputs
1-5-6	Lubricate Landing Gear Shock Struts
1-5-7	Lubricate Stabilator Actuator Assembly
1-5-8	Lubricate Windshield Wiper Pivot Studs
1-5-9	Lubricate Windshield Wiper Converter
1-5-10	Lubricate Tail Wheel Lockpin
1-5-11	Lubricate Tail Wheel Bearings
1-5-12	Lubricate Door Locks
1-5-13	Lubricate Ejector Rack, BRU-22A/A

1-5-1 LUBRICATION REQUIREMENTS.

1-5-1.1 Scope.

This section contains lubrication procedures. Lubrication intervals may be dependent upon environmental conditions or specific time intervals. When unusual local conditions of environment, utilization, mission, experience of flight crew and maintenance personnel, or periods of inactivity, etc., are encountered, the scope and/or frequency of

lubrication may need to be increased to make sure of safe flight.

1-5-1.2 Work Platforms.

Work platforms located on engine cowlings provide access to the engines. Each platform is capable of supporting a static weight of 400 pounds.

1-5-1-1/(1-5-1-2 Blank)

1-5-2 LUBRICATE BIFILAR.

PARAGRAPH BREAKDOWN

	PAGE
1-5-2.1 Lubricate Bifilar	1-5-2-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Grease, Item 151A, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D

1-5-2.1 Lubricate Bifilar.

1-5-2.1.1. Every 30 Hours/42 Days.

- Clean old lubricant from bifilar with machinery towel, Item 211, Appendix D.

seriously degrade the life of the rubber in the bearing. If contact occurs, rinse immediately with clean soapy water.

- Apply grease, Item 151A, Appendix D, to bifilar bushings, pins, and washers (Figure 1-5-2-1).



Do not allow any lubricant to contact the elastomeric bearings. Any contact may

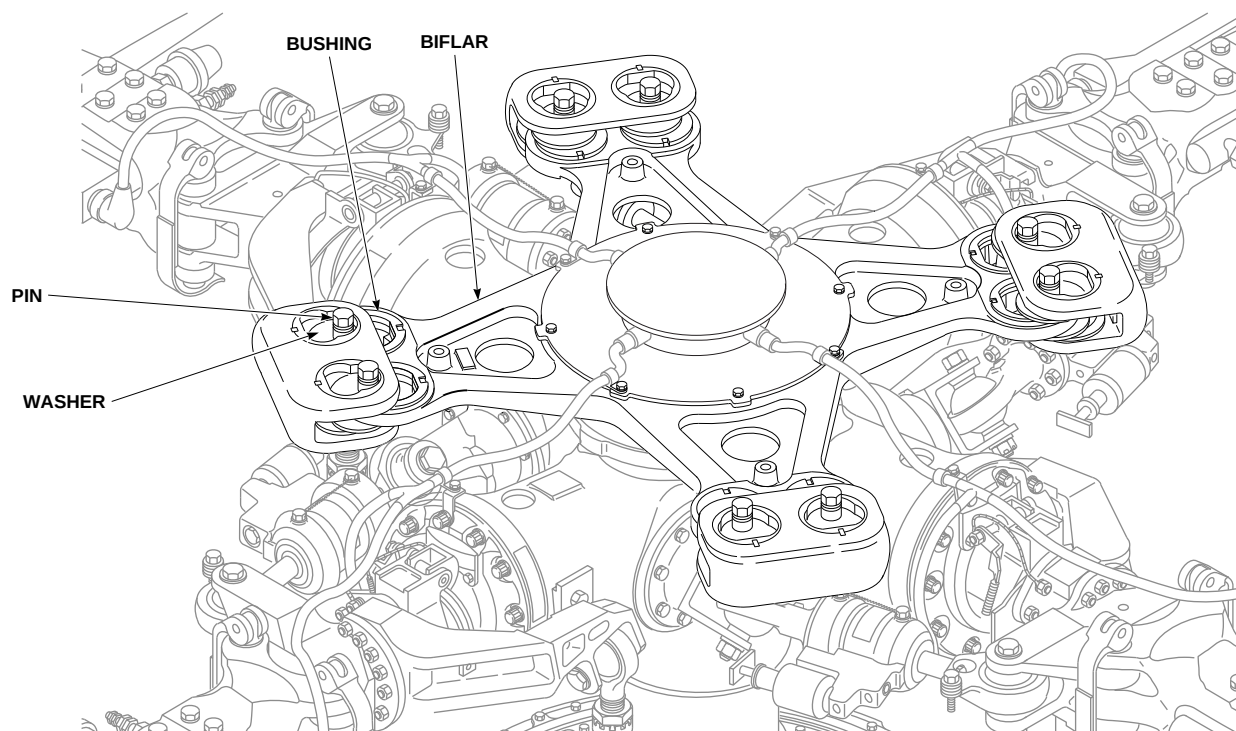
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FIGURE 1-5-2-1. LUBRICATE BIFILAR

1-5-3 LUBRICATE SWASHPLATE.

PARAGRAPH BREAKDOWN

	PAGE
1-5-3.1 Lubricate Swashplate	1-5-3-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Lubricating Gun, Hand-Operated
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Cloth, Cleaning, Item 102, Appendix D
- Grease, Item 149, Appendix D

References

- Appendix D
- PARA 1-6-18

1-5-3.1 Lubricate Swashplate.

1-5-3.1.1 Every 500 Hours/12 Months.

CAUTION

- Lubricate swashplate every 500 hours or 12 months, whichever comes first. More frequent lubrication, or using too much grease, will not improve bearing life and will cause helicopter to be sprayed with grease.
- Do not try to remove the grease shield from the bearing retaining ring. Damage to the grease shield/bearing retaining ring will result. The shield and ring are bonded together to form one part.
 - Turn main rotor head until red index marks on rotating and stationary swashplates line up (Figure 1-5-3-1, Detail B).

- If not already done, engage gust lock or rotor brake (PARA 1-6-18).

NOTE

Thirty-three bolts must be removed from inner bearing retaining ring in order to gain access to bearing seal.

- Remove bolts and washers securing inner bearing retaining ring to swashplate (Figure 1-5-3-1, Detail C).

CAUTION

Do not disturb shims when sliding retaining ring. Shimming is performed at depot to make sure of proper clamp up on bearing.

- Slide inner bearing retaining ring to one side so upper bearing seal can be seen.

CAUTION

Damage to bearings will result if pressure lubrication system is used to lubricate bearings. Use only hand-operated lubricating gun.

NOTE

For replacement swashplate assemblies built-up over 12 months prior to installation, lubricate swashplate until approximately 5 cubic inches of grease has been purged from upper and lower bearing seal at each grease fitting (approximately 20 cubic inches total).

- e. Using grease, Item 149, Appendix D, lubricate bearings through four grease fittings on rotat-

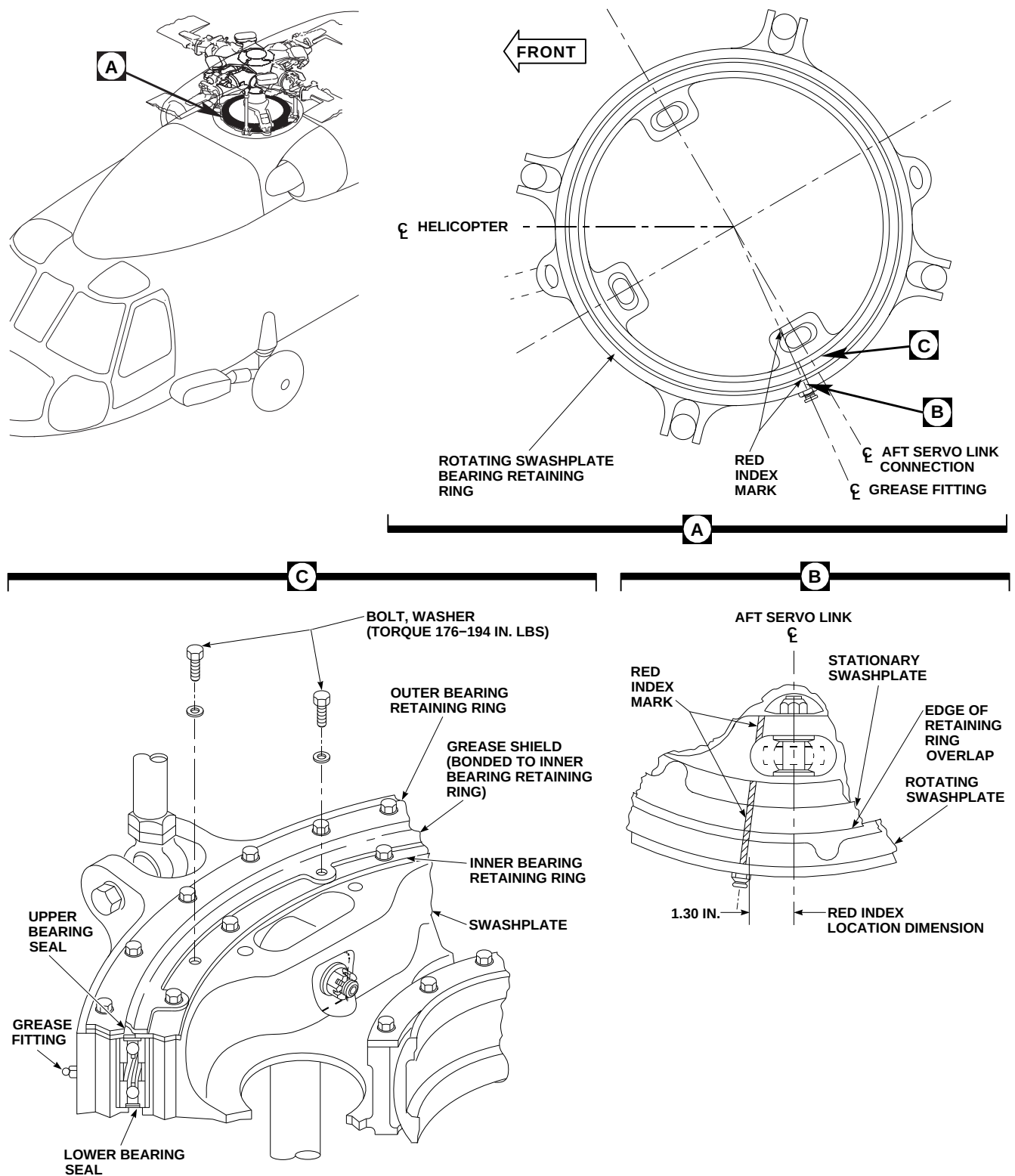
ing swashplate with hand-operated lubricating gun.

- f. Discontinue lubricating when grease seeps out of either upper or lower bearing seal (normally within 10-inches of fitting).
- g. Using cleaning cloth, Item 102, Appendix D, wipe off extra grease in seal area.

NOTE

Make sure all nine tabs of grease shield receive a bolt and washer.

- h. Install inner bearing retainer ring with bolts and washers, making sure red index marks are lined up. **TORQUE BOLTS TO 176 - 194 INCH-POUNDS.**
- i. If required, disengage gust lock or rotor brake (PARA 1-6-18).



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FIGURE 1-5-3-1. LUBRICATE SWASHPLATE

1-5-4 LUBRICATE MIXER BELLCRANKS.

PARAGRAPH BREAKDOWN

		PAGE
1-5-4.1	Lubricate Mixer Bellcranks	1-5-4-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Lubricating Gun

Materials/Parts

- Grease, Item 149, Appendix D

References

- Appendix D

1-5-4.1 Lubricate Mixer Bellcranks.

1-5-4.1.1. Every 500 Hours.

NOTE

Lubricate mixer bellcranks every 500 hours, if grease fittings are installed in bellcranks.

- Open main rotor pylon sliding cover.
- Slide work platform as far to front of helicopter as possible.

- Lubricate bearings on mixer bellcranks with hand-operated lubricating gun using grease, Item 149, Appendix D (Figure 1-5-4-1, Detail A). Stop lubricating when grease seeps out of bearings.
- Slide work platform to rear of helicopter, install quick-release pins in stop position.
- Close and latch main rotor pylon sliding cover.

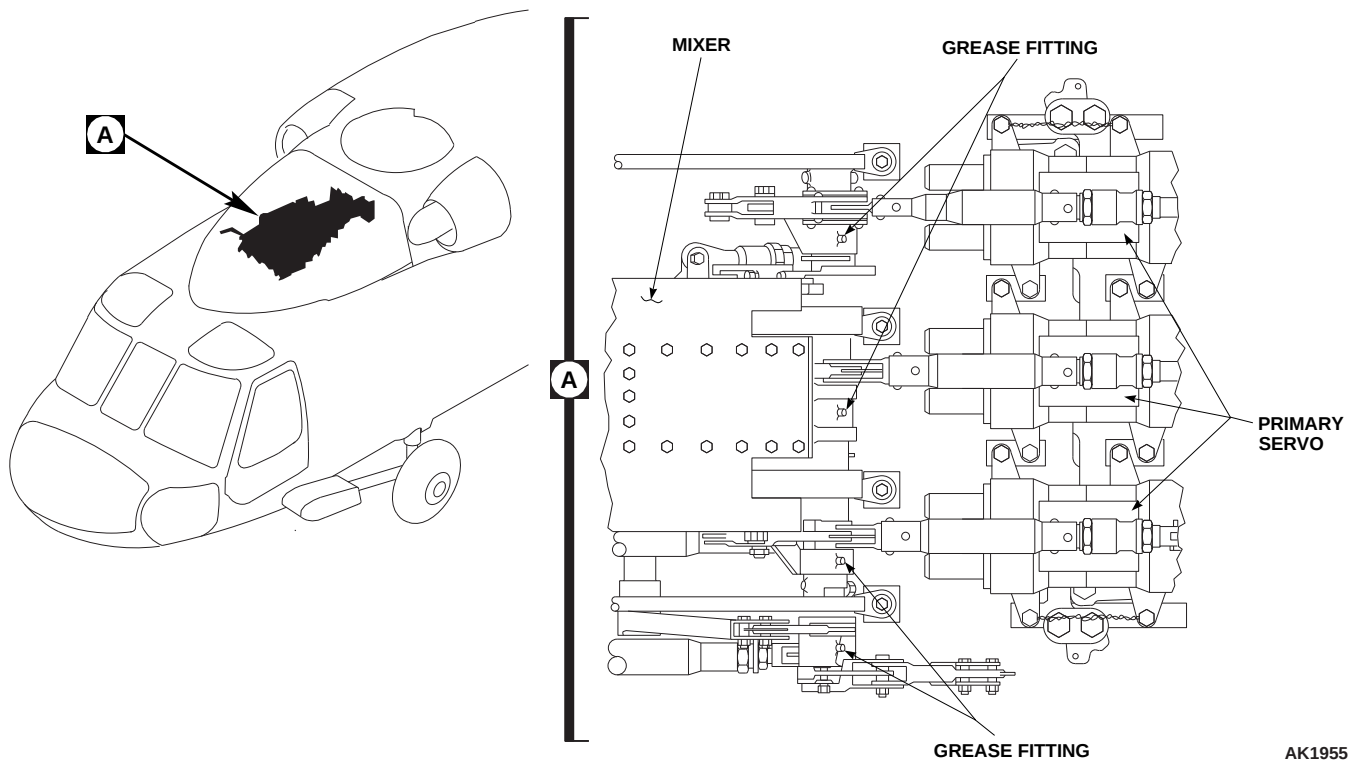


FIGURE 1-5-4-1. LUBRICATE MIXER BELLCRANKS

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1-5-5 LUBRICATE ROTARY INPUTS.

PARAGRAPH BREAKDOWN

	PAGE
1-5-5.1 Lubricate Rotary Inputs	1-5-5-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Grease, Item 150, Appendix D

References

- Appendix D

1-5-5.1 Lubricate Rotary Inputs.

Lubricate all moving parts with grease, Item 150, Appendix D.

1-5-6 LUBRICATE LANDING GEAR SHOCK STRUTS.

PARAGRAPH BREAKDOWN

		PAGE
1-5-6.1	Lubricate Landing Gear Shock Struts	1-5-6-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D
- PARA 2-4-96

1-5-6.1 Lubricate Landing Gear Shock Struts.

1-5-6.1.1. Every 10 Hours/14 Days.

- If required, remove upper main landing gear fairing to gain access to main landing gear shock strut piston (PARA 2-4-96).



Damage to main and tail landing gear shock strut seals will result if shock strut

pistons are not kept clean and lubricated. Shock struts will leak fluid if seals are damaged. Make sure shock strut pistons are clean and lubricated.

- Wipe exposed portion of main and tail landing gear shock strut pistons with machinery towel, Item 211, Appendix D, dampened with hydraulic fluid, Item 154, Appendix D.
- If removed, install upper main landing gear fairing (PARA 2-4-96).

1-5-7 LUBRICATE STABILATOR ACTUATOR ASSEMBLY.

PARAGRAPH BREAKDOWN

		PAGE
1-5-7.1	Lubricate Stabilator Ac- tuator Assembly	1-5-7-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D
- PARA 2-4-146

1-5-7.1 Lubricate Stabilator Actuator As-
sembly.

1-5-7.1.1. Every 10 Hours/14 Days.

Wipe exposed portion of lower actuator output shaft with machinery towel, Item 211, Appendix D, dampened with hydraulic fluid, Item 154, Appendix D.

1-5-7.1.2. Every 500 Hours.

- Remove stabilator actuator (PARA 2-4-146).

- Fully extend both output shafts (PARA 2-4-146).
- Wipe exposed portion of shafts with machinery towel, Item 211, Appendix D, dampened with hydraulic fluid, Item 154, Appendix D.
- Install stabilator actuator (PARA 2-4-146).

1-5-8 LUBRICATE WINDSHIELD WIPER PIVOT STUDS.

PARAGRAPH BREAKDOWN

		PAGE
1-5-8.1	Lubricate Windshield Wiper Pivot Studs	1-5-8-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Cup

Materials/Parts

- Brush, Item 83, Appendix D
- Hydraulic Fluid, Item 155, Appendix D

References

- Appendix D

1-5-8.1 Lubricate Windshield Wiper Pivot Studs.

1-5-8.1.1. Every 10 Hours/14 Days.

- Pour a small quantity of hydraulic fluid, Item 155, Appendix D, in clean 1-cup container.

- Using brush, Item 83, Appendix D, apply hydraulic fluid on pilot’s and copilot’s windshield wiper pivot studs link attachment area.

1-5-9 LUBRICATE WINDSHIELD WIPER CONVERTER.

PARAGRAPH BREAKDOWN

		PAGE
1-5-9.1	Lubricate Windshield Wiper Converter	1-5-9-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Cup
- Hypodermic Syringe

Materials/Parts

- Grease, Item 149, Appendix D
- Hydraulic Fluid, Item 155, Appendix D

References

- Appendix D
- PARA 12-4-14

1-5-9.1 Lubricate Windshield Wiper Converter.

1-5-9.1.1. Every 10 Hours/14 Days.

- Pour a small quantity of hydraulic fluid, Item 155, Appendix D, in clean 1-cup container.
- Using hypodermic syringe, apply hydraulic fluid on converter shaft below splines to lubricate internal packing.

1-5-9.1.2. Every 500 Hours.

- Remove, disassemble, and clean windshield wiper converter (PARA 12-4-14).

- Pack housing with grease, Item 149, Appendix D.
- Pack gears and bearings on end of converter linkage with grease, Item 149, Appendix D.
- Assemble and install windshield wiper converter (PARA 12-4-14).

1-5-10 LUBRICATE TAIL WHEEL LOCKPIN.

PARAGRAPH BREAKDOWN

		PAGE
1-5-10.1	Lubricate Tail Wheel Lockpin	1-5-10-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Grease, Item 149, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D

1-5-10.1 Lubricate Tail Wheel Lockpin.

1-5-10.1.1. Every 500 hours.

- Manually unlock tail wheel lockpin.
- Clean old grease from lockpin with a clean machinery towel, Item 211, Appendix D.

- Lightly coat lockpin with grease, Item 149, Appendix D. Actuate lockpin by hand to make sure it is completely coated with grease.

1-5-11 LUBRICATE TAIL WHEEL BEARINGS.

PARAGRAPH BREAKDOWN

		PAGE
1-5-11.1	Lubricate Tail Wheel Bearings	1-5-11-1

INITIAL SETUP

References

- PARA 3-4-28
-

1-5-11.1 Lubricate Tail Wheel Bearings.

1-5-11.1.1. Every 250 Hours/6 Months.

Remove tail wheel and lubricate bearings (PARA 3-4-28).

1-5-12 LUBRICATE DOOR LOCKS.

PARAGRAPH BREAKDOWN

	PAGE
1-5-12.1 Lubricate Door Locks	1-5-12-1

INITIAL SETUP

Materials/Parts

- Lubricating Oil, Item 209, Appendix D

References

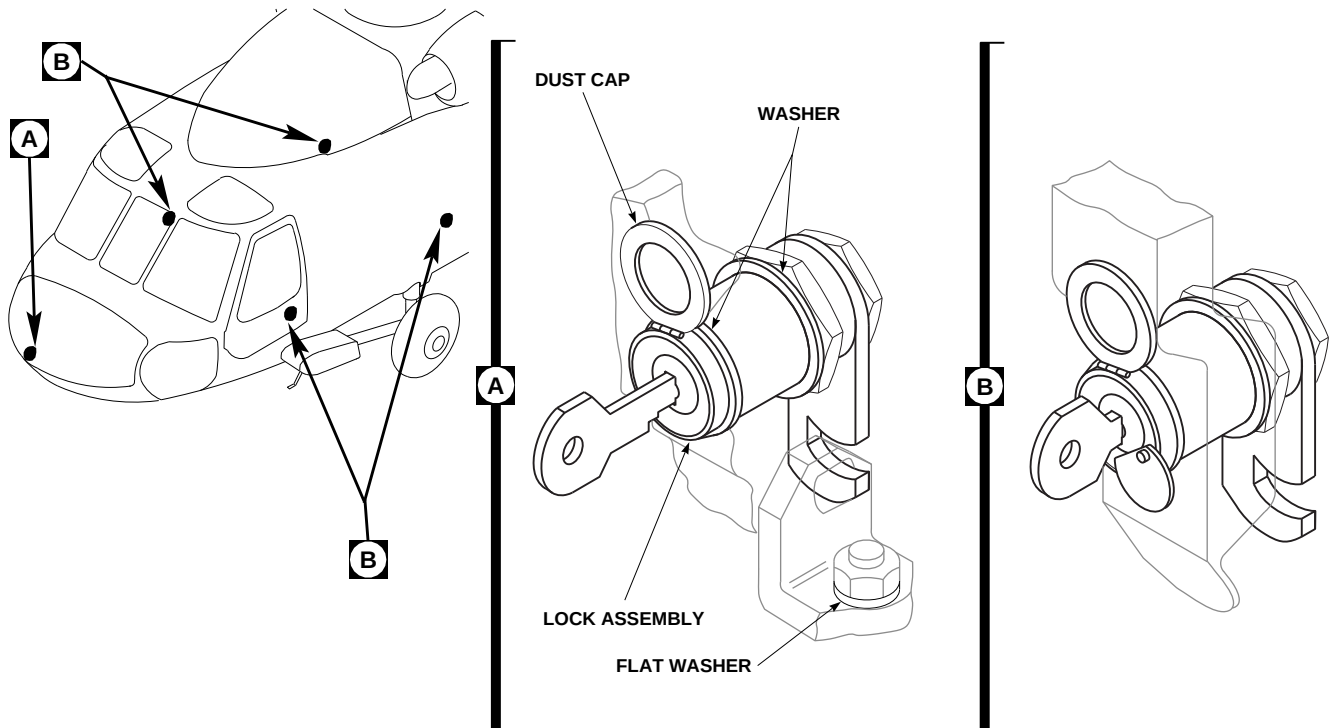
- Appendix D

1-5-12.1 Lubricate Door Locks.

1-5-12.1.1. Every 500 Hours.

Lightly coat keys, lock sets, and lock hooks with lubricating oil, Item 209, Appendix D. Actuate

locks using key to make sure there is no binding or other interference (Figure 1-5-12-1, Detail A and Detail B).



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FIGURE 1-5-12-1. LUBRICATE DOOR LOCKS

1-5-13 LUBRICATE EJECTOR RACK, BRU-22A/A.

PARAGRAPH BREAKDOWN

		PAGE
1-5-13.1	Lubricate Ejector Rack, BRU-22A/A	1-5-13-1

INITIAL SETUP

Materials/Parts

- Lubricating Oil, Item 209, Appendix D

References

- Appendix D

1-5-13.1 Lubricate Ejector Rack, BRU-22A/A.

1-5-13.1.1. Every 500 Hours.

Lubricate mechanical arming unit plunger using lubricating oil, Item 209, Appendix D.

SECTION 1-6

GROUND HANDLING

SECTION OVERVIEW

PARAGRAPH	TITLE
1-6-1	Ground Handling General
1-6-2	Tow Helicopter
1-6-3	Jack Helicopter
1-6-4	Tie Down and Moor Helicopter
1-6-5	Protective Covers
1-6-6	Park Helicopter
1-6-7	Fold/Spread Main Rotor Blades
1-6-8	Fold/Spread Tail Rotor Blades
1-6-9	Fold/Unfold Tail Rotor Pylon
1-6-10	Kneel Helicopter
1-6-11	Cabin Tiedown Fittings
1-6-12	Tiedown Plates
1-6-13	Tail Cone Tow Fittings
1-6-14	Fuselage and Tail Cone Tiedown Rings
1-6-15	Apply Hydraulic Power
1-6-16	Apply Electrical Power
1-6-17	Hydraulic Cart Connect and Checkout
1-6-18	Engage/Disengage Gust Lock and Rotor Brake

1-6-1 GROUND HANDLING GENERAL.

1-6-1.1 Scope.

This section contains ground handling procedures.
Procedures in this section include, but are not

limited to, towing, jacking, mooring, hoisting,
leveling, and external power connection.

1-6-2 TOW HELICOPTER.

PARAGRAPH BREAKDOWN

	PAGE
1-6-2.1 Tow Helicopter	1-6-2-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Chocks
- Tow Bar, AA1730-1251

References

- PARA 1-6-4

1-6-2.1 Tow Helicopter.

1-6-2.1.1 Tow Helicopter.

CAUTION

- Damage to tail landing gear will result if helicopter over 16,825 pounds gross weight is towed with tail landing gear swiveled 180° underneath tail cone. Do not tow helicopter with tail landing gear tail wheel swiveled 180° underneath tail cone, when gross weight exceeds 16,825 pounds.
- Damage to tow bar or helicopter can result if too sharp an angle exists between tow vehicle and tow bar. The angle between tow vehicle and tow bar should not exceed 90° to avoid damage.

NOTE

- The helicopter may be either pushed or pulled during towing.
- Battery power may be used to level stabilator. However, if battery low light is

on, battery power will only be available for APU starting, and external power should be used to operate stabilator.

- Cabin floor panel installation is not required for towing operations.
- a. Level stabilator so tow operator can see all around helicopter.

CAUTION

Damage to helicopter will result if tow bar becomes separated from tail wheel. Make sure tow bar is properly secured to tail wheel.

- b. Unpin tow bar crossbeam from tube to allow positioning of tow bar on tail wheel.
- c. Insert axle pins into tail wheel axle holes, and re-pin crossbeam to tube (Figure 1-6-2-1, Detail A).
- d. Connect tow bar to tow vehicle.
- e. Engage tailwheel lockpin using either of the following methods:

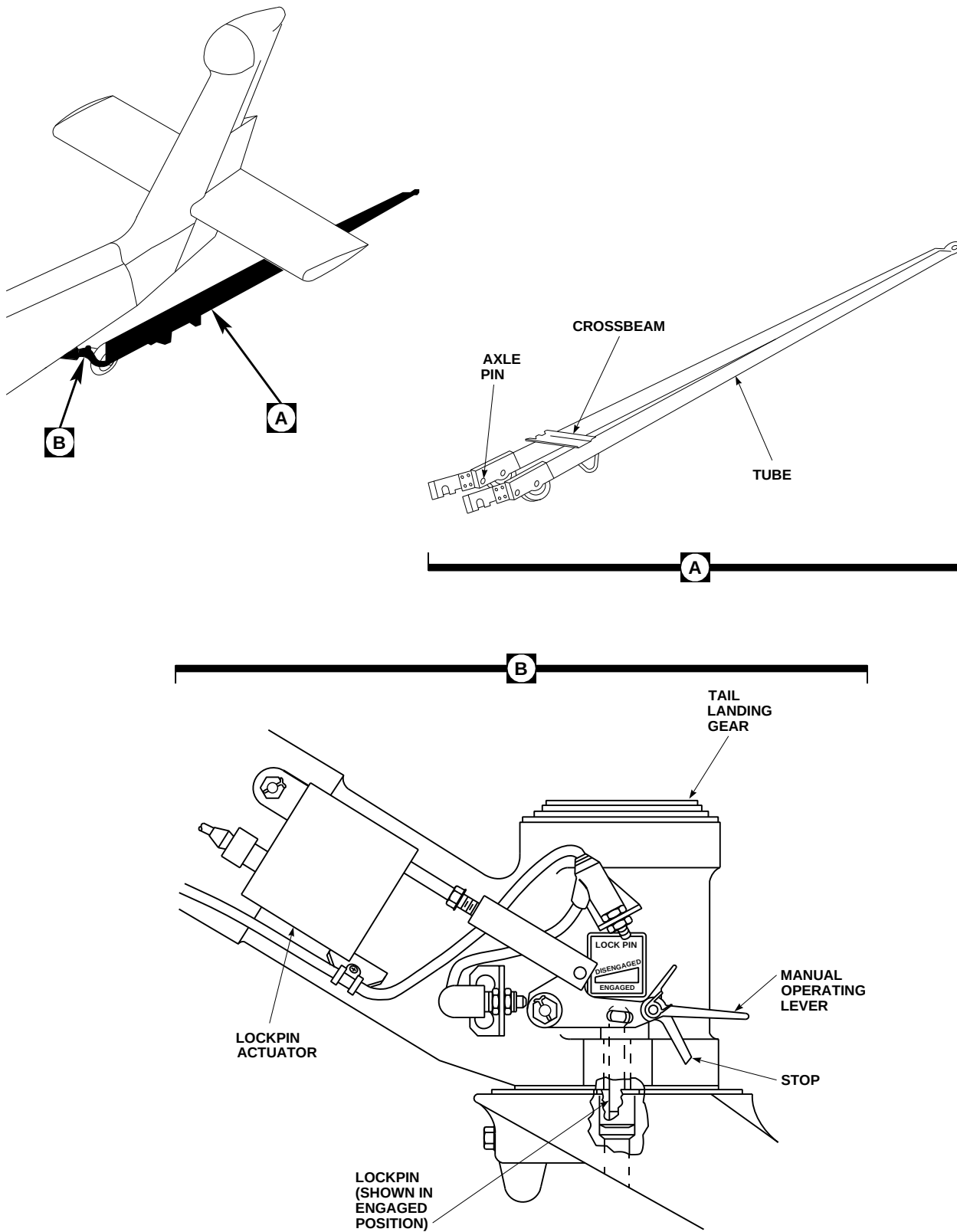


FIGURE 1-6-2-1. TOW HELICOPTER

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- (1) On tail landing gear, place tailwheel lockpin in ENGAGED position using manual operating lever (Figure 1-6-2-1, Detail B).
 - (2) In cockpit on upper console, place BATT switch ON. On miscellaneous switch panel, push TAILWHEEL switch to engage tailwheel lock. Place BATT switch OFF.
- f. If towing helicopter in a congested area, install main rotor blade tiedown ropes (PARA 1-6-4).
- g. Disconnect any tiedown lines or grounding straps.
- h. Have an assistant release parking brakes and ride in cockpit to work brakes, if required.
- i. Prior to towing, review the following precautions to tow helicopter safely without damage:
- (1) If possible, wait 20 minutes after removing electrical power from vertical gyro before towing to prevent gyro damage.
 - (2) Make sure all external power sources are disconnected and clear of helicopter.
- (3) Make sure all engines are off, and both main and tail rotors are stopped.
- (4) Do not release tail wheel lockpin until tow bar is attached to tow vehicle.
- (5) Do not release parking brake or remove chocks from wheels until after tow bar is installed, and tow vehicle is attached.
- (6) Check operation of brakes.
- (7) Tow only on level surface and do not go over 5 miles-per-hour.
- (8) Do not brake to sudden stop when turning; do not make quick starts and stops.
- (9) Have personnel on both sides of helicopter to watch main rotor blades, tail rotor blades, and airframe clearance when towing and turning helicopter.
- (10) If helicopter is to be towed in a congested area, main rotor blades can be moved using tiedown ropes.
- j. Remove chocks from main wheels and tow helicopter.

1-6-3 JACK HELICOPTER.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-6-3.1	Jack Main Landing Gear	1-6-3-1	1-6-3.3	Jack Entire Helicopter	1-6-3-5
1-6-3.2	Jack Tail Landing Gear	1-6-3-5			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Chocks
- Jack(s)

- Plumb Bob

References

- PARA 1-6-6

1-6-3.1 Jack Main Landing Gear.

1-6-3.1.1 Jack.

NOTE

- Normal load for all jackpads is 7,492 pounds.
 - Floor panel installation is not required when jacking helicopter.
- Park helicopter on as solid and level an area as possible (PARA 1-6-6).
 - Chock helicopter.

WARNING

Injury to personnel and damage to equipment may occur if tailwheel lockpin is not engaged before jacking helicopter. Make sure to engage tailwheel lockpin.

- Engage tailwheel lockpin using either of the following methods:
 - On tail landing gear, place tailwheel lockpin in ENGAGED position using manual operating lever (Figure 1-6-3-1, Detail A).
 - In cockpit on upper console, place BATT switch ON. On miscellaneous switch panel, push TAILWHEEL switch to engage tailwheel lock. Place BATT switch OFF.
- Position jacks beneath jackpads on left and right drag beams (Figure 1-6-3-1, Detail B and Detail C).
- Slowly jack landing gear until tire just clears ground.

1-6-3.1.2 Lower.

- Slowly lower landing gear to ground (Figure 1-6-3-1, Detail B and Detail C).
- Remove jacks.

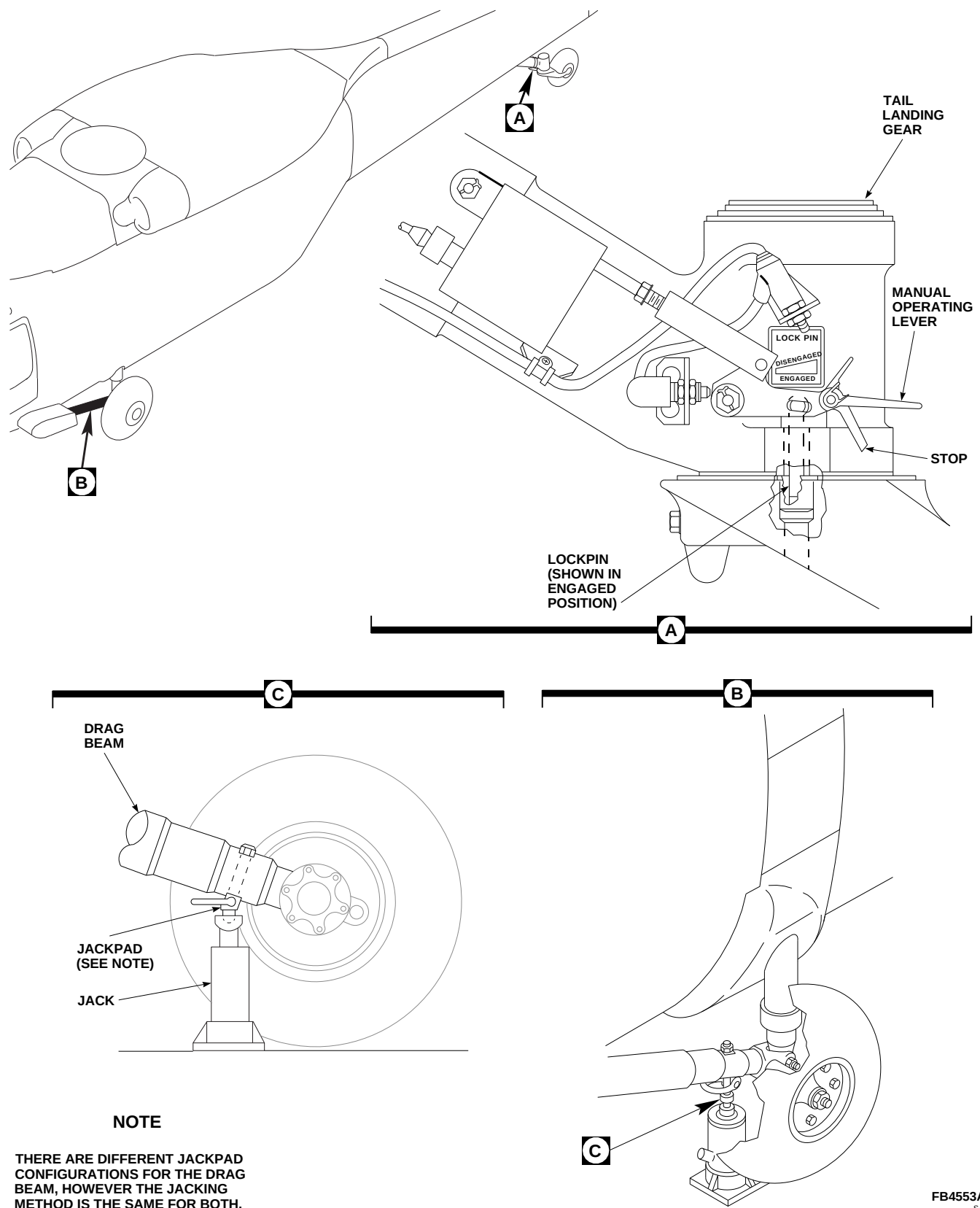
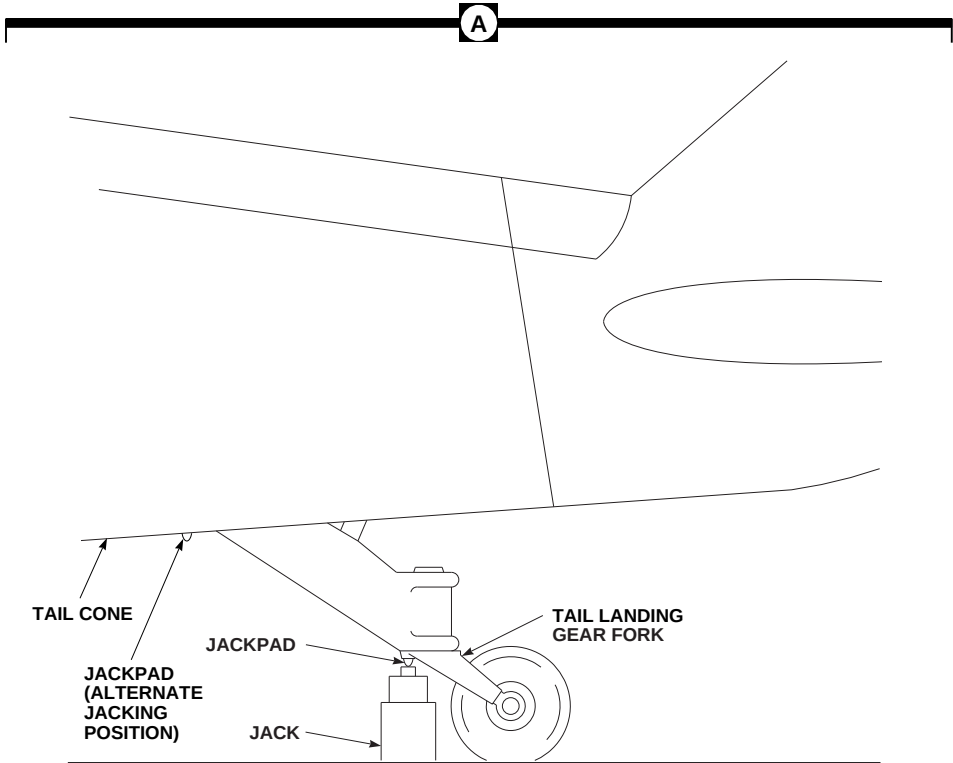
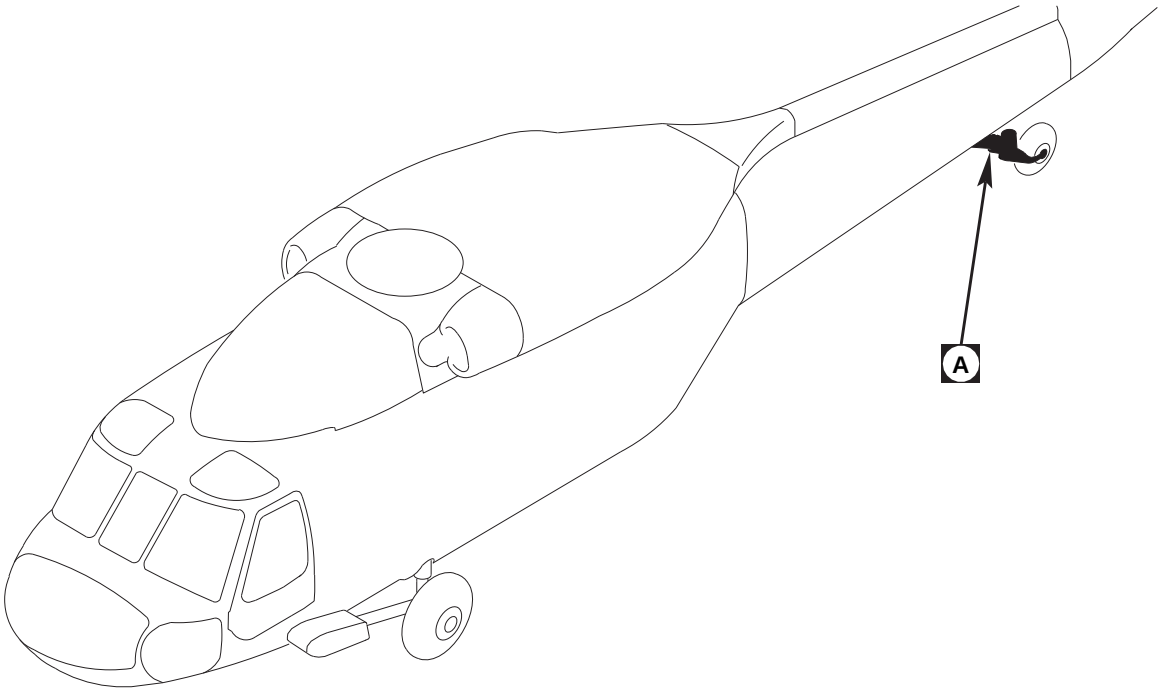
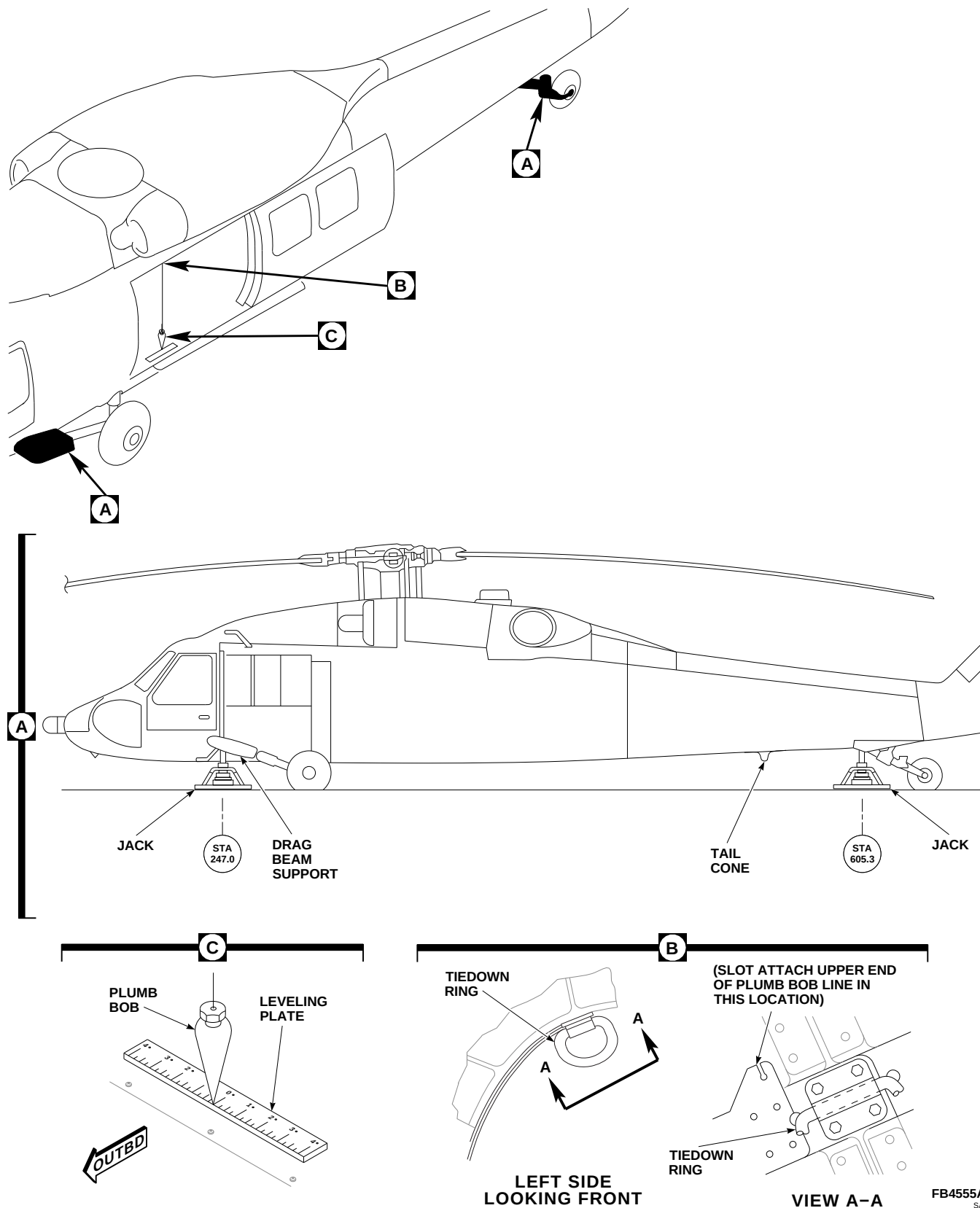


FIGURE 1-6-3-1. JACK MAIN LANDING GEAR



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FIGURE 1-6-3-2. JACK TAIL LANDING GEAR

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- c. Disengage tailwheel lockpin using either of the following methods:
 - (1) On tail landing gear, place tailwheel lockpin in DISENGAGED position using manual operating lever (Figure 1-6-3-1, Detail A).
 - (2) In cockpit on upper console, place BATT switch ON. On miscellaneous switch panel, push TAILWHEEL switch to release tailwheel lock. Place BATT switch OFF.

1-6-3.2 Jack Tail Landing Gear.

1-6-3.2.1 Jack.

NOTE

- Normal load for all jackpads is 7,492 pounds.
- Floor panel installation is not required when jacking helicopter.
- a. Park helicopter on as solid and level an area as possible (PARA 1-6-6).
- b. Chock helicopter.
- c. Position jack beneath jackpad on tail landing gear fork or beneath jackpad on tail cone (Figure 1-6-3-2, Detail A).
- d. Slowly jack landing gear on tail cone until tire just clears ground.

1-6-3.2.2 Lower.

- a. Slowly lower landing gear to ground (Figure 1-6-3-2, Detail A).
- b. Remove jack.

1-6-3.3 Jack Entire Helicopter.

1-6-3.3.1 Jack.

NOTE

- Normal load for all jackpads is 7,492 pounds.

- Floor panel installation is not required when jacking helicopter.

- a. Park helicopter indoors on a solid and level area as possible (PARA 1-6-6).

CAUTION

To prevent damage to helicopter, check overhead clearance before jacking.

- b. Check for adequate clearance above helicopter.
- c. Position jack beneath jackpad on bottom front of each drag beam support, and beneath jackpad on tail cone (Figure 1-6-3-3, Detail A).
- d. If not already done, open left cabin door.
- e. Hang plumb bob from slot just inside left cabin door above leveling plate (Figure 1-6-3-3, Detail B and Detail C).

WARNING

To prevent helicopter from rolling off jacks, make sure parking brake is off, and that chocks are moved as helicopter is jacked.

- f. Release parking brake. Do not remove chocks.
- g. Using four assistants, one for each jack, and one observer, slowly and evenly jack helicopter while adjusting height of jacks to line up plumb bob with intersection of "0" marked on outboard edge of leveling plate until desired height is reached, and helicopter is level (Figure 1-6-3-3, Detail C). As helicopter is jacked, move chocks to keep landing gear secured until tires are off ground.

1-6-3.3.2 Lower.

- a. Using four assistants, one for each jack, and one observer, slowly and evenly lower helicopter (Figure 1-6-3-3, Detail A).

- b. As helicopter is lowered, move chocks to keep landing gear secured as weight of helicopter comes off jacks.
- c. Lock parking brake.
- d. Remove plumb bob (Figure 1-6-3-3, Detail B).

1-6-4 TIE DOWN AND MOOR HELICOPTER.

PARAGRAPH BREAKDOWN

	PAGE
1-6-4.1 Tie Down and Moor Helicopter	1-6-4-2

INITIAL SETUP

Tools

- Aircraft Cable, ½-inch
- Aircraft Mechanic’s Toolkit
- Anchor, Deadman
- Anchor Kits
- Chain
- Chocks
- Chocks, Ice Grip
- W/O HIRSS

 Flyaway Equipment Kit, 70700-10000-011, consisting of:
 - APU Exhaust Plug, 70700-20509-041
 - Blade Fold Set, 70700-20412-011, consisting of:
 - Alignment Pin, 70700-20394-111
 - Blade Fold Support, 70700-20417-041
 - Blade Fold Support, 70700-20417-042
 - Blade Handling Pole, 70700-20395-041
 - Retainer, 70700-20428-041
 - Retainer, 70700-20428-042
 - Support Poles, 70700-20427-041 through 70700-20427-044
 - Engine Air Inlet Plug, 70700-20507-042
 - Engine Exhaust Plug, 70700-20508-041
 - Lock Assembly, 70700-20369-041
 - Oil Cooler, IR Jammer, Screen Cover, 70700-20511-041
 - Pitot Tube Cover, 70700-20460-041
- HIRSS

 Flyaway Equipment Kit, 70700-10000-014, consisting of:

- APU Exhaust Plug, 70700-20509-041
 - Blade Fold Set, 70700-20412-011, consisting of:
 - Alignment Pin, 70700-20394-111
 - Blade Fold Support, 70700-20417-041
 - Blade Fold Support, 70700-20417-042
 - Blade Handling Pole, 70700-20395-041
 - Retainer, 70700-20428-041
 - Retainer, 70700-20428-042
 - Support Poles, 70700-20427-041 through 70700-20427-044
 - Engine Air Inlet Plug, 70700-20507-042
 - Forward and Rear Tiedown Fitting, 70700-20433-045
 - Lock Assembly, 70700-20369-041
 - Oil Cooler, IR Jammer, Screen Cover, 70700-20511-041
 - Pitot Tube Cover, 70700-20460-041
 - HIRSS Cover, LH 89SDSCC-D-0162-1
 - HIRSS Cover, RH 89SDSCC-D-0162-2
 - HIRSS Intake, Plug Assembly, LH 70700-20514-041
 - HIRSS Intake, Plug Assembly, RH 70700-20514-042
 - HIRSS Exhaust, Plug Assembly, 70700-20515-043
-
- Rope, Manila, 1 ½-inch
 - Rope, Wire
 - Tiedown Chain Assembly, 627728A
 - Tiedown Tensioner, CGU-4/E
 - Wooden Cleats

References

- PARA 1-3-17
- PARA 1-3-18
- PARA 1-3-19
- PARA 1-6-5
- PARA 1-6-6
- PARA 1-6-18

1-6-4.1 Tie Down and Moor Helicopter.

CAUTION

Structural damage can occur from high velocity surface winds. When possible, helicopter shall be evacuated to a safe weather area when tornado, hurricane, or winds above 75 knots are expected.

NOTE

- Mooring hardware is not flyaway equipment. Each mooring point in use will be equipped with mooring hardware.
 - Cabin floor panel installation is not required when tying down or mooring helicopter.
- a. Park helicopter (PARA 1-6-6).
 - b. Fill fuel cells (PARA 1-3-17, PARA 1-3-18, or PARA 1-3-19).
 - c. Secure chocks by nailing wooden cleats from chock to chock on each side of each wheel. Use rope to secure chocks if wooden cleats are not available, or when using ice grip chocks.

CAUTION

To prevent damage, it is recommended that blades be folded if wind speeds above 60 knots are expected. Tie down of the main rotor blades should be done when actual or projected wind conditions are 45 - 60 knots.

- d. Turn rotor head and position a blade over center of cockpit.

CAUTION

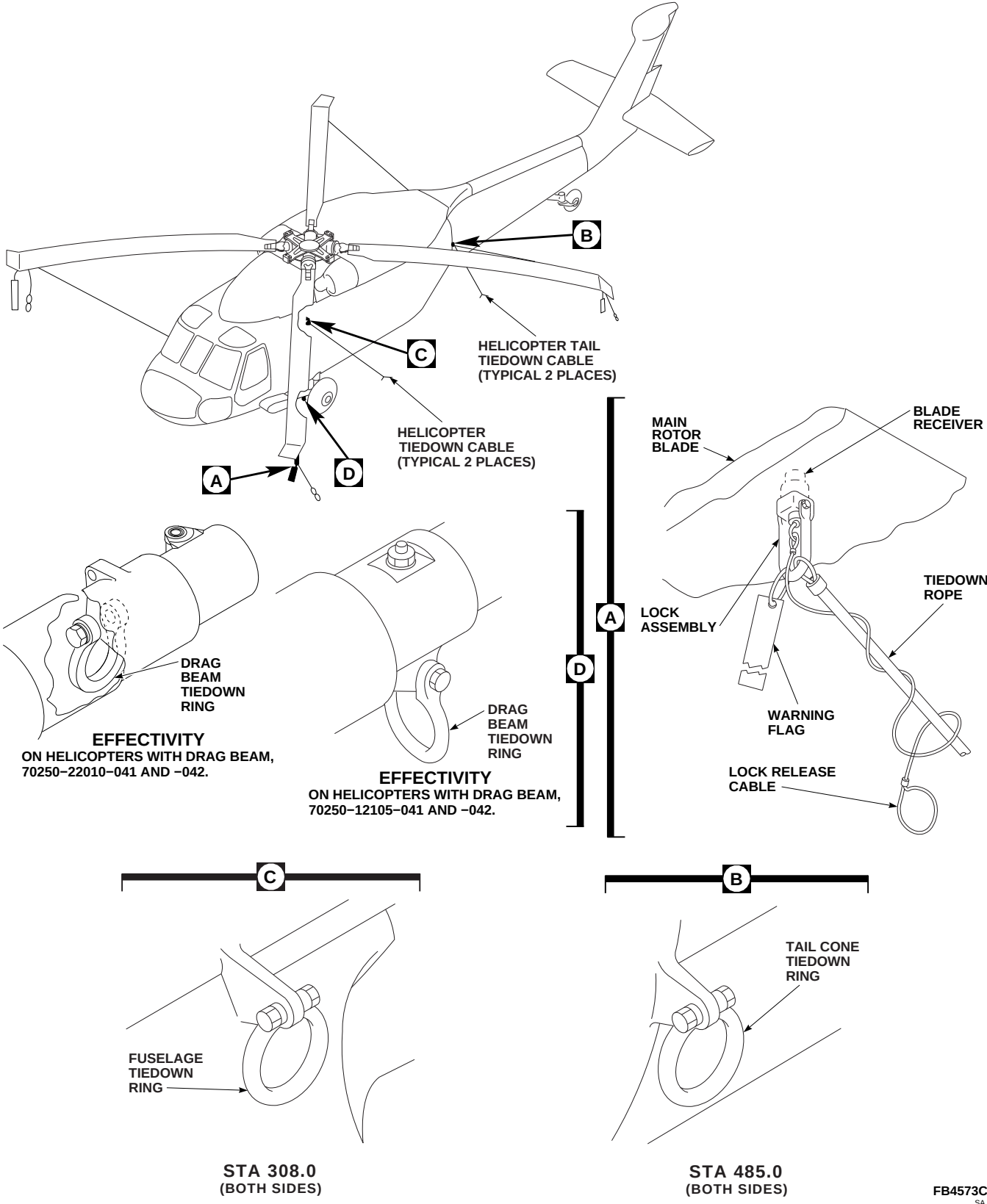
Damage to main rotor blades will occur if end of blade is pulled down more than 6-inches below normal drooped position. Do not pull blade down more than 6-inches when installing tiedown ropes.

- e. Install lock assembly into blade receiver on tip end of blade. Pull down on lock assembly lock release cable, insert lock assembly in blade receiver, and release lock release cable (Figure 1-6-4-1, Detail A).
- f. Pull down on lock assembly to make sure it is locked into main rotor blade.

CAUTION

Damage to main rotor blade will occur if lock release cable is not secure. Strong winds will cause lock release cable to strike blade. Make sure lock release blade is secure when tiedown ropes are attached to blade.

- g. Secure lock release cable by wrapping it several times around tiedown rope, and pass end loop through last wrap.
- h. Turn rotor head, and install lock assemblies in other blades in same manner.
- i. Position blades at an angle of about 45° from centerline of helicopter, and engage gust lock (PARA 1-6-18).



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FIGURE 1-6-4-1. TIE DOWN MAIN ROTOR BLADES

CAUTION

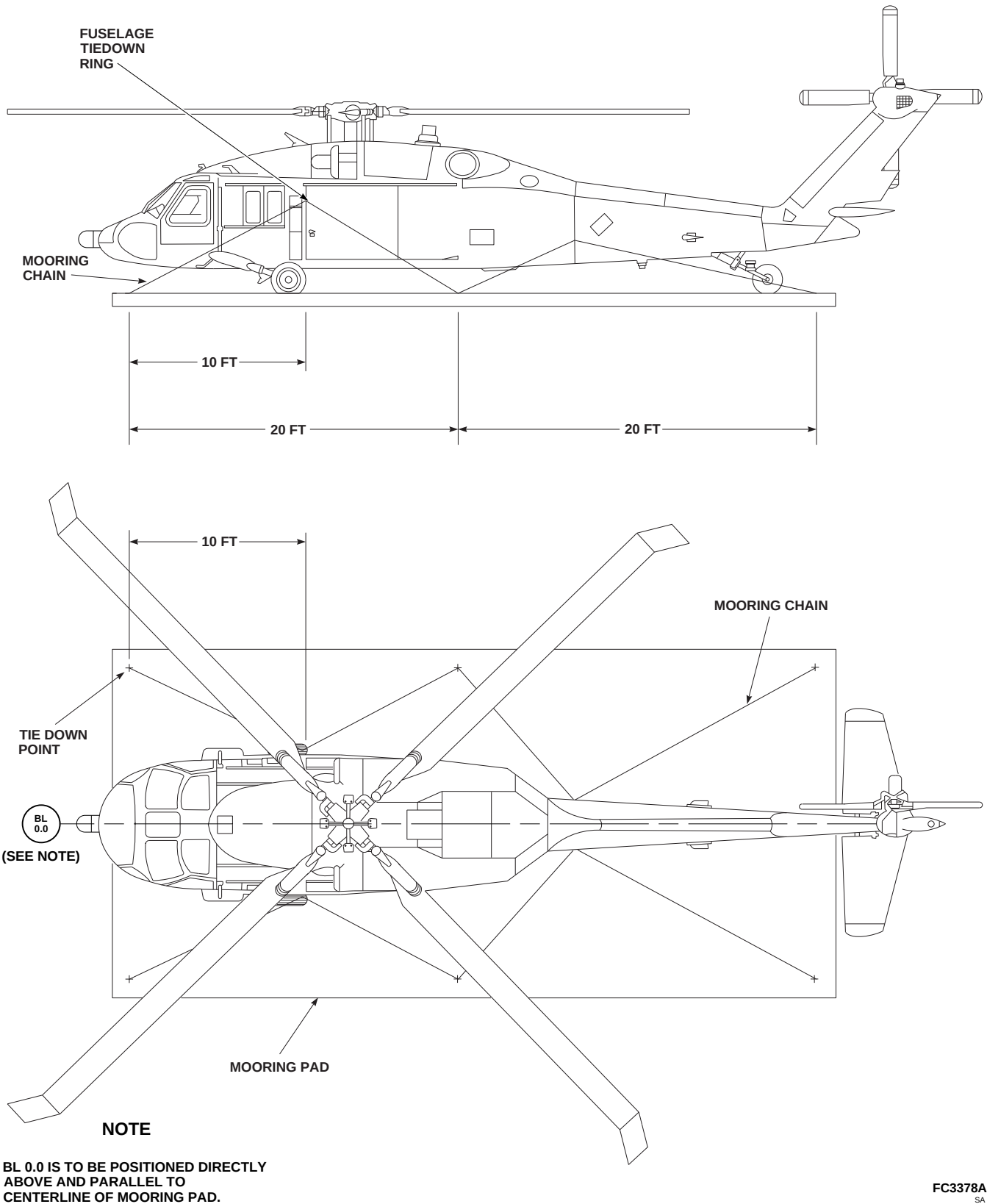
Damage to main rotor blades will occur if end of blade is pulled down more than 6-inches below normal drooped position. Do not pull blade down more than 6-inches when installing tiedown ropes.

- j. Secure tiedown ropes, from two rear blades to tail cone tiedown rings (Figure 1-6-4-1, Detail B).
- k. Secure tiedown ropes, from two front blades to main landing gear drag beams tiedown rings (Figure 1-6-4-1, Detail D).
- l. If helicopter is to be moored on paved surface, perform the following:
 - (1) Position helicopter on mooring pad with BL 0.0 directly above and parallel to centerline of mooring pad (Figure 1-6-4-2).
 - (2) Position fuselage tiedown rings an equal distance between two forward pairs of pad mooring points, 10 feet behind forward pair of mooring points (Figure 1-6-4-1, Detail C and Figure 1-6-4-2).
 - (3) Connect hook-ends of tiedown chain assemblies to helicopter forward mooring rings.
 - (4) Connect tiedown chain assemblies to mooring pad tiedown points using tiedown tensioners.

- (5) Adjust length of chains to take slack from chains.
- (6) Repeat procedure for helicopter rear mounting points.

NOTE

- When ground tiedown rings are not available, move helicopter to an area where anchor kits or deadman-type anchors, capable of sustaining a 12,500 pound per cable pull test, may be used.
 - Chain must be capable of sustaining a 12,500 pound pull test.
- m. If helicopter is to be moored on an unpaved surface, perform the following:
 - (1) Secure 1/2-inch aircraft cables or wire rope to tiedown rings, and extend outward to ground mooring points at 45° angles (Figure 1-6-4-2).
 - (2) Use chain or 1 1/2-inch manila rope if cable is not available. Provide enough slack in cable, chain, or rope between anchor and tiedown ring to prevent tightening due to moisture, distortion, or tire deflation on the opposite side.
 - (3) Use anti-slip knots in rope, such as the square or bowline.
 - n. Install protective covers (PARA 1-6-5).



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FIGURE 1-6-4-2. PAVED SURFACE MOORING CONFIGURATION (TYPICAL)

1-6-5 PROTECTIVE COVERS.

PARAGRAPH BREAKDOWN

	PAGE
1-6-5.1 Protective Covers	1-6-5-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- W/O HIRSS

 Flyaway Equipment Kit, 70700-10000-011, consisting of:
 - APU Exhaust Plug, 70700-20509-041
 - Blade Fold Set, 70700-20412-011, consisting of:
 - Alignment Pin, 70700-20394-111
 - Blade Fold Support, 70700-20417-041
 - Blade Fold Support, 70700-20417-042
 - Blade Handling Pole, 70700-20395-041
 - Retainer, 70700-20428-041
 - Retainer, 70700-20428-042
 - Support Poles, 70700-20427-041 through 70700-20427-044
 - Engine Air Inlet Plug, 70700-20507-042
 - Engine Exhaust Plug, 70700-20508-041
 - Lock Assembly, 70700-20369-041
 - Oil Cooler, IR Jammer, Screen Cover, 70700-20511-041
 - Pitot Tube Cover, 70700-20460-041
- HIRSS

 Flyaway Equipment Kit, 70700-10000-014, consisting of:
 - APU Exhaust Plug, 70700-20509-041
 - Blade Fold Set, 70700-20412-011, consisting of:
 - Alignment Pin, 70700-20394-111
 - Blade Fold Support, 70700-20417-041
 - Blade Fold Support, 70700-20417-042
 - Blade Handling Pole, 70700-20395-041
 - Retainer, 70700-20428-041

- Retainer, 70700-20428-042
 - Support Poles, 70700-20427-041 through 70700-20427-044
 - Engine Air Inlet Plug, 70700-20507-042
 - Forward and Rear Tiedown Fitting, 70700-20433-045
 - Lock Assembly, 70700-20369-041
 - Oil Cooler, IR Jammer, Screen Cover, 70700-20511-041
 - Pitot Tube Cover, 70700-20460-041
 - HIRSS Cover, LH 89SDSCC-D-0162-1
 - HIRSS Cover, RH 89SDSCC-D-0162-2
 - HIRSS Intake, Plug Assembly, LH 70700-20514-041
 - HIRSS Intake, Plug Assembly, RH 70700-20514-042
 - HIRSS Exhaust, Plug Assembly, 70700-20515-043
 -
 - Main Rotor Blade Cover, 1730BH025-1, Locally-Made, Figure H-187, Appendix H
 - Main Rotor Head Cover, 1730BH025-3, Locally-Made, Figure H-189, Appendix H
 - Mast Cover, 1730BH025-6, Locally-Made, Figure H-192, Appendix H
 - Tail Rotor Blade Cover, 1730BH025-2, Locally-Made, Figure H-188, Appendix H
 - Tail Rotor Gear Box Cover, 1730BH025-4, Locally-Made, Figure H-190, Appendix H
- References
- Appendix H

1-6-5.1 Protective Covers.**1-6-5.1.1. Install.**

- a. Install covers, 70700-20460-041, over pitot tubes (Figure 1-6-5-1).
- b. Install plugs, 70700-20507-042, in engine air inlets.
- c. **HIRSS** Install HIRSS protective covers as follows:
 - (1) Install right hand HIRSS intake plugs, 70700-20514-042.
 - (2) Install left hand HIRSS intake plugs, 70700-20514-041.
 - (3) Install HIRSS exhaust plugs, 70700-20515-043, on both sides.
 - (4) Install left hand HIRSS cover, 89SDSCC-D-0162-1.
 - (5) Install right hand HIRSS cover, 89SDSCC-D-0162-2.
- d. **W/O HIRSS** Insert engine exhaust plugs, 70700-20508-041, two places.
- e. Install plug, 70700-20509-041, in APU exhaust duct.
- f. Install cover, 70700-20511-041, over oil cooler, IR jammer, if installed, and screens.

NOTE

Use extreme care while installing covers.

- g. Install the following covers for storage, shipping, and adverse weather conditions.
 - (1) Locally make main rotor blade covers, 1730BH025-1, and install over main rotor blades (Figure H-187, Appendix H).
 - (2) Locally make tail rotor blade covers, 1730BH025-2, and install over tail rotor blades (Figure H-188, Appendix H).

- (3) Locally make mast cover, 1730BH025-6, and install over mast (Figure H-192, Appendix H).
- (4) Locally make main rotor head cover, 1730BH025-3, and install over main rotor head (Figure H-189, Appendix H).
- (5) Locally make tail rotor gear box cover, 1730BH025-4, and install over tail rotor gear box (Figure H-190, Appendix H).

1-6-5.1.2. Remove.

- a. Remove cover from oil cooler, IR jammer, if installed, and screens (Figure 1-6-5-1).
- b. Remove plug from APU and engine exhaust ducts.
- c. **W/O HIRSS** Remove plugs from engine exhaust ducts.
- d. **HIRSS** Remove covers from left and right HIRSS, engine exhaust inlet, and engine exhaust plug.
- e. Remove plug from engine air inlets.
- f. Remove covers from pitot tubes.

NOTE

Use extreme care while removing covers.

- g. If installed, remove the following covers:
 - (1) Main rotor blade covers.
 - (2) Tail rotor blade covers.
 - (3) Main rotor head cover.
 - (4) Mast cover.
 - (5) Tail rotor gear box cover.

1-6-5-2

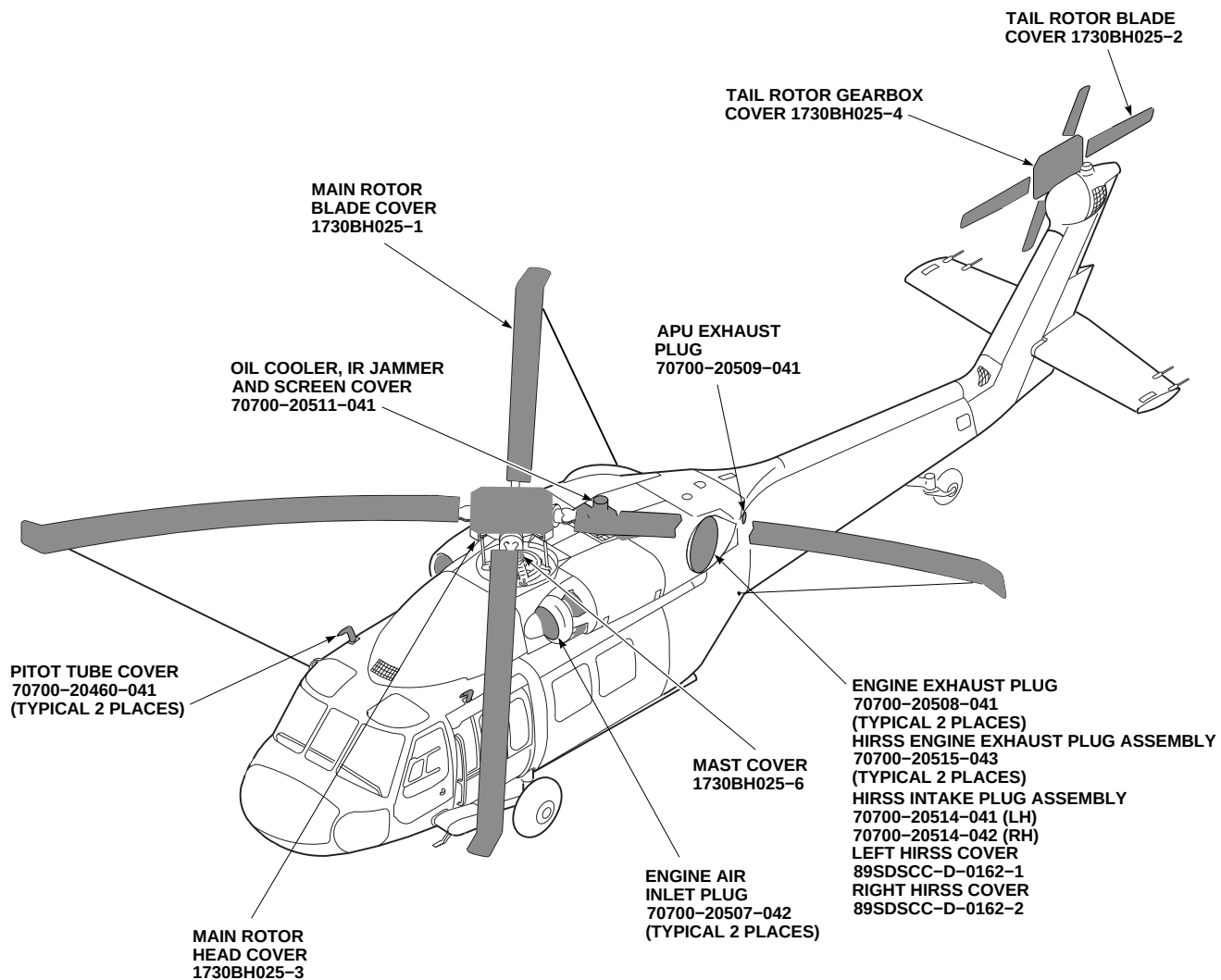


FIGURE 1-6-5-1. PROTECTIVE COVERS

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1-6-6 PARK HELICOPTER.

PARAGRAPH BREAKDOWN

	PAGE
1-6-6.1 Park Helicopter	1-6-6-1

INITIAL SETUP

- PARA 1-6-7

Tools

- PARA 1-6-18

- Aircraft Mechanic’s Toolkit
- Chocks

References

- PARA 1-6-4
- PARA 1-6-5

1-6-6.1 Park Helicopter.

- a. Make sure stabilator is in neutral position, 0°.

1-6-6.1.1 General.

- b. Engage gust lock (PARA 1-6-18).

- a. Locate helicopter more than rotor blade distance from other helicopters or objects.
- b. Park helicopter for ease of maintenance and servicing helicopter.

- c. Center tail wheel and place tail wheel lockpin in ENGAGED position.
- d. Depress toe brake pedals and pull up on parking brake handle to lock parking brakes. Chock main wheels.

1-6-6.1.2 Short-Term Parking.

NOTE

- e. Install protective covers (PARA 1-6-5).
- f. Connect low-resistance ground wire to static ground wire on left main landing gear.

- During short-term parking, helicopter should be attended or monitored at all times.
- With winds 20 knots or more, helicopter shall face into wind.
- You may use battery power to position stabilator. If battery low light is on, battery power will be available only for APU starting, and external power should be used to operate stabilator.

1-6-6.1.3 Long-Term Parking.

NOTE

During long-term parking, the helicopter will not be attended or monitored.

- a. Park helicopter same as short-term parking.

NOTE

If 60-knot winds or more are expected, main rotor blades must be folded.

- b. If necessary, fold main rotor blades (PARA 1-6-7).
- c. Tie down helicopter (PARA 1-6-4).

1-6-7 FOLD/SPREAD MAIN ROTOR BLADES.

PARAGRAPH BREAKDOWN

		PAGE
1-6-7.1	Fold/Spread Main Rotor Blades	1-6-7-2

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Bracket Link, 70700-20415-041
- Bracket Link, 70700-20415-042
- **W/O HIRSS** Flyaway Equipment Kit, 70700-10000-011, consisting of:
 - APU Exhaust Plug, 70700-20509-041
 - Blade Fold Set, 70700-20412-011, consisting of:
 - Alignment Tool, 70700-20394-111
 - Blade Fold Bracket, 70700-20416-042
 - Blade Fold Support, 70700-20417-041
 - Blade Fold Support, 70700-20417-042
 - Blade Handling Pole, 70700-20395-041
 - Retainer, 70700-20428-041
 - Retainer, 70700-20428-042
 - Support Poles, 70700-20427-041 through 70700-20427-044
 - Engine Air Inlet Plug, 70700-20507-042
 - Engine Exhaust Plug, 70700-20508-041
 - Lock Assembly, 70700-20369-041
 - Oil Cooler, IR Jammer, Screen Cover, 70700-20511-041
 - Pitot Tube Cover, 70700-20460-041
- **HIRSS** Flyaway Equipment Kit, 70700-10000-014 (HIRSS), consisting of:
 - APU Exhaust Plug, 70700-20509-041
 - Blade Fold Set, 70700-20412-011, consisting of:
 - Alignment Tool, 70700-20394-111
 - Blade Fold Support, 70700-20417-041
 - Blade Fold Support, 70700-20417-042
 - Blade Handling Pole, 70700-20395-041
 - Retainer, 70700-20428-041
 - Retainer, 70700-20428-042
 - Support Poles, 70700-20427-041 through 70700-20427-044
 - Engine Air Inlet Plug, 70700-20507-042
 - Engine Exhaust Plug, 70700-20508-041
 - Lock Assembly, 70700-20369-041
 - Oil Cooler, IR Jammer, Screen Cover, 70700-20511-041
 - Pitot Tube Cover, 70700-20460-041
- Grease Pencil
- Main Rotor Blade Holder, Locally-Made, Figure H-193, Appendix H (Alternate for Blade Handling Pole)
- Maintenance Platform (Alternate for Small Workstand)
- Padding
- Pin, 70700-20394-045
- Protective Covering
- Small Workstand (Alternate for Maintenance Platform)

References

- Appendix H
- PARA 1-6-18
- PARA 1-7-42
- PARA 5-3-9
- PARA 5-4-5
- PARA 5-4-18
- PARA 5-4-31
- PARA 12-4-26

1-6-7.1 Fold/Spread Main Rotor Blades.

WARNING

- Make sure there is no equipment or personnel in path of folding blades.
- Do not release rotor brake or gust lock while main rotor blades are folded.
- To prevent injury to personnel or damage to equipment, do not block swash-plate during blade folding.

CAUTION

- To prevent damage to equipment, do not start engines or pressurize hydraulic systems while main rotor blades are folded.
- When folding/spreading any blade, watch for possible interference between main rotor blade pins of another blade and blade being folded.
- When folding blade, make sure trim tab does not come in contact with other blades.
- To prevent damage to trim tab, do not support blade in trim tab area when using main rotor blade holder.

NOTE

- If pylon is to be folded, fold main rotor blades first.

- If any blade is dropped during fold/spread sequence, inspect elastomeric bearing (PARA 5-4-5). If necessary, inspect per PARA 1-7-42.

- Main rotor blade holder (Figure H-193, Appendix H) may be used in place of blade handling pole.

1-6-7.1.1 Fold Main Rotor Blades.

- a. If blade deice kit is installed, disconnect electrical connectors from main rotor blades (PARA 12-4-26).

WARNING

To prevent injury to personnel or damage to equipment, pitch control rods must be disconnected prior to starting blade fold procedures.

- b. Disconnect pitch control rods from spindle horns (PARA 5-4-18).
- c. Reinstall hardware in rod ends.

CAUTION

To prevent damage to rod ends and/or main shaft extension, use suitable protective covering when securing rod ends.

- d. Install protective covering and secure rod ends to main shaft extension.

- e. Attach retainer, 70700-20428-041, on left and right side at STA 585.0 (Figure 1-6-7-1, Sheet 1, Detail B).
- f. Attach retainer, 70700-20428-042, on left and right side at STA 624.0.
- g. Turn main rotor head until yellow index marks on rotating and stationary swashplates line up. If main rotor head is without yellow index marks, perform the following:
 - (1) Line up red index marks.
 - (2) Mark rotating and stationary swashplate with grease pencil (Figure 1-6-7-1, Sheet 1, Detail A).
 - (3) Line up grease pencil marks.
- h. Turn main rotor head 360° as many times as required to have inboard tail rotor blade approximately horizontal so that it does not extend above top of tail rotor pylon. Engage gust lock or rotor brake (PARA 1-6-18).

NOTE

- Main rotor blades may be secured by two different kinds of pins, main rotor blade expandable pins, 70103-08107-102, or main rotor blade solid pins, 70103-08108-041. In the following procedures both types of pins are referred to as main rotor blade pins, except where noted.
- i. Unlock all main rotor blade pins.
 - j. Fold red color-coded blade as follows:

WARNING

- Main rotor blade pins contain critical surfaces which must not be damaged during and after removal/installation of pin. Use extreme caution when removing/installing and provide protective covering for pins after removal, if not immediately installed in spindle.

NOTE

- If using blade handling pole, make sure pivot at upper fitting of pole is oriented so as not to pry fitting from blade. Rotate fitting before pulling on pole.
 - Lead and lag the main rotor blade gently to ease main rotor blade pin removal and installation.
 - On helicopters with main rotor blade expandable pins, pins must be installed in the same hole from which they were removed. Note and record pin hole and blade location of all pins removed.
- (1) With aid of assistants, connect blade handling pole to blade lock receiver near end of blade. Make sure lock is fully engaged. If main rotor blade holder is used, slide over end of blade (Figure 1-6-7-1, Sheet 1).
 - (2) Relieve load on main rotor blade pins on main rotor sleeve by lifting blade tip while pulling up on pin.
 - (3) Remove leading edge main rotor blade pin from blade cuff and main rotor sleeve. If main rotor blade pin is stuck, remove stuck pin (PARA 5-4-31).
 - (4) Using blade handling pole, walk blade forward over nose of helicopter.
 - (5) Install blade fold bracket, 70700-20416-042, between blade cuff with pin, 70700-20394-045, and sleeve with main rotor blade pin (Figure 1-6-7-1, Sheet 2, Detail D).
 - (6) Raise or lower the blade to line up pin holes. Alignment tool may be used to line up holes when installing pins.
 - (7) Remove trailing edge main rotor blade pin. If main rotor blade pin is stuck, remove stuck pin (PARA 5-4-31).
 - (8) Walk blade to rear of helicopter until fold bracket link, 70700-20415-042, can be in-

stalled between sleeve with main rotor blade pin, and fold bracket with quick-release pin.

- (9) Continue walking blade to rear of helicopter until blade is above front retainer, 70700-20428-041.

NOTE

When installing support pole, make sure pivot at upper fitting of pole is oriented so as not to pry fitting from blade. Rotate fitting before pulling on pole.

- (10) Using maintenance platform, have assistant manually hold blade at tip while blade handling pole or main rotor blade holder is removed. Immediately install support pole, 70700-20427-041, color-coded red, in blade lock receiver (Figure 1-6-7-1, Sheet 2, Detail E).

NOTE

Blade may be guided by assistant on maintenance platform/small workstand.

- (11) Place ball on lower end of support pole in socket of retainer, 70700-20428-041. Hold ball in socket by installing quick-release pin.
- (12) Attach fork on lower end of support pole to rod end on retainer, using quick-release pin.

- k. Fold blue color-coded blade as follows:

WARNING

- Main rotor blade pins contain critical surfaces which must not be damaged during and after removal/installation of pin. Use extreme caution when removing/installing and provide protective covering for pins after removal, if not immediately installed in spindle.

NOTE

- If using blade handling pole, make sure pivot at upper fitting of pole is oriented so as not to pry fitting from blade. Rotate fitting before pulling on pole.
 - Lead and lag main rotor blade gently to ease main rotor blade pin removal on installation and removal.
 - On helicopters with main rotor blade expandable pins, pins must be installed in the same hole from which they were removed. Note and record pin hole and blade location of all pins removed.
- (1) With aid of assistants, connect blade handling pole to blade lock receiver near end of blade. Make sure lock is fully engaged. If main rotor blade holder is used, slide over end of blade (Figure 1-6-7-1, Sheet 1).
 - (2) Relieve load on main rotor blade pins on main rotor sleeve by lifting blade tip while pulling up on pin.
 - (3) Remove trailing edge main rotor blade pin. If main rotor blade pin is stuck, remove stuck pin (PARA 5-4-31).
 - (4) Using blade handling pole, walk blade back until it is above support fitting on tail cone.

NOTE

When installing support pole, make sure pivot at upper fitting of pole is oriented so as not to pry fitting from blade. Rotate fitting before pulling on pole.

- (5) Using maintenance platform or small workstand, have assistant manually hold blade while blade handling pole or main rotor blade holder is removed. Immediately install support pole, 70700-20427-043, color-coded blue, in blade lock receiver (Figure 1-6-7-1, Sheet 3, Detail F).

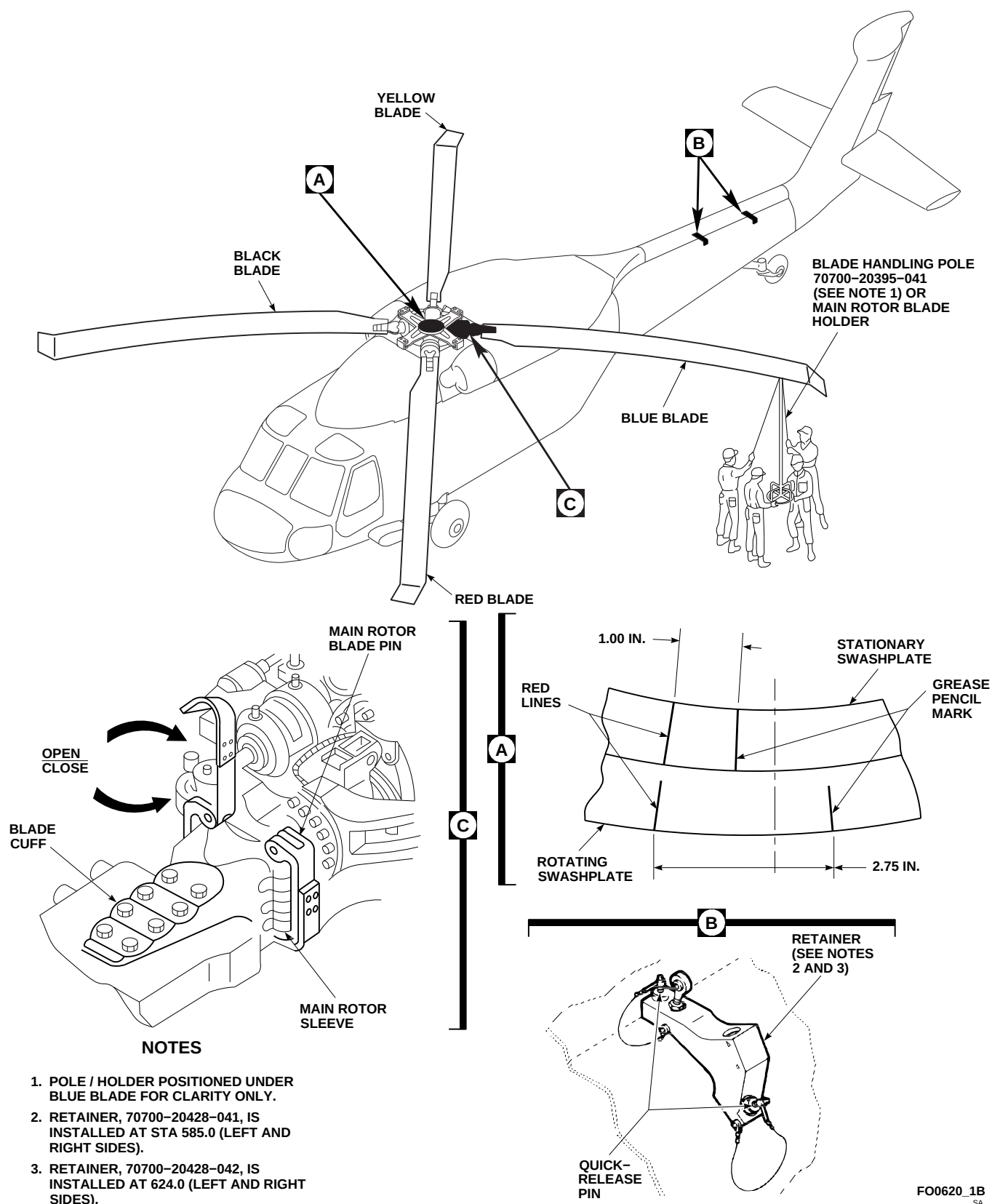


FIGURE 1-6-7-1. FOLD/SPREAD MAIN ROTOR BLADES (SHEET 1 OF 7)

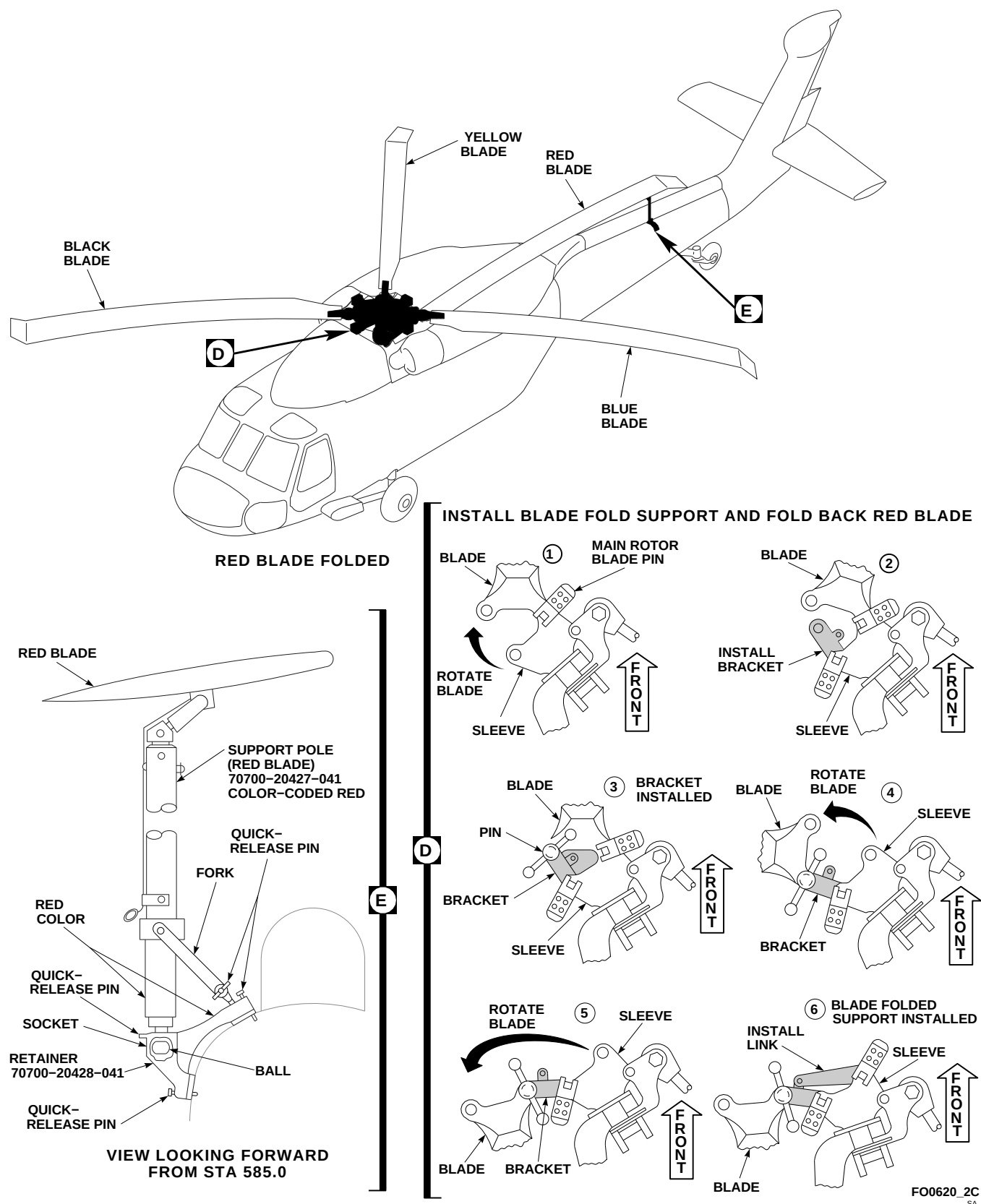
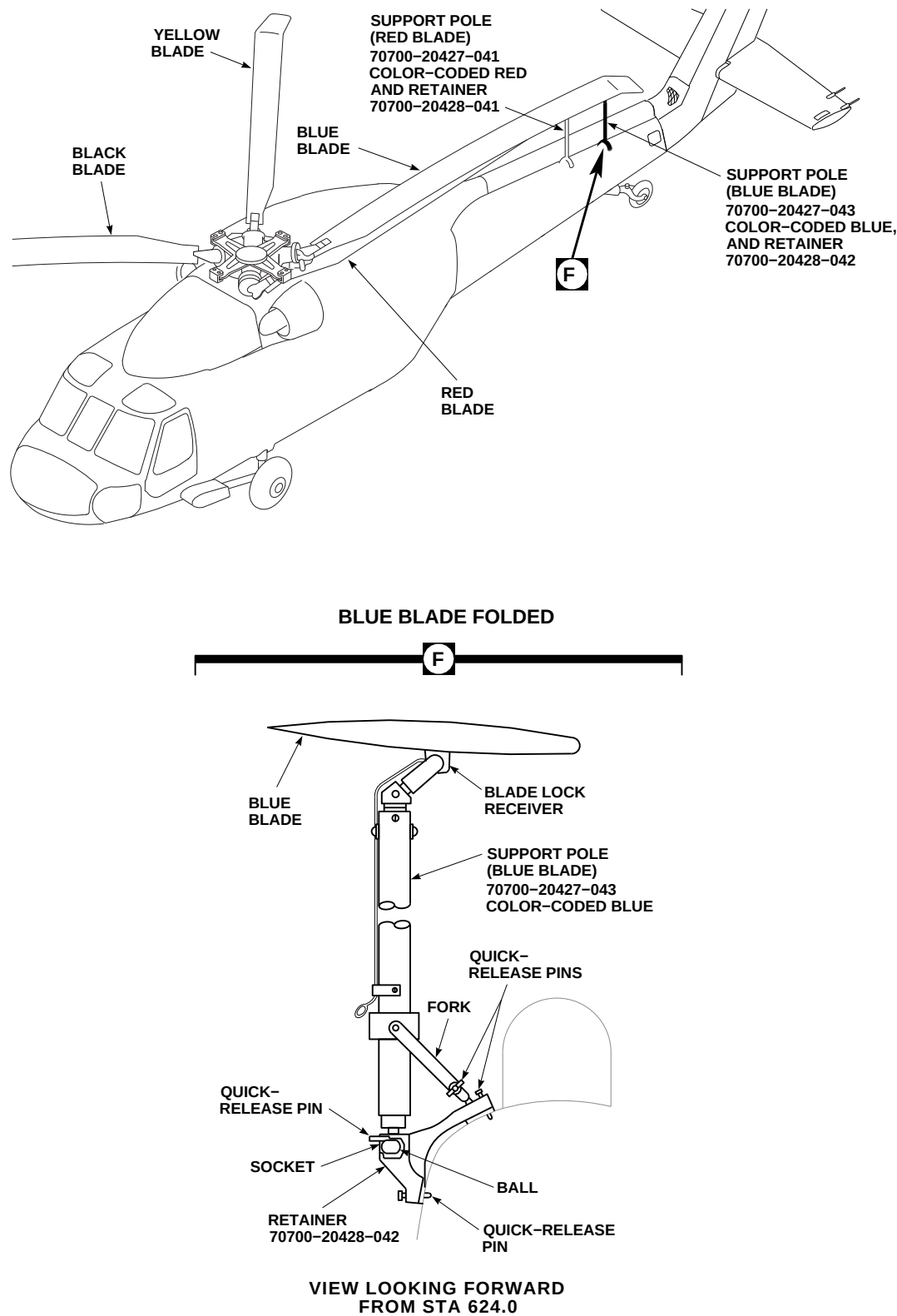


FIGURE 1-6-7-1. FOLD/SPREAD MAIN ROTOR BLADES (SHEET 2 OF 7)



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FIGURE 1-6-7-1. FOLD/SPREAD MAIN ROTOR BLADES (SHEET 3 OF 7)

NOTE

Blade may be guided by assistant on maintenance platform/small workstand.

- (6) Place ball on lower end of support pole in socket of retainer, 70700-20428-042. Hold ball in socket by installing quick-release pin.
- (7) Using quick-release pin, attach fork on lower end of support pole to rod end of retainer.
- (8) Install padding between red and blue blades.

1. Fold black color-coded blade as follows:

WARNING

- Main rotor blade pins contain critical surfaces which must not be damaged during and after removal/installation of pin. Use extreme caution when removing/installing and provide protective covering for pins after removal, if not immediately installed in spindle.

NOTE

- When installing blade handling pole, make sure pivot at upper fitting of pole is oriented so as not to pry fitting from blade. Rotate fitting before pulling on pole.
- Lead and lag main rotor blade gently to ease main rotor blade pin removal on installation and removal.
- On helicopters with main rotor blade expandable pins, pins must be installed in the same hole from which they were removed. Note and record pin hole and blade location of all pins removed.

- (1) Install blade handling pole in blade lock receiver near end of blade. Make sure

lock is fully engaged. If main rotor blade holder is used, slide over end of blade (Figure 1-6-7-1, Sheet 1).

- (2) Relieve load on main rotor blade pins on main rotor sleeve by lifting blade tip while pulling up on pin.
- (3) Remove trailing edge main rotor blade pin. If main rotor blade pin is stuck, remove stuck pin (PARA 5-4-31).
- (4) Walk blade forward over nose of helicopter and install blade fold bracket, 70700-20416-041 (part of blade fold support, 70700-20417-041), in blade cuff with pin, 70700-20394-045, and sleeve, using main rotor blade pin (Figure 1-6-7-1, Sheet 4, and Detail G).
- (5) Remove leading edge main rotor blade pin. If main rotor blade pin is stuck, remove stuck pin (PARA 5-4-31).
- (6) Walk blade to rear of helicopter until fold bracket link, 70700-20415-041, can be installed between sleeve with main rotor blade pin, and fold bracket, using quick-release pin. Continue walking blade to rear of helicopter.

NOTE

When installing support pole, make sure pivot at upper fitting of pole is oriented so as not to pry fitting from blade. Rotate fitting before pulling on pole.

- (7) Using maintenance platform or small workstand, have assistant manually hold blade while blade handling pole or main rotor blade holder is removed. Immediately install support pole, 70700-20427-042, color-coded black, in blade lock receiver (Figure 1-6-7-1, Sheet 4, Detail H).

NOTE

Blade may be guided by assistant on maintenance platform/small workstand.

- (8) Place ball on lower end of support pole in socket of retainer, 70700-20428-041.

Hold ball in socket by installing quick-release pin.

- (9) Using quick-release pin, attach fork on lower end of support pole to rod end on retainer.

m. Fold yellow color-coded blade as follows:

WARNING

- Main rotor blade pins contain critical surfaces which must not be damaged during and after removal/installation of pin. Use extreme caution when removing/installing and provide protective covering for pins after removal, if not immediately installed in spindle.
- To prevent damage to trim tab, do not support blade in trim tab area when using main rotor blade holder.

NOTE

- If using blade handling pole, make sure pivot at upper fitting of pole is oriented so as not to pry fitting from blade. Rotate fitting before pulling on pole.
 - Lead and lag main rotor blade gently to ease main rotor blade pin removal on installation and removal.
 - On helicopters with main rotor blade expandable pins, pins must be installed in the same hole from which they were removed. Note and record pin hole and blade location of all pins removed.
- (1) Install blade handling pole in blade lock receiver near end of blade. Make sure

lock is fully engaged. If main rotor blade holder is used, slide over end of blade (Figure 1-6-7-1, Sheet 1).

- (2) Relieve load on main rotor blade pins on main rotor sleeve by lifting blade tip while pulling up on pin.
- (3) Remove leading edge main rotor blade pin. If main rotor blade pin is stuck, remove stuck pin (PARA 5-4-31).
- (4) Using blade handling pole, walk blade to rear of helicopter until blade is above rear retainer, 70700-20428-042, on tail cone.

NOTE

When installing support pole, make sure pivot at upper fitting of pole is oriented so as not to pry fitting from blade. Rotate fitting before pulling on pole.

- (5) Using maintenance platform or small workstand, have assistant manually hold blade while blade handling pole or main rotor blade holder is removed. Immediately install support pole, 70700-20427-044, in blade lock receiver (Figure 1-6-7-1, Sheet 5, Detail J).

NOTE

Blade may be guided by assistant on maintenance platform/small workstand.

- (6) Place ball on lower end of support pole in socket of retainer. Hold ball in socket by installing quick-release pin.
- (7) Using quick-release pin, attach fork on lower end of support pole to rod end on retainer.
- (8) Install padding between yellow and black blades.

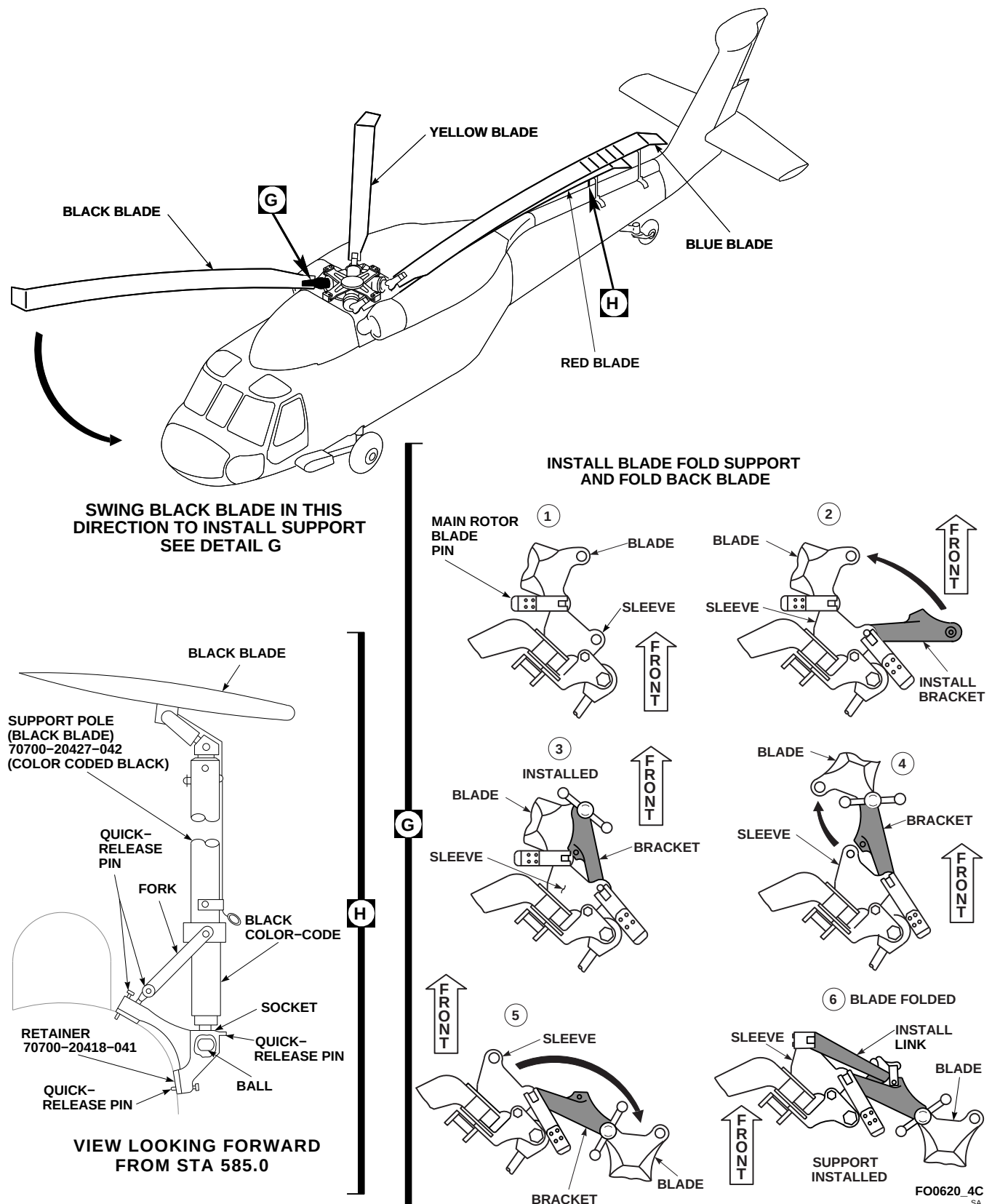


FIGURE 1-6-7-1. FOLD/SPREAD MAIN ROTOR BLADES (SHEET 4 OF 7)

1-6-7-10 Change 10

This Document contains technical data subject to ITAR. See WARNING and classifications on first page.

1-6-7.1.2 Spread Main Rotor Blades.**WARNING**

- Main rotor blade pins contain critical surfaces which must not be damaged during and after removal/installation of pin. Use extreme caution when removing/installing and provide protective covering for pins after removal, if not immediately installed in spindle.

NOTE

- Lead and lag the main rotor blade gently to ease main rotor blade pin installation.
 - Main rotor blades may be secured by two different kinds of pins, main rotor blade expandable pins, 70103-08107-102, or main rotor blade solid pins, 70103-08108-041. In the following procedures, both types of pins are referred to as main rotor blade pins, except where noted.
 - On helicopters with main rotor blade expandable pins, pins must be installed in the same hole noted and recorded during fold procedure.
- a. Spread yellow color-coded blade as follows:
- (1) Remove padding from between blades.
 - (2) Using maintenance platform or small workstand, have assistant manually hold blade.
 - (3) Remove quick-release pins from retainer, 70700-20428-042 (Figure 1-6-7-1, Sheet 5, Detail J).

NOTE

If using blade handling pole, make sure pivot at upper fitting of pole is oriented so as not to pry fitting from blade. Rotate fitting before pulling on pole.

- (4) Remove yellow color-coded support pole, 70700-20427-044, from blade and from retainer. Immediately install blade handling pole in blade lock receiver near end of blade. Make sure lock is fully engaged. If main rotor blade holder is used, slide over end of blade.
- (5) Using blade handling pole, walk blade forward until main rotor blade expandable pin holes line up, and blade is in flight position.
- (6) Inspect main rotor blade pins (PARA 5-3-9).

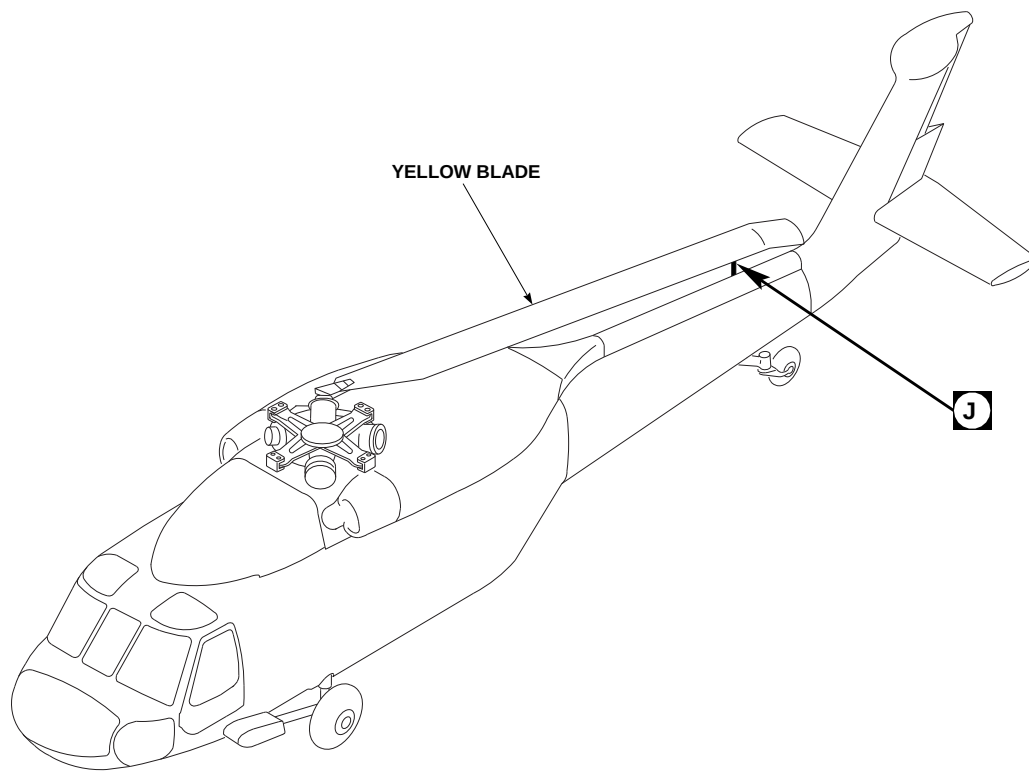
WARNING

- Do not intermix expandable pins with solid pins on same main rotor blade. Intermixing expandable and solid pins on same blade may cause fatigue damage to spindle, cuff, and main rotor blade pins.
 - It is permissible to use expandable pins or solid pins on any main rotor blade, as long as both pins on any one blade are the same.
- (7) Insert main rotor blade pins through blade/spindle cuff lugs.
 - (a) Align pin handle with decal on cuff.

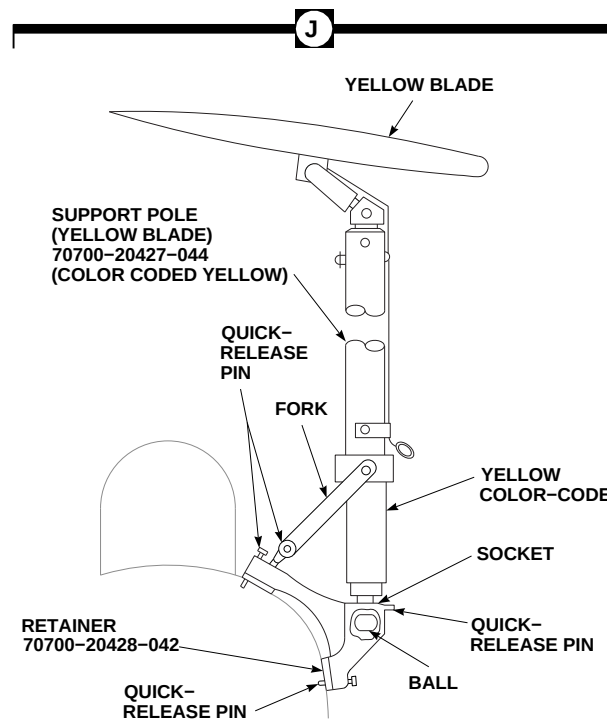
NOTE

Do not hit or force main rotor pins into position.

- (b) Gently lead/lag blade to get proper alignment for main rotor blade pin to drop in.
- (8) If installing main rotor blade expandable pins, perform spring clip/handle tension check (PARA 5-3-9).
 - (9) Close and lock main rotor blade pin handles as follows:



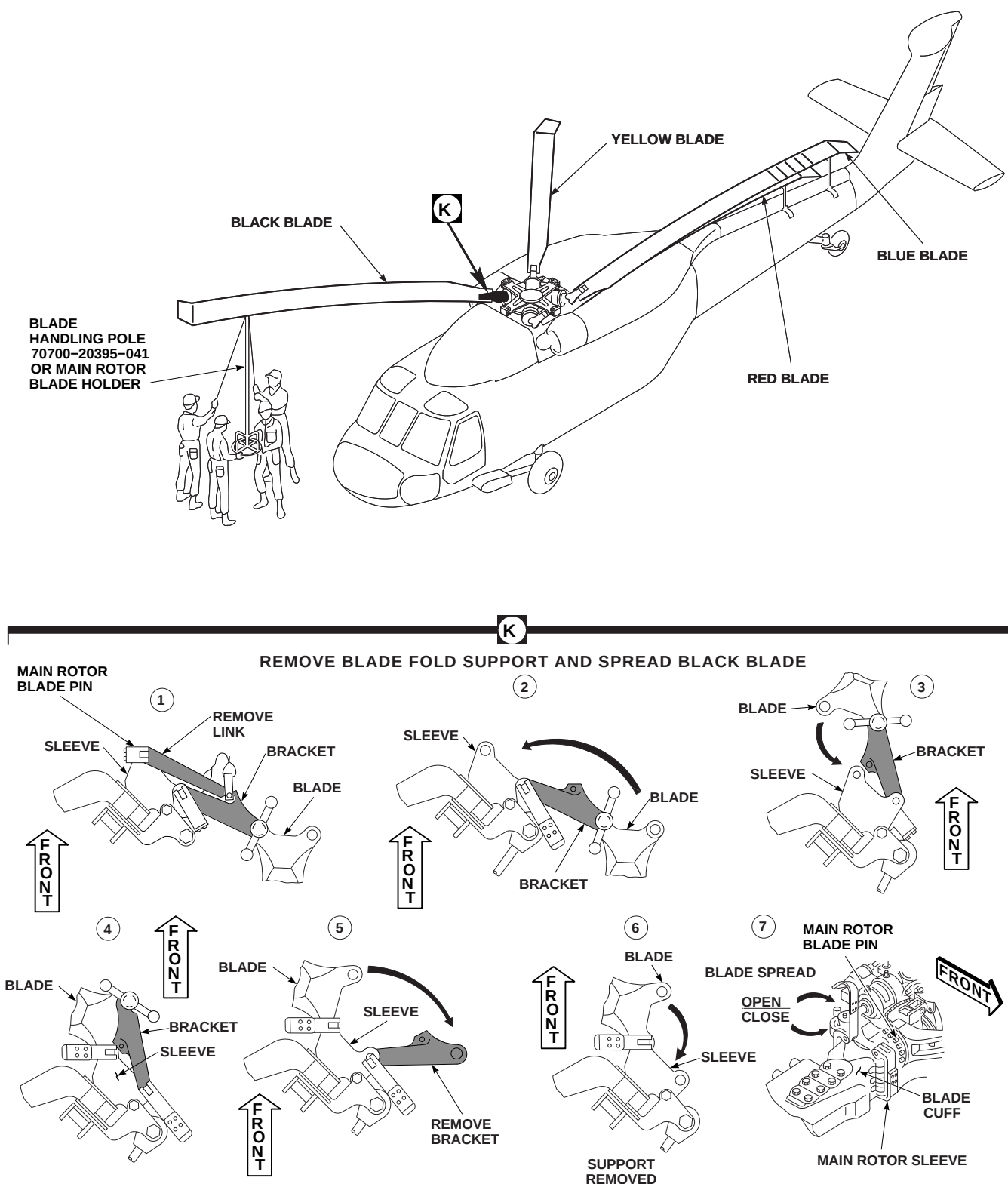
**YELLOW BLADE FOLDED
RED, BLUE, AND BLACK BLADES NOT SHOWN**



**VIEW LOOKING FORWARD
FROM STA 624.0**

FO0620_5B
SA

FIGURE 1-6-7-1. FOLD/SPREAD MAIN ROTOR BLADES (SHEET 5 OF 7)



FO0620_6B
SA

FIGURE 1-6-7-1. FOLD/SPREAD MAIN ROTOR BLADES (SHEET 6 OF 7)

- (a) On helicopters with main rotor blade expandable pins, close handle, making sure spring handle of pin snaps over fully and engages hex on nut (Figure 1-6-7-1, Sheet 1, Detail C). Visually check for expansion of lockring on end of pin. Make sure slots in spring clip line up with points of nut. Repeat for other main rotor blade expandable pin.
 - (b) On helicopters with main rotor blade solid pins, close handle, making sure retaining clip fork engages groove in pin and that bottom of pin protrudes through hole in spring clip. Install cotter pin from side of pin closest to main rotor hub (Figure 1-6-7-1, Sheet 1, Detail C). Repeat for other main rotor blade solid pin.
- (10) Remove blade handling pole or main rotor blade holder from blade.

WARNING

- Main rotor blade pins contain critical surfaces which must not be damaged during and after removal/installation of pin. Use extreme caution when removing/installing and provide protective covering for pins after removal, if not immediately installed in spindle.

NOTE

- Lead and lag the main rotor blade gently to ease main rotor blade pin installation.
- Main rotor blades may be secured by two different kinds of pins, main rotor blade expandable pins, 70103-08107-102, or main rotor blade solid pins, 70103-08108-041. In the following procedures both types of pins are referred to as main rotor blade pins, except where noted.

- On helicopters with main rotor blade expandable pins, pins must be installed in the same hole noted and recorded during fold procedure.
- b. Spread black color-coded blade as follows:
- (1) Using maintenance platform or small workstand, have assistant hold blade.
 - (2) Remove quick-release pins from retainer, 70700-20428-041, holding black color-coded support pole, 70700-20427-042, in place (Figure 1-6-7-1, Sheet 4, Detail H).

NOTE

When installing blade handling pole, make sure pivot at upper fitting of pole is oriented so as not to pry fitting from blade. Rotate fitting before pulling on pole.

- (3) Remove support pole from blade and from retainer. Immediately install blade handling pole in blade lock receiver near end of blade. Make sure lock is fully engaged. If main rotor blade holder is used, slide over end of blade.
- (4) Using blade handling pole, walk blade forward, remove fold bracket link, 70700-20415-041, from fold bracket, 70700-20416-041, and sleeve. Continue walking blade forward over nose of helicopter until leading edge of blade cuff can be secured to sleeve with a main rotor blade pin. If necessary, lift blade to install pin. Do not lock pin.
- (5) Remove bracket from blade and sleeve, walk blade to rear of helicopter until trailing edge of blade cuff can be secured to sleeve with an expandable pin. If necessary, lift blade to install pin. Close and lock both main rotor blade pins.
- (6) Inspect main rotor blade pins (PARA 5-3-9).

WARNING

- Do not intermix expandable pins with solid pins on same main rotor blade. Intermixing expandable and solid pins on same blade may cause fatigue damage to spindle, cuff, and main rotor blade pins.
 - It is permissible to use expandable pins or solid pins on any main rotor blade, as long as both pins on any one blade are the same.
- (7) Insert main rotor blade pins through blade/spindle cuff lugs.
 - (a) Align pin handle with decal on cuff.

NOTE

Do not hit or force main rotor pins into position.

- (b) Gently lead/lag blade to get proper alignment for main rotor blade pin to drop in.
- (8) If installing main rotor blade expandable pins, perform spring clip/handle tension check (PARA 5-3-9).
- (9) Close and lock main rotor blade pin handles as follows:
 - (a) On helicopters with main rotor blade expandable pins, close handle, making sure spring handle of pin snaps over fully and engages hex on nut (Figure 1-6-7-1, Sheet 1, Detail C). Visually check for expansion of lockring on end of pin. Make sure slots in spring clip line up with points of nut. Repeat for other main rotor blade expandable pin.
 - (b) On helicopters with main rotor blade solid pins, close handle, making sure retaining clip fork engages groove in

pin and that bottom of pin protrudes through hole in spring clip. Install cotter pin from side of pin closest to main rotor hub (Figure 1-6-7-1, Sheet 1, Detail C). Repeat for other main rotor blade solid pin.

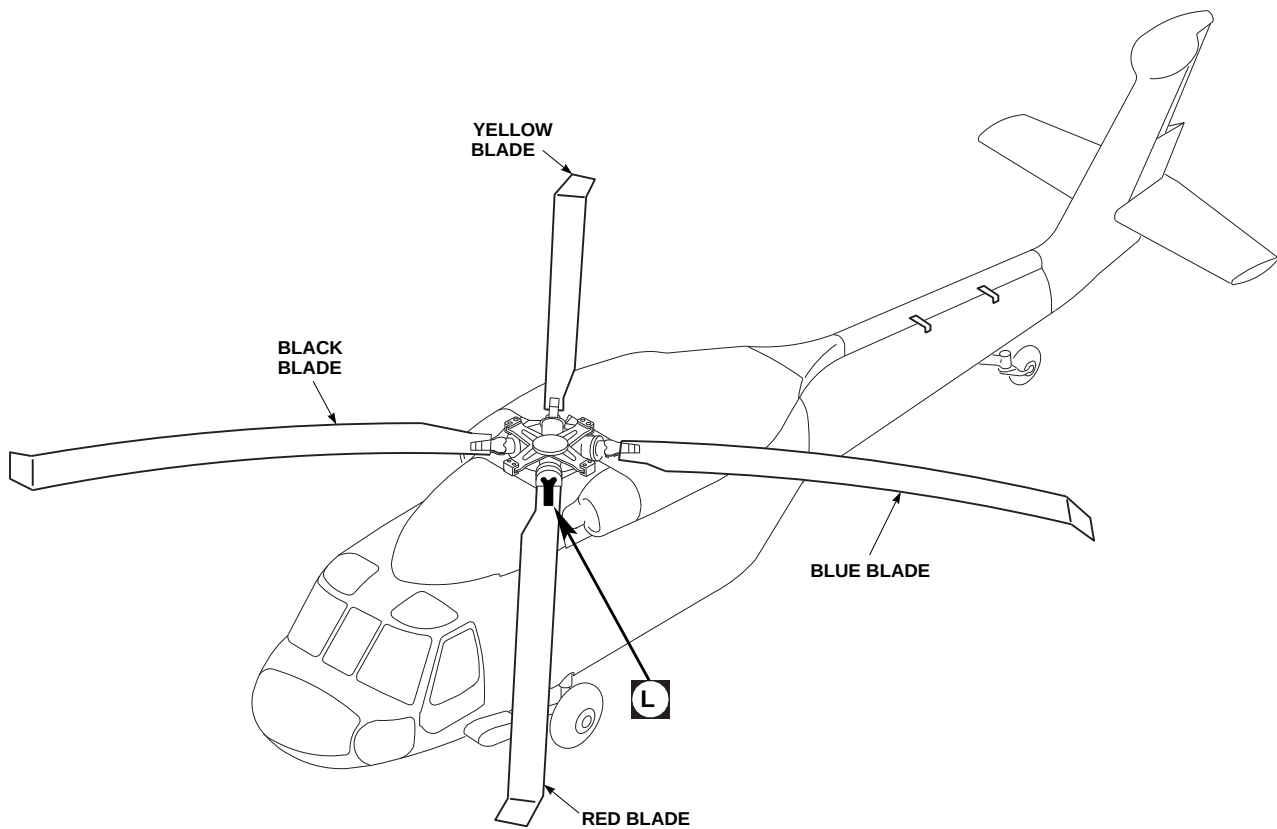
- (10) Remove blade handling pole or main rotor blade holder from blade.

WARNING

- Main rotor blade pins contain critical surfaces which must not be damaged during and after removal/installation of pin. Use extreme caution when removing/installing and provide protective covering for pins after removal, if not immediately installed in spindle.
- To prevent damage to trim tab, do not support blade in trim tab area when using main rotor blade holder.

NOTE

- Lead and lag the main rotor blade gently to ease main rotor blade pin installation.
 - Main rotor blades may be secured by two different kinds of pins, main rotor blade expandable pins, 70103-08107-102, or main rotor blade solid pins, 70103-08108-041. In the following procedures, both types of pins are referred to as main rotor blade pins, except where noted.
- c. Spread blue color-coded blade as follows:
 - (1) Remove padding from between blades.
 - (2) Using maintenance platform or small workstand, have assistant hold blade.
 - (3) Remove quick-release pins from retainer, 70700-20428-042, holding support pole, 70700-20427-043, color-coded blue (Figure 1-6-7-1, Sheet 3, Detail F).



L

REMOVE BLADE FOLD SUPPORT AND SPREAD RED BLADE

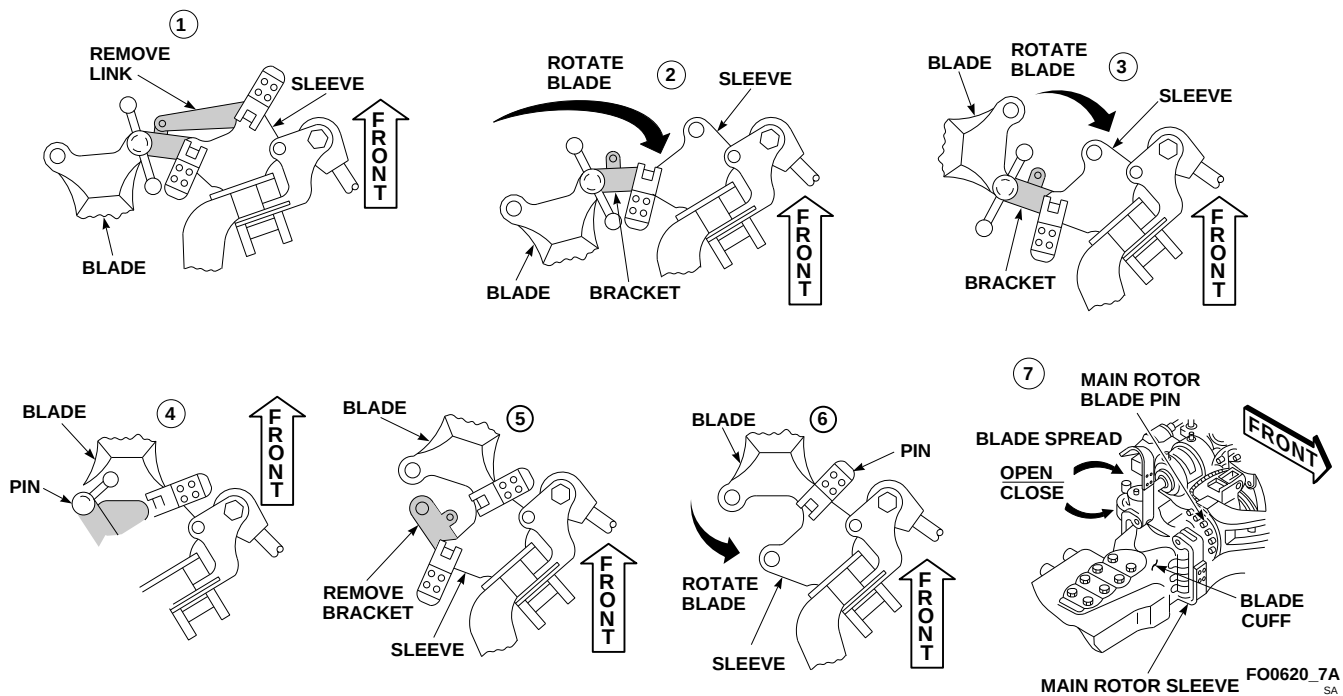


FIGURE 1-6-7-1. FOLD/SPREAD MAIN ROTOR BLADES (SHEET 7 OF 7)

NOTE

When installing blade handling pole, make sure pivot at upper fitting of pole is oriented so as not to pry fitting from blade. Rotate fitting before pulling on pole.

- (4) Remove support pole from blade and retainer. Immediately install blade handling pole in blade lock receiver near end of blade. Make sure lock is fully engaged. If main rotor blade holder is used, slide over end of blade.
- (5) Walk blade forward until leading edge of blade cuff can be secured to sleeve with main rotor blade pin. If necessary, lift blade to install pin.
- (6) Inspect main rotor blade pins (PARA 5-3-9).

WARNING

- The outside surface of blade cuffs are critical surfaces which must not be damaged during blade fold/spread procedure. Use extreme caution when folding/spreading main rotor blades.
 - Do not intermix expandable pins with solid pins on same main rotor blade. Intermixing expandable and solid pins on same blade may cause fatigue damage to spindle, cuff, and main rotor blade pins.
 - It is permissible to use expandable pins or solid pins on any main rotor blade as long as both pins on any one blade are the same.
- (7) Insert main rotor blade pins through blade/spindle cuff lugs.
 - (a) Align pin handle with decal on cuff.

NOTE

Do not hit or force main rotor pins into position.

- (b) Gently lead/lag blade to get proper alignment for main rotor blade pin to drop in.

- (8) If installing main rotor blade expandable pins, perform spring clip/handle tension check (PARA 5-3-9).
- (9) Close and lock main rotor blade pin handles as follows:
 - (a) On helicopters with main rotor blade expandable pins, close handle, making sure spring handle of pin snaps over fully and engages hex on nut (Figure 1-6-7-1, Sheet 1, Detail C). Visually check for expansion of locking on end of pin. Make sure slots in spring clip line up with points of nut. Repeat for other main rotor blade expandable pin.
 - (b) On helicopters with main rotor blade solid pins, close handle, making sure retaining clip fork engages groove in pin and that bottom of pin protrudes through hole in spring clip. Install cotter pin from side of pin closest to main rotor hub (Figure 1-6-7-1, Sheet 1, Detail C). Repeat for other main rotor blade solid pin.
- (10) Remove blade handling pole or main rotor blade holder from blade.

WARNING

- Main rotor blade pins contain critical surfaces which must not be damaged during and after removal/installation of pin. Use extreme caution when removing/installing and provide protective covering for pins after removal, if not immediately installed in spindle.

NOTE

- Lead and lag the main rotor blade gently to ease main rotor blade pin installation.

- Main rotor blades may be secured by two different kinds of pins, main rotor blade expandable pins, 70103-08107-104, or main rotor blade solid pins, 70103-08108-041. In the following procedures both types of pins are referred to as main rotor blade pins, except where noted.
 - On helicopters with main rotor blade expandable pins, pins must be installed in the same hole noted and recorded during fold procedure.
- d. Spread red color-coded blade as follows:
- (1) Using maintenance platform or small workstand, have assistant hold blade.
 - (2) Remove quick-release pins from retainer, 70700-20428-041, holding support pole, 70700-20427-041, color-coded red (Figure 1-6-7-1, Sheet 2, Detail E).

NOTE

When installing blade handling pole, make sure pivot at upper fitting of pole is oriented so as not to pry fitting from blade. Rotate fitting before pulling on pole.

- (3) Remove support pole from blade and retainer. Immediately install blade handling pole in blade lock receiver near end of blade. Make sure lock is fully engaged. If main rotor blade holder is used, slide over end of blade.
- (4) Using blade handling pole, walk blade forward, remove fold bracket link, 70700-20415-042, from fold bracket, 70700-20416-042, and sleeve. Continue walking blade forward over nose of helicopter until leading edge of blade cuff can be secured to sleeve with a main rotor blade pin. If necessary, lift blade to install pin. Do not lock pin.
- (5) Remove bracket from blade and sleeve, walk blade to rear of helicopter until trail-

ing edge of blade cuff can be secured to sleeve with an expandable pin. If necessary, lift blade to install pin.

- (6) Inspect main rotor blade pins (PARA 5-3-9).

WARNING

- The outside surface of blade cuffs are critical surfaces which must not be damaged during blade fold/spread procedure. Use extreme caution when folding/spreading main rotor blades.
- Do not intermix expandable pins with solid pins on same main rotor blade. Intermixing expandable and solid pins on same blade may cause fatigue damage to spindle, cuff, and main rotor blade pins.
- It is permissible to use expandable pins or solid pins on any main rotor blade as long as both pins on any one blade are the same.

CAUTION

On helicopters with main rotor blade expandable pins, pins must be installed in the same hole noted and recorded during blade fold procedure.

- (7) Insert main rotor blade pins through blade/spindle cuff lugs.

- (a) Align pin handle with decal on cuff.

NOTE

Do not hit or force main rotor pins into position.

- (b) Gently lead/lag blade to get proper alignment for main rotor blade pin to drop in.

- (8) If installing main rotor blade expandable pins, perform spring clip/handle tension check (PARA 5-3-9).
- (9) Close and lock main rotor blade pin handles as follows:
 - (a) On helicopters with main rotor blade expandable pins, close handle, making sure spring handle of pin snaps over fully and engages hex on nut (Figure 1-6-7-1, Sheet 1, Detail C). Visually check for expansion of locking on end of pin. Make sure slots in spring clip line up with points of nut. Repeat for other main rotor blade expandable pin.
 - (b) On helicopters with main rotor blade solid pins, close handle, making sure retaining clip fork engages groove in pin and that bottom of pin protrudes through hole in spring clip. Install cotter pin from side of pin closest to main rotor hub (Figure 1-6-7-1, Sheet 1, Detail C). Repeat for other main rotor blade solid pin.
- (10) Remove blade handling pole or main rotor blade holder from blade.
 - e. Remove retainers, 70700-20428-041 and -042, from tail cone (Figure 1-6-7-1, Sheet 1, Detail B).
 - f. If blade deice kit is installed, connect electrical connectors to main rotor blades (PARA 12-4-26).
 - g. Connect pitch control rods to spindle horns (PARA 5-4-18).
 - h. If required, disengage gust lock or rotor brake (PARA 1-6-18).

1-6-8 FOLD/SPREAD TAIL ROTOR BLADES.

PARAGRAPH BREAKDOWN

		PAGE
1-6-8.1	Fold/Spread Tail Rotor Blades	1-6-8-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Grease Pencil
- Maintenance Platform
- Protective Covering
- Tail Rotor Blade Spacers, 70700-20453-041

Materials/Parts

- Tiedown Strap, Item 333, Appendix D

References

- Appendix D
- Appropriate MTF
- PARA 1-6-9
- PARA 5-4-38
- PARA 5-4-39
- PARA 12-4-52

1-6-8.1 Fold/Spread Tail Rotor Blades.

1-6-8.1.1 Fold Tail Rotor Blades.

WARNING

Injury to personnel and damage to equipment will result if tail rotor pylon is folded to 90° incorrectly.

NOTE

Tail rotor blades should be folded with pylon spread using maintenance platform. If maintenance platform is not available, fold tail rotor pylon 90° using pylon lock strut. The pylon lock strut can then be used as a step for access to tail rotor blades.

- If required, fold tail rotor pylon 90° (PARA 1-6-9).
- If blade deice kit is installed, disconnect tail rotor slipping electrical connectors from tail rotor blade electrical connectors (PARA 12-4-52).

NOTE

Do not change length of pitch control rods.

- Disconnect tail rotor pitch control rods from pitch beam (PARA 5-4-39).
- If color bands have worn off, use grease pencil and put index marks on blades, retention plates, pitch control rods, and pitch beam before separating retention plates, to maintain balance integrity during buildup.

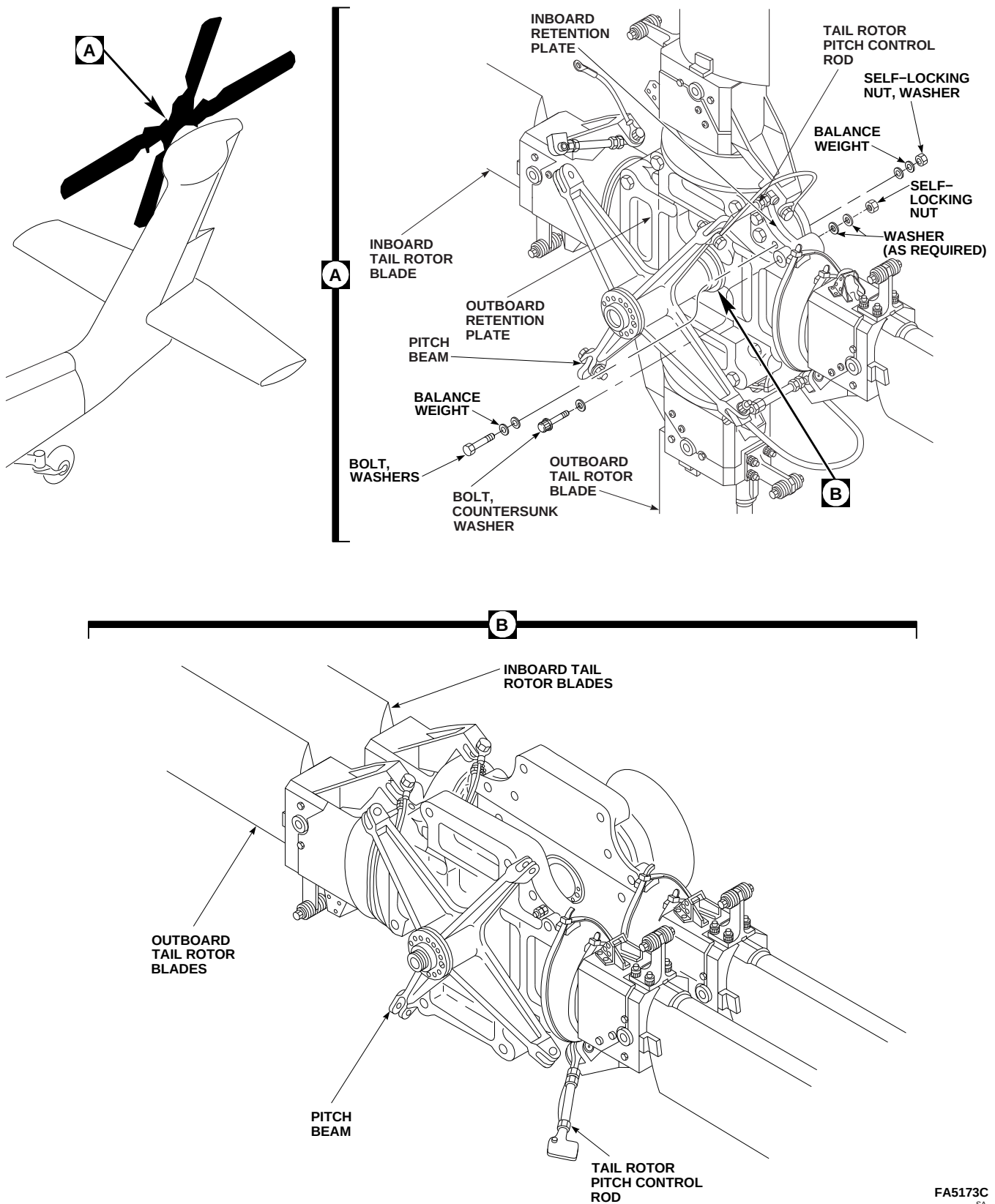


FIGURE 1-6-8-1. FOLD/SPREAD TAIL ROTOR BLADES

NOTE

- Record location and stack-ups of all hardware removed.
- Retention plates may contain hardware for balance purposes. Reinstall hardware in the same location from which they were removed.
- e. Remove outboard retention plate bolts, washers, balance weights, and nuts. Save hardware for use when spreading blades (Figure 1-6-8-1, Detail A).
- f. Slide outboard tail rotor blades towards pitch beam (Figure 1-6-8-1, Detail B).
- g. If blade deice kit is installed, secure slipring rotor, spacers, and shims to inner retention plate with tiedown straps, Item 333, Appendix D.
- h. Turn outboard tail rotor blades clockwise on pitch control shaft until both blades (inboard and outboard) are parallel with each other.
- i. Place tail rotor blade spacer between blades near blade tip, and tie blades together.

CAUTION

To prevent damage to rod ends and/or blades, use suitable protective covering when securing pitch control rods.

- j. Install protective covering and secure loose ends of pitch control rods to pitch beam or blades.

1-6-8.1.2 Spread Tail Rotor Blades.

- a. Remove protective covering and free pitch control rod ends from pitch beam or blades.
- b. Untie blades and remove tail rotor blade spacers.
- c. Turn outboard tail rotor blade counterclockwise on pitch control shaft until blade is in normal flight position.

CAUTION

Make sure color bands or index marks on blades, retention plates, pitch control rods, and pitch beam line up.

- d. Slide outboard blade in toward inboard blade and line up boltholes of inboard and outboard retention plates. Outboard blade should line up with deep slot in outboard retention plate (Figure 1-6-8-1, Detail A).
- e. If blade deice kit is installed, remove tiedown straps securing slipring rotor and hardware in place. Keep spacers and laminated shims next to their holes in retention plate.

NOTE

Retention plates may contain hardware for balance purposes. Reinstall hardware in the same location from which they were removed.

- f. Secure outboard retention plate to inboard retention plate using bolts, washers, balance weights, and nuts (PARA 5-4-38).

NOTE

Do not change lengths of pitch control rods.

- g. Connect tail rotor pitch control rods to pitch beam (PARA 5-4-39).
- h. If folded, unfold tail rotor pylon (PARA 1-6-9).
- i. Connect tail rotor slipring electrical connectors to tail rotor blade electrical connectors (PARA 12-4-52).
- j. If blade deice kit is installed, check for proper clearance between tail rotor slipring rotor and stator (PARA 12-4-52).
- k. Perform a limited maintenance test flight (Appropriate MTF).

1-6-9 FOLD/UNFOLD TAIL ROTOR PYLON.

PARAGRAPH BREAKDOWN

		PAGE
1-6-9.1	Fold/Unfold Tail Rotor Pylon	1-6-9-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Crowfoot Adapter, 7⁄8-inch
- Feeler Gage
- Flight Control Quick-Disconnect Wrench, Locally-Made, Figure H-48, Appendix H
- Fluorescent Penetrant Inspection Kit
- Padding
- Pylon Lock Strut, 70700-20385-041
- Tail Rotor Pylon Strut Assembly, 70700-20448-044
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 700 - 1600 in. lbs
- Vise, Soft-Padded

Materials/Parts

- Bolt, AN9-13A
- Fibrous Cord, Item 139, Appendix D
- Lockwire, Item 197, Appendix D

- Nut, MS21083N9
- Protective Caps and Plugs
- Tiedown Strap, Item 333, Appendix D
- Washer, NAS1149D0963K

References

- Appendix D
- Appendix H
- ASTM E1417
- PARA 1-6-7
- PARA 1-6-8
- PARA 1-7-11
- PARA 1-7-20
- PARA 2-4-139
- PARA 2-4-142
- PARA 2-6-5
- PARA 2-6-42
- PARA 6-4-40
- PARA 11-4-109
- PARA 11-4-118

1-6-9.1 Fold/Unfold Tail Rotor Pylon.**1-6-9.1.1 Fold Tail Rotor Pylon.****NOTE**

- When folding tail rotor pylon, tail rotor blades may or may not be folded.
 - If main rotor blades are to be folded, fold pylon last.
- a. If required, fold main rotor blades (PARA 1-6-7) and/or tail rotor blades (PARA 1-6-8).

NOTE

Stabilator does not have to be removed if pylon is to be folded 90°.

- b. If required, remove stabilator (PARA 2-4-142).
- c. Install flight control cable lockout blocks (PARA 11-4-109). If lockout blocks are not available, disconnect tail rotor quadrant spring cylinders from support (PARA 11-4-118). Tie down cylinders to tail rotor pylon with tiedown straps, Item 333, Appendix D.
- d. Open rear tail rotor drive shaft cover.
- e. Remove intermediate gear box fairing (PARA 2-4-139).
- f. Remove screws and washers securing tail cone flight controls access panel to tail cone at STA 640.0 (Figure 1-6-9-1, Sheet 1, Detail A).

NOTE

Use either crowfoot adapter or locally-made flight control quick-disconnect wrench to disconnect cables.

- g. If necessary, locally make flight control quick-disconnect wrench (Figure H-48, Appendix H).
- h. Back off jamnut securing quick-disconnect and retainer. Separate quick-disconnect and retainer (Figure 1-6-9-1, Sheet 1, Detail B).
- i. Disconnect electrical connectors near intermediate gear box (Figure 1-6-9-1, Sheet 1, Detail C).

- j. Install protective caps and plugs on electrical connectors.
- k. Remove bolts, washers, and nuts connecting tail rotor drive shaft section III to flexible coupling (Figure 1-6-9-1, Sheet 2, Detail D). After removing each bolt, keep washers with that bolt. Put hardware in suitable bag and tie to drive shaft.
- l. Working within tail cone, back off tapered pins from pylon (Figure 1-6-9-1, Sheet 2, Detail E).

WARNING

Make sure personnel secure pylon to prevent inadvertent folding when bolts are removed.

NOTE

To prevent excessive hydraulic fluid leaks from tail rotor quick-disconnects, make sure hydraulic system pressure is reduced to static.

- m. Remove pylon-to-tail cone bolts, countersunk washers, and washers.
- n. Position padding under tail rotor drive shaft section III to support drive shaft (Figure 1-6-9-1, Sheet 3, Detail F).
- o. Slowly fold pylon and watch that tail rotor blades clear main rotor blades. If tail rotor blades need repositioning, turn at intermediate gear box input flange.
- p. If pylon is to be folded 90°, install pylon lock strut as follows (Figure 1-6-9-1, Sheet 3, Detail G):
 - (1) Position pylon lock strut between tail rotor pylon and tail cone upper mounting holes.
 - (2) With assistant holding tail cone side in place, install bolt, AN9-13A, and washer, NAS1149D0963K, securing pylon lock strut to tail rotor pylon, picking up existing barrel-nut in pylon.

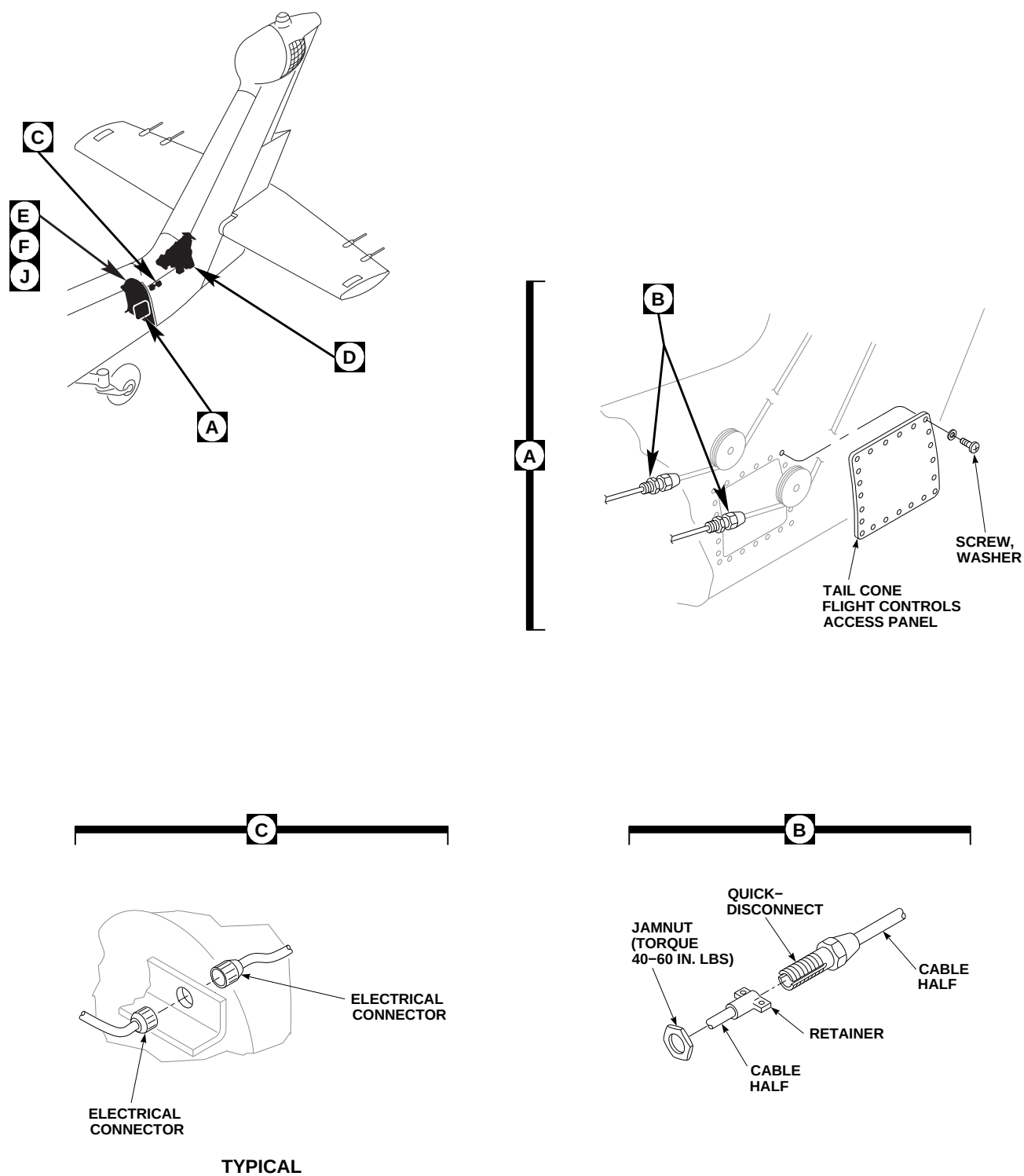


FIGURE 1-6-9-1. FOLD/UNFOLD TAIL ROTOR PYLON (SHEET 1 OF 4)

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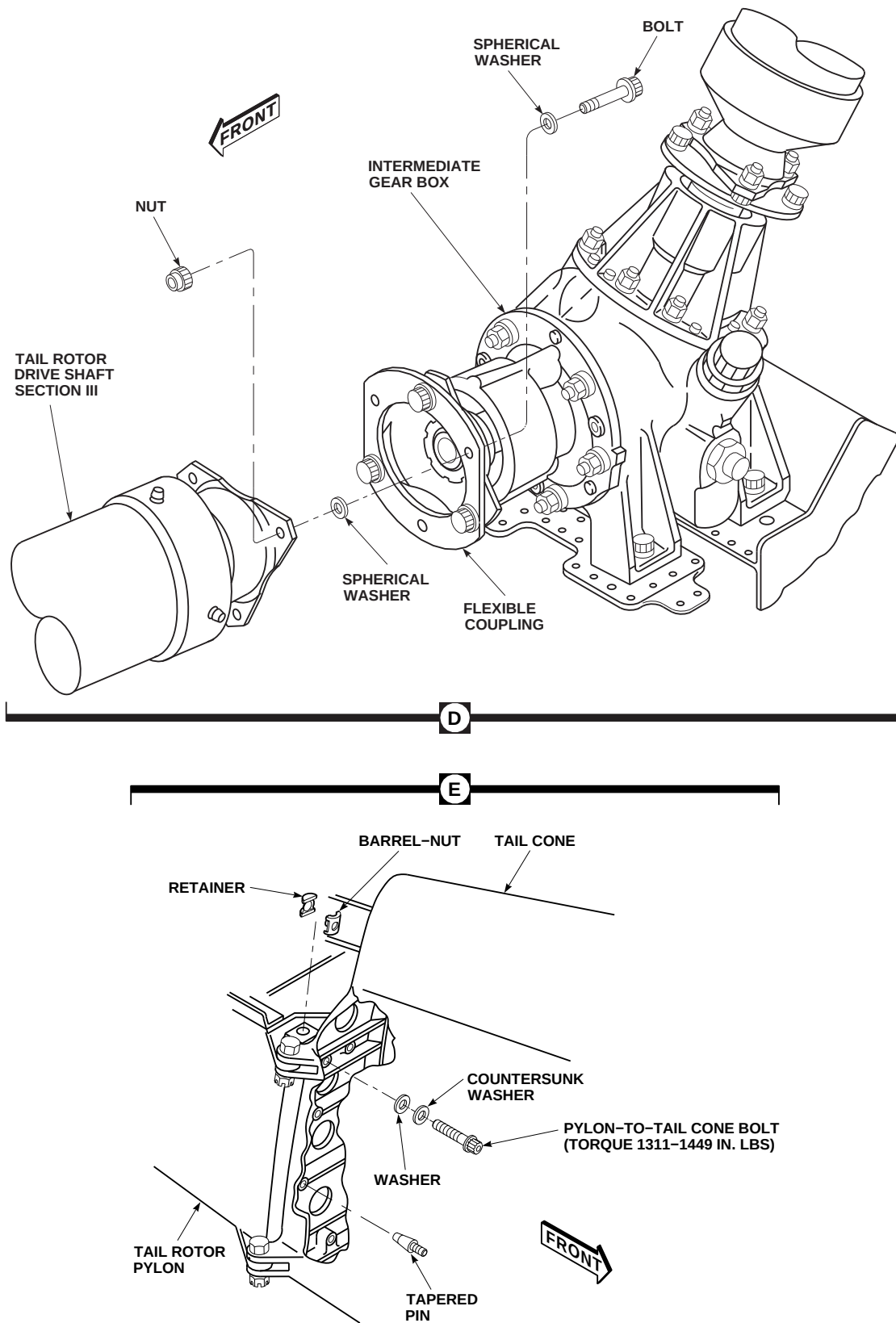


FIGURE 1-6-9-1. FOLD/UNFOLD TAIL ROTOR PYLON (SHEET 2 OF 4)

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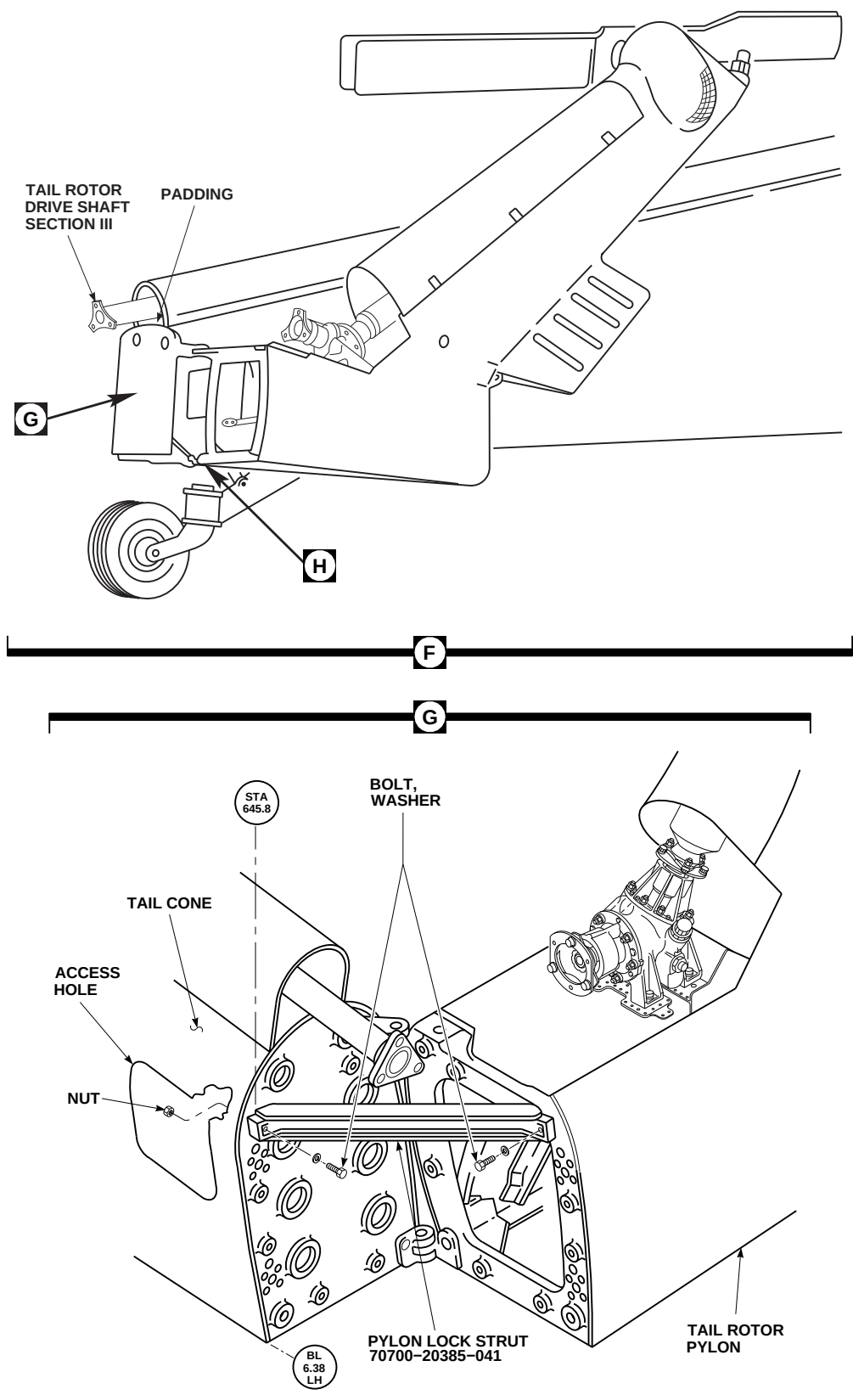


FIGURE 1-6-9-1. FOLD/UNFOLD TAIL ROTOR PYLON (SHEET 3 OF 4)

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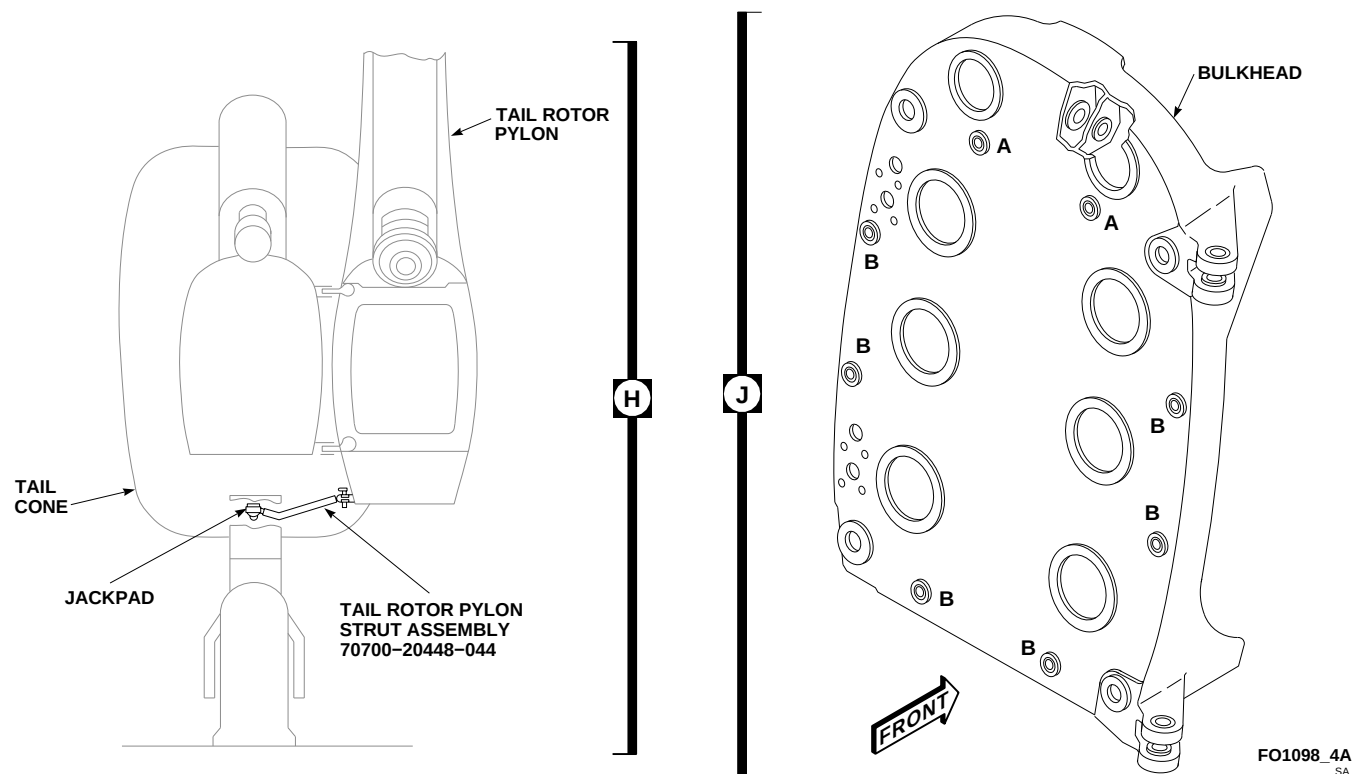


FIGURE 1-6-9-1. FOLD/UNFOLD TAIL ROTOR PYLON (SHEET 4 OF 4)

- (3) Install bolt, AN9-13A, and washer, NAS1149D0963K, securing pylon lock strut to tail cone. Reach through access hole in tail cone and install nut, MS21083N9, on bolt.

WARNING

FSCAP

Verification of barrel-nut and retainer installation and breakaway torque are critical characteristics. Verification of correct installation and torque is required.

NOTE

Make sure only square barrel-nuts are used.

- q. If pylon is to be folded 180°, install tail rotor pylon strut assembly between pylon and jackpad on tail cone (Figure 1-6-9-1, Sheet 4, Detail H).
- r. Position padding under tips of tail rotor blades, and tie tail rotor blades to main rotor blades or blade support poles, using fibrous cord, Item 139, Appendix D.
- s. Remove tapered pins from tail cone (Figure 1-6-9-1, Sheet 2, Detail E).
- t. Remove barrel-nuts and retainers from pylon.
- u. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect pylon attachment bolts and washers, tapered pins, and barrel-nuts (ASTM E1417).

- v. Check all four pylon-to-tail cone bolts and barrel-nuts for minimum breakaway torque as follows:
 - (1) Place barrel-nut in soft-padded vise.
 - (2) Install bolt into barrel-nut until two threads protrude past nut.

- (3) Check breakaway torque by backing off bolt. **IF TORQUE NEEDED TO REMOVE BOLT IS LESS THAN 24 INCH-POUNDS, REPLACE BARREL-NUT AND RECHECK TORQUE. IF TORQUE IS STILL LESS THAN 24 INCH-POUNDS, REPLACE BOLT.**

WARNING

FSCAP

Radius pad installation is a critical characteristic. Make sure installation inspection is performed.

- w. Inspect canted bulkhead for damage and radius pad installation (PARA 2-6-42).

1-6-9.1.2 Unfold Tail Rotor Pylon.

- a. Make sure all inspections as noted in PARA 1-6-9.1.1 have been performed prior to unfolding pylon.
- b. Install barrel-nuts and retainers in pylon (Figure 1-6-9-1, Sheet 2, Detail E). Barrel side of nut must be toward bolthead.
- c. Install tapered pins in tail cone. Make sure pins are backed off to prevent interference when closing pylon.

NOTE

When unfolding pylon, tail rotor blades may or may not be folded.

- d. Remove padding and untie tail rotor blades.
- e. If pylon is folded 180°, remove tail rotor pylon strut assembly from pylon and tail cone (Figure 1-6-9-1, Sheet 4, Detail H).
- f. If pylon is folded 90°, remove bolts, washers, and nut securing pylon lock strut between pylon and tail cone (Figure 1-6-9-1, Sheet 3, Detail G). Remove pylon lock strut.

CAUTION

To prevent crushing, make sure all electrical connectors and control cables are out of the way.

NOTE

To prevent excessive hydraulic fluid leaks from tail rotor quick-disconnects, make sure hydraulic system is reduced to static.

- g. Slowly unfold pylon and watch that tail rotor blades clear main rotor blades. If tail rotor blades need repositioning, turn at intermediate gear box input flange.
- h. Remove padding from under drive shaft (Figure 1-6-9-1, Sheet 3, Detail F).
- i. Before closing pylon, make sure area is clean and free of foreign material.
- j. Install countersunk washers (countersunk side facing bolthead) and washers on pylon-to-tail cone bolts (Figure 1-6-9-1, Sheet 2, Detail E).
- k. Install four pylon-to-tail cone bolts, countersunk washers, and washers securing pylon to tail cone. Run bolts down until pylon evenly contacts tail cone. Do not torque bolts now.
- l. Turn all tapered pins into mating bushings in pylon.

WARNING

FSCAP

Installation of pylon-to-tail cone bolts and washers and bolt torque are critical characteristics. Verification of correct installation and torque is required.

NOTE

Be careful not to damage tail rotor servo lines when torquing bolts.

- m. **TORQUE PYLON-TO-TAIL CONE BOLTS TO 1311 - 1449 INCH-POUNDS.** Apply torque stripe to bolts (PARA 2-6-5).
- n. Perform tail cone/tail rotor pylon gap check as follows (Figure 1-6-9-1, Sheet 4, Detail J):

NOTE

No gaps may exceed the stated maximum allowable gap.

- (1) Working within tail cone, back off tapered pins from pylon.
- (2) Using feeler gage, measure gap between tail pylon and tail cone bosses marked "A". **MAXIMUM ALLOWABLE GAP TO BE 0.012-INCH.**
- (3) Using feeler gage, measure gap between tail pylon and tail cone bosses marked "B". **MAXIMUM ALLOWABLE GAP TO BE 0.030-INCH. NO MORE THAN TWO NON-ADJACENT LOCATIONS CAN BE GREATER THAN 0.015-INCH.**
- (4) If gap exceeds limits, contact Sikorsky ILS for recommended disposition.
- (5) Turn all tapered pins into mating bushings in pylon.
- o. Remove protective caps and plugs from electrical connectors.
- p. Inspect and treat electrical connectors (PARA 1-7-20).
- q. Connect electrical connectors near intermediate gear box (Figure 1-6-9-1, Sheet 1, Detail C).
- r. Remove hardware from bag tied to drive shaft.
- s. Connect tail rotor drive shaft section III to flexible coupling (PARA 6-4-40).

WARNING

Verification of proper cable routing between pulleys and retaining pins or guards is a critical characteristic.

NOTE

- Either crowfoot adapter or locally-made flight control quick-disconnect wrench, Figure H-48, Appendix H, may be used to connect cables.
- Quick-disconnects have two configurations.
- Extension half must bottom in slot half (2.44-inch slot).
- Ball-end bottoms before extension half (2.63-inch slot).
- Both assemblies are acceptable when torqued to specifications.
- t. Connect cable halves at retainer and quick-disconnect, and secure with jamnut (two places) (Figure 1-6-9-1, Sheet 1, Detail B). **TORQUE JAMNUT TO 40 - 60 INCH-POUNDS.** Using lockwire, Item 197, Appendix D, lockwire jamnut to retainer.

CAUTION

Make sure cable halves are connected at quick-disconnects before removing lockout blocks from quadrant.

- u. Remove flight control cable lockout blocks (PARA 11-4-109). If required, remove tiedown straps securing spring cylinders to pylon and connect tail rotor quadrant spring cylinders to support (PARA 11-4-118).
- v. Before closing cover and installing fairings, make sure area is clean and free of foreign material.

CAUTION

To prevent damage to equipment, do not start engines or pressurize hydraulic systems while blades are folded.

- w. Close rear tail rotor drive shaft cover.

- x. Install tail cone flight controls access panel on tail cone and secure with screws and washers (Figure 1-6-9-1, Sheet 1, Detail A). 30 flight hours after installation (PARA 1-7-11).
- y. Install intermediate gear box fairing (PARA 2-4-139).
 - aa. If removed, install stabilator (PARA 2-4-142).
 - ab. If required, spread main rotor blades (PARA 1-6-7) and/or tail rotor blades (PARA 1-6-8).
- z. Check helicopter logbook to make sure that pylon to tail cone torque check is shown as due

1-6-10 KNEEL HELICOPTER.

PARAGRAPH BREAKDOWN

	PAGE
1-6-10.1 Kneel Helicopter	1-6-10-1

INITIAL SETUP

References

- PARA 1-6-17

1-6-10.1 Kneel Helicopter.

Kneel helicopter, using instructions given in PARA 1-6-17. Pay particular attention to clearance

beneath helicopter under cargo hook, LF/ADF loop antenna, AN/ARQ-31 antenna, and lower anticollision light, and drag beam wire strike cutters.

1-6-11 CABIN TIEDOWN FITTINGS.

PARAGRAPH BREAKDOWN

	PAGE
1-6-11.1 Replace Cabin Tiedown Fittings	1-6-11-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Pliers
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 0 - 600 ft lbs

Materials/Parts

- Lockwire, Item 198, Appendix D
- Machinery Towel, Item 211, Appendix D
- Sealing Compound, Item 292, Appendix D
- Trichloroethylene, Technical, Item 343, Appendix D

References

- Appendix D
- PARA 2-4-45
- PARA 2-4-69
- PARA 2-4-70

1-6-11.1 Replace Cabin Tiedown Fittings.

1-6-11.1.1 Install.

- Remove tiedown cover plates from soundproofing panel in cabin ceiling.
- Thread new bolt into barrel-nut. **TORQUE MUST BE 32 - 300 INCH-POUNDS.** If run-on torque is not 32 - 300 inch-pounds, replace barrel-nut as follows:
 - If installed, remove crew chief’s/ passenger’s seats (PARA 2-4-45).

NOTE

Removal of main transmission drip pan is required only when replacing barrel-nuts at STA 343.58.

- If required, remove main transmission drip pan (PARA 2-4-70).
- Remove soundproofing panels as required (PARA 2-4-69).
- At STA 312.3, remove barrel-nut from transmission beam using pliers.
- At STA 343.58, perform the following:
 - Remove sealing compound using nonmetallic scraper.
 - Remove plastic plug in transmission beam.
 - Remove barrel-nut from transmission beam.

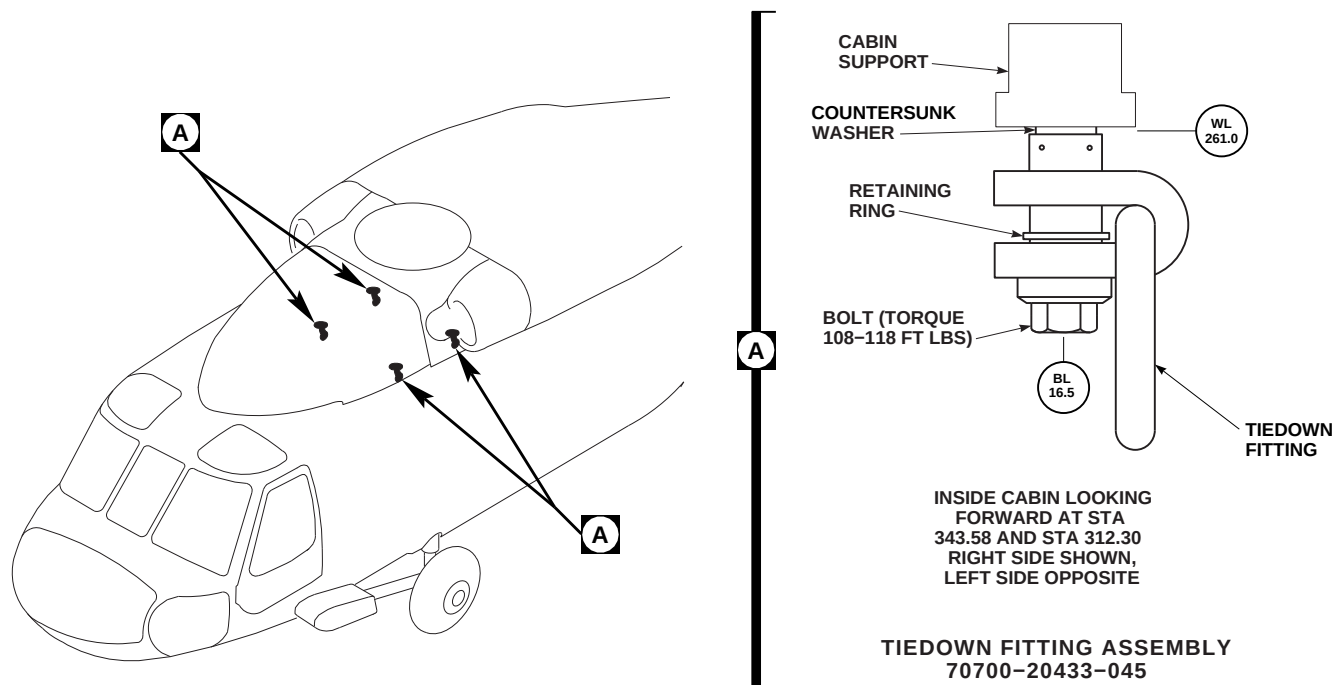
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FIGURE 1-6-11-1. REPLACE CABIN TIEDOWN FITTINGS

CAUTION

Damage to barrel-nut recess will occur if barrel-nut is improperly installed. Make sure barrel side of nut is toward bolthead.

- (6) Install retainer over barrel-nut.
- (7) At STA 312.3, push barrel-nut into transmission beam with barrel side toward bolthead.
- (8) At STA 343.58, perform the following:
 - (a) Push barrel-nut into transmission beam with barrel side toward bolt-head.
 - (b) Install plastic plug in transmission beam.
 - (c) Clean surface of plastic plug and transmission beam area using machinery towel, Item 211, Appendix D, wet with technical trichloroethylene, Item 343, Appendix D.
 - (d) Apply a bead of sealing compound, Item 292, Appendix D, around plastic plug.
- (9) Install soundproofing panels as required (PARA 2-4-69).
- (10) If removed, install main transmission drip pan (PARA 2-4-70).
- (11) If removed, install crew chief's/passenger's seats (PARA 2-4-45).

WARNING

Injury to personnel and damage to equipment will result if improper tiedown fitting is installed. Tiedown fitting, 70700-20433-045, with an "H" etched in bolthead, 70700-20472-101, is the only approved tiedown fitting.

NOTE

Install countersunk washer with countersunk side facing down against shoulder of bolt.

- c. Install countersunk washer on bolt and install tiedown fitting (Figure 1-6-11-1, Detail A) to

cabin support. **TORQUE BOLT TO 108 - 118 FOOT-POUNDS.**

- d. Lockwire, Item 198, Appendix D, bolt to drilled-head bolt in passenger's seat support fitting.

1-6-11.1.2 Remove.

NOTE

Keep bolt and washer with fitting.

- a. Remove bolt and countersunk washer securing tiedown fitting assembly to cabin support (Figure 1-6-11-1, Detail A).
- b. Install tiedown cover plates on soundproofing panel in cabin ceiling.

1-6-12 TIEDOWN PLATES.

PARAGRAPH BREAKDOWN

	PAGE
1-6-12.1 Replace Tiedown Plates	1-6-12-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Tiedown Plate, 70700-20414-041
- Tiedown Plate, 70700-20414-042

- Torque Wrench, 700 - 1600 in. lbs

References

- PARA 2-4-36

1-6-12.1 Replace Tiedown Plates.

1-6-12.1.1 Install.

- Remove left and right roll vibration absorbers (PARA 2-4-36).

NOTE

- Tiedown plate, 70700-20414-042, is installed on left landing gear support.
 - Tiedown plate, 70700-20414-041, is installed on right landing gear support.
- Install tiedown plate on landing gear support and secure with bolts and washers provided

with plate (Figure 1-6-12-1, Detail A).
TORQUE BOLTS TO 732 - 808 INCH-POUNDS.

1-6-12.1.2 Remove.

NOTE

- Keep hardware together and stow with kit.
- Remove bolts and washers securing tiedown plates to landing gear supports. Remove tiedown plate (Figure 1-6-12-1, Detail A).
 - Install left and right roll vibration absorbers (PARA 2-4-36).

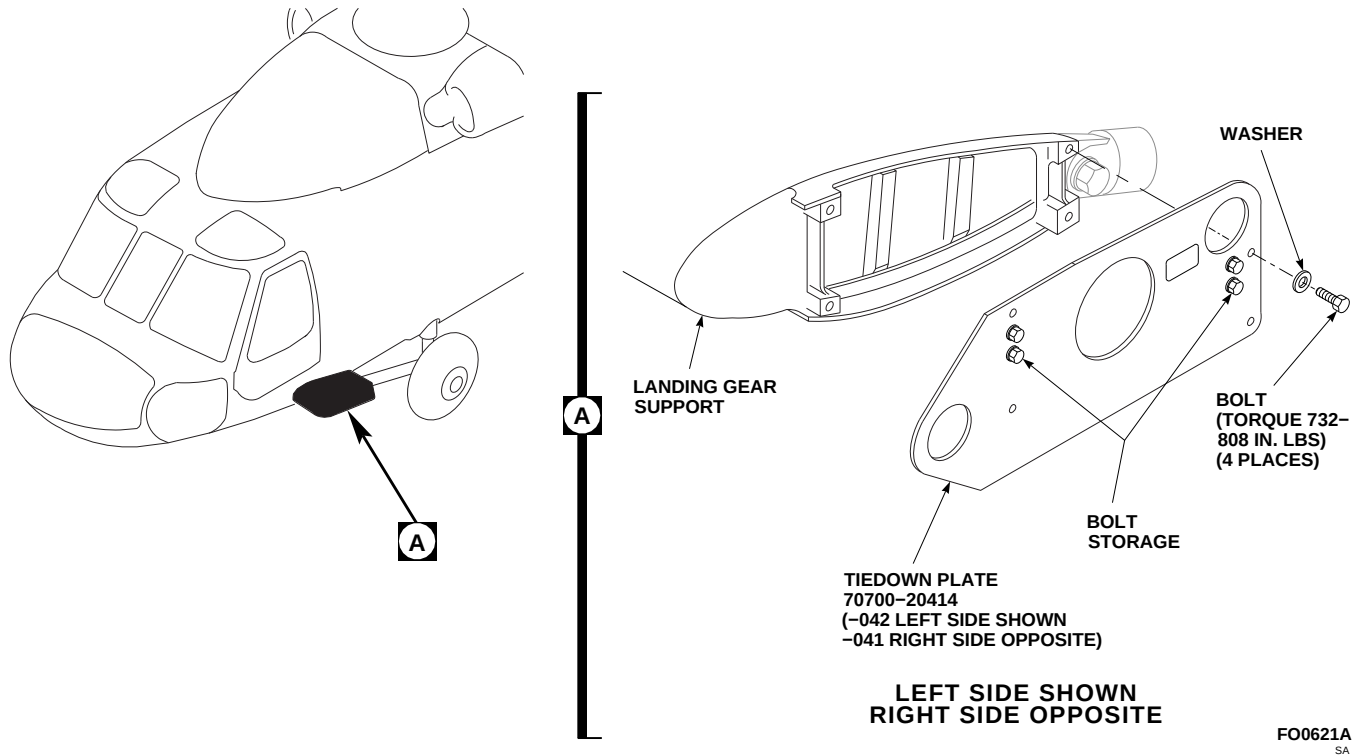


FIGURE 1-6-12-1. REPLACE TIEDOWN PLATES

1-6-13 TAIL CONE TOW FITTINGS.

PARAGRAPH BREAKDOWN

	PAGE
1-6-13.1 Install/Remove Tail Cone Tow Fittings	1-6-13-1

INITIAL SETUP

Tools

- Tail Rotor Pylon Strut Assembly, 70700-20448-044
- Tow Fitting, 70700-20433-042
- Aircraft Mechanic’s Toolkit

1-6-13.1 Install/Remove Tail Cone Tow Fittings.

NOTE

- Tail cone tow fittings, 70700-20433-042, are used for air transport loading and tiedown of helicopter.
- Installation and removal of left and right tail cone tow fittings is the same.

1-6-13.1.1 Install.

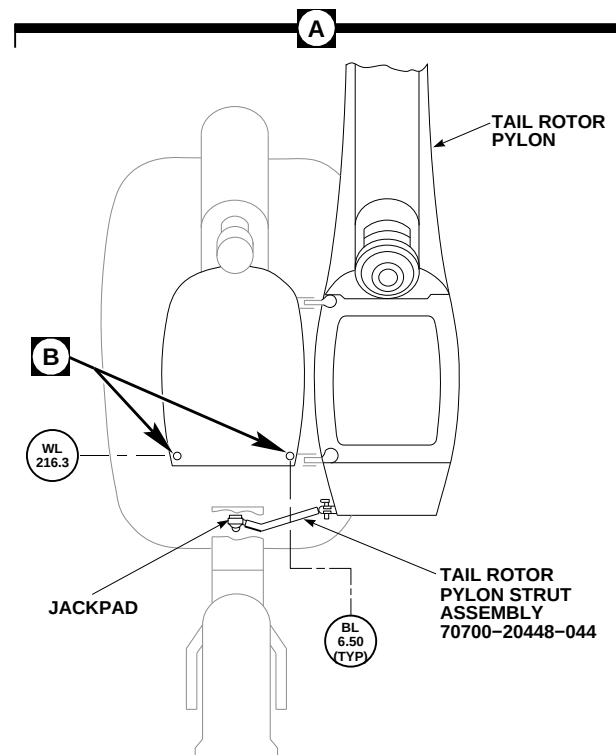
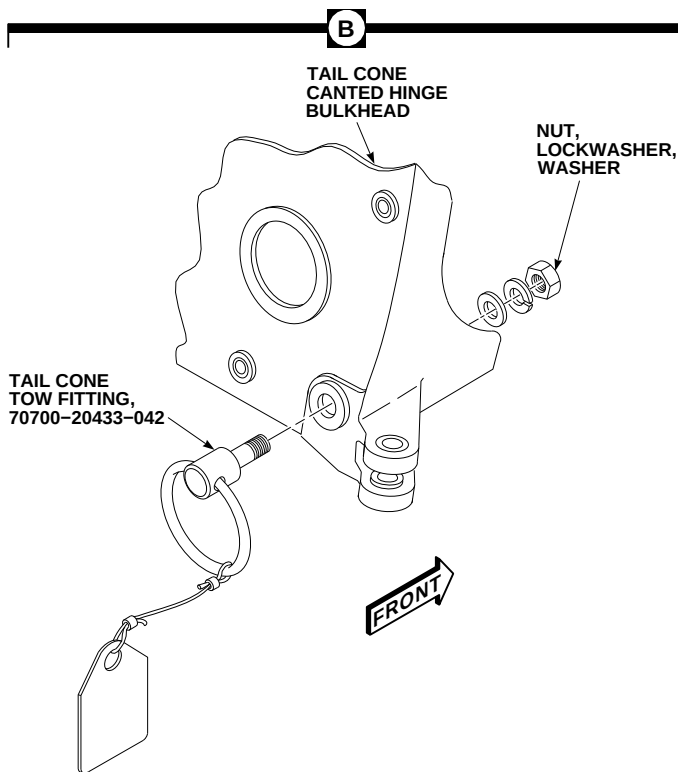
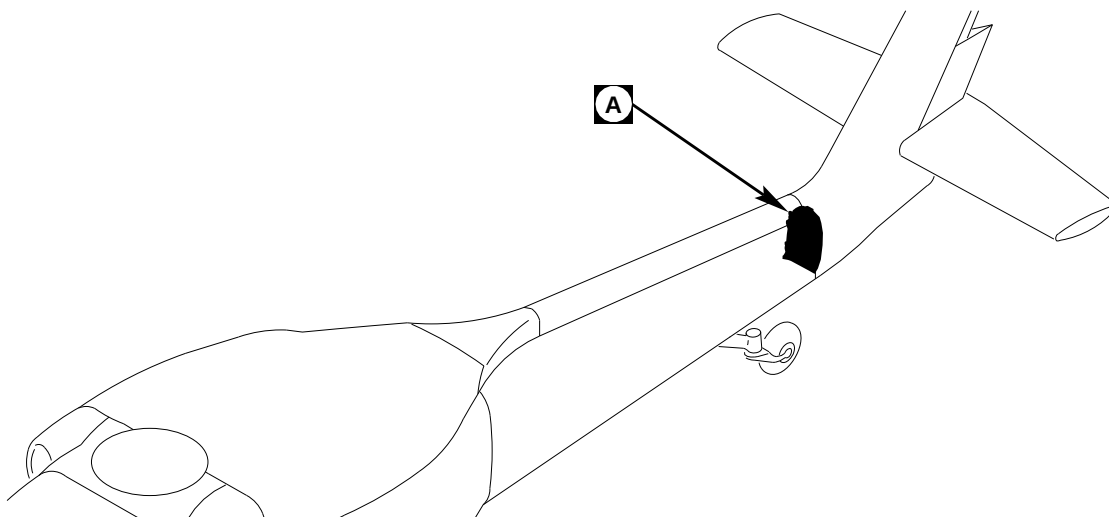
- Fold tail rotor pylon 180° (PARA 1-6-9).
- Check that tail rotor pylon strut assembly, 70700-20448-044, is installed between tail ro-

tor pylon and jackpad (Figure 1-6-13-1, Detail A).

- Install tow fittings on tail cone canted hinge bulkhead at STA 646, BL 6.50, and secure with nuts, lockwashers, and washers (Figure 1-6-13-1, Detail B).

1-6-13.1.2 Remove.

- Remove nuts, lockwashers, and washers securing tow fittings to tail cone canted hinge bulkhead (Figure 1-6-13-1, Detail B). Remove tow fittings.
- Unfold tail rotor pylon 180° (PARA 1-6-9).



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FIGURE 1-6-13-1. INSTALL/REMOVE TAIL CONE TOW FITTINGS

1-6-14 FUSELAGE AND TAIL CONE TIEDOWN RINGS.

PARAGRAPH BREAKDOWN

	PAGE
1-6-14.1 Replace Fuselage and Tail Cone Tiedown Rings	1-6-14-1

INITIAL SETUP

- Torque Wrench, 150 - 750 in. lbs

Tools

- Aircraft Mechanic’s Toolkit

1-6-14.1 Replace Fuselage and Tail Cone Tiedown Rings.

1-6-14.1.1. Remove.

- Remove nut, washers, and eyebolt attaching tiedown ring to fitting at STA 308.0. Remove tiedown ring (Figure 1-6-14-1, Detail A).
- Remove nut, washers, and bolt attaching tiedown ring to fitting at STA 485.0. Remove tiedown ring (Figure 1-6-14-1, Detail B).
- Remove nut, washers, and eyebolt from fitting at STA 308.0 (Figure 1-6-14-1, Detail C).

1-6-14.1.2. Install.

- Install tiedown ring to fitting at STA 308.0 with eyebolt, washers, and nut. **TORQUE NUT TO 150 - 300 INCH-POUNDS.** Install cotter pin (Figure 1-6-14-1, Detail A).
- Install tiedown ring to fitting at STA 485.0 with bolt, washers, and nut. **TORQUE NUT TO 170 - 340 INCH-POUNDS.** Install cotter pin (Figure 1-6-14-1, Detail B).
- Install eyebolt in fitting at STA 308.0 with washers and nut. **TORQUE NUT TO 150 - 300 INCH-POUNDS.** Install cotter pin (Figure 1-6-14-1, Detail C).

NOTE

Eyebolt, 70219-02136-102, is to be used.

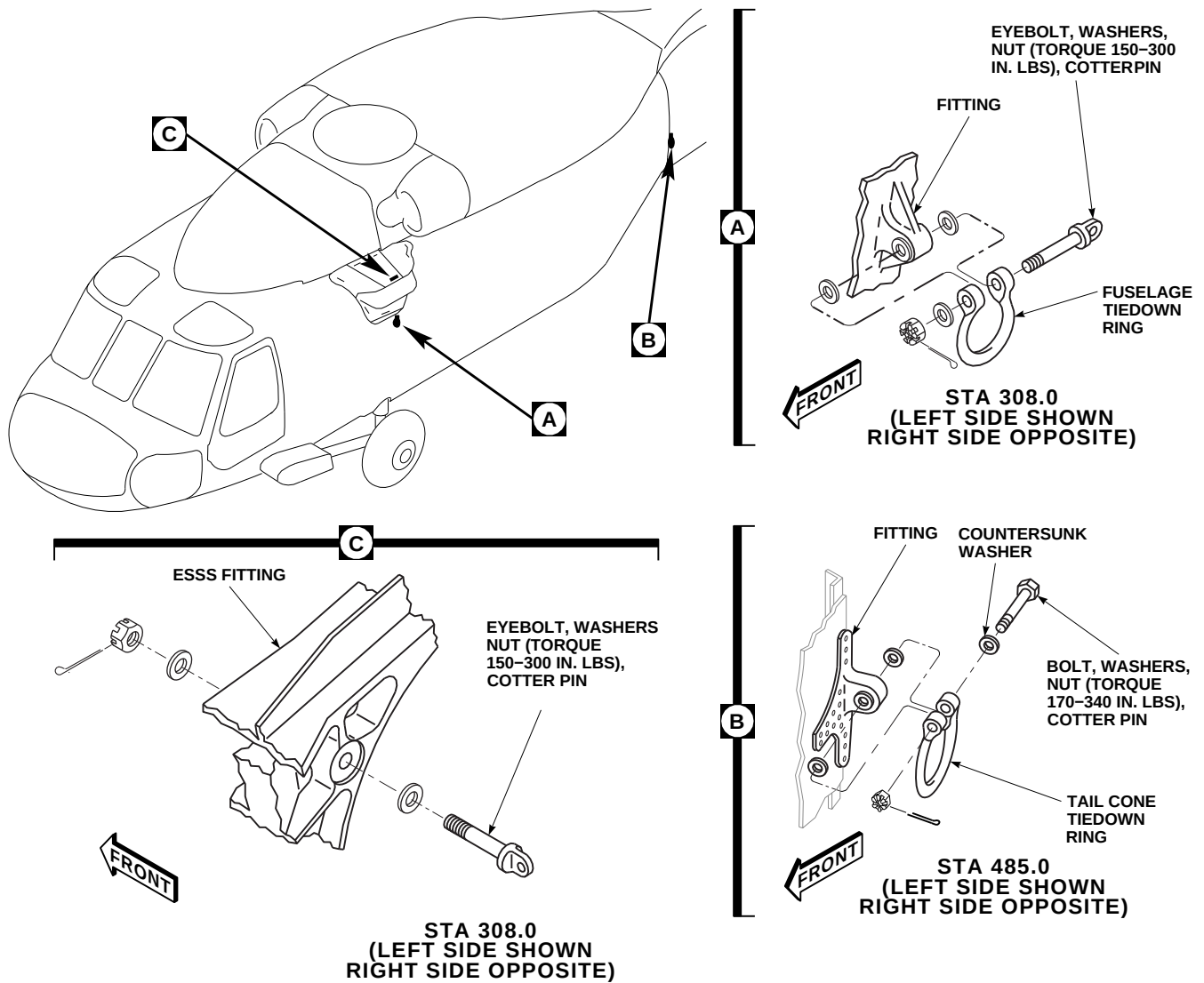
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FIGURE 1-6-14-1. REPLACE FUSELAGE AND TAIL CONE TIEDOWN RINGS

1-6-15 APPLY HYDRAULIC POWER.

PARAGRAPH BREAKDOWN

	PAGE		PAGE
1-6-15.1 External Hydraulic Power	1-6-15-1	1-6-15.2 Internal Hydraulic Power	1-6-15-4

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Hydraulic Power Source

References

- PARA 1-3-8
- PARA 1-6-16

- PARA 11-4-90
- PARA 11-4-91
- PARA 11-4-92

1-6-15.1 External Hydraulic Power.

1-6-15.1.1 General.



- To prevent accidental engagement of backup hydraulic pump when external hydraulic power is applied to helicopter, pull out BACKUP PUMP PWR circuit breaker (NO. 1 DC PRI BUS - copilot’s circuit breaker panel).
- Operation of the backup hydraulic pump while external hydraulic power is applied to the helicopter backup system can cause damage to the backup pump.

NOTE

- Whenever ground return lines from external hydraulic power source are con-

nected to helicopter, the pump module reservoir will deplete, causing its respective reservoir low capsule on the caution/advisory panel to go on.

- Make sure helicopter logbook and maintenance documentation is reviewed prior to applying hydraulic power.
 - a. Review helicopter logbook and maintenance documentation prior to applying hydraulic power.
 - b. Check with maintenance personnel in charge of helicopter, prior to applying hydraulic power.

1-6-15.1.2 Connect/Turn On.

- a. Connect hydraulic pressure source pressure and return lines as follows:
 - (1) Open main rotor pylon sliding cover.

CAUTION

To prevent damage to flight control system, do not move flight controls while blades are folded and hydraulic power is applied. If movement of flight controls is required while blades are folded, forward, aft, and lateral control rods must be disconnected.

- (2) If required, disconnect forward, aft, and lateral control rods (PARA 11-4-90, PARA 11-4-91, PARA 11-4-92).
- (3) Remove caps from pressure and return line quick-disconnects.
- (4) Inspect quick-disconnect male couplings for damage prior to installation.

NOTE

Make sure to loop pressure line over high point on helicopter to prevent air from entering hydraulic system. Loop line over engine cowling or main rotor pylon transmission cover.

- (5) Connect pressure line from hydraulic power source to pressure quick-disconnect on No. 1, No. 2, or backup hydraulic pump (Figure 1-6-15-1, Detail A and Detail B).

NOTE

Make sure to loop return line over high point on helicopter to prevent air from entering hydraulic system. Loop line over engine cowling or main rotor pylon transmission cover.

- (6) Connect return line from hydraulic power source to return quick-disconnect on No. 1, No. 2, or backup return port.

b. Turn on hydraulic power as follows:

- (1) Start hydraulic power source.
- (2) Adjust flow rate and pressure to 6 gpm at 3000 psi.
- (3) Open hydraulic power source outlet and inlet valves.

1-6-15.1.3 Turn Off/Disconnect.

a. Turn off hydraulic power as follows:

- (1) Close hydraulic power source outlet and inlet valves.
- (2) Shut down hydraulic power source.

b. Disconnect hydraulic pressure source pressure and return lines as follows:

- (1) Disconnect hydraulic power source pressure and return lines from No. 1, No. 2, or backup hydraulic pump and No. 1, No. 2, or backup return port (Figure 1-6-15-1, Detail A and Detail B).
- (2) Install caps on pressure and return line quick-disconnects.
- (3) Service hydraulic pump modules (PARA 1-3-8).
- (4) If required, connect forward, aft, and lateral control rods (PARA 11-4-90, PARA 11-4-91, PARA 11-4-92).
- (5) Make sure area is clean and free of foreign material.

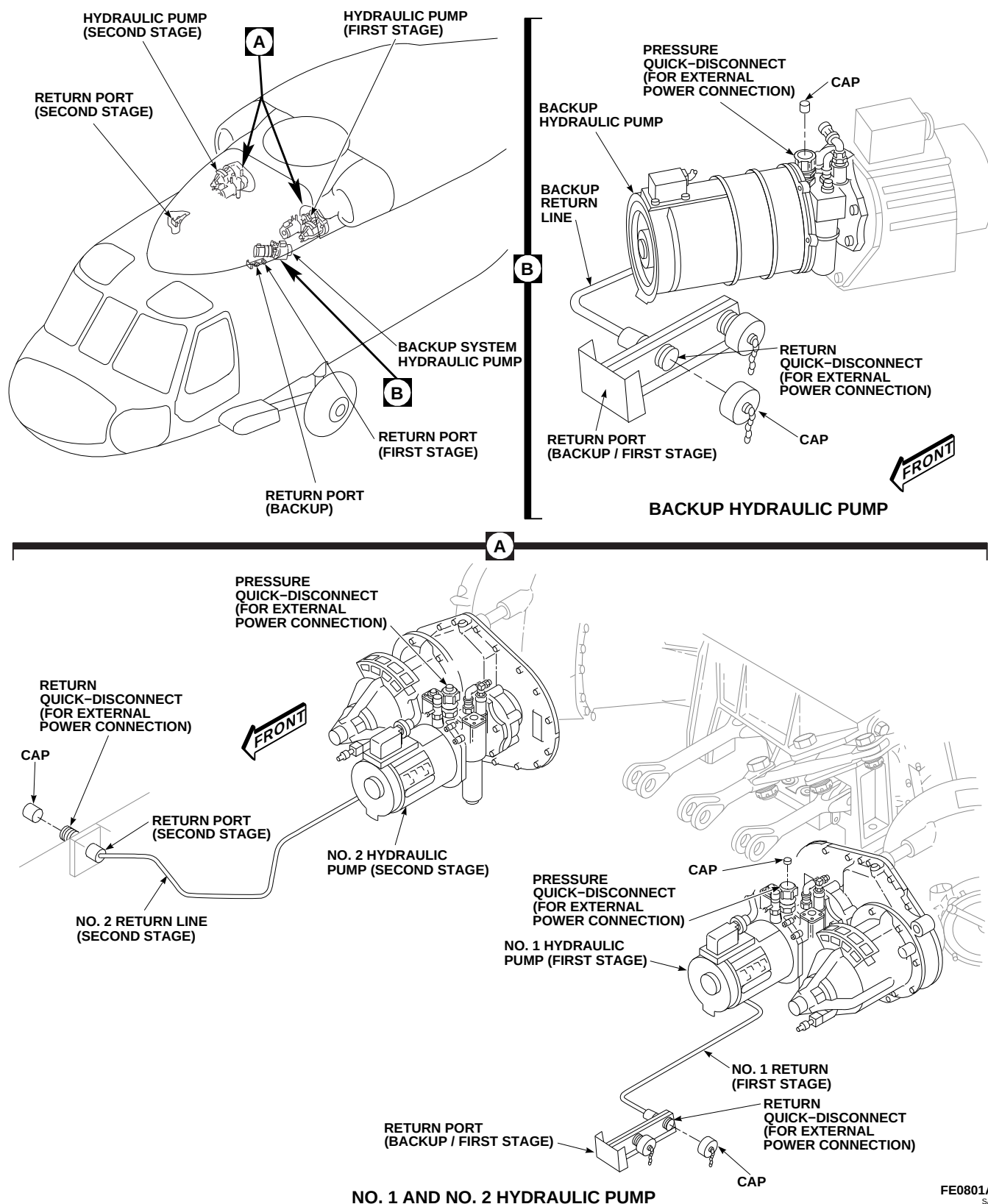


FIGURE 1-6-15-1. APPLY EXTERNAL HYDRAULIC POWER

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- (6) Close main rotor pylon sliding cover.

1-6-15.2 Internal Hydraulic Power.

1-6-15.2.1 General.

CAUTION

- Damage to backup hydraulic system will result if backup pump is run for an extended period of time when surrounding temperature is above 90°F (32°C). To prevent overheating backup hydraulic system, use this chart for limits when running backup pump.
- If weight on wheels switch is bypassed, overheating of backup pump module can occur.

Surrounding Temperature F° (C°)	Operating Time (Minutes)	Cooldown Time, Pump Off (Minutes)
90° (32°) max	Unlimited	-
91° - 100° (33° - 38°)	24	72
101° - 125° (39° - 52°)	16	48

NOTE

- Make sure flight controls are free to move.
- Make sure helicopter logbook and maintenance documentation is reviewed prior to applying hydraulic power.

- Review helicopter logbook and maintenance documentation prior to applying hydraulic power.
- Check with maintenance personnel in charge of helicopter, prior to applying hydraulic power.

1-6-15.2.2 Connect/Turn On.

CAUTION

To prevent damage to flight control system, do not move flight controls while blades are folded and hydraulic power is applied. If movement of flight controls is required while blades are folded, forward, aft, and lateral control rods must be disconnected.

- If required, disconnect forward, aft, and lateral control rods from (PARA 11-4-90, PARA 11-4-91, PARA 11-4-92).
- Connect/turn on electrical power (PARA 1-6-16).
- On upper console, place BACKUP HYD PUMP switch to ON.

1-6-15.2.3 Turn Off/Disconnect.

- On upper console, place BACKUP HYD PUMP switch to OFF.
- Turn off/disconnect electrical power (PARA 1-6-16).
- If required, connect forward, aft, and lateral control rods (PARA 11-4-90, PARA 11-4-91, PARA 11-4-92).

1-6-16 APPLY ELECTRICAL POWER.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-6-16.1	External Electrical Power	1-6-16-1	1-6-16.2	Internal Electrical Power	1-6-16-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Generator Set, MD-3M (or equivalent)

References

- Appropriate Flight Manual

1-6-16.1 External Electrical Power.

1-6-16.1.1 General.

WARNING

- The AC electrical supply system of helicopter can be lethal under certain conditions.
- Before applying electrical power to helicopter, make sure area around stabilator is clear of personnel and equipment. Should stabilator be in any position other than trailing edge down, it will reposition automatically to full down when electrical power is applied.
- Backup pump may start if APU accumulator is below 2,600 psi when AC power is applied to helicopter.

NOTE

Make sure helicopter logbook and maintenance documentation is reviewed prior to applying electrical power.

- Review helicopter logbook and maintenance documentation prior to applying electrical power.
- Check with maintenance personnel in charge of helicopter, prior to applying electrical power.
- Make sure all hydraulic and flight components are secured for operation.
- Make sure all personnel are clear of flight controls.

1-6-16.1.2 Connect/Turn On.

- Make sure all switches and circuit breakers are in the appropriate position (Appropriate Flight Manual).
- Open access cover (4B-26).
- Connect generator set to external power receptacle, using equipment operation instructions (Figure 1-6-16-1, Detail A).
- Start generator set.
- On upper console, place EXT PWR switch to RESET, then ON (Figure 1-6-16-1, Detail B).
- On upper console, place BATT switch to ON.

1-6-16.1.3 Turn Off/Disconnect.

- a. On upper console, place BATT switch to OFF (Figure 1-6-16-1, Detail B).
- b. On upper console, place EXT PWR switch to OFF.

WARNING

Make sure generator set is off, prior to disconnecting.

- c. Shut down generator set.
- d. Shut down and disconnect generator set from external power receptacle (Figure 1-6-16-1, Detail A).
- e. Close access cover (4B-26).

1-6-16.2 Internal Electrical Power.**1-6-16.2.1 General.****WARNING**

- The AC electrical supply system of helicopter can be lethal under certain conditions.
- Before applying electrical power to helicopter, make sure area around stabilator is clear of personnel and equipment. Should stabilator be in any position other than trailing edge down, it will reposition automatically to full down when electrical power is applied.
- Backup pump may start if APU accumulator is below 2,600 psi when AC power is applied to helicopter.

NOTE

Make sure helicopter logbook and maintenance documentation is reviewed prior to applying electrical power.

- a. Review helicopter logbook and maintenance documentation prior to applying electrical power.
- b. Check with maintenance personnel in charge of helicopter, prior to applying electrical power.
- c. Make sure all hydraulic and flight components are secured for operation.
- d. Make sure all personnel are clear of flight controls.

1-6-16.2.2 Connect/Turn On.**WARNING**

To prevent injury to personnel and damage to helicopter, have fire guard check APU fuel drain. Wait at least two minutes after APU shut down before attempting a restart. DO NOT RESTART APU until fuel drainage has stopped.

- a. Have fire guard check APU fuel drain to make sure there is no fuel draining.
- b. With qualified and authorized personnel at controls, start APU.

1-6-16.2.3 Turn Off/Disconnect.

With qualified and authorized personnel at controls, shutdown APU.

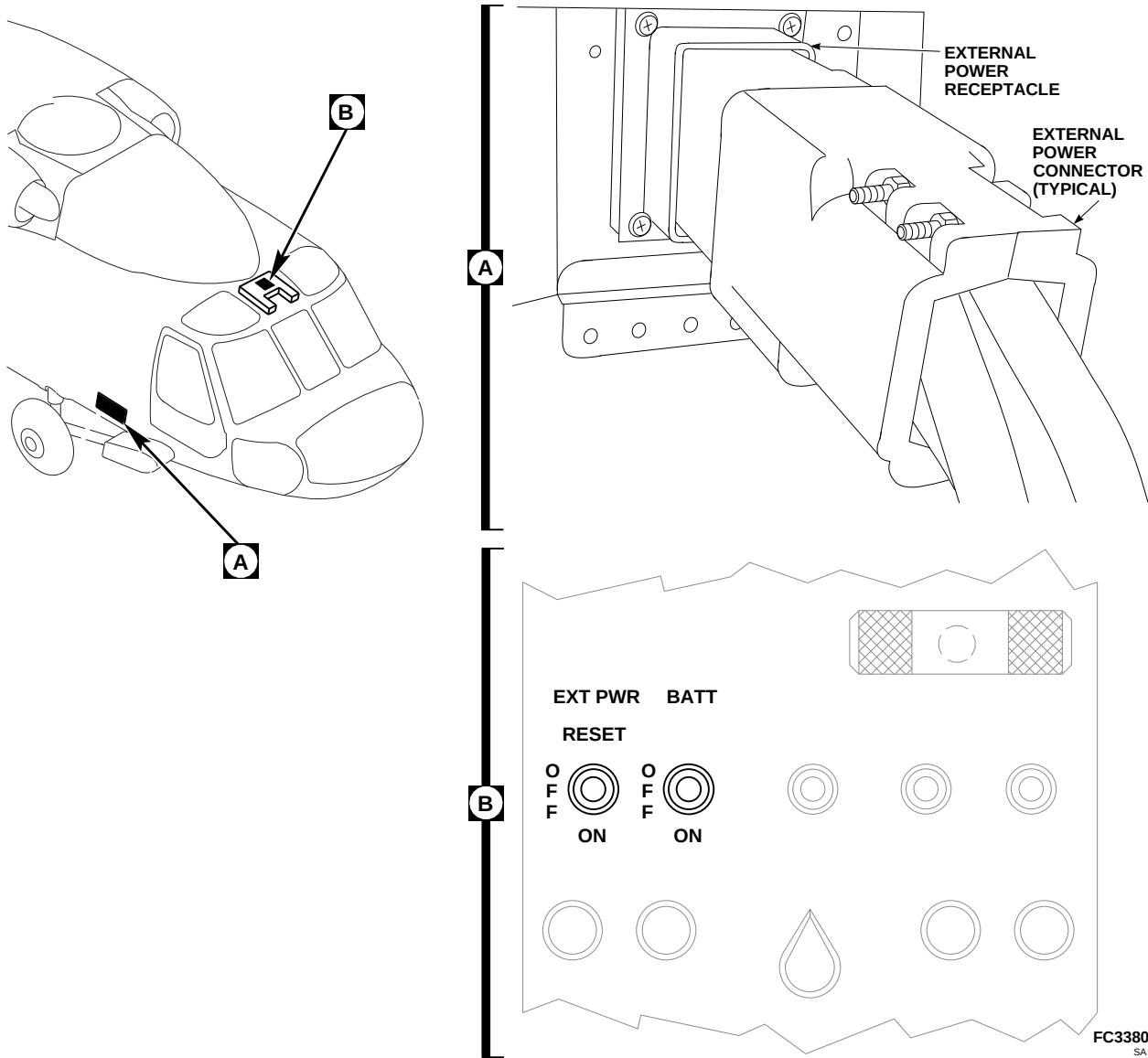


FIGURE 1-6-16-1. APPLY EXTERNAL ELECTRICAL POWER

1-6-17 HYDRAULIC CART CONNECT AND CHECKOUT.

PARAGRAPH BREAKDOWN

	PAGE
1-6-17.1 Connect and Check Out Hydraulic Cart	1-6-17-1

INITIAL SETUP

Tools

- Hydraulic Cart Assembly, 70700-20437-047

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D

References

- Appendix D
- PARA 2-4-95
- PARA 2-4-96
- PARA 2-4-97

1-6-17.1 Connect and Check Out Hydraulic Cart.

1-6-17.1.1 Connect.

- Unfasten and fold down left and right main gear shock strut upper fairings (PARA 2-4-96).
- Remove left and right lower main landing gear fairings (PARA 2-4-95).
- Remove left and right tail cone covers from tail cone.
- Remove tail landing gear fairing (PARA 2-4-97).

NOTE

Take necessary precautions to prevent hydraulic cart and lines from soiling interior furnishings.

- Position hydraulic cart assembly in cabin and tie down (Figure 1-6-17-1, Sheet 1, Detail B).

- Measure and record exposed shiny portion of upper and lower shock strut for all three struts. Tape recorded dimension to each strut.
- Check hydraulic cart quick-disconnects for cleanness; if needed, flush ends of quick-disconnects with hydraulic fluid, Item 154, Appendix D.
- Plug ground wire from hydraulic cart in ground receptacle on helicopter fuselage.
- Service hydraulic cart with hydraulic fluid, Item 154, Appendix D. Do not let level in reservoir fall below 1 gallon, or rise above 2 gallons.
- Connect hose, 70700-20368-048, to upper and lower kneeling valve quick-disconnects of left main wheel shock strut. Connect other end of left gear hose to LEFT MAIN connection on cart (Figure 1-6-17-1, Sheet 1, Detail B and Detail C).

NOTE

- Hose is connected to cart, on lower left side of cart, just above hose connection for tail landing gear shock strut.
 - If necessary, loosen jamnut securing disconnect to valve, and release trapped hydraulic pressure.
- k. Connect hose, 70700-20368-048, to upper and lower kneeling valve quick-disconnects of right main wheel shock strut. Connect other end of right gear hose to RIGHT MAIN connection on cart.

NOTE

Hose is connected to cart on lower right side of cart.

- l. Connect hose, 70700-20368-047, to upper and lower kneeling valve quick-disconnects on tail wheel shock strut. Connect other end of hose to TAIL connection on cart (Figure 1-6-17-1, Sheet 1, Detail A and Detail B).
- m. Open upper and lower kneeling valves of three shock struts.
- n. Connect power cable, 70700-20437-044, to connector, J2 on back of cart (Figure 1-6-17-1, Sheet 1, Detail B and Sheet 2, Detail D). Connect other end of cable to 28 VDC source.
- o. Connect electrical connector, P4, of remote control pendant, 70700-20443-045, to receptacle, J4 on control box, 70700-20444-041 (Figure 1-6-17-1, Sheet 2, Detail D and Detail E).

1-6-17.1.2 Check Out.

- a. Push in CONTROL POWER and MOTOR/PUMP POWER circuit breakers on hydraulic cart control box (Figure 1-6-17-1, Sheet 2, Detail D).
- b. Press START switch, on control box. POWER ON light control box and light on pendant will go on.

WARNING

Before raising or lowering helicopter check that there are no obstructions overhead, and that the area under helicopter is clear of personnel/equipment and tools; kneeling valves are opened, and that there is enough hydraulic fluid in cart.

NOTE

The hydraulic cart operator should place himself where he can see all areas of the helicopter affected.

- c. Press START button on pendant.

NOTE

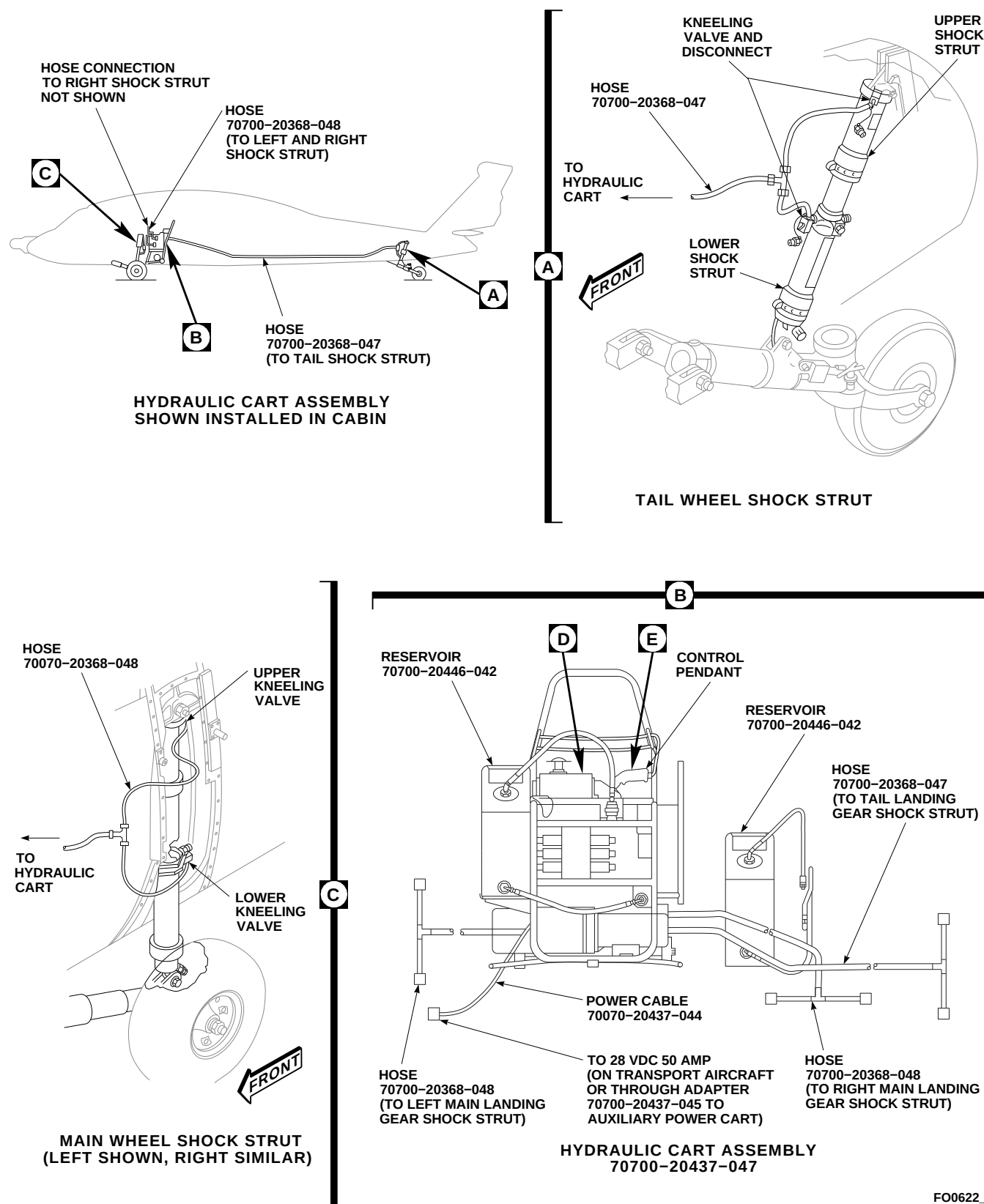
MOTOR STOP light on control box will go on when and if return line to reservoir is disconnected or when pressure (hydraulic) is over 50 psi.

- d. Place RIGHT switch, on pendant, to UP. Right gear should lengthen. Release switch. Place switch to DOWN. Right gear should shorten. Release switch.
- e. Repeat procedure for left and tail struts, using LEFT and TAIL switches on pendant.
- f. Partly lower helicopter, by lowering struts using switches on pendant.
- g. Push EMERGENCY STOP switch, on top of control box. Pump will stop running and all lights will go off.

CAUTION

To prevent damage to underside of helicopter, lower main and tail struts together. Do not lower helicopter any lower than $\frac{3}{8}$ -inch from floor.

- h. Repeat check out without pushing in circuit breakers. Lower helicopter by lowering struts, using switches on pendant.



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FIGURE 1-6-17-1. CONNECT AND CHECK OUT HYDRAULIC CART (SHEET 1 OF 2)

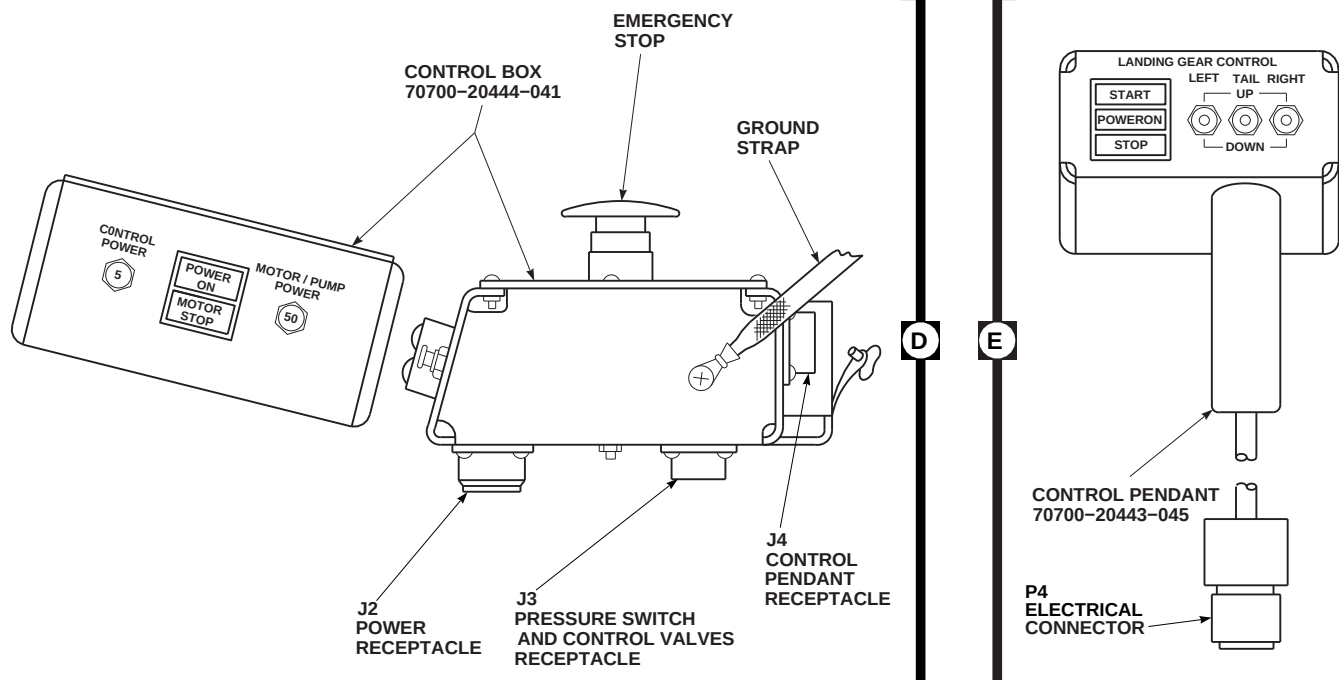
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FIGURE 1-6-17-1. CONNECT AND CHECK OUT HYDRAULIC CART (SHEET 2 OF 2)

- i. Push STOP switch on pendant. Pump will stop running and all lights will go off. Pull out CONTROL POWER AND MOTOR/PUMP POWER circuit breakers on control box.
- j. Close kneeling valves on all shock struts.
- k. Install tail landing gear fairing (PARA 2-4-97).
- l. Install left and right tail cone covers to tail cone.
- m. Install left and right lower main landing gear fairings (PARA 2-4-95).
- n. Fold up and secure left and right main landing gear shock strut upper fairings (PARA 2-4-96).

1-6-18 ENGAGE/DISENGAGE GUST LOCK AND ROTOR BRAKE.

PARAGRAPH BREAKDOWN

		PAGE	NOTE
1-6-18.1	Gust Lock	1-6-18-1	When pulling on the handle to verify engagement, do not depress the release button.
1-6-18.2	Rotor Brake	1-6-18-1	
1-6-18.1	Gust Lock.		(2) Verify positive engagement by pulling down on gust lock handle to ensure it does not move.
1-6-18.1.1	Engage.		

CAUTION

Damage to gust lock and rotor brake will result if engine(s) are operated with gust lock engaged on helicopters equipped with rotor brake. Do not operate an engine with gust lock engaged on helicopters equipped with rotor brake.

- a. While grasping gust lock handle using index and middle fingers, depress release button using thumb to unlock gust lock handle (Figure 1-6-18-1, Detail A).

NOTE

As gust lock handle is being pushed up, release the release button.

- b. Push gust lock handle to full up position. Release button will pop out when positive engagement is made, when pawl on the gust lock lever meshes with a tooth on main transmission tail takeoff flange (Figure 1-6-18-1, Detail B).
- c. If release button does not pop out, indicating that positive engagement was not achieved, perform the following:
 - (1) Manually turn the rotor head until engagement is obtained.

1-6-18.1.2 Disengage.

- a. While grasping gust lock handle using index and middle fingers, depress release button using thumb to unlock gust lock handle (Figure 1-6-18-1, Detail A).

NOTE

As gust lock handle is being pulled down, release the release button.

- b. Pull gust lock handle to full down position. The release button will pop out when gust lock is unlocked.
- c. Push up on gust lock handle when locked down to confirm that gust lock is in unlocked position.

1-6-18.2 Rotor Brake.

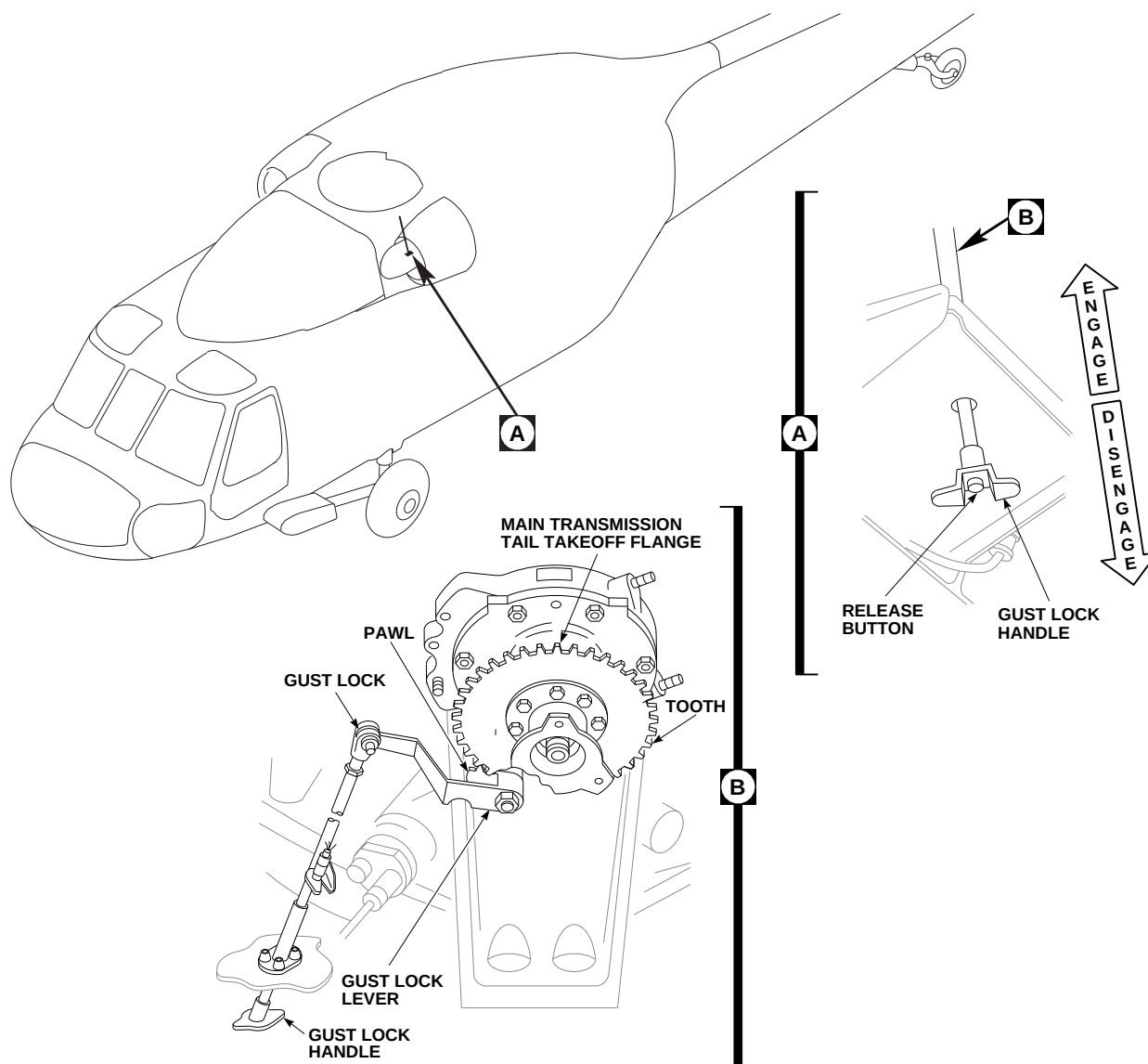
1-6-18.2.1 Engage.

CAUTION

Damage to rotor brake will result if engaged with engine(s) operating and rotor head turning. Do not engage rotor brake with engine(s) operating and rotor head turning.

NOTE

One or two pumps of the master cylinder pump handle are authorized

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SA**FIGURE 1-6-18-1. ENGAGE/DISENGAGE GUST LOCK**

providing the pressure meets the normal system operating pressure required for rotor brake engagement.

- a. In cockpit above pilot's seat, pull master cylinder pump handle down and forward (Figure 1-6-18-2, Detail A).

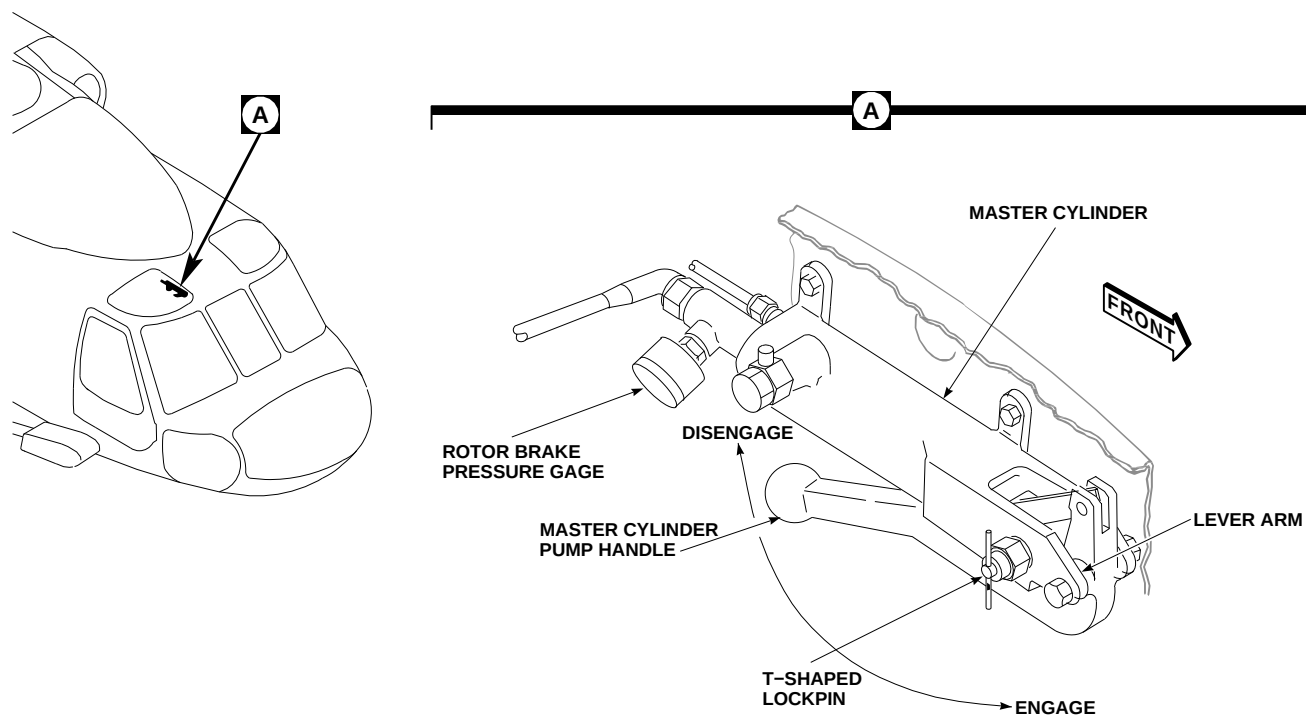
NOTE

The rotor brake handle lockpin prevents inadvertent release once the rotor brake has been engaged.

- b. Pull T-shaped lockpin out, rotate 90°, then push back in until seated into the lever arm.

1-6-18.2.2 Disengage.

- a. Pull T-shaped lockpin out, rotate 90°, then push back in until seated into the lever arm (Figure 1-6-18-2, Detail A).



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FIGURE 1-6-18-2. ENGAGE/DISENGAGE ROTOR BRAKE

NOTE

An audible click is produced when master cylinder pump handle is positively engaged in master cylinder detent.

- b. Push master cylinder pump handle up and aft until handle makes positive engagement with master cylinder detent and rotor brake pressure gage indicates 0 psi.

SECTION 1-7

PERIODIC/SPECIAL INSPECTIONS

SECTION OVERVIEW

PARAGRAPH	TITLE
1-7-1	Special Inspection Requirements
1-7-2	Engine Output Shaft Inspection - 1 to 3 Hours
1-7-3	Gear Boxes, Mounting Bolts Torque Check - 9 to 11 Hours
1-7-4	Main Rotor Blade Tip Cap Screws - 9 to 11 Hours
1-7-5	Main Rotor Head Torque Check - 9 to 11 Hours
1-7-6	Spindle Sleeve Bearing Inspection - 9 to 11 Hours
1-7-7	Outboard Retention Plate Nuts and Tail Rotor Pitch Beam Retaining Nut Torque Check - 9 to 11 Hours
1-7-8	Tail Rotor Drive Shaft Torque Check - 9 to 11 Hours
1-7-9	Tail Gear Box Inboard Retention Plate Torque Check - 9 to 11 Hours
1-7-10	Tail Rotor Cable Tension Check - 9 to 11 Hours
1-7-11	Every 30 Hours
1-7-12	Every 50 Hours
1-7-13	Every 60 Hours
1-7-14	Every 100 Hours
1-7-15	Every 150 Hours
1-7-16	Every 250 Hours
1-7-17	Every 1000 Hours
1-7-18	Before First Flight of Day
1-7-19	Every 30 Days

PARAGRAPH	TITLE
1-7-20	Every 90 Days
1-7-21	Every 120 Days
1-7-22	Every 6 Months
1-7-23	Every 12 Months
1-7-24	After Dual-Engine Operation with Gust Lock Engaged
1-7-25	After Exceeding 145 KIAS with Cabin Doors Opened
1-7-26	After Fire Extinguishing System Discharge (Halon-Type)
1-7-27	After Firing Chaff Dispenser
1-7-28	After Firing Ejector Rack, BRU-22A/A
1-7-29	After Engine Output Shaft Maintenance
1-7-30	Deleted
1-7-31	After Main Rotor Blade Contact with Tail Rotor Pylon or Tail Cone
1-7-32	After Main Rotor Blade Contact with IR Jammer, ALQ-144
1-7-33	After Operating Transmission at High Temperature or Low/No Oil Pressure
1-7-34	After Single-Engine Operation Above Idle with Gust Lock Engaged
1-7-35	Before Loading Stores - Inspect Ejector Rack, BRU-22A/A
1-7-36	Before Every Flight - Inspect Hydraulic System Components
1-7-37	Engine Output Shaft Inspection - After Disconnect
1-7-38	Engine Whine/High Frequency Vibration Inspection
1-7-39	Hard Landing
1-7-40	Helicopter Subject to Excessive Spin Rate
1-7-41	Helicopter Subject to Salt Water Immersion or Dry Chemical and Foam-Type Fire Extinguishing Agents
1-7-42	Main Rotor Blades Dropped During Blade Fold

PARAGRAPH	TITLE
1-7-43	Main Rotor Blades Subject to High Winds
1-7-44	Main Rotor Droop Stop Pounding
1-7-45	Operating Helicopter in Erosive Conditions
1-7-46	Operating Helicopter in Heavy Rainfall 70583-702374
1-7-47	Post Lightning Strike Inspection
1-7-48	Hydraulic Rescue Hoist Inspection
1-7-49	Rotor Brake Inspection
1-7-50	Sudden Stoppage
1-7-51	Tail Rotor Out of Balance - Loss of Material
1-7-52	Transmission System - Overspeed
1-7-53	Transmission System - Overtemperature S70-I
1-7-54	Transmission System - Overtemperature S70-II
1-7-55	Transmission System - Overtorque S70-I
1-7-56	Transmission System - Overtorque S70-II
1-7-57	After Tail Rotor Blade Removal
1-7-58	Electric Rescue Hoist Inspections HOIST BL-29900-30
1-7-59	Every 60 Months
1-7-60	Every 500 Hours
1-7-61	Every 120 Hours
1-7-62	Severe or Sudden Onset of Low-Frequency Main Rotor Vibration or Out-of-Track Condition

1-7-1 SPECIAL INSPECTION REQUIREMENTS.

1-7-1.1 Scope.

This section contains requirements for standards of serviceability applicable to the helicopter. The inspection requirements are in these manuals: TM 1-70-PMS-1 10 Hour/14 Day Inspection, TM 1-70-PMS-2 Periodic Inspections, and TM 1-70-23, Chapter 1, Section 7, Special Inspections.

1-7-1.2 Special Inspection Description.

The special inspections contained herein supplement the scheduled inspections in Preventive Maintenance Services Checklists, TM 1-70-PMS-1 and TM 1-70-PMS-2. These inspections are required at intervals that are not the same as the scheduled inspections. The special inspections include, but are not limited to:

NOTE

When unusual local conditions of environment, utilization, mission, experience of flight crew and maintenance personnel, or periods of inactivity etc. are encountered, the scope and/or frequency of inspections may increase as necessary to make sure of safe flight.

- a. Inspections which depend upon specific incidents, such as a hard landing or overspeed. Immediate inspection is required for further flight.
- b. Inspections which depend upon specific conditions, such as main rotor head torque check.
- c. Inspections based on calendar time.

1-7-1.3 Inspection Requirements.

Inspection requirements establish when certain equipment is to be inspected and the condition to be sought. Compliance with the inspections is required to make sure that latent problems are discovered and corrected before malfunctioning or serious trouble results. Record all inspections in helicopter log book.

1-7-1.4 Standards of Serviceability.

Standards of serviceability to be used in the day-to-day inspection and maintenance of the helicopter can be found as fits, tolerances, wear limits, and specifications in the helicopter maintenance manuals.

1-7-1.5 Scheduling.

- a. Generally, special inspections are to be accomplished within, but not to exceed $\pm 10\%$ from the scheduled time, not to exceed five flight hours or 30 calendar days.
- b. Special inspections are divided into the following four categories, describing their specific tolerances. These tolerances apply only to the special inspections found in this section:
 - (1) Special inspections pertaining to "specific incidents" (e.g., Hard Landing, Post Lightning Strike Inspection) are required to be performed immediately for further flight.
 - (2) Special inspections pertaining to "specific conditions" (e.g., Main Rotor Head Torque Check - 9 to 11 Hours, Engine Output Shaft Inspection - 3 to 13 Hours) are required to be performed in their designated period. Special inspections pertaining to torque checks are non-recurring. Once the requirements of the torque check are met, the torque check is not to be performed at its stated hourly interval (e.g., 9 to 11 Hours). Torque check requirements are stated in the necessary maintenance procedures.
 - (3) Special inspections pertaining to "flight hours" (e.g., Every 30 Hours, Every 500 Hours) are to be accomplished within, but not to exceed $\pm 10\%$ from the scheduled time, and not to exceed five flight hours.
 - (4) Special inspections pertaining to "calendar time" (e.g., Every 90 Days, Every 12 Months) are to be accomplished within,

but not to exceed $\pm 10\%$ from the scheduled time, or a ± 30 day tolerance is allowed. In the case of the Every 30 Day special inspection, only the $\pm 10\%$ applies.

1-7-1.6 Work Platforms.



Damage can occur if limits noted on engine cowlings are exceeded. Maxi-

mum weight, engine cowlings can support including equipment, is 400 pounds. No more than two people are permitted on engine cowlings at any one time, with no one person weighing over 250 pounds. Do not apply load to vertical surface.

Work platforms located on engine cowlings provide access to the engines. Each platform is capable of supporting a static weight of 400 pounds.

1-7-2 ENGINE OUTPUT SHAFT INSPECTION.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-2.1	Engine Output Shaft Inspection - 1 to 3 Hours 70361-08004-042, -043	1-7-2-1	1-7-2.2	Engine Output Shaft Inspection - 3 to 13 Hours, 70361-08014-041	1-7-2-3

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Powerplant Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Epoxy Primer Coating Kit, Item 135, Appendix D
- Sealing Compound, Item 286, Appendix D

References

- Appendix D
- PARA 4-4-1
- PARA 4-4-4
- TM 1-70-VIB

1-7-2.1 Engine Output Shaft Inspection - 1 to 3 Hours 70361-08004-042, -043.

NOTE

Whenever engine output shaft is replaced, perform a torque check of bolts or studs connecting flexible coupling to input module flange after the first 1 to 3 flight-hours.

- Remove engine air inlet (PARA 4-4-4).

WARNING

FSCAP

Torque stabilization is a critical characteristic.

NOTE

- Perform engine output shaft vibration check whenever engine output shaft is removed or disconnected (TM 1-70-VIB).
- Check torque on three bolts or studs connecting flexible coupling to input module flange while holding nuts as follows:
 - Initially check torque of bolts or studs in tightening direction. If input module flange, 70351-08206-102, is used, check torque while holding nuts.**TORQUE MUST NOT BE LESS THAN 90 INCH-POUNDS.**
 - IF TORQUE ON ANY ONE BOLT OR STUD IS BELOW 90 INCH-POUNDS, REPLACE ALL BOLTS OR STUDS**

AND ASSEMBLY HARDWARE CONNECTING DISC PACK TO INPUT FLANGE (PARA 4-4-1). IF INPUT MODULE FLANGE, 70351-08206-041, WITH NUTPLATES IS USED, REPLACE WITH INPUT MODULE FLANGE, 70351-08206-102, IF AVAILABLE. IF INPUT MODULE FLANGE, 70351-08206-041 IS USED, AFTER TORQUE HAS STABILIZED, PLACE A DROP OF SEALING COMPOUND, ITEM 286, APPENDIX D, TO THREADS OF BOLTS ADJACENT TO NUTPLATES. MAKE SURE COMPOUND SEEPS INTO NUTPLATES. Perform engine output shaft vibration check if shaft was disconnected (TM 1-70-VIB).

- (3) Check bolts or studs in tightening direction. If input module flange, 70351-08206-102, is used, check torque while holding nut. **TORQUE MUST NOT BE LESS THAN 130 INCH-POUNDS.**
- (4) **IF TORQUE ON ANY BOLT OR STUD IS BELOW 130 INCH-POUNDS, RETORQUE TO 130 - 140 INCH-POUNDS.** Repeat torque check after each 1 to 3 flight hours until torque stabilizes.
- (5) If torque does not stabilize after third consecutive torque check, perform the following:
 - (a) Replace three bolts or studs and assembly hardware connecting disc pack to input module flange (PARA 4-4-1).
 - (b) If input module flange, 70351-08206-041 with nutplates is used, replace with input module flange, 70351-08206-102, if available.
 - (c) If input module flange, 70351-08206-041 is used, after torque has stabilized, place a drop of sealing compound, Item 286, Appendix D, to threads of bolts adjacent to nutplates. Make sure compound seeps into nutplates.

- (d) Balance engine output shaft (TM 1-70-VIB).
- c. Check torque on three nuts securing flexible coupling to engine output shaft by torquing nuts in tightening direction only while holding bolts as follows:
 - (1) Initially check nuts in tightening direction while holding bolts. **TORQUE MUST NOT BE LESS THAN 90 INCH-POUNDS.**
 - (2) **IF TORQUE ON ANY ONE NUT IS BELOW 90 INCH-POUNDS, REPLACE ENGINE OUTPUT SHAFT.** Perform an engine output shaft balance check if shaft was disconnected (TM 1-70-VIB).
 - (3) Check nuts in tightening direction while holding bolts. **TORQUE MUST NOT BE LESS THAN 130 INCH-POUNDS.**
 - (4) **IF TORQUE ON ANY NUT IS BELOW 130 INCH-POUNDS, RETORQUE TO 130 - 140 INCH-POUNDS.** Repeat torque check after each 1 to 3 flight hours until torque stabilizes.
 - (5) If torque does not stabilize after third consecutive torque check, replace engine output shaft assembly. Balance engine output shaft (TM 1-70-VIB).
- d. If bolts were loose during torque check, perform the following:
 - (1) Check helicopter logbook to make sure torque check of bolts connecting disc pack coupling to input module flange is due after initial ground run and after 1 to 3 flight hours.
 - (2) Check helicopter logbook to make sure torque check of bolts connecting disc pack coupling to engine output shaft is due after initial ground run and after 1 to 3 flight hours.
- e. Install engine air inlet (PARA 4-4-4).

1-7-2.2 Engine Output Shaft Inspection - 3 to 13 Hours, 70361-08014-041.

NOTE

Check torque on six bolts securing flange/coupling to engine output shaft in tightening direction only after first 3 to 13 flight hours.

- a. Initially check bolts in tightening direction. **TORQUE MUST NOT BE LESS THAN 90 INCH-POUNDS. IF TORQUE ON ANY BOLT IS BELOW 90 INCH-POUNDS, REPLACE ALL SIX BOLTS AND FLANGE/COUPLING.**

- b. Check bolts in tightening direction. **TORQUE MUST NOT BE LESS THAN 130 INCH-POUNDS. IF TORQUE ON ANY BOLT IS BELOW 130 INCH-POUNDS, RETORQUE TO 130 - 140 INCH-POUNDS.** Repeat torque check after each 3 to 13 flight hours until torque stabilizes.
- c. If torque does not stabilize after third consecutive torque check, replace all six bolts and flange/coupling.
- d. Apply epoxy primer coating kit, Item 135, Appendix D, to flange in area around boltheads.

1-7-3 GEAR BOXES, MOUNTING BOLTS TORQUE CHECK - 9 TO 11 HOURS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-3.1	Main Transmission	1-7-3-1	1-7-3.3	Tail Gear Box	1-7-3-2
1-7-3.2	Intermediate Gear Box	1-7-3-2			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Powerplant Repairer’s Toolkit
- Wrench

Materials/Parts

- Acetone, Item 25, Appendix D
- Corrosion Preventive Compound, Item 113, Appendix D

- Epoxy Primer Coating Kit, Item 135, Appendix D
- Paint, Polyurethane, Item 224, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 6-3-1
- PARA 6-4-31
- PARA 6-4-52

1-7-3.1 Main Transmission.

NOTE

- Whenever main transmission is replaced or the torque on mounting bolts is broken, perform a mounting bolt torque check after the first 9 to 11 flight hours.
 - Turn torque wrench in tightening direction to check torques.
- Check main transmission mounting bolts for proper torque (PARA 6-3-1).



FSCAP

- Torque stabilization check is a critical characteristic.
- If any torques were below minimums, retorque bolts.
 - Repeat mounting bolts torque check every 9 to 11 flight hours until torque stabilizes. If torque has not stabilized after three consecutive torque

checks, replace hardware and repeat torque stabilization check.

- d. After torque has stabilized, perform the following:

- (1) Using acetone, Item 25, Appendix D, remove corrosion preventive compound.
- (2) Apply sealing compound, Item 292, Appendix D, to joints around boltheads, nuts, and washers. Allow to dry.
- (3) Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to joints around boltheads, nuts, and washers. Allow to dry.
- (4) Apply one coat of polyurethane paint, Item 224, Appendix D, to joints around boltheads, nuts, and washers. Allow to dry.
- (5) Apply two coats of corrosion preventive compound, Item 113, Appendix D, to gear box attachment feet.

1-7-3.2 Intermediate Gear Box.

NOTE

- Whenever intermediate gear box is replaced or the torque on mounting bolts is broken, perform a mounting bolt torque check after the first 9 to 11 flight hours.
- Turn torque wrench in tightening direction to check torques.

- a. Check intermediate gear box mounting bolts for proper torque (PARA 6-4-31).

WARNING

FSCAP

Torque stabilization check is a critical characteristic.

- b. If any torques were below minimums, retorquer bolts.

- c. Repeat mounting bolts torque check every 9 to 11 flight hours until torque stabilizes. If torque has not stabilized after three consecutive torque checks, replace hardware and repeat torque stabilization check.

- d. After torque has stabilized, perform the following:

- (1) Using acetone, Item 25, Appendix D, remove corrosion preventive compound.
- (2) Apply sealing compound, Item 292, Appendix D, to joints around boltheads, nuts, and washers. Allow to dry.
- (3) Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to joints around boltheads, nuts, and washers. Allow to dry.
- (4) Apply one coat of polyurethane paint, Item 224, Appendix D, to joints around boltheads, nuts, and washers. Allow to dry.
- (5) Apply two coats of corrosion preventive compound, Item 113, Appendix D, to gear box attachment feet.

1-7-3.3 Tail Gear Box.

NOTE

- Whenever a tail gear box is replaced or the torque on mounting bolts is broken, perform a mounting bolt torque check after the first 9 to 11 flight hours.
- Turn torque wrench in the tightening direction to check torques.

- a. Check tail gear box mounting bolts for proper torque (PARA 6-4-52).

WARNING**FSCAP**

Torque stabilization check is a critical characteristic.

- b. If any torques were below minimums, retorque bolts.
- c. Repeat mounting bolts torque check every 9 to 11 flight hours until torque stabilizes. If torque has not stabilized after three consecutive torque checks, replace hardware and repeat torque stabilization check.
- d. After torque has stabilized, perform the following:

- (1) Using acetone, Item 25, Appendix D, remove corrosion preventive compound.
- (2) Apply sealing compound, Item 292, Appendix D, to joints around boltheads, nuts, and washers. Allow to dry.
- (3) Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to joints around boltheads, nuts, and washers. Allow to dry.
- (4) Apply one coat of polyurethane paint, Item 224, Appendix D, to joints around boltheads, nuts, and washers. Allow to dry.
- (5) Apply two coats of corrosion preventive compound, Item 113, Appendix D, to tail gear box attachment feet.

1-7-4 MAIN ROTOR BLADE TIP CAP SCREWS - 9 TO 11 HOURS.

PARAGRAPH BREAKDOWN

		PAGE
1-7-4.1	Main Rotor Blade Tip Cap Screws - 9 to 11 Hours	1-7-4-1

INITIAL SETUP

- Torque Wrench, 30 - 150 in. lbs

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 0 - 30 in. lbs

1-7-4.1 Main Rotor Blade Tip Cap Screws - 9 to 11 Hours.



FSCAP

Torque stabilization is a critical characteristic.

- a. Check torque on tip cap screws in tightening direction. **TORQUE MUST NOT BE LESS THAN 25 INCH-POUNDS.**

- b. **IF TORQUE IS 25 INCH-POUNDS OR GREATER, TORQUE HAS STABILIZED. TORQUE CHECK IS COMPLETE.**
- c. **IF TORQUE IS BELOW 25 INCH-POUNDS, TORQUE HAS NOT STABILIZED. TORQUE SCREW TO 29 - 31 INCH-POUNDS. REPEAT TORQUE CHECK EVERY 3 TO 5 FLIGHT HOURS UNTIL SCREW STABILIZES AT 25 INCH-POUNDS MINIMUM TORQUE.**

1-7-5 MAIN ROTOR HEAD TORQUE CHECK - 9 TO 11 HOURS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-5.1	General	1-7-5-2	1-7-5.6	Damper Attachment Nuts Retorque	1-7-5-6
1-7-5.2	Hub-to-Shaft Extension Bolts	1-7-5-2	1-7-5.7	Main Rotor Head Shaft Bolts	1-7-5-6
1-7-5.3	Hub-to-Shaft Expandable Pins	1-7-5-4	1-7-5.8	Spindle Attachment Bolts Retorque	1-7-5-7
1-7-5.4	Pressure Plate Bolts	1-7-5-4	1-7-5.9	Pitch Control Rod Lock-nuts Retorque	1-7-5-7
1-7-5.5	Spindle-to-Horn Attachment Bolts	1-7-5-5			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Feeler Gage
- Hex Wrench
- Nonmetallic Scraper, Locally-Made
- Powerplant Repairer’s Toolkit
- Spindle Horn Nut Torque Adapter, Locally-Made, Figure H-205, Appendix H
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 500 - 2500 in. lbs
- Torque Wrench, 100 - 500 ft lbs

Materials/Parts

- Corrosion Preventive Compound, Item 113, Appendix D
- Lockwire, Item 197, Appendix D
- Sealing Compound, Item 292, Appendix D
- Trichloroethylene, Technical, Item 343, Appendix D

References

- Appendix D
- Appendix H
- PARA 5-3-9
- PARA 5-4-1
- PARA 5-4-28
- PARA 12-4-26

1-7-5.1 General.**NOTE**

Whenever a main rotor head (new or used) is installed or the main rotor head is lowered and raised, perform a main rotor head torque check every 9 to 11 flight hours until stabilized. The expression "main rotor head torque check" as used in this procedure refers to the items and torque values shown in Figure 1-7-5-1. This figure also defines the two zones referred to in this step.

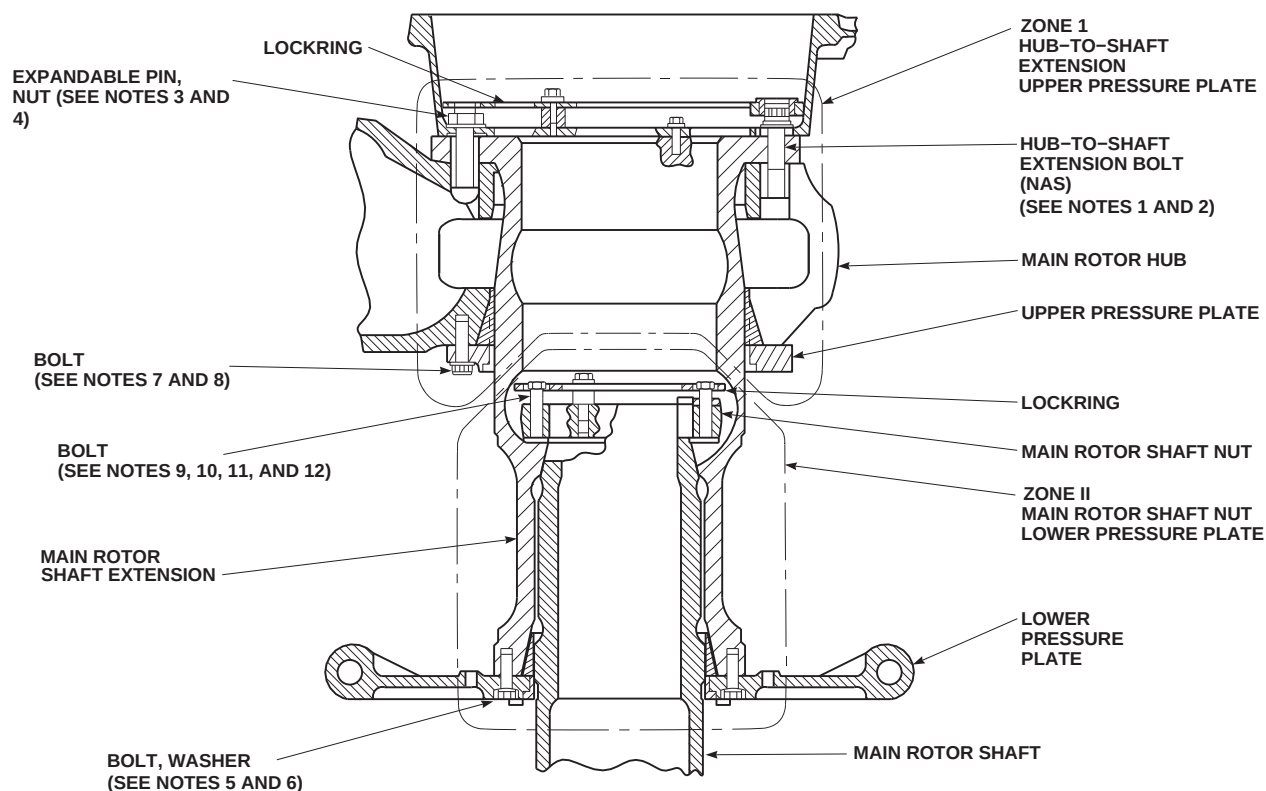
- a. Whenever spindle attachment bolts are installed, retorque them after 9 to 11 flight hours.
- b. Whenever damper attachment bolts are installed, retorque them after 9 to 11 flight hours.
- c. Whenever any spindle to horn attachment bolts are removed or replaced, or when a replacement spindle assembly or spindle horn is installed, or when antifiap assembly is removed or replaced, perform a torque check on spindle-to-horn attachment bolts every 9 to 11 flight hours until torque has stabilized.
- d. During checks, record number of bolts at each location below minimum torque to move each bolt. Turn torque wrench in tightening direction to check. Also, follow numbered sequence on main rotor shaft nut and upper and lower pressure plate when checking torque.
- e. To stabilize a fastener at required torque level, repeat specified torquing sequence until bolts do not move at required torque.
- f. The 11 hour flight interval is the maximum and shall not be exceeded. Torque checks at less than 9 flight hours are permissible in order to avoid mission interruption. Torque checks done at less than 9 flight hours, however, are not valid for purposes of stabilization. For stabilization a minimum flight time of 9 hours is required.
- g. Torque ranges for components inspected during 9 to 11 flight hours main rotor head torque

check are different from the components' installation torque range.

- h. Torque check of zone I and zone II are independent. When bolts in either zone are stabilized above minimum torque value, torque check is complete for that zone.

1-7-5.2 Hub-to-Shaft Extension Bolts.

- a. If blade deice kit is not installed, remove bifilar cover (PARA 5-4-28).
- b. If blade deice kit is installed, remove main rotor blade deice distributor (PARA 12-4-26).
- c. Remove lockring (PARA 5-4-1).
- d. Check torque on NAS hub-to-shaft extension bolts. **TORQUE MUST NOT BE LESS THAN 228 FOOT-POUNDS** (Figure 1-7-5-1).
- e. **IF TORQUE IS 228 FOOT-POUNDS OR GREATER, TORQUE HAS STABILIZED. TORQUE ALL BOLTS TO 262 FOOT-POUNDS. TORQUE CHECK IS COMPLETE.**
- f. **IF TORQUE IS LESS THAN 228 FOOT-POUNDS, TORQUE HAS NOT STABILIZED. TORQUE ALL BOLTS TO 262 FOOT-POUNDS. CHECK TORQUE ON ALL ZONE I COMPONENTS. REPEAT TORQUE CHECK AFTER 9 TO 11 FLIGHT HOURS.**
- g. If torque has not stabilized on fourth consecutive torque check, perform the following:
 - (1) Remove main rotor head (PARA 5-4-1).
 - (2) Inspect split cones, main rotor shaft extension, cone seats, bolts, upper pressure plate, lower pressure plate, and main rotor shaft nut for nicks, gouges, dents, or distortion which might prevent correct installation and torquing of parts.
 - (3) Replace parts as required and repeat torque check after 9 to 11 flight hours.
- h. Install lockring (PARA 5-4-1).



NOTES

1. MINIMUM TORQUE 228 FT LBS.
2. TORQUE 262 FT LBS MAXIMUM.
3. MINIMUM TORQUE 900 IN. LBS.
4. TORQUE 1207 IN. LBS MAXIMUM.
5. MINIMUM TORQUE 336 IN. LBS.
6. TORQUE 425 IN. LBS MAXIMUM.
7. MINIMUM TORQUE 708 IN. LBS.
8. TORQUE 945 IN. LBS MAXIMUM.
9. BREAKING TORQUE LESS THAN 288 IN. LBS ON THREE OR MORE BOLTS NOT PERMITTED.
10. BREAKING TORQUE LESS THAN 576 IN. LBS ON EIGHT OR MORE BOLTS NOT PERMITTED.
11. MINIMUM TORQUE 672 IN. LBS.
12. TORQUE 787 IN. LBS MAXIMUM.

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FIGURE 1-7-5-1. MAIN ROTOR HEAD TORQUE CHECK

- i. If blade deice kit is not installed, install bifilar cover (PARA 5-4-28).
- j. If blade deice kit is installed, install main rotor blade deice distributor (PARA 12-4-26).
- k. Make sure area is clean and free of foreign material.

1-7-5.3 Hub-to-Shaft Expandable Pins.

- a. Open main rotor pylon sliding cover.
- b. If blade deice kit is not installed, remove bifilar cover (PARA 5-4-28).
- c. If blade deice kit is installed, remove main rotor blade deice distributor (PARA 12-4-26).
- d. Remove lockring (PARA 5-4-1).
- e. Check torque on nuts on expandable pins. **TORQUE MUST NOT BE LESS THAN 900 INCH-POUNDS.** Use hex wrench to prevent pins from turning (Figure 1-7-5-1).
- f. **IF TORQUE IS 900 INCH-POUNDS OR GREATER, TORQUE HAS STABILIZED. TORQUE ALL NUTS TO 1207 INCH-POUNDS. TORQUE CHECK IS COMPLETE.**
- g. **IF TORQUE IS LESS THAN 900 INCH-POUNDS, TORQUE HAS NOT STABILIZED. TORQUE ALL NUTS TO 1207 INCH-POUNDS. CHECK TORQUE ON ALL ZONE I COMPONENTS. REPEAT TORQUE CHECK AFTER 9 TO 11 FLIGHT HOURS.**
- h. If torque has not stabilized on fourth consecutive torque check, perform the following:
 - (1) Remove main rotor blade pins and inspect for cleanness and damage (PARA 5-3-9).
 - (2) Remove main rotor head (PARA 5-4-1).
 - (3) Inspect split cones, main rotor shaft extension, cone seats, bolts, upper pressure plate, lower pressure plate, and main

rotor shaft nut for nicks, gouges, dents, or distortion, which might prevent correct installation and torquing of bolts.

- (4) Replace parts as required and repeat torque check after 9 to 11 flight hours.

- i. Install lockring (PARA 5-4-1).
- j. If blade deice kit is not installed, install bifilar cover (PARA 5-4-28).
- k. If blade deice kit is installed, install main rotor blade deice distributor (PARA 12-4-26).
- l. Make sure area is clean and free of foreign material.
- m. Close and latch main rotor pylon sliding cover.

1-7-5.4 Pressure Plate Bolts.

- a. Every three days, inspect area where corrosion preventive compound, Item 113, Appendix D, was applied and reapply as required. After each torque check, reapply two coats of corrosion preventive compound, Item 113, Appendix D.
- b. Check torque on lower pressure plate bolts in numerical sequence. **TORQUE MUST NOT BE LESS THAN 336 INCH-POUNDS** (Figure 1-7-5-1).
- c. **IF TORQUE IS LESS THAN 336 INCH-POUNDS ON TWO OR LESS BOLTS, TORQUE HAS STABILIZED. TORQUE ALL BOLTS TO 425 INCH-POUNDS.**
- d. Measure gap between lower pressure plate and main rotor shaft extension using feeler gage. **MINIMUM GAP IS 0.001-INCH. A GAP DISTANCE OF NOT OVER 0.035-INCH IS ALLOWED BETWEEN SMALLEST AND LARGEST GAP AREAS.** If gap is beyond limits, lower pressure plate and repeat torquing sequence (PARA 5-4-1).
- e. **IF TORQUE IS LESS THAN 336 INCH-POUNDS ON THREE OR MORE BOLTS, TORQUE HAS NOT STABILIZED.** Measure gap between lower pressure plate and main rotor

shaft extension using feeler gage. **MINIMUM GAP IS 0.001-INCH. A GAP DISTANCE OF NOT OVER 0.035-INCH IS ALLOWED BETWEEN SMALLEST AND LARGEST GAP AREAS.** If gap is beyond limits, lower pressure plate and repeat torquing sequence (PARA 5-4-1). **IF GAP IS WITHIN LIMITS, TORQUE ALL BOLTS TO 425 INCH-POUNDS. CHECK TORQUE ON ALL ZONE II COMPONENTS. REPEAT TORQUE CHECK AFTER 9 TO 11 FLIGHT HOURS.**

- f. If torque has not stabilized on fourth consecutive torque check, perform the following:
 - (1) Remove main rotor head (PARA 5-4-1).
 - (2) Inspect split cones, main rotor shaft extension, cone seats, bolts, upper pressure plate, lower pressure plate and main rotor shaft nut for nicks, gouges, dents, or distortion which might prevent correct installation and torquing of parts.
 - (3) Replace parts as required and repeat torque check after 9 to 11 flight hours.
- g. After torque has stabilized, remove corrosion preventive compound using technical trichloroethylene, Item 343, Appendix D.
- h. Seal gap between pressure plate and main rotor shaft extension, and areas around lockring and lower pressure plate boltheads with sealing compound, Item 292, Appendix D.
- i. Check torque on upper pressure plate bolts in numerical sequence. **TORQUE MUST NOT BE LESS THAN 708 INCH-POUNDS.**
- j. **IF TORQUE IS 708 INCH-POUNDS OR GREATER, OR IF TORQUE IS LESS THAN 708 INCH-POUNDS ON ONE BOLT, TORQUE HAS STABILIZED. TORQUE ALL BOLTS TO 945 INCH-POUNDS. TORQUE CHECK IS COMPLETE.**
- k. **IF TORQUE IS LESS THAN 708 INCH-POUNDS ON TWO OR MORE BOLTS, TORQUE HAS NOT STABILIZED.**

TORQUE ALL BOLTS TO 945 INCH-POUNDS. CHECK TORQUE ON ALL ZONE I COMPONENTS. REPEAT TORQUE CHECK AFTER 9 TO 11 FLIGHT HOURS.

- l. If torque has not stabilized on fourth consecutive torque check, perform the following:
 - (1) Remove main rotor head (PARA 5-4-1).
 - (2) Inspect split cones, main rotor shaft extension, cone seats, bolts, upper pressure plate, lower pressure plate, and main rotor shaft nut for nicks, gouges, dents, or distortion which might prevent correct installation and torquing of parts.
 - (3) Replace parts as required and repeat torque check after 9 to 11 flight hours.

1-7-5.5 Spindle-to-Horn Attachment Bolts.

WARNING

FSCAP

Torque stabilization is a critical characteristic.

NOTE

Sealing compound may not be present.

- a. If required, remove sealing compound using nonmetallic scraper.
- b. Using spindle horn nut torque adapter (Figure H-205, Appendix H) check torque of nuts. **TORQUE MUST NOT BE LESS THAN 281 INCH-POUNDS.** Proceed as follows:
 - (1) **IF TORQUE IS 281 INCH-POUNDS OR GREATER, TORQUE ALL NUTS TO 304 - 336 INCH-POUNDS. TORQUE HAS STABILIZED.**
 - (2) **IF TORQUE IS LESS THAN 281 INCH-POUNDS, RETORQUE ALL LOOSE**

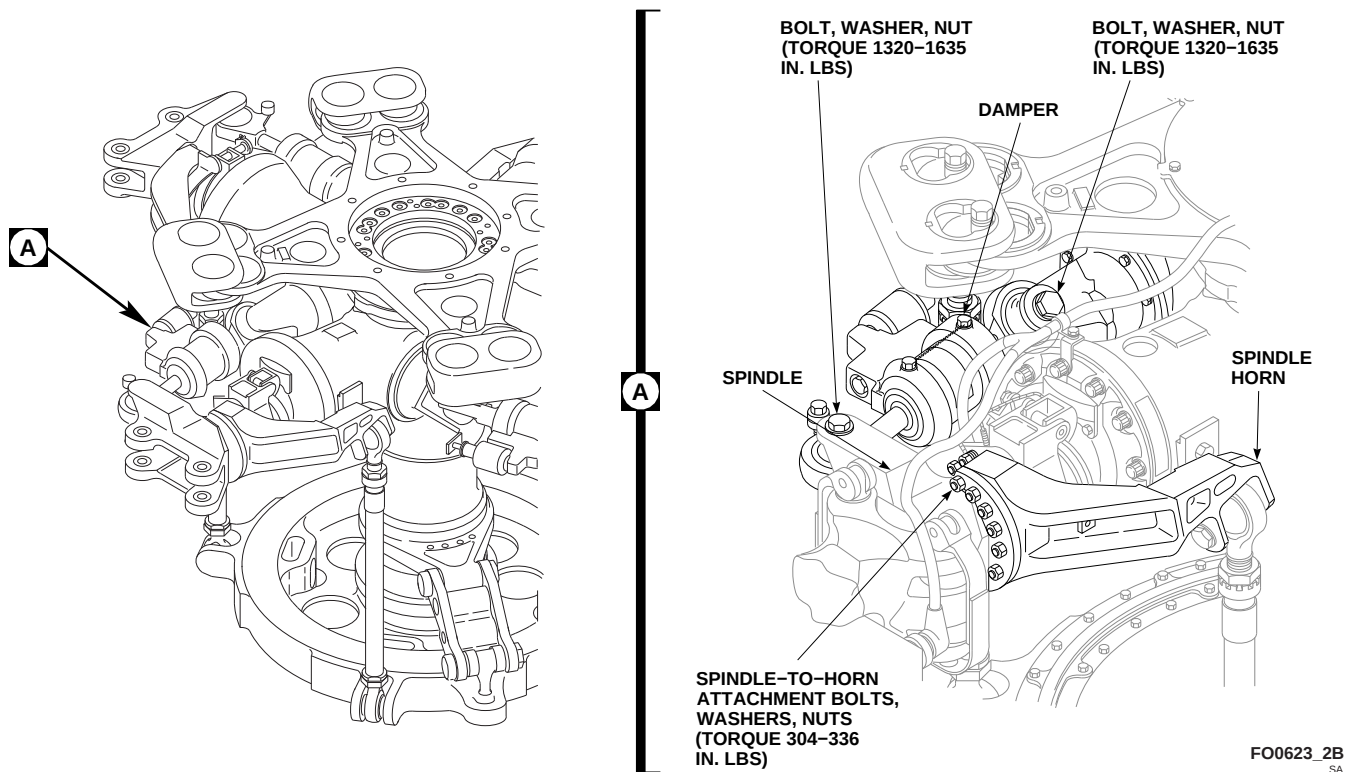


FIGURE 1-7-5-2. MAIN ROTOR HEAD TORQUE CHECK/RETORQUE

NUTS TO 304 - 336 INCH-POUNDS.

Repeat torque check at 9 to 11 flight hours until stabilized (Figure 1-7-5-2, Detail A).

- c. If, on the fourth torque check, any bolt has a breaking torque less than 281 inch-pounds, disassemble attachment horn and visually inspect clamped parts and bolts for gouges, nicks, dents, or distortion which might interfere with clamp up. Replace parts as required. Repeat torque check every 9 to 11 flight hours until stabilized.
- d. If torque has stabilized, apply sealing compound, Item 292, Appendix D, to bolt-heads, washers, and nuts.

1-7-5.6 Damper Attachment Nuts Retorque.

NOTE

Sealing compound may not be present.

- a. If required, remove sealing compound and cotter pins from damper attachment bolts.

- b. **TORQUE INBOARD AND OUTBOARD DAMPER ATTACHMENT NUTS TO 1320 - 1635 INCH-POUNDS** (Figure 1-7-5-2, Detail A).
- c. Install cotter pins. Seal area around boltheads and nuts with sealing compound, Item 292, Appendix D.

1-7-5.7 Main Rotor Head Shaft Bolts.

- a. If blade deice kit is not installed, remove bifilar cover (PARA 5-4-28).
- b. If blade deice kit is installed, remove main rotor blade deice distributor (PARA 12-4-26).
- c. Remove lockring (PARA 5-4-1).
- d. Turn torque wrench in tightening direction to check torque on bolts securing main rotor shaft nut in numerical sequence as follows (Figure 1-7-5-1):

- (1) **CHECK BREAKING TORQUE. IF BREAKING TORQUE IS LESS THAN**

288 INCH-POUNDS ON THREE OR MORE BOLTS, GO TO STEP d. (5).

- (2) **CHECK BREAKING TORQUE. IF BREAKING TORQUE IS LESS THAN 576 INCH-POUNDS ON EIGHT OR MORE BOLTS, TORQUE HAS NOT STABILIZED. GO TO STEP d. (5).**
- (3) **IF TORQUE IS LESS THAN 672 INCH-POUNDS, TORQUE HAS NOT STABILIZED. TORQUE ALL BOLTS TO 787 INCH-POUNDS. CHECK TORQUE ON ALL ZONE II COMPONENTS. REPEAT TORQUE CHECK AFTER 9 TO 11 FLIGHT HOURS.**
- (4) **IF TORQUE IS 672 INCH-POUNDS OR GREATER, TORQUE HAS STABILIZED. TORQUE ALL BOLTS TO 787 INCH-POUNDS. TORQUE CHECK IS COMPLETE.**
- (5) If torque has not stabilized after fourth consecutive torque check, or if bolts failed torque checks in step d. (1) or step d. (2), perform the following:
 - (a) Remove main rotor head (PARA 5-4-1).
 - (b) Inspect split cones, main rotor shaft extension, cone seats, bolts, upper pressure plate, lower pressure plate, and main rotor shaft nut for nicks, gouges, dents or distortion which might prevent correct installation and torquing of parts.
 - (c) Replace parts as required and repeat torque check after 9 to 11 flight hours.
- e. The lower pressure plate bolts may have dropped in torque value, as a result of tightening main rotor shaft nut bolts. Recheck torque on lower pressure plate bolts (PARA 1-7-5.4). Reseal bolts with sealing compound, Item 292, Appendix D, after torque check.

- f. Install lockring (PARA 5-4-1).
- g. If blade deice kit is not installed, install bifilar cover (PARA 5-4-28).
- h. If blade deice kit is installed, install main rotor blade deice distributor (PARA 12-4-26).
- i. Make sure area is clean and free of foreign material.

1-7-5.8 Spindle Attachment Bolts Retorque.

NOTE

Sealing compound may not be present.

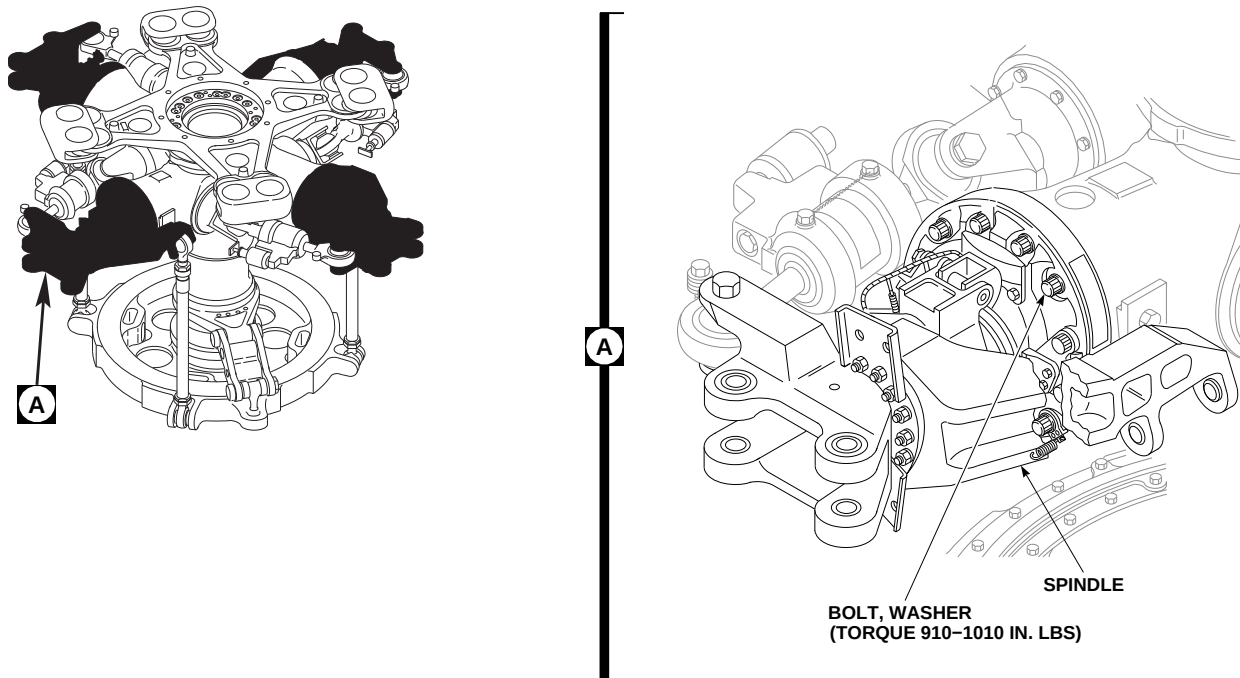
- a. If required, remove sealing compound from spindle attachment boltheads using nonmetallic scraper.
- b. **TORQUE SPINDLE ATTACHMENT BOLTS TO 910 - 1010 INCH-POUNDS IN CRISSCROSS SEQUENCE** (Figure 1-7-5-3, Detail A).
- c. Apply sealing compound, Item 292, Appendix D, around boltheads.

1-7-5.9 Pitch Control Rod Locknuts Retorque.

NOTE

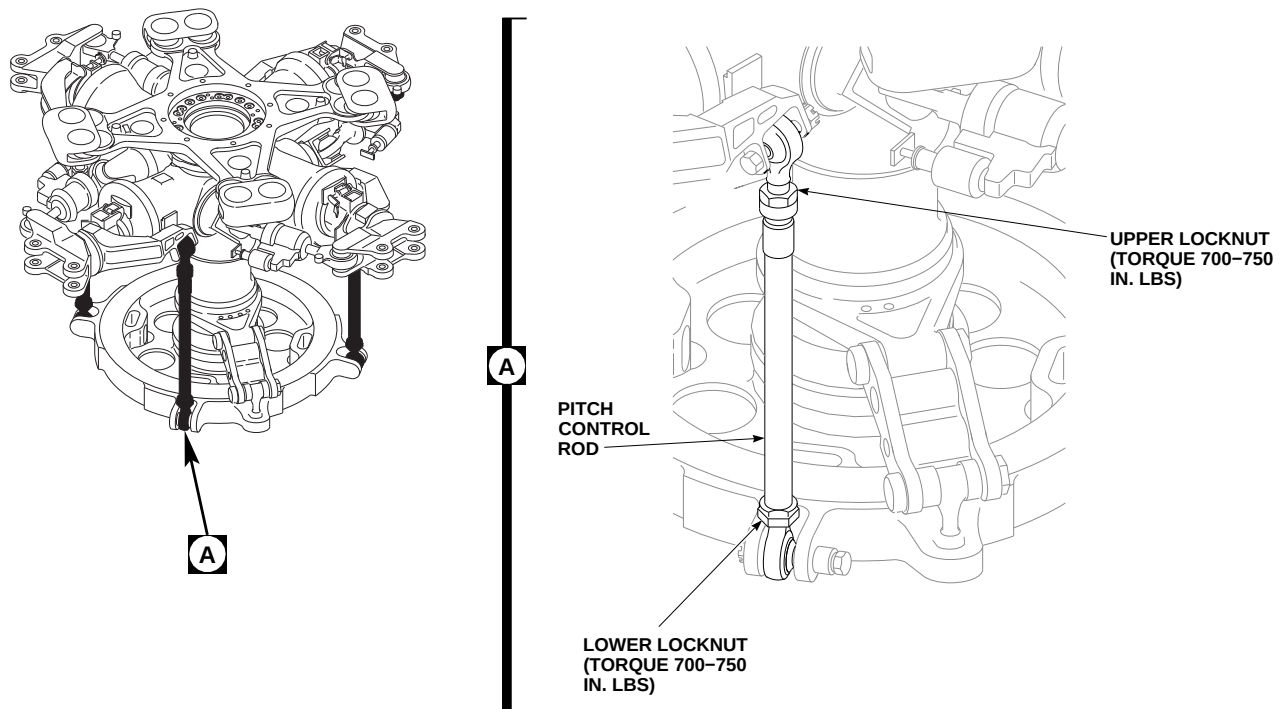
Pitch control rod locknut retorque is only required for pitch control rods equipped with elastomeric rod ends.

- a. Remove lockwire from upper and lower locknuts.
- b. **TORQUE UPPER AND LOWER LOCKNUTS TO 700 - 750 INCH-POUNDS** (Figure 1-7-5-4, Detail A).
- c. Lockwire, Item 197, Appendix D, lower locknut to link, and then back to lower locknut (double-safety).
- d. Lockwire, Item 197, Appendix D, upper locknut to lock key, and then back to opposite side of upper locknut (double-safety).



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FIGURE 1-7-5-3. SPINDLE ATTACHMENT BOLTS RETORQUE



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FIGURE 1-7-5-4. PITCH CONTROL ROD LOCKNUTS RETORQUE

1-7-6 SPINDLE SLEEVE BEARING INSPECTION - 9 TO 11 HOURS.

PARAGRAPH BREAKDOWN

		PAGE
1-7-6.1	Inspect Spindle Sleeve Bearing - 9 to 11 Hours	1-7-6-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Flashlight
- Inspection Mirror

References

- PARA 5-4-5

1-7-6.1 Inspect Spindle Sleeve Bearing - 9 to 11 Hours.

- a. Using inspection mirror and flashlight, visually check spindle sleeve bearing for metal-to-metal contact in and around outboard end face of sleeve bearing. **NONE ALLOWED.** Remove, disassemble, and inspect spindle sleeve bearing inner and outer sleeve if metal to metal contact is found (PARA 5-4-5).
- b. Using inspection mirror and flashlight, visually check spindle sleeve bearing for Teflon fabric extrusion in and around outboard end face of sleeve bearing. **NONE ALLOWED.** Remove, disassemble, and inspect spindle sleeve bearing inner and outer sleeve if Teflon fabric extrusion is found (PARA 5-4-5).

1-7-7 OUTBOARD RETENTION PLATE NUTS AND TAIL ROTOR PITCH BEAM RETAINING NUT TORQUE CHECK - 9 TO 11 HOURS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-7.1	Outboard Retention Plate Nuts Retorque - 9 to 11 Hours	1-7-7-1	1-7-7.2	Tail Rotor Pitch Beam Retaining Nut Torque Check - 9 to 11 Hours	1-7-7-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Spanner Wrench
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs

Materials/Parts

- Corrosion Preventive Compound, Item 115, Appendix D

- Lockwire, Item 197, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 5-4-40
- PARA 11-4-113

1-7-7.1 Outboard Retention Plate Nuts Retorque - 9 to 11 Hours.

NOTE

- Inboard retention plate bolts torque must be stabilized prior to performing torque check on outboard retention plate nuts.
- Whenever any outboard retention plate nut torque is broken, a torque check on all nuts is required after the first 9 to 11 flight hours.

TORQUE OUTBOARD RETENTION PLATE NUTS TO 627 - 693 INCH-POUNDS IN A CRISSCROSS PATTERN (Figure 1-7-7-1).

1-7-7.2 Tail Rotor Pitch Beam Retaining Nut Torque Check - 9 to 11 Hours.

NOTE

- Inboard retention plate nut and outboard retention plate nut torques must be stabilized prior to performing torque check on tail rotor pitch beam retaining nut.
 - Whenever tail rotor pitch beam retaining nut torque is broken, a torque check on nut is required after the first 9 to 11 flight hours.
- Remove screws and washers from retaining nut (Figure 1-7-7-1).

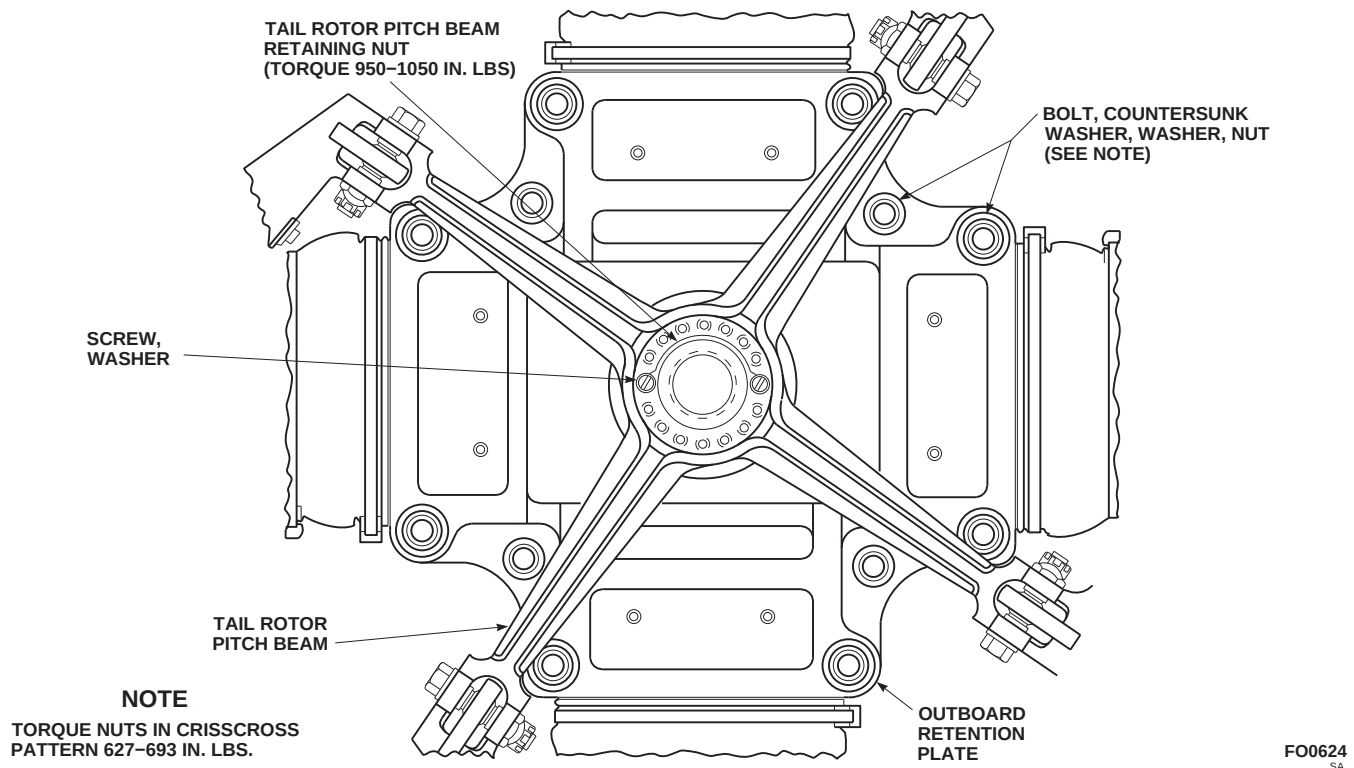
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FIGURE 1-7-7-1. OUTBOARD RETENTION PLATE NUTS RETORQUE AND TAIL ROTOR PITCH BEAM RETAINING NUT TORQUE CHECK

- b. Check torque on pitch beam retaining nut in tightening direction. **TORQUE MUST NOT BE LESS THAN 950 INCH-POUNDS.**
- c. **IF TORQUE IS 950 INCH-POUNDS OR GREATER, TORQUE HAS STABILIZED. TORQUE CHECK IS COMPLETE.**
- d. Install screws and washers to retaining nut. **TIGHTEN SCREWS SNUG. DO NOT TORQUE.** Lockwire, Item 197, Appendix D, screwheads. Proceed to step h. to apply corrosion preventive compound.
- e. If torque is below 950 inch-pounds, torque has not stabilized. Torque nut as follows:
 - (1) **TORQUE NUT TO 950 - 1050 INCH-POUNDS USING SPANNER WRENCH. DO NOT EXCEED 1050 INCH-POUNDS TORQUE LIMIT WHEN ATTEMPTING TO ALIGN HOLES IN NUT AND WASHER.**
 - (2) If holes in retaining nut do not align with holes in washer when 1050 inch-pounds is reached, perform the following:
 - (a) Remove tail rotor pitch beam (PARA 5-4-40). Check pitch beam conical seat and washer mating surface, retaining nut, washer, and pitch control shaft tapered surface for damage or debris which may prevent alignment of holes. Replace components as required.
 - (b) Reinstall tail rotor pitch beam (PARA 5-4-40). Install washer 180° from original position. Line up pitch beam lands with slot in washer. Turn pitch beam as necessary.
 - (3) Install screws and washers to retaining nut. **TIGHTEN SCREWS SNUG, DO NOT TORQUE.** Lockwire, Item 197, Appendix D.

- (4) Repeat torque check after 9 to 11 flight hours.
- f. If torque does not stabilize after three consecutive torque checks, perform the following:
- (1) Remove tail rotor pitch beam (PARA 5-4-40).
 - (2) Inspect pitch beam conical seat and pitch control shaft tapered surface for excessive corrosion preventive compound buildup.
 - (3) Inspect pitch beam, retaining nut, and washer for damage or debris which might prevent correct installation and torquing of parts. Replace parts as required (PARA 5-4-40).
 - (4) Inspect pitch control shaft for damage or debris which might prevent correct installation and torquing of parts. Replace pitch control shaft as required (PARA 11-4-113).
- (5) Clean pitch beam, retaining nut, washer, and pitch control shaft and install tail rotor pitch beam (PARA 5-4-40). Repeat torque check after 9 to 11 flight hours.
- g. If torque does not stabilize after 3 more consecutive torque checks, replace tail rotor pitch beam, retaining nut, washer, and pitch control shaft (PARA 5-4-40 and PARA 11-4-113). Repeat torque check.
- h. Apply corrosion preventive compound, Item 115, Appendix D, to pitch beam retaining nut, washer, and pitch change shaft thread mating surfaces and inboard pitch beam to pitch change shaft mating surface. After torque has stabilized, apply sealing compound, Item 292, Appendix D, to pitch beam retaining nut, washer, pitch change shaft threaded mating surfaces and inboard pitch beam to pitch shaft mating surfaces.

1-7-8 TAIL ROTOR DRIVE SHAFT TORQUE CHECK - 9 TO 11 HOURS.

PARAGRAPH BREAKDOWN

		PAGE
1-7-8.1	Tail Rotor Drive Shaft Torque Check - 9 to 11 Hours	1-7-8-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 150 - 750 in. lbs

References

- PARA 6-4-38 through PARA 6-4-41

1-7-8.1 Tail Rotor Drive Shaft Torque Check - 9 to 11 Hours.



FSCAP

Torque stabilization check is a critical characteristic.



Tail rotor drive shaft section I, 70361-23001-047, is a dynamically balanced assembly. Do not disturb the hardware attaching adapters and inner flexible couplings to drive shaft.

NOTE

- Perform a torque stabilization check if tail rotor drive shafts are removed, or if the installation torque is broken.

- Check tail rotor drive shaft torque stabilization after 9 to 11 flight hours and every 9 to 11 flight hours thereafter until no loss of torque is noted.
- If inner flexible coupling on tail rotor drive shaft section I is damaged, drive shaft must be replaced.
 - a. Check torque on all self-locking nuts on tail rotor drive shaft couplings. **TORQUE MUST NOT BE LESS THAN 328 INCH-POUNDS. IF TORQUE IS 328 INCH-POUNDS OR GREATER, TORQUE HAS STABILIZED. TORQUE CHECK IS COMPLETE.**
 - b. **IF TORQUE IS LESS THAN 328 INCH-POUNDS, TORQUE HAS NOT STABILIZED. TORQUE NUT TO 328 - 362 INCH-POUNDS.** Repeat torque check after 9 to 11 flight hours.
 - c. If torque has not stabilized after three consecutive torque checks, perform the following:
 - (1) Remove bolt, self-locking nut, and spherical washers. Inspect for damage, corrosion, and fretting.

- (2) Replace any damaged parts.
- (3) The self-locking nut may be reused, if replacement part is not available, providing the nut meets a minimum run-on torque of 9.5 inch-pounds.
- (4) Install bolt, spherical washers, and self-locking nut (PARA 6-4-38, PARA 6-4-39, PARA 6-4-40, and PARA 6-4-41). Repeat torque check after 9 to 11 flight hours.

1-7-9 TAIL GEAR BOX INBOARD RETENTION PLATE TORQUE CHECK - 9 TO 11 HOURS.

PARAGRAPH BREAKDOWN

		PAGE
1-7-9.1	Tail Gear Box Inboard Retention Plate Torque Check - 9 to 11 Hours	1-7-9-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Ink Marker
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 5-4-38
- PARA 6-4-52
- PARA 6-4-54

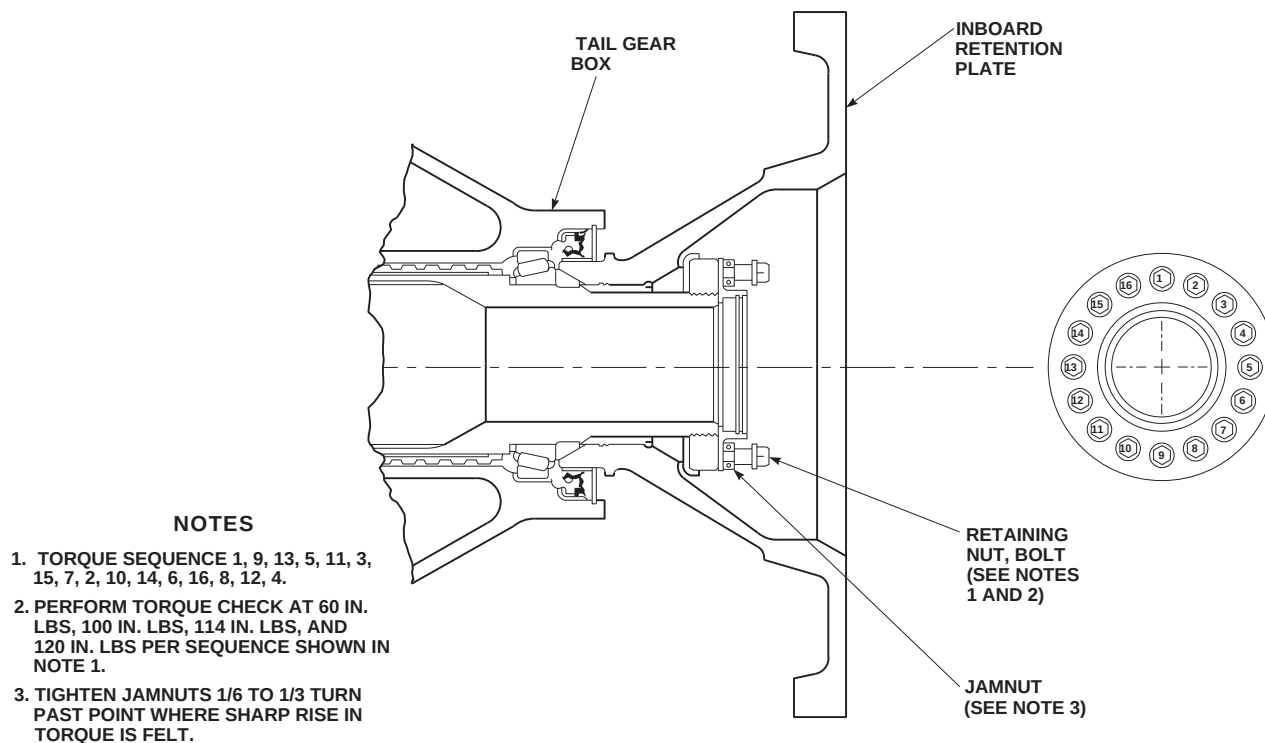
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FIGURE 1-7-9-1. TAIL GEAR BOX INBOARD RETENTION PLATE BOLTS - TORQUE CHECK

1-7-9.1 Tail Gear Box Inboard Retention Plate Torque Check - 9 to 11 Hours.



NOTE

Whenever tail gear box (new or used) is replaced, or bolts shown in Figure 1-7-9-1 are untorqued for maintenance such as inboard retention plate or seal replacement, check inboard retention plate retaining nut bolts for loss of torque 9 to 11 flight hours after installation, and every 9 to 11 flight hours thereafter, until no loss of torque is noted.

- Remove tail rotor blades (PARA 5-4-38).

Damage to retention plate will result if tools come in contact when loosening jamnuts. Do not allow tools to come in contact with retention plate when loosening jamnuts.

- Using ink marker, mark one of the sixteen bolts securing the inboard retention plate to bevel gear as a reference point and loosen jamnuts on inboard retention plate bolts.
- Check torque on tail gear box inboard retention plate bolts in the following sequence: 1, 9, 13, 5, 11, 3, 15, 7, 2, 10, 14, 6, 16, 8, 12, 4.

NOTE

Check torque in a tightening direction only.

- Using reference bolt as number one, apply torque in four steps and record bolt number and torque range at first movement as follows:

WARNING

FSCAP

Torque stabilization check is a critical characteristic.

- (1) Check torque on all 16 bolts at 60 inch-pounds using the sequence specified in step c. Record readings.
- (2) Check torque on all 16 bolts at 100 inch-pounds using the sequence specified in step c. Record readings.
- (3) Check torque on all 16 bolts at 114 inch-pounds using the sequence specified in step c. Record readings.
- (4) Check torque on all 16 bolts at 120 inch-pounds using the sequence specified in step c. Record readings.

e. On completion of torque check, perform the following:

- (1) **IF NO BOLTS MOVED BELOW 120 INCH-POUNDS, TORQUE CHECK IS COMPLETE. PROCEED TO STEP F.**
- (2) **IF NO BOLTS MOVED BELOW 114 INCH-POUNDS, BUT ONE OR MORE MOVED BETWEEN 114 - 120 INCH-POUNDS, TIGHTEN ALL BOLTS TO 120 INCH-POUNDS USING THE SAME SEQUENCE SPECIFIED IN STEP C. REPEAT SEQUENCE UNTIL ALL BOLTS HAVE STABILIZED AND NO MOTION IS FELT AT 120 INCH-POUNDS. TORQUE STABILIZATION IS COMPLETE. PROCEED TO STEP F.**
- (3) **IF ONE OR MORE BOLTS MOVED BELOW 114 INCH-POUNDS, BUT NO MORE THAN TWO BOLTS MOVED BELOW 100 INCH-POUNDS, TIGHTEN ALL BOLTS TO 120 INCH-POUNDS USING THE SAME SEQUENCE SPECIFIED IN STEP C. REPEAT SEQUENCE UNTIL ALL BOLTS HAVE STABILIZED AND NO**

MOTION IS FELT AT 120 INCH-POUNDS. CHECK HELICOPTER LOGBOOK TO MAKE SURE RETENTION PLATE TORQUE CHECK IS PERFORMED WITHIN THE NEXT 9 TO 11 FLIGHT HOURS. PROCEED TO STEP F.

- (4) If eight or more bolts moved below 60 inch-pounds, replace tail gear box (PARA 6-4-52).
- (5) If three or more bolts moved below 100 inch-pounds, but less than eight bolts moved below 60 inch-pounds, remove inboard retention plate (PARA 6-4-54) and perform the following:
 - (a) Inspect split cones, inboard retention plate sleeve, splines on inboard retention plate and bevel gear, split cone races on inboard retention plate and bevel gear, threads on bevel gear and large nut and lipped washer (PARA 6-4-52).
 - (b) Install inboard retention plate (PARA 6-4-54).

CAUTION

Damage to retention plate will result if tools come in contact while tightening jamnuts. Do not allow tools to come in contact with retention plate when tightening jamnuts.

- f. **TIGHTEN JAMNUTS $\frac{1}{6}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- g. Lockwire, Item 197, Appendix D, all bolts and jamnuts.
- h. Install tail rotor blades (PARA 5-4-38).

1-7-10 TAIL ROTOR CABLE TENSION CHECK - 9 TO 11 HOURS.

PARAGRAPH BREAKDOWN

	PAGE
1-7-10.1 Tail Rotor Cable Tension Check - 9 to 11 Hours	1-7-10-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Rigging Pin, ¼ x 6-inch
- Rigging Pin, 70700-20380-062
- Tensiometer
- Thermometer

References

- PARA 1-6-15
- PARA 11-4-6
- PARA 11-4-104
- PARA 11-4-105
- PARA 11-4-106

1-7-10.1 Tail Rotor Cable Tension Check - 9 to 11 Hours.

CAUTION

Damage to flight control components will result if hydraulic power is shut down during rigging with rigging pins still installed. Make sure all rigging pins are removed before shutting down hydraulic power.

NOTE

- Tail rotor cable tension must be stabilized on newly delivered helicopter, or on those that have had recent cable replacement. A tail rotor cable tension check must be done every 9 to 11 flight hours, until tension stabilizes (PARA 11-4-104, PARA 11-4-105, PARA 11-4-106, and Table 11-4-104-1).

- Check cable tension only after helicopter has been at an even temperature for 1 hour. Do not check tension with helicopter parked in hot sun.
- a. If using external hydraulic power, pull out BACKUP HYD CONTR circuit breaker (DC ESNTL BUS - copilot’s side upper console circuit breaker panel).
- b. Pressurize backup hydraulic system (PARA 1-6-15).
- c. On STABILATOR CONTROL/AUTO FLT CONT panel, disengage SAS 1, SAS 2, and TRIM switches. Engage BOOST switch.
- d. Using thermometer, take air temperature reading inside tail cone area close to where tensiometer readings are to be taken.
- e. Install rigging pin, 70700-20380-062, in yaw boost servo.
- f. Install rigging pin, 70700-20380-062, in collective boost servo.

- g. Take cable tensiometer readings on both left and right cables, at least 10-inches away from any cable supporting mechanism in front tail cone section (PARA 11-4-104).

NOTE

If tail rotor cable check is being performed on newly delivered helicopter or on helicopter that has had recent cable replacement, proceed to step l. There will be no earlier recorded tension readings to calculate.

- h. Record air temperature readings and cable tension readings in helicopter logbook.
- i. Add left and right cable tension readings from earlier record. Divide total by 2.
- j. Add new left and right cable tension readings. Divide total by 2 (see example).
- k. Compare temperature at which tension readings were taken. Compensate readings for temperature as follows:
- (1) Subtract lower temperature from higher temperature, to get range.
 - (2) Multiply range by 1.25 if temperature is in °F (2.25 if temperature is °C).
 - (3) Add result to average tension recorded with lower temperature. Compare this adjusted tension reading to average tension recorded for higher temperature.
READINGS MUST BE WITHIN 5 POUNDS OF EACH OTHER.

EXAMPLE:

Reading	Temperature	Tensiometer Reading (Left + Right)
1	50°F (10°C)	(117 + 121) ÷ 2 = 119 average
2 (9 to 11 flight hours later)	68°F (20°C)	(143 + 148) ÷ 2 = 145.5 average

COMPENSATE FOR TEMPERATURE

$$20^{\circ}\text{C} - 10^{\circ}\text{C} = 10^{\circ}\text{C}$$

or

$$68^{\circ}\text{F} - 50^{\circ}\text{F} = 18^{\circ}\text{F}$$

$$10^{\circ}\text{C} \times 2.25 = 22.5$$

or

$$18^{\circ}\text{F} \times 1.25 = 22.5$$

$$119 + 22.5 = 141.5$$

(tension compensated for temperature)

- l. Adjust cable tension as necessary to get equal tension readings on left and right cables (PARA 11-4-106).
- m. After cable tension checks and adjustments have been completed, attempt to install rigging pin (¼ x 6-inch) in tail rotor servo. If pin does not slide in easily, perform a tail rotor rig

check (PARA 11-4-6). If pin slides in easily, no tail rotor rig check is required.

- n. Remove rigging pin from tail rotor servo.

NOTE

When earlier and new tension readings, compensated for temperature, are within 5 pounds of each other, consider tail rotor cable tension as stabilized, and no further cable tension checks are required.

- o. Remove rigging pin, 70700-20380-062, from yaw boost servo.
- p. Remove rigging pin, 70700-20380-062, from collective boost servo.
- q. Turn off all hydraulic power (PARA 1-6-15).
- r. Push in BACKUP HYD CONTR circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel).

1-7-11 EVERY 30 HOURS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-11.1	Biflar Weights	1-7-11-2	1-7-11.7	Stabilator Hinge Fitting Torque Check	1-7-11-4
1-7-11.2	Horizontal Stores Support (HSS) Pneumatic Check Valve	1-7-11-2	1-7-11.8	Tail Rotor Pitch Control Rods with Spherical Bearings	1-7-11-5
1-7-11.3	Main Rotor Head Dampers	1-7-11-2	1-7-11.9	Tail Rotor Pylon-to-Tail Cone Bolts Torque Check	1-7-11-5
1-7-11.4	Main Rotor Pitch Control Rod Spherical Bearings	1-7-11-4	1-7-11.10	Transition Section-to-Tail Cone Bolt Torque Check	1-7-11-6
1-7-11.5	Rotating Scissors	1-7-11-4	1-7-11.11	Oil Cooler	1-7-11-6
1-7-11.6	Spindle Elastomeric Bearing Assembly	1-7-11-4			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Teflon Lined Spherical Bearing Plastic Feeler Gage, Figure H-209, Appendix H
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs

Materials/Parts

- Abrasive Cloth, Item 12, Appendix D
- Abrasive Wheel, Item 22, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- Appendix H

- PARA 1-2-1
- PARA 2-3-12
- PARA 2-6-5
- PARA 2-6-42
- PARA 5-3-1
- PARA 5-3-4
- PARA 5-3-5
- PARA 5-3-6
- PARA 5-3-7
- PARA 5-3-12
- PARA 5-4-29
- PARA 6-3-15
- PARA 16-4-27

1-7-11.1 Bifilar Weights.

- a. Inspect bifilar weights for nicks, gouges, and cracks in AREA A, AREA B, and AREA C as follows (Figure 1-7-11-1, Detail A and Detail B):

NOTE

Visually inspect each weight for cracks between and adjacent to the bushing bores on upper and lower surfaces of each flange. **NONE ALLOWED.**

- (1) AREA A - **NICKS AND GOUGES SHALL NOT EXCEED 0.150-INCH DEEP.**
- (2) AREA B - **CORNER NICKS AND GOUGES SHALL NOT EXCEED 0.10-INCH DEEP. SURFACE NICKS AND GOUGES SHALL NOT EXCEED 0.030-INCH DEEP. NO CRACKS ALLOWED.**
- (3) AREA C - **NO NICKS OR GOUGES ALLOWED.**
- (4) Using abrasive cloth, Item 12, Appendix D, or abrasive wheel, Item 22, Appendix D, blend nicks and gouges in Area A and Area B to a 0.50-inch radius.
- (5) If damage is more than limits, replace bifilar weight (PARA 5-4-29).

NOTE

Cracked paint in area C of weights with "UT" or "X" marked in area of part number is acceptable.

- b. Inspect weight marked "UT" or "X" following the part number. If the letters "UT" or "X" are present, no further inspection is required. If the letters "PL" or no part marking after the part number, visually inspect each bifilar weight for cracked paint in the bonded joint area (AREA C). If cracks in paint are found, perform the following:
 - (1) Remove paint with methyl ethyl ketone, Item 217, Appendix D.

- (2) Inspect AREAS A and B for cracks. **NONE ALLOWED.** If cracks are found, replace bifilar weight (PARA 5-4-29).
- (3) Inspect AREA C for bond separation. **NONE ALLOWED.** If bond separation is found, replace bifilar weight (PARA 5-4-29).
- (4) Touch up paint on bifilar weight as required (PARA 2-6-5).

1-7-11.2 Horizontal Stores Support (HSS) Pneumatic Check Valve.**NOTE**

The following procedure is required when operating in dry, dusty and/or sandy environment.

Clean pneumatic check valves (PARA 16-4-27).

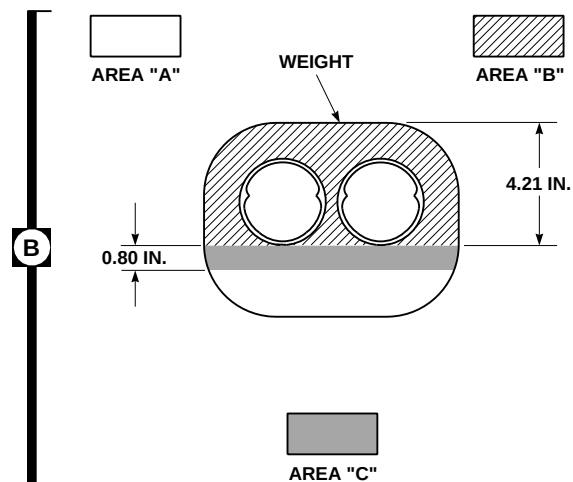
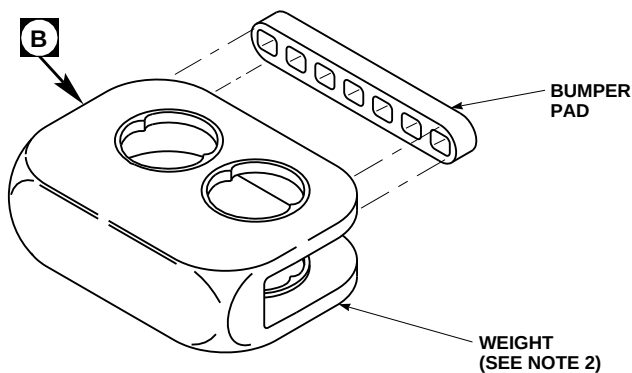
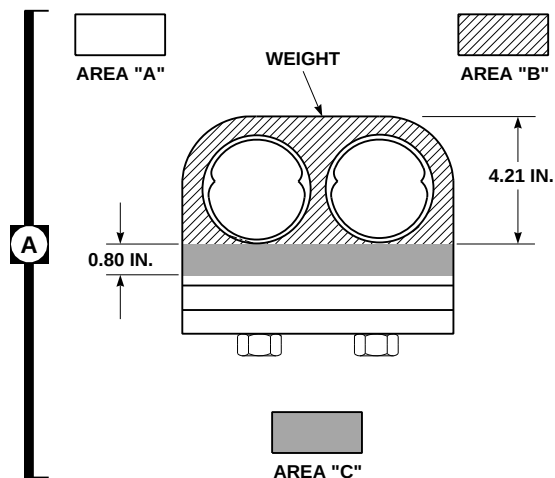
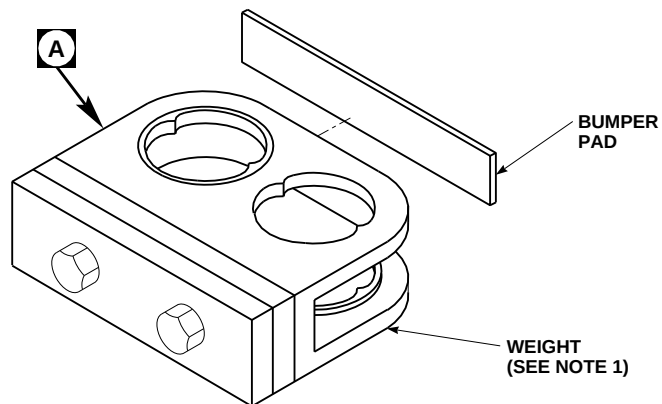
1-7-11.3 Main Rotor Head Dampers.**NOTE**

Removal of damper is not required for feeler gage inspection.

- a. Inspect inboard and outboard main rotor head damper rod end bearings for wear using 0.010-inch side of Teflon lined spherical bearing plastic feeler gage (Figure H-209, Appendix H) (PARA 5-3-4).

NOTE

- Perform the following inspections every 15 flight hours only on bearings that have failed the 0.010-inch feeler gage inspection.
 - Removal of damper assembly is not required for feeler gage inspection.
- b. Inspect inboard and outboard main rotor head damper rod end bearings for wear using 0.015-inch side of Teflon lined spherical bearing plastic feeler gage (Figure H-209, Appendix H) (PARA 5-3-4).



NOTES

1. WEIGHT, 70107-08404-042.
2. WEIGHTS, 70107-08404-043 AND 70107-08404-044.

FIGURE 1-7-11-1. INSPECT BIFILAR WEIGHTS

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SA

1-7-11.4 Main Rotor Pitch Control Rod Spherical Bearings.

- a. Inspect main rotor head upper pitch control rod end bearings for wear using 0.010-inch side of locally made, spherical bearing, plastic feeler gage (Figure H-209, Appendix H) (PARA 5-3-5).
- b. Inspect main rotor head lower pitch control rod end bearing for wear using 0.010-inch side of locally made, spherical bearing, plastic feeler gage (Figure H-209, Appendix H) (PARA 5-3-5).
- c. Inspect main rotor pitch control rods for any obvious damage or deformation (PARA 5-3-5).

NOTE

Perform the following inspections every 15 flight hours only on bearings that have failed the 0.010-inch feeler gage inspection.

- d. Inspect main rotor head upper pitch control rod end bearings for wear using 0.015-inch side of locally made, spherical bearing, plastic feeler gage (Figure H-209, Appendix H) (PARA 5-3-5).
- e. Inspect main rotor head lower pitch control rod end bearing for wear using 0.015-inch side of locally made, spherical bearing, plastic feeler gage (Figure H-209, Appendix H) (PARA 5-3-5).

1-7-11.5 Rotating Scissors.

- a. Inspect scissors attachment bearings on swash-plate for play (PARA 5-3-6).
- b. Inspect rotating scissors bearings for play (PARA 5-3-7).

1-7-11.6 Spindle Elastomeric Bearing Assembly.

Inspect elastomeric spherical and thrust bearings (PARA 5-3-1).

1-7-11.7 Stabilator Hinge Fitting Torque Check.

NOTE

Whenever a stabilator hinge fitting is replaced, removed and reinstalled, or torque on mounting bolts is broken, the following check is required each 30 flight hours until torque is stabilized for three consecutive 30 hour torque checks.

- a. Remove screws and washers securing right and left stabilator/pylon fairings. Remove fairings.
- b. Remove screws and washers securing inspection covers to underside of stabilator. Remove covers.
- c. Check torque on top and bottom $\frac{5}{16}$ -inch corner bolts which attach each hinge fitting to front spar of stabilator. **TORQUE MUST NOT BE LESS THAN 180 INCH-POUNDS.**
- d. **IF TORQUE IS LESS THAN 110 INCH-POUNDS ON ANY BOLT, REPLACE BOLT AND NUT. TORQUE BOLT TO 214 - 236 INCH-POUNDS. REPEAT TORQUE CHECK UNTIL BOLT STABILIZES AT 180 INCH-POUNDS MINIMUM FOR THREE CONSECUTIVE 30 HOUR TORQUE CHECKS.**
- e. **IF TORQUE IS 110 - 179 INCH-POUNDS, RETORQUE TOP AND BOTTOM CORNER BOLTS IN A DIAGONAL PATTERN (TOP LEFT CORNER, BOTTOM RIGHT CORNER, TOP RIGHT CORNER, BOTTOM LEFT CORNER) TO 214 - 236 INCH-POUNDS. REPEAT TORQUE CHECK UNTIL BOLT STABILIZES AT 180 INCH-POUNDS MINIMUM FOR THREE CONSECUTIVE 30 HOUR TORQUE CHECKS.**
- f. **IF TORQUE IS 110 - 179 INCH-POUNDS FOR THREE CONSECUTIVE 30 HOUR TORQUE CHECKS, REPLACE BOLT AND NUT. TORQUE BOLT TO 214 - 236 INCH-POUNDS. REPEAT TORQUE CHECK UNTIL BOLT STABILIZES AT 180 INCH-**

POUNDS MINIMUM FOR THREE CONSECUTIVE 30 HOUR TORQUE CHECKS.

- g. **IF TORQUE IS MORE THAN 180 INCH-POUNDS FOR THREE CONSECUTIVE 30 HOUR TORQUE CHECKS, TORQUE HAS STABILIZED. TORQUE CHECK IS COMPLETE.**
- h. Install inspection covers to underside of stabilator with washers and screws.
- i. Install right and left stabilator/pylon fairings with washers and screws.

1-7-11.8 Tail Rotor Pitch Control Rods with Spherical Bearings.

NOTE

Rod end spherical bearings that fail the 0.010-inch feeler gage inspection, are required to perform the 0.015-inch feeler gage inspection every 15 hours.

Inspect tail rotor blade rod end spherical bearings for wear (PARA 5-3-12).

1-7-11.9 Tail Rotor Pylon-to-Tail Cone Bolts Torque Check.

NOTE

Perform this check each time tail rotor pylon is separated from tail cone.

- a. Remove tail cone access covers (PARA 1-2-1).
- b. Check bolts and nuts securing pylon to tail cone. **TORQUE MUST NOT BE LESS THAN 1311 INCH-POUNDS.**
- c. **IF TORQUE IS GREATER THAN 1311 INCH-POUNDS, TORQUE HAS STABILIZED. TORQUE CHECK IS COM-**

plete. Apply torque stripe to bolts (PARA 2-6-5).

WARNING

FSCAP

Verification of proper hardware installation and torque of bolts is a critical characteristic.

- d. **IF TORQUE IS BELOW 1311 INCH-POUNDS, TORQUE HAS NOT STABILIZED. TORQUE BOLTS TO 1311 - 1449 INCH-POUNDS. REPEAT TORQUE CHECK EVERY 30 FLIGHT HOURS UNTIL TORQUE STABILIZES AT 1311 INCH-POUNDS MINIMUM TORQUE.** Apply torque stripe to bolts after torque stabilizes (PARA 2-6-5).
- e. If torque does not stabilize after three consecutive torque checks, perform the following:
 - (1) Replace discrepant bolt, nut, and washers.

WARNING

FSCAP

Verification of proper hardware installation and torque of bolts is a critical characteristic.

- (2) **TORQUE BOLTS TO 1311 - 1449 INCH-POUNDS. REPEAT TORQUE CHECK EVERY 30 FLIGHT HOURS UNTIL TORQUE STABILIZES AT 1311 INCH-POUNDS MINIMUM TORQUE.** Apply torque stripe to bolts after torque stabilizes (PARA 2-6-5).

WARNING**FSCAP**

Installation of radius pads is a critical characteristic. Make sure radius pads are bonded securely to tail cone.

- f. Check tail cone attachment points for four installed radius pads. Replace missing or damaged radius pads (PARA 2-6-42).
- g. Make sure area is clean and free of foreign material.

- h. Install tail cone access covers (PARA 1-2-1).

1-7-11.10 Transition Section-to-Tail Cone Bolt Torque Check.**NOTE**

Perform this check each time tail cone is separated from transition section.

Perform transition section to tail cone bolt torque check (PARA 2-3-12).

1-7-11.11 Oil Cooler.

Inspect oil cooler viscous damper bearing for debris, damage, or overheating (PARA 6-3-15).

1-7-12 EVERY 50 HOURS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-12.1	Aircraft Battery	1-7-12-1	1-7-12.3	Main Gear Box Planetary Carrier Assembly	1-7-12-2
1-7-12.2	Deswirl Duct Clamp	1-7-12-1			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electric Etcher
- Fluorescent Penetrant Inspection Kit

References

- ASTM E1417
- NAS 410

- PARA 4-4-2
- PARA 9-2-22
- PARA 9-2-23
- Special Service Instructions (SSI) No. 70-109

1-7-12.1 Aircraft Battery.

Refer to PARA 9-2-22 or PARA 9-2-23 for 50 hour battery inspection.

1-7-12.2 Deswirl Duct Clamp.

NOTE

- Perform this inspection every 50 flight hours, or before first flight following completion of 50 flight hours.
- The following inspection is applicable only to helicopters with deswirl duct clamp, 12J72-1568, installed. This

- inspection does not apply if deswirl duct clamp, 15J65-1568, 70300-30800-101, or 70300-30800-102, is installed.
- Remove clamp and deswirl duct (PARA 4-4-2).
 - Using fluorescent penetrant inspection kit, inspect clamp in area of T-bolt strap to V-retainer welds for cracks (ASTM E1417). **NONE ALLOWED.**
 - Remove torque markings, 50 - 70 inch-pounds on clamp. Remark clamp, with electric etcher to read 35 - 40 inch-pounds.

- d. Install deswirl duct and clamp (PARA 4-4-2).

1-7-12.3 Main Gear Box Planetary Carrier Assembly.

NOTE

- The following inspection is applicable to main gear box (2800 series) assemblies, 70351-08100 and 70351-78100, containing planetary carrier, 70351-08175-043, 70351-08175-044, or 70351-08175-045.
- The following inspection is to be performed on main gear box planetary carrier assemblies after accumulating 1,400 flight hours and initial ultrasonic test inspection.
- The following procedure is to be performed only by personnel certified as Level II UT technicians (or equivalent, per NAS 410) with a minimum of eight hours of Sikorsky Aircraft administered training specific to use of this procedure.
- If required, contact Sikorsky ILS for information regarding this procedure.
- a. Perform ultrasonic test inspection of main gear box planetary carrier (special service instructions (SSI) No. 70-109).
- b. Check helicopter logbook to make sure main gear box planetary carrier assembly inspection is shown due in 50 flight hours.

1-7-13 EVERY 60 HOURS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-13.1	Tail Rotor Blade Horn Assembly	1-7-13-1	1-7-13.2	Swashplate Links	1-7-13-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Allen Wrench, 1/4 inch
- Magnifying Glass, 10X
- Feeler Gage
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Alcohol, Denatured, Item 70, Appendix D
- Soap

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 5-3-13
- PARA 11-4-95

1-7-13.1 Tail Rotor Blade Horn Assembly.

- Clean surfaces of four tail rotor blade horn assemblies with soap and water or denatured alcohol, Item 70, Appendix D.
- Visually examine painted surfaces for cracks with 10X magnifying glass.
- If no cracks are found, tail rotor blade assembly may continue in service.

- If crack is suspected, refer to PARA 5-3-13.

1-7-13.2 Swashplate Links.

- Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

CAUTION

Any swashplate link assembly which exceeds or does not conform to accept/reject criteria listed in this inspection is to be replaced.

NOTE

- The following inspection is required on helicopters which have the swashplate link assemblies specified in step b. If these swashplate links are not installed, this inspection is not required.
 - Cracks of spherical bearing will be found on uniball surface usually starting at the outer lip and running parallel to the bolt axis across the bearing.
- b. With expandable pin installed, visually inspect forward longitudinal swashplate link assemblies, 70400-08110-047, 70400-08110-049, and 70400-08110-053; aft longitudinal swashplate link assemblies, 70400-08110-048, 70400-08110-050, and 70400-08110-054; and lateral swashplate link assemblies, 70400-08151-046, 70400-08151-048, and 70400-08151-050 for cracks. **A MAXIMUM OF ONE CRACK IS ALLOWED.** If no cracks are found and expandable pin is properly installed, inspection is complete.
 - c. If more than one crack is found, replace swashplate link (PARA 11-4-95).
 - d. If one crack is found, do this:
 - (1) Check Teflon liner for deterioration or extrusion between spherical bearing outer

race. **NONE ALLOWED.** Replace swashplate link if liner deterioration or extrusion is found.

- (2) Check spherical bearing for radial play. **NONE ALLOWED.** Replace swashplate link if radial play is found (PARA 11-4-95).
- (3) Check gap under expandable pin washer using feeler gage, while pin is installed. **GAP SHALL NOT EXCEED 0.002 - INCH.** If gap exceeds 0.002 inch, reinstall expandable pin to obtain correct gap (PARA 11-4-95).
- (4) If expandable pin has passed gap check, perform torque check while pin is installed as follows:
 - (a) Remove cotter pin from expandable pin.
 - (b) Insert 1/4 inch Allen wrench into end of expandable pin.
 - (c) While holding Allen wrench firmly, apply 300 inch - pounds of torque to expandable pin nut. **MINIMUM TORQUE OF 300 INCH - POUNDS MUST BE ACHIEVED WITHOUT MOVEMENT OF NUT.** If nut moves before minimum torque is reached, replace swashplate link (PARA 11-4-95)
 - (d) If minimum torque is achieved, remove Allen wrench and install new cotter pin.
- (5) If all of the above conditions are met, swashplate link may remain in use for an additional 10 flight hours before replacement is required.

1-7-14 EVERY 100 HOURS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-14.1	Engine Output Shaft, Input Module Input Flange, Elastomeric Gimbal, and Adjacent Components Inspection	1-7-14-2	1-7-14.6	Oil Cooler	1-7-14-4
			1-7-14.7	Tail Gear Box Chip Detector/Temperature Sensor	1-7-14-4
1-7-14.2	Following First External Lift Mission Exceeding 8,000 Pounds	1-7-14-4	1-7-14.8	Tail Rotor Pitch Horn Fairing	1-7-14-4
			1-7-14.9	Aircraft Battery	1-7-14-5
1-7-14.3	Intermediate Gear Box Chip Detector/ Temperature Sensor	1-7-14-4	1-7-14.10	Rotor Brake	1-7-14-5
			1-7-14.11	Accessory Module Oil Return Screen	1-7-14-6
1-7-14.4	Main Module Sump Chip Detector/Temperature Sensor	1-7-14-4	1-7-14.12	Rescue Hoist	1-7-14-6
			1-7-14.13	Main Rotor Pitch Control Rods	1-7-14-6
1-7-14.5	Main Transmission Support Beams and Upper Deck Skin	1-7-14-4			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Drill Bit, No. 60 (0.040-inch diameter)
- Flashlight
- Inspection Mirror
- Straight Edge
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 700 - 1600 in. lbs

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Sealing Compound, Item 286, Appendix D

References

- Appendix D
- PARA 1-6-18
- PARA 2-4-70
- PARA 2-6-1
- PARA 4-4-1
- PARA 4-4-4
- PARA 4-4-6
- PARA 5-4-18
- PARA 5-4-38
- PARA 6-2-2

- PARA 6-3-8
- PARA 6-4-15
- PARA 6-4-25
- PARA 6-4-28
- PARA 6-4-59
- PARA 6-4-71
- PARA 6-4-73
- PARA 9-2-22
- PARA 9-2-23
- PARA 14-3-1
- PARA 14-3-2
- PARA 14-4-1
- PARA 14-4-32
- TM 1-70-VIB

1-7-14.1 Engine Output Shaft, Input Module Input Flange, Elastomeric Gimbal, and Adjacent Components Inspection.

- a. Remove engine air inlet (PARA 4-4-4).
- b. Visually inspect accessible portion of engine output shaft with flashlight and inspection mirror through inspection hole in cowl for obvious damage. **IF DAMAGE FOUND EXCEEDS ALLOWABLE LIMITS, REPLACE ENGINE OUTPUT SHAFT** (PARA 4-4-6).
- c. Visually inspect engine output shaft for obvious damage. **IF DAMAGE FOUND EXCEEDS ALLOWABLE LIMITS, REPLACE ENGINE OUTPUT SHAFT** (PARA 4-4-6).
- d. Visually inspect disc pack coupling for cracks, obvious damage, buckling, and lamination spread (PARA 4-4-6).

WARNING

FSCAP

Verification of minimum torque of bolts is a critical characteristic.

NOTE

Whenever the engine output shaft is removed or replaced, an engine output

shaft vibration level check must be performed (TM 1-70-VIB).

- e. Check torque on three bolts connecting flexible coupling to input module input flange while securing nuts as follows:

- (1) Initially check bolts in tightening direction. If input module flange, 70351-08206-102, is used, check torque while securing nut. **TORQUE MUST NOT BE LESS THAN 90 INCH-POUNDS.**

- (2) **IF TORQUE ON ANY ONE BOLT IS BELOW 90 INCH-POUNDS, REPLACE ALL BOLTS AND ASSEMBLY HARDWARE CONNECTING DISC PACK TO INPUT FLANGE** (PARA 4-4-1). If shaft was disconnected, perform engine output shaft vibration check (TM 1-70-VIB).

- (3) **IF INPUT FLANGE, 70351-08206-041, IS USED, PERFORM THE FOLLOWING:**

- (a) **IF INPUT FLANGE, 70351-08206-041, WITH NUTPLATES IS USED, REPLACE WITH INPUT FLANGE, 70351-08206-102, IF AVAILABLE.**

- (b) **IF INPUT FLANGE, 70351-08206-041, IS USED, AFTER TORQUE HAS STABILIZED, PLACE A**

DROP OF SEALING COMPOUND, ITEM 286, APPENDIX D, TO THREADS OF BOLTS ADJACENT TO NUTPLATES. MAKE SURE COMPOUND SEEPS INTO NUTPLATES. If shaft was disconnected, perform engine output shaft vibration check (TM 1-70-VIB).

- (4) Check bolts in tightening direction. If input flange, 70351-08206-102, is used, check torque while holding nut. **TORQUE MUST NOT BE LESS THAN 130 INCH-POUNDS.**
- (5) **IF TORQUE ON ANY BOLT IS BELOW 130 INCH-POUNDS, RETORQUE TO 130 - 140 INCH-POUNDS.** Repeat torque check after each 1 to 3 flight hours until torque stabilizes.
- (6) If torque does not stabilize after third consecutive torque check, replace three bolts and assembly hardware connecting disc pack to input flange (PARA 4-4-1).
- (7) If input flange, 70351-08206-041, is used, perform the following:
 - (a) If input flange, 70351-08206-041, with nutplates is used, replace with input flange, 70351-08206-102, if available.
 - (b) If input flange, 70351-08206-041, is used, after torque has stabilized, place a drop of sealing compound, Item 286, Appendix D, to threads of bolts adjacent to nutplates. Make sure compound seeps into nutplates. Balance engine output shaft (TM 1-70-VIB).
- f. Check torque on three nuts securing flexible coupling to engine output shaft by torquing nuts in tightening direction only while securing bolts as follows:
 - (1) Initially check nuts in tightening direction while securing bolts. **TORQUE MUST**

NOT BE LESS THAN 90 INCH-POUNDS.

- (2) **IF TORQUE ON ANY ONE NUT IS BELOW 90 INCH-POUNDS, REPLACE ENGINE OUTPUT SHAFT.** If shaft was disconnected, perform an engine output shaft balance check (TM 1-70-VIB).
- (3) Check nuts in tightening direction while securing bolts. **TORQUE MUST NOT BE LESS THAN 130 INCH-POUNDS.**
- (4) **IF TORQUE ON ANY NUT IS BELOW 130 INCH-POUNDS, RETORQUE TO 130 - 140 INCH-POUNDS.** Repeat torque check after each 1 to 3 flight hours until torque stabilizes.
- (5) If torque does not stabilize after third consecutive torque check, perform the following:
 - (a) Replace engine output shaft (PARA 4-4-6).
 - (b) Balance engine output shaft (TM 1-70-VIB).
- g. Visually inspect elastomeric gimbal for bond separation and bolt breakage (PARA 6-3-8).
- h. Visually inspect for evidence of oil leakage from input module seal (PARA 6-3-8).
- i. Perform engine output shaft vibration level check (TM 1-70-VIB).
- j. Check helicopter logbook to make sure torque check of bolts connecting disc pack coupling to input module input flange is due after initial ground run and after 1 to 3 flight hours, if found loose during the torque check. Make sure torque check of bolts connecting disc pack coupling to engine output shaft is due after initial ground run and after 1 to 3 flight hours, if found loose during the torque check.
- k. Install engine air inlet (PARA 4-4-4).

1-7-14.2 Following First External Lift Mission Exceeding 8,000 Pounds.

NOTE

The following inspection is to be performed every 100 hours following first external lift mission exceeding 8,000 pounds.

Inspect cargo hook well area between STA 343.0 and STA 363.0 and BL 10.0 left and right for cracks. **NONE ALLOWED.** Pay special attention to lower caps located on STA 343.0 and STA 363.0, inboard of BL 10.0 left and right.

1-7-14.3 Intermediate Gear Box Chip Detector/Temperature Sensor.

NOTE

The following inspection is only required on helicopters without a functional built-in test system installed.

- Remove chip detector/temperature sensor and inspect self-closing valve (PARA 6-4-15).
- Perform operational check of intermediate gear box chip detector system (PARA 6-2-2).

1-7-14.4 Main Module Sump Chip Detector/Temperature Sensor.

NOTE

The following inspection is only required on helicopters without a functional built-in test system installed.

Perform operational check on chip detector in left and right accessory module, left and right input module, and main module sump (PARA 6-2-2).

1-7-14.5 Main Transmission Support Beams and Upper Deck Skin.

- Remove main transmission drip pan (PARA 2-4-70).
- Inspect left and right main transmission support beams at BL 16.5 for cracks in upper cap,

front and rear of main transmission mounting pad at STA 343.0. Inspect web stiffener attaching STA 343.0 frames to left and right support beams at BL 16.5 for cracks. **NONE ALLOWED.** If cracks are found contact, Sikorsky Aircraft ILS manager.

- Inspect upper deck skin in vicinity of gusset doubler on left and right main transmission support beams for cracks. **NONE ALLOWED.** If cracks are present in skin and not in support beam, repair skin (PARA 2-6-1).
- Inspect upper deck skin in vicinity of gusset doubler on left and right support beams for working fasteners. **NON-WORKING FASTENERS NOT ALLOWED.** If non-working fasteners are detected, replace fasteners.
- Install main transmission drip pan (PARA 2-4-70).

1-7-14.6 Oil Cooler.

Perform vibration check on oil cooler (TM 1-70-VIB).

1-7-14.7 Tail Gear Box Chip Detector/Temperature Sensor.

NOTE

The following inspection is only required on helicopters without a functional built-in test system installed.

- Remove chip detector/temperature sensor and inspect self-closing valve (PARA 6-4-59).
- Perform operational check of tail gear box chip detector system (PARA 6-2-2).

1-7-14.8 Tail Rotor Pitch Horn Fairing.

NOTE

Tail rotor blade prior to serial number A009-02083 will be inspected at 100 hour intervals, or when replacing tail rotor blades or fairing.

- Remove tiedown strap from boot and slide boot inboard.

NOTE

Interference between tail rotor blade fairing assembly and tail rotor blade spar is measured on specific tail rotor blades and are inspected for delamination and/or splintering.

- b. Check for interference between tail rotor blade fairing assembly inner lip and tail rotor blade spar.
- c. Inspect rotor blades for delamination and/or splintering. **NONE ALLOWED.** If damage is evident, repair damage or replace tail rotor blade assembly (PARA 5-4-38).

NOTE

Pay particular attention to spar corner-to-fairing clearance.

- d. Measure fairing clearance between inner lip and tail rotor blade spar. Evaluate as follows:
 - (1) If clearance is 0.350-inch or more at leading and trailing edges, slide boot back over fairing and install tiedown strap (PARA 5-4-38).
 - (2) If clearance is less than 0.350-inch at leading and trailing edges, and spar contact is evident, measure and record spar damage as follows:
 - (a) Place No. 60 (0.040-inch diameter) drill bit into damage notch.
 - (b) Place straight edge parallel to spar surface over drill bit. If straight edge touches drill bit, stop measuring and record in log, "spar damage is less than 0.040-inch in depth."
 - (c) If straight edge touches spar on each side of drill bit but does not touch drill bit, use a drill bit large enough to fill damage notch.
 - (d) Place straight edge parallel to spar surface over drill bit. If straight edge

touches drill bit, repeat measurement check using progressively smaller drill bit until straight edge touches spar on each side of drill bit but does not touch drill bit.

- (e) Record in log, "spar damage is (drill bit size) inches in depth."

NOTE

Maximum contact damage is 0.070-inch.

- (f) Repair or replace tail rotor blade (PARA 5-4-38).
- (g) Slide boot back over fairing and install tiedown strap (PARA 5-4-38).

1-7-14.9 Aircraft Battery.

Refer to PARA 9-2-22 or PARA 9-2-23 for 100 hour inspection.

1-7-14.10 Rotor Brake.

- a. Engage rotor brake (PARA 1-6-18).
- b. Clean top of brake linings and exposed piston housing with dry-cleaning solvent, Item 124, Appendix D.
- c. Check wear pin extension. **PINS SHOULD EXTEND $\frac{3}{16}$ -INCH MINIMUM BEYOND ROTOR BREAK HOUSING.** If pin extension is less than $\frac{3}{16}$ -inch, replace brake pucks (PARA 6-4-71).
- d. If brake pucks are to be replaced, check rotor brake disc thickness. **THICKNESS MUST NOT BE LESS THAN 0.410-INCHES.** If thickness is less than limit, replace rotor brake disc (PARA 6-4-73).
- e. Check rotor brake housing and disc for warping, scoring, corrosion, or cracks.
- f. Disengage rotor brake (PARA 1-6-18).

1-7-14.11 Accessory Module Oil Return Screen.**NOTE**

Helicopters with main generator serial numbers through 774 will be inspected every 100 flight hours for 500 flight hours. Thereafter, inspections shall be performed at 500 flight hour intervals as long as generator serial numbers through 774 are installed.

- a. Open main rotor pylon sliding cover.
- b. Visually inspect accessory module and input module for discoloration. If housing paint is found to be discolored, from aluminized gray to brown, remove the affected module (PARA 6-4-25 and PARA 6-4-28), and perform the following:
 - (1) Remove oil drain tubes from left and right accessory module.
 - (2) Remove and inspect oil screens for any foreign material.
 - (3) Clean oil screens with dry-cleaning solvent, Item 124, Appendix D.

- (4) Install oil screens and drain tubes.

- c. Make sure area is clean and free of foreign material.
- d. Close and latch main rotor pylon sliding cover.

1-7-14.12 Rescue Hoist.

- a. Inspect hoist support assembly and airframe fitting for cracks, corrosion, or deformation (PARA 14-4-1 or PARA 14-4-32).
- b. Check electrical cables and hydraulic lines/hoses for damage, chafing, leakage, and wear (PARA 14-3-1 or PARA 14-3-2).

1-7-14.13 Main Rotor Pitch Control Rods.

Check breakaway torque on upper main rotor pitch control rod attach bolt only with torque wrench. Apply torque in tightening direction to bolthead. If bolthead moves at torque below 700 inch-pounds, remove and inspect bolt for wear and washers for stackup and deformation (washers must be flat). Replace bolt and/or washer(s) as required (PARA 5-4-18).

1-7-15 EVERY 150 HOURS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-15.1	Tail Rotor Drive Shaft Section I Flexible Couplings	1-7-15-1	1-7-15.2	Cargo Door Upper Track Installation	1-7-15-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Magnifying Glass, 10X

References

- PARA 2-6-15
- PARA 6-4-38
- PARA 6-4-42

1-7-15.1 Tail Rotor Drive Shaft Section I Flexible Couplings.

NOTE

- This procedure is applicable to helicopters equipped with rotor brake.
- Red oxide stains on coupling discs are acceptable.
- Using 10X magnifying glass, inspect the double tail drive shaft flexible couplings on both ends of tail drive shaft section I for cracks on both sides of coupling, especially in area of conical washers. **NONE ALLOWED.**

- If outer coupling is cracked, replace tail rotor drive shaft flexible coupling (PARA 6-4-42).
- If inner coupling is cracked, replace tail rotor drive shaft section I (PARA 6-4-38).

1-7-15.2 Cargo Door Upper Track Installation.

- Check cargo door upper track fasteners for corrosion, cracks, and wear.
- If corrosion is found, clean and repair (PARA 2-6-15).
- If cracks or wear to 0.040-inch is found, replace track or hangers (PARA 2-6-15).

1-7-16 EVERY 250 HOURS.

PARAGRAPH BREAKDOWN

	PAGE
1-7-16.1 Inspect Stabilator	1-7-16-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 2-3-15
- PARA 2-4-142
- PARA 2-6-47

1-7-16.1 Inspect Stabilator.

NOTE

If stabilator is removed and reinstalled on same serial number helicopter and no adjustments or repairs have been made, no test flight is required.

- Remove stabilator (PARA 2-4-142).
- Inspect stabilator attachment bolts (PARA 2-3-15).
- Check torque on 1/4-inch center bolts which attach each hinge fitting to front spar of stabilator. **TORQUE MUST NOT BE LESS THAN 100 INCH-POUNDS.**
- IF TORQUE IS 100 INCH-POUNDS OR GREATER, FOR THREE CONSECUTIVE 250 HOUR TORQUE CHECKS, TORQUE HAS STABILIZED. TORQUE CHECK COMPLETE.**
- IF TORQUE IS LESS THAN 65 INCH-POUNDS ON ANY BOLT, REPLACE BOLT AND NUT. TORQUE BOLTS TO 128 - 142 INCH-POUNDS. REPEAT TORQUE CHECK UNTIL BOLT STABILIZES AT 100**

INCH-POUNDS MINIMUM FOR THREE CONSECUTIVE 250 HOUR TORQUE CHECKS.

- IF TORQUE IS 65 - 99 INCH-POUNDS, RE-TORQUE CENTER BOLTS IN A DIAGONAL PATTERN (TOP LEFT CENTER, BOTTOM RIGHT CENTER, TOP RIGHT CENTER, BOTTOM LEFT CENTER, THEN TWO MIDDLE BOLTS) TO 128 - 142 INCH-POUNDS. REPEAT TORQUE CHECK UNTIL BOLT STABILIZES AT 100 INCH-POUNDS MINIMUM FOR THREE CONSECUTIVE 250 HOUR TORQUE CHECKS.**
- IF TORQUE IS 65 - 99 INCH-POUNDS FOR THREE CONSECUTIVE 250 HOUR TORQUE CHECKS, REPLACE BOLT AND NUT. TORQUE BOLT TO 128 - 142 INCH-POUNDS. REPEAT TORQUE CHECK UNTIL BOLT STABILIZES AT 100 INCH-POUNDS MINIMUM FOR THREE CONSECUTIVE 250 HOUR TORQUE CHECKS.**
- Inspect double row ball bearings, mounted in pylon fitting, for cracked inner races. Replace bearings if inner races are cracked (PARA 2-6-47).

- i. Install stabilator (PARA 2-4-142).

1-7-17 EVERY 1000 HOURS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-17.1	Cabin Tub Inspection	1-7-17-1	1-7-17.3	Transition Section Tub Inspection	1-7-17-3
1-7-17.2	Flight Control Rods	1-7-17-2			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Corrosion Preventive Compound, Item 112, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- PARA 2-4-71
- PARA 2-6-12
- PARA 2-6-27
- PARA 2-6-30

- PARA 11-4-63
- PARA 11-4-64
- PARA 11-4-65
- PARA 11-4-66
- PARA 11-4-73
- PARA 11-4-75
- PARA 11-4-76
- PARA 11-4-80
- PARA 11-4-81
- PARA 11-4-82
- PARA 11-4-84
- PARA 11-4-112
- PARA 11-5-11

1-7-17.1 Cabin Tub Inspection.

NOTE

Inspection of cabin tub is to be performed every 1000 flight hours, or 36 month interval, whichever occurs first.

- Remove cabin floor (PARA 2-4-71).

- Inspect tub section for corrosion, loose or missing fasteners, damage to stringers, and general condition. If necessary, repair cabin tub section (PARA 2-6-12).
- Inspect floor panels for missing, damaged, and security of fasteners. If necessary, repair or replace missing or damaged fasteners (PARA 2-6-27).

- d. Install cabin floor (PARA 2-4-71).

1-7-17.2 Flight Control Rods.

NOTE

- Inspection of flight control rods is to be performed every 1000 flight hours, or 36 month interval, whichever occurs first.
- Inspection interval may be adjusted to coincide with the nearest 500 hour inspection, providing the 36 month interval is not exceeded.

- a. Remove the following flight control rods:

- (1) Torque shaft to servo control rods:

- (a) Collective input control rod (PARA 11-4-63).
- (b) Pitch/trim input control rod (PARA 11-4-64).
- (c) Lateral input control rod (PARA 11-4-65).
- (d) Yaw boost servo input control rod (PARA 11-4-66).

- (2) Boost servo to mixer control rods:

- (a) Collective servo to mixer control rod (PARA 11-4-73).
- (b) Roll trim assembly to mixer control rod (PARA 11-4-75).
- (c) Yaw boost servo to mixer control rod (PARA 11-4-76).

- (3) Mixer to primary servo control rods:

- (a) Mixer to forward primary servo control rod (PARA 11-4-80).
- (b) Mixer to aft primary servo control rod (PARA 11-4-81).
- (c) Mixer to lateral primary servo control rod (PARA 11-4-82).

- (4) Yaw lever to forward quadrant control rod (PARA 11-4-84).

- (5) Tail rotor servo to tail rotor quadrant pushrod (PARA 11-4-112).

NOTE

When performing bench level inspection/maintenance of control rods, do not disassemble any riveted connections when inspecting control rods. Inspect only the internal surfaces of the adapter tube that are accessible and exposed after disassembly of threaded components. **DO NOT REMOVE PLUG OR ANY CAST RESIN FROM ADAPTER TO FACILITATE INSPECTION.**

- b. Perform bench level maintenance of control rods (PARA 11-5-11).

- c. Install flight control rods:

- (1) Torque shaft to servo control rods:

- (a) Collective input control rod (PARA 11-4-63).
- (b) Pitch/trim input control rod (PARA 11-4-64).
- (c) Lateral input control rod (PARA 11-4-65).
- (d) Yaw boost servo input control rod (PARA 11-4-66).

- (2) Boost servo to mixer control rods:

- (a) Collective servo to mixer control rod (PARA 11-4-73).
- (b) Roll trim assembly to mixer control rod (PARA 11-4-75).
- (c) Yaw boost servo to mixer control rod (PARA 11-4-76).

- (3) Mixer to primary servo control rods:

- (a) Mixer to forward primary servo control rod (PARA 11-4-80).
- (b) Mixer to aft primary servo control rod (PARA 11-4-81).
- (c) Mixer to lateral primary servo control rod (PARA 11-4-82).
- (4) Yaw lever to forward quadrant control rod (PARA 11-4-84).
- (5) Tail rotor servo to tail rotor quadrant pushrod (PARA 11-4-112).

1-7-17.3 Transition Section Tub Inspection.

NOTE

Inspection of transition section tub is to be performed every 1000 flight hours, or 36 month interval, whichever occurs first.

- a. Remove fuel cell liner (PARA 2-6-30).
- b. Thoroughly clean tub structure and fuel cell liner bonding surfaces, using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.
- c. Inspect tub section for corrosion, loose or missing fasteners, damage to stringers, and general condition. If necessary, repair transition section tub (PARA 2-6-30).
- d. Coat cleaned area of tub with corrosion preventive compound, Item 112, Appendix D. **DO NOT ALLOW CORROSION PREVENTIVE COMPOUND TO CONTACT ANY AREA USED FOR BONDING FOAM BLOCKS AND FUEL CELL LINERS TO FRAME.** Allow sufficient time for corrosion preventive compound to dry completely.
- e. Install fuel cell liner (PARA 2-6-30).

1-7-18 BEFORE FIRST FLIGHT OF DAY.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-18.1	Inspect Collective and Yaw Boost Servos	1-7-18-1	1-7-18.4	Inspect Main Landing Gear Drag Beam	1-7-18-2
1-7-18.2	Inspect Stabilator Assembly	1-7-18-2	1-7-18.5	Inspect Pitch Control Rods	1-7-18-2
1-7-18.3	Inspect Main Rotor Spindle Blade Attachment Lugs	1-7-18-2	1-7-18.6	Inspect ESSS External 230 Gallon Fuel Tank Stabilizer Fin	1-7-18-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Fluorescent Penetrant Inspection Kit
- Magnifying Glass, 10X
- Nonmetallic Scraper, Locally-Made

References

- ASTM E1417
- PARA 1-7-60

- PARA 2-4-142
- PARA 2-6-5
- PARA 3-4-1
- PARA 3-4-5
- PARA 5-3-1
- PARA 5-3-5
- PARA 11-4-67
- PARA 11-4-70

1-7-18.1 Inspect Collective and Yaw Boost Servos.

- Open main rotor pylon sliding cover.
- Pull on collective stick with about 10 - 20 pounds of force. Release grip but do not push down.
- Push left pedal with about 20 - 30 pounds of force. Release load but do not push right pedal.

- Inspect both collective and yaw boost servo pistons as follows:
 - Visually inspect where output clevis attaches. Check for cracks or a break where clevis butts against piston. **NONE ALLOWED.**
 - Check for broken piston by grabbing top end of the output link between thumb and forefinger and attempt to move the link

vertically. **NO VERTICAL MOVEMENT ALLOWED.**

- (3) **REPLACE SERVO HAVING CRACKED OR BROKEN PISTON** (PARA 11-4-67 or PARA 11-4-70).

- e. Make sure area is clean and free of foreign material.
- f. Close main rotor pylon sliding cover.

1-7-18.2 Inspect Stabilator Assembly.

NOTE

Perform the following inspection on helicopters that have stabilators without inspection covers in lower skin of leading edge.

- a. Inspect upper and lower exterior surfaces of stabilator between BL 47.0 and BL 52.0 for cracks and loose rivets. **NONE ALLOWED.** If crack(s) or loose rivets are found, replace stabilator (PARA 2-4-142).
- b. Send damaged stabilator to Sikorsky Aircraft Corporation for repair (PARA 2-4-142).

1-7-18.3 Inspect Main Rotor Spindle Blade Attachment Lugs.

Inspect main rotor spindle blade attachment lugs (PARA 5-3-1).

1-7-18.4 Inspect Main Landing Gear Drag Beam.

- a. Inspect left and right main landing gear drag beams for cracks, corrosion, and blistering or flaking paint.
- b. On helicopters with main landing gear drag beam, 70250-12105-041 and -042, inspect area around jackpad and tiedown ring mounting holes as follows:

NOTE

Make sure area is clean prior to inspection. Sealing compound is not to extend more than 0.380-inch beyond edge of washers found under jackpad and retaining nut.

- (1) If excess sealing compound is found, remove using nonmetallic scraper.

NOTE

A crack will appear as a thin line in the painted surface radiating outwards from under the bead of sealing compound.

- (2) Using 10X magnifying glass, visually inspect 3-inch radius around upper and lower jackpad holes for evidence of cracking.
- (3) If crack is suspected, fluorescent penetrant inspect using fluorescent penetrant inspection kit (ASTM E1417). **NONE ALLOWED.** If cracks are found, replace drag beam (PARA 3-4-1).
- (4) Inspect jackpad nut torque stripe for slippage. If slippage has occurred, retorque nut (PARA 3-4-5), and reapply torque stripe (PARA 2-6-5).

1-7-18.5 Inspect Pitch Control Rods.

Perform pitch control rod movement inspection (PARA 5-3-5).

1-7-18.6 Inspect ESSS External 230 Gallon Fuel Tank Stabilizer Fin.

Check for looseness between stabilizer fin and tank by applying no more than 10 pounds of force to cone on the rear end of stabilizer fin. **NONE ALLOWED.** If looseness is found, perform stabilizer fin inspection (PARA 1-7-60).

1-7-19 EVERY 30 DAYS.

PARAGRAPH BREAKDOWN

		PAGE
1-7-19.1	Inspect Forward Looking Infrared Radar (FLIR)	1-7-19-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- TM 11-70-23

1-7-19.1 Inspect Forward Looking Infrared Radar (FLIR).

- a. Inspect desiccant cartridge indicator. Replace cartridge when center dot of indicator is pink (TM 11-70-23).

- b. Purge and fill nitrogen (TM 11-70-23).

1-7-20 EVERY 90 DAYS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-20.1	General	1-7-20-2	1-7-20.5	Treat Electrical/Coaxial Connectors	1-7-20-8
1-7-20.2	Inspect - Every 90 Days	1-7-20-2			
1-7-20.3	Clean Buildup of Corrosion Preventive Compound	1-7-20-6	1-7-20.6	Inspect Main Transmission Forward Bridge Drain Hole	1-7-20-9
1-7-20.4	Clean - Corrosion Removal Procedures	1-7-20-6			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Brush, Small

Materials/Parts

- Abrasive Mat, Item 14, Appendix D
- Acetone, Item 25, Appendix D
- Alcohol, Isopropyl, Item 72, Appendix D
- Brush, Acid, Item 84, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Corrosion Compound, Item 108, Appendix D
- Corrosion Preventive Compound, Item 113, Appendix D
- Corrosion Preventive Compound, Item 113A, Appendix D
- Corrosion Preventive Compound, Item 114, Appendix D
- Corrosion Preventive Compound, Item 115, Appendix D

- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D
- PARA 2-6-27
- PARA 2-6-31
- PARA 3-3-1
- PARA 3-4-1
- PARA 3-4-26
- PARA 3-4-27
- PARA 5-3-10
- PARA 5-5-14
- PARA 6-4-11
- PARA 6-4-36
- PARA 6-4-38

- PARA 6-4-39
- PARA 6-4-40
- PARA 6-4-41
- PARA 6-4-43
- PARA 6-4-46
- PARA 6-4-50
- PARA 6-4-57
- PARA 11-3-16
- PARA 11-3-17
- PARA 11-3-19
- PARA 11-3-20
- PARA 11-3-21
- PARA 11-3-22

1-7-20.1 General.**NOTE**

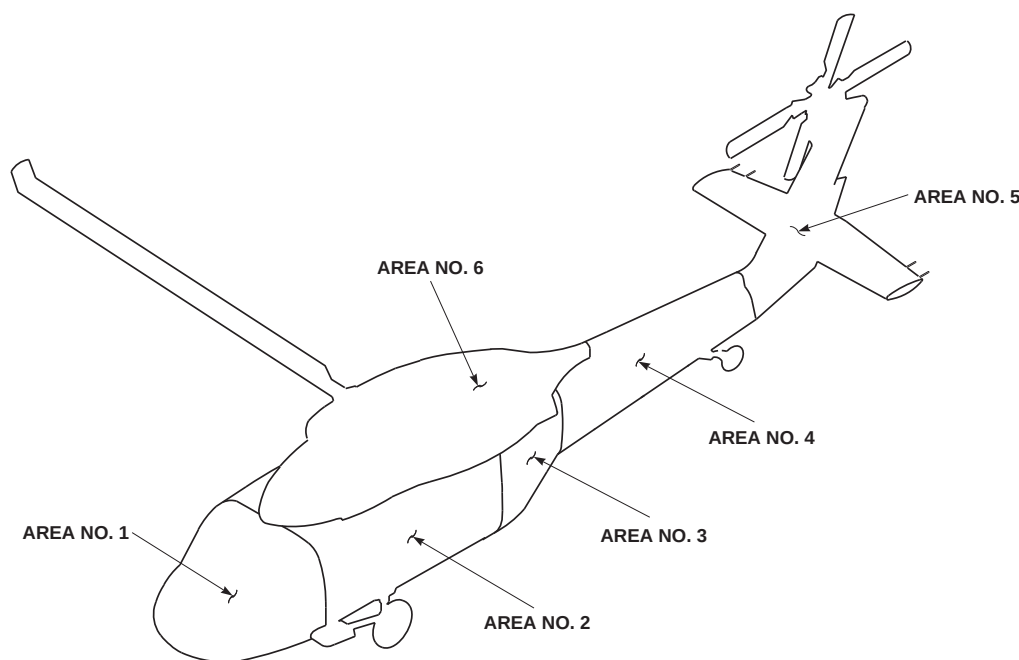
- Components listed in PARA 1-7-20.2 are to be inspected and have corrosion preventive compounds applied.
- Electrical/coaxial connectors must be treated whenever disconnected, even if corrosion is not present (PARA 1-7-20.5).

1-7-20.2 Inspect - Every 90 Days.**NOTE**

- The following areas (Figure 1-7-20-1) require inspection and application of corrosion preventive compounds to the indicated components at 90 day intervals.
- When helicopters are operating in highly corrosive environments (continued exposure to salt water or salt air), the Commander or Maintenance Officer may, at his/her discretion, increase the scope and/or frequency of the application of corrosion preventive compounds to 30 days.
- Inspection/application interval can be adjusted either way by 20% to coincide with the nearest Periodic or 100 hour inspection.

1-7-20.2.1 Area No. 1 (Cockpit Section).

- a. Clean corrosion preventive compound (PARA 1-7-20.3) from the following components and inspect for corrosion:
 - (1) Door hinge bearing rod end.
 - (2) Windshield wiper pivot stud.
 - (3) Nose door latch assembly and strut.
 - (4) Nose door hinge bolts.
 - (5) Cockpit door hinge, lever assemblies and hardware, jettison assembly.
 - (6) Parking brake control rod.
 - (7) Avionics securing hardware.
 - (8) Avionics bay connectors (internal components).
 - (9) Avionics bay connectors (external surfaces).
- b. If corrosion is found on items a. (1) through a. (7) refer to Corrosion Removal Procedure No. 1 (PARA 1-7-20.4.1).
- c. If corrosion is found on item a. (8) refer to Corrosion Removal Procedure No. 2 (PARA 1-7-20.4.2).
- d. If corrosion is found on item a. (9) refer to Corrosion Removal Procedure No. 3 (PARA 1-7-20.4.3).



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FIGURE 1-7-20-1. LOCATION OF INSPECTION AREAS

1-7-20.2.2 Area No. 2 (Cabin Section).

- a. Clean corrosion preventive compound (PARA 1-7-20.3) from the following components and inspect for corrosion:

- (1) Cabin door track.
- (2) Bolt, nut, and washer on folding step.
- (3) Main landing gear bolts at both ends of strut.
- (4) Cargo hook load beam and release lever.
- (5) Brake assembly hydraulic fittings.
- (6) Floor attachment screws.
- (7) Cargo tiedown rings and floor pans.
- (8) Drag beam pivot inside diameter.
- (9) Drag beam exterior surfaces.
- (10) 70583-702374 Interior of battery conditioner/analyzer connector.◀
- (11) Interior of battery power connector.

- (12) Exterior of main gear switch connector.
- (13) Exterior of cargo hook connectors.
- (14) 70583-702374 Exterior of battery conditioner/analyzer connector.◀
- (15) Crew chief's window latch assembly, receiver hardware, and rollers.
- (16) Cabin door latch assembly and receiver assembly.
- (17) Unpainted mounting points on jack points.

- b. If corrosion is found on items a. (1) through a. (5) refer to Corrosion Removal Procedure No. 1 (PARA 1-7-20.4.1).
- c. If corrosion is found on items a. (6) and a. (7) refer to (PARA 2-6-27).
- d. If corrosion is found on item a. (8) refer to (PARA 3-4-1).
- e. If corrosion is found on item a. (9) refer to (PARA 3-3-1).

- f. If corrosion is found on items a. (10) and a. (11) refer to Corrosion Removal Procedure No. 2 (PARA 1-7-20.4.2).
- g. If corrosion is found on items a. (12) through a. (14) refer to Corrosion Removal Procedure No. 3 (PARA 1-7-20.4.3).
- h. If corrosion is found on items a. (15) and a. (16) refer to Corrosion Removal Procedure No. 5 (PARA 1-7-20.4.5).
- i. If corrosion is found on item a. (17) refer to Corrosion Removal Procedure No. 4 (PARA 1-7-20.4.4).

1-7-20.2.3 Area No. 3 (Transition Section).

- a. Clean corrosion preventive compound (PARA 1-7-20.3) from the following components and inspect for corrosion:
 - (1) Fuel sump door hardware.
 - (2) Pressure refueling fuel access door hinges and hardware and pneumatic ground start door hardware.
 - (3) Tiedown rings.
 - (4) Doppler antenna support hinges and hardware.
 - (5) Exterior of environmental control system connectors P3, P4, P7, P456, and P457 (if applicable).
 - (6) External electrical connectors over fuel cell.
 - (7) Avionics access door, latch assembly, hinges, and hardware (if applicable).
- b. If corrosion is found on items a. (1) through a. (4) refer to Corrosion Removal Procedure No. 1 (PARA 1-7-20.4.1).
- c. If corrosion is found on items a. (5) and a. (6) refer to Corrosion Removal Procedure No. 3 (PARA 1-7-20.4.3).
- d. If corrosion is found on item a. (7) refer to (PARA 2-6-31).

1-7-20.2.4 Area No. 4 (Tail Cone Section).

- a. Clean corrosion preventive compound (PARA 1-7-20.3) from the following components and inspect for corrosion:
 - (1) Pivot bolts at both ends of oleo.
 - (2) Tail landing gear fork retaining ring.
 - (3) Anticollision light mounting hardware.
 - (4) Antenna mount screws.
 - (5) Drive shaft coupling bolts.
 - (6) Tail wheel lock connector exterior.
 - (7) Exterior of all electrical connectors.
 - (8) Tail rotor drive shaft cover hinges, nut-plates, and quarter-turn fasteners.
 - (9) Pylon pivot bolts.
 - (10) Low points inside tail cone.
 - (11) Tail wheel lock actuator, attaching hardware, and bellcrank mechanism.
 - (12) Tail wheel axle bushings.
 - (13) Tail drive shaft spherical washers.
 - (14) Tail drive shaft flange nut.
 - (15) Tail drive shaft tube to flange juncture.
- b. If corrosion is found on items a. (1) through a. (5) refer to Corrosion Removal Procedure No. 1 (PARA 1-7-20.4.1).
- c. If corrosion is found on items a. (6) and a. (7) refer to Corrosion Removal Procedure No. 3 (PARA 1-7-20.4.3).
- d. If corrosion is found on items a. (8) and a. (9) refer to Corrosion Removal Procedure No. 6 (PARA 1-7-20.4.6).
- e. If corrosion is found on item a. (10) refer to Corrosion Removal Procedure No. 7 (PARA 1-7-20.4.7).

- f. If corrosion is found on item a. (11) refer to (PARA 3-4-27).
- g. If corrosion is found on item a. (12) replace (PARA 3-4-26).
- h. If corrosion is found on item a. (13) refer to PARA 6-4-38, PARA 6-4-39, and PARA 6-4-40.
- i. If corrosion is found on item a. (14) refer to (PARA 6-4-43).
- j. If corrosion is found on item a. (15) refer to (PARA 6-4-46).

1-7-20.2.5 Area No. 5 (Tail Rotor Pylon Section).

- a. Clean corrosion preventive compound (PARA 1-7-20.3) from the following components and inspect for corrosion:
 - (1) Stabilator sensor pushrod and attaching hardware.
 - (2) Stabilator actuator shaft nut and lock-washer.
 - (3) Toggle link on tail rotor quadrant.
 - (4) Quadrant control rod including redundant bolts and bushings.
 - (5) Tail rotor pitch beam bushings.
 - (6) Tail rotor retention plate bolts.
 - (7) Tail rotor drive shaft coupling bolts.
 - (8) Exterior of all connectors.
 - (9) Tail pylon drive shaft cover hardware.
 - (10) Intermediate gear box housing.
 - (11) Tail gear box housing.
 - (12) Tail rotor drive shaft flange nuts.
 - (13) Tail rotor drive shaft spherical washers.

(14) Tail rotor pitch horn bushings.

(15) Tail rotor drive shaft to flange juncture.

- b. If corrosion is found on items a. (1) through a. (7) refer to Corrosion Removal Procedure No. 1 (PARA 1-7-20.4.1).
- c. If corrosion is found on item a. (8) refer to Corrosion Removal Procedure No. 3 (PARA 1-7-20.4.3).
- d. If corrosion is found on item a. (9) refer to Corrosion Removal Procedure No. 6 (PARA 1-7-20.4.6).
- e. If corrosion is found on item a. (10) refer to (PARA 6-4-36).
- f. If corrosion is found on item a. (11) refer to (PARA 6-4-57).
- g. If corrosion is found on items a. (12) and a. (13) refer to (PARA 6-4-41).
- h. If corrosion is found on item a. (14) refer to (PARA 5-5-14).
- i. If corrosion is found on item a. (15) refer to (PARA 6-4-46).

1-7-20.2.6 Area No. 6 (Main Rotor Pylon Section).

- a. Clean corrosion preventive compound (PARA 1-7-20.3) from the following components and inspect for corrosion:
 - (1) Main rotor pylon sliding cover latch and receiver.
 - (2) Main rotor pylon sliding cover tapered pins and receivers.
 - (3) Engine cover latch, bolts, washers, nuts, tapered lockpins, and receivers.
 - (4) Redundant flight control bolts, bushings.
 - (5) If blade deice kit is installed, unpainted clamps on blade deice.

- (6) Drive shaft coupling bolts.
- (7) Exterior of all electrical connectors.
- (8) Interior attachment of fairing above skin at STA 470.0 along stringer inside fairing.
- (9) Bifilar washers and bushings.
- (10) Main transmission housing around bolts and studs.
- (11) Tail rotor drive shaft tube to flange juncture.
- (12) Tail rotor drive shaft flange nut.
- (13) Scratched, worn painted areas on control rods, bellcranks, and tie rod attachment to gear box.
- (14) Oil cooler flange nut(s).
- (15) Boost servo plate alignment pins.
- b. If corrosion is found on items a. (1) through a. (6) refer to Corrosion Removal Procedure No. 1 (PARA 1-7-20.4.1).
- c. If corrosion is found on item a. (7) refer to Corrosion Removal Procedure No. 3 (PARA 1-7-20.4.3).
- d. If corrosion is found on item a. (8) refer to Corrosion Removal Procedure No. 7 (PARA 1-7-20.4.7).
- e. If corrosion is found on item a. (9) refer to (PARA 5-3-10).
- f. If corrosion is found on item a. (10) refer to (PARA 6-4-11).
- g. If corrosion is found on item a. (11) refer to (PARA 6-4-50).
- h. If corrosion is found on item a. (12) refer to (PARA 6-4-43).
- i. If corrosion is found on item a. (13) refer to (PARA 11-3-16, PARA 11-3-17, PARA 11-3-19, PARA 11-3-20, PARA 11-3-21, and PARA 11-3-22).

- j. If corrosion is found on item a. (14) apply corrosion preventive compound, Item 114, Appendix D, using acid brush, Item 84, Appendix D.
- k. If corrosion is found on item a. (15) apply corrosion preventive compound, Item 114, Appendix D, using acid brush, Item 84, Appendix D.

1-7-20.3 Clean Buildup of Corrosion Preventive Compound.

NOTE

Continued application of corrosion preventive compound may cause excessive buildup.

- a. Remove corrosion preventive compound using machinery towel, Item 211, Appendix D, dampened with isopropyl alcohol, Item 72, Appendix D.
- b. Wipe dry with clean machinery towel, Item 211, Appendix D.

1-7-20.4 Clean - Corrosion Removal Procedures.

1-7-20.4.1 Corrosion Removal - Procedure No. 1.

- a. Remove surface corrosion using abrasive mat, Item 14, Appendix D. Remove as much corrosion as possible without removing part. Do not remove any more material than is necessary to remove corrosion. Do not remove sealant to permit cleanup.
- b. Clean repaired area using isopropyl alcohol, Item 72, Appendix D, and acid brush, Item 84, Appendix D. Dry using clean machinery towel, Item 211, Appendix D.
- c. Inspect repaired area. **IF AFTER REWORK, LESS THAN 80% OF THE CORROSION WAS REMOVED, REPLACE COMPONENT.**

CAUTION

Damage to equipment will result if corrosion preventive compound contacts Teflon-lined spherical bearings. Teflon linings will degrade. Use care when applying corrosion preventive compound.

NOTE

Do not spray corrosion preventive compound on components with Teflon-lined spherical bearings. Use brush only.

- d. Using acid brush, Item 84, Appendix D, apply corrosion preventive compound, Item 114, Appendix D, to repaired area.

1-7-20.4.2 Corrosion Removal - Procedure No. 2.

- a. If necessary, disconnect electrical connector.
- b. Remove surface corrosion from electrical connector.
- c. Inspect repaired area. **IF AFTER REWORK, LESS THAN 80% OF THE CORROSION WAS REMOVED, REPLACE COMPONENT.**

NOTE

Avoid corrosion preventive compound, Item 115, Appendix D, from entering internal sections of electrical connector. Corrosion preventive compound, Item 115, Appendix D, deposits a thin, non-conductive film which must be removed from electrical connectors and mounting surfaces for proper operation of electrical and avionic components.

- d. Using acid brush, Item 84, Appendix D, apply corrosion preventive compound, Item 115, Appendix D, to repaired area.
- e. Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, remove corrosion preventive

compound from electrical contact points and mounting surfaces. If necessary, use small brush to remove corrosion preventive compound from inside electrical connectors.

- f. Treat electrical/coaxial connector (PARA 1-7-20.5).
- g. If necessary, connect electrical connector.

1-7-20.4.3 Corrosion Removal - Procedure No. 3.

- a. If necessary, disconnect electrical connector.
- b. Remove surface corrosion from electrical connector.
- c. Inspect repaired area. **IF AFTER REWORK, LESS THAN 80% OF THE CORROSION WAS REMOVED, REPLACE COMPONENT.**
- d. Using acid brush, Item 84, Appendix D, apply corrosion preventive compound, Item 113, Appendix D, to repaired area.
- e. Treat electrical/coaxial connector (PARA 1-7-20.5).
- f. If necessary, connect electrical connector.

1-7-20.4.4 Corrosion Removal - Procedure No. 4.

- a. Remove surface corrosion using abrasive mat, Item 14, Appendix D. Remove as much corrosion as possible without removing part. Do not remove any more material than is necessary to remove corrosion.
- b. Clean repaired area using isopropyl alcohol, Item 72, Appendix D, and acid brush, Item 84, Appendix D. Dry using clean machinery towel, Item 211, Appendix D.
- c. Inspect repaired area. **IF AFTER REWORK, CORROSION IS DEEPER THAN 0.010-INCH, REPLACE COMPONENT.**
- d. Using acid brush, Item 84, Appendix D, apply corrosion compound, Item 108, Appendix D, to repaired area.

- e. Using acid brush, Item 84, Appendix D, apply corrosion preventive compound, Item 114, Appendix D, to repaired area.

1-7-20.4.5 Corrosion Removal - Procedure No. 5.

- a. Remove surface corrosion using abrasive mat, Item 14, Appendix D. Remove as much corrosion as possible without removing part. Do not remove any more material than is necessary to remove corrosion. Do not remove sealant to permit cleanup.
- b. Clean repaired area using isopropyl alcohol, Item 72, Appendix D, and acid brush, Item 84, Appendix D. Dry using clean machinery towel, Item 211, Appendix D.
- c. Using acid brush, Item 84, Appendix D, apply corrosion preventive compound, Item 114, Appendix D, to repaired area.

1-7-20.4.6 Corrosion Removal - Procedure No. 6.

- a. Remove surface corrosion using abrasive mat, Item 14, Appendix D. Remove as much corrosion as possible without removing part. Do not remove any more material than is necessary to remove corrosion. Do not remove sealant to permit cleanup.
- b. Clean repaired area using isopropyl alcohol, Item 72, Appendix D, and acid brush, Item 84, Appendix D. Dry using clean machinery towel, Item 211, Appendix D.
- c. Inspect repaired area. **IF AFTER REWORK, LESS THAN 80% OF THE CORROSION WAS REMOVED, REPLACE COMPONENT.**
- d. Using acid brush, Item 84, Appendix D, apply corrosion compound, Item 108, Appendix D, to repaired area.
- e. Using acid brush, Item 84, Appendix D, apply corrosion preventive compound, Item 114, Appendix D, to repaired area.

1-7-20.4.7 Corrosion Removal - Procedure No. 7.

- a. Remove surface corrosion. Do not remove any more material than is necessary to remove corrosion.
- b. Clean repaired area using isopropyl alcohol, Item 72, Appendix D, and acid brush, Item 84, Appendix D. Dry using clean machinery towel, Item 211, Appendix D.
- c. Inspect repaired area. **PITS SHALL NOT EXCEED 0.005-INCH DEEP.**
- d. Treat repaired aluminum surfaces.
- e. Using epoxy primer coating kit, Item 135, Appendix D, touch up repaired area with epoxy primer.
- f. Using acid brush, Item 84, Appendix D, apply corrosion preventive compound, Item 114, Appendix D, to repaired area.

1-7-20.5 Treat Electrical/Coaxial Connectors.

NOTE

- Electrical/coaxial connectors must be treated whenever disconnected, even if corrosion is not present.
- The application of corrosion preventive compound to internal sections of plugs and receptacles will assist in displacing any moisture present in the internal sections of connector.

1-7-20.5.1 Treat Electrical Connectors.

- a. Apply a light film of corrosion preventive compound, Item 114, Appendix D, to internal sections of plugs and receptacles. Avoid excessive application of preservative.
- b. If possible, tilt plug and receptacle down to drain excess corrosion preventive compound. Wipe up extra corrosion preventive compound with cleaning cloth, Item 102, Appendix D.

- c. Lubricate threads of plug, receptacle, and back-shell with corrosion preventive compound, Item 114, Appendix D.
- d. Apply a light coating of corrosion preventive compound, Item 114, Appendix D, to outer shell of connectors, plugs, and receptacles.

1-7-20.5.2 Treat Coaxial Connectors.

NOTE

Coaxial connectors require special steps in order to avoid water intrusion. In most cases, moisture, contaminants, and corrosion in fuel quantity, and similar capacitive-type connectors, will cause erroneous quantity indications in the cockpit indicating systems. Antenna coaxial connectors have similar moisture problems.

- a. Clean internal and external sections of connector by spraying with acetone, Item 25, Appendix D. Use acid brush, Item 84, Appendix D, to clean internal sections.
- b. Wipe off excess acetone with cleaning cloth, Item 102, Appendix D.

NOTE

Do not apply corrosion preventive compound, Item 114, Appendix D, to internal areas of coaxial connectors. Application of preservative to internal sections of coaxial connectors may cause erroneous indications in system performances.

- c. Assemble connector sections and preserve external areas of connectors with a thin coating of corrosion preventive compound, Item 114, Appendix D.

1-7-20.6 Inspect Main Transmission Forward Bridge Drain Hole.

- a. Inspect drain hole on forward bridge mounting pad of main module for obstruction or corrosion (Figure 1-7-20-2, Detail B).
- b. If any evidence of obstruction or corrosion is found, remove as necessary (PARA 6-4-11).
- c. Apply corrosion preventive compound, Item 113A, Appendix D.

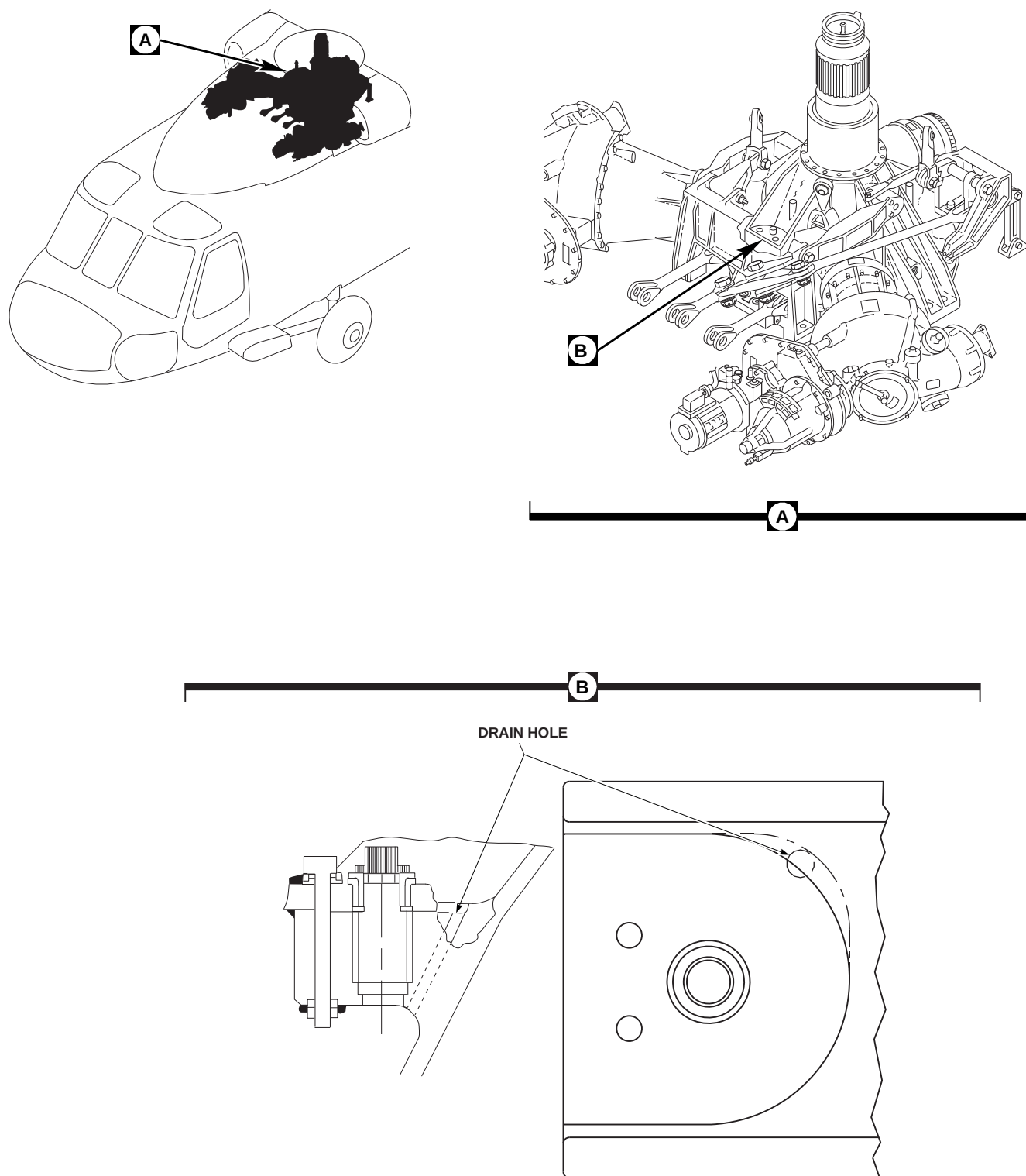
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FIGURE 1-7-20-2. INSPECT MAIN TRANSMISSION FORWARD BRIDGE DRAIN HOLE

1-7-21 EVERY 120 DAYS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-21.1	Inspect Ejector Rack, BRU-22A/A	1-7-21-1	1-7-21.2	Inspect Rescue Hoist Spring Pin	1-7-21-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 1-4-16

1-7-21.1 Inspect Ejector Rack, BRU-22A/A.

Clean ejector rack, BRU-22A/A, and firing lead cable (PARA 1-4-16).

1-7-21.2 Inspect Rescue Hoist Spring Pin.

- Visually inspect rescue hoist spring pin for signs of corrosion or cracks. **NO CORROSION OR CRACKS ALLOWED.** Replace spring pin with correct spring pin, if corrosion or cracks are detected (refer to step b.).



Damage to equipment and/or injury to personnel may occur if rescue hoist hook assembly is equipped with spring pin, MS171500. This spring pin is subject to stress corrosion under conditions of severe salt water exposure.

- Verify part number of spring pin. If part number is MS171500 or part number is unknown, replace as follows:

NOTE

If replacement spring pin is unavailable, it is acceptable to install a new spring pin, MS171500, until a replacement spring pin is obtained.

- If equipped with a Lucas Aerospace manufactured rescue hoist and hook assembly, replace spring pin, MS171500, with spring pin, MS39086-494.

NOTE

- The replacement spring pin, BL-14792-2, has been painted orange along its entire length for ease of identification.
- If replacement spring pin is unavailable, it is acceptable to install a new spring pin, until a replacement spring pin is obtained.
- If equipped with a Breeze-Eastern manufactured rescue hoist, replace spring pin, MS171500, with spring pin, BL-14792-2.

1-7-22 EVERY 6 MONTHS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-22.1	Main Rotor Blade	1-7-22-1	1-7-22.4	Overload Clutch Margin Validation Load Check	1-7-22-2
1-7-22.2	Ground Receptacle	1-7-22-1		HOIST, 42315-10	
1-7-22.3	Portable Fire Extinguish- ers	1-7-22-2			

INITIAL SETUP

Tools

- Box Wrench
- Load Check Tool, 49900-890

References

- PARA 2-6-39
- PARA 5-3-8

- PARA 14-3-1
- PARA 14-4-4
- PARA 14-4-5

1-7-22.1 Main Rotor Blade.

Check main rotor blade spar pressure (PARA 5-3-8).

1-7-22.2 Ground Receptacle.

Inspect and test ground receptacle (PARA 2-6-39).

1-7-22.3 Portable Fire Extinguishers.**NOTE**

Fire extinguishers are weighed with cylinder head and horn assembly installed.

Weigh portable fire extinguishers located in cockpit and cabin. **GROSS WEIGHT OF FIRE EXTINGUISHER MUST NOT HAVE DECREASED 4 OUNCES OR MORE FROM GROSS WEIGHT STAMPED ON DATA RING.** If weight is below limit, replace fire extinguisher.

1-7-22.4 Overload Clutch Margin Validation Load Check HOIST, 42315-10 .**NOTE**

- This procedure shall be performed every 6 months, 300 cycles, 400 lifts, or if the operator suspects that the overload clutch has been compromised by, either the full-in limit switch or partial peel-out due to hook snag or flight maneuvering. These intervals may be reduced based on test results as defined in Figure 1-7-22-3.
- If flight operations have just been completed with rescue hoist operation, wait for a period of 5 hours to allow oil temperature of hoist gear box to return to ambient conditions.

- a. Perform a cable conditioning lift as follows:

NOTE

A cable conditioning lift is required prior to performing load check to prevent rescue hoist cable from being pulled between wraps on outer layer of the cable drum.

- (1) From a hovering helicopter, fully extend the hoist cable to the down limit stop.
- (2) On the ground, have assistant carefully attach a load of 500 - 600 pounds to the

rescue hoist hook. Using hand signals, have the rescue hoist operator fully raise the load for the full length of the cable.

- (3) After performing the cable conditioning lift, allow 2 hours to allow rescue hoist temperature to return to ambient temperatures before performing load check.

NOTE

It is recommended that the operator contact Goodrich for load check tool availability and technical support.

- b. Reel out sufficient amount of cable to be able to remove cable hook.
- c. Remove rescue hoist hook (PARA 14-4-5).
- d. If installed, remove the hook adapter from the test tool and install on the cable and turn on the tool load cell meter.
- e. Gently reel in cable until body of load check tool contacts cable guide portion of rescue hoist.

NOTE

- Performing the remaining steps of test at temperatures below 0° (32 °F) or greater than 50° (120 °F) is not authorized. Record air temperatures on the test record form.

- If possible, perform testing in a controlled environment to minimize difference between ambient temperature and temperature of rescue hoist gear box.

- f. Engage the Peak button on the load cell meter.
- g. Using the provided box wrench, tighten hexnut on load check tool at a rate not exceeding 90° per second to increase load on cable until clutch slips.
- h. Record Peak load as captured on the load cell meter.

- i. Reset Peak hold value by pressing the Peak and Reset buttons simultaneously.

WARNING

If a rescue hoist cable assembly has a load applied in excess of 1,550 pounds with the field load check tool, the cable must be replaced prior to returning the rescue hoist to service.

NOTE

Clutch slippage will be indicated by a sudden slippage and loss of load, or a failure of the clutch to reach the required load. Do not exceed 1,500 pounds on the field load check tool as this may cause damage to the tool. Care must be taken to perform testing per instructions provided for the field load check tool to prevent possible damage. There is potential damage to the rescue hoist cable if subjected to loads greater than 1,500 pounds with current field load check tool.

- j. Repeat the test four additional times, resetting the Peak hold after each slip.

NOTE

As testing progresses, it may be necessary to stop between tests to unscrew the

hexnut down threaded shaft and reseal the load tool against the rescue hoist using pendant.

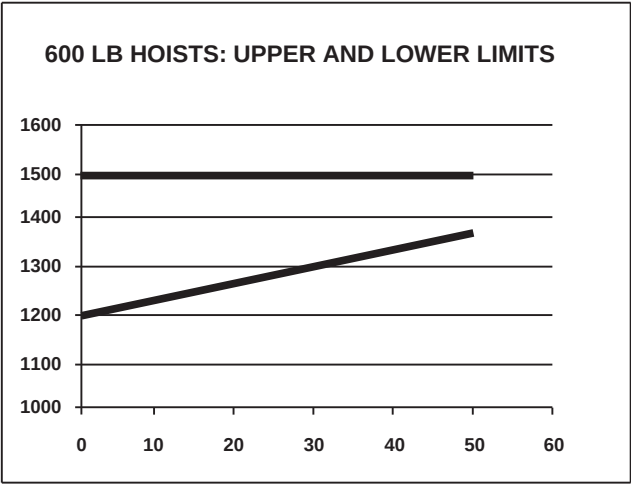
- k. Average the five slip values to obtain the final value to be recorded. Record test results on the rescue hoist log card and evaluate the results in accordance with the values in Figure 1-7-22-1 and Figure 1-7-22-2, based on ambient temperature at the time of testing.
- l. Loosen hexnut or reel out the cable to remove load on the cable. Remove tool from rescue hoist.
- m. Inspect first 30 feet of cable for damage and size in accordance with the method and criteria found in the 28 day corrosion control inspection for the rescue hoist cable.
- n. Inspect first 18-inches of cable after load check has been completed to make sure there are no broken wires or necked down sections of cable within the region. The minimum allowable cable diameter is 0.185-inch. If damage is found, replace cable immediately (PARA 14-4-4) and return cable to Goodrich.
- o. Install rescue hoist hook (PARA 14-4-5).
- p. If test results indicate a value within the allowable limits per Figure 1-7-22-1 and Figure 1-7-22-2, refer to Figure 1-7-22-3 for next action and testing.
- q. If test results indicate a value below acceptable lower limits per Figure 1-7-22-1 and Figure 1-7-22-2, contact Goodrich.

SLIP LOAD LIMIT vs. TEMPERATURE FOR 600 LB CAPACITY HOISTS

TEMP [°F]	TEMP [°C]	MIN SLIP LOAD [LBS]	MAX SLIP LOAD [LBS]	TEMP [°C]	TEMP [°C]	MIN SLIP LOAD [LBS]	MAX SLIP LOAD [LBS]
32	0	1200	1500	78.8	26	1284	1500
33.8	1	1203	1500	80.6	27	1287	1500
35.6	2	1206	1500	82.4	28	1291	1500
37.4	3	1210	1500	84.2	29	1294	1500
39.2	4	1213	1500	86	30	1297	1500
41	5	1216	1500	87.8	31	1300	1500
42.8	6	1219	1500	89.6	32	1304	1500
44.6	7	1223	1500	91.4	33	1307	1500
46.4	8	1226	1500	93.2	34	1310	1500
48.2	9	1229	1500	95	35	1313	1500
50	10	1232	1500	96.8	36	1317	1500
51.8	11	1236	1500	98.6	37	1320	1500
53.6	12	1239	1500	100.4	38	1323	1500
55.4	13	1242	1500	102.2	39	1326	1500
57.2	14	1245	1500	104	40	1330	1500
59	15	1249	1500	105.8	41	1333	1500
60.8	16	1252	1500	107.6	42	1336	1500
62.6	17	1255	1500	109.4	43	1339	1500
64.4	18	1258	1500	111.2	44	1343	1500
66.2	19	1262	1500	113	45	1346	1500
68	20	1265	1500	114.8	46	1349	1500
69.8	21	1268	1500	116.6	47	1352	1500
71.6	22	1271	1500	118.4	48	1356	1500
73.4	23	1275	1500	120.2	49	1359	1500
75.2	24	1278	1500	122	50	1362	1500
77	25	1281	1500				

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FIGURE 1-7-22-1. SLIP LOAD/TEMPERATURE CAPACITY



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FIGURE 1-7-22-2. UPPER AND LOWER ACCEPTABLE LIMITS

TEST RESULT	CHANGE FROM INITIAL RESULT	ACTION	NEXT TEST
BELOW LOWER LIMIT AS DEFINED IN FIGURE 1 AND FIGURE 2	N/A	REMOVE RESCUE HOIST FROM SERVICE AND CONTACT SIKORSKY	N/A
WITHIN LIMITS AS DEFINED IN FIGURE 1 AND FIGURE 2	<200 LBS	CONTINUE IN SERVICE	6 MONTHS, 300 CYCLES, OR 400 LIFTS
	200 – 300 LBS	CONTINUE IN SERVICE	3 MONTHS, 150 CYCLES, 200 LIFTS
	>300 LBS	CONTACT GOODRICH	N/A

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FIGURE 1-7-22-3. EVALUATION OF TEST RESULTS

1-7-23 EVERY 12 MONTHS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-23.1	Compass Indicators	1-7-23-2	1-7-23.5	Pivot Bolts Inspection	1-7-23-2
1-7-23.2	Pitot-Static System Inspection	1-7-23-2	1-7-23.6	Engine Cowling Inspection	1-7-23-5
1-7-23.3	Free-Air Thermometer (FAT) Inspection	1-7-23-2	1-7-23.7	Main Rotor Blade Skin-to-Core Inspection	1-7-23-5
1-7-23.4	Primary Servo Inspection	1-7-23-2			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Crowfoot Attachment
- Flashlight
- Inspection Mirror
- Magnifying Glass, 10X
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1,600 in. lbs
- Wood Blocks, 1 ½-inch thick
- Wrench

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Corrosion Preventive Compound, Item 109, Appendix D

- Corrosion Preventive Compound, Item 114, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-107
- PARA 2-4-118
- PARA 2-4-122
- PARA 2-6-5
- PARA 2-6-29
- PARA 5-3-11
- PARA 8-2-1
- PARA 8-4-7
- PARA 8-4-30

- PARA 11-4-87
- PARA 11-4-90 through PARA 11-4-98
- TM 11-70-23

1-7-23.1 Compass Indicators.

- Check gyromagnetic compass indicator for correct reading on all cardinal headings (TM 11-70-23).
- Check magnetic standby compass indicator for correct reading on all cardinal headings (PARA 8-4-7).

1-7-23.2 Pitot-Static System Inspection.**NOTE**

If helicopter is operated in inclement weather, drain ports should be checked more frequently.

- From under cockpit, remove caps from Pitot and Static drain ports underneath pilot's and copilot's seats.
- Drain water from lines. Check drain holes for blockage.
- Coat caps with antiseize compound, Item 74, Appendix D.
- Install caps. **TORQUE TO 40 - 60 INCH-POUNDS.**
- Perform inspection and operational check of the Pitot-Static System to include the altimeter, vertical velocity indicator, and the airspeed indicator as applicable (PARA 8-2-1).

1-7-23.3 Free-Air Thermometer (FAT) Inspection.**NOTE**

Inspect and test FAT any time the FAT gage is replaced.

Inspect and test FAT (PARA 8-4-30).

1-7-23.4 Primary Servo Inspection.

- Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- Open main rotor pylon sliding cover.
- Rotate main rotor head to position blades to allow best working conditions.
- Using clean machinery towel, Item 211, Appendix D, thoroughly clean output end of each servo assembly.
- With servos installed on helicopter, using flashlight and inspection mirror, visually inspect the bottom side of rod end for cracks. **NONE ALLOWED.** Pay particular attention to area under piston nuts and output lug area. If damage is discovered, replace servo (PARA 11-4-87).
- Close main rotor pylon sliding cover.

1-7-23.5 Pivot Bolts Inspection.

- Pivot bolts, 70400-08159, listed in Table 1-7-23-1, are to be inspected.

Table 1-7-23-1. Pivot Bolt, 70400-08159, Inspection

Bolt Dash Number	Quantity	Location
-101	8	70400-08155 control rod ends

Table 1-7-23-1. Pivot Bolt, 70400-08159, Inspection (Cont)

Bolt Dash Number	Quantity	Location
-102	2	70400-08110 swashplate link lower pivot
-103	1	70400-08104 walking beam pivot
-104	1	70400-08151 swashplate link lower pivot
-105	1	70400-08150/08166 lateral bellcrank pivot
-106	1	70400-08102 aft bellcrank pivot
-107	1	70400-08101 forward bellcrank pivot (when 70400-08116 bellcrank support is installed)
-108	1	70400-08101 forward bellcrank pivot (when 70400-08162 bellcrank support is installed)

- b. Open main rotor pylon sliding cover.
- c. Remove main rotor pylon transmission cover (PARA 2-4-107).
- d. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- e. Raise collective stick to raise swashplate.
- f. Install three 1 1⁄2-inch thick wood blocks equally spaced between swashplate and main transmission, lower swashplate onto wood blocks to support swashplate while removing pivot bolt(s).



- Damage to swashplate will result if collective stick is lowered while wood blocks are installed. Do not apply down collective with wood blocks installed.
- g. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- h. Remove cotter pin and mating nut.
- i. Clean bolt using cleaning cloth, Item 102, Appendix D, moistened with dry-cleaning solvent, Item 124, Appendix D. Inspect bolt as follows:

NOTE

Use wrench on bolthead to rotate bolt a full 360° during inspection.

- (1) Inspect bolt for evidence of corrosion in the area of retaining balls, threads, cotter pin hole, retaining pin hole in bolthead, and hole in threaded end of bolt. **NONE ALLOWED.** If corrosion is visible, replace bolt (PARA 11-4-90, PARA 11-4-91, PARA 11-4-92, PARA 11-4-93, PARA 11-4-94, PARA 11-4-95, PARA 11-4-97, and PARA 11-4-98).
- (2) Visually inspect bolt for cracks in area of retaining balls and threads. **NONE ALLOWED.** If crack(s) is visible, replace bolt (PARA 11-4-90, PARA 11-4-91, PARA 11-4-92, PARA 11-4-93, PARA 11-4-94, PARA 11-4-95, PARA 11-4-97, and PARA 11-4-98).

CAUTION

If resistance or binding of the retaining ball mechanism is observed when pin is actuated, do not apply corrosion preventive compound to free the mechanism. The inner core of the bolt is corroded and must be rejected.

- j. Release retaining balls by holding retaining pin in. If resistance or binding of the retaining ball mechanism is observed when pin is actuated, discard bolt. If no resistance or binding of the retaining ball mechanism is observed when pin is actuated, apply corrosion preventive compound, Item 114, Appendix D, to bolt as follows:
 - (1) Work corrosion preventive compound through one retaining ball hole until excess is seen at opposite ball hole. Rotate bolt 180° and repeat procedure.
 - (2) While working retaining pin, spray corrosion preventive compound, Item 114, Appendix D, into hole in bolthead and hole in end of bolt.

- (3) Wipe excessive corrosion preventive compound from bolt using cleaning cloth, Item 102, Appendix D, prior to installation.
- (4) Apply corrosion preventive compound, Item 109, Appendix D, to the exterior retaining ball area and threads.

k. Install nut. Torque nut as follows:

- (1) **ON BOLTS -101 THROUGH -106, TORQUE INSTALLED NUT TO 600 INCH-POUNDS. IF SLOT IN NUT DOES NOT LINE UP WITH COTTER PIN HOLE, SET TORQUE WRENCH TO 900 INCH-POUNDS AND TIGHTEN NUT UNTIL SLOT IN NUT AND COTTER PIN HOLE LINE UP. DO NOT EXCEED MAXIMUM TORQUE.** Install cotter pin.
- (2) **ON BOLTS -107 AND -108 BOLTS, TORQUE INSTALLED NUT ON TO 925 INCH-POUNDS. IF SLOT IN NUT DOES NOT LINE UP WITH COTTER PIN HOLE, SET TORQUE WRENCH TO 1350 INCH-POUNDS AND TIGHTEN NUT UNTIL SLOT IN NUT AND COTTER PIN HOLE LINE UP. DO NOT EXCEED MAXIMUM TORQUE.** Install cotter pin.

NOTE

If unable to torque bolthead, use a crowfoot attachment to torque nut, without disturbing cotter pin, and without restraining bolthead.

- l. Apply tightening torque (proof load) as follows:
 - (1) **ON BOLTS -101 THROUGH -106, TORQUE INSTALLED NUT IN TIGHTENING DIRECTION TO 400 - 420 INCH-POUNDS.**
 - (2) **ON BOLTS -107 AND -108, TORQUE INSTALLED NUT IN TIGHTENING DIRECTION TO 630 - 650 INCH-POUNDS.**

- (3) If bolt rotates, remove special self-retaining bolt, washer(s), nut, and cotter pin (PARA 11-4-90, PARA 11-4-91, PARA 11-4-92, PARA 11-4-93, PARA 11-4-94, PARA 11-4-95, PARA 11-4-97, and PARA 11-4-98).
- (4) Adjust washer thicknesses or add washers and reinstall special self-retaining bolt, washer(s), nut, and cotter pin (PARA 11-4-90, PARA 11-4-91, PARA 11-4-92, PARA 11-4-93, PARA 11-4-94, PARA 11-4-95, PARA 11-4-96, PARA 11-4-97, and PARA 11-4-98).
- m. Apply torque stripe (PARA 2-6-5).
- n. If any bolt is removed for suspected cracks, the bolt/nut and all applicable information (aircraft model, nut installed, bolt dash number, and TSN (calendar time) for the bolt) shall be submitted to Sikorsky Aircraft Corporation.
- o. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- p. Raise collective stick and remove wood blocks from between swashplate and main transmission.
- q. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- r. Make sure area is clean and free of foreign material.
- s. Install main rotor pylon transmission cover (PARA 2-4-107).
- t. Close main rotor pylon sliding cover.

1-7-23.6 Engine Cowling Inspection.

- a. Open engine cowling.
- b. Using 10X magnifying glass, inspect fiberglass top ply for damage at hinge-rib location on

both ends of engine cowling (Figure 1-7-23-1, Detail A). Proceed as follows:

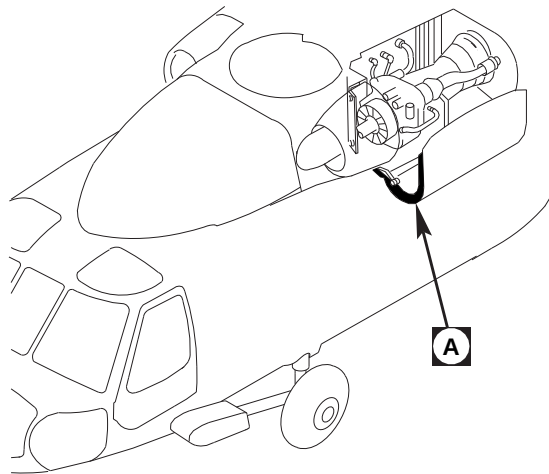
- (1) On engine cowling assemblies, 70302-02141-041, -042, -044, -045, -048, and -049, **NO DAMAGE ALLOWED**. If cracks are found through either of the first two fiberglass layers or there is evidence of exposed fiber, replace fittings (PARA 2-4-122), and repair cowling (PARA 2-6-29). If damage extends into third layer, replace cowling (PARA 2-4-118).
- (2) On engine cowling assemblies, 70070-30308-041 through -046, and 70302-02141-052 and -053, **DAMAGE IS ALLOWED THROUGH FIRST TWO FIBERGLASS LAYERS. NO DAMAGE ALLOWED TO THIRD LAYER**. If damage does not extend into third layer, repair cowling (PARA 2-6-29). If damage extends into third layer, replace cowling (PARA 2-4-118).

- c. Close engine cowling.

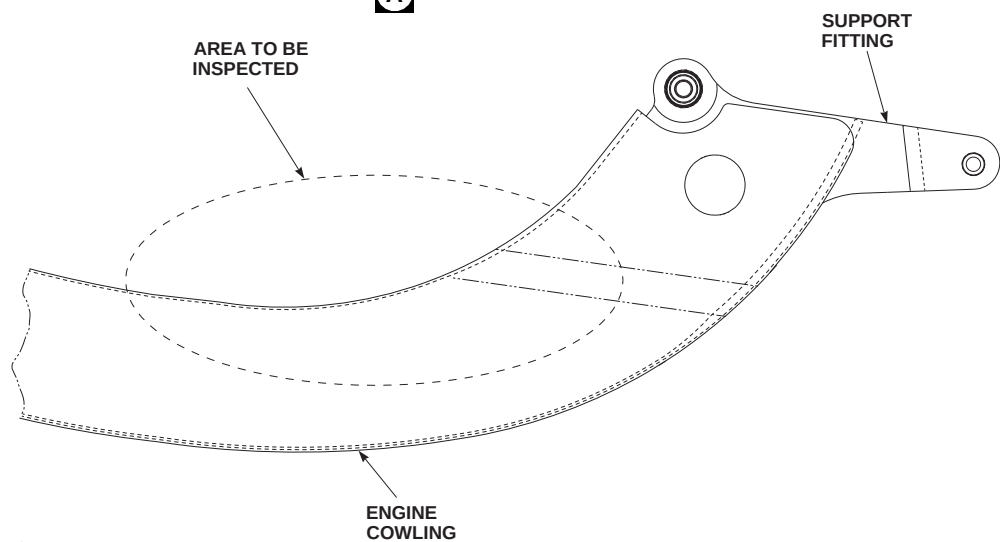
1-7-23.7 Main Rotor Blade Skin-to-Core Inspection.

NOTE

- If main rotor blade has been removed from service when 12 month inspection is due, inspection can be delayed until main rotor blade is installed on helicopter.
- Main rotor blades may be inspected on or off helicopter.
- a. Using coin tapping inspection method (PARA 5-3-11), inspect all the skin-to-core region on upper airfoil of main rotor blade for disbonds and core crush.
- b. Refer to PARA 5-3-11 for acceptance/rejection criteria.

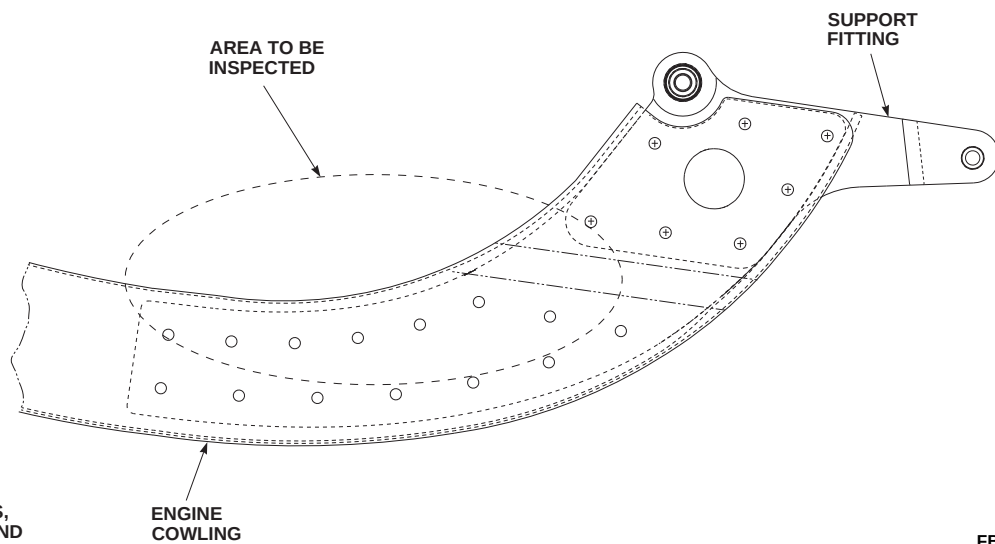


A



EFFECTIVITY

ENGINE COWLING ASSEMBLIES,
70302-02141-041, -042, -044, -045,
-048, AND -049.



EFFECTIVITY

ENGINE COWLING ASSEMBLIES,
70070-30308-041 THRU -046, AND
70702-02141-052 AND -053.

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FIGURE 1-7-23-1. ENGINE COWLING INSPECTION

1-7-24 AFTER DUAL-ENGINE OPERATION WITH GUST LOCK ENGAGED.

PARAGRAPH BREAKDOWN

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1-7-24.1 After Dual-Engine Operation with Gust Lock Engaged	1-7-24-1

INITIAL SETUP

- PARA 2-4-113

Tools

- Aircraft Mechanic’s Toolkit
- Fluorescent Penetrant Inspection Kit

- PARA 6-4-2
- PARA 6-4-8

References

- ASTM E1417
- PARA 1-6-16

1-7-24.1 After Dual-Engine Operation with Gust Lock Engaged.

transmission tail takeoff flange (PARA 6-4-2) or gust lock rod and lever (PARA 6-4-8).

- Turn off all electrical power (PARA 1-6-16).
- Remove oil cooler compartment access door (PARA 2-4-113).
- Inspect tail takeoff flange for broken or deformed teeth, check pawl and surrounding areas for cracks. If required, replace main

- Using fluorescent penetrant inspection kit, fluorescent penetrant inspect gust lock and output housing attaching points, on main module, for cracks (ASTM E1417).
- Install oil cooler compartment access door (PARA 2-4-113).

1-7-25 AFTER EXCEEDING 145 KIAS WITH CABIN DOORS OPENED.

PARAGRAPH BREAKDOWN

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1-7-25.1	After Exceeding 145 KIAS with Cabin Doors Opened	1-7-25-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

1-7-25.1 After Exceeding 145 KIAS with Cabin Doors Opened.

NOTE

Do the following before next flight after exceeding 145 KIAS with cabin doors open.

- Check doors and windows for distortion and damage.
- Check door tracks, rollers and latching mechanism for damage.
- Check door for proper operation.

1-7-26 AFTER FIRE EXTINGUISHING SYSTEM DISCHARGE (HALON-TYPE).

PARAGRAPH BREAKDOWN

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1-7-26.1	After Fire Extinguishing System Discharge (Halon-Type)	1-7-26-1

INITIAL SETUP

- PARA 12-4-4

Tools

- PARA 12-4-6

- Aircraft Mechanic’s Toolkit
- Protection, Eye

References

- PARA 12-2-2
- PARA 12-4-1

1-7-26.1 After Fire Extinguishing System Discharge (Halon-Type).

all personnel are wearing safety glasses when compressed air is being used.

- Remove discharged extinguishing agent container (PARA 12-4-1).
- If extinguishing agent was discharged through directional valve to No. 1 engine or APU, remove directional valve (PARA 12-4-6).

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure

CAUTION

Damage to equipment will result if system is flushed with water. Halon becomes corrosive when in contact with water. Use filtered, dry compressed air when purging system.

- Purge directional valve using filtered, dry compressed air. Check flapper for freedom of movement.
- Purge discharge tubes using filtered, dry compressed air.
- If necessary, install directional valve (PARA 12-4-6).

- f. Purge engine compartments and APU compartment where extinguishing agent has been discharged using filtered, dry compressed air.

WARNING

Injury to personnel and damage to equipment will result, unless stresses, thermal effects, and other damage, to which parts are subjected as a result of a fire, can be evaluated and determined to be within design parameters. Any parts of the component that may affect safe operation shall be assumed to be damaged beyond acceptable limits and will not be reused on airworthy helicopters.

- g. Inspect engine compartments and components for damage. **NONE ALLOWED.** If damaged, replace parts as required.
- h. Inspect rubber materials and insulations for damage. **NONE ALLOWED.** If damaged, replace parts as required.
- i. Replace thermal disc (PARA 12-4-4).
- j. Replace discharged extinguishing agent container with a fully charged container.
- k. Install extinguishing agent container (PARA 12-4-1).
- l. Perform operational check of fire extinguishing system (PARA 12-2-2).

1-7-27 AFTER FIRING CHAFF DISPENSER.

PARAGRAPH BREAKDOWN

		PAGE
1-7-27.1	After Firing Chaff Dispenser	1-7-27-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- PARA 1-4-15
- PARA 12-4-52 through PARA 12-4-55

1-7-27.1 After Firing Chaff Dispenser.

- Inspect tail rotor deice slipring and brush assembly for accumulation of chaff.
- If disassembly of slipring is required to remove chaff, refer to PARA 12-4-52, PARA 12-4-53, PARA 12-4-54, and PARA 12-4-55.

- Clean any accumulated chaff from slipring and brush holder with methyl ethyl ketone, Item 217, Appendix D, and machinery towel, Item 211, Appendix D (PARA 1-4-15).

1-7-28 AFTER FIRING EJECTOR RACK, BRU-22A/A.

PARAGRAPH BREAKDOWN

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1-7-28.1 After Firing Ejector Rack, BRU-22A/A	1-7-28-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Corrosion Preventive Compound, Item 114, Appendix D

References

- Appendix D
- PARA 1-4-16
- PARA 16-4-36

1-7-28.1 After Firing Ejector Rack, BRU-22A/A.



Breech cannot be removed or installed with stores loaded on ejector rack. If breech is removed with stores on ejector rack, stores will be released.

- Clean breech, gun, and piston assembly (PARA 1-4-16).
- Visually inspect breech, gun, and piston assembly for cracks or breaks (PARA 16-4-36).
- Visually inspect breech and piston assembly for missing or broken rings.

- Visually inspect piston assembly for straightness.
- Replace breech, gun, or piston assembly as required.
- Replace missing or broken bearing ring and pressure rings on piston or breech as required (PARA 16-4-36).
- Spray breech, gun, and piston assembly lightly using corrosion preventive compound, Item 114, Appendix D.

NOTE

Make certain breech is fully forward and locked. Close hooks and leave latched at all times when ejector unit is not in use.

- Assemble breech, gun, and piston assembly (PARA 16-4-36).

1-7-29 AFTER ENGINE OUTPUT SHAFT MAINTENANCE.

PARAGRAPH BREAKDOWN

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1-7-29.1 After Engine Output Shaft Maintenance	1-7-29-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Powerplant Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Sealing Compound, Item 286, Appendix D

References

- Appendix D
- PARA 4-4-1
- PARA 4-4-4
- PARA 4-4-6
- TM 1-70-VIB

1-7-29.1 After Engine Output Shaft Maintenance.

- a. Remove engine air inlet (PARA 4-4-4).

WARNING

FSCAP

Verification of minimum torque of bolts is a critical characteristic.

- b. Check torque on three bolts or studs connecting flexible coupling to input module flange while holding nuts as follows:
 - (1) Initially check bolts in tightening direction. If input module flange, 70351-08206-102, is used, check torque while holding nuts. **TORQUE MUST NOT BE LESS THAN 90 INCH-POUNDS.**
 - (2) **IF TORQUE ON ANY ONE BOLT OR STUD IS BELOW 90 INCH-POUNDS,**

REPLACE ALL BOLTS OR STUDS AND ASSEMBLY HARDWARE CONNECTING DISC PACK TO INPUT FLANGE (PARA 4-4-6). IF INPUT MODULE FLANGE, 70351-08206-041 WITH NUTPLATES IS USED, REPLACE WITH INPUT MODULE FLANGE, 70351-08206-102, IF AVAILABLE. IF INPUT MODULE FLANGE, 70351-08206-041 IS USED, AFTER TORQUE HAS STABILIZED, PLACE A DROP OF SEALING COMPOUND, ITEM 286, APPENDIX D, TO THREADS OF BOLTS ADJACENT TO NUTPLATES. MAKE SURE COMPOUND SEEPS INTO NUTPLATES. Perform engine output shaft vibration check if shaft was disconnected (TM 1-70-VIB).

- (3) Check bolts or stud in tightening direction. If input module flange, 70351-08206-102, is used, check torque while holding nut. **TORQUE MUST NOT BE LESS THAN 130 INCH-POUNDS.**

- (4) **IF TORQUE ON ANY BOLT OR STUD IS BELOW 130 INCH-POUNDS, RETORQUE BOLTS TO 130 - 140 INCH-POUNDS.** Repeat torque check after each 1 to 3 flight hours until torque stabilizes.
- (5) If torque does not stabilize after third consecutive torque check, replace three bolts or studs and assembly hardware connecting disc pack to input module flange (PARA 4-4-1). If input module flange, 70351-08206-041 with nutplates is used, replace with input module flange, 70351-08206-102, if available. If input module flange, 70351-08206-041 is used, after torque has stabilized, place a drop of sealing compound, Item 286, Appendix D, to threads of bolts adjacent to nutplates. Make sure compound seeps into nutplates. Balance engine output shaft (TM 1-70-VIB).
- c. Check torque on three nuts securing flexible coupling to engine output shaft by torquing nuts in tightening direction only while holding bolts as follows:
 - (1) Initially check nuts in tightening direction while holding bolts. **TORQUE MUST NOT BE LESS THAN 90 INCH-POUNDS.**
 - (2) **IF TORQUE ON ANY ONE NUT IS BELOW 90 INCH-POUNDS, REPLACE ENGINE OUTPUT SHAFT** (PARA 4-4-6). If shaft was disconnected, perform an engine output shaft balance check (TM 1-70-VIB).
 - (3) Check nuts in tightening direction while holding bolts. **TORQUE MUST NOT BE LESS THAN 130 INCH-POUNDS.**
 - (4) **IF TORQUE ON ANY BOLT OR STUD IS BELOW 130 INCH-POUNDS, RETORQUE TO 130 - 140 INCH-POUNDS.** Repeat torque check after each 1 to 3 flight hours until torque stabilizes.
 - (5) If torque does not stabilize after third consecutive torque check, replace engine output shaft. Balance engine output shaft (TM 1-70-VIB).
- d. Install engine air inlet (PARA 4-4-4).

1-7-30 DELETED.

1-7-31 AFTER MAIN ROTOR BLADE CONTACT WITH TAIL ROTOR PYLON OR TAIL CONE.

PARAGRAPH BREAKDOWN

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1-7-31.1 After Main Rotor Blade Contact with Tail Rotor Pylon or Tail Cone	1-7-31-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Tags

References

- PARA 1-7-50
- PARA 5-3-11
- PARA 5-4-5

1-7-31.1 After Main Rotor Blade Contact with Tail Rotor Pylon or Tail Cone.

NOTE

- The following inspection is required when any turning main rotor blades come in contact with tail rotor pylon or tail cone, this includes tail drive shaft and covers, and intermediate gear box and fairing.
- Other inspections may apply depending on circumstances.
- Inspect spindle bearings on spindle being replaced to determine if bearing is serviceable. If bearing is serviceable, it may be used on replacement spindle.
- a. **REPLACE ALL SPINDLES REGARDLESS OF THE EXTENT OF DAMAGE TO BLADE(S), TAIL PYLON OR TAIL CONE** (PARA 5-4-5). Clearly tag spindles "DAMAGE OCCURRED DUE TO BLADE STRIKING

TAIL PYLON/TAIL CONE". Also, explain the severity of damage to tail cone or tail pylon.

- b. Inspect the hub arm inside diameter (4 places) for damage caused by spindle retention nut contact. If evidence of contact is found, replace hub. Clearly tag hub "DAMAGE OCCURRED DUE TO BLADE STRIKING TAIL PYLON/ TAIL CONE". Also, explain the severity of damage to tail cone or tail pylon.
- c. Inspect main rotor blade(s) paying particular attention to the inboard five foot section. If damage to blade(s) is less than accept/reject criteria, repair blade(s) and return to service. If damage is greater than accept/reject criteria remove blade(s) and return to an authorized overhaul facility. Clearly tag blade(s) "BLADE DAMAGE OCCURRED DUE TO BLADE STRIKING TAIL PYLON/TAIL CONE". Also, explain the severity of damage to tail cone or tail pylon (PARA 5-3-11).
- d. Perform sudden stoppage inspection (PARA 1-7-50).

1-7-32 AFTER MAIN ROTOR BLADE CONTACT WITH IR JAMMER, ALQ-144.

PARAGRAPH BREAKDOWN

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1-7-32.1 After Main Rotor Blade Contact with IR Jammer, ALQ-144	1-7-32-1
INITIAL SETUP	• PARA 5-4-1
Tools	• PARA 5-4-5
• Aircraft Mechanic’s Toolkit	• PARA 5-4-31
Materials/Parts	• PARA 5-5-4
• Tags	• TM 11-70-23
References	• PARA 1-7-50 • PARA 5-3-11

1-7-32.1 After Main Rotor Blade Contact with IR Jammer, ALQ-144.

NOTE

The following inspection is required when main rotor blades come in contact with IR jammer, ALQ-144, for any reason other than hard landing.

- a. Inspect main rotor blades for damage as follows:

(1) Inspect main rotor blades (PARA 5-3-11).
 (2) If damage to blades is within limits, repair blades (PARA 5-4-31), and continue this inspection.
- (3) If damage to blades exceeds limits, perform sudden stoppage inspection (PARA 1-7-50).

b. Inspect droop stop ears for cracks. If cracks are found, replace spherical bearing (PARA 5-5-4). Clearly tag bearing "DAMAGE OCCURRED DUE TO EXCESSIVE BLADE FLAPPING." Note on tag what components were hit and severity of damage to helicopter parts.
 c. Inspect hub arm inside diameter for damage caused by spindle retention nut contact. If evidence of contact is found, replace spindle and hub (PARA 5-4-5 and PARA 5-4-1). Clearly tag spindle "DAMAGE OCCURRED DUE TO EXCESSIVE BLADE FLAPPING." Note on tag what components were hit and severity of damage to helicopter parts.

- d. Remove IR jammer, ALQ-144, and perform functional check (TM 11-70-23).

1-7-33 AFTER OPERATING TRANSMISSION AT HIGH TEMPERATURE OR LOW/NO OIL PRESSURE.

PARAGRAPH BREAKDOWN

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1-7-33.1 After Operating Transmission at High Temperature or Low/No Oil Pressure	1-7-33-1

INITIAL SETUP

- PARA 6-4-1

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 6-2-2
- PARA 6-3-5

1-7-33.1 After Operating Transmission at High Temperature or Low/No Oil Pressure.

- Before replacing component(s), troubleshoot the appropriate indicating system(s) to make sure of proper operation (PARA 6-2-2).
- Transmission operated:
 - With oil temperature at or above 284°F (140°C) for less than 30 seconds, perform serviceability inspection (PARA 6-3-5).
 - With no oil pressure for more than 30 seconds in flight, or no oil pressure for

more than 1 minute during ground run, replace all five main transmission modules (PARA 6-4-1).

- With no oil pressure for less than 30 seconds in flight, or no oil pressure for less than 1 minute during ground run, perform serviceability inspection (PARA 6-3-5).
- With oil pressure below 20 psi but greater than 0 psi, perform serviceability inspection (PARA 6-3-5).

1-7-34 AFTER SINGLE-ENGINE OPERATION ABOVE IDLE WITH GUST LOCK ENGAGED.

PARAGRAPH BREAKDOWN

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1-7-34.1 After Single-Engine Operation Above Idle with Gust Lock Engaged	1-7-34-1

INITIAL SETUP

- PARA 6-4-2

Tools

- PARA 6-4-8

- Aircraft Mechanic’s Toolkit
- Fluorescent Penetrant Inspection Kit

- PARA 6-4-38
- PARA 6-4-39

References

- ASTM E1417
- PARA 1-3-15
- PARA 1-6-16
- PARA 2-4-113
- PARA 5-3-4

- PARA 6-4-40
- PARA 6-4-41
- PARA 6-4-42
- PARA 6-4-50
- TM 1-70-VIB

1-7-34.1 After Single-Engine Operation Above Idle with Gust Lock Engaged.

output housing attaching points, on main module, for cracks (ASTM E1417).

- | | |
|--|---|
| <ul style="list-style-type: none"> a. Turn off all electrical power (PARA 1-6-16). b. Remove oil cooler compartment access doors (PARA 2-4-113). c. Inspect tail takeoff flange for broken or deformed teeth, check pawl and surrounding areas for cracks. If required, replace main transmission tail takeoff flange (PARA 6-4-2) or gust lock rod and lever (PARA 6-4-8). d. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect gust lock and | <ul style="list-style-type: none"> e. Visually inspect oil cooler shaft couplings (PARA 6-4-42). f. Perform runout check on oil cooler (PARA 6-4-50). g. Visually inspect tail rotor drive shaft (PARA 6-4-38, PARA 6-4-39, PARA 6-4-40, and PARA 6-4-41). h. Visually inspect and service main rotor head dampers (PARA 1-3-15). |
|--|---|

- i. Perform vibration check on oil cooler drive shaft and engine output shaft (TM 1-70-VIB).
- j. Inspect main rotor head dampers for leakage (PARA 5-3-4).

1-7-35 BEFORE LOADING STORES - INSPECT EJECTOR RACK, BRU-22A/A.

PARAGRAPH BREAKDOWN

	PAGE
1-7-35.1 Before Loading Stores - Inspect Ejector Rack, BRU-22A/A	1-7-35-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Protection, Eye

References

- PARA 16-4-36

1-7-35.1 Before Loading Stores - Inspect Ejector Rack, BRU-22A/A.

- Place ejector unit safety stop lever to unlocked position. Open rear suspension hook and close front suspension hook. Turn safety stop lever toward locked position. **SAFETY STOP LEVER MUST NOT GO TO LOCKED POSITION.** If safety stop lever moves into the locked position, repair ejector rack (PARA 16-4-36).
- Place safety stop lever to unlocked position. Open forward suspension hook and close rear suspension hook. Turn safety stop lever toward locked position. **SAFETY STOP LEVER MUST NOT GO TO LOCKED POSITION.** If safety stop lever moves into the locked position, repair ejector rack (PARA 16-4-36).
- Place safety stop lever to unlocked position. Close both suspension hooks. Turn safety stop lever toward locked position. **SAFETY STOP**

LEVER MUST GO TO LOCKED POSITION (SAFETY STOP LEVER BALL PLUNGER DROPS INTO SAFETY BUSHING DETENT). If safety stop lever does not move into the locked position, repair ejector rack (PARA 16-4-36).

- Pull forward on manual release lever. **SUSPENSION HOOKS MUST NOT OPEN.** If suspension hooks open, repair ejector rack (PARA 16-4-36).
- Place safety stop lever to unlocked position. Pull forward on manual release lever. **BOTH SUSPENSION HOOKS SHALL SNAP OPEN.** If suspension hooks do not open, repair ejector rack (PARA 16-4-36).
- Inspect all visible ejector unit wiring for frayed or burned insulation and broken wires. **NONE ALLOWED.** If frayed or burned insulation or broken wires are found, repair ejector rack (PARA 16-4-36).

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- g. Inspect electrical connectors and receptacles for dirt, moisture, and other contaminants. **NONE ALLOWED.** Using dry, compressed air, remove dirt, moisture, or contamination.
- h. Inspect electrical connectors and receptacles for broken or damaged pins, thread damage, cracked or deformed shell, cracked or loose potting, and corrosion. **NONE ALLOWED.** If

broken or damaged pins, thread damage, cracked or deformed shell, cracked or loose potting, or corrosion is found, repair ejector rack (PARA 16-4-36).

- i. Inspect ejector unit assembly for deep scratches, dents, or fractures to housing and swaybraces that may affect structural integrity of the ejector unit. **NONE ALLOWED.** If deep scratches, dents, or fractures to housing and swaybraces are found to affect the structural integrity of the unit, repair ejector rack (PARA 16-4-36).
- j. Inspect ejector unit for damage that may affect proper latching and unlatching of hooks. **NONE ALLOWED.** If damage is found that may affect proper latching and unlatching of hooks, repair ejector rack (PARA 16-4-36).
- k. Inspect ejector unit for loose swaybrace attaching hardware. **NONE ALLOWED.** If loose swaybrace/attaching hardware is found, **TIGHTEN HARDWARE.**

1-7-36 BEFORE EVERY FLIGHT - INSPECT HYDRAULIC SYSTEM COMPONENTS.

PARAGRAPH BREAKDOWN

		PAGE
1-7-36.1	Inspect Hydraulic System Components	1-7-36-1

INITIAL SETUP

- PARA 7-4-51

Tools

- Aircraft Mechanic’s Toolkit

- PARA 7-4-65

References

- PARA 1-3-8
- PARA 7-4-12

1-7-36.1 Inspect Hydraulic System Components.

NOTE

The following inspection is for helicopters operating in desert environment or for helicopters whose temperature label shows black to 270°F (132°C)/Tempilabel shows black to 275°F (135°C).

- Inspect bleed valve drain line on hydraulic pump. If drain line shows signs of melting, replace drain line and hydraulic pump (PARA 7-4-12 and PARA 7-4-51).
- Inspect hydraulic fluid as follows:

- Drain a small amount of hydraulic fluid from pump module bleed/relief valve. Check hydraulic fluid for discoloration.
- If hydraulic fluid is red in color, add same amount of fluid removed to hydraulic reservoir (PARA 1-3-8).
- If hydraulic fluid is brownish, drain entire reservoir. Flush fluid from affected system, and reservice reservoir (PARA 1-3-8 and PARA 7-4-65).
- Check hydraulic system components for leakage. Replace seals or components as required if leakage is excessive.

1-7-37 ENGINE OUTPUT SHAFT INSPECTION - AFTER DISCONNECT.

PARAGRAPH BREAKDOWN

	PAGE
1-7-37.1 Engine Output Shaft Inspection - After Disconnect	1-7-37-1

INITIAL SETUP

- Torque Wrench, 150 - 750 in. lbs

Tools

- Aircraft Mechanic’s Toolkit
- Hardware Kit, 70000-30100-043
- Powerplant Repairer’s Toolkit

References

- PARA 4-4-6
- PARA 6-4-27
- TM 1-70-VIB

1-7-37.1 Engine Output Shaft Inspection - After Disconnect.

NOTE

NOTE

The following step will be done whenever the engine output shaft is removed or bolts are untorqued.

- Remove engine output shaft from input module (PARA 4-4-6).
- Inspect input module input flange (PARA 6-4-27).
- PERFORM TORQUE CHECK ON THE INPUT MODULE INPUT FLANGE RETAINING NUT. TORQUE RETAINING NUT IN TIGHTENING DIRECTION. MINIMUM TORQUE IS 646 INCH-POUNDS. IF TORQUE IS LESS THAN 646 INCH-POUNDS, TORQUE RETAINING NUT TO 646 - 714 INCH-POUNDS.**

Hardware used to secure disc pack coupling to input module flange is a balanced set. If any individual piece of hardware is to be replaced, replace all hardware with the correct balanced hardware kit. If kit is not available, hardware is to be selected in sets of three so that difference in weight between individual bolts, individual washers, and individual nuts per set is not over 0.025-gram.

- Inspect engine output shaft installation bolts as follows:
 - Visually inspect bolts for signs of burrs or evidence of binding. **NONE ALLOWED.** Replace all bolts and washers with correct hardware kit if burrs or evidence of binding is found.
 - Inspect bolts for binding in disc pack coupling by inserting bolts into disc pack

only and checking for freedom of rotation. **BOLTS MUST ROTATE FREELY WHEN TURNED BY HAND.** Replace all bolts and washers with correct hardware kit if bolts do not rotate freely. If replacement bolts still do not rotate freely, replace engine output shaft assembly (PARA 4-4-6).

- e. If input module input flange, 70351-08206-102, is installed, inspect engine output shaft hardware as follows:

- (1) Thread bolt into nut.
- (2) If any one of the three bolts can be fully threaded by hand into nut, replace bolts and nuts.
- f. Perform engine output shaft vibration level check prior to next flight (TM 1-70-VIB).
- g. Record vibration data for each shaft on component log card (SA Form 7343-21).

1-7-38 ENGINE WHINE/HIGH FREQUENCY VIBRATION INSPECTION.

PARAGRAPH BREAKDOWN

		PAGE
1-7-38.1	Engine Whine/High Frequency Vibration Inspection	1-7-38-1

INITIAL SETUP

- SEI 671

Tools

- Aircraft Mechanic’s Toolkit

- TM 1-2840-248-23

References

- PARA 4-4-6
- PARA 6-4-27

1-7-38.1 Engine Whine/High Frequency Vibration Inspection.

- If a loud whine or high frequency vibration can be identified as coming from No. 1 or No. 2 gear box input area, do this:
 - Reduce the suspected engine power to ground idle with opposite engine driving the main rotor at 100% Nr. If whine increases in volume at about 85% Ng, replace the suspected engine output shaft assembly (PARA 4-4-6) and input module flange (PARA 6-4-27).

- Make entry on appropriate forms.

- If whine does not increase, shutdown opposite engine. Increase throttle of the affected engine to fly position. Increase and decrease collective pitch. If the whine changes pitch at the same time Ng speed changes, perform borescope inspection (SEI 671 and TM 1-2840-248-23).
- If borescope inspection shows no signs of foreign object damage to engine compressor, replace the engine output shaft assembly (PARA 4-4-6) and input module flange (PARA 6-4-27).

1-7-39 HARD LANDING.

PARAGRAPH BREAKDOWN

	PAGE
1-7-39.1 Hard Landing	1-7-39-2

INITIAL SETUP	• PARA 1-3-4
Tools	• PARA 2-4-71
• Aircraft Mechanic’s Toolkit	• PARA 2-4-94
• Dial Indicating Scale, 643J	• PARA 2-4-97
• Flashlight	• PARA 2-4-131
• Fluorescent Penetrant Inspection Kit	• PARA 2-6-12
• Inspection Mirror	• PARA 2-6-40
• Powerplant Repairer’s Toolkit	• PARA 5-4-1
• Torque Wrench, 30 - 150 in. lbs	• PARA 5-4-2
	• PARA 5-4-5
Materials/Parts	• PARA 5-5-4
• Soap	• PARA 6-4-39
• Tags	• PARA 6-4-50
References	• PARA 10-4-18
• ASTM E1417	• PARA 16-2-2
• PARA 1-3-2	• PARA 16-4-21

1-7-39.1 Hard Landing.**WARNING**

Injury to personnel and damage to equipment will result unless stresses, thermal effects, and other damage to which parts are subjected as a result of a hard landing can be evaluated and determined to be within design parameters, any parts of the component that may affect safe operation shall be assumed to be damaged beyond acceptable limits and shall not be reused on airworthy helicopters.

- a. Inspect main landing gear attachment, hardware, and surrounding structure.
- b. Visually inspect the axle, wheel, drag beam assembly, drag beam support fitting, bolts, and pins for cracks, deformation, or loose hardware.
- c. Remove drag beam support fairings (PARA 2-4-94) and cabin floor panels (PARA 2-4-71).
- d. Inspect skins and stringers in the area of the drag beam support fitting for cracks, deformation, and loose or sheared rivets.
- e. Inspect tub interior, especially in area of landing gear, for damage, distortion, foreign matter, and corrosion.
- f. Repair or replace components that show evidence of cracks, looseness, or permanent deformation (PARA 2-6-12).
- g. Inspect tail landing gear attachment points and surrounding structure.
- h. Visually inspect the axle, wheel, fork assembly, shock strut, yoke support fitting, shock strut support fitting, bolts, and pins for cracks, deformation, or loose hardware.
- i. Remove tail landing gear fairing (PARA 2-4-97).

- j. Remove drag beam support fairing (PARA 2-4-94).
- k. Remove screws and washers that secure access covers to rear section of tail cone at STA 640.0 (Figure 1-7-39-1, Sheet 1, Detail A).
- l. Inspect skins and stringers in the area of the yoke support and shock strut support fittings for cracks, nicks, dents, corrosion, deformation, and loose, missing, or sheared fasteners.
- m. Inspect tail cone skins and stringers for cracks, nicks, dents, corrosion, deformation, and loose, missing, or sheared fasteners.
- n. Open tail cone drive shaft covers and inspect drive shaft supports (PARA 2-4-131).
- o. Inspect lower shear deck web and corner longerons between yoke support and fold point as follows (Figure 1-7-39-1, Sheet 1, Detail B and Sheet 2, Detail C):

NOTE

Damage could be very fine and difficult to see. Inspection areas may have to be cleaned.

- (1) Clean inspection areas as necessary using soap and water, and/or appropriate degreaser, if necessary.

NOTE

Make sure nested areas are carefully inspected (longerons, -103 and -105).

- (2) Using flashlight and inspection mirror as necessary, inspect tail landing gear shear deck assembly, 70211-05010-041, for cracks, nicks, dents, corrosion, and loose or missing fasteners. Pay particular attention to web, -101, channels, -107 and -108, and longerons, -103, -104, -105, and -106. If web, -101, is damaged, repair web. If channels are damaged, repair channels. If damage is severe in these areas, contact Sikorsky Aircraft Corporation engineering for disposition. Damage

to longerons, -103 through -106, is a serious matter. Contact Sikorsky Aircraft Corporation engineering for disposition.

- p. Inspect tail cone canted hinge bulkhead fitting tension joint ears, 70219-05001-107 or -113, as follows (Figure 1-7-39-1, Sheet 2, Detail D):

NOTE

Damage could be very fine and difficult to see. Inspection areas may have to be cleaned.

- (1) Clean inspection areas as necessary using soap and water, and/or appropriate degreaser, if necessary.
- (2) Using flashlight and inspection mirror as necessary, inspect tail cone canted hinge bulkhead fitting tension joint ears for cracks, nicks, dents, corrosion, and loose or missing fasteners. **DAMAGE TO BULKHEAD FITTING IS A SERIOUS MATTER.** Contact Sikorsky Aircraft Corporation engineering for disposition.
- q. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect tail cone fold fitting aft side between the centerlines of upper shear pins and upper lightening holes (ASTM E1417).
- r. Repair or replace components that show evidence of cracks, leakage, or deformation (PARA 2-6-40).
- s. Install access covers on rear section of tail cone at STA 640.0 with screws and washers (Figure 1-7-39-1, Sheet 1, Detail A).
- t. Service main and tail landing gear struts (PARA 1-3-2 and PARA 1-3-4).
- u. Inspect inside of main rotor hub arm for signs of contact with spindle retention nut. If evidence of contact is found, replace spindles (PARA 5-4-5) and main rotor hub (PARA 5-4-1). Tag components "DAMAGE OCCURRED DUE TO EXCESSIVE BLADE FLAPPING."
- v. Inspect droop stop ears for cracks. If ear is cracked, remove spindle (PARA 5-4-5), and re-

place elastomeric spherical bearing (PARA 5-5-4). Tag removed elastomeric spherical bearing "DAMAGE OCCURRED DUE TO EXCESSIVE BLADE FLAPPING."

- w. Inspect radar warning system blade antenna, AN/APR-39, for damage and distortion.
- x. Inspect fuel system breakaway valves for visible yellow line around center of valve. If yellow line is visible, replace valve (PARA 10-4-18).
- y. If main rotor blades contacted tail cone, pylon, or IR jammer with or without blade tip damage, perform the following:
 - (1) Replace spindles (PARA 5-4-5). Tag spindle "POSSIBLE DAMAGE OCCURRED DUE TO BLADE STRIKE." Also note what blade contacted.
 - (2) Inspect hub for damage (PARA 5-4-2). If hub is to be replaced, tag damaged hub "DAMAGE OCCURRED DUE TO EXCESSIVE BLADE FLAPPING."
- z. Check oil cooler drive shaft for preloading, as follows:
 - (1) Disconnect tail rotor drive shaft section II near No. 1 viscous damper bearing (PARA 6-4-39).
 - (2) Remove bolts attaching rear oil cooler support.
 - (3) Using dial indicating scale, measure force needed to align rear oil cooler support boltholes with boltholes in bearing flange. **FORCE MUST NOT EXCEED 10 POUNDS.** Evaluate as follows:
 - (a) If preload is more than 10 pounds, replace oil cooler (PARA 6-4-50).
 - (b) If preload is 10 pounds or less, install oil cooler rear support to bearing support with bolts, washers, shims and nuts. **TORQUE NUTS TO 115 - 125 INCH-POUNDS.**

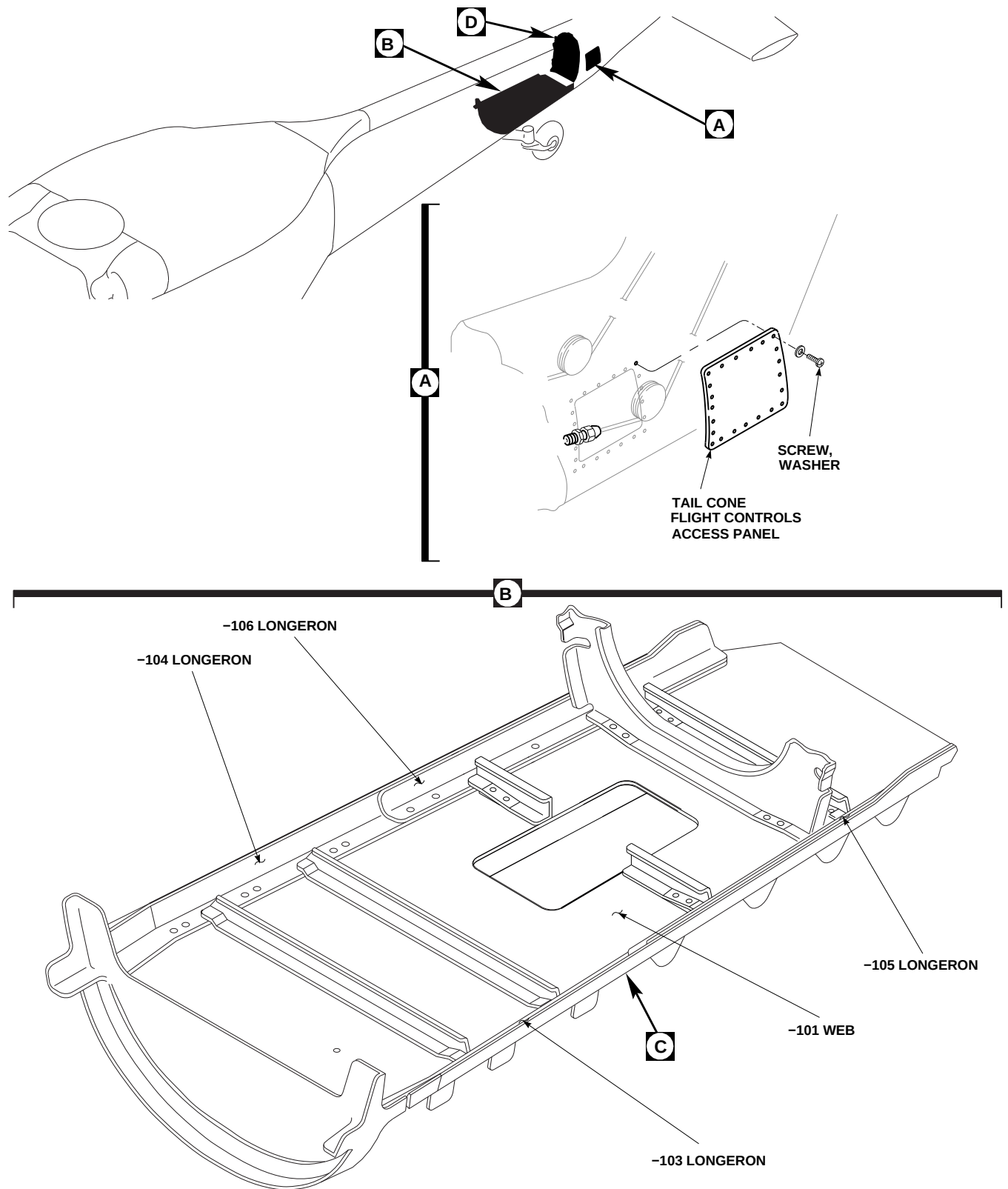
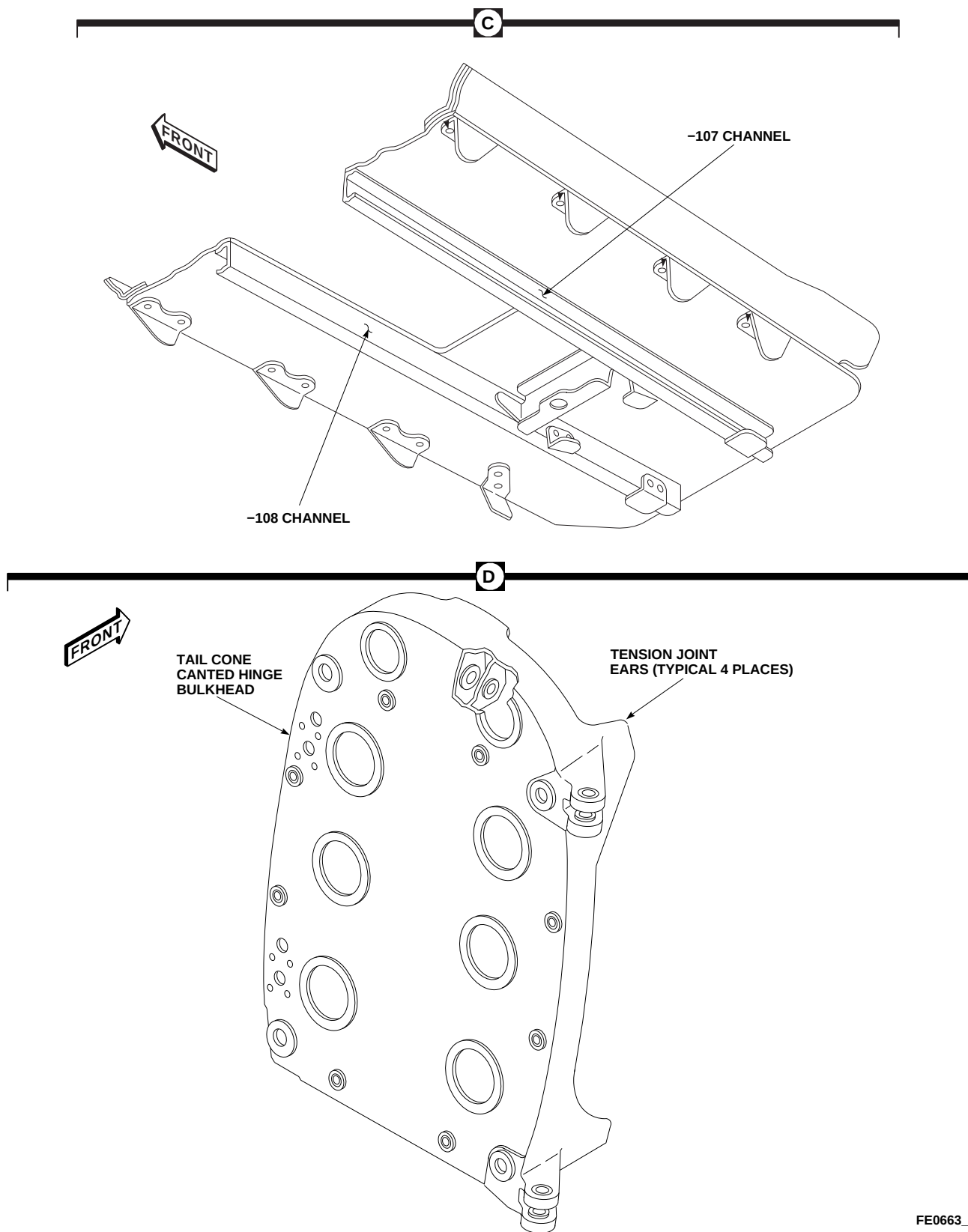
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FIGURE 1-7-39-1. TAIL LANDING GEAR SHEAR DECK ASSEMBLY, 70211-05010-041, AND TAIL CONE CANTED HINGE BULKHEAD, 70219-05001-044 OR -046 (SHEET 1 OF 2)



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FIGURE 1-7-39-1. TAIL LANDING GEAR SHEAR DECK ASSEMBLY, 70211-05010-041, AND TAIL CONE CANTED HINGE BULKHEAD, 70219-05001-044 OR -046 (SHEET 2 OF 2)

Make sure shimming procedures in PARA 6-4-50 are followed.

- (4) Reconnect tail rotor drive shaft section II (PARA 6-4-39).

aa. If ESSS was installed, inspect as follows:

- (1) Remove upper and lower ESSS fairings (PARA 16-4-21).
- (2) Inspect upper and lower attachment fittings, surrounding structure, and struts for damage, deformation, and cracks.
- (3) If 230 gallon external fuel tanks were installed, inspect as follows:
 - (a) Inspect external tank suspension lugs, suspension lug bushings, and

the structure around lugs for damage, deformation, delamination, or any indication of movement.

- (b) **IF DAMAGE IS LOCATED WITHIN 6-INCHES OF THE LUGS, TANK MUST BE RETURNED TO CONTRACTOR.**

- (c) If no damage is found, perform functional check (PARA 16-2-2). If tank passes functional check, it may be returned to service.

- (4) Install upper and lower ESSS fairings (PARA 16-4-21).

ab. Evaluate operational readiness and recommend corrective action.

1-7-40 HELICOPTER SUBJECT TO EXCESSIVE SPIN RATE.

PARAGRAPH BREAKDOWN

		PAGE
1-7-40.1	Helicopter Subject to Excessive Spin Rate	1-7-40-1

INITIAL SETUP

- PARA 6-4-54

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 5-4-38
- PARA 6-4-52

1-7-40.1 Helicopter Subject to Excessive Spin Rate.

- a. At next 10 hour/14 day inspection, visually inspect the following:
 - (1) Pylon/tail cone fold frame at hinge points, plus longeron attachment adjacent to fittings.
 - (2) Tail gear support structure at STA 605.0.

- b. If helicopter has spun more than two complete turns, and did so at a rate exceeding normal hovering turn rate, replace the following at next 500 hour periodic inspection:
 - (1) Tail rotor blades (PARA 5-4-38).
 - (2) Inboard retention plate (PARA 6-4-54).
 - (3) Tail gear box (PARA 6-4-52).

1-7-41 HELICOPTER SUBJECT TO SALT WATER IMMERSION OR DRY CHEMICAL AND FOAM-TYPE FIRE EXTINGUISHING AGENTS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-41.1	General	1-7-41-2	1-7-41.5	Foam-Type Fire Extinguishing Agents	1-7-41-7
1-7-41.2	Salt Water Immersion - Primary Method	1-7-41-3	1-7-41.6	Main Transmission or Gear Boxes Internal Surfaces Contaminated with Salt Water or Fire Extinguishing Agents	1-7-41-8
1-7-41.3	Salt Water Immersion - Alternate Method	1-7-41-5			
1-7-41.4	Dry Chemical Fire Extinguishing Agents	1-7-41-5			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Drill
- Protection, Eye
- Static Ground Cable
- Wet/Dry Vacuum

Materials/Parts

- Brush, Acid, Item 84, Appendix D
- Brush, Bristle
- Cleaning Compound, Solvent, Item 100, Appendix D
- Corrosion Preventive Compound, Item 114, Appendix D
- Corrosion Preventive Compound, Item 115, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D

- Foam Liquid, Fire Extinguishing, Item 141A, Appendix D
- Machinery Towel, Item 211, Appendix D
- Sponge
- Tiedown Straps

References

- Appendix D
- PARA 1-5-1
- PARA 1-7-26
- PARA 2-4-44
- PARA 2-6-2
- PARA 4-4-1
- PARA 6-4-1
- PARA 6-4-31
- PARA 6-4-52
- PARA 15-4-1

• SEI 671

• TM 1-2840-248-23

1-7-41.1 General.**WARNING**

Injury to personnel and damage to equipment will result unless stresses, thermal effects, and other damage to which parts are subjected as a result of a fire can be evaluated and determined to be within design parameters, any parts of the component that may affect safe operation shall be assumed to be damaged beyond acceptable limits and shall not be reused on airworthy helicopters.

- a. This paragraph describes decontamination procedures to be followed after incidents involving exposure to large amounts of salt water or fire extinguishing agents other than Halon. Refer to PARA 1-7-26 for decontamination of areas exposed to Halon-type fire extinguishing agents.
- b. Helicopters exposed to salt water, dry chemical and foam-type fire extinguishing agents require immediate action to prevent serious corrosion damage.

CAUTION

Magnesium parts such as main transmission and gear box housings are very susceptible to corrosion attack when exposed to salt water or fire extinguishing agents. Magnesium parts will begin to corrode very rapidly after contamination. Removal of these components is recommended to make sure decontamination is thorough.

NOTE

If salt water or fire extinguishing agents get inside main transmission or gear box

housings, contact Sikorsky ILS. Immediately decontaminate these internal surfaces (PARA 1-7-41.6), pending Sikorsky ILS recommendations.

- c. Removal of components other than those listed below may be accomplished depending upon severity of exposure to salt water or fire extinguishing agents. Remove the following components to make sure decontamination is thorough:
 - (1) Main transmission (PARA 6-4-1).
 - (2) Intermediate gear box (PARA 6-4-31).
 - (3) Tail gear box (PARA 6-4-52).
 - (4) Engines (PARA 4-4-1).
 - (5) APU (PARA 15-4-1).
 - (6) Pilot's and copilot's seats (PARA 2-4-44).
 - (7) Avionics and electrical equipment.
- d. A priority list shall be created to aid in recovery and treatment of salvageable equipment. The priority is recommended as follows:
 - (1) Main rotor head, main transmission, intermediate gear box, tail gear box.
 - (2) Engines and accessories, main and tail rotor components, hydraulic pumps, generators.
 - (3) Instruments.
 - (4) Airframe structure.
 - (5) Fuel and hydraulic systems (after systems are drained).
 - (6) Main and tail landing gear.
 - (7) Safety and survival equipment.

- (8) Avionic and electrical equipment.
- (9) Rescue equipment.
- (10) Fixed equipment (seats, etc.).
- (11) Miscellaneous equipment.

WARNING

Injury to personnel and damage to equipment will result if helicopter is not grounded properly before starting decontamination procedures. In instances where fuel cells have ruptured and fuel or fuel vapor are present, make sure that fuel spills are cleaned up and vapors are purged. Make sure that helicopter has been certified as fire and explosion safe by qualified personnel.

- e. Connect static ground cable to main landing gear lower stage air valve and ground.

CAUTION

Damage to equipment will result if areas contaminated with Halon-type fire extinguishing agents are flushed with water. Halon becomes corrosive when contacted by water. If Halon-type fire extinguishing agents were used, refer to PARA 1-7-26 for decontamination procedures.

- f. Determine which fire extinguishing agent was used. Traces of foam-type agents may not be present, however, moist residues often indicate foam application. Dry chemical agents may be present in joints, faying surfaces, etc. Halon will leave no visible trace. Carbon dioxide will leave no visible trace and does not require decontamination. After determining which fire extinguishing agent was used, refer to the appropriate decontamination procedures as listed below:
 - (1) Dry chemical agents refer to PARA 1-7-41.4.

- (2) Foam-type agents refer to PARA 1-7-41.5.
- (3) Halon-type agents refer to PARA 1-7-26.
- g. Decontaminate engines (SEI 671 and TM 1-2840-248-23).

1-7-41.2 Salt Water Immersion - Primary Method.

NOTE

If salt water gets inside main transmission or gear box housings, internal surfaces must be decontaminated, in addition to performing the following procedures.

- a. If necessary, decontaminate internal surfaces (PARA 1-7-41.6).
- b. Open or remove access and inspection panels, doors, floor panels, soundproofing panels, etc. Allow salt water to drain off where possible. If salt water is trapped in undrainable areas inside cockpit or cabin tub, perform the following:
 - (1) Drill out fasteners at low points.
 - (2) If salt water is still trapped, drill hole in skin at low points.
- c. Remove tiedown straps and clamps from wire bundles and cables to make sure flushing is thorough.

NOTE

- Electrical switches and circuit breakers are normally enclosed in a sealed case. Decontamination of switches and circuit breakers is not possible. Contaminated switches and circuit breakers shall be replaced.
- Terminal boards, junction boxes, relay panels, and circuit breaker panels not normally removed from helicopter shall be decontaminated in their installed position.

- Disassemble electrical connectors and receptacles prior to decontamination.
- d. Flush all areas thoroughly with clean fresh water. Components which have been removed should be immersed in fresh water and then thoroughly flushed with clean fresh water. Allow to drain.
- e. Dry areas by vacuuming excess water and blotting with machinery towels, Item 211, Appendix D. Complete drying in a room with temperature of 130°F (54°C) maximum or in a low humidity, well ventilated room by one of the following methods:

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- (1) Compressed air, not to exceed 10 psi.
- (2) Electric fans or hand fans.
- (3) Hot air blowers.

- f. Inspect area for evidence of salt. If salt still remains, perform the following:

- (1) Mix a solution of 1 part cleaning compound solvent, Item 100 (Type I or II), Appendix D, with 9 parts clean, fresh water.
- (2) Apply solution to affected areas and scrub with bristle brush and sponge.
- (3) Rinse thoroughly with clean, fresh water. Allow to drain.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- (4) Dry areas by vacuuming excess water and blotting with machinery towels, Item 211, Appendix D. Complete drying by blowing compressed air, not to exceed 10 psi.
- (5) Inspect for evidence of salt. If salt still remains, repeat scrubbing procedure.

NOTE

Do not apply corrosion preventive compound to electrical connectors or electrical and avionic components.

- g. Apply generous amounts of corrosion preventive compound, Item 114, Appendix D, to all areas that cannot be properly drained or contain recesses which are difficult to reach. Make sure all surfaces are coated.

NOTE

Corrosion preventive compound deposits a thin, nonconductive film which must be removed from electrical connectors and mounting surfaces for proper operation of electrical and avionic components.

- h. Treat electrical and avionic components as follows:
 - (1) Apply corrosion preventive compound, Item 115, Appendix D, to electrical connectors and electrical and avionic components.

- (2) Remove corrosion preventive compound from electrical contact points and mounting surfaces using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D. Use a small brush to remove corrosion preventive compound from inside electrical connectors.
- i. Repair any holes drilled in skin (PARA 2-6-2) or reinstall fasteners that were removed to drain salt water.
- j. Replace tiedown straps and clamps to wire bundles and cables.
- k. Lubricate affected areas (PARA 1-5-1). If component has pressure fittings, apply enough lubricant to displace all contaminated lubricant.

NOTE

To make sure of future inspection, all affected areas must be entered in logbook and noted as corrosion prone.

- l. Enter affected areas in logbook.
- m. Close or install access and inspection panels, doors, floor panels, soundproofing panels, etc.

1-7-41.3 Salt Water Immersion - Alternate Method.

NOTE

This procedure is to be used only when fresh water supply is in limited supply or not available, or when time prevents immediate flushing with water. Primary method of decontamination shall be performed when time permits and/or fresh water is available.

- a. Open or remove access and inspection panels, doors, floor panels, soundproofing panels, etc. Allow salt water to drain off where possible.

- b. Method I - Apply generous amounts of corrosion preventive compound, Item 114, Appendix D, to all affected areas and components. Make sure all surfaces are coated.
- c. Method II:
 - (1) Mix a solution of 1 part cleaning compound, Item 100 (Type I or II), Appendix D, with 9 parts clean, fresh water.
 - (2) Apply solution to affected areas and scrub with bristle brush and sponge. Scrub until contaminants become intermixed with cleaning solution.
 - (3) Wipe area thoroughly with clean towels.
 - (4) Apply generous amounts of corrosion preventive compound, Item 114, Appendix D, to all affected areas and components. Make sure all surfaces are coated.
- d. Lubricate affected areas (PARA 1-5-1). If component has pressure fittings, apply enough lubricant to displace all contaminated lubricant.
- e. Close or install access and inspection panels, doors, floor panels, soundproofing panels, etc.

1-7-41.4 Dry Chemical Fire Extinguishing Agents.

NOTE

If dry chemical fire extinguishing agents get inside main transmission or gear box housings, decontaminate internal surfaces (PARA 1-7-41.6) in addition to the following procedures.

- a. Open or remove access and inspection panels, doors, floor panels, soundproofing panels, etc.
- b. Remove tiedown straps and clamps from wire bundles and cables to make sure powder residue is completely removed.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- c. Vacuum powder residue from affected areas. Use a soft bristle brush and compressed air, not to exceed 10 psi, to dislodge powder from recesses and close-fitting areas.
- d. Mix a solution of 1 part solvent, cleaning compound, Item 100 (Type I or II), Appendix D, with 9 parts clean, fresh water.

NOTE

- Terminal boards, junction boxes, relay panels, and circuit breaker panels not normally removed from helicopter shall be decontaminated in their installed position.
- Disassemble electrical connectors and receptacles prior to decontamination.
- e. Apply solution to affected areas and scrub with bristle brush and sponge.
- f. Rinse thoroughly with clean, fresh water. Allow to drain.
- g. Dry areas by vacuuming excess water and blotting with machinery towels, Item 211, Appendix D. Complete drying in a room with temperature of 130°F (54°C) maximum or in a low humidity, well ventilated room by one of the following methods:

WARNING

Injury to personnel will result if precautions are not taken while using com-

pressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- (1) Compressed air, not to exceed 10 psi.
- (2) Electric fans or hand fans.
- (3) Hot air blowers.

NOTE

Do not apply corrosion preventive compound to electrical connectors or electrical and avionic components.

- h. Apply generous amounts of corrosion preventive compound, Item 114, Appendix D, to all areas that cannot be properly drained or contain recesses which are difficult to reach. Make sure all surfaces are coated.
- i. Treat electrical and avionic components as follows:

NOTE

Corrosion preventive compound deposits a thin, nonconductive film which must be removed from electrical connectors and mounting surfaces for proper operation of electrical and avionic components.

- (1) Using acid brush, Item 84, Appendix D, apply corrosion preventive compound, Item 115, Appendix D, to electrical connectors and electrical and avionic components.
- (2) Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, remove corrosion preventive compound from electrical contact points and mounting surfaces. Use small brush to remove corrosion preventive compound from inside electrical connectors.

- j. Replace tiedown straps and clamps to wire bundles and cables.
- k. Lubricate affected areas (PARA 1-5-1). If component has pressure fittings, apply enough lubricant to displace all contaminated lubricant.

NOTE

To make sure of future inspection, all affected areas must be entered in logbook and noted as corrosion prone.

- l. Enter affected areas in logbook.
- m. Close or install access and inspection panels, doors, floor panels, soundproofing panels, etc.

1-7-41.5 Foam-Type Fire Extinguishing Agents.

NOTE

If foam-type fire extinguishing agents get inside main transmission or gear box housings, decontaminate internal surfaces (PARA 1-7-41.6) in addition to the following procedures.

- a. Open or remove access and inspection panels, doors, floor panels, soundproofing panels, etc.
- b. Remove tiedown straps and clamps from wire bundles and cables to make sure flushing is thorough.

NOTE

- Electrical switches and circuit breakers are normally enclosed in a sealed case. Decontamination of switches and circuit breakers is not possible. Contaminated switches and circuit breakers shall be replaced.
- Terminal boards, junction boxes, relay panels, and circuit breaker panels not normally removed from helicopter shall

be decontaminated in their installed position.

- Disassemble electrical connectors and receptacles prior to decontamination.
- c. If a mixture of salt water and fire extinguishing foam liquid, Item 141A, Appendix D, was used, perform the following:
 - (1) Flush all areas thoroughly with clean fresh water. Components which have been removed should be immersed in fresh water and then thoroughly flushed with clean fresh water. Allow to drain.
 - (2) Mix a solution of 1 pint of foam with 10 gallons of fresh water.
 - (3) Apply solution to affected areas using spray equipment. Keep areas wet for 3 to 5 minutes.
 - (4) Proceed to step e.
- d. If a mixture of fresh water and fire extinguishing foam liquid, Item 141A, Appendix D, was used, perform the following:
 - (1) Flush all areas thoroughly with clean fresh water. Components which have been removed should be immersed in fresh water and then thoroughly flushed with clean fresh water. Allow to drain.
 - (2) Proceed to step e.
- e. Mix a solution of 1 part cleaning compound, Item 100 (Type I or II), Appendix D, with 9 parts clean, fresh water.
- f. Apply solution to affected areas and scrub with bristle brush and sponge.
- g. Dry areas by vacuuming excess water and blotting with machinery towels, Item 211, Appendix D. Complete drying in a room with temperature of 130°F (54°C) maximum or in a

low humidity, well ventilated room by one of the following methods:

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- (1) Compressed air, not to exceed 10 psi.
- (2) Electric fans or hand fans.
- (3) Hot air blowers.

NOTE

Do not apply corrosion preventive compound to electrical connectors or electrical and avionic components.

- h. Apply generous amounts of corrosion preventive compound, Item 114, Appendix D, to all areas that cannot be properly drained or contain recesses which are difficult to reach. Make sure all surfaces are coated.

NOTE

Corrosion preventive compound deposits a thin, nonconductive film which must be removed from electrical connectors and mounting surfaces for proper operation of electrical and avionic components.

- i. Treat electrical and avionic components as follows:
 - (1) Apply corrosion preventive compound, Item 115, Appendix D, to electrical connectors and electrical and avionic components.

- (2) Remove corrosion preventive compound from electrical contact points and mounting surfaces using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D. Use a small brush to remove corrosion preventive compound from inside electrical connectors.

- j. Replace tiedown straps and clamps to wire bundles and cables.
- k. Lubricate affected areas (PARA 1-5-1). If component has pressure fittings, apply enough lubricant to displace all contaminated lubricant.

NOTE

To make sure of future inspection, all affected areas must be entered in logbook and noted as corrosion prone.

- l. Enter affected areas in logbook.
- m. Close or install access and inspection panels, doors, floor panels, soundproofing panels, etc.

1-7-41.6 Main Transmission or Gear Boxes Internal Surfaces Contaminated with Salt Water or Fire Extinguishing Agents.

NOTE

Step b. through step j. make up an interim decontamination process. This interim decontamination process is to be used as temporary measure until Sikorsky Engineering can evaluate extent of damage.

- a. If internal surfaces of main transmission or gear boxes are contaminated, immediately contact Sikorsky ILS.
- b. Drain all fluids. Partial disassembly not normally performed at field level maintenance may be done to make sure that all contaminants are drained.
- c. Flush all internal surfaces and passages with fresh water.

- d. Mix a solution of 1 part solvent, cleaning compound, Item 100 (Type I or II), Appendix D, with 9 parts clean, fresh water.
- e. Flush all internal surfaces and passages with cleaning solution.
- f. Rinse thoroughly with fresh water. Allow to drain.
- g. Fill housing with corrosion preventive compound, Item 114, Appendix D.
- h. Rotate main transmission shaft or gear box input shaft five turns.
- i. Allow corrosion preventive compound to remain in housing for 4 hours.
- j. Drain corrosion preventive compound.

1-7-42 MAIN ROTOR BLADES DROPPED DURING BLADE FOLD.

PARAGRAPH BREAKDOWN

	PAGE
1-7-42.1 Main Rotor Blades Dropped During Blade Fold	1-7-42-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 5-3-1
- PARA 5-4-18
- PARA 5-4-22
- PARA 11-4-8
- PARA 11-4-11
- PARA 11-4-13
- PARA 11-4-14
- PARA 11-4-16
- PARA 11-4-27

- PARA 11-4-28
- PARA 11-4-42 through PARA 11-4-48
- PARA 11-4-50
- PARA 11-4-51 through PARA 11-4-58
- PARA 11-4-63 through PARA 11-4-70
- PARA 11-4-73 through PARA 11-4-75
- PARA 11-4-77
- PARA 11-4-80 through PARA 11-4-87
- PARA 11-4-90 through PARA 11-4-98
- PARA 11-4-100
- PARA 11-4-101
- PARA 11-4-110
- PARA 11-4-112
- PARA 11-4-113

1-7-42.1 Main Rotor Blades Dropped During
Blade Fold.

NOTE

This inspection only pertains if the pitch control rods were connected when blade folding was attempted and the blade fell from the spindle.

- Inspect elastomeric bearing (PARA 5-3-1).
- Inspect the following parts for looseness, distortion, misalignment, cracks, nicks, bearing abnormality, and elongated rivets. Special attention should be given to linkages between stationary swashplate. Replace damaged parts.
 - Pitch control rods (PARA 5-4-18).

- (2) Swashplate (PARA 5-4-22).
- (3) Pilot's and copilot's yaw control pedals (PARA 11-4-8).
- (4) Pilot's and copilot's yaw control pedal adjuster (PARA 11-4-11).
- (5) Pilot's and copilot's cyclic stick (PARA 11-4-16).
- (6) Pilot's and copilot's collective stick (PARA 11-4-27 and PARA 11-4-28).
- (7) Yaw trim and roll trim servos (PARA 11-4-52 and PARA 11-4-53).
- (8) Yaw boost servo, pitch/trim assembly, collective boost assembly, and roll actuator (PARA 11-4-67, PARA 11-4-68, PARA 11-4-69, and PARA 11-4-70).
- (9) Mixer (PARA 11-4-77).
- (10) Primary servos (PARA 11-4-87).
- (11) Collective, pitch, lateral, and yaw torque shaft and levers (PARA 11-4-55, PARA 11-4-56, PARA 11-4-57, and PARA 11-4-58).
- (12) Midsection bellcranks and shaft (PARA 11-4-50).
- (16) Pilot's and copilot's directional and cyclic control rods (PARA 11-4-42 and PARA 11-4-43).
- (17) Pilot's and copilot's collective control rods (PARA 11-4-44 and PARA 11-4-45).
- (18) Cabin control rods (PARA 11-4-46).
- (19) Collective, pitch/trim, lateral, and yaw boost input control rods (PARA 11-4-63, PARA 11-4-64, PARA 11-4-65, and PARA 11-4-66).
- (20) Trim servo control rod (PARA 11-4-54).
- (21) Collective, roll trim, and yaw boost servo to mixer control rods (PARA 11-4-73, PARA 11-4-75, and PARA 11-4-85).
- (22) Pitch/trim to mixer link (PARA 11-4-74).
- (23) Mixer to forward, aft, and lateral primary servo control rods (PARA 11-4-80, PARA 11-4-81, and PARA 11-4-82).
- (24) Mixer to yaw lever pushrod (PARA 11-4-83).
- (25) Yaw lever to forward quadrant control rod (PARA 11-4-84).
- (26) Forward, aft, lateral, and long control rods (PARA 11-4-90, PARA 11-4-91, PARA 11-4-92, and PARA 11-4-94).
- (27) Swashplate links (PARA 11-4-95).
- (28) Cockpit bellcranks and levers (PARA 11-4-47).
- (29) Cabin bellcranks and bellcrank support (PARA 11-4-48).
- (30) Upper cabin links and levers (PARA 11-4-51).
- (31) Walking beam (PARA 11-4-93).
- (32) Forward, lateral, and aft bellcranks (PARA 11-4-96, PARA 11-4-97, and PARA 11-4-98).

NOTE

If damage is found in area of yaw lever adjustable pedal stops, inspect tail rotor servo to tail rotor quadrant pushrod (PARA 11-4-112), tail rotor servo (PARA 11-4-113), and tail rotor quadrant (PARA 11-4-110).

- (13) Yaw lever and support (PARA 11-4-85).
- (14) Forward quadrant and supports (PARA 11-4-86).
- (15) Pilot's and copilot's yaw pedals and adjuster control rods (PARA 11-4-13 and PARA 11-4-14).

- (33) Forward and aft bellcrank supports
(PARA 11-4-100 and PARA 11-4-101).

1-7-43 MAIN ROTOR BLADES SUBJECT TO HIGH WINDS.

PARAGRAPH BREAKDOWN

	PAGE
1-7-43.1 Main Rotor Blades Subject to High Winds	1-7-43-1

INITIAL SETUP

- PARA 5-3-11

Tools

- Aircraft Mechanic’s Toolkit
- Fluorescent Penetrant Inspection Kit

- PARA 5-4-2

- PARA 5-4-5

Materials/Parts

- Tags

- PARA 5-4-9

- PARA 5-5-4

References

- ASTM E1417
- PARA 5-3-1

- PARA 11-4-4

1-7-43.1 Main Rotor Blades Subject to High Winds.

- If main rotor blades are subjected to winds great enough to break antifiap bracket, perform the following:

- Inspect elastomeric bearing assembly end-plate for nicks and gouges (PARA 5-5-4).

- Inspect elastomeric bearing for rubber disbond.

NOTE

Refer to PARA 5-4-9 for maintenance procedures and damage limitations.

- Inspect droop stop attachment ears for cracks. **NONE ALLOWED.** If crack is found, replace elastomeric bearing assembly with spindle (PARA 5-4-5). Tag bearing "DAMAGE OCCURRED DUE TO EXCESSIVE WINDS."

- Remove and replace spindle, without elastomeric bearings (PARA 5-4-5), and return to approved overhaul facility with tag stating "DAMAGE OCCURRED DUE TO EXCESSIVE WINDS."
- Inspect inside of main rotor hub for evidence of contact by spindle nut (PARA 5-4-2).

- Inspect main rotor blades for abnormal twists and bends (pay particular attention to inboard 5-foot section of blade).
- Coin-tap inboard 5-foot section of blade over spar to check bonding (PARA 5-3-11).

(8) Visually inspect flight control rods and linkages between main rotor head and pilot-assist servos (pay particular attention to mixer, mixer limiter bearings, and cams).

(9) Perform main rotor rig check (PARA 11-4-4).

b. If main rotor blades are subjected to winds great enough to roll over and/orpeen anti flap stop, but not break anti flap bracket, perform the following:

NOTE

Refer to PARA 5-4-9 for maintenance procedures and damage limitations.

- (1) Remove anti flap bracket.
- (2) Inspect inside of main rotor hub for evidence of contact by spindle nut (PARA 5-4-2).
- (3) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type II, Method C, anti flap bracket for cracks (ASTM E1417). If anti flap bracket is broken, remove and replace spindle, without elastomeric bearing (PARA 5-4-5), and anti flap bracket assembly. Tag

parts "DAMAGE OCCURRED DUE TO EXCESSIVE WINDS."

(4) Inspect elastomeric bearing endplate for nicks and gouges.

(5) Inspect elastomeric bearing for rubber disbond (PARA 5-3-1).

(6) Inspect droop stop attachment ears for cracks. **NONE ALLOWED.** If crack is found, replace elastomeric bearing assembly with spindle (PARA 5-4-5). Tag bearing "DAMAGE OCCURRED DUE TO EXCESSIVE WINDS."

(7) Inspect main rotor blades for abnormal twists and bends (pay particular attention to inboard 5-foot section of blade).

(8) Coin tap inboard 5-foot section of blade over spar to check bonding (PARA 5-3-11).

(9) Visually inspect flight control rods and linkages between main rotor head and pilot-assist servos (pay particular attention to mixer, mixer limiter bearings, and cams).

(10) Perform main rotor rig check (PARA 11-4-4).

1-7-44 MAIN ROTOR DROOP STOP POUNDING.

PARAGRAPH BREAKDOWN

		PAGE
1-7-44.1	Main Rotor Droop Stop Pounding	1-7-44-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Tag

1-7-44.1 Main Rotor Droop Stop Pounding.

- If droop stop pounding is experienced, inspect droop stop ears (on elastomeric bearing end plate) for cracks. If one or two ears are cracked, remove and replace elastomeric bearing.
- Inspect inside of hub arm for signs of contact with spindle retention nut. If contact is found,

remove and scrap spindle and return hub to an authorized overhaul facility with tag stating "DAMAGE DUE TO EXCESSIVE SPINDLE FLAPPING".

- If main rotor blades contact tail cone/pylon, remove and scrap spindle and return hub to an authorized overhaul facility with tag stating "DAMAGE DUE TO EXCESSIVE BLADE FLAPPING".

1-7-45 OPERATING HELICOPTER IN EROSIIVE CONDITIONS.

PARAGRAPH BREAKDOWN

	PAGE
1-7-45.1 Erosive Conditions	1-7-45-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 1-4-3
- PARA 1-4-4

- PARA 1-4-6
- PARA 1-4-7
- PARA 5-3-6

1-7-45.1 Erosive Conditions.

NOTE

Items in this section are minimum requirements for helicopters operating in erosive environments. Areas to be inspected and frequency of inspection will depend on severity of erosive environment.

- Clean swashplate guide and spherical bearing (PARA 1-4-7).

- Inspect swashplate guide and spherical bearing (PARA 5-3-6).
- Clean main rotor blade pins (PARA 1-4-6).
- Clean helicopter exterior (PARA 1-4-3).

NOTE

Wash engines of aircraft operated within 200 miles of an active volcano, or operated within 10 miles of salt water after last flight of the day.

- Clean engines (PARA 1-4-4).

1-7-46 OPERATING HELICOPTER IN HEAVY RAINFALL 70583–702374 .

PARAGRAPH BREAKDOWN

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1-7-46.1	Operating Helicopter in Heavy Rainfall	1-7-46-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 9-4-34

1-7-46.1 Operating Helicopter in Heavy Rainfall.

NOTE

Perform this inspection after helicopter has flown in heavy rainfall.

- Inspect battery vent tubes for collection of water.

- Inspect battery case for collection of water (PARA 9-4-34).
- Remove all water that collected in battery case and battery vent tubes.

1-7-47 POST LIGHTNING STRIKE INSPECTION.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-47.1	Preliminary Inspection	1-7-47-3	1-7-47.8	Intermediate Gear Box, Tail Gear Box, and Tail Rotor Drive Shaft	1-7-47-7
1-7-47.2	Main Rotor and Flight Controls	1-7-47-4			
1-7-47.3	Main Transmission and Tail Rotor Drive Shaft	1-7-47-5	1-7-47.9	Oil Cooler	1-7-47-7
			1-7-47.10	Engine	1-7-47-7
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1-7-47.7	Tail Rotor Pylon	1-7-47-6	1-7-47.14	External Fuel Tank	1-7-47-8

INITIAL SETUP

- PARA 3-4-7

Tools

- PARA 3-4-22

- Aircraft Mechanic’s Toolkit

- PARA 3-4-25

Materials/Parts

- PARA 4-2-1

- Tags

- PARA 4-4-5

References

- PARA 4-4-6

- PARA 1-3-2

- PARA 5-3-11

- PARA 1-3-4

- PARA 5-3-12

- PARA 1-6-9

- PARA 5-4-5

- PARA 2-3-14

- PARA 5-4-13

- PARA 2-3-15

- PARA 5-4-18

- PARA 2-4-137

- PARA 5-4-22

- PARA 2-4-139

- PARA 5-4-27

- PARA 3-4-1

- PARA 5-4-38

- PARA 6-4-1
- PARA 6-4-25
- PARA 6-4-31
- PARA 6-4-38
- PARA 6-4-39
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- PARA 9-1-1
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- PARA 11-4-100
- PARA 11-4-101
- PARA 11-4-106
- PARA 11-4-113
- PARA 15-2-1
- PARA 15-2-2
- PARA 15-4-1
- PARA 16-4-71
- SEI 671
- TM 1-2840-248-23
- TM 11-70-23

1-7-47.1 Preliminary Inspection.

NOTE

- Following a known or suspected lightning strike, the visual and operational checks listed below must be done before next flight.
 - Lightning strike damage can be distinguished from other damage, such as physical gouging, denting, or erosion, because it normally appears as a blackened surface, pit, or hole with a burned rim. Since lightning is attracted to sharp objects, pay close attention to protrusions, sharp corners, outer surfaces such as main and tail rotor blade tips and trailing edges, pitot heads, main and tail landing gear, electrostatic discharger, and antennas.
- a. Inspect visible surfaces of main rotor blades and main rotor hub assembly including pitch horns, spindles, hub, dampers, bifilar, main rotor shaft extension, controls, and swashplate assembly for evidence of lightning damage.
 - b. Inspect visible surfaces of tail rotor blades and tail rotor assembly, including inboard and outboard retention plates, pitch horn, pitch change beam and shaft, and control linkages for evidence of lightning damage.
 - c. Visually inspect helicopter flight controls for evidence of lightning damage. Check for freedom of movement of all flight controls.
 - d. Visually inspect entire helicopter surface for arc burns, burn holes, burn runners, paint discoloration, lost or eroded metal, torn metal, bulged or delaminated skin, pits, or punctures, no matter how slight.
 - e. Visually inspect exposed electrical wiring, including wiring in nose compartment and under main rotor pylon fairing.
 - f. Inspect all electrical, AFCS, and avionic components, including antennas for damage.
 - g. Visually inspect gravity refueling ports, pressure refueling port, and fuel cell vent port.
 - h. Visually inspect APU for damage.
 - i. If no visible damage is found, perform the following:
 - (1) Perform a complete operational check of the electrical, avionics, and automatic flight controls systems, and all helicopter subsystems with electrical controls and/or wiring (e.g., engine fire detection and extinguishing systems, caution/advisory warning system, and hydraulic system). Refer to paragraph in the appropriate maintenance manual.
 - (2) Include measurement of voltage standing-wave ratio (VSWR) on all antenna systems and a check of the magnetic compass accuracy.
 - j. If no damage was found after completing operational checks, lightning energy level was very low or was a near miss, then no further inspection is required, inspection is complete.

- k. If any type of damage was found after completing operational checks, refer to appropriate paragraph below for a detailed damage inspection. Document fully all damage found and list all components needing replacement.

1-7-47.2 Main Rotor and Flight Controls.

- a. Tag all components being sent to an authorized overhaul facility with "LIGHTNING STRIKE".
- b. Inspect entire top and bottom length of main rotor blades for the following (PARA 5-3-11):

NOTE

If minor external damage is found, major internal damage may have occurred.

- (1) Entire blade length and trailing edge for arc burns, burn holes, burn runners, arcing, paint discoloration, voids, delamination and puncture marks. **NONE ALLOWED.** If damage is found, return blade to an authorized overhaul facility.
- (2) Tip cap, tip cap nickel abrasion strip, tip cap attaching screws, and trailing edge of trim tab for arc burns or discoloration. **NONE ALLOWED.**
- (3) Ends of blade leading edge nickel abrasion strip or titanium strip for arc burns. **NONE ALLOWED.**
- (4) Remove one main rotor blade pin at a time from blade and check for discoloration. **NONE ALLOWED.** If discoloration is found, replace pin. Partially fold blade and check for damage.
- c. Remove spindle assembly of damaged blade and send to an authorized overhaul facility (PARA 5-4-5). After removal of spindle assembly, inspect the following:
 - (1) Check liner on main rotor hub in spindle attach area for damage and disbonding. **NONE ALLOWED.** If damage or disbonding is found, send hub assembly to an authorized overhaul facility.
- (2) Check spindle attachment bolts for damage and discoloration. **NONE ALLOWED.** If damage or discoloration is found, replace spindle attachment bolts.
- (3) Check outboard damper attaching bolt and upper pitch control rod attaching bolt for damage or discoloration. **NONE ALLOWED.** If damage is found, replace bolt(s).
- (4) Disconnect pitch control rod end from swashplate (PARA 5-4-18). Inspect upper and lower pitch control rod end bearings for arc burns. **NONE ALLOWED.** If arc burns are found, replace control rod (PARA 5-4-18).
- (5) Inspect rotating scissors for damage. **NONE ALLOWED.** If damage is found, replace scissors (PARA 5-4-27).
- (6) Disconnect servo control rods (PARA 11-4-90, PARA 11-4-91, and PARA 11-4-92). Check swashplate for proper rotation and cyclic and collective movement. **REPLACE SWASHPLATE IF ROTATING OR STATIONARY PARTS ARE DAMAGED AND IF PITCH CONTROL ROD OR DAMPER ATTACHING BOLTS WERE DAMAGED** (PARA 5-4-22).
- (7) Inspect entire hub assembly for direct lightning attachment.
- d. After main rotor blade lightning strike, remove damper (PARA 5-4-13). Inspect as follows:
 - (1) Check damper attaching bolts for arcing burns or pitting. **NONE ALLOWED.** If damage is found, replace bolts.
 - (2) Inspect bearings for arc burns or pitting. Fully extend damper control rod and check for arc burns. Check for leaks around indicator and control rods. **NO DAMAGE OR LEAKS ALLOWED.** If arc burns, pitting, or leaks are found, send damper to an authorized overhaul facility.
 - (3) Check damper bracket lugs and bushings for arc burns. **NONE ALLOWED.** If

damage is found, send hub assembly to an authorized overhaul facility.

e. After blade lightning strike, check the following:

- (1) Check bearings, bushings and attaching bolts on flight control rods, bellcranks, and levers between servo output rods and swashplate for arc burns. **NONE ALLOWED.** If arc burns are found, replace part (PARA 5-4-22, PARA 11-4-90, PARA 11-4-91, PARA 11-4-92, PARA 11-4-93, PARA 11-4-94, PARA 11-4-95, PARA 11-4-96, PARA 11-4-97, and PARA 11-4-98).
- (2) Check forward and aft bellcrank support attaching fittings for arc burns. **NONE ALLOWED.** If arc burns are found, replace bellcrank supports (PARA 11-4-100 and PARA 11-4-101).

f. After any lightning strike, check the following:

- (1) Check primary servo input and output linkage bearings for arc burns. Fully extend pistons and check for roughness, pitting, or arc burns. Check rubber bellows for damage. **NO DAMAGE ALLOWED.** If damage is found, send servo to an authorized overhaul facility (PARA 11-4-87).
- (2) If burn marks were found on input linkage, check mixer, yaw boost servo output rod, pitch/trim output rod, collective boost output rod, roll actuator output rod, yaw boost servo, pitch/trim servo, collective boost servo, and roll actuator for arc burns. **NONE ALLOWED.** If arc burns are found, send servo to an authorized overhaul facility and replace burned components (PARA 11-4-67, PARA 11-4-68, PARA 11-4-69, PARA 11-4-70, PARA 11-4-73, PARA 11-4-74, PARA 11-4-75, PARA 11-4-76, and PARA 11-4-77).
- (3) Check yaw boost servo input rod, pitch/trim input rod, collective boost input rod, roll actuator input rod, and torque shafts for arc burns. If arc burns are found,

check upper cabin control rods, control rods on right and left sides of cabin, all bellcranks and control rods in cockpit for arc burns. **NONE ALLOWED.** If arc burns are found, replace damaged components (PARA 11-4-13, PARA 11-4-14, PARA 11-4-42, PARA 11-4-43, PARA 11-4-44, PARA 11-4-45, PARA 11-4-46, PARA 11-4-47, PARA 11-4-48, PARA 11-4-51, PARA 11-4-54, PARA 11-4-55, PARA 11-4-56, PARA 11-4-57, PARA 11-4-58, PARA 11-4-63, PARA 11-4-64, and PARA 11-4-65).

1-7-47.3 Main Transmission and Tail Rotor Drive Shaft.

- a. Visually inspect main gear box mounting points, attaching bolts, and exposed portion of main rotor shaft for arc burns or leakage. **NONE ALLOWED.** If arc burns or leakage are found, remove main transmission and send to an authorized overhaul facility (PARA 6-4-1).
- b. If any main rotor component shows signs of lightning strike, remove main module and send to an authorized overhaul facility (PARA 6-4-1).
- c. Visually inspect input drive shafts, gimbals, and support tubes for arc burns or leakage. **NONE ALLOWED.** If arc burns or leakage are found, send components to an authorized overhaul facility (PARA 6-4-25).

1-7-47.4 Tail Rotor System.

- a. Inspect entire inboard and outboard surfaces of tail rotor blades for the following (PARA 5-3-12):
 - (1) Entire blade length for arc burns, burn holes, burn runners, paint discoloration, voids, and delaminations.
 - (2) Blade tip and trailing edge for arc burns, disbonding, and loose rivets in tip cap.
 - (3) Leading edge abrasion strip and skin-to-horn joint for arc burns.

- (4) Pull boot from horn and check spar and pivot bearing for arc burns.

- (5) **NO DAMAGE ALLOWED.**

b. If damage was found, perform the following:

- (1) Remove outboard retention plate and outboard tail rotor blade (PARA 5-4-38). Check spar retention plate area for fuzzy appearance and/or loose fibers. **NONE ALLOWED.** If damage is found, return blade to an authorized overhaul facility.
- (2) Check pitch control rod end bearings, pitch change beam clevis ends and inside taper, and outboard retention plate corners for damage. **NONE ALLOWED.** If damage is found, return components to an authorized overhaul facility.
- (3) Check pitch control rod and retention plate attaching bolts for damage. **NONE ALLOWED.** If damage is found, replace bolt(s).
- (4) Check inboard retention plate (paying particular attention to corners) for damage. **NONE ALLOWED.** If damage is found, send plate to an authorized overhaul facility.

1-7-47.5 Main Landing Gear.

- a. Check strut lower stage piston, axle, drag beam, and attaching parts of main landing gear for damage. Check landing gear dimension for proper servicing and leaks (PARA 1-3-2).
- b. If damage was found, remove main landing gear drag beam and shock strut (PARA 3-4-1 and PARA 3-4-7), and perform the following:
 - (1) Check drag beam attaching bolt and pivot pins for arc burns. **NONE ALLOWED.** If arc burns are found, replace part (PARA 3-4-1).
 - (2) Check ground wire (left side only) and bonding jumper from fuselage to drag beam for damage. **NONE ALLOWED.** If damaged, replace components (PARA 3-4-1).

- (3) Fully extend lower stage shock strut and check for pitting and roughness on chrome-plated surface. **NONE ALLOWED.** If pitting or roughness is found, send shock strut to an authorized overhaul facility.

1-7-47.6 Tail Landing Gear.

- a. Check tail landing gear for damage, proper servicing dimension, and leaks (PARA 1-3-4).
- b. If damage was found, remove tail landing gear shock strut and yoke (PARA 3-4-22 and PARA 3-4-25), and perform the following:
 - (1) Check yoke and landing gear shock strut attaching bolts for arc burns. **NONE ALLOWED.** If arc burns are found, replace bolts.
 - (2) Check inside diameter of bushings in yoke for arc burns. **NONE ALLOWED.** If arc burns are found, replace bushings.
 - (3) Check bearings in shock strut for arc burns. **NONE ALLOWED.** If arc burns are found, send shock strut to an authorized overhaul facility.
 - (4) Fully extend lower stage shock strut and check for pitting and roughness on chrome-plated surface. **NONE ALLOWED.** If pitting or roughness is found, send shock strut to an authorized overhaul facility.

1-7-47.7 Tail Rotor Pylon.

- a. Check surface of tail rotor pylon, pylon fold joint, stabilator, and electrostatic dischargers for lightning passage.
- b. If damage was found, perform the following:
 - (1) Remove stabilator and inspect attaching bolts, bearings, attachment fittings, actuator attaching bolts, and bearings for damage (PARA 2-3-15). **NONE ALLOWED.** If damage is found, send stabilator to an authorized overhaul facility. Replace attaching hardware.

- (2) Check stabilator attachment fittings, bearings, bushings, and stabilator actuator attachment bearing and fitting on tail pylon for arc burns. **NONE ALLOWED.** If arc burns are found on stabilator attachment fitting or actuator fitting, send stabilator and tail rotor pylon to an authorized overhaul facility.
- (3) Fold tail rotor pylon (PARA 1-6-9) and check fold joint, attaching bolts, surface of joints, shearpins, barrel-nuts, and hinge fittings on tail rotor pylon for arc burns (PARA 2-3-14). **NONE ALLOWED.** If arc burns are found, replace attaching hardware and send pylon to an authorized overhaul facility.
- (4) Check hinge fittings on tail cone for arc burns. **NONE ALLOWED.**
- (5) Remove tail gear box fairings (PARA 2-4-137) and intermediate gear box fairings (PARA 2-4-139). Check tail rotor control cables for blown plastic covering. **NONE ALLOWED.** If damage is found, check all sections of cable and replace any damaged section (PARA 11-4-106).

1-7-47.8 Intermediate Gear Box, Tail Gear Box, and Tail Rotor Drive Shaft.

- a. Visually inspect intermediate and tail gear box mounting points and attaching hardware for arc burns or leakage (PARA 6-4-52). **NONE ALLOWED.** If arc burns or leakage are found, remove tail gear box and send to an authorized overhaul facility. Replace attaching hardware.
- b. If no visual damage was found on intermediate gear box, but any one of the couplings, viscous damper bearings, hardware, or shafts adjacent to gear box are damaged, remove intermediate gear box (PARA 6-4-31) and send to an authorized overhaul facility.
- c. Inspect tail rotor drive shafts, flexible couplings, and viscous damper bearings for arc burns or leakage. **NONE ALLOWED.** If arc burns or leakage are found, remove component and send to an authorized overhaul facility

(PARA 6-4-38, PARA 6-4-39, PARA 6-4-40, PARA 6-4-41, PARA 6-4-42, PARA 6-4-43).

- d. If no visual damage was found on tail rotor gear box, but there was evidence of a lightning strike on tail rotor, remove tail gear box (PARA 6-4-52) and tail rotor servo (PARA 11-4-113) and send to an authorized overhaul facility.

1-7-47.9 Oil Cooler.

Visually inspect oil cooler for arc burns or leakage (PARA 6-4-47). **NONE ALLOWED.** If arc burns or leakage are found, remove oil cooler fan (PARA 6-4-50) and send to an authorized overhaul facility.

1-7-47.10 Engine.

- a. Visually inspect engine compartment, output drive shaft, and engine mounting points for arc burns. **NONE ALLOWED.** If arc burns or leakage are found, replace damaged components (PARA 4-4-5 and PARA 4-4-6).
- b. Inspect engines (SEI 671 and TM 1-2840-248-23).
- c. If no visual damage is found, perform operational check (PARA 4-2-1).

1-7-47.11 APU.

- a. Visually inspect APU for damage. **NONE ALLOWED.** If damage is found, remove APU (PARA 15-4-1).
- b. Send APU to an authorized overhaul facility. If no damage is found, perform operational check (PARA 15-2-2 and PARA 15-2-1).

1-7-47.12 Electrical/Avionics.

- a. Visually inspect electrical (PARA 9-1-1 and PARA 9-1-5), AFCS, and avionic components (TM 11-70-23) for damage. **NONE ALLOWED.**
- b. Remove and bench check components that are damaged. Replace all damaged components found during bench test or inspection.

1-7-47.13 Fuel System.

- a. Visually inspect exposed electrical wiring, including wiring in nose compartment, and under main rotor pylon, for burns or charring. **NONE ALLOWED.** If damage is found, repair or replace wiring.
- b. Visually inspect gravity refueling ports, pressure refuel port, and fuel cell vent port for arc burns. **NONE ALLOWED.** If arc burns are found, replace component. During next refueling, check pressure refuel/defuel high-level shutoff valves and vent valve for proper operation (PARA 10-2-4).

1-7-47.14 External Fuel Tank.

- a. Inspect refueling port, air, fuel, and electrical interfaces for arc burns.
- b. Inspect for any indication of damage to composite structure (PARA 16-4-71).
- c. **IF THE DAMAGE EXCEEDS MAXIMUM ALLOWABLE TOLERANCES, THE TANK MUST BE RETURNED TO THE CONTRACTOR.**

1-7-48 HYDRAULIC RESCUE HOIST INSPECTION.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-48.1	Before Each Rescue Hoist Mission	1-7-48-1	1-7-48.3	Hoist Overload	1-7-48-4
1-7-48.2	After Each Rescue Hoist Mission Use	1-7-48-3	1-7-48.4	After Every 300 Cycles or 400 Lifts	1-7-48-4
				HOIST, 42315-10	

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Chocks
- Gloves

Materials/Parts

- Alcohol, Isopropyl, Item 72, Appendix D
- Barrier Material, Item 80, Appendix D
- Cheesecloth, Item 90, Appendix D
- Corrosion Preventive Compound, Item 114, Appendix D

- Lockwire, Item 195, Appendix D

References

- Appendix D
- PARA 1-6-6
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-6-18
- PARA 1-7-22
- PARA 14-4-4

1-7-48.1 Before Each Rescue Hoist Mission.

NOTE

This inspection is only required if rescue hoist is going to be used during a mission.

- Prior to performing inspection, perform the following:
 - Park helicopter on a flat, level surface (PARA 1-6-6).

- Lock tailwheel.
- Set parking brake.
- Engage gust lock or rotor brake (PARA 1-6-18).
- Chock wheels.
- Connect external hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- On upper console, set RESCUE HOIST switch to ALL.

CAUTION

If hoist system is under backup power, system limit switches will be bypassed. Check that system will operate in normal power before reeling cable.

- d. In cabin, check that cabin RESCUE HOIST control box NORMAL POWER light is on. If BACKUP CONTROL POWER switch light is on, open switch guard assembly, and press switch. BACKUP CONTROL POWER switch light should go off and NORMAL POWER light should come on. Close switch guard assembly and using lockwire, Item 195, Appendix D, lockwire switch guard assembly to screws.

WARNING

- To prevent injury to personnel or damage to equipment, rescue hoist cable must be kept clear of all parts of the aircraft and free from other external obstacles. Cable abrasion during hoist operation can lead to cable failure. If cable contact or snagging occurs, inspect cable for damage immediately.
- Injury to personnel may occur if protective gloves are not worn when reeling cable in or out. Use protective gloves when reeling cable in or out.

CAUTION

Damage to cable may occur if cable is kinked when reeling cable in or out. Use care not to kink cable when reeling cable in or out.

- e. Place barrier material, Item 80, Appendix D, under hoist area to lay cable on during hoist operation.

- f. In cabin, operate rescue hoist and cargo hook pendant, while assistant walks out the first 10 to 20 feet of cable.
- g. While running gloved hand slowly and very lightly over cable, inspect entire cable for separated strands. **NO MORE THAN 0.005-INCH SEPARATION STANDING OUT FROM BODY OF CABLE IS ALLOWED.** If required, replace hoist cable (PARA 14-4-4).
- h. Inspect cable for kinks, broken wire, or shock load (strong jerk) to cable. **NONE ALLOWED.** If required, replace hoist cable (PARA 14-4-4).

CAUTION

Do not run backup control with hoist in down position. Doing so will result in back winding cable on drum.

NOTE

While cable is reeling in, visually inspect cable for kinks, separated strands beyond specified tolerance, or broken wires. Any evidence of the listed damage is cause for cable replacement.

- i. Slowly reel hoist cable onto cable drum, while assistant applies firm drag on cable with gloved hand. On Breeze-Eastern hoist (BL27100 series), wipe cable with cheesecloth, Item 90, Appendix D, dampened with isopropyl alcohol, Item 72, Appendix D. On Lucas Aerospace hoist (42315 series), wipe cable with cheesecloth, Item 90, Appendix D, dampened with corrosion preventive compound, Item 114, Appendix D.
- j. Turn off all switches.
- k. Disconnect external hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- l. Make sure area is clean and free of foreign material.

1-7-48.2 After Each Rescue Hoist Mission Use.

CAUTION

This inspection is required after each use of the rescue hoist. The entire length of cable must be inspected.

- a. Prior to performing inspection, perform the following:
 - (1) Park helicopter on a flat, level surface (PARA 1-6-6).
 - (2) Lock tailwheel.
 - (3) Set parking brake.
 - (4) Engage gust lock or rotor brake (PARA 1-6-18).
 - (5) Chock wheels.
- b. Connect external electrical and hydraulic power (PARA 1-6-15 and PARA 1-6-16).
- c. On upper console, set RESCUE HOIST switch to ALL.

CAUTION

If hoist system is under backup power, system limit switches will be bypassed. Check that system will operate in normal power before reeling cable.

- d. In cabin, check that cabin RESCUE HOIST control box NORMAL POWER light is on. If BACKUP CONTROL POWER switch light is on, open switch guard assembly, and press switch. BACKUP CONTROL POWER switch light should go off and NORMAL POWER light should come on. Close switch guard assembly and using lockwire, Item 195, Appendix D, lockwire switch guard assembly to screws.

WARNING

- To prevent injury to personnel or damage to equipment, rescue hoist cable must be kept clear of all parts of the aircraft and free from other external obstacles. Cable abrasion during hoist operation can lead to cable failure. If cable contact or snagging occurs, inspect cable for damage immediately.
- Injury to personnel may occur if protective gloves are not worn when reeling cable in or out. Use protective gloves when reeling cable in or out.

CAUTION

Damage to cable may occur if cable is kinked when reeling cable in or out. Use care not to kink cable when reeling cable in or out.

- e. Place barrier material, Item 80, Appendix D, under hoist area to lay cable on during hoist operation.

NOTE

If hoist was used in salt water conditions, rinse cable with cold, clean fresh water.

- f. In cabin, operate rescue hoist and cargo hook pendant, while assistant walks out entire length of cable.
- g. While running gloved hand slowly and very lightly over cable, inspect entire length of cable for separated strands. **NO MORE THAN 0.005-INCH SEPARATION STANDING OUT FROM BODY OF CABLE IS ALLOWED.** If required, replace hoist cable (PARA 14-4-4).
- h. Inspect cable for kinks, broken wire, or shock load (strong jerk) to cable. **NO DAMAGE AL-**

LOWED. If required, replace hoist cable (PARA 14-4-4).

CAUTION

Do not run backup control with hoist in down position. Doing so will result in back winding cable on drum.

NOTE

While cable is reeling in, visually inspect cable for kinks, separated strands beyond specified tolerance, or broken wires. Any evidence of the listed damage is cause for cable replacement.

- i. Slowly reel hoist cable onto cable drum, while assistant applies firm drag on cable with gloved hand. On Breeze-Eastern hoist (BL27100 series), wipe cable with cheesecloth, Item 90, Appendix D, dampened with isopropyl alcohol, Item 72, Appendix D. On Lucas Aerospace hoist (42315 series), wipe cable with cheesecloth, Item 90, Appendix D, dampened with corrosion preventive compound, Item 114, Appendix D.
- j. Turn off all switches.
- k. Disconnect external hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- l. Make sure area is clean and free of foreign material.

1-7-48.3 Hoist Overload.

- a. When rescue hoist weight limits are exceeded, inspect rescue hoist backup structure and fittings for cracks or loose attach bolts. Repair or replace components as required.

NOTE

- The rescue hoist incorporates a slip clutch designed to slip at not less than 1,200 pounds but before 1,800 pounds. If an overload occurs, it will cause slip-page of the cable hoist clutch.
 - If an overload occurs while cable is extended to extreme position, the cable will unreel to end of terminal on drum and fall free from winch.
- b. If cable hoist clutch slips due to overload, no corrective action is required.
 - c. If cable unreels and falls free from rescue hoist, perform the following:
 - (1) Inspect cable terminal on drum for obvious damage.
 - (2) Install new cable on rescue hoist drum (PARA 14-4-4).

1-7-48.4 After Every 300 Cycles or 400 Lifts **HOIST, 42315-10** .

Perform overload clutch margin validation load check (PARA 1-7-22).

1-7-49 ROTOR BRAKE INSPECTION.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-49.1	Rotor Brake Slippage on Start-Up	1-7-49-1	1-7-49.2	Rotor Brake Nut Torque Stabilization Check	1-7-49-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Rotor Brake Wrench, 70700-77117-041
- Torque Reactor, 70700-20688-041
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs

Materials/Parts

- Alcohol, Isopropyl, Item 72, Appendix D
- Cheesecloth, Item 90, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-6-18
- PARA 6-4-2
- PARA 6-4-38
- PARA 6-4-70

1-7-49.1 Rotor Brake Slippage on Start-Up.

NOTE

Use the following procedures when rotor brake is properly broken-in and slippage still occurs during engine start-up.

- Clean rotor brake disc using cheesecloth, Item 90, Appendix D, saturated with isopropyl alcohol, Item 72, Appendix D.
- If brake still slips after cleaning, check puck clearance (PARA 6-4-70).
- If brake still slips, check that static pressure is at least 480 psi.

1-7-49.2 Rotor Brake Nut Torque Stabilization Check.

NOTE

Every 5 to 15 rotor brake stops, nut torques must be checked until stabilized.

- Turn off all electrical power (PARA 1-6-16).
- Engage gust lock and disengage rotor brake (PARA 1-6-18).
- Open engine cowlings.
- Open oil cooler compartment access doors.
- Remove tail rotor drive shaft section I (PARA 6-4-38).

- f. Remove bolt, washers, and nuts securing shield to rotor brake and main transmission. Remove shield (Figure 1-7-49-1, Detail A).

NOTE

Check torque in tightening direction only.

- g. Perform torque stabilization check of retaining nut, SS5081-10, every 5 to 15 rotor brake stops, until torque stabilizes as follows:
 - (1) Install torque reactor, and check retaining nut torque. **CHECK TORQUE AT 840 INCH-POUNDS.** If torque is at or below 840 inch-pounds, remove main transmission tail takeoff flange and inspect splines and mating surfaces for wear (PARA 6-4-2).
 - (2) **CHECK TORQUE AT 1,344 INCH-POUNDS.** If torque is at 1,344 inch-pounds or more, no further action is required.
- h. Using rotor brake wrench, check torque on rotor brake mounting bracket nuts, SS5086-06. **TORQUE MUST NOT BE LESS THAN 261 INCH-POUNDS. IF TORQUE IS BELOW 261 INCH-POUNDS, RETORQUE TO 261 - 289 INCH-POUNDS.** Repeat torque check every 5 to 15 rotor brake stops until torque stabilizes.
- i. Check torque on rotor brake housing nuts, SS5086-08. **TORQUE MUST NOT BE LESS THAN 912 INCH-POUNDS. IF TORQUE IS BELOW 912 INCH-POUNDS, RETORQUE TO 912 - 1,008 INCH-POUNDS.** Repeat torque check every 5 to 15 rotor brake stops until torque stabilizes.
- j. Check torque on rotor brake disc nut, SS5089-06. **TORQUE MUST NOT BE LESS THAN 385 INCH-POUNDS. IF TORQUE IS BELOW 385 INCH-POUNDS, RETORQUE TO 385 - 425 INCH-POUNDS.** Repeat torque check every 5 to 15 rotor brake stops until torque stabilizes.
- k. Inspect rotor brake disc for contamination. **NONE ALLOWED.** If disc is contaminated, clean using cheesecloth, Item 90, Appendix D, saturated with isopropyl alcohol, Item 72, Appendix D.
- l. Install tail rotor drive shaft section I (PARA 6-4-38).
- m. Install shield on studs and shim as required using washers. Install unused washers under nuts. **TORQUE NUTS TO 95 - 105 INCH-POUNDS.**
- n. Install shield to upper bracket with bolt, washers and nut. **TORQUE NUT TO 105 - 115 INCH-POUNDS.**
- o. Make sure area is clean and free of foreign material.
- p. Close oil cooler compartment access doors.
- q. Close engine cowlings.
- r. If required, disengage gust lock or engage rotor brake (PARA 1-6-18).

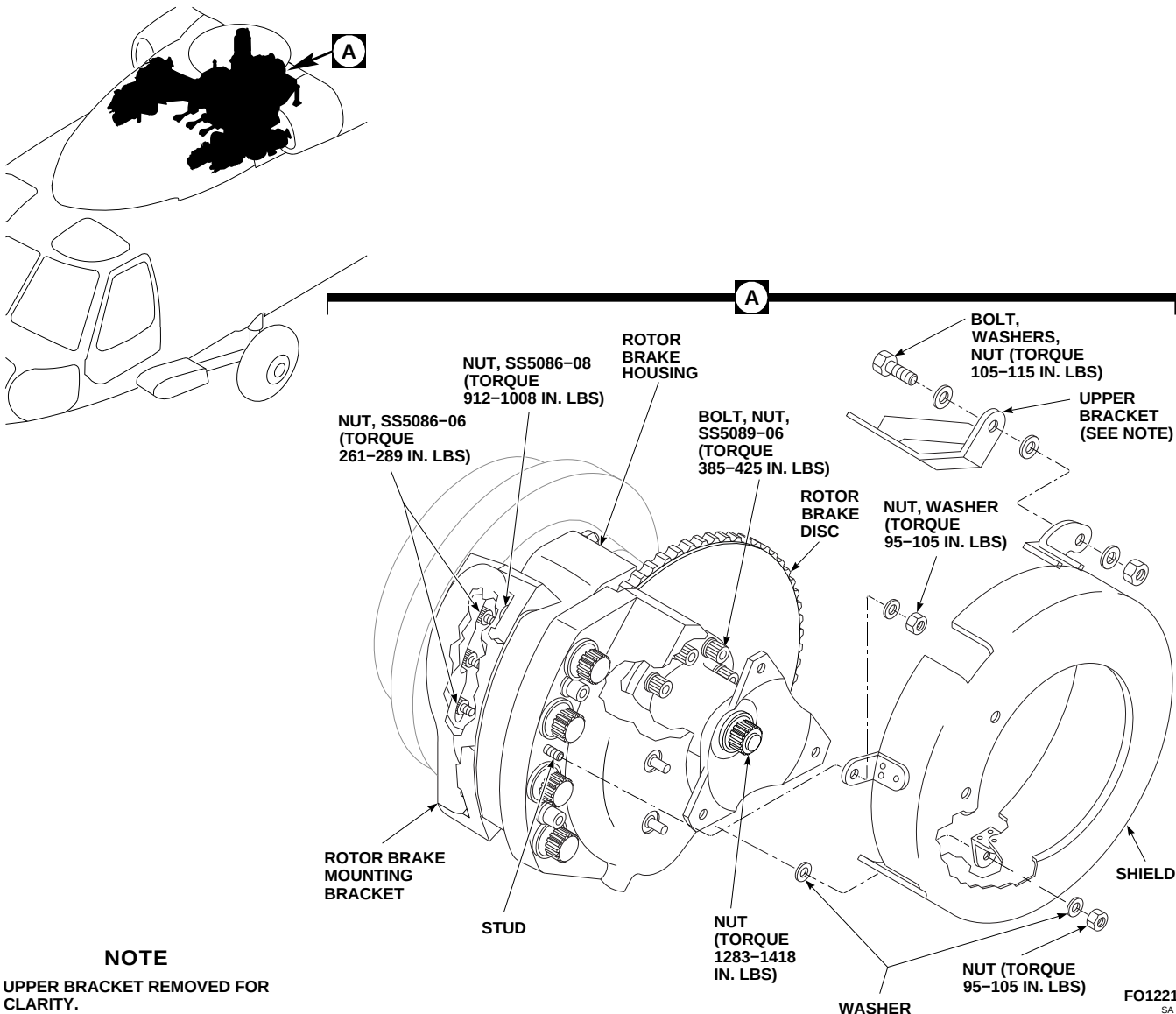


FIGURE 1-7-49-1. ROTOR BRAKE INSPECTION

1-7-50 SUDDEN STOPPAGE.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-50.1	Main Rotor Blades	1-7-50-1	1-7-50.2	Tail Rotor Blades	1-7-50-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Tags

References

- PARA 5-3-11
- PARA 5-3-13
- PARA 5-4-5
- PARA 5-4-31
- PARA 5-4-38

- PARA 5-4-40
- PARA 6-3-15
- PARA 6-4-1
- PARA 6-4-31
- PARA 6-4-38 through PARA 6-4-41
- PARA 6-4-50
- PARA 6-4-52
- PARA 6-4-54
- SEI 671
- TM 1-2840-248-23

1-7-50.1 Main Rotor Blades.

NOTE

The following inspections are required whenever a blade is damaged by striking an object.

- a. If damage to blades is less than the accept/reject criteria per PARA 5-3-11, repair blades and return to service (PARA 5-4-31).
- b. Inspect droop stop ears for cracks. If ear is cracked, replace spindle (PARA 5-4-5). Tag removed spindle "DAMAGE OCCURRED DUE TO EXCESSIVE BLADE FLAPPING".
- c. If damage to blades includes severing blades inboard of tip cap, remove transmission and rotor head assembly (PARA 6-4-1) and send to overhaul facility. Clearly tag all components "BLADE DAMAGE OCCURRED WHILE ROTOR HEAD IN MOTION". Also explain what the blades hit, if known, and how badly helicopter parts were damaged.
- d. If damage to blades is greater than main rotor blade accept/reject criteria per PARA 5-3-11, and includes damage to primary structural components (cuff, spacer assembly, and spar assembly), do this:

(1) Remove blades and spindles (PARA 5-4-31 and PARA 5-4-5). Clearly tag all

components "BLADE DAMAGE OCCURRED WHILE ROTOR HEAD IN MOTION". Also explain what the blades hit, if known, and how badly helicopter parts were damaged.

- (2) Inspect elastomeric bearings for damage (PARA 5-4-5).
 - (3) Inspect blade attachment lugs and damper lugs on spindle for damage. Cracking of paint on lugs is evidence of damage.
 - (4) Inspect hub arm inside diameter (4 places) for damage caused by elastomeric bearing contact.
 - (5) Inspect dampers, control horns, pitch control rods, and swashplate for damage.
 - (6) Inspect airframe attachment points for deformation and damage.
 - (7) Visually inspect input drive shafts, tail drive shafts, oil cooler shaft area, all associated flexible couplings, bearings, and supports for cracks and distortion.
 - (8) Send removed blades and spindles to overhaul an authorized overhaul facility.
- e. If damage to blades is greater than main rotor accept/reject criteria per PARA 5-3-11, but damage is limited to pocket skin assembly, trim tab, abrasion strip, and balance block and does not include damage to primary structural components, remove blades and send to approved overhaul facility. Clearly tag all components "BLADE DAMAGE OCCURRED WHILE ROTOR HEAD IN MOTION". Also explain what blades hit, if known, and how badly helicopter parts were damaged.
- f. If damage to tip cap is greater than tip cap accept/reject criteria per PARA 5-3-11, remove tip cap (PARA 5-4-31) and inspect adjacent blade structure (tip block, tip block Rosan inserts, tip weights, weight studs, tip rib and skin adjacent tip rib) for damage. Disposition of blade shall be based on repairability of these components.

- g. Perform sudden stoppage inspection for engines (SEI 671 and TM 1-2840-248-23).

1-7-50.2 Tail Rotor Blades.

- a. If damage to blades is less than tail rotor blade accept/reject criteria per PARA 5-3-13, do this:
- (1) Inspect all four tail gear box mounting lugs.
 - (2) Visually inspect intermediate gear box mounting, all drive shaftings, flexible couplings, drive shaft bearings, oil cooler axial fan, and drive shaft supports for cracks and distortion.
 - (3) Perform oil cooler spline wear inspection (PARA 6-3-15).
 - (4) Inspect tail rotor blade damage per accept/reject criteria (PARA 5-3-13), and repair as necessary (PARA 5-4-38).
- b. If damage to blades is greater than tail rotor blade accept/reject criteria (PARA 5-3-13) and includes damage to spar and/or severing blade inboard of tip weights, dents or gouges in leading edge over 1/4 inch deep in either polyurethane or nickel abrasion strips, or damage to attachment block, do this:
- (1) Remove the following components:
 - (a) Tail rotor pitch beam (PARA 5-4-40).
 - (b) Tail rotor pitch control rods (PARA 5-4-38).
 - (c) Tail rotor blades (PARA 5-4-38).
 - (d) Inboard retention plate (PARA 6-4-54).
 - (e) Tail gear box (PARA 6-4-52).
 - (f) Intermediate gear box (PARA 6-4-31).
 - (g) Oil cooler fan (PARA 6-4-50).

- (h) Tail rotor drive shafts (PARA 6-4-38 through PARA 6-4-41).
- (2) Send components, with all attaching hardware, to overhaul facility for detail inspection and overhaul. Clearly tag all components "DAMAGE OCCURRED WHILE TAIL ROTOR IN MOTION". Explain what the blades hit, if known, and how badly the helicopter parts were damaged.
- (3) Visually inspect tail pylon for possible structural damage such as loose rivets, cracks, etc. Inspect gear box attachment points for deformation and other signs of damage.
- c. If damage to blades is greater than tail rotor blade accept/reject criteria per PARA 5-3-13, but is limited to damage to tip weights or tip block studs, and/or skin and honeycomb in Region A, Region B, or Region C, do this:
 - (1) Inspect all four tail gear box mounting lugs.
 - (2) Visually inspect intermediate gear box mounting, all drive shafting, flexible couplings, drive shaft bearings, oil cooler axial fan, and drive shaft supports for cracks and distortion.
 - (3) Perform oil cooler spline wear inspection (PARA 6-3-15).
 - (4) Remove blades from helicopter and send to overhaul facility. Clearly tag all components "DAMAGE OCCURRED WHILE TAIL ROTOR IN MOTION". Also explain what blades hit, if known, and how badly helicopter parts were damaged.
- d. Perform sudden stoppage inspection for engines (SEI 671 and TM 1-2840-248-23).

1-7-51 TAIL ROTOR OUT OF BALANCE - LOSS OF MATERIAL.

PARAGRAPH BREAKDOWN

	PAGE
1-7-51.1 Tail Rotor Out of Balance - Loss of Material	1-7-51-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 150 - 750 in. lbs

References

- PARA 1-6-9
- PARA 6-4-41
- PARA 6-4-52

1-7-51.1 Tail Rotor Out of Balance - Loss of Material.

NOTE

Loss of material may be the result of lightning strike, ballistic damage, or loss of rotor heater mat/abrasion strip.

- If tail rotor becomes unbalanced causing severe vibration through loss of material or loosening of rotor plug (centering bushing), or inboard boot support touches outboard retention plate, do inspections as follows:
 - Check torque on nuts and studs that hold both sections of tail gear box together. Do not loosen nuts before checking torque. **TORQUE MUST BE 171 - 189 INCH-POUNDS.** Replace gear box if torque is below minimum (PARA 6-4-52).

- Check tail gear box mounting bolts for torque (PARA 6-4-52).
- Visually inspect tail gear box fitting inside and outside. Remove drive shaft to get through access hole (PARA 6-4-41). Carefully examine airframe for cracks.
- Inspect both sides of tail cone and pylon hinge fittings. Fold pylon to examine inside surfaces for cracks (PARA 1-6-9).
- Inspect stringers near barrel-nuts for cracks. **NO CRACKS ALLOWED.**
- Inspect intermediate gear box mounting fitting for cracks. **NO CRACKS ALLOWED.**

1-7-52 TRANSMISSION SYSTEM - OVERSPEED.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-52.1	Up to 126% Nr	1-7-52-1	1-7-52.3	Between 137% to 141% Nr	1-7-52-2
1-7-52.2	Between 127% to 136% Nr	1-7-52-1	1-7-52.4	Above 141% Nr	1-7-52-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Tags

References

- PARA 1-3-29
- PARA 5-3-11
- PARA 5-3-13
- PARA 5-4-1
- PARA 5-4-22
- PARA 5-4-31

- PARA 5-4-38
- PARA 5-4-41
- PARA 6-3-6
- PARA 6-3-11
- PARA 6-3-17
- PARA 6-4-38
- PARA 6-4-39
- PARA 6-4-40
- PARA 6-4-41
- PARA 6-4-42
- PARA 6-4-50
- PARA 7-4-12

1-7-52.1 Up to 126% Nr.

No special inspection required.

1-7-52.2 Between 127% to 136% Nr

- Inspect main rotor blade tip caps for security, cracks, bond separations, elongation of holes, erosion, or other damage (PARA 5-3-11).
- Inspect main rotor blade cover skins for cracks, bond separations, crazing of paint, erosion, or

other damage. Pay close attention to area around tip rib for damaged conditions (PARA 5-3-11).

- Inspect main rotor blade root laminated areas for cracks, bond separations, delamination of plies, and crazing of paint at bond interface with blade skin (PARA 5-3-11).
- Inspect main rotor blade cuffs and attaching hardware for signs of movement from attach-

ing hardware. Check sealing compound around boltheads for damage (PARA 5-3-11).

- e. Test main rotor blade pressure indicator (PARA 1-3-29). Check indicator for damage and security.
- f. Inspect main rotor head elastomeric bearings, spindle modules, hub, and bifilar for general condition.
- g. Inspect tail rotor blades for signs of cracks, bond separations, crazing, or skin delaminations (PARA 5-3-13).
- h. Inspect tail rotor blade from mid-span to control horn for bond separations and delaminations of trailing edge skins. Check areas between control horn and trailing edge skins for above condition (PARA 5-3-13).
- i. Remove tail rotor boots and inspect for security of pivot retainer (PARA 5-4-41). Check boot support for above condition (PARA 5-3-13).
- j. Inspect engine drive shaft assemblies for loose fasteners and for yielding flexible couplings.
- k. Inspect ends of tail drive shaft sections for signs of yielding, indicated by loose lockbolts, elongated lockbolt holes, and paint flaking near or between lockbolts (PARA 6-4-38 through PARA 6-4-41).
- l. Inspect flexible couplings for yielding and distortion of segments (PARA 6-4-42).
- m. Inspect drive shaft tubes for any signs of bending.

1-7-52.3 Between 137% to 141% Nr.

- a. Remove main and tail rotor blades from helicopter and send to overhaul facility whenever rotor speed readings are above 136%. Tag all components, explaining reason for removal. Record amount of Nr overspeed on tags.
- b. Inspect main rotor head elastomeric bearings, spindle modules, hub, and bifilar, for general condition.

- c. Inspect tail rotor blades for evidence of crazing or delamination of skin (PARA 5-3-13).
- d. Inspect engine drive shaft assemblies for loose fasteners and for yielding flexible couplings.
- e. Inspect ends of tail drive shaft sections for signs of yielding, indicated by loose lockbolts, elongated lockbolt holes, and paint flaking near or between lockbolts (PARA 6-4-38 through PARA 6-4-41).
- f. Inspect flexible couplings for yielding and distortion of segments (PARA 6-4-42).
- g. Inspect drive shaft tubes for any signs of bending.

1-7-52.4 Above 141% Nr.

- a. Remove main (PARA 5-4-31) and tail rotor blades (PARA 5-4-38) from helicopter and send to overhaul facility. Tag all components, explaining reason for removal. Record amount of Nr overspeed on tags.
- b. Remove main rotor head (PARA 5-4-1), swashplate (PARA 5-4-22), and tail rotor assembly from helicopter and send to overhaul facility.
- c. Remove all tail drive shafting, Thomas couplings, and support bearings from helicopter and send to overhaul facility (PARA 6-4-38 through PARA 6-4-41). Tag all components, explaining reason for removal. Record amount of Nr overspeed on tags.
- d. Remove both engine drive shaft assemblies from helicopter and send to overhaul facility.
- e. Remove oil cooler assembly from helicopter and send to overhaul facility (PARA 6-4-50). Tag components, explaining reason for removal. Record amount of Nr overspeed on tag.
- f. Visually inspect main transmission chip detectors and screens for contamination (PARA 6-3-6).

- g. Visually inspect intermediate and tail gear box chip detectors and screens for contamination (PARA 6-3-11 and PARA 6-3-17).
- h. Remove both main generators and send to overhaul facility. Tag components explaining reason for removal. Record amount of Nr overspeed on tags.
- i. Remove No. 1 and No. 2 hydraulic pump modules and return to overhaul facility (PARA 7-4-12). Tag components explaining reason for removal. Record amount of Nr overspeed on tags.

1-7-53 TRANSMISSION SYSTEM - OVERTEMPERATURE **S70-I** .

PARAGRAPH BREAKDOWN

	PAGE
1-7-53.1 Transmission System - Overtemperature	1-7-53-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Borescope Kit, 6650-T700-003

Materials/Parts

- Tags

References

- PARA 1-3-12
- PARA 6-3-5

- PARA 6-3-6
- PARA 6-4-1
- PARA 6-4-20
- PARA 6-4-25
- PARA 6-4-28

1-7-53.1 Transmission System - Overtem-
perature.

- MAIN TRANSMISSION OIL TEMPERA-
TURE MUST NOT EXCEED 280°F (140°C),
OR MUST NOT BE BETWEEN 248°F
(120°C) AND 280°F (140°C) FOR OVER 60
MINUTES.** If oil temperature exceeds limits, do
this:
 - Inspect oil cooler, lines, and interior of
duct for blockage and debris. Remove any
blockage or debris (Figure 1-7-53-1).
 - Replace all five main transmission
modules.
- If main transmission oil temperature exceeds
248°F (120°C) and main transmission oil tem-
perature caution light is illuminated, do this:

- Inspect oil cooler, lines, and interior of
duct for blockage and debris. Remove any
blockage or debris.
- Replace all five main transmission
modules (PARA 6-4-1).
- If main transmission oil temperature was 248° -
280°F (120° - 140°C) for less than 60 minutes
and MAIN XMSN OIL TEMP caution light
did not illuminate, do this:
 - Perform serviceability inspection. Inspect
all strainers, chip detectors, and oil filter
(PARA 6-3-5, PARA 6-4-1, PARA 6-4-
25, and PARA 6-4-28).
 - Inspect both input and accessory modules
for evidence of temperature discoloration
or paint blistering. **NONE ALLOWED.** If
discoloration or paint blistering is found,

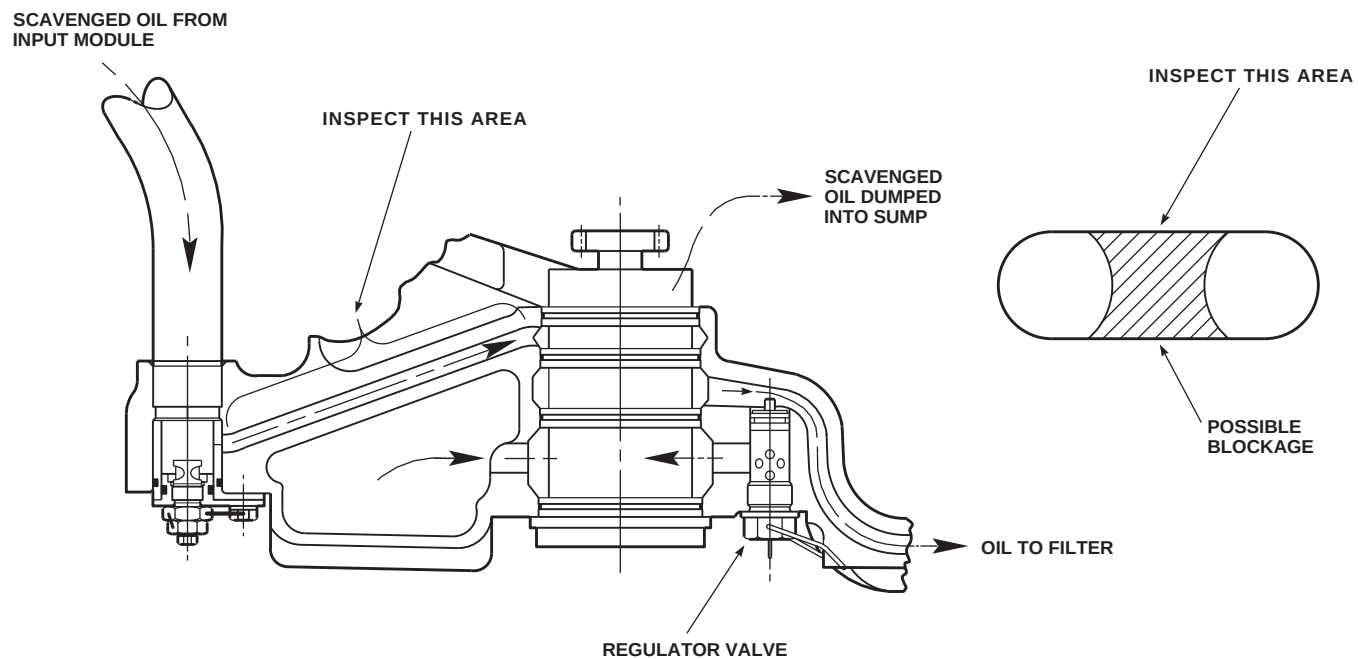
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FIGURE 1-7-53-1. TRANSMISSION CORE HOLE INSPECTION

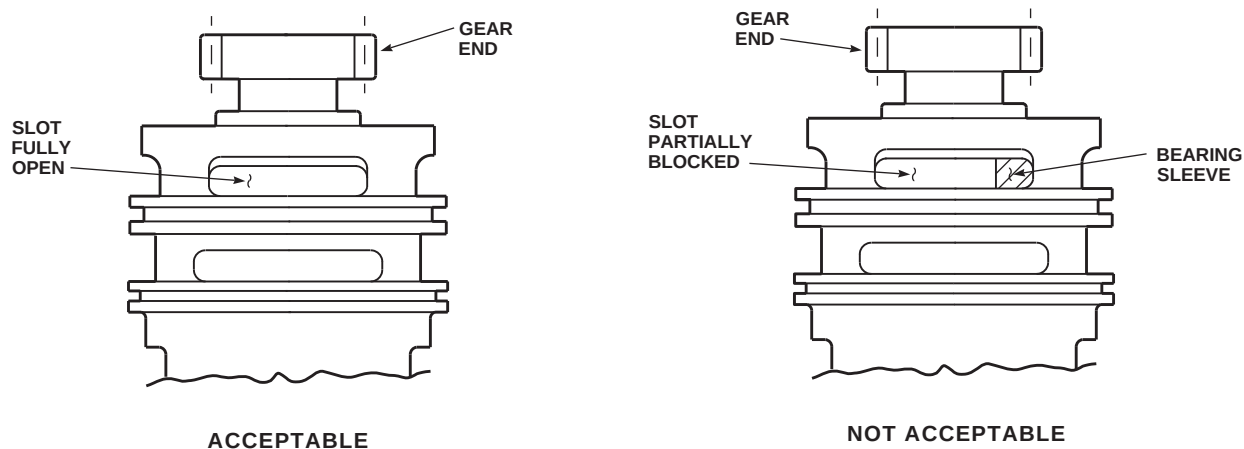


FIGURE 1-7-53-2. TRANSMISSION BEARING SLEEVE INSPECTION

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replace that sides input and accessory modules (PARA 6-4-25 and PARA 6-4-28).

NOTE

Bearing sleeve is bronze in color.

- (3) Drain main gear box oil (PARA 1-3-12).
 - (4) Remove both input module chip detectors and strainer assemblies. Using borescope with light source, inspect both strainer sleeves for blockage of the core hole oil passage in the sump housing. If blockage exists, measure amount chip detector sleeve overlaps core hole oil passage. **MAXIMUM OVERLAP 0.030 INCH** (PARA 6-3-6).
 - (5) Using borescope with light source, inspect core hole oil passage between input module chip detectors and pumps for blockage (Figure 1-7-53-2). **NO BLOCKAGE ALLOWED.** If blockage is found, replace main module (PARA 6-4-1).
 - (6) Remove oil pumps (PARA 6-4-20).
 - (7) Check for damaged packings. Replace damaged packings.
 - (8) Check both upper slots to make sure bearing sleeve has not rotated and caused partial blockage. **NO BLOCKAGE ALLOWED.** If blockage is found, replace oil pump (PARA 6-4-20).
 - (9) If no cause has been found, replace both input modules.
 - (10) **IF OVERTEMP CONTINUES, REPLACE MAIN MODULE AND RETURN TO DEPOT.** Clearly tag components with removal reason, maximum oil temperature reached, and duration of time at high temperature.
- d. If main transmission oil temperature was 221° - 248°F (105° - 120°C), do this:
- (1) Inspect oil cooler, lines and interior of duct for blockage and debris. Remove blockage and debris.

- (2) Inspect input module drain tubes for blockage and debris. Remove blockage and debris.
- (3) Perform serviceability inspection. Inspect all strainers, chip detectors, and oil filter. **IF CHIPS ARE BEYOND ACCEPTABLE LIMITS, REPLACE MODULES SHOWING SIGNS OF CHIPS** (PARA 6-3-5, PARA 6-4-1, PARA 6-4-25 and PARA 6-4-28).
- (4) Inspect input and accessory modules for evidence of temperature discoloration or paint blistering. **NONE ALLOWED.** If discoloration or paint blistering is found, replace that sides input and accessory modules (PARA 6-4-25 and PARA 6-4-28).
- (5) Drain main gear box oil (PARA 1-3-12).
- (6) Remove both input module chip detectors and strainer assemblies (PARA 6-3-6). Using borescope with light source, inspect both strainer sleeves for blockage of the core hole oil passage in the sump housing. If blockage exists, measure amount chip detector sleeve overlaps core hole oil passage. **CHIP DETECTOR SLEEVE SHALL NOT OVERLAP CORE HOLE OIL PASSAGE BY MORE THAN 0.030 INCH.**

- (7) Using borescope with light source, inspect core hole oil passage between input module chip detectors and pumps for blockage. **NO BLOCKAGE ALLOWED.** If blockage is found, replace main module (PARA 6-4-1).
- (8) Remove oil pumps (PARA 6-4-20).
- (9) Check for damaged packings. Replace damaged packings.

NOTE

Bearing sleeve is bronze in color.

- (10) Check both upper slots to make sure bearing sleeve has not rotated and caused partial blockage (Figure 1-7-53-2). **NO BLOCKAGE ALLOWED.** If blockage is found, replace oil pump (PARA 6-4-20).
- (11) If no cause has been found, replace both input modules.
- (12) **IF OVERTEMP CONTINUES, REPLACE MAIN MODULE AND SEND TO DEPOT.** Clearly tag components with removal reason, maximum oil temperature reached, and duration of time at high temperature.

1-7-54 TRANSMISSION SYSTEM - OVERTEMPERATURE **S70-II** .

PARAGRAPH BREAKDOWN

	PAGE
1-7-54.1 Transmission System - Overtemperature	1-7-54-1

INITIAL SETUP

- PARA 6-4-1

Tools

- Aircraft Mechanic’s Toolkit

- PARA 6-4-25

References

- PARA 6-4-28

- PARA 6-2-2
- PARA 6-3-5

1-7-54.1 Transmission System - Overtem-
perature.

NOTE

Prior to replacing any component, perform operational check of the oil temperature indicating system.

- If required, perform operational check of the oil temperature indicating system (PARA 6-2-2).
- MAIN TRANSMISSION OIL TEMPERATURE MUST NOT EXCEED 280°F (140°C), OR MUST NOT REMAIN IN PRECAUTIONARY RANGE 221° - 280°F (105 - 140°C), FOR LONGER THAN 60 MINUTES.** If oil temperature exceeds limits, do this:
 - Inspect oil cooler, lines, and interior of duct for blockage and debris. Remove any blockage or debris.
 - Replace all five main transmission modules (PARA 6-4-1).

- If main transmission oil temperature is in precautionary range 221° - 280°F (105° - 140°C), for less than 60 minutes, do this:
 - Remove oil return tubes from input modules and main transmission sump (PARA 6-4-25).
 - Inspect tubes for blockage or debris. **NONE ALLOWED.** Remove any blockage or debris.
 - Inspect oil cooler, lines, and interior of duct for blockage and debris. Remove any blockage or debris.
 - Perform a serviceability inspection. Inspect all strainers, chip detectors, and oil filter (PARA 6-3-5, PARA 6-4-1, PARA 6-4-25, and PARA 6-4-28).
 - Inspect both sides of input and accessory modules for evidence of temperature discoloring or paint blistering. **NONE ALLOWED.** If discoloration or blistering is found, replace both input and accessory

modules on that side (PARA 6-4-25 and PARA 6-4-28).

- (6) If no cause for overtemp can be found, replace both input modules (PARA 6-4-25).

- (7) If overtemp condition persists, replace main module (PARA 6-4-1).

1-7-55 TRANSMISSION SYSTEM - OVERTORQUE **S70-I** .

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-55.1	Between 107% - 127%, More Than 10 Seconds, Both Engines Operating	1-7-55-1	1-7-55.3	Between 127% - 144%, More Than 10 Seconds, One or Both Engines Operating	1-7-55-1
1-7-55.2	Between 121% - 127%, More Than 10 Seconds, One Engine Operating	1-7-55-1	1-7-55.4	Above 144%, One or Both Engines Operating	1-7-55-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Feeler Gage

Materials/Parts

- Tags

References

- PARA 4-4-1
- PARA 6-3-5

- PARA 6-4-1
- PARA 6-4-25
- PARA 6-4-28
- TM 1-70-VIB

1-7-55.1 Between 107% - 127%, More Than 10 Seconds, Both Engines Operating.

Perform serviceability inspection on main transmission. Inspect all strainers, chip detectors, and oil filter (PARA 6-3-5, PARA 6-4-1, PARA 6-4-25, and PARA 6-4-28).

1-7-55.2 Between 121% - 127%, More Than 10 Seconds, One Engine Operating.

Perform serviceability inspection on main transmission. Inspect all strainers, chip detectors, and oil filter (PARA 6-3-5, PARA 6-4-1, PARA 6-4-25, and PARA 6-4-28).

1-7-55.3 Between 127% - 144%, More Than 10 Seconds, One or Both Engines Operating.

- Replace main module (PARA 6-4-1).
- Visually inspect input shafts and flexible couplings for cracks or buckling (PARA 4-4-1). **USING FEELER GAGE, A GAP OF NO MORE THAN 0.030 INCH BETWEEN COUPLING LAMINATIONS IS ACCEPTABLE.**
- Perform engine output shaft vibration level check on No. 1 and No. 2 engines (TM 1-70-VIB).

1-7-55.4 Above 144%, One or Both Engines Operating.

Remove main transmission and input shafting from helicopter and send to overhaul facility (PARA

6-4-1). Accessory modules do not have to be replaced. Tag components stating reason for removal. Record amount of overtorque on tag.

1-7-56 TRANSMISSION SYSTEM - OVERTORQUE **S70-II** .

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-56.1	121% to 150%, More Than 10 Seconds, Both Engines Operating	1-7-56-1	1-7-56.3	150%, One or Both Engines Operating	1-7-56-2
1-7-56.2	135% to 150%, More Than 10 Seconds, One Engine Operating	1-7-56-1			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Tags

References

- PARA 6-3-5
- TM 1-70-VIB

1-7-56.1 121% to 150%, More Than 10 Seconds, Both Engines Operating.

- Perform serviceability check on main transmission whenever transient torque readings are between 121% to 150% for more than 10 seconds (PARA 6-3-5).
- Visually inspect engine output shafts and flexible couplings for cracks or buckling whenever transient torque readings are between 121% to 150% for more than 10 seconds. **A GAP OF NO MORE THAN 0.030-INCH BETWEEN COUPLING LAMINATIONS IS ACCEPTABLE.**
- Visually inspect all tail gear box mounting feet, intermediate gear box mounting feet, tail rotor drive shafts, flexible couplings, flanges, and drive shaft supports for cracks or buckling. **A GAP OF NO MORE THAN 0.030-INCH BETWEEN COUPLING LAMINATIONS IS ACCEPTABLE.**

- Perform engine output shaft vibration level check on No. 1 and No. 2 engines (TM 1-70-VIB).

1-7-56.2 135% to 150%, More Than 10 Seconds, One Engine Operating.

- Perform serviceability check on main transmission whenever transient torque readings are between 135% to 150% for more than 10 seconds (PARA 6-3-5).
- Visually inspect engine output shafts and flexible couplings for cracks or buckling whenever transient torque readings are between 135% to 150% for more than 10 seconds. **A GAP OF NO MORE THAN 0.030-INCH BETWEEN COUPLING LAMINATIONS IS ACCEPTABLE.**
- Perform engine output shaft vibration level check on No. 1 and No. 2 engines (TM 1-70-VIB).

1-7-56.3 150%, One or Both Engines Operating.

- a. Remove main module, input modules, and engine output shafts from helicopter and send to an authorized overhaul facility whenever transient torque readings go above 150% regardless of time. Accessory modules do not have to be replaced. Tag components stating reason for removal. Record amount of over-torque on tag.
- b. Visually inspect all tail gear box mounting feet, intermediate gear box mounting feet, tail rotor drive shafts, flexible couplings, flanges, and drive shaft supports for cracks or buckling. **A GAP OF NO MORE THAN 0.030-INCH BETWEEN COUPLING LAMINATIONS IS ACCEPTABLE.**

1-7-57 AFTER TAIL ROTOR BLADE REMOVAL.

PARAGRAPH BREAKDOWN

		PAGE
1-7-57.1	Tail Rotor Boot Support Inspection	1-7-57-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 5-4-38

1-7-57.1 Tail Rotor Boot Support Inspection.

Perform tail rotor boot support inspection (PARA 5-4-38).

NOTE

This inspection must be performed whenever the tail rotor blades are removed from aircraft.

1-7-58
 ELECTRIC RESCUE HOIST INSPECTIONS
 HOIST BL-29900-30

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PARAGRAPH BREAKDOWN

		PAGE
1-7-58.1	Before Each Mission Use	1-7-58-2
1-7-58.2	Inspection and Maintenance	1-7-58-3

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 55-Gallon

Materials/Parts

- Cloth, Cleaning, Item 102, Appendix D

References

- Appendix D
- PARA 14-3-2

1-7-58.1 Before Each Mission Use.**1-7-58.1.1 Preflight Check.****NOTE**

- This inspection is only required if rescue hoist is going to be used during a mission.
- Preflight check requires AC power applied.
- a. On pilot's hoist control panel, turn HOIST POWER switch ON.

WARNING

- Injury to personnel and damage to equipment will occur if work gloves are not used whenever handling rescue hoist cable with broken cable strands. Care should be taken not to damage cable by kinking when handling on ground.
- Electronic controller contains internal circuitry that monitors status of "up slow" limit switches. The "up slow" limit switch, when actuated, slows retracting cable speed to 50 feet per minute fixed. This limit switch is actuated when retracting cable is 10 ± 2 feet from fully reeled in position. If electronic controller detects a "full in" switch actuation with no "up slow" actuation the electronic controller inhibits hoist operation. This feature is needed because of the high speed capabilities of hoist and will prevent possibility of breaking cable. It is important that ground testing full in limit switch be done after "up slow" switch is actuated. In the event that "up stop" switch is actuated manually with no "up slow" actuation the system can be reset by turning master hoist switch "OFF", and then "ON". This will reset system.
- b. Using rescue hoist pendant, rotate thumbwheel downward to maximum position and hold. Let

cable payout on a clean ground without interference and proceed with the following:

NOTE

Make sure cable is wrapping smoothly and evenly without birdcages or gaps.

- (1) Allow cable to reel out at full speed.
- (2) Midway of total payout, operate pilot control in lower direction. Cable speed shall reduce to approximately 200 feet per minute. Verify override function by rotating pendant thumbwheel in both directions with no change.
- (3) Set pilot control switch to off.
- (4) Hoist shall decelerate, then stop.
- (5) Operate pendant in lower direction.
- (6) Cable shall reduce speed to approximately 33 feet per minute at 10 ± 2 feet.
- (7) Cable stops with $4 \pm \frac{1}{2}$ wraps on drum.
- (8) Verify cable payout display reads 290 ± 5 and full-out lamp indicators illuminate.
- (9) Perform load cable inspection (PARA 1-7-58.2).
- (10) Repeat above procedure in hoisting direction.
- (11) Repeat pilot control functions as described in step b. (2) to verify override function in hoisting mode.
- (12) Be sure that cable reduces speed and stops at full in position.
- (13) Check cable payout displays, both should read $000 +2/-0$ and full-in lamp indicators illuminate.
- c. Visually check for oil leaks.
- d. Visually check wiring and electrical connectors for damage.

- e. Check hook assembly for freeness of swivel and latch. If hook was exposed to salt water or if swivel does not operate freely, disassemble and clean hook assembly.
- f. Check crushable bumper and spring (PARA 14-3-2).

1-7-58.1.2 Flightline Cable Handling Techniques.

- a. The following ground procedures are vital in achieving reliable rescue hoist operation:

WARNING

Do not wind load cable from hoist drum onto a take-up reel. This can induce twist into the load cable and cause miswrapping when winding back onto the drum.

(1) Reeling In:

It is important that cable is reeled in under a reasonably heavy, even pull so that it does not wrap loosely on the drum. A drag load must be applied using a gloved hand or clean heavy cloth on the cable to achieve tight, even layers on the drum. Reeling in should be accomplished at a slow speed so that when loops form they can be straightened out and not form kinks. A kink in the cable is caused by a loop in the cable being pulled up tight, resulting in a sharp, permanent bend in the cable. Kinks can lead to fouling failures as they cause a retarding force on the cable when the kink gets hung up going through the cable guide, especially when reeling out with no load. If it is required to reel in the cable at full speed for acceptance test purposes, the cable should be reeled out in a straight line for its full length so that no kinks or permanent cable damage occur reeling in. If the purpose of the test is to measure hoist speed, timing of a specified shorter length is advisable.

WARNING

It is important that there is no retarding load on the cable while reeling out. If the action of the tension roller is retarded, the cable will immediately loosen on the drum and foul.

(2) Reeling Out:

The rescue hoist unit incorporates a tension roller system, the purpose of which is to make sure that there is always tension on the cable. During reeling out, the peripheral speed of the tension roller is slightly higher than that of the drum so the cable is always under tension between the roller and the drum. The load is in the order of seventeen pounds. When reeling out, the cable may be allowed to coil on the ground. It may also be coiled in a drum, however, caution must be exercised while reeling in to avoid kinks.

1-7-58.1.3 Postflight Check.

- a. Check hoist structure for cracks, missing mounting hardware, hardware slippage marks, or material surface imperfection.
- b. Check all covers, plugs, and guards for proper attachment and tie wire installation.
- c. Check for oil leaks.
- d. Check wiring and electrical connectors for damage.
- e. Check hook assembly for freeness of swivel and latch. If hook was exposed to salt water or if swivel does not operate freely, disassemble and clean hook assembly.
- f. Check crushable bumper and spring (PARA 14-3-2).

1-7-58.2 Inspection and Maintenance.

- a. In order to be effective, cable inspection must be performed on a regularly scheduled basis.

Cable used for personnel lifting should be inspected carefully and frequently and rigid standards of acceptance and rejection requirements must be adhered to. Cable inspection should include careful visual checks for the following defects:

(1) Evidence of Dragging Over Obstacles or Worn Cable Guides:

Abrasion of cable is caused by contact with other cable sections or any material that may contact the cable such as aircraft structural members, auxiliary equipment, or almost any abrasive surface. Abrasion removes metal from the external wires in the outer strands and thus weakens the cable. Nicking, scarring, and scrubbing are also referred to as abrasion. Severe abrasion resulting from dragging a cable across an aircraft structural member can result in catastrophic cable failure and possibly loss of life. Severe abrasion as evidenced by a worn or damaged rough surface cable texture resulting in multiple broken wires in an outer strand in any section requires immediate cable replacement.

(2) Improperly Adjusted or Defective Cable Guide:

Cables evidencing damage such as broken wires at the turnaround or at cable crossovers are an indication that the hoist needs to have the levelwind readjusted or repaired. Smiles (gaps between adjacent turns of cable) are the result of the cable not being tightly and evenly wrapped and/or the levelwind leading the cable too much. If the levelwind is adjusted to lag the cable too much, excessive abrasion of the cable as it wraps on the drum occurs, particularly at the crossovers. If loud snapping is heard, shimming of the levelwind shafts relative to the drum may be improper. All of the above inspections are subjective and require an experienced inspector to make the inspection and readjustment if required. If the condition is corrected prior to excessive damage oc-

curing, the cable can be continued in service if no more than one wire break is discovered and snipped and the necessary adjustments are properly carried out.

(3) Severe Overload:

A cable that has been severely overloaded may or may not show immediate results of the overload. Broken strands are the most obvious defects. However, broken core wires may also result from overload. "Necking Down" of a section of cable is an indication of broken core wires. A cable that has been knowingly overloaded should be replaced immediately even if no obvious physical damage is detected. Any obvious reduction in diameter that is noted is cause for immediate cable replacement.

(4) Kinking:

Kinks are man-made defects in cables caused by a loop in a cable being pulled up tight, resulting in a sharp, permanent bend in the cable. Kinks are identified as "open" or "closed". An "open" kink is caused by pulling a cable loop tight, creating a sharp permanent bend which tends to open the lay of the cable. A "closed" kink is caused in the same manner, but creates a sharp permanent bend which tends to close the lay of the cable. Continued use of a kinked cable can result in the kink being caught in the cable guide assembly upon paying out and will result in cable fouling. A kinked cable should not be run back onto a hoist for any reason.

(5) Internal Wear Caused by Grit or Dirt Penetration:

Internal wear caused by grit, dirt, sand, etc., embedded between cable strands is extremely difficult to detect. Preventive maintenance is the best procedure. The cable should be carefully wiped to remove gritty material as it is reeled in during inspection. Prolonged use of a cable

loaded with gritty material will result in broken wires.

(6) Improperly Attached or Defective Fittings:

Carefully inspect the cable in the area of the hook ball end attachment. Do not bend the cable in the process of the inspection. The ball end of the cable must not evidence any broken wires at the swaged ball fitting. Any necking down, broken wires, loose wires in the immediate vicinity of the ball end is a cause for cable replacement. This condition may have resulted from the cable not being snugged up against the cable guide bell mouth properly or a full-in limit switch that is activating too fast. The continual vibration of a helicopter and a loosely "homed" hook can result in a fatigue failure of the cable immediately above the ball end fitting. This inspection should be performed prior to every flight. If a loose hook is noted, the necessary adjustment to the full-in limit switch shall be performed after the cable inspection.

(7) Broken Wires:

The number of broken wires on the outside of a wire rope is an index of its general condition and whether or not it must be considered for replacement. Tests have shown that a large number of broken wires can exist without significant reduction on the cable's braking strength. Wires that are broken may cause some other problems that may lead to more severe situations. It is for this reason that no broken wires are allowed for the hoist cable. If broken wires are detected, the cable shall be removed and replaced as soon as possible. If it is required to operate with broken wires, the operator should pay extra attention to the hoist operation

with broken protruding wires from the cable.

(8) Bird Caging:

Bird caging results from torsional imbalance that occurs because of abuse such as sudden stops, the rope being worked through tight sheaves, or a hook bearing that is not rotating. Any evidence of bird caging requires immediate replacement of the load cable.

(9) Breaking in a New Wire Rope:

In the event a wire rope needs to be replaced, the new rope requires careful installation and close adherence to all the appropriate procedures previously noted. After a new cable has been installed, the hoist should be run through a cycle of operation with no load at slow speed. During this trial operation, a very close watch should be kept of all working parts to make sure the cable runs freely, the levelwind is adjusted properly, and the tensioning system is operating as it should. One may witness some tendency for the cable to twist, which may result in a smile(s) developing. This may not be a problem, since the cable is not broken in yet and is still inflexible. If no other problems are noted, the next step is to run several cycles under a light load at a reduced speed. This procedure allows the component parts of the new rope to make a gradual adjustment to the actual operating conditions of the hoist.

(10) After Each Use in Salt Water Environments:

Cable will be reeled out and washed with fresh water. Cable may be reeled out into a 55-gallon container filled with fresh water, and dried by passing cable through

a cleaning cloth, Item 102, Appendix D, when reeling in.

WARNING

Injury to personnel and damage to equipment will occur if lubricant is applied to cable or hoist. Tension roller system works on basis of friction. If oil or grease gets on cable, clean it with a cleaning cloth and fresh water. Solvents may be detrimental to tension roller, so a cable cleaned in this manner must be dry before reeling in.

NOTE

Cloth should be held firmly to impart a steady, reasonably heavy tension load to cable.

- (11) During final rewind of hoist cable, a cleaning cloth, Item 102, Appendix D, should be held firmly around cable. Cloth will help remove foreign particles (grit, dirt, etc.) and, in event a broken wire was overlooked during visual inspection, wire end will snag on cloth, preventing break from being overlooked.

1-7-59 EVERY 60 MONTHS.

PARAGRAPH BREAKDOWN

	PAGE
1-7-59.1 Inspect Fire Extinguishing System Agent Container	1-7-59-1

INITIAL SETUP

References

- Manufacturer Maintenance Manual
-

1-7-59.1 Inspect Fire Extinguishing System Agent Container.

Perform hydrostatic test of fire extinguishing system agent container in accordance with manufacturer’s maintenance manual.

1-7-60 EVERY 500 HOURS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-60.1	Pitch Control Rod Locknut Torque Check	1-7-60-2	1-7-60.4	Pilot's/Copilot's Seat Trunnion Torque Stripe Inspection	1-7-60-12
1-7-60.2	Tail Gear Box Output Bevel Gear Ultrasonic Inspection	1-7-60-2	1-7-60.5	Adjustable Control Rod Inspection	1-7-60-15
1-7-60.3	ESSS External 230 Gallon Fuel Tank Stabilizer Fin Inspection	1-7-60-12			

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Calibration Ring, EWR203026 TECH 5047
- Feeler Gage, 0.010-inch
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Ultrasonic Inspection Tool, Olympus HB 2344 (or equivalent, as approved by Sikorsky) (for Tail Gear Box Output Bevel Gear)

Materials/Parts

- Adhesive, Item 44B, Appendix D
- Brush
- Cloth, Cleaning, Item 102, Appendix D
- Grease Pencil
- Ink Marker
- Lockwire, Item 197, Appendix D

- Lubricating Oil, Item 205, Appendix D (Alternate)
- Lubricating Oil, Item 206, Appendix D (Preferred)
- Q-Tip
- Tag

References

- Appendix D
- Nondestructive Testing/Inspection Technique, No. 5047 (Latest Revision)
- PARA 2-4-44
- PARA 5-4-20
- PARA 6-4-52
- PARA 11-4-63
- PARA 11-4-64
- PARA 11-4-65
- PARA 11-4-66
- PARA 11-4-73
- PARA 11-4-75

- PARA 11-4-76
- PARA 11-4-80
- PARA 11-4-81
- PARA 11-4-82
- PARA 11-4-84

- PARA 11-4-112
- PARA 16-4-54
- PARA 16-4-62
- PARA 16-4-71

1-7-60.1 Pitch Control Rod Locknut Torque Check.

- a. Remove lockwire from upper and lower locknuts.

NOTE

To prevent improper torque of upper locknut on pitch control rod, make sure torque wrench is not in contact with lock key when torquing.

- b. Check torque on upper and lower locknuts. **TORQUE MUST NOT BE LESS THAN 650 INCH-POUNDS.** Perform the following:
 - (1) If torque is equal to or greater than 650 inch-pounds, **TORQUE LOCKNUTS TO 700 - 750 INCH-POUNDS** (Figure 1-7-60-1, Detail A). Torque check is complete.
 - (2) If torque is less than 650 inch-pounds, inspect upper and/or lower rod end, link, and locknut(s) (PARA 5-4-20).
- c. Using lockwire, Item 197, Appendix D, lockwire lower locknut to link, and then back to lower locknut (double-safety).
- d. Using lockwire, Item 197, Appendix D, lockwire upper locknut to lock key, and then back to opposite side of upper locknut (double-safety).

1-7-60.2 Tail Gear Box Output Bevel Gear Ultrasonic Inspection.

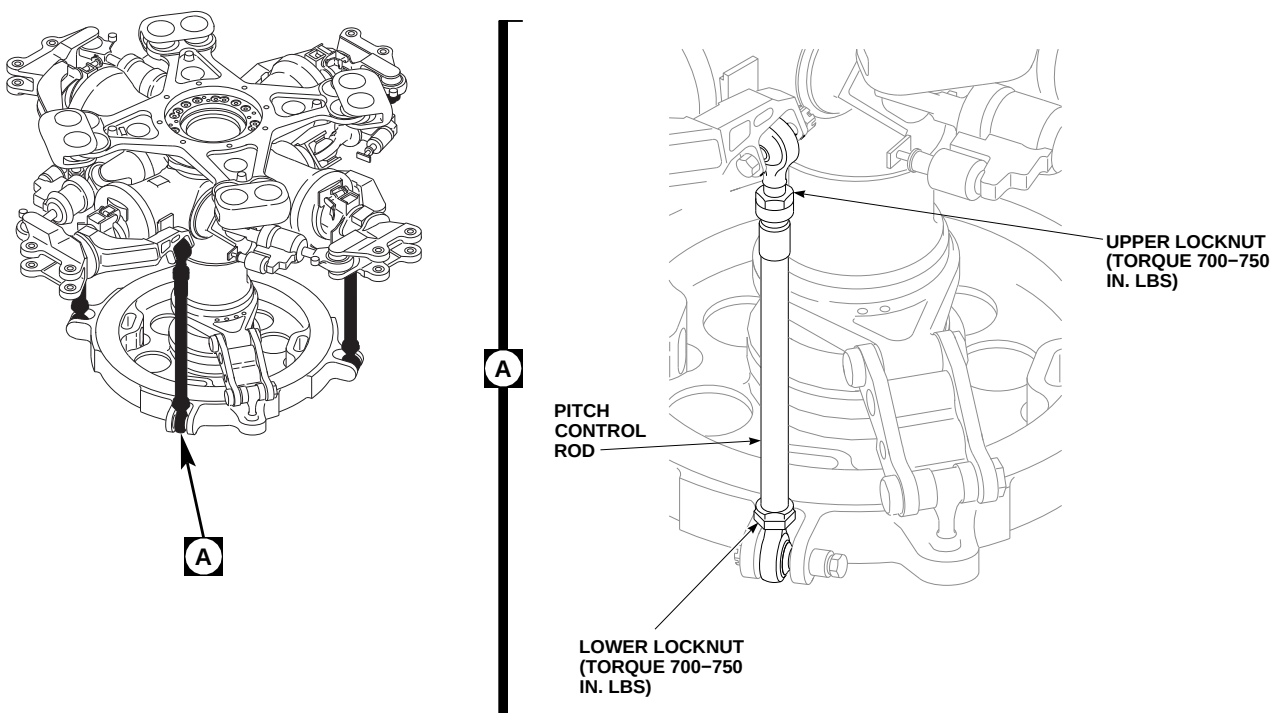
1-7-60.2.1 Prepare to Inspect.

NOTE

- Only personnel certified as Level II (or equivalent) ultrasonic inspectors (with Sikorsky training specific to this procedure), are qualified to perform this inspection.
- If necessary, Nondestructive Testing/Inspection Technique, No. 5047 (latest revision), may be referenced for further information.
- For technical questions regarding this procedure, contact:

Sikorsky Field Service Representative
Attention: Nondestructive Inspection
(NDI) Group

- Tail rotor servo and pitch control shaft are removed as an assembly. Unless servo or shaft is being replaced, it is not necessary to separate shaft from servo after removal.
- a. Remove tail rotor servo and pitch control shaft (PARA 11-4-113).



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FIGURE 1-7-60-1. PITCH CONTROL ROD LOCKNUT TORQUE CHECK

1-7-60.2.2 Clean.

NOTE

Refrain from using solvents or cleaners that could contaminate tail gear box.

- Thoroughly clean flange of tail gear box and inboard bore of output bevel gear with cleaning cloth, Item 102, Appendix D.
- Count number of cleaning cloths used to make sure they are all removed upon completion.
- To avoid pushing dirt into tail gear box, push cloths in from inboard bore of bevel gear to outboard bore.
- Verify all cleaning clothes are removed from tail gear box.
- Do not touch cleaned surfaces.

1-7-60.2.3 Calibrate.

- Calibrate ultrasonic inspection transducer and testing instrument as follows:

NOTE

If multiple output bevel gears are to be inspected, calibration of inspection tool and testing instrument must be checked prior to each inspection.

- Frequency: 10 MHz
- Range: 1-inch
- Dampening: 50 Ohms
- Velocity: 0.1254-inch/microsecond
- Delay/Zero: Exit point at "0" graticule (0.266-inch) or 3.998 μ sec
- Reject: Off
- Mode: Pulse-Echo
- Sensitivity Level: Set signal amplitude from calibration notch to 80% full screen height (FSH), using calibration ring (Figure 1-7-60-2). (Add 6 dB when scanning).



**CALIBRATING SENSITIVITY LEVEL
SCREEN VIEW**

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FIGURE 1-7-60-2. CALIBRATING SENSITIVITY LEVEL

1-7-60.2.4 Check/Set Ultrasonic Inspection Tool.

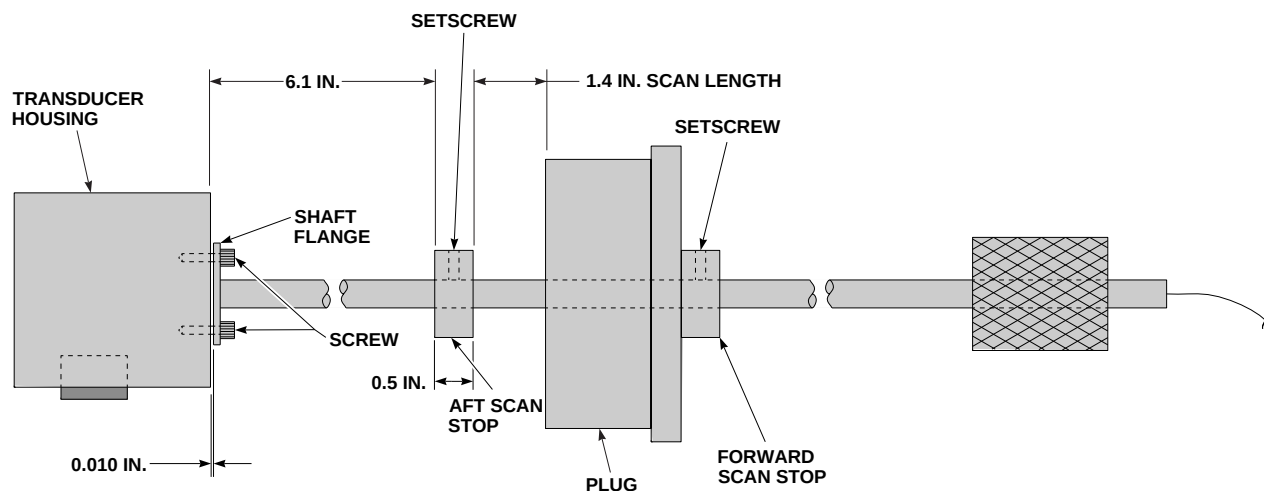
- a. Check inspection tool for proper scan position and length (Figure 1-7-60-3). If necessary, set tool as follows:
 - (1) Loosen setscrew in aft scan stop. Position front of aft scan stop 6.1-inches from transducer housing. Secure stop by tightening setscrew.
 - (2) Position plug 1.4-inches from rear of aft scan stop and firmly hold in place.
 - (3) While firmly holding plug in place, loosen setscrew in forward scan stop and position against plug. Secure stop by tightening setscrew.
 - (4) Adjust transducer housing/shaft flange clearance as follows:

- (a) Loosen screws securing shaft flange $\frac{3}{8}$ -inch.

NOTE

Apply adhesive sparingly. Over application could cause adhesive to flow into other areas of inspection tool. Do not apply more than one drop to each screw.

- (b) Apply adhesive, Item 44B, Appendix D, to threads of screws.
- (c) Insert 0.010-inch feeler gage between transducer housing and shaft flange, and position housing against feeler gage.
- (d) While holding feeler gage between transducer housing shaft flange, tighten screws to set clearance.



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FIGURE 1-7-60-3. CHECK/SET ULTRASONIC INSPECTION TOOL

- (e) Remove feeler gage. Allow 1 minute for adhesive to cure.



1-7-60.2.5 Inspect/Scan.

NOTE

Lubricating oil, Item 206, Appendix D, is preferred for use on output bevel gear. If lubricating oil, Item 206, Appendix D, is not available, lubricating oil, Item 205, Appendix D, may be used as an alternate.

- Using suitable brush, lubricate outboard bore of output bevel gear with lubricating oil, Item 206, Appendix D.
- Inspect all surfaces of inspection tool to make sure it is clean, and that all components and hardware of tool are secure.

Damage to equipment may result if inspection tool is not carefully inserted into output bevel gear. Transducer wedge on tool must be tipped forward to allow it to be inserted into bore.

- Pull cable at rear of inspection tool to tip transducer wedge and insert tool from inboard bore of tail gear box, into inboard bore of output bevel gear. Locate transducer of tool in spline area of gear (Figure 1-7-60-4, Sheet 1, Section A-A).
- Rotate inspection tool 360° to spread lubricating oil. Make sure there is adequate coverage of lubricating oil.

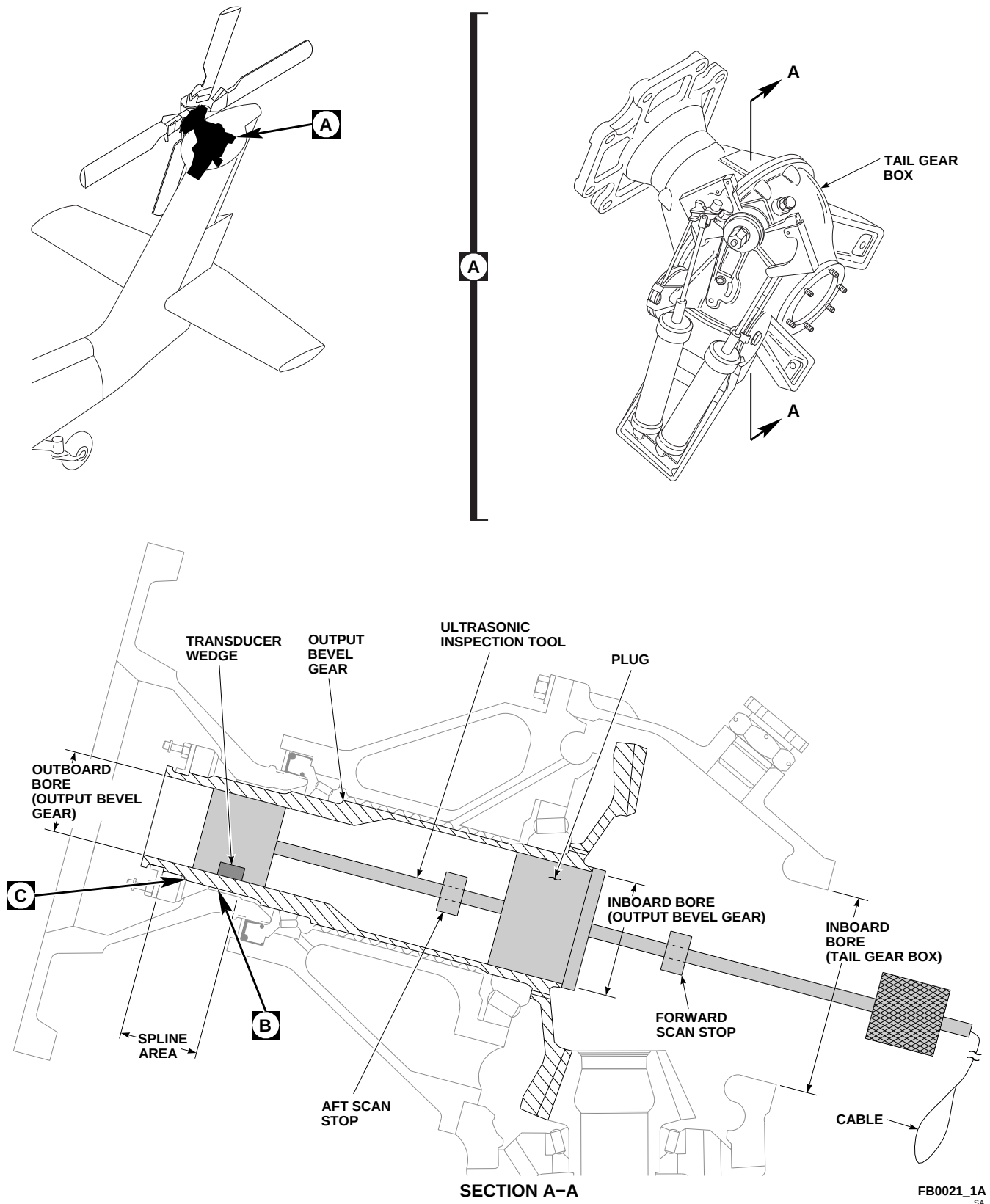
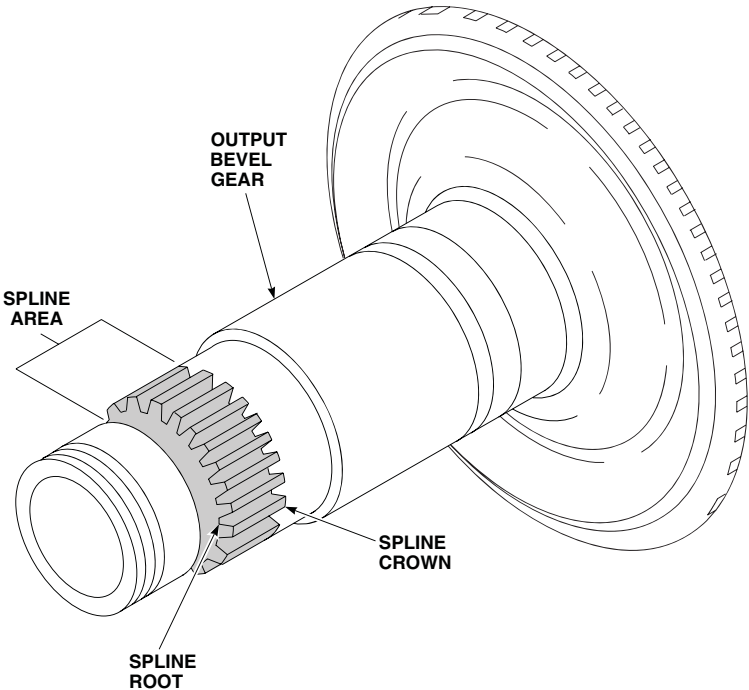
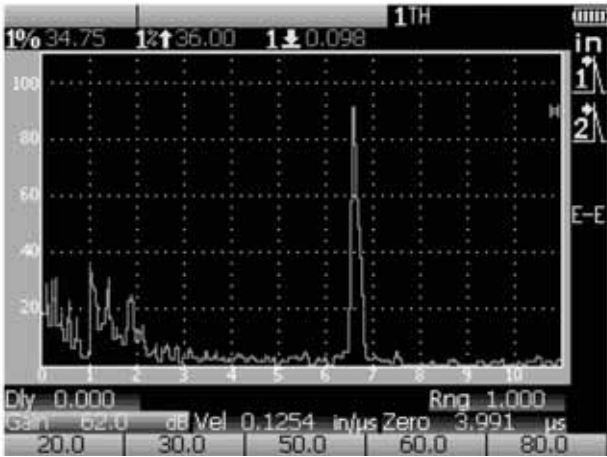


FIGURE 1-7-60-4. TAIL GEAR BOX OUTPUT BEVEL GEAR ULTRASONIC INSPECTION (SHEET 1 OF 3)

B



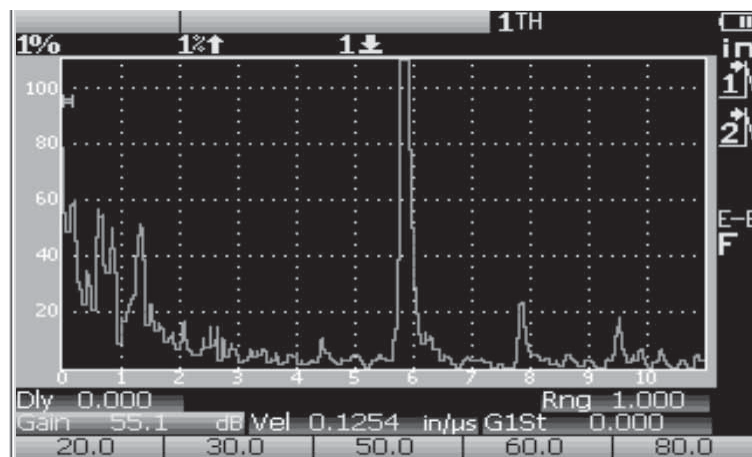
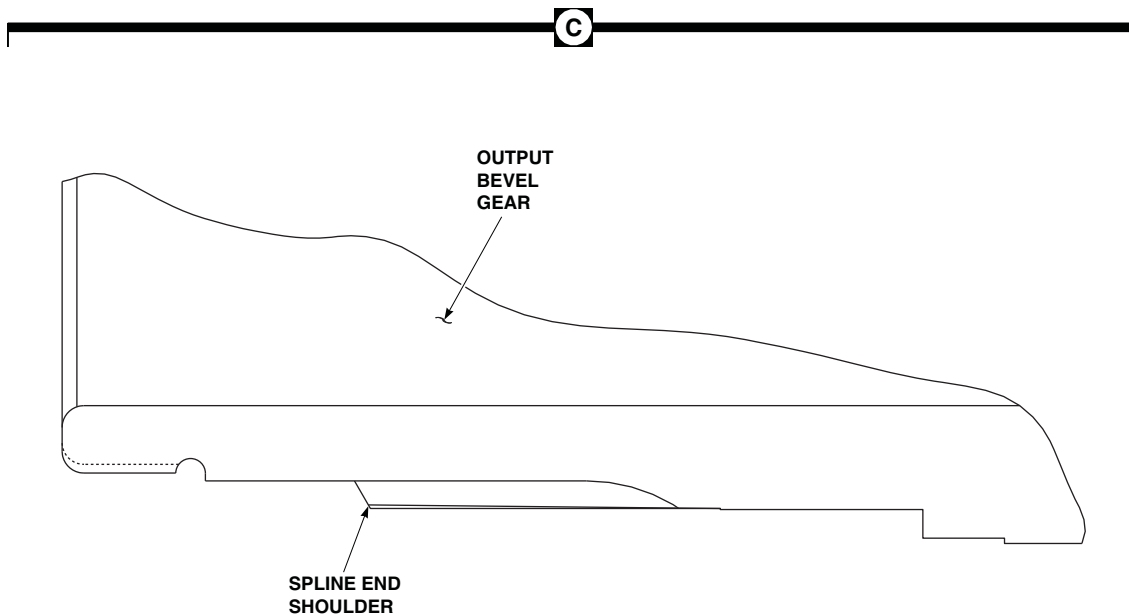
SPLINE CROWN ANOMALY SIGNAL
SCREEN VIEW



SPLINE ROOT ANOMALY SIGNAL
SCREEN VIEW

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FIGURE 1-7-60-4. TAIL GEAR BOX OUTPUT BEVEL GEAR ULTRASONIC INSPECTION
(SHEET 2 OF 3)



**SPLINE END SHOULDER SIGNAL
SCREEN VIEW**

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SA

**FIGURE 1-7-60-4. TAIL GEAR BOX OUTPUT BEVEL GEAR ULTRASONIC INSPECTION
(SHEET 3 OF 3)**

- e. Position plug in inboard bore of output bevel gear, lining up keyway in plug with keyway in bore. Seat plug until secure in bore. Hold plug in place during scan.

NOTE

Make sure inspection tool is adequately lubricated though out the scans. A dry bore is usually indicated by increased tool friction, reduction in amplitude of front spline signal, and/or baseline noise.

- f. If a dry bore is experienced throughout the scans, perform the following:
 - (1) Remove inspection tool from output bevel gear/tail gear box.
 - (2) Repeat step a. through step e.
 - (3) Re-scan area.
- g. If forward or aft scan stop loosens during a scan, perform the following:
 - (1) Remove inspection tool from output bevel gear/tail gear box.
 - (2) Set inspection tool to proper scan position and length (PARA 1-7-60.2.4).
 - (3) Repeat step a. through step e.
 - (4) Re-scan area.

NOTE

There are two scan methods for inspecting output bevel gear: axial and circumferential. Both are to be performed prior to investigating/evaluating any suspect signal(s).

- h. Before scanning, review PARA 1-7-60.2.6 for signal ranges and their required action, if any.

NOTE

- Use grease pencil to mark location of any suspect signal(s) identified during

scan. Index mark on knob indicates orientation of transducer wedge. Marks are to be applied to outboard end of output bevel gear.

- Signal from spline crown anomaly will be further out in time than one from a spline root anomaly. Screen views in Figure 1-7-60-4, Sheet 2, Detail B, show spline crown anomaly signal coming in after 7, and spline root anomaly signal coming in after 6.
- Screen views in Figure 1-7-60-4, Sheet 2, Detail B, show examples of signals from a simulated crack (electrical discharge machined (EDM) notches).
- Screen view in Figure 1-7-60-4, Sheet 3, Detail C, shows the signal response from the spline end. Note the two low amplitude signals from the spline end that follow the main spline end signal.

- i. Perform axial scan as follows:
 - (1) Position transducer at inboard of spline area, by gently pulling tool backward, until stop contacts plug (Figure 1-7-60-4, Sheet 1, Section A-A).
 - (2) Begin scanning toward outboard of spline area, by gently pushing inspection tool forward, until signal from the spline end shoulder is visible and stop contacts plug.
 - (3) Rotate inspection tool clockwise approximately $\frac{1}{16}$ -inch. Scan toward inboard of spline area, by gently pulling tool backward, until stop contacts plug.
 - (4) Rotate inspection tool clockwise approximately $\frac{1}{16}$ -inch.
 - (5) Repeat step i. (2) through step i. (4) until entire circumference of spline area has been scanned.

- j. Perform circumferential scan as follows:

- (1) While inspection tool is still in output bevel gear, perform circumferential scan

by pulling inspection tool backwards to aft scan stop.

- (2) Rotate tool 360° and push tool outboard approximately $\frac{1}{16}$ -inch.
- (3) Repeat circumferential scan in increments of $\frac{1}{16}$ -inch until tool reaches forward scan stop.

NOTE

Sensitivity level of inspection tool must be checked after scans are complete.

- k. Check sensitivity level of inspection tool with calibration ring as follows:
 - (1) Remove tool from tail gear box.
 - (2) Inspect all components and hardware of tool for security.
 - (3) Reduce instrument sensitivity to initial calibration level (subtract 6 dB), insert tool into calibration ring.
 - (4) If calibration is off by more than -5%, recalibrate tool (PARA 1-7-60.2.3, step a. (8)). Re-scan output bevel gear, repeat axial scan (step i.) and circumferential scan (step j.).
- l. If scan revealed no suspect signal(s), output bevel gear has passed inspection, proceed to PARA 1-7-60.2.7. If scan revealed suspect signal(s), signal(s) must be investigated and evaluated, proceed to PARA 1-7-60.2.6.

1-7-60.2.6 Investigate/Evaluate Suspect Signals.

- a. If scan revealed suspect signal(s), investigate and evaluate as follows:

NOTE

Ultrasonic testing personnel must use experience in interpreting signal, and visual inspection, when investigating and evaluating signal.

- (1) **Signal:** 5 - 10% FSH.
Action: None (Not a suspect signal. This is the baseline material signal in shaft).
- (2) **Signal:** 11 - 19% FSH.
Action: None (Not a suspect signal).
- (3) **Signal:** 20 - 24% FSH.
Action: Re-scan of suspect area required, perform the following:

NOTE

Lubricating oil, Item 206, Appendix D, is preferred for use on output bevel gear. If lubricating oil, Item 206, Appendix D, is not available, lubricating oil, Item 205, Appendix D, may be used as an alternate.

- (a) Using suitable brush, reapply lubricating oil, Item 206, Appendix D, to outboard bore of output bevel gear.
- (b) Re-scan suspect area (PARA 1-7-60.2.5).
- (c) If a signal 20 - 24% FSH results again, signal has passed inspection. If a signal more than 24% FSH results, signal has failed inspection.

NOTE

- Zone 1: Areas within the entire circumference of splines (Figure 1-7-60-5).
- Zone 2: Tips of all splines.
- The **Action** described in suspect signal (4) is also applicable for suspect signals (5) (i.e., signals 35% FSH or greater in Zone 2).
- (4) **Signal:** 25% FSH or greater in Zone 1.
Action: Re-scan and visually inspect as follows:

NOTE

- The following scan is to be performed at calibration sensitivity (subtract 6 dB from scan sensitivity).
- Tail gear box could possibly fail ultrasonic inspection based on the re-scan and visual inspection. A tail gear box failing inspection requires a torque check of the bolts securing seal housing to large nut. Due to this conditional measure, torque check must be performed prior to removing inboard retention plate.

- (a) Locally reproduce Tail Gear Box Worksheet (Figure 1-7-60-6).
- (b) Using ink marker, mark one of the sixteen bolts securing seal housing to large nut, as a reference point.

NOTE

Torque sequence for all the following torques is the same. Torque sequence is as follows: 1, 9, 13, 5, 11, 3, 15, 7, 2, 10, 14, 6, 16, 8, 12, and 4.

- (c) **TORQUE BOLTS TO 60 INCH-POUNDS** in the above noted sequence. Complete section 4 on Worksheet, entering P (Pass) or F (Fail) as each bolt is torqued.
- (d) **TORQUE BOLTS TO 100 INCH-POUNDS** in the above noted sequence. Complete section 4 on Worksheet, entering P (Pass) or F (Fail) as each bolt is torqued.
- (e) **TORQUE BOLTS TO 114 INCH-POUNDS** in the above noted sequence. Complete section 4 on Worksheet, entering P (Pass) or F (Fail) as each bolt is torqued.
- (f) **TORQUE BOLTS TO 120 INCH-POUNDS** in the above noted sequence. Complete section 4 on

Worksheet, entering P (Pass) or F (Fail) as each bolt is torqued.

- (g) Remove inboard retention plate (PARA 6-4-54).
- (h) Calibrate inspection tool to calibration sensitivity (PARA 1-7-60.2.3).

NOTE

- It is possible that the source of the signal (e.g., grease, dirt, etc.) may have been removed when inboard retention plate was removed.
- In the below step, substitute "35%" for "25%" when investigating/evaluating suspect signals (5) (i.e., signals 35% FSH or greater in Zone 2).
 - (i) Perform an axial re-scan of suspect area to try and obtain suspect signal (PARA 1-7-60.2.6). If suspect signal cannot be obtained or if signal is less than 25% FSH, signal has passed inspection. If suspect signal is detected, do not move inspection tool and maintain signal on screen.

NOTE

Lubricating oil, Item 206, Appendix D, is preferred for use on output bevel gear. If lubricating oil, Item 206, Appendix D, is not available, lubricating oil, Item 205, Appendix D, may be used as an alternate.

- (j) Soak Q-tip in lubricating oil, Item 206, Appendix D.
- (k) Lightly rub suspect spline(s) with oil-soaked Q-tip while viewing signal on screen. Once signal drops, the general location (source) of the signal has been identified.
- (l) Visually inspect area for the source of the signal. If the source of the signal can be removed (grease, dirt,

etc.) and the signal drops to within acceptable limits, signal has passed inspection. If the cause of the signal cannot be removed (e.g., crack, etc.), signal has failed inspection.

- (5) **Signal:** 35% FSH or greater in Zone 2.
Action: Same as PARA 1-7-60.2.6, step a. (4).

- b. If output bevel gear has passed all investigated/evaluated signals, proceed to PARA 1-7-60.2.7. If output bevel gear has failed any investigated/evaluated signal, proceed to PARA 1-7-60.2.8.

1-7-60.2.7 Complete (Passed Inspection).

NOTE

Refrain from using solvents or cleaners that could contaminate tail gear box.

- a. Clean flange of tail gear box and bore of output bevel gear (PARA 1-7-60.2.2).

NOTE

If inboard retention plate was removed for re-scan/visual inspection, re-verification is not required after retention plate installation.

- b. If removed for visual inspection, install inboard retention plate (PARA 6-4-54), and discard Tail Gear Box Worksheet.
 c. Install tail rotor servo and pitch control shaft (PARA 11-4-113).

1-7-60.2.8 Complete (Failed Inspection).

- a. If output bevel gear has failed inspection, tail gear box must be rejected. Perform the following:
- (1) Complete sections 1 through 3 on Tail Gear Box Worksheet (Figure 1-7-60-6).

- (2) Install inboard retention plate (PARA 6-4-54).
 (3) Install tail rotor servo and pitch control shaft (PARA 11-4-113).
 (4) Remove tail gear box (PARA 6-4-52).

NOTE

Section 4 of Worksheet was completed prior to removal of inboard retention plate.

- (5) Complete section 5 on Worksheet.

NOTE

The split cones, inboard retention plate, nut, and hardware must be returned with tail gear box for investigation.

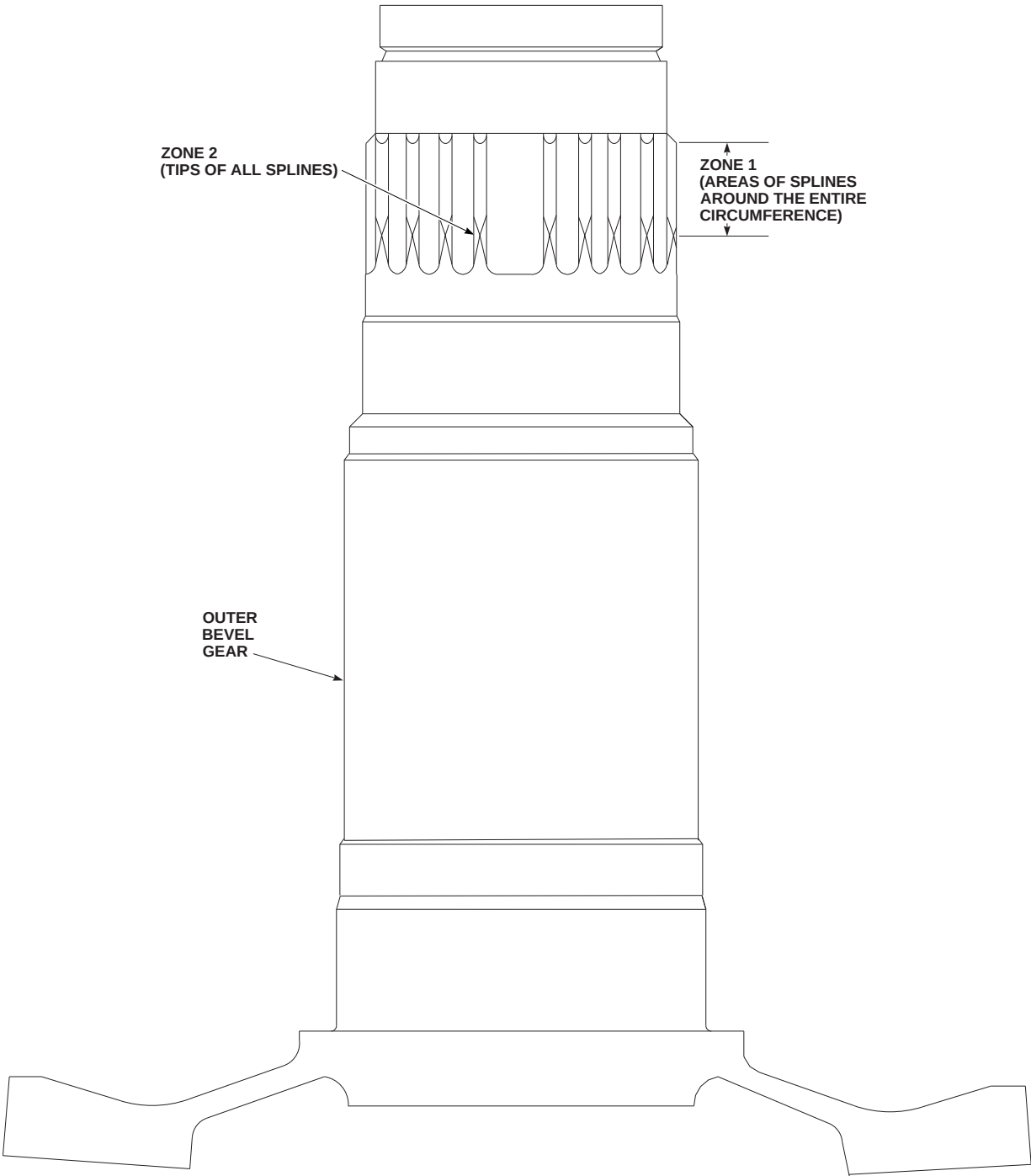
- (6) Send tail gear box, with associated components, and Worksheet to Sikorsky.
 (7) Contact Sikorsky Safety and Engineering informing them of tail gear box return.

1-7-60.3 ESSS External 230 Gallon Fuel Tank Stabilizer Fin Inspection.

- a. Remove stabilizer fin (PARA 16-4-62).
 b. Inspect and repair stabilizer fin and mounting flange area (PARA 16-4-71).
 c. If repairs were performed or if no damage was found, install stabilizer fin (PARA 16-4-62).
 d. If damage exceeds limits, replace ESSS external fuel tank (PARA 16-4-54).

1-7-60.4 Pilot's/Copilot's Seat Trunnion Torque Stripe Inspection.

Inspect trunnion torque stripe for slippage. **NONE ALLOWED.** If slippage has occurred, replace threaded insert (PARA 2-4-44).



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FIGURE 1-7-60-5. OUTPUT BEVEL GEAR - ZONE 1 AND ZONE 2

TAIL GEAR BOX WORKSHEET

SECTION 1: TAIL GEAR BOX

P / N: 70358-06600-

S / N: _____

TIME SINCE NEW: _____

TIME SINCE OVERHAUL: _____

SECTION 2: INBOARD RETENTION PLATE

P / N: 70358-06612-

S / N: _____

TIME SINCE NEW: _____

TIME SINCE OVERHAUL: _____

SECTION 3: OUTPUT BEVEL GEAR

NOTE: ONLY FOR MODELS WHERE THIS IS A PART RECORDED ON TAIL GEAR BOX LOG CARD

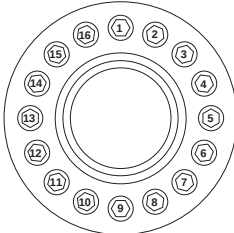
P / N: 70358-06620-

S / N: _____

TIME SINCE NEW: _____

TIME SINCE OVERHAUL: _____

SECTION 4: TORQUE CHECK

			BOLT NUMBER (IN TORQUEING ORDER) P (PASS) / F (FAIL)																
			1	9	13	5	11	3	15	7	2	10	14	6	16	8	12	4	
	TORQUE (INCH-POUNDS)	60																	
		100																	
		114																	
120																			

SECTION 5: REJECT AND RETURN

TAG REJECTED TAIL GEAR BOX AS FOLLOWS:

THIS TGB HAS BEEN REJECTED AND BEING RETURNED FOR OUTPUT SHAFT
SPLINE SUSPECT CRACK INDICATION DURING PHASE UT INSPECTION.

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**FIGURE 1-7-60-6. TAIL GEAR BOX WORKSHEET
(LOCALLY REPRODUCE)**

1-7-60.5 Adjustable Control Rod Inspection.

NOTE

Inspections below apply to the following adjustable control rods:

- Torque shaft to boost servo control rods (PARA 11-4-63 (Collective Input), PARA 11-4-64 (Pitch/Trim Input), PARA 11-4-65 (Lateral Input), and PARA 11-4-66 (Yaw Input)).
 - Boost servo to mixer control rods (PARA 11-4-73 (Collective), PARA 11-4-75 (Lateral), and PARA 11-4-76 (Yaw)).
 - Mixer to primary servo control rods (PARA 11-4-80 (Forward), PARA 11-4-81 (Aft), and PARA 11-4-82 (Lateral)).
 - Yaw lever to forward quadrant control rod (PARA 11-4-84).
 - Tail rotor servo to tail rotor quadrant pushrod (PARA 11-4-112).
- a. Perform the following thread engagement checks on adjustable control rods:

- (1) Adjustable end of control rods are to be inspected for minimum thread engagement. Check that adjustable rod end is threaded beyond inspection hole such that an 0.20-inch diameter wire cannot be inserted through inspection hole (Figure 1-7-60-7).
- (2) If applicable, check that witness hole in threaded rod end cannot be seen outside rod end jamnut (Figure 1-7-60-7, Detail A).
- (3) If adjustable control rod(s) do not pass either of the thread engagement checks, adjust control rod per applicable maintenance procedure.

NOTE

Do not turn tube assembly to determine looseness of jamnut. Only apply finger pressure to the jamnut itself.

- b. Check both jamnuts on adjustable control rods for looseness, using only finger pressure. **NONE ALLOWED.** If a loose jamnut is found, replace adjustable control rod per applicable maintenance procedure.

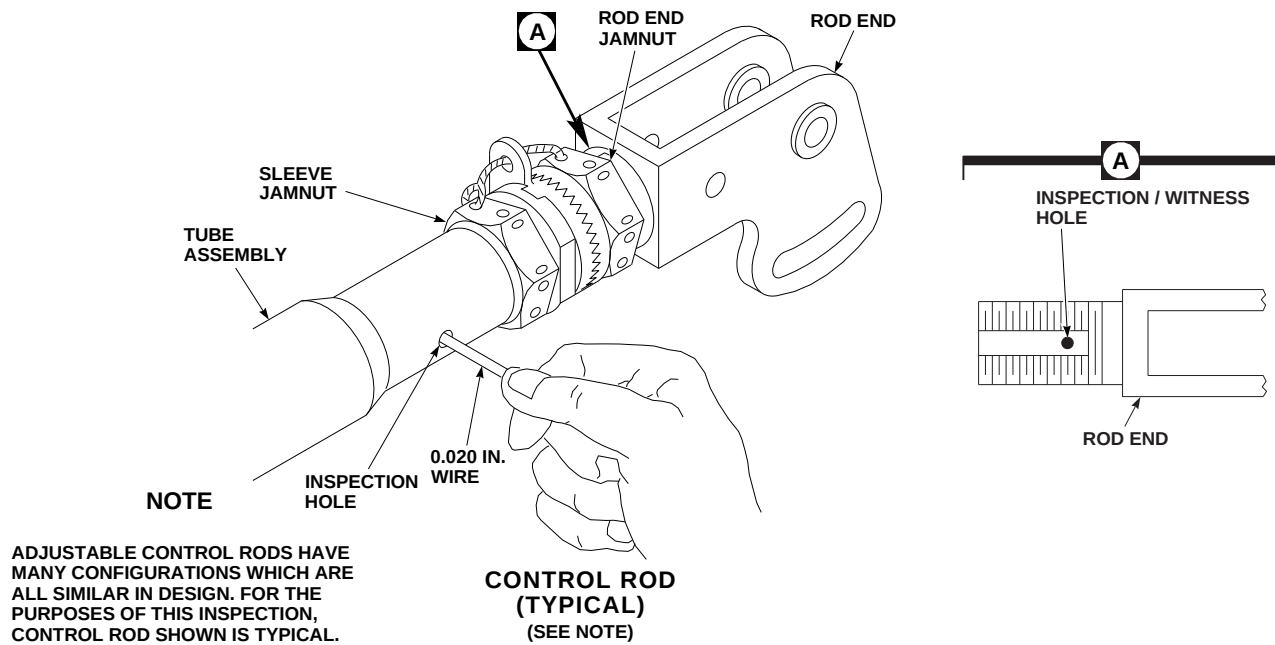


FIGURE 1-7-60-7. THREAD ENGAGEMENT AND LOCKWIRE (TYPICAL)

1-7-61 EVERY 120 HOURS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-61.1	Damper Attachment Torque Check	1-7-61-1	1-7-61.2	External Rescue Hoist Fitting Inspection	1-7-61-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Flashlight
- Fluorescent Penetrant Inspection Kit
- Magnifying Glass, 10X
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 500 - 2500 in. lbs

Materials/Parts

- Paint Remover, Item 230, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- MIL-STD-6866
- PARA 2-6-11
- PARA 5-4-13

1-7-61.1 Damper Attachment Torque Check.

- Using nonmetallic scraper, remove sealing compound and cotter pin from damper attachment bolts and nuts.

NOTE

Check torque in a tightening direction only.

- Check torque of inboard and outboard damper attachment nuts. **TORQUE MUST NOT BE LESS THAN 1255 INCH-POUNDS.** Perform the following:
 - If torque is 1255 inch-pounds or greater, **TORQUE NUT(S) TO 1320 - 1635**

INCH-POUNDS (Figure 1-7-61-1, Detail A). Torque check is complete.

- If torque is less than 1255 inch-pounds, perform the following:
 - Disconnect or remove damper as required (PARA 5-4-13), and inspect bolt(s) for wear. **NONE ALLOWED.** If wear is found, replace bolt(s).
 - Connect or install damper as required (PARA 5-4-13).
- Install cotter pin. Seal area around boltheads and nuts with sealing compound, Item 292, Appendix D.

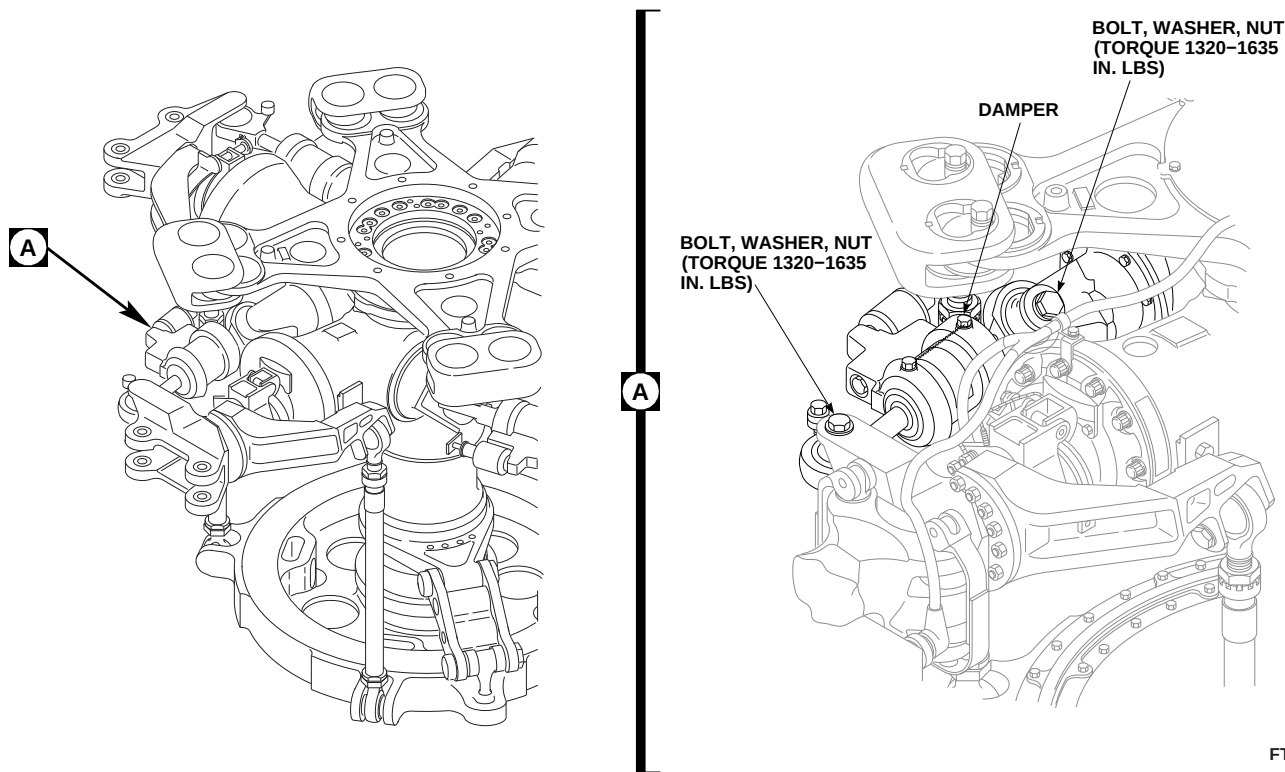
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FIGURE 1-7-61-1. DAMPER ATTACHMENT TORQUE CHECK

1-7-61.2 External Rescue Hoist Fitting Inspection.**NOTE**

Removal of fairing to perform inspection is not required unless a crack is found during inspection.

- a. Using 10X magnifying glass and flashlight, visually inspect entire rescue hoist fitting for cracks, working fasteners, corrosion, and damage. Inspect the top flange of the fitting (outboard of the BL 34.5 beam) for cracks. Disposition as follows:

- (1) If cracks, working fasteners, corrosion, or any other damage is found anywhere other than the top flange of the fitting (outboard of BL 34.5), disposition in accordance with current maintenance and repair procedures (PARA 2-6-11).
- (2) If cracks are found or suspected in the top flange of the fitting (outboard of BL 34.5), perform the following:

- (a) Remove fairing (PARA 14-4-37).
- (b) Using paint remover, Item 230, Appendix D, remove paint from the top and bottom of the flange in a 1-inch area around the crack and any adjacent nutplate.
- (c) Using fluorescent penetrant kit, fluorescent penetrant inspect, Type 1, Method C, Sensitivity Level 3, Grade B, both sides of the top flange for cracks (MIL-STD-6866). **NONE ALLOWED.**
- (d) Note the position and length of any cracks.
- (e) Mark the cracks, if any, on the aircraft in order to determine future growth.
- (f) Note the location of all holes, all hole sizes, all countersunk depths, and the surface finish of all countersunk holes.

- (g) Discontinue use of the rescue hoist.
- (h) Contact Sikorsky Field Service Representative, with the following information:
 - Sketch with location of discrepant crack.

- Location and size of all holes.
- Size and surface finish of all countersinks, and all photos.
- (i) Install fairing (PARA 14-4-37).

1-7-62 SEVERE OR SUDDEN ONSET OF LOW-FREQUENCY MAIN ROTOR VIBRATION OR OUT-OF-TRACK CONDITION.

PARAGRAPH BREAKDOWN

		PAGE
1-7-62.1	Severe or Sudden Onset of Low-Frequency Main Rotor Vibration or Out-of-Track Condition Inspection.	1-7-62-1

INITIAL SETUP

- PARA 5-3-5

References

- Appendix D
 - PARA 1-3-15
 - PARA 1-7-5
 - PARA 1-7-11
 - PARA 5-3-1
 - PARA 5-3-4
- PARA 5-3-6
 - PARA 5-3-7
 - PARA 5-3-10
 - PARA 5-3-11
 - PARA 5-4-5
 - PARA 6-3-1
 - PARA 11-2-3

1-7-62.1 Severe or Sudden Onset of Low-Frequency Main Rotor Vibration or Out-of-Track Condition Inspection.

WARNING

- Severe low frequency (1 per main rotor revolution) vibration, a sudden onset of vibrations, or out-of-track conditions can indicate a serious problem in the main rotor head or blades.
- A "bang", or other unusual noises, or transient helicopter motions may also

occur with the sudden onset of vibrations. Low frequency vibrations can indicate a potential safety problem in the main rotor system including blades, rotors, transmissions, and flight controls.

- Ground and flight operations shall be discontinued until the cause of the vibration has been positively identified and corrective action has been taken.
- a. Perform inspection of main rotor blades as follows:

WARNING

Injury to personnel will result if blade is lifted without proper support equipment. Do not attempt to raise blade by hand or with any other part of body. Use blade clamp assembly, hoist (or maintenance crane), hoisting sling, and adapter as necessary.

NOTE

Blade must be raised from droop stop during this inspection.

- (1) Raise blade off droop stop using blade clamp assembly, hoist (or maintenance crane), hoisting sling, and adapter as necessary.
- (2) Inspect upper and lower surfaces of blade cuff for cracks (PARA 5-3-11).
- (3) Inspect overall condition of blade (PARA 5-3-11).
- (4) Inspect skin-to-core and skin-to-spar bonds for integrity (PARA 5-3-11).
- (5) Inspect tip caps for security and condition (PARA 5-3-11).
- (6) Inspect trim tabs for security and condition (PARA 5-3-11).
- (7) Inspect abrasion strips for presence and security (PARA 5-3-11).
- b. Perform inspection of main rotor head as follows:
 - (1) Inspect pitch control rod ends for excessive wear and security to link (PARA 1-7-11 and PARA 5-3-5).
 - (2) Inspect elastomeric bearings for delaminations (PARA 5-3-1).
 - (3) Inspect bifilar weights, washers, and arms for security and condition (PARA 1-7-11).
- (4) Inspect bifilar washers, pins, and bushings for wear (PARA 5-3-10).
- (5) Inspect dampers for condition and security (PARA 1-7-11 and PARA 5-3-4).
- (6) Inspect inboard and outboard damper rod ends for excessive wear and binding (PARA 1-7-11 and PARA 5-3-4).
- (7) Inspect dampers for proper servicing (PARA 1-3-15).
- (8) Inspect swashplate uniball for excessive wear (PARA 5-3-6).
- (9) Inspect rotating scissors for security and looseness (PARA 5-3-6 and PARA 5-3-7).
- (10) Inspect antilap assemblies for security and proper function as follows:
 - (a) Check antilap assembly for damage.
 - (b) Visually inspect for interference or contact between antilap cam and antilap stop.
 - (c) Visually inspect for contact of droop stop cam and spindle horn attachment bolts.
- (11) Perform main rotor head torque check on main rotor head shaft extension nut bolts, lockring bolts, main rotor shaft nut bolts, upper and lower pressure plate bolts. Check lower pressure plate for unequal gap (PARA 1-7-5).
- (12) Inspect spindles for fractured retaining rod and delaminated elastomeric bearings (PARA 5-4-5).
- c. Inspect main transmission mounting bolts for looseness and condition (PARA 6-3-1).
- d. Perform flight control noise checkout operational troubleshooting procedure (PARA 11-2-3).

1-7-63 EVERY 120 HOURS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
1-7-63.1	Damper Attachment Torque Check	1-7-63-1	1-7-63.2	External Rescue Hoist Fitting Inspection	1-7-63-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Flashlight
- Fluorescent Penetrant Inspection Kit
- Magnifying Glass, 10X
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 500 - 2500 in. lbs

Materials/Parts

- Paint Remover, Item 230, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- MIL-STD-6866
- PARA 2-6-11
- PARA 5-4-12
- **AGFI** ▶ TMP AAP 7210.015-7 **◀**

1-7-63.1 Damper Attachment Torque Check.

NOTE

AGFI ▶ Unless specified in the Sikorsky maintenance procedures, the hour/calendar requirement is per TMP AAP 7210.015-7. **◀**

- Using nonmetallic scraper, remove sealing compound and cotter pin from damper attachment bolts and nuts.

NOTE

Check torque in a tightening direction only.

- Check torque of inboard and outboard damper attachment nuts. **TORQUE MUST NOT BE**

LESS THAN 1255 INCH-POUNDS. Perform the following:

- If torque is 1255 inch-pounds or greater, **TORQUE NUT(S) TO 1320 - 1635 INCH-POUNDS** (Figure 1-7-63-1, Detail A). Torque check is complete.
- If torque is less than 1255 inch-pounds, perform the following:
 - Disconnect or remove damper as required (PARA 5-4-12), and inspect bolt(s) for wear. **NONE ALLOWED.** If wear is found, replace bolt(s).
 - Connect or install damper as required (PARA 5-4-12).

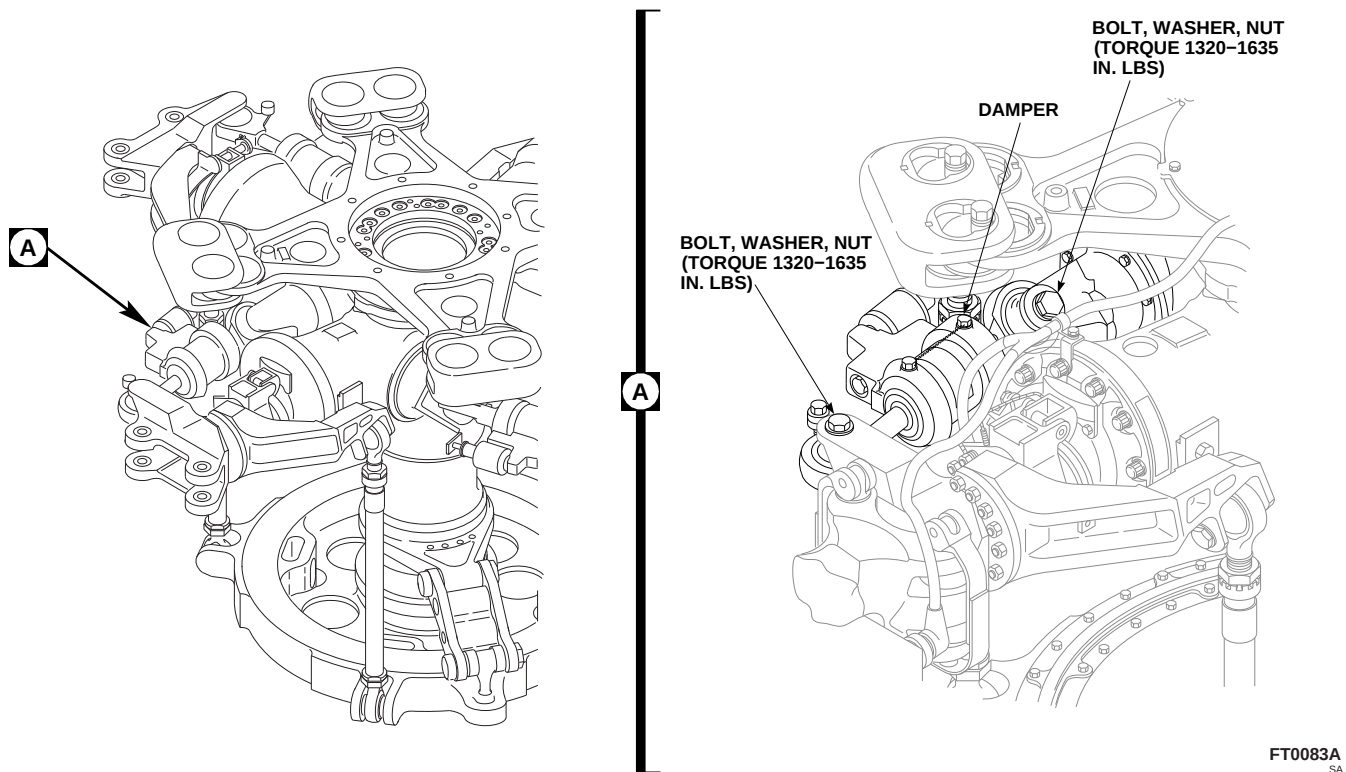


FIGURE 1-7-63-1. DAMPER ATTACHMENT TORQUE CHECK

- c. Install cotter pin. Seal area around boltheads and nuts with sealing compound, Item 292, Appendix D.

1-7-63.2 External Rescue Hoist Fitting Inspection.

NOTE

- **AGFI** Unless specified in the Sikorsky maintenance procedures, the hour/calendar requirement is per TMP AAP 7210.015-7.■
 - Removal of fairing to perform inspection is not required, unless a crack is detected or suspected during inspection.
- a. Using 10X magnifying glass and flashlight, visually inspect entire rescue hoist fitting for crack(s), working fasteners, corrosion, and damage. Inspect top flange of fitting (outboard of BL 34.5 beam) for cracks. Evaluate as follows:
- (1) If crack(s), working fasteners, corrosion, or any other damage is found anywhere other than top flange of fitting (outboard of BL 34.5), evaluate per current maintenance and repair procedures (PARA 2-6-11).
 - (2) If crack(s) are found or suspected in top flange of fitting (outboard of BL 34.5), proceed to step b.
- b. If crack(s) are found or suspected in top flange of fitting (outboard of BL 34.5), perform the following:
- (1) Remove fairing (PARA 14-4-16).
 - (2) Using paint remover, Item 230, Appendix D, remove paint from top and bottom of flange in a 1-inch area around crack and any adjacent nutplate.
 - (3) Using fluorescent penetrant kit, fluorescent penetrant inspect, Type 1, Method C, Sensitivity Level 3, Grade B, both sides of top flange for cracks (ASTM E 1417). **NONE ALLOWED.** If no crack(s) are found, proceed to step b.(4). If crack(s) are found, perform the following:

- (a) Note the position and length of crack(s).
- (b) Mark crack(s), on helicopter in order to determine future growth.
- (c) Note location of all holes, all hole sizes, all countersunk depths, and surface finish of all countersunk holes.
- (d) Discontinue use of rescue hoist.
- (e) Contact Sikorsky Field Service Representative, with the following

information:

- Sketch with location of discrepant crack.
- Location and size of all holes.
- Size and surface finish of all countersinks, and all photos.

- (4) Install fairing (PARA 14-4-16).

SECTION 1-8

RETIREMENT SCHEDULE

SECTION OVERVIEW

PARAGRAPH	TITLE
1-8-1	Calendar Retirement
1-8-2	Life Limited Components

1-8-1/(1-8-2 Blank)

1-8-1 CALENDAR RETIREMENT.

1-8-1.1 Retirement Schedule.

NOTE

- Installed life is the period of time an impulse cartridge, explosive cartridge, or cartridge activated device (CAD) is allowed to be used after its hermetically-sealed container is opened; however, the installed life expiration date shall never exceed the shelf life expiration date. Installed life is never added to shelf life.
- The installed life expiration date is computed from the date the hermetically-sealed container is opened

and is always computed to the last day of the month involved.

- If the date the sealed container was opened is not available, the installed life shall be computed from the date of manufacture as determined from the lot number.

The calendar retirement schedule, Table 1-8-1-1, lists units of operating equipment that are to be retired at the period specified.

1-8-1.2 Retirement Schedule Definition.

The operating time or calendar interval specified for removal, condemnation, and disposal of parts.

Table 1-8-1-1. Calendar Retirement Schedule

Part No.	Nomenclature	Installed Life (Mo)	Shelf Life (Yrs)	Notes
ARD863-1	Ejector Rack Impulse Cartridge	15	10	1 through 4
BL9280, MK44 MOD 0	Rescue Hoist Impulse Cartridge	60	10	1 through 4
CCU-44/B	Ejector Rack Impulse Cartridge	36	10	1 through 4
CCU-90/A, 1512AS105, 30903824-1	Fire Extinguis- her Cartridge	36	12	1 through 4
CCU-92/A, FE7590-95, FE7590-168, 103377-3, 103377-107	Cargo Hook Impulse Cartridge	54	12	1 through 4
FH-900-2	Hand Held Fire Extinguisher	60	12	1 through 4

Table 1-8-1-1. Calendar Retirement Schedule (Cont)

Part No.	Nomenclature	Installed Life (Mo)	Shelf Life (Yrs)	Notes
MARK 44 MOD 0	Cable Cutter Cartridge	66	11	1 through 4
MK 2 MOD 1	Ejector Rack Impulse Cartridge	12	21.5	1 through 4
SS30DF07F322000, SS30DN15F206000, SS30DN15F242000, SS30DF07F214000, SS30DN41G276000, SS30DN40G252000, SS30DN40G243000	Fuel System Hoses, Nonmetallic	On Condition	8	
T04/51	Battery, Digital Clock	12	10	
300078-1	Internal Rescue Hoist Cartridge	36	6	1 through 4
3208-0013-3	Pressure Cartridge (Explosive Valve)	28	5	10
70250-35801-101, 70250-35801-102,	Flotation Cylinder and Valve As- sembly	120	5	8, 9
70310-03101-109, 897899-1	Fire Extinguis- her Cartridge (Walter Kidde)	120	10	1 through 5
70400-06660-045, 70400-06660-048	Spring Cylinder	180	N/A	6, 7
931160 M796	Chaff Dispenser Impulse Cartridge	12	11.5	1 through 4

Table 1-8-1-1. Calendar Retirement Schedule (Cont)

Part No.	Nomenclature	Installed Life (Mo)	Shelf Life (Yrs)	Notes
70850-22111-132	Rescue Hoist	24	N/A	11

NOTES

1. Shelf life is the maximum calendar time a cartridge can reach before it must be retired, whether installed or not. Container storage life shall not be exceeded.
2. The computed cartridge replacement date, installation date, and lot number shall be recorded in helicopter logbook for each installed cartridge.
3. Age life retired cartridges shall be returned to the supporting ammunition supply activity in the container used to transport the replacement cartridge.
4. Unserviceable cartridges shall be tagged with the reason for removal, installation date, lot number, helicopter serial number, and Aviation Unit designation.
5. Installed life and shelf life are identical (120 months/10 years).
6. If no aircraft maintenance records can be found indicating date of installation, the installed cylinders shall be assigned the age of the basic airframe since delivery.
7. The computed cylinder replacement and installation date shall be recorded in helicopter logbook for each installed cylinder.
8. Flotation cylinder and valve assembly, 70250-35801-101, has pressure cartridge installed. Flotation cylinder and valve assembly, 70250-35801-102, does not have pressure cartridge installed. Pressure cartridge (explosive valve) has installed and shelf lives as noted.
9. Flotation cylinder and valve assembly installed life time is counted from date of initial cylinder pressurization.
10. Pressure cartridge (explosive valve), 3208-0013-3, installed life time is counted from date of opening the hermetically sealed container. Installed life is 28 months beginning anytime within the 5 year shelf life period (i.e., valve may be on the shelf for a maximum of 5 years and then placed in service for an additional 28 months).
11. Overhaul time for the external-mounted rescue hoist is 24 months, 1,200 hoist operating cycles, or 1,600 hoist lifts starting from December 4, 2013, and then every 24 months, 1,200 hoist operating cycles, or 1,600 hoist lifts thereafter.

1-8-2 LIFE LIMITED COMPONENTS.

Life limited components are identified in TM 1-70-23AW-1 for Sikorsky Quality Assurance Board certified helicopters and TM 1-70-23AW-2 for Federal Aviation Administration certified helicopters.

